AutoLISP / Visual LISP Language Specification

HyperSpec-Style Draft Reference for clautolisp

Codex

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Contents

1 1 Introduction

1.1 Overview

This document is a HyperSpec-style reference specification for the AutoLISP / Visual LISP language family targeted by ‘clautolisp‘.

The short name of this document is **AutoLISP Spec**.

The current document version is **0.8.2**.

Any modification of this document must increment its version number.

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It is intended to be usable as a local reference document, not merely as a planning memo. It therefore combines:

- a chapter structure modeled on the Common Lisp HyperSpec,
- chapter-level normative text,
- dictionary-style entries for defined names and syntax classes,
- explicit source notes for each specification element.

Because Autodesk and Bricsys do not publish a single unified standard, this document distinguishes documented, inferred, under-specified, tested, and implementation-defined material.

1.2 Scope

This specification covers:

- core AutoLISP syntax and evaluation,
- Visual LISP language extensions,
- host-visible runtime types,
- CAD interaction surfaces that are language-defining in practice,
- version and vendor divergence that must be tracked by ‘clautolisp‘.

This specification does not attempt to reproduce full external standards for:

- DXF object schemas,
- COM type libraries,
- .NET, ObjectARX, or BRX APIs,
- every interactive CAD command.

1.3 Conformance Model

‘clautolisp‘ should distinguish at least:

- a core AutoLISP profile,
- an AutoCAD-oriented profile,
- a BricsCAD-oriented profile,
- strict and lax acceptance modes,
- selectable host-environment profiles independent from the native OS.

When an AutoLISP or Visual LISP function family is specified as a wrapper over an underlying host API that is documented elsewhere, the underlying API documentation is part of the normative reference corpus for that family, subject to the calling conventions and restrictions documented by the AutoLISP layer.

This applies in particular to:

- ‘vla-*‘ wrappers over the AutoCAD ActiveX object model,
- ‘vlax-*‘ property, method, variant, safearray, and object-creation functions that expose COM / OLE Automation semantics.

It does **not** follow that every Windows-only or Visual LISP extension function is merely a generic foreign-function binding; some families, such as ‘vlr-*‘ reactors, remain primarily specified by Autodesk’s own AutoLISP / ActiveX documentation.

1.4 Source Notes

- Primary chapter model adapted from the Common Lisp HyperSpec chapter index: [CLHS Chapter Index](`https://www.lispworks.com/documentation/HyperSpec/Front/Contents.htm`)
- AutoLISP language history: [AutoLISP and Visual LISP](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-49AAEA0E-C422-48C4-87F0-52FCA491BF2C.htm`)
- BricsCAD developer overview: [BricsCAD Developer Reference Overview](`https://developer.bricsys.com/bricscad/help/en_US/V24/DevRef/source/DevRef_Overview.htm`)

- Status: chapter structure adapted by design; language scope documented and inferred.

1.5 Brief History

The historical genealogy of AutoLISP / Visual LISP shapes several otherwise-arbitrary corners of the language. The chapter-level normative rules — Lisp-1 semantics, dynamic scope, silent-NIL on unbound reference, no tail-call optimization, no multiple values, upper-case folding — are downstream consequences of the choices made by the implementations enumerated below.

- **XLISP (David Betz, 1980s).** AutoLISP was derived from a slimmed XLISP. XLISP itself targeted personal computers as a compact, embeddable Lisp dialect with a single-cell symbol model and dynamic scope. AutoLISP inherited that core.

- **AutoLISP shipped in AutoCAD 2.18, January 1986.** Released as Autodesk's API for customising AutoCAD. Successive AutoCAD releases extended it through to Release 13 (February 1995).

- **Vital LISP (Basis Software).** A third-party superset of AutoLISP adding an IDE, debugger, compiler, ActiveX (COM) bindings, reactors, and additional general-purpose Lisp functions.

- **Visual LISP (Autodesk, 1997).** Autodesk acquired Vital LISP and rebranded it. Initially shipped as an add-on to AutoCAD R14.

- **Merger into AutoCAD 2000 (March 1999).** Visual LISP replaced the standalone AutoLISP runtime and IDE inside AutoCAD. From this point onward, "AutoLISP" and "Visual LISP" name the same runtime; the documentation distinguishes "core" AutoLISP from the Visual LISP extensions (`vl-*`, `vla-*`, `vlax-*`, `vlr-*`, etc.).

- **BricsCAD AutoLISP.** Bricsys's BricsCAD product implements AutoLISP and large parts of Visual LISP from scratch as a vendor-independent compatibility layer. BricsCAD V25 introduced lax-mode tokenizer and `atof` extensions; V26 (this work's reference target) preserves them. Bricsys maintains its own bug tracker and release notes; defects such as `SR44723` are normative-grade evidence for behaviours that Autodesk's reference is silent on.

- **clautolisp (this project).** A Common-Lisp-based reimplementation of the AutoLISP runtime designed to match the Lisp-1, dynamically-scoped, host-product-conformant semantics documented in this specification.

The genealogy explains, among other things, why `lambda` produces a function value but does not lexically capture (XLISP roots), why integer division truncates (C-influenced numerics from PC-era XLISP), and why `values` does not exist (the XLISP core never had it).

1.5.1 Source Notes

- [AutoLISP — Wikipedia](`https://en.wikipedia.org/wiki/AutoLISP`)
- [History of AutoLISP — Ron Leigh](`http://ronleigh.com/autolisp/ahistory.htm`)
- [AutoLISP and Visual LISP (Autodesk)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-49AAEA0E-C422-48C4-87F0-52FCA htm`)
- BricsCAD release notes (per-version, Bricsys help center).

2 2 Syntax

2.1 Overview

AutoLISP uses parenthesized Lisp syntax. Vendor documentation specifies the syntax mostly through examples, concept pages, and individual function documentation rather than through a single formal grammar.

2.2 Definitions

An AutoLISP source contains expressions. Expression classes include:

- symbols,
- integers,
- reals,
- strings,
- lists,
- dotted pairs,
- quoted expressions,
- calls,
- special forms.

2.3 Reader Syntax Entry: Integer Literal

2.3.1 Name

Integer Literal

2.3.2 Class

Reader Syntax

2.3.3 Concrete Syntax

```
integer ::= sign? digit+
sign    ::= "+" | "-"
digit   ::= "0" | "1" | ... | "9"
```

2.3.4 Constituents

- optional sign
- one or more decimal digits

2.3.5 Description

An integer literal denotes a signed 32-bit integer value when its magnitude lies in the documented integer range.

2.3.6 Acceptance Rules

- documented examples include ‘2‘ and ‘-56‘
- no decimal point appears
- literals beyond the maximum integer range may be read as reals instead of integers

2.3.7 Strict/Lax Policy

- strict: accept only the conservative grammar above
- lax: may accept additional host-confirmed integer token forms if tests justify them

2.3.8 Examples

- ‘0‘
- ‘12‘
- ‘-42‘
- ‘+7‘

2.3.9 Compatibility

- Autodesk documents the range and prose semantics
- exact lexical grammar is inferred, not fully formalized

2.3.10 Notes

Some older or host-specific readers may accept additional token spellings, but those are not standardized here.

2.3.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.3.12 Source Notes

- [About Integers (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-MAC-AutoLisp/files/GUID-EF6114FC-F1E4-4C71-91CC-07D01E6C8ABB.htm`)
- Status: documented in prose, formal syntax inferred.

2.4 Reader Syntax Entry: Real Literal

2.4.1 Name

Real Literal

2.4.2 Class

Reader Syntax

2.4.3 Concrete Syntax

```
real ::= sign? digit+ "." digit* exponent?
       | sign? digit+ exponent
exponent ::= ("e" | "E") sign? digit+
```

2.4.4 Constituents

- optional sign
- decimal digits
- decimal point and/or scientific exponent

2.4.5 Description

A real literal denotes a double-precision floating-point number.

2.4.6 Acceptance Rules

- documented examples include ‘3.1‘, ‘0.23‘, ‘-56.123‘, ‘4.1e-6‘
- numbers between ‘-1‘ and ‘1‘ must contain a leading zero according to Autodesk prose

2.4.7 Strict/Lax Policy

- strict: reject ambiguous forms such as ‘.5‘ and ‘1.‘ until product tests confirm them
- lax: may accept host-confirmed forms beyond the conservative grammar

2.4.8 Examples

- ‘1.0‘
- ‘0.5‘
- ‘-3.25‘
- ‘4.1e-6‘

2.4.9 Compatibility

- AutoCAD 2021+ Unicode changes do not directly alter numeric tokens, but reader changes should still be version-qualified

2.4.10 Notes

The published documentation leaves ‘.5‘, ‘1.‘, and some exponent variants under-specified.

2.4.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.4.12 Source Notes

- [About Reals (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP/files/GUID-F2BFD097-295B-4499-BBD9-0A785C377070.htm`)
- Status: partially documented, partially inferred.

2.5 Reader Syntax Entry: String Literal

2.5.1 Name

String Literal

2.5.2 Class

Reader Syntax

2.5.3 Concrete Syntax

```
string ::= "\"" string-char* "\""
```

2.5.4 Constituents

- opening double quote
- zero or more string characters
- closing double quote

2.5.5 Description

A string literal denotes a string object.

2.5.6 Acceptance Rules

- a string is delimited by double quotes
- backslash introduces escape/control sequences

2.5.7 Strict/Lax Policy

- strict: accept only documented or tested escapes
- lax: accept additional escapes only when they do not conflict with documented behavior and compatibility requires them

2.5.8 Examples

- ‘”abc”‘
- ‘”A string with spaces”‘

2.5.9 Compatibility

- string interpretation changed materially with Autodesk Unicode/LISPSYS changes beginning in 2021
- BricsCAD also documents Unicode-aware file text handling

2.5.10 Notes

The complete escape repertoire remains under-specified in the current source corpus.

2.5.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.5.12 Source Notes

- [About Strings (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-D7C33A49-9715-4E9B-9D8A-4C2E7BFC0FE5.htm`)
- [LISPSYS](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-Core/files/GUID-1853092D-6E6D-4A06-8956-AD2C3DF203A3.htm`)
- [read + write text files (BricsCAD)](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/readwritetextfiles.htm`)
- Status: documented in principle, incomplete in detail.

2.6 Reader Syntax Entry: Symbol

2.6.1 Name

Symbol Token

2.6.2 Class

Reader Syntax

2.6.3 Concrete Syntax

A symbol token is any token not otherwise read as a number or syntax delimiter, subject to the character restrictions below.

2.6.4 Constituents

- one or more non-delimiter characters

2.6.5 Description

AutoLISP symbol names are case-insensitive. Autodesk documents that they may contain alphanumeric and notation characters, except at least:

- ‘(‘
- ‘)‘
- ‘.‘
- ‘’‘
- ‘”‘
- ‘;‘

A symbol name cannot consist solely of numeric characters.

2.6.6 Acceptance Rules

- blanks, newlines, tabs, and parentheses are documented token delimiters for ‘read‘

2.6.7 Strict/Lax Policy

- strict: reject tokens that fall outside documented delimiter and reserved-character rules
- lax: may accept host-confirmed punctuation or edge cases when needed for compatibility

2.6.8 Examples

- ‘FOO‘
- ‘bar‘
- ‘c:my-command‘

2.6.9 Compatibility

- print convention is uppercase in Autodesk examples
- exact escaping and extended symbol syntax remain under-specified

2.6.10 Notes

This specification does not standardize CL-style package syntax or vertical-bar escaping.

2.6.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.6.12 Source Notes

- [About Symbols and Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP/files/GUID-1855E7B0-2474-49E6-9EE3-8BC54 htm`)
- [read (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5B50BB3E-C244-46E8-85D8-6A2D48B1FE51.htm`)

- Status: partially documented.

2.7 Reader Syntax Entry: List

2.7.1 Name

List

2.7.2 Class

Reader Syntax

2.7.3 Concrete Syntax

```
list ::= "(" element* ")"
```

2.7.4 Description

A list is a sequence of zero or more expressions enclosed in parentheses.

2.7.5 Acceptance Rules

- elements are separated by delimiters and nested syntax

2.7.6 Strict/Lax Policy

- strict and lax modes agree for ordinary list syntax

2.7.7 Examples

- ‘()‘
- ‘(A B C)‘
- ‘((A B) C)‘

2.7.8 Compatibility

Stable across vendors.

2.7.9 Notes

The empty list is semantically ‘nil‘.

2.7.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.7.11 Source Notes

- [About Lists (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-F8074AA9-F361-4960-B628-681465B74229.htm`)
- Status: documented.

2.8 Reader Syntax Entry: Dotted Pair

2.8.1 Name

Dotted Pair

2.8.2 Class

Reader Syntax

2.8.3 Concrete Syntax

```
dotted-pair ::= "(" expression "." expression ")"
```

2.8.4 Description

A dotted pair is a two-part cons-style representation written with a dot separator.

2.8.5 Acceptance Rules

- used heavily in association lists and entity data

2.8.6 Strict/Lax Policy

- strict and lax modes agree for ordinary dotted-pair syntax

2.8.7 Examples

- ‘(A . 2)‘
- ‘(8 . ”WALLS”)‘

2.8.8 Compatibility

Language-central across AutoCAD and BricsCAD.

2.8.9 Notes

Dotted pairs are semantically more important in AutoLISP than in many other Lisp environments because of CAD data conventions.

2.8.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.8.11 Source Notes

- [About Dotted Pairs (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/PLK/AutoCAD-AutoLISP/files/GUID-877B87BA-6FA2-4894-9F94-184E7DF25B7D.htm`)
- Status: documented.

2.9 Reader Syntax Entry: Quote

2.9.1 Name

Quote Syntax

2.9.2 Class

Reader Syntax

2.9.3 Concrete Syntax

```
quoted-expression ::= "'" expression
```

2.9.4 Description

Leading apostrophe is shorthand for ‘quote‘.

2.9.5 Acceptance Rules

- the following expression is not evaluated

2.9.6 Strict/Lax Policy

- strict and lax modes agree

2.9.7 Examples

- ‘’A‘
- ‘’(1 2 3)‘

2.9.8 Compatibility

Stable across vendors.

2.9.9 Notes

This specification does not standardize backquote/comma syntax yet.

2.9.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

2.9.11 Source Notes

- [quote (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-18F7E287-CB2F-4150-9A07-CE23C3F9E604.htm`)
- Status: documented.

2.10 Other Reader Syntax

The following remain under-specified and are therefore strict/lax configuration points:

- backquote
- comma
- comma-at
- vertical-bar symbol escaping
- sharpsign dispatch
- package-like syntax

2.11 Source Notes

- Current vendor corpus sampled in ‘autolisp-spec/documentation/autolisp-reference-notes.org‘
- Status: under-specified pending further research and testing.

3 3 Evaluation and Execution

3.1 Overview

AutoLISP evaluation is Lisp-like but not identical to Common Lisp evaluation. This chapter defines the general evaluation model needed by the rest of the specification.

3.2 Normative Rules

- numbers and strings self-evaluate
- ‘nil‘ and ‘T‘ are predefined values
- symbols evaluate to their bound values
- ordinary calls evaluate arguments before invocation unless the operator is a special form
- ‘nil‘ is false and every non-‘nil‘ value is true in conditional contexts

Evaluation order is historically version-sensitive in error-reporting contexts. Autodesk explicitly notes that modern AutoLISP evaluates all arguments before checking argument types in at least some cases, unlike older releases.

3.3 Normative Rules: Argument Evaluation Order

3.3.1 Statement

For every ordinary call `(operator e1 e2 e3 ...)` — i.e. every call whose operator is **not** a special form — the argument expressions `e1, e2, e3, ...` are evaluated **strictly left-to-right**, with each argument's side effects visible to the next.

Special forms (`setq`, `if`, `cond`, `and`, `or`, `while`, `repeat`, `foreach`, `lambda`, `function`, `defun`, `defun-q`, `progn`, `quote`, `command`, `trace`, `untrace`, `vlax-for`) decide their own evaluation order; their per-entry rules override this default.

3.3.2 Vendor Evidence

BricsCAD V26 / macOS direct probe: a side-effecting `note` function records each argument's invocation order. `(+ (note 1) (note 2) (note 3))`, `(list (note 1) (note 2) (note 3))`, and an immediately-applied lambda `((lambda (a b c) ...) (note 1) (note 2) (note 3))` all yield the recorded sequence `(1 2 3)`. The order is identical for builtin operators, ordinary user functions, and lambda calls.

3.3.3 Cross-references

- Compare **CLHS 3.1.2.1.2.3** — Common Lisp specifies left-to-right argument evaluation.
- Compare **R7RS 4.1.3** — Scheme specifies that argument evaluation order is unspecified, only that each is fully evaluated before any procedure is applied. AutoLISP picks the stricter contract.

3.3.4 Source Notes

Phase-6 BricsCAD V26 / macOS general-semantics probe — `results/bricscad/macos/20260426T*-s` section A.

3.4 Normative Rules: Variable Scope (Dynamic, Not Lexical)

3.4.1 Statement

AutoLISP variables are **dynamically scoped**. A binding established at function call entry — through a positional parameter, through a `(/ locals)` declaration, or through any subsequent `setq` on those names — is visible to every function called transitively from the body. The binding is restored to its prior value when the function returns. There is no lexical capture: a `lambda` does **not** close over the names referenced by its body. When the lambda is later called, every free reference is resolved against the **active** dynamic-frame chain, then the active namespace.

3.4.2 Concrete Consequences

- `(defun make-adder (/ x) (setq x 10) (lambda (y) (+ x y)))` — calling `(make-adder)` then later applying the returned lambda outside `make-adder`'s extent fails: `x` has no binding any more, so the body's reference to `x` resolves to `nil` (see "Unbound-Variable Reference" below) and the `(+ ... y)` call surfaces **bad argument type <NIL>; expected <NUMBER> at** [+].

- A counter built with `(defun with-counter (/ k) (setq k 0) (lambda () (setq k (+ k 1)) k))` does **not** accumulate state across calls; each call from outside `with-counter` sees `k` unbound (and thus `nil`).

- A lambda referencing a **global** (top-level) symbol picks up that global's **current** value at every call, not the value at lambda creation. Mutating the global between two calls of the same lambda changes what subsequent calls return.

3.4.3 Vendor Evidence

BricsCAD V26 / macOS direct probe (section C):

- Captured-local closure call: error **bad argument type <NIL>; expected <NUMBER> at** [+] — the local was popped on return; the lambda saw `nil`.

- Top-level closure: `(c2)` first returns `7` (set inside maker2), then `42` after a top-level `(setq x 42)`.

- Local counter: every call from outside the maker fails with the same NIL-at-[+] error — no accumulation.

3.4.4 Strict / Lax Policy

Dynamic scoping is **strict** across all supported dialects. No host product is known to expose lexical-capture semantics in its AutoLISP layer; any such extension would require a separate dialect knob.

3.4.5 Cross-references

- Compare **CLHS 3.1.1** — Common Lisp is lexical by default; dynamic scope requires ‘defvar‘ / ‘(declare (special …))‘.
- Compare **R7RS 5.2** — Scheme is unconditionally lexical.
- Special Form Entry: LAMBDA, FUNCTION (this chapter) — produce function values; do not capture.
- Single-Cell Symbol Binding (chapter 7) — the dynamic frame chain consults the same per-symbol binding cell.

3.4.6 Source Notes

- [About Local and Global Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-A8045F62-1CE3-4022-ACC1-CC0E2 htm`) — describes shadowing pragmatically.
- Phase-6 BricsCAD V26 / macOS semantics probe section C.

3.5 Normative Rules: Unbound-Variable Reference

3.5.1 Statement

Evaluating a symbol with no active dynamic binding and no namespace binding yields `nil` rather than signalling an error. Referencing a never-set name is therefore not directly observable; the typical surface symptom is a downstream type error when `nil` is passed to an arithmetic or other type-checking primitive.

This contrasts with both Common Lisp (`unbound-variable` is a condition) and R7RS Scheme (the result is an error). The silent-NIL behaviour is **strict** across every supported AutoLISP host: there is no language-level way for portable AutoLISP source to distinguish "bound to nil" from "never bound" at reference time, and `boundp` — the only predicate exposed for that question — additionally **creates** the symbol and binds it to `nil` as a side effect of being asked (chapter 7, BOUNDP entry). Programmer-visible state therefore collapses every never-set name into a `nil`-bound name on first observation.

The consequence: there is no compliant AutoLISP dialect in which a bare reference to an unset symbol can signal an error without breaking deployed code. Strict mode in this specification means **common to every conforming AutoLISP runtime**, not **the most pedantic**; silent-NIL is therefore the strict rule.

3.5.2 Concrete Consequences

- `(setq y (+ never-set-symbol 1))` does not signal **unbound variable**; it signals **bad argument type <NIL>; expected <NUMBER> at** [+].
- `(boundp 'never-set)` returns `nil` but additionally creates the symbol and binds it to `nil` (chapter 7, BOUNDP). After this call `never-set` is observably `nil`-bound.
- A program cannot reliably detect "never assigned" once any code path has either referenced or `boundp`-tested the name.

3.5.3 Vendor Evidence

BricsCAD V26 / macOS direct probe section C — every closure test that referenced an out-of-extent local surfaced its error **at the consuming primitive** (‘+‘), not at the variable reference. The error message wrapped the value ‘<NIL>‘, which is the expected payload after the silent-NIL evaluation.

3.5.4 Strict / Lax Policy

Silent-NIL on unbound reference is **strict** across the `strict`, `autocad-2026`, and `bricscad-v26` dialects. clautolisp exposes a non-conforming diagnostic knob, `unbound-variable-mode :strict-error`, that programs can opt into through a custom-built dialect descriptor; doing so produces source code that does not run on any real AutoLISP host and is intended only for static-analysis tooling and unit tests.

3.5.5 Cross-references

- Function Entry: BOUNDP (chapter 7) — the only language-level distinction between bound-to-nil and unbound, with its documented binding side effect.

3.5.6 Source Notes

- Phase-6 BricsCAD V26 / macOS semantics probe section C (indirect evidence; explicit probe pending due to launch-flake on this run).

3.6 Normative Rules: Tail-Call Optimization (None Required)

3.6.1 Statement

AutoLISP does **not** guarantee tail-call optimization. Recursion is stack-bound on every observed host. Programs that rely on bounded space for tail-recursive loops must be rewritten to use the iterative special forms `while`, `repeat`, or `foreach`.

3.6.2 Vendor Evidence

BricsCAD V26 / macOS direct probe section B: a deliberately-written tail-recursive `(count-down n)` succeeds at `n=1000`, then fails at `n=10000, 100000, 1000000` with **LISP - system call stack limit exceeded**. The boundary is well below typical Scheme implementations and is consistent with ordinary stack-frame consumption per call.

3.6.3 Cross-references

- Compare **R7RS 5.11.4** — Scheme requires **proper tail recursion** (constant space).
- Compare **CLHS 3.5.1.1** — Common Lisp does **not** require it; behaviour matches AutoLISP.
- Special Form Entry: WHILE, REPEAT, FOREACH (this chapter) — bounded-space iteration alternatives.

3.6.4 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe section B.

3.7 Normative Rules: Multiple Values (None)

3.7.1 Statement

AutoLISP has no concept of multiple return values. Every form returns exactly one value. There is no `values` primitive, no `multiple-value-bind`, no `call-with-values`. Functions that conceptually return more than one piece of data return a list, a dotted pair, or a host-defined structure (e.g. an entity-data dotted-pair list).

3.7.2 Vendor Evidence

BricsCAD V26 / macOS direct probe: `(values 1 2 3)` raises **no function definition <VALUES>; expected FUNCTION at** [**eval**].

3.7.3 Cross-references

- Compare **CLHS 3.1.7** and **R7RS 6.10** — both Common Lisp and Scheme have first-class multiple-value mechanisms.

3.7.4 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe section D.

3.8 Normative Rules: Symbol Case Folding

3.8.1 Statement

The reader case-folds every unquoted symbol to **upper case** at read time. The case-folding applies uniformly to source text, the result of `(read string)`, and the canonical name returned by `vl-symbol-name`. There is no escape syntax (such as Common Lisp's `|...|` or `\`) for preserving mixed case in symbol names.

3.8.2 Vendor Evidence

BricsCAD V26 / macOS direct probe section G:

- `'foo` prints `FOO`.
- `'FoO` prints `FOO`.
- `(eq 'foo 'FOO)` returns `T`.
- `(vl-symbol-name 'mixed-Symbol)` returns the string `"MIXED-SYMBOL"`.
- `(read "mixedFromRead")` returns the symbol `MIXEDFROMREAD`.

3.8.3 Strict / Lax Policy

Strict in every supported dialect. The `bricscad-v26` dialect's lax token mode does not relax case folding; it relaxes only token classification (e.g. lone trailing identifier characters on integers). String literals are unaffected.

3.8.4 Cross-references

- Compare **CLHS 23.1** — CL also case-folds by default, but provides `readtable-case` and the `|...|` escape.
- Compare **R7RS 7.1.1** — Scheme is case-sensitive by default since R7RS.

3.8.5 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe section G.

3.9 Normative Rules: Number Tower and Division

3.9.1 Statement

AutoLISP recognises two numeric types: **integer** and **real** (IEEE-754 double-precision binary floating-point). Mixed-type arithmetic is permitted; the result is **real** whenever any argument is real, otherwise **integer**. Division follows these rules:

- `(/ INT INT)` — integer division, **truncating toward zero**. `(/ 7 2) = 3`; `(/ -7 2)` truncates toward zero to `-3` (subject to per-vendor confirmation; toward-zero is the consistent reading on modern hosts).
- `(/ REAL ANY)` or `(/ ANY REAL)` — real-valued division. `(/ 7.0 2) = 3.5`; `(/ 7 2.0) = 3.5`.
- `(/ INT 0)` — signals **divide by zero**.
- `(/ REAL 0.0)` — undefined-behaviour territory on some hosts (BricsCAD V26 / macOS aborts the application); programs must avoid it.

Integer width on modern AutoLISP is at least 64 bits in BricsCAD V26: `(+ 2147483640 100) = 2147483740` exceeds the historical 31-bit range without promotion or overflow. AutoCAD's per-version width remains 32-bit signed int historically; verification against AutoCAD 2026 is deferred (no licensed seat available).

Floating-point underflow goes to `0.0` rather than to a denormal indicator: `(* 1.0e-300 1.0e-300) = 0.0`.

3.9.2 Vendor Evidence

BricsCAD V26 / macOS direct probe section F.

3.9.3 Strict / Lax Policy

- Truncating integer division and the integer/real promotion rule are **strict** across all dialects.
- Real division by zero is **strict-error** in clautolisp; the `bricscad-v26` dialect's behaviour mirrors the host's documented "divide by zero" error rather than the application abort observed in this build.

- Integer width — clautolisp persists 32-bit signed integers as the strict baseline (BricsCAD release-history compatibility). Wider-integer support is a future dialect knob if AutoCAD evidence demands it.

3.9.4 Cross-references

- Compare **CLHS 12** — CL has a numeric tower with rationals, separates `/` (exact) from `truncate~/~floor~/~round`. AutoLISP does not.
- Compare **R7RS 6.2** — Scheme distinguishes exact and inexact numbers.

3.9.5 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe section F.

3.10 Normative Rules: Equality Predicates

3.10.1 Statement

AutoLISP defines the following equality predicates with the semantics observed in BricsCAD V26 / macOS:

- `(eq a b)` — pointer-level identity for cons cells, distinguishes freshly-allocated objects, but returns `T` for two textually-identical string literals (the host **interns** string literals), and `T` for two textually-identical symbol literals (canonical case-folded name).
- `(equal a b)` — structural equality over cons cells; numerically compares an integer to a real (`(equal 1 1.0)` returns `T`); compares strings by content.
- Truthiness: a value is **true** iff it is not `nil`. `0`, `0.0`, `""` are **true**. `nil` is the unique false value.

3.10.2 Vendor Evidence

BricsCAD V26 / macOS direct probe section H–J:

- `(equal "abc" "abc") = T` ; `(eq "abc" "abc") = T` — string literals are interned.
- `(equal 1 1.0) = T` — cross-type numeric equality.
- `(equal '(1 2) (list 1 2)) = T` ; `(eq '(1 2) (list 1 2)) = nil` — structural vs identity.
- `(if 0 'yes 'no) = YES` ; `(if 0.0 'yes 'no) = YES` ; `(if "" 'yes 'no) = YES` ; `(if '() 'yes 'no) = NO`.
- `(eq nil '()) = T` ; `(null nil) = T` ; `(null '()) = T` — `nil` and the empty list are the same object.

3.10.3 Cross-references

- Compare **CLHS 14.5.1** — CL's `eq` is finer-grained (does not intern strings).
- Compare **R7RS 6.1** — Scheme distinguishes `eqv?`, `eq?`, and `equal?`.

3.10.4 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe sections H, I, J.

3.11 Normative Rules: NIL and the Empty List

3.11.1 Statement

The symbol `nil`, the literal `()`, and the literal `'()` all denote the **same** object — the empty list and the false value. `(car nil)` and `(cdr nil)` both return `nil`, **not** an error.

3.11.2 Vendor Evidence

BricsCAD V26 / macOS direct probe section I:

- `(eq nil '()) = T`
- `(car nil) = nil` ; `(cdr nil) = nil`

3.11.3 Cross-references

- Compare **CLHS 26.1.97** — CL also identifies `nil` with the empty list and accepts `(car nil) = nil`.
- Compare **R7RS 6.4** — in Scheme `nil` is a separate symbol; `()` is the empty list; `(car '())` is an error.

3.11.4 Source Notes

Phase-6 BricsCAD V26 / macOS semantics probe section I.

3.12 Special Form Entry: QUOTE

3.12.1 Name

‘quote‘

3.12.2 Class

Special Form

3.12.3 Syntax

```
(quote expr)
'expr
```

3.12.4 Arguments and Values

- ‘expr‘: any expression; not evaluated

3.12.5 Description

Returns ‘expr‘ without evaluating it.

3.12.6 Evaluation Rules

No subform evaluation occurs.

3.12.7 Return Values

Returns ‘expr‘.

3.12.8 Side Effects

None.

3.12.9 Exceptional Situations

Malformed syntax is erroneous.

3.12.10 Examples

- ‘(quote A) => A‘
- ‘’(1 2) => (1 2)‘

3.12.11 Compatibility

Stable across vendors.

3.12.12 Notes

Quote shorthand is defined in chapter 2.

3.12.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.12.14 Source Notes

- [quote (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-18F7E287-CB2F-4150-9A07-CE23C3F9E604.htm`)
- Status: documented.

3.13 Special Form Entry: IF

3.13.1 Name

‘if‘

3.13.2 Class

Special Form

3.13.3 Syntax

`(if testexpr thenexpr [elseexpr])`

3.13.4 Arguments and Values

- ‘testexpr‘: an expression
- ‘thenexpr‘: an expression
- ‘elseexpr‘: an expression; optional

3.13.5 Description

Conditionally evaluates one of two branches.

3.13.6 Evaluation Rules

- evaluate ‘testexpr‘
- if result is non-‘nil‘, evaluate ‘thenexpr‘
- otherwise evaluate ‘elseexpr‘ if supplied

3.13.7 Return Values

- returns the selected branch value
- if ‘elseexpr‘ is absent and the test is false, returns ‘nil‘

3.13.8 Side Effects

Side effects are those of:

- the evaluation of ‘testexpr‘, which always occurs,
- the evaluation of ‘thenexpr‘ when ‘testexpr‘ is non-‘nil‘,
- the evaluation of ‘elseexpr‘ when ‘testexpr‘ is ‘nil‘ and ‘elseexpr‘ is present.

The non-selected branch is not evaluated and has no side effects.

3.13.9 Exceptional Situations

Malformed syntax is erroneous.

3.13.10 Examples

- ‘(if T 1 2) => 1‘
- ‘(if nil 1 2) => 2‘

3.13.11 Compatibility

Stable across vendors.

3.13.12 Notes

Conditional truth is based on ‘nil‘ versus non-‘nil‘.

3.13.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.13.14 Source Notes

- [if (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-916F1A5C-FD70-4D66-897E-6DCD666DCB39.htm)
- Status: documented.

3.14 Special Form Entry: COND

3.14.1 Name

‘cond‘

3.14.2 Class

Special Form

3.14.3 Syntax

```
(cond [(test result ...) ...])
```

3.14.4 Arguments and Values

- each clause contains a test and zero or more result forms

3.14.5 Description

Sequentially selects the first successful clause.

3.14.6 Evaluation Rules

- tests are evaluated in order
- when a test yields non-‘nil‘:
  - if result forms exist, they are evaluated sequentially and the last value is returned
  - otherwise the test value itself is returned

3.14.7 Return Values

- the selected clause value
- ‘nil‘ if no clause succeeds

3.14.8 Side Effects

Side effects arise from:

- evaluation of each test expression up to and including the first successful one,

- evaluation of the result forms of the selected clause, if any.

No later clause is evaluated after a successful clause is found.

3.14.9 Exceptional Situations

Malformed clause structure is erroneous.

3.14.10 Examples

- ‘(cond ((> 2 1) ’YES) (T ’NO)) => YES‘

3.14.11 Compatibility

Stable across vendors.

3.14.12 Notes

‘T‘ may be used in the last clause as an always-true test.

3.14.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.14.14 Source Notes

- [cond (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7BA45202-D95F-4F2D-8D83-965024826074.htm`)
- Status: documented.

3.15 Special Form Entry: AND

3.15.1 Name

‘and‘

3.15.2 Class

Special Form

3.15.3 Syntax

`(and [expr ...])`

3.15.4 Arguments and Values

- zero or more expressions

3.15.5 Description

Computes the logical conjunction of a sequence of expressions.

3.15.6 Evaluation Rules

- expressions are evaluated left to right
- evaluation stops as soon as one expression evaluates to ‘nil‘
- if no expression evaluates to ‘nil‘, all expressions are evaluated

3.15.7 Return Values

Vendor summary documentation describes ‘and‘ as returning the logical ‘AND‘ of the expressions.

The exact return-value convention still needs dedicated per-function confirmation in this draft and therefore remains mildly under-specified.

3.15.8 Side Effects

Side effects are those of the expressions evaluated before short-circuiting stops evaluation, or of all expressions if none evaluates to ‘nil‘.

3.15.9 Exceptional Situations

Malformed syntax is erroneous.

3.15.10 Examples

- ‘(and T T) => T‘
- ‘(and T nil)‘ yields false

3.15.11 Compatibility

Autodesk classifies ‘and‘ as a special form.

3.15.12 Notes

Because Autodesk’s summary page is brief, the exact value returned in non-trivial truthy cases should still be confirmed by dedicated vendor pages or tests.

3.15.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.15.14 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- [Equality and Conditional Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C6A7FC58-7A07-4650-9D59-12CAFD194FDA.htm`)
- Status: documented as a special form and by summary; exact return convention still needs tighter confirmation.

3.16 Special Form Entry: SETQ

3.16.1 Name

‘setq‘

3.16.2 Class

Special Form

3.16.3 Syntax

`(setq sym expr [sym expr] ...)`

3.16.4 Arguments and Values

- ‘sym‘: a symbol; not evaluated
- ‘expr‘: an expression

3.16.5 Description

Assigns values to variables.

3.16.6 Evaluation Rules

- symbols are taken literally
- expressions are evaluated left to right
- assignments occur left to right

3.16.7 Return Values

Returns the value of the last ‘expr‘.

3.16.8 Side Effects

Establishes or changes variable bindings. Writes the supplied value into the symbol’s **single binding cell** (see chapter 7, ”Single-Cell Symbol Binding”); a previous `defun` on the same symbol is therefore overwritten.

3.16.9 Exceptional Situations

Malformed syntax or invalid assignment targets are erroneous.

3.16.10 Examples

- ‘(setq A 10) => 10‘
- ‘(setq A 1 B 2) => 2‘
- ‘(defun foo (x) x) (setq foo 42)‘ — ‘foo‘ evaluates to ‘42‘; ‘(foo 5)‘ then signals **no function definition**.

3.16.11 Compatibility

Stable across vendors.

3.16.12 Notes

‘set‘ differs by evaluating the symbol designator. AutoLISP is **Lisp-1**: `setq` and `defun` share the same per-symbol binding cell; the most recent write wins (chapter 7).

3.16.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.16.14 Source Notes

- [setq (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-LT-AutoLISP-Reference/files/GUID-2F4B7A7B-7B6F-4E1C-B32E-677506094EAA.htm`)
- Status: documented.

3.17 Special Form Entry: SET

3.17.1 Name

‘set‘

3.17.2 Class

Function-like assignment operator

3.17.3 Syntax

`(set sym expr)`

3.17.4 Arguments and Values

- ‘sym‘: an expression whose value designates a symbol
- ‘expr‘: an expression

3.17.5 Description

Assigns a value to the symbol designated by the first argument.

3.17.6 Return Values

Returns the assigned value.

3.17.7 Side Effects

Changes the binding of the designated symbol.

3.17.8 Exceptional Situations

Erroneous if the first evaluated argument does not designate a valid assignable symbol.

3.17.9 Examples

- ‘(set ’A 3) => 3‘

3.17.10 Compatibility

Stable across vendors.

3.17.11 Notes

Unlike ‘setq‘, the symbol argument is evaluated.

3.17.12 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.17.13 Source Notes

- [set (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-041BE20B-CFA6-465B-A98B-1BCBDC810881.htm`)
- Status: documented.

3.18 Special Form Entry: WHILE

3.18.1 Name

‘while‘

3.18.2 Class

Special Form

3.18.3 Syntax

`(while testexpr [expr ...])`

3.18.4 Description

Repeatedly evaluates body forms while the test remains non-‘nil‘.

3.18.5 Evaluation Rules

- evaluate ‘testexpr‘
- if the result is non-‘nil‘, evaluate the body forms in order and then repeat
- stop when ‘testexpr‘ evaluates to ‘nil‘

3.18.6 Return Values

Returns the value of the last body expression of the last iteration, or ‘nil‘ if the body is never run.

3.18.7 Side Effects

Side effects are those of:

- each evaluation of ‘testexpr‘,
- each evaluation of the body forms during iterations that occur.

3.18.8 Exceptional Situations

Malformed syntax is erroneous.

3.18.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.18.10 Source Notes

- [while (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E7C900DB-8B66-4109-BEF6-B0A18E8CF6B6.htm`)
- Status: documented in core behavior; concise here.

3.19 Special Form Entry: REPEAT

3.19.1 Name

‘repeat‘

3.19.2 Class

Special Form

3.19.3 Syntax

```
(repeat int [expr ...])
```

3.19.4 Description

Evaluates body forms a fixed number of times.

3.19.5 Evaluation Rules

- evaluate the repetition count
- if the count permits iteration, evaluate body forms in order for each iteration

3.19.6 Return Values

Returns the final body value or ‘nil‘ if no body exists.

3.19.7 Side Effects

Side effects are those of:

- evaluation of the repetition count,
- repeated evaluation of the body forms.

3.19.8 Exceptional Situations

Erroneous if the repetition count is not acceptable to the active host semantics.

3.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.19.10 Source Notes

- [repeat (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/ENU/AutoCAD-AutoLISP-Reference/files/GUID-413F72B4-BA37-4E5E-9D51-A0091130A317.htm`)
- Status: documented in core behavior; concise here.

3.20 Special Form Entry: FOREACH

3.20.1 Name

‘foreach‘

3.20.2 Class

Special Form

3.20.3 Syntax

```
(foreach name list [expr ...])
```

3.20.4 Description

Iterates over each element of ‘list‘, binding it to ‘name‘.

3.20.5 Evaluation Rules

- evaluate ‘list‘
- for each element of the resulting list, bind it to ‘name‘
- evaluate the body forms in order

3.20.6 Return Values

Returns the final body value or ‘nil‘ if no body exists.

3.20.7 Side Effects

Side effects are those of:

- evaluation of ‘list‘,
- rebinding ‘name‘ for each iteration,
- evaluation of the body forms for each iteration.

3.20.8 Exceptional Situations

Erroneous if the list argument is not acceptable to the active host semantics.

3.20.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.20.10 Source Notes

- [foreach (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-BE6A23C4-4E18-45A6-854E-2DE9574A6925.htm`)
- Status: documented in core behavior; concise here.

3.21 Special Form Entry: OR

3.21.1 Name

‘or‘

3.21.2 Class

Special Form

3.21.3 Syntax

`(or [expr ...])`

3.21.4 Description

Tests expressions sequentially and returns ‘T‘ if any is non-‘nil‘, otherwise ‘nil‘.

3.21.5 Evaluation Rules

Short-circuits on the first non-‘nil‘ result.

3.21.6 Return Values

Returns ‘T‘ or ‘nil‘.

3.21.7 Side Effects

Side effects are those of the expressions evaluated up to and including the first non-‘nil‘ expression, or of all expressions if all are ‘nil‘.

Expressions not reached because of short-circuiting are not evaluated and have no side effects.

3.21.8 Compatibility

This differs from Common Lisp ‘or‘, which returns the first non-‘nil‘ value. AutoLISP ‘or‘ returns only ‘T‘ or ‘nil‘. Confirmed against BricsCAD V26 by direct citation: the BricsCAD V26 ‘or‘ reference page documents the return as ”T if any provided argument evaluates as non-NIL; if all provided arguments evaluate as NIL, or if no argument is provided, NIL is returned” with examples ‘(or 123 nil) -> T‘. Idioms ported from Common Lisp such as ‘(setq x (or x default))‘ **clobber** ‘x‘ with ‘T‘ rather than preserving the existing value; portable AutoLISP code must use ‘(if (not x) (setq x default))‘ instead.

3.21.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: documented (BricsCAD V26 dedicated per-symbol page, return: ”T if any provided argument evaluates as non-NIL; ... NIL otherwise”).
- Status: documented in both vendors. Phase-5 closure: behaviour confirmed identical (T / nil only) across both vendors; surfaced as a real cross-vendor **and** cross-language hazard for code ported from Common Lisp.

3.21.10 See Also

- probe-core.lsp — uses an ‘if‘-guarded init pattern explicitly because AutoLISP’s ‘or‘ does not return the first non-‘nil‘ value (the original ‘(setq x (or x default))‘ idiom was discovered to clobber ‘x‘ with ‘T‘ against BricsCAD V26 during Phase 5).

3.21.11 Source Notes

- [or (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-64ECF4D5-0714-45FD-8E27-284E9D2FC8B2.htm`)
- [or (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/or.htm`)
- Status: documented (Phase 5 closure; cross-vendor behaviour confirmed).

3.22 Special Form Entry: PROGN

3.22.1 Name

‘progn‘

3.22.2 Class

Special Form

3.22.3 Syntax

`(progn [expr ...])`

3.22.4 Arguments and Values

- zero or more expressions

3.22.5 Description

Evaluates expressions sequentially.

3.22.6 Evaluation Rules

- evaluate each expression left to right

3.22.7 Return Values

- returns the value of the last expression
- returns ‘nil‘ if no expressions are supplied

3.22.8 Side Effects

Side effects are those of the evaluated expressions.

3.22.9 Exceptional Situations

Malformed syntax is erroneous.

3.22.10 Examples

- ‘(progn (setq a 1) (setq b 2) b) => 2‘

3.22.11 Compatibility

Autodesk classifies ‘progn‘ as a special form.

3.22.12 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.22.13 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- Status: documented as a special form; detailed entry still partly inferred from stable Lisp usage.

3.23 Special Form Entry: LAMBDA

3.23.1 Name

‘lambda‘

3.23.2 Class

Special Form

3.23.3 Syntax

`(lambda ([arguments] [/ locals ...]) expr ...)`

3.23.4 Arguments and Values

- AutoLISP parameter and local syntax
- body expressions

3.23.5 Description

Creates an anonymous function object.

3.23.6 Evaluation Rules

The body expressions are not evaluated at lambda-creation time. They are evaluated when the resulting function is invoked.

3.23.7 Return Values

Returns an anonymous callable object.

3.23.8 Side Effects

None at creation time, apart from effects of malformed usage signaling errors.

3.23.9 Exceptional Situations

Malformed parameter/local syntax is erroneous.

3.23.10 Examples

- use in functional position
- use with ‘function‘ in mapped or callback contexts

3.23.11 Compatibility

Autodesk classifies ‘lambda‘ as a special form.

3.23.12 Notes

The exact object model of the resulting anonymous function is not formally specified in the vendor documentation sampled here.

3.23.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.23.14 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- Status: documented as a special form; detailed semantics partly inferred.

3.24 Special Form Entry: FUNCTION

3.24.1 Name

‘function‘

3.24.2 Class

Special Form

3.24.3 Syntax

`(function arg)`

3.24.4 Arguments and Values

- ‘arg‘: a function name (symbol) or a lambda expression; **not** evaluated.

3.24.5 Description

Returns ‘arg‘ without evaluating it, as ‘quote‘ does. The form additionally signals intent to the Visual LISP byte-compiler, which uses the hint to link and optimise ‘arg‘ as if it were a built-in operator or ‘defun‘-named function (Autodesk: ”The function function is identical to the quote function, except it tells the Visual LISP compiler to link and optimize the argument as if it were a built-in function or defun.”).

3.24.6 Evaluation Rules

No subform evaluation occurs. The returned value is the unevaluated ‘arg‘; resolution to a callable object — symbol lookup, lambda capture — happens at the point the value is **used** (typically inside ‘apply‘, ‘mapcar‘, ‘vl-some‘, ‘vl-every‘, or any other higher-order operator), not at the point the ‘function‘ form was reached.

3.24.7 Return Values

- For a symbol ‘arg‘: the symbol itself, dynamically resolved by the consumer.
- For a ‘(lambda ...)‘ form: the lambda form, reified to a function object when first applied.

3.24.8 Side Effects

None.

3.24.9 Exceptional Situations

- ‘arg‘ neither a symbol nor a lambda form: ‘:invalid-function-designator‘.
- An undefined function name only surfaces when the value is actually applied — ‘(function missing)‘ itself does not signal.

3.24.10 Examples

- ‘(setq v (function +)) (apply v ’(1 2)) => 3‘
- ‘(setq lam (function (lambda (x) (* x x)))) (apply lam ’(5)) => 25‘
- ‘(mapcar (function 1+) ’(1 2 3)) => (2 3 4)‘

3.24.11 Portable Higher-Order-Function Idiom

AutoLISP is nominally a Lisp-1 with dynamic scope (chapter 7), so a parameter bound to a function value should be callable as ‘(v arg)‘. The two reference implementations diverge on this point:

- AutoCAD 2022/2026 warns (”parameter used as a function”) and may refuse to dispatch through a parameter binding.
- BricsCAD V25/V26 silently ignores the parameter shadow at call position and resolves the name to the surrounding (global) binding — including when the caller passed a lambda anonymously, which then becomes unreachable.

Code that must run on both vendors therefore treats functions as in a Lisp-2:

Goal	Portable form
Define a named function	‘(defun foo (x) …)‘
Obtain the function value of a name	‘(function foo)‘
Anonymous function value	‘(function (lambda (x) …))‘
Call a function held in a **variable** (parameter…)	‘(apply v (list arg1 …))‘
Pass a function as parameter	pass ‘(function f)‘, apply via ‘apply‘

Avoid:

```
(defun hof (object op)
  (op object))           ; non-portable direct call through a parameter
```

Prefer:

```
(defun hof (object op)
  (apply op (list object)))
```

3.24.12 Compiled-Code Caveats

Two subtle effects fall out of the compiler hint:

1. Visual LISP may **link** the argument of ‘(function foo)‘ at compile time. After such compilation, a later ‘(defun foo …)‘ from elsewhere does **not** necessarily replace the call site’s binding. Example:

   ```
   ;; foobar.lsp
   (defun foo () 'foo)
   (defun bar () (foo))

   ;; --- separately ---
   (load (vlisp-compile 'st "foobar.lsp"))
   (defun foo () 'new-foo)
   (bar) ; => foo or new-foo, depending on compile-time linking
   ```

   Programs that need to override ‘foo‘ at runtime should avoid early-bind hints in the compiled producer (use ‘(quote foo)‘ rather than ‘(function foo)‘ for symbols that are intended to be redefinable from a host), or use ‘apply‘ with a symbol passed at runtime.

2. For lambda forms, Autodesk recommends ‘function‘ over ‘quote‘ so the byte-compiler can optimise the body once, rather than rebuilding it on every call. Source: Lee Mac, **The Apostrophe and the Quote Function**.

3.24.13 Compatibility

Autodesk classifies ‘function‘ as a special form. Both vendors document the ”identical to quote with compiler hint” semantics. clautolisp implements the same late-resolution semantic: the form returns its unevaluated argument.

3.24.14 Notes

- The portable HOF rule lives at the language level: it does not depend on whether the producer is interpreted, byte-compiled (FAS/VLX), or hosted by clautolisp.

- ‘function‘ is widely used with ‘mapcar‘, ‘vl-some‘, ‘vl-every‘, ‘vl-remove-if‘, ‘vl-member-if‘, and other Visual LISP higher-order utilities. Whether the argument is a symbol or a lambda, the consumer resolves it via the same dispatch path used by ‘apply‘.

3.24.15 Availability

- AutoCAD 2026: documented (per Autodesk reference page).

- BricsCAD V26: presumed compatible (no contradicting page found).

- clautolisp: implemented (eval-function-form returns arg unevaluated; apply/mapcar/vl-* honour late resolution against the dynamic scope chain).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.24.16 Source Notes

- [function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-MAC-AutoLISP-Reference/files/GUID-CF7E5870-561F-42DB-B134-CCD41EF93A htm`)

- [apply (AutoLISP 2023)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0574ADA0-0950-456A-9330-A25184215 htm`)

- [Passing and calling functions (Autodesk Community)](`https://forums.autodesk.com/t5/visual-lisp-autolisp-and-general/passing-and-calling-functions-h td-p/2463166`)

- Lee Mac, **The Apostrophe and the Quote Function** (`https://www.lee-mac.com/quote.html`)

- TN003 — **Valeurs fonctions en AutoLISP** (Pascal Bourguignon, 2026-05-22)

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- Status: documented.

3.25 Special Form Entry: COMMAND

3.25.1 Name

‘command‘

3.25.2 Class

Special Form

3.25.3 Syntax

`(command [args ...])`

3.25.4 Arguments and Values

- a sequence of command name and command input arguments

3.25.5 Description

Executes an AutoCAD command through the host command processor.

3.25.6 Evaluation Rules

Autodesk classifies ‘command‘ as a special form because its argument handling and command-loop interaction differ from ordinary function calls.

3.25.7 Return Values

Host-dependent command-execution result; exact return-value conventions are not fully formalized in the current source set.

3.25.8 Side Effects

- executes a host command
- may modify the drawing, system variables, user interaction state, and command stack

3.25.9 Affected By

- current document state
- command versioning
- localization
- ‘PAUSE‘
- ‘*error*‘ mode

3.25.10 Exceptional Situations

Command failure, invalid input, and command-loop state errors are host-dependent.

3.25.11 Compatibility

- central AutoLISP host interaction facility
- related to ‘command-s‘

3.25.12 Notes

This is one of the clearest examples of an AutoLISP construct that cannot be reduced to pure Lisp evaluation semantics.

3.25.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.25.14 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)

- [Query and Command Functions Reference](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-F9ABE09D-9A71-4C00-8E5D-5BA7C` `htm`)
- Status: documented in role and category; further host details remain chapter-level.

3.26 Special Form Entry: TRACE

3.26.1 Name

‘trace‘

3.26.2 Class

Special Form

3.26.3 Syntax

`(trace [sym ...])`

3.26.4 Description

Enables tracing of function calls for debugging.

3.26.5 Compatibility

Autodesk classifies ‘trace‘ as a special form.

3.26.6 Notes

Detailed debugger/tracer output semantics are not fully developed in this draft.

3.26.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.26.8 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- Status: category documented; detailed semantics still sparse.

3.27 Special Form Entry: UNTRACE

3.27.1 Name

‘untrace‘

3.27.2 Class

Special Form

3.27.3 Syntax

`(untrace [sym ...])`

3.27.4 Description

Disables tracing of function calls for debugging.

3.27.5 Compatibility

Autodesk classifies ‘untrace‘ as a special form.

3.27.6 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.27.7 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)
- Status: category documented; detailed semantics still sparse.

3.28 Special Form Entry: VLAX-FOR

3.28.1 Name

‘vlax-for‘

3.28.2 Class

Special Form

3.28.3 Syntax

```
(vlax-for var collection [expr ...])
```

3.28.4 Description

Iterates over an ActiveX collection object.

3.28.5 Compatibility

- Autodesk classifies ‘vlax-for‘ as a special form
- Windows-only in Autodesk documentation because it depends on ActiveX support

3.28.6 Notes

This construct belongs to the Visual LISP / ActiveX layer, not the core portable AutoLISP layer.

3.28.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.28.8 Source Notes

- [About Special Forms (AutoLISP)](https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm)
- [AutoLISP Extensions Reference (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-07C3599E-EC87-4EC0-8F0B-8DE4BD150A64.htm)
- Status: category documented; detailed semantics tied to ActiveX collections.

3.29 Dictionary Coverage Index: Special Forms

The standardized special forms identified in Autodesk documentation are:

- ‘and‘
- ‘command‘
- ‘cond‘
- ‘defun‘
- ‘defun-q‘
- ‘foreach‘
- ‘function‘
- ‘if‘
- ‘lambda‘
- ‘or‘
- ‘progn‘
- ‘quote‘
- ‘repeat‘
- ‘setq‘
- ‘trace‘

- ‘untrace‘
- ‘vlax-for‘
- ‘while‘

3.30 Special Form Entry: DEFUN

3.30.1 Name

‘defun‘

3.30.2 Class

Special Form

3.30.3 Syntax

```
(defun sym ([arguments] [/ locals ...]) expr ...)
```

3.30.4 Arguments and Values

- ‘sym‘: function name
- ‘arguments‘: parameter names
- ‘/ locals‘: local variable declarations within the AutoLISP parameter syntax
- ‘expr‘: body forms

3.30.5 Description

Defines a named function. Names beginning with ‘C:‘ define CAD commands callable at the command line.

3.30.6 Evaluation Rules

The parameter/local syntax is AutoLISP-specific and must not be replaced with Common Lisp lambda-list semantics.

3.30.7 Return Values

Under-specified in the current vendor corpus sampled here; implementations commonly return the function name, but this should be version-tested before being treated as normative.

3.30.8 Side Effects

Writes the resulting function value into the symbol's **single binding cell** (see chapter 7, "Single-Cell Symbol Binding"). Any previous binding on 'sym' — variable value, prior function, or built-in — is overwritten and is no longer reachable through the symbol.

3.30.9 Compatibility

- local variable syntax is vendor-documented
- duplicate-name behavior is uniform across vendors at the per-symbol level: the most recent `setq` or `defun` on a symbol wins. This was confirmed for BricsCAD V26 by direct probe (`results/bricscad/macos/20260426T155521Z-namesp` and is consistent with the BricsCAD release-note defect SR44723 fix.

3.30.10 Notes

The slash separating arguments and locals must be surrounded by spaces in Autodesk documentation. AutoLISP is **Lisp-1**: `defun` and `setq` share the same per-symbol binding cell.

3.30.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.30.12 Source Notes

- [About Defining a Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP/files/GUID-9EDF48C2-1678-4DC3-BFD6-9D` `htm`)
- Status: documented in practical syntax, partially under-specified semantically.

3.31 Special Form Entry: DEFUN-Q

3.31.1 Name

‘defun-q‘

3.31.2 Class

Special Form

3.31.3 Syntax

```
(defun-q sym ([arguments] [/ variables ...]) expr ...)
```

3.31.4 Description

Defines a function in a backward-compatible representation that can be accessed as a list structure.

Historically, older non-compiled AutoLISP implementations represented ‘defun‘ definitions as list structure. Autodesk documents that this changed in AutoCAD 2000. After that change, ordinary ‘defun‘ definitions were no longer safely manipulable as lists.

‘defun-q‘ exists specifically to preserve compatibility with older code that:

- inspects function definitions as list data,
- appends to or rewrites function bodies as lists,
- expects to retrieve a function definition structurally rather than only call it.

Accordingly, ‘defun-q‘ is not just another spelling of ‘defun‘. It defines a callable function while also preserving a list-form definition that can be retrieved and manipulated by compatibility utilities such as ‘defun-q-list-ref‘ and ‘defun-q-list-set‘.

Autodesk explicitly says that ‘defun-q‘ is provided strictly for backward compatibility and should not be used for other purposes.

3.31.5 Evaluation Rules

- the defining form establishes a global function binding
- the defined function is callable in the usual way

- the definition is also retained in list form for reflective compatibility operations

3.31.6 Return Values

Returns the function name.

3.31.7 Side Effects

- defines or redefines a global function
- stores a list-structured representation of the definition for later access by compatibility functions

3.31.8 Compatibility

Autodesk documents duplicate names as tolerated with first occurrence used and later duplicates ignored.

An ordinary ‘defun‘ definition should not be assumed to be accessible as list structure in modern AutoCAD semantics. Attempting to use a ‘defun‘ function as a list is documented to produce an error, with Autodesk explicitly suggesting ‘defun-q‘ instead for old compatibility patterns.

3.31.9 Examples

- define a compatibility-visible function:
  - ‘(defun-q my-startup (x) (print (list x)))‘
- retrieve its stored list structure:
  - ‘(defun-q-list-ref 'my-startup) => ((X) (PRINT (LIST X)))‘
- older code patterns that splice or append function bodies are the intended use case for ‘defun-q‘

3.31.10 Notes

Conceptually, ‘defun-q‘ separates two concerns that were historically conflated:

- defining a callable function,
- treating the function definition as ordinary list data.

Modern ‘defun‘ covers the first concern. ‘defun-q‘ exists only for the second, compatibility-driven concern.

3.31.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.31.12 Source Notes

- [defun-q (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP-Reference/files/GUID-5EE138EF-D531-441B-9F12-8D5645C540FE.htm`)
- [About Compatibility of Defun with Earlier Releases of AutoCAD (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ITA/AutoCAD-AutoLISP/files/GUID-16246E02-459F-4408-9CB4-6F2CB306FD9F.htm`)
- [defun-q-list-ref (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-DAF4A76E-9346-4A68-A0C8-17695 htm`)
- Status: documented.

3.32 Function Entry: DEFUN-Q-LIST-REF

3.32.1 Name

‘defun-q-list-ref‘

3.32.2 Class

Function

3.32.3 Syntax

```
(defun-q-list-ref 'function)
```

3.32.4 Arguments and Values

- ‘function‘: a symbol naming a function

3.32.5 Description

Returns the stored list structure of a function defined with ‘defun-q‘.

This is the reflective accessor that makes the special purpose of ‘defun-q‘ concrete: code can inspect the compatibility-preserved structural definition rather than only call the function.

3.32.6 Return Values

- the function’s list definition if the named function was defined as a list
- ‘nil‘ if the argument does not designate a list-defined function

Autodesk’s example shows a result of the form:

```
((X) (PRINT (LIST X)))
```

which corresponds to:

- the argument/local-variable specification
- the body forms

3.32.7 Side Effects

None specified.

3.32.8 Affected By

- whether the target function was defined with ‘defun-q‘

3.32.9 Exceptional Situations

The sampled Autodesk reference does not document additional exceptional situations beyond normal function-call validity requirements.

3.32.10 Examples

```
(defun-q my-startup (x) (print (list x)))
(defun-q-list-ref 'my-startup)
=> ((X) (PRINT (LIST X)))
```

3.32.11 Compatibility

- intended only for the backward-compatibility workflow surrounding ‘defun-q‘

- documented by Autodesk on Windows, Mac OS, and Web

3.32.12 Notes

‘defun-q-list-ref‘ should not be understood as a general reflection API over arbitrary function definitions. It is specifically tied to the compatibility representation established by ‘defun-q‘.

3.32.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).

- BricsCAD V26: presumed compatible (no contradicting page found).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.32.14 Source Notes

- [defun-q-list-ref (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-DAF4A76E-9346-4A68-A0C8-17695` `htm`)

- [Function-Handling Functions Reference](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6804DF2D-E5C4-466A-9869 htm)
- Status: documented.

3.33 Function Entry: DEFUN-Q-LIST-SET

3.33.1 Name

‘defun-q-list-set‘

3.33.2 Class

Function

3.33.3 Syntax

```
(defun-q-list-set 'sym list)
```

3.33.4 Arguments and Values

- ‘sym‘: a symbol naming the function to define or redefine
- ‘list‘: a list-structured function definition

3.33.5 Description

Defines or redefines a function from a list-structured compatibility representation.

Together, ‘defun-q‘, ‘defun-q-list-ref‘, and ‘defun-q-list-set‘ form the compatibility triad for programs that manipulate function definitions as list data.

3.33.6 Return Values

Under-specified in the current source set.

The function-handling summary identifies it as a compatibility function that ”defines a function as a list”, but the dedicated page was not collected in this pass.

3.33.7 Side Effects

- defines or redefines the named function
- installs a list-based compatibility definition

3.33.8 Affected By

- validity of the supplied list structure
- compatibility behavior of the active dialect

3.33.9 Exceptional Situations

Under-specified in the current source set.

At minimum, invalid list structure should be expected to be erroneous, but the exact contract still needs the dedicated Autodesk page.

3.33.10 Examples

The current source set does not yet contain a fully documented vendor example for this function.

3.33.11 Compatibility

- documented by Autodesk as part of the function-handling family
- intended for backward compatibility only, like ‘defun-q‘

3.33.12 Notes

This function is important because it implies that the ‘defun-q‘ list representation is not only inspectable but writable.

That makes the preserved list representation semantically active, not merely descriptive.

3.33.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

3.33.14 Source Notes

- [Function-Handling Functions Reference](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6804DF2D-E5C4-466A-9869 htm)
- [D Functions Reference](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-76EE3904-E89B-441A-84FD-C0AB3289742C.htm)
- Status: partially documented; dedicated per-function page still needed.

3.34 Source Notes

- [About Special Forms (AutoLISP)](https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm)
- [About Error Handling (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-AutoLISP/files/GUID-027AD2E0-5AC5-48DA-B451-112B7EECB htm)
- Status: mixed documented and under-specified in certain details.

4 4 Types and Runtime Classes

4.1 Overview

AutoLISP exposes a practical runtime type system through the ‘type‘ function and through function contracts.

4.2 Distinguished Object Entry: NIL

4.2.1 Name

‘nil‘

4.2.2 Class

Distinguished object

4.2.3 Related Types

- ‘LIST‘
- generalized boolean false

4.2.4 Description

‘nil‘ is a distinguished object in AutoLISP.

In documented AutoLISP behavior:

- ‘nil‘ denotes falsity in conditional and predicate contexts,
- ‘nil‘ is also the empty list,
- ‘(type nil)‘ returns ‘nil‘, not a distinct runtime type designator such as ‘NIL‘,
- ‘listp‘ returns ‘T‘ for ‘nil‘,
- ‘null‘ and ‘not‘ return ‘T‘ for expressions that evaluate to ‘nil‘.

Accordingly, this specification does not treat ‘nil‘ as a separate runtime type in the same sense as ‘INT‘, ‘REAL‘, ‘STR‘, ‘LIST‘, or ‘SYM‘.

4.2.5 Representation

It is printed as ‘nil‘ in value position and corresponds semantically to the empty list ‘()‘.

4.2.6 Valid Operations

- ‘listp‘ returns ‘T‘
- ‘vl-symbolp‘ returns ‘nil‘
- ‘type‘ returns ‘nil‘
- conditional contexts treat it as false

4.2.7 Compatibility

This is a core language distinction and must not be collapsed to generic Common Lisp symbol or type behavior.

In particular, the Common Lisp type named ‘nil‘ is not an appropriate model for AutoLISP’s documented ‘type‘ behavior.

4.2.8 Source Notes

- [type (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-506C9CC8-B0BD-4A4C-B4C2-006750504509.htm`)
- [listp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-8755F083-D3BA-45F0-BBAF-D60D3A73240C.htm`)
- [vl-symbolp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-F51AFE0A-CA6C-428D-8E8D-3607BF55199 htm`)
- [null (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/DEU/AutoCAD-AutoLISP/files/GUID-2CA2E5F1-297F-4ECA-9500-5FFCD4877126.htm`)
- [not (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-MAC-AutoLISP-Reference/files/GUID-991AD6A0-61AA-45C9-8C27-CADF9F36BE71.htm`)
- Status: documented by behavior; explicitly not treated here as a separate runtime type.

4.3 Type Entry: Symbol

4.3.1 Name

Symbol

4.3.2 Class

Type

4.3.3 Related Types

- variable names
- function names
- command names

4.3.4 Description

Symbols identify variables, functions, commands, and symbolic data.

4.3.5 Representation

Symbols are case-insensitive in the source language. Printed examples use uppercase.

4.3.6 Valid Operations

- ‘type‘
- ‘vl-symbolp‘
- ‘atoms-family‘
- binding operations

4.3.7 Compatibility

Do not model AutoLISP symbols using raw Common Lisp package semantics.

4.3.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

4.3.9 Source Notes

- [About Symbols and Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP/files/GUID-1855E7B0-2474-49E6-9EE3-8BC54` `htm`)
- [type (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-506C9CC8-B0BD-4A4C-B4C2-006750504509.htm`)
- Status: documented in practice.

4.4 Type Entry: ENAME

4.4.1 Name

‘ENAME‘

4.4.2 Class

Host Type

4.4.3 Description

An entity name designates an object in the host drawing database.

4.4.4 Representation

Opaque host-managed object identity.

4.4.5 Valid Operations

- ‘entget‘
- ‘entmod‘
- ‘entlast‘
- ‘ssname‘
- ‘namedobjdict‘

4.4.6 Compatibility

Core to host interaction in both Autodesk and BricsCAD environments.

4.4.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

4.4.8 Source Notes

- [namedobjdict (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-A1E43B30-EF8C-452E-97F5-8AC201 htm)
- Status: documented by host APIs.

4.5 Type Entry: PICKSET

4.5.1 Name

‘PICKSET‘

4.5.2 Class

Host Type

4.5.3 Description

A selection set is a host-managed collection of selected entities.

4.5.4 Representation

Opaque host-managed selection object.

4.5.5 Valid Operations

- ‘ssget‘
- ‘ssname‘
- ‘sslength‘

4.5.6 Compatibility

Selection-set indexing and limits are version-sensitive and must be tested.

4.5.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

4.5.8 Source Notes

- [ssname (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-LT-AutoLISP-Reference/files/GUID-EFB83751-9AC5-4C52-AD3D-D971BC560C1 htm)
- Status: documented in usage.

4.6 Type Entry: FILE

4.6.1 Name

‘FILE‘

4.6.2 Class

Type

4.6.3 Description

A file descriptor returned by ‘open‘.

4.6.4 Valid Operations

- ‘read-char‘
- ‘read-line‘
- output functions
- ‘close‘

4.6.5 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

4.6.6 Source Notes

- [open (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-089A323F-21FF-4337-99A9-375758E23BA4.htm`)
- Status: documented.

4.7 Source Notes

- [type (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-506C9CC8-B0BD-4A4C-B4C2-006750504509.htm)
- Status: document chapter still incomplete but structure is stable.

5 5 Data and Control Flow

5.1 Overview

AutoLISP data and control flow are built around symbols, numbers, strings, lists, dotted pairs, and control operators such as ‘if‘, ‘cond‘, ‘while‘, ‘repeat‘, and ‘foreach‘.

5.2 Normative Rules

- ‘nil‘ is both false and the empty list
- points are ordinary lists of two or three numbers
- association lists are central host data structures
- dotted pairs are first-class data

5.3 Variable Entry: NIL

5.3.1 Name

‘nil‘

5.3.2 Class

Predefined value

5.3.3 Value Type

Distinguished object.

5.3.4 Initial Value

Itself.

5.3.5 Description

‘nil‘ is the canonical false value and the canonical empty-list value.

Autodesk documentation sampled here does not describe ‘nil‘ as a separate ‘type‘ designator. Instead, it describes ‘nil‘ through its behavior:

- as the value recognized by ‘null‘ and ‘not‘,
- as the empty list accepted by ‘listp‘,
- as a value for which ‘(type nil)‘ returns ‘nil‘.

5.3.6 Affected By

- conditional evaluation
- list-processing functions
- type and predicate functions

5.3.7 Exceptional Situations

None specific to ‘nil‘ as a predefined value are documented in the sampled corpus.

5.3.8 Examples

- ‘(null nil) => T‘
- ‘(not nil) => T‘
- ‘(listp nil) => T‘
- ‘(type nil) => nil‘

5.3.9 Compatibility

This entry is intentionally written as a predefined value entry, not a type entry.

5.3.10 Notes

This specification avoids importing the Common Lisp distinction between the ‘nil‘ type and the ‘null‘ type into AutoLISP unless vendor evidence later requires it.

5.3.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

5.3.12 Source Notes

- [type (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-506C9CC8-B0BD-4A4C-B4C2-006750504509.htm`)
- [listp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-8755F083-D3BA-45F0-BBAF-D60D3A73240C.htm`)
- [null (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/DEU/AutoCAD-AutoLISP/files/GUID-2CA2E5F1-297F-4ECA-9500-5FFCD4877126.htm`)

- [not (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-MAC-AutoLISP-Reference/files/GUID-991AD6A0-61AA-45C9-8C27-CADF9F36BE71.htm`)
- Status: documented by behavior.

5.4 Source Notes

- [About Point Lists (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP/files/GUID-BFED8707-AB01-49BC-B0A9-D872794311DF.htm`)
- [About Dotted Pairs (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/PLK/AutoCAD-AutoLISP/files/GUID-877B87BA-6FA2-4894-9F94-184E7DF25B7D.htm`)

5.5 Function Entry: APPLY

5.5.1 Name

‘apply‘

5.5.2 Class

Function

5.5.3 Syntax

```
(apply 'function list)
```

5.5.4 Arguments and Values

- ‘’function‘: a symbol (defun name) or a lambda expression.
- ‘list‘: a list of arguments, or ‘nil‘ if the function takes no arguments.

5.5.5 Description

Passes a list of arguments to, and executes, the specified function. The function may be a built-in operator, a user-defined function, or a lambda expression.

5.5.6 Return Values

Returns whatever the called function returns: integer, real, string, list, ename, T, or nil.

5.5.7 Side Effects

Side effects are those of the called function.

5.5.8 Examples

- ‘(apply ’+ ’(1 2 3))‘ returns ‘6‘.
- ‘(apply ’strcat ’(”a” ”b” ”c”))‘ returns ‘”abc”‘.

5.5.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

5.5.10 Source Notes

- [apply (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0574ADA0-0950-456A-9330-A25184215` `htm`)
- Status: documented (Phase 3 body fill).

5.6 Function Entry: EVAL

5.6.1 Name

‘eval‘

5.6.2 Class

Function

5.6.3 Syntax

`(eval expr)`

5.6.4 Arguments and Values

- ‘expr‘: any AutoLISP expression. Documented value types are integer, real, string, list, symbol, ename, ‘T‘, or ‘nil‘.

5.6.5 Description

Returns the result of evaluating an AutoLISP expression. Enables dynamic evaluation, including resolving a symbol bound through a value-cell to its current value.

5.6.6 Return Values

Returns the value produced by evaluating ‘expr‘. The return type follows the expression type and is one of integer, real, string, list, ename, ‘T‘, or ‘nil‘.

5.6.7 Side Effects

Side effects are those of the evaluated expression.

5.6.8 Examples

Given ‘(setq a 123)‘ and ‘(setq b ’a)‘:

- ‘(eval 4.0)‘ returns ‘4.0‘.
- ‘(eval (abs -10))‘ returns ‘10‘.
- ‘(eval a)‘ returns ‘123‘.

- ‘(eval b)‘ returns ‘123‘, because ‘b‘ holds the symbol ‘a‘ which itself names the value ‘123‘.

5.6.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

5.6.10 Source Notes

- [eval (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D9B3E6CC-A982-4040-AE6E-6FD63D6C54D0.htm`)
- Status: documented (Phase 3 body fill).

5.7 Function Entry: EQ

5.7.1 Name

‘eq‘

5.7.2 Class

Function

5.7.3 Syntax

```
(eq expr1 expr2)
```

5.7.4 Arguments and Values

- ‘expr1‘, ‘expr2‘: integer, real, string, list, ename, ‘T‘, or ‘nil‘.

5.7.5 Description

Determines whether two expressions are **identical** — that is, whether both arguments reference the same object in memory, not merely equivalent values. For lists this is object identity; two structurally equal but distinct lists are not ‘eq‘.

5.7.6 Return Values

Returns ‘T‘ when the expressions reference identical objects, otherwise ‘nil‘.

5.7.7 Side Effects

None.

5.7.8 Examples

Given ‘(setq f1 ’(a b c))‘, ‘(setq f2 ’(a b c))‘, ‘(setq f3 f2)‘:

- ‘(eq f1 f3)‘ returns ‘nil‘ — ‘f1‘ and ‘f3‘ denote distinct list objects, even though they are structurally equal.
- ‘(eq f3 f2)‘ returns ‘T‘ — ‘f3‘ and ‘f2‘ refer to the same list object.

5.7.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

5.7.10 Source Notes

- [eq (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-BC81B391-48BF-401E-80D0-05A1B578A0FF.htm`)
- Status: documented (Phase 3 body fill).

5.8 Function Entry: EQUAL

5.8.1 Name

‘equal‘

5.8.2 Class

Function

5.8.3 Syntax

```
(equal expr1 expr2 [fuzz])
```

5.8.4 Arguments and Values

- ‘expr1‘, ‘expr2‘: integer, real, string, list, ename, ‘T‘, or ‘nil‘.
- ‘fuzz‘ (optional): integer or real tolerance value used for inexact numeric comparison.

5.8.5 Description

Determines whether two expressions evaluate to the same value. Unlike ‘eq‘, ‘equal‘ compares structurally: two distinct but element-wise equal lists are ‘equal‘. The optional ‘fuzz‘ argument controls tolerance for real-number comparison and for lists of reals such as point coordinates; two reals computed by different methods can differ slightly, and ‘fuzz‘ lets such values compare equal when the absolute difference is within tolerance.

5.8.6 Return Values

Returns ‘T‘ if the expressions are equal under the rules above, otherwise ‘nil‘.

5.8.7 Side Effects

None.

5.8.8 Examples

- Structural equality on lists: ‘(equal ’(a b c) ’(a b c))‘ returns ‘T‘.
- Numeric tolerance: with ‘(setq a 1.123456)‘ and ‘(setq b 1.123457)‘, ‘(equal a b)‘ returns ‘nil‘, while ‘(equal a b 0.000001)‘ returns ‘T‘.

5.8.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

5.8.10 Source Notes

- [equal (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7E85CB8F-B4DA-42F3-ABD3-89342A11E` `htm`)
- Status: documented (Phase 3 body fill).

6 6 Iteration

6.1 Overview

The primary standardized iteration constructs explicitly documented in the sampled corpus are ‘while‘, ‘repeat‘, and ‘foreach‘.

6.2 Dictionary Entries

See chapter 3 entries for ‘while‘, ‘repeat‘, and ‘foreach‘.

6.3 Source Notes

- [About Special Forms (AutoLISP)](`https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP/files/GUID-F992EAA3-AFFC-425D-B7C9-6A9CB0EA2768.htm`)

7 7 Symbols and Bindings

7.1 Overview

Symbols name values and functions. AutoLISP variable management is documented pragmatically rather than through a formal environment calculus.

7.2 Normative Rule: Single-Cell Symbol Binding (Lisp-1)

7.2.1 Statement

Each AutoLISP symbol carries exactly **one** binding cell. `setq` and `defun` both write to that single cell; either operation overwrites whatever the other wrote previously. There is no separate function cell distinct from the variable cell. In every position — value position or function-call position — the evaluator dispatches on the same stored value.

This places AutoLISP / Visual LISP in the **Lisp-1** family (a single namespace shared by variables and functions, like Scheme), not the **Lisp-2** family (separate cells per role, like Common Lisp).

7.2.2 Concrete Consequences

- After `(setq foo 42)` followed by `(defun foo (x) (+ 1 x))`, evaluating the bare symbol `foo` returns the function object — not `42`. The `setq`-supplied integer is gone.
- After `(defun bar (x) ...)` followed by `(setq bar 42)`, calling `(bar 5)` raises **no function definition** — the function is gone.
- `(setq myfunc eq)` followed by `(myfunc "1" "1")` must succeed: the variable's stored value (the `eq` subroutine) is dispatched as the call's function. Bricsys logged this as defect **SR44723** and shipped the fix accordingly.
- `defun-q` obeys the same single-cell rule, with the additional property that the function value remains list-accessible through `defun-q-list-ref` / `defun-q-list-set`.
- A `(defun ... (/ x ...) ...)` local — i.e. a name appearing after the slash in a lambda list — establishes a **dynamic shadowing** of the same single cell during the call's extent, then restores the prior binding on exit. Local declarations therefore work uniformly for symbols whose outer binding is a value, a function, or both successively.

7.2.3 Vendor Evidence

- BricsCAD V26 release-note defect **SR44723** explicitly documents the expected behaviour: `(setq myfunc eq)` followed by `(myfunc "1" "1")` must run the `eq` implementation. The fix landed in mainline V25 and is preserved through V26.

- BricsCAD V26 / macOS direct probe (this repository, `results/bricscad/macos/20260426T1555`
  - TEST1: `(setq probe-foo 42)` then `(defun probe-foo (x) (+ 1 x))` — evaluating the bare symbol `probe-foo` returns #<<FUNCTION> ...>, **not** `42`. `(probe-foo 5)` returns `6`.
  - TEST2: after `(setq probe-bar 42)` on a previously `defun`'d symbol, calling `(probe-bar 5)` raises **no function definition <PROBE-BAR>; expected FUNCTION at [eval]**. The function value was overwritten by the integer.

- Autodesk reference, **About Using Defun to Define a Function**: "Never use the name of a built-in function or symbol for the sym argument to defun, as this overwrites the original definition and makes the built-in function or symbol inaccessible." The wording does not distinguish a function role from a value role; it speaks of a single overridable definition.

- Autodesk reference, **About Local and Global Variables**: "If a variable name is added to the local variables list of a function, the global variable with the same name is ignored." This describes single-cell dynamic shadowing.

7.2.4 Strict / Lax Policy

The single-cell rule is **strict** in every supported dialect (`strict`, `autocad-2026`, `bricscad-v26`). No host product is known to expose Lisp-2 semantics; the conformance contract treats any Lisp-2-style behaviour as a regression.

7.2.5 Cross-references

- Special Form Entry: SETQ (chapter 3) — assigns a value to the single binding cell.

- Special Form Entry: DEFUN (chapter 3) — assigns a function value to the single binding cell.

- Special Form Entry: DEFUN-Q (chapter 3) — single-cell write plus list-accessible compatibility metadata.

- Special Form Entry: LAMBDA, FUNCTION (chapter 3) — produce function values that participate in the same cell.

- Function Entry: VL-SYMBOL-VALUE (this chapter) — returns the single cell's value.
- Function Entry: BOUNDP (this chapter) — predicates the single cell's bound state.

7.2.6 Source Notes

- [About Symbols and Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP/files/GUID-1855E7B0-2474-49E6-9EE3-8BC54` `htm`)
- [About Local and Global Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-A8045F62-1CE3-4022-ACC1-CC0E2` `htm`)
- [About Using Defun to Define a Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-9EDF48C2-1678-4DC3-B` `htm`)
- BricsCAD V26 release-note defect SR44723 (`(setq myfunc eq)` callable as function).
- Phase-6 BricsCAD V26 / macOS namespace probe — `results/bricscad/macos/20260426T1555`

7.3 Variable Entry: T

7.3.1 Name

'T'

7.3.2 Class

Predefined Variable

7.3.3 Value Type

Distinguished true value.

7.3.4 Initial Value

Itself.

7.3.5 Description

Canonical non-'nil' truth value.

7.3.6 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.3.7 Source Notes

- [About Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP/files/GUID-CC89231D-E10F-47D0-A1FD-A989D24C7105.htm`)
- Status: documented in practical terms.

7.4 Variable Entry: PI

7.4.1 Name

‘PI‘

7.4.2 Class

Predefined Variable

7.4.3 Value Type

Real number.

7.4.4 Initial Value

Approximation of pi.

7.4.5 Description

Predefined mathematical constant.

7.4.6 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.4.7 Source Notes

- [About Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP/files/GUID-CC89231D-E10F-47D0-A1FD-A989D24C7105.htm`)
- Status: documented.

7.5 Variable Entry: PAUSE

7.5.1 Name

‘PAUSE‘

7.5.2 Class

Predefined Variable

7.5.3 Description

Special interaction token used in command invocation contexts.

7.5.4 Compatibility

Host-interaction-specific.

7.5.5 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.5.6 Source Notes

- [About Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP/files/GUID-CC89231D-E10F-47D0-A1FD-A989D24C7105.htm`)
- Status: documented at variable overview level.

7.6 Source Notes

- [About Symbols and Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP/files/GUID-1855E7B0-2474-49E6-9EE3-8BC54`
`htm`)
- [About Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP/files/GUID-CC89231D-E10F-47D0-A1FD-A989D24C7105.htm`)

7.7 Function Entry: ATOM

7.7.1 Name

‘atom‘

7.7.2 Class

Function

7.7.3 Syntax

`(atom item)`

7.7.4 Arguments and Values

- ‘item‘: any value

7.7.5 Description

Returns ‘T‘ if ‘item‘ is not a list; otherwise returns ‘nil‘.

7.7.6 Return Values

- ‘T‘ if ‘item‘ is not a list
- ‘nil‘ otherwise

7.7.7 Side Effects

None.

7.7.8 Affected By

- the AutoLISP distinction between lists and non-lists

7.7.9 Exceptional Situations

None specific are documented.

7.7.10 Examples

- ‘(atom 1) => T‘
- ‘(atom ’(a b)) => nil‘
- ‘(atom nil) => T‘

7.7.11 Compatibility

Autodesk explicitly notes that some other Lisp dialects use different conventions, so this behavior should not be silently replaced by non-AutoLISP semantics.

7.7.12 Notes

Because ‘nil‘ is also the empty list in other AutoLISP contexts, the ‘atom‘ result on ‘nil‘ is particularly important and should be tested against target hosts.

7.7.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.7.14 Source Notes

- [atom (AutoLISP)](`https://help.autodesk.com/cloudhelp/2019/KOR/AutoCAD-AutoLISP-Reference/files/GUID-3B74E28D-7C3B-4409-9DFE-0305E9881560.htm`)
- Status: documented.

7.8 Function Entry: BOUNDP

7.8.1 Name

‘boundp‘

7.8.2 Class

Function

7.8.3 Syntax

`(boundp sym)`

7.8.4 Arguments and Values

- ‘sym‘: symbol

7.8.5 Description

Tests whether a symbol has a value bound to it.

7.8.6 Return Values

- ‘T‘ if the symbol has a value
- ‘nil‘ otherwise

Autodesk explicitly states that if no value is bound to ‘sym‘, or if ‘sym‘ has been bound to ‘nil‘, ‘boundp‘ returns ‘nil‘.

It also states that if ‘sym‘ is an undefined symbol, it is automatically created and assigned ‘nil‘.

7.8.7 Side Effects

None.

7.8.8 Exceptional Situations

Erroneous if the argument is not acceptable as a symbol designator under host semantics.

7.8.9 Examples

- ‘(boundp ’a)‘

7.8.10 Compatibility

Useful for distinguishing unassigned symbols from symbols intentionally set to ‘nil‘.

7.8.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.8.12 Source Notes

- [boundp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2021/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F8491426-8505-4390-825F-CACE15EBB48F.htm`)
- Status: documented.

7.9 Function Entry: NULL

7.9.1 Name

‘null‘

7.9.2 Class

Function

7.9.3 Syntax

`(null expr)`

7.9.4 Arguments and Values

- ‘expr‘: expression

7.9.5 Description

Tests whether ‘expr‘ evaluates to ‘nil‘.

7.9.6 Return Values

- ‘T‘ if the expression evaluates to ‘nil‘
- ‘nil‘ otherwise

7.9.7 Side Effects

Side effects are those of evaluating ‘expr‘.

7.9.8 Exceptional Situations

Any error arises from evaluating ‘expr‘.

7.9.9 Examples

- ‘(null nil) => T‘
- ‘(null 1) => nil‘

7.9.10 Compatibility

Closely related to ‘not‘, but documented separately.

7.9.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.9.12 Source Notes

- [null (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/DEU/AutoCAD-AutoLISP/files/GUID-2CA2E5F1-297F-4ECA-9500-5FFCD4877126.htm`)
- Status: documented.

7.10 Function Entry: NOT

7.10.1 Name

‘not‘

7.10.2 Class

Function

7.10.3 Syntax

```
(not expr)
```

7.10.4 Arguments and Values

- ‘expr‘: expression

7.10.5 Description

Tests whether ‘expr‘ evaluates to ‘nil‘.

7.10.6 Return Values

- ‘T‘ if the expression evaluates to ‘nil‘
- ‘nil‘ otherwise

7.10.7 Side Effects

Side effects are those of evaluating ‘expr‘.

7.10.8 Exceptional Situations

Any error arises from evaluating ‘expr‘.

7.10.9 Examples

- ‘(not nil) => T‘
- ‘(not 1) => nil‘

7.10.10 Compatibility

Behaviorally aligned with ‘null‘ in documented AutoLISP usage.

7.10.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.10.12 Source Notes

- [not (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-MAC-AutoLISP-Reference/files/GUID-991AD6A0-61AA-45C9-8C27-CADF9F36BE71.htm`)
- Status: documented.

7.11 Function Entry: TYPE

7.11.1 Name

‘type‘

7.11.2 Class

Function

7.11.3 Syntax

```
(type item)
```

7.11.4 Arguments and Values

- ‘item‘: any value

7.11.5 Description

Returns a symbol describing the runtime type category of ‘item‘, or ‘nil‘ for items that evaluate to ‘nil‘.

7.11.6 Return Values

Documented return values include symbols such as:

- ‘INT‘
- ‘REAL‘
- ‘STR‘
- ‘LIST‘
- ‘SYM‘
- ‘FILE‘
- ‘ENAME‘
- ‘PICKSET‘
- ‘SUBR‘
- ‘USUBR‘

- ‘SAFEARRAY‘
- ‘VARIANT‘
- ‘VLA-object‘

and ‘nil‘ for values that evaluate to ‘nil‘.

7.11.7 Side Effects

None.

7.11.8 Exceptional Situations

No special exceptional situations are documented.

7.11.9 Examples

- ‘(type 1) => INT‘
- ‘(type ”x”) => STR‘
- ‘(type nil) => nil‘

7.11.10 Compatibility

This function is one of the strongest direct sources for the practical runtime type model of AutoLISP.

7.11.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.11.12 Source Notes

- [type (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-506C9CC8-B0BD-4A4C-B4C2-006750504509.htm`)
- Status: documented.

7.12 Function Entry: VL-SYMBOLP

7.12.1 Name

‘vl-symbolp‘

7.12.2 Class

Visual LISP Function

7.12.3 Syntax

`(vl-symbolp obj)`

7.12.4 Arguments and Values

- ‘obj‘: any value

7.12.5 Description

Tests whether ‘obj‘ is a symbol.

7.12.6 Return Values

- ‘T‘ if ‘obj‘ is a symbol
- ‘nil‘ otherwise

Autodesk explicitly documents that:

- ‘(vl-symbolp t) => T‘
- ‘(vl-symbolp nil) => nil‘

7.12.7 Side Effects

None.

7.12.8 Compatibility

This entry is particularly important because its treatment of ‘nil‘ is easy to get wrong if one projects Common Lisp assumptions into AutoLISP.

7.12.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.12.10 Source Notes

- [vl-symbolp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-F51AFE0A-CA6C-428D-8E8D-3607BF55199` `htm`)
- Status: documented.

7.13 Function Entry: VL-SYMBOL-NAME

7.13.1 Name

‘vl-symbol-name‘

7.13.2 Class

Visual LISP Function

7.13.3 Syntax

`(vl-symbol-name sym)`

7.13.4 Arguments and Values

- ‘sym‘: symbol

7.13.5 Description

Returns the string name of a symbol.

7.13.6 Return Values

String corresponding to the symbol name.

7.13.7 Side Effects

None.

7.13.8 Exceptional Situations

Erroneous if the argument is not a symbol acceptable to the active host semantics.

7.13.9 Compatibility

Belongs to the Visual LISP reflection/introspection layer.

7.13.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.13.11 Source Notes

- [Symbol-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-796EFF94-E904-4F5B-B120-C6D2DCD8646A.htm`)
- Status: documented by family summary; dedicated page still desirable.

7.14 Function Entry: VL-SYMBOL-VALUE

7.14.1 Name

‘vl-symbol-value‘

7.14.2 Class

Visual LISP Function

7.14.3 Syntax

```
(vl-symbol-value sym)
```

7.14.4 Arguments and Values

- ‘sym‘: symbol

7.14.5 Description

Returns the current value of the symbol.

7.14.6 Return Values

The symbol’s value.

7.14.7 Side Effects

None beyond any host-defined lookup effects.

7.14.8 Exceptional Situations

Under-specified in the current source set for unbound-symbol behavior; this should be test-qualified.

7.14.9 Compatibility

Useful as a reflective counterpart to ordinary variable evaluation.

7.14.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.14.11 Source Notes

- [vl-symbol-value (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-5DAAABCF-7210-4A90-809D-6C27CCEF6 htm)
- Status: documented, with some behavioral edge cases still to be tested.

7.15 Dictionary Coverage Index: Symbol-Handling Functions

The Autodesk symbol-handling function family includes at least:

- ‘atom‘
- ‘atoms-family‘
- ‘boundp‘
- ‘not‘
- ‘null‘
- ‘numberp‘
- ‘quote‘
- ‘set‘
- ‘setq‘
- ‘type‘
- ‘vl-symbol-name‘
- ‘vl-symbol-value‘
- ‘vl-symbolp‘

Source:

- [Symbol-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-796EFF94-E904-4F5B-B120-C6D2DCD8646A.htm)

7.16 Function Entry: ATOMS-FAMILY

7.16.1 Name

‘atoms-family‘

7.16.2 Class

Function

7.16.3 Syntax

```
(atoms-family format [symlist])
```

7.16.4 Arguments and Values

- ‘format‘: integer; ‘0‘ to return symbols as a list of symbol objects, ‘1‘ to return them as a list of strings.
- ‘symlist‘ (optional): a list of strings naming symbols to look up. When supplied, the result has the same length as ‘symlist‘; for each name, the corresponding result element is the symbol (or its string name) if defined, or ‘nil‘ if undefined.

7.16.5 Description

Returns a list of currently defined symbols in the active document namespace. With only ‘format‘, returns every defined symbol. With ‘format‘ and ‘symlist‘, returns only entries for the named symbols, in the same order, with ‘nil‘ for any name that is not currently defined.

7.16.6 Return Values

A list whose elements are either symbols or strings, depending on ‘format‘.

7.16.7 Side Effects

None.

7.16.8 Examples

- ‘(atoms-family 0)‘ returns every defined symbol as a list of symbols.
- ‘(atoms-family 1 ’(”CAR” ”CDR” ”XYZ”))‘ returns ‘(”CAR” ”CDR” nil)‘ if ‘XYZ‘ is not defined.

7.16.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

7.16.10 Source Notes

- [atoms-family (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0B35850D-9E62-446B-BAA2-0598A4BD htm`)
- Status: documented (Phase 3 body fill).

8 8 Numbers

8.1 Overview

AutoLISP numbers are divided primarily into integers and reals.

8.2 Normative Rules

- integers are 32-bit signed values in documented modern Autodesk semantics
- reals are double-precision floating-point values
- integer overflow in literal reading may produce a real
- arithmetic overflow on integers is invalid

8.3 Source Notes

- [About Integers (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-MAC-AutoLisp/files/GUID-EF6114FC-F1E4-4C71-91CC-07D01E6C8ABB.htm`)
- [About Reals (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP/files/GUID-F2BFD097-295B-4499-BBD9-0A785C377070.htm`)

8.4 Function Entry: +

8.4.1 Name

‘+‘

8.4.2 Class

Function

8.4.3 Syntax

`(+ [number ...])`

8.4.4 Arguments and Values

- zero or more numeric arguments

8.4.5 Description

Returns the sum of the arguments.

8.4.6 Return Values

Numeric sum.

8.4.7 Side Effects

None.

8.4.8 Exceptional Situations

Erroneous for non-numeric arguments or host-specific numeric overflow conditions.

8.4.9 Examples

- ‘(+ 1 2 3) => 6‘

8.4.10 Compatibility

Core arithmetic primitive across dialects.

8.4.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.4.12 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary; detailed corner cases still to be tightened.

8.5 Function Entry: -

8.5.1 Name

'-'

8.5.2 Class

Function

8.5.3 Syntax

`(- number [number ...])`

8.5.4 Description

Subtracts the second and following arguments from the first.

8.5.5 Return Values

Numeric difference.

8.5.6 Side Effects

None.

8.5.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.5.8 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary.

8.6 Function Entry: *

8.6.1 Name

‘*‘

8.6.2 Class

Function

8.6.3 Syntax

`(* [number ...])`

8.6.4 Description

Returns the product of the arguments.

8.6.5 Return Values

Numeric product.

8.6.6 Side Effects

None.

8.6.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.6.8 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary.

8.7 Function Entry: /

8.7.1 Name

‘/‘

8.7.2 Class

Function

8.7.3 Syntax

```
(/ number [number ...])
```

8.7.4 Description

Divides the first argument by the product of the remaining arguments.

8.7.5 Return Values

Numeric quotient.

8.7.6 Side Effects

None.

8.7.7 Exceptional Situations

Division by zero and invalid numeric arguments are erroneous.

8.7.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.7.9 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary.

8.8 Function Entry: =

8.8.1 Name

‘=‘

8.8.2 Class

Function

8.8.3 Syntax

`(= [expr ...])`

8.8.4 Arguments and Values

- one or more expressions acceptable to numeric/string comparison semantics

8.8.5 Description

Compares arguments for equality under AutoLISP comparison semantics.

8.8.6 Return Values

- ‘T‘ if all arguments compare equal
- ‘nil‘ otherwise

8.8.7 Side Effects

Side effects are those of evaluating the argument expressions.

8.8.8 Compatibility

Autodesk comparison functions accept strings as well as numbers in documented summaries, so this should not be reduced to a purely numeric predicate without tighter source confirmation.

8.8.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.8.10 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- [Equality and Conditional Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C6A7FC58-7A07-4650-9D59-12CAFD194FDA.htm`)
- Status: documented by family summary; exact domain rules should be tightened.

8.9 Function Entry: /=

8.9.1 Name

‘/=‘

8.9.2 Class

Function

8.9.3 Syntax

```
(/= [expr ...])
```

8.9.4 Description

Compares arguments for inequality under AutoLISP comparison semantics.

8.9.5 Return Values

- ‘T‘ if the arguments are pairwise unequal under the active semantics
- ‘nil‘ otherwise

8.9.6 Side Effects

Side effects are those of evaluating the arguments.

8.9.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.9.8 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary.

8.10 Function Entry: <

8.10.1 Name

‘<‘

8.10.2 Class

Function

8.10.3 Syntax

```
(< [expr ...])
```

8.10.4 Description

Returns ‘T‘ if each argument is less than the argument to its right.

8.10.5 Return Values

- ‘T‘ if the ordered comparison succeeds
- ‘nil‘ otherwise

8.10.6 Side Effects

Side effects are those of evaluating the arguments.

8.10.7 Examples

- ‘(< 1 2 3) => T‘

8.10.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.10.9 Source Notes

- [< (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-67C8900F-0491-4D91-AA05-9136639E7424.htm`)
- Status: documented.

8.11 Function Entry: <=

8.11.1 Name

‘<=‘

8.11.2 Class

Function

8.11.3 Syntax

```
(<= [expr ...])
```

8.11.4 Description

Returns ‘T‘ if each argument is less than or equal to the argument to its right.

8.11.5 Return Values

- ‘T‘ on success
- ‘nil‘ otherwise

8.11.6 Side Effects

Side effects are those of evaluating the arguments.

8.11.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.11.8 Source Notes

- [<= (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-66B7B146-E279-4C22-9C0A-1F312166C15B.htm`)
- Status: documented.

8.12 Function Entry: >

8.12.1 Name

‘>‘

8.12.2 Class

Function

8.12.3 Syntax

`(> [expr ...])`

8.12.4 Description

Returns ‘T‘ if each argument is greater than the argument to its right.

8.12.5 Return Values

- ‘T‘ on success
- ‘nil‘ otherwise

8.12.6 Side Effects

Side effects are those of evaluating the arguments.

8.12.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.12.8 Source Notes

- [> (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ITA/AutoCAD-LT-AutoLISP-Reference/files/GUID-DAB2660E-AB9D-425D-B0C5-C5774164ADCD.htm`)
- Status: documented.

8.13 Function Entry: >=

8.13.1 Name

‘>=‘

8.13.2 Class

Function

8.13.3 Syntax

`(>= [expr ...])`

8.13.4 Description

Returns ‘T‘ if each argument is greater than or equal to the argument to its right.

8.13.5 Return Values

- ‘T‘ on success
- ‘nil‘ otherwise

8.13.6 Side Effects

Side effects are those of evaluating the arguments.

8.13.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.13.8 Source Notes

- [Non-alphabetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1.htm`)
- Status: documented by family summary.

8.14 Function Entry: ABS

8.14.1 Name

‘abs‘

8.14.2 Class

Function

8.14.3 Syntax

`(abs number)`

8.14.4 Description

Returns the absolute value of a number.

8.14.5 Return Values

Non-negative number of the same general magnitude.

8.14.6 Side Effects

None.

8.14.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.14.8 Source Notes

- [A Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2 htm`)
- Status: documented by alphabetic summary.

8.15 Function Entry: FIX

8.15.1 Name

‘fix‘

8.15.2 Class

Function

8.15.3 Syntax

`(fix number)`

8.15.4 Description

Truncates a real to an integer.

8.15.5 Return Values

Integer value.

8.15.6 Compatibility

The documentation on integers and reals makes ‘fix‘ especially relevant to the integer-range model.

8.15.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.15.8 Source Notes

- [fix (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-93B5F13B-348E-49E0-A116-0861684506D5.htm`)
- Status: documented.

8.16 Function Entry: FLOAT

8.16.1 Name

‘float‘

8.16.2 Class

Function

8.16.3 Syntax

`(float number)`

8.16.4 Description

Converts a numeric value to a real.

8.16.5 Return Values

Real value.

8.16.6 Side Effects

None.

8.16.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.16.8 Source Notes

- [zerop (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3D509EE1-A809-4B78-915C-28ECF483BB72.htm`)
- Status: documented.

8.17 Function Entry: ZEROP

8.17.1 Name

‘zerop‘

8.17.2 Class

Function

8.17.3 Syntax

`(zerop number)`

8.17.4 Description

Tests whether a number is zero.

8.17.5 Return Values

- ‘T‘ if the number is zero
- ‘nil‘ otherwise

8.17.6 Side Effects

None.

8.17.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.17.8 Source Notes

- [Arithmetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-FC14467C-63B9-4400-` `htm`)
- Status: documented by Autodesk’s arithmetic reference table.

8.18 Function Entry: MINUSP

8.18.1 Name

‘minusp‘

8.18.2 Class

Function

8.18.3 Syntax

`(minusp number)`

8.18.4 Description

Tests whether a number is negative.

8.18.5 Return Values

- ‘T‘ if the number is negative
- ‘nil‘ otherwise

8.18.6 Side Effects

None.

8.18.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.18.8 Source Notes

- [Arithmetic Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-FC14467C-63B9-4400-` `htm`)
- Status: documented by Autodesk’s arithmetic reference table.

8.19 Dictionary Coverage Index: Arithmetic and Numeric Functions

The Autodesk arithmetic function family includes at least:

- ‘+‘
- ‘-‘
- ‘*‘
- ‘/‘
- ‘~‘
- ‘1+‘
- ‘1-‘
- ‘abs‘
- ‘atan‘
- ‘cos‘
- ‘fix‘
- ‘float‘
- ‘gcd‘
- ‘log‘
- ‘logand‘
- ‘logior‘
- ‘lsh‘
- ‘max‘
- ‘min‘
- ‘minusp‘
- ‘rem‘
- ‘sin‘

- ‘sqrt‘
- ‘zerop‘

The equality and conditional family contributes additional numeric predicates and comparisons:

- ‘=‘
- ‘/=‘
- ‘<‘
- ‘<=‘
- ‘>‘
- ‘>=‘
- ‘boole‘
- ‘eq‘
- ‘equal‘

Sources:

- [Arithmetic Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-FC14467C-63B9-4400-8C6A-266D21C846AE.htm)
- [Equality and Conditional Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C6A7FC58-7A07-4650-9D59-12CAFD194FDA.htm)

8.20 Function Entry: ATAN

8.20.1 Name

‘atan‘

8.20.2 Class

Function

8.20.3 Syntax

```
(atan num1 [num2])
```

8.20.4 Arguments and Values

- ‘num1‘: integer or real.
- ‘num2‘ (optional): integer or real.

8.20.5 Description

Returns the arctangent in radians. With one argument, returns the arctangent of ‘num1‘. With two arguments, returns the arctangent of ‘num1/num2‘ and uses the signs of both arguments to choose the correct quadrant. If ‘num2‘ is zero, the result is + /2 or - /2 depending on the sign of ‘num1‘.

8.20.6 Return Values

A real number in the range [- /2, + /2] for the one-argument form, and in (- , +] for the two-argument form.

8.20.7 Side Effects

None.

8.20.8 Examples

- ‘(atan 1)‘ returns ‘0.785398‘.
- ‘(atan 0.5)‘ returns ‘0.463648‘.
- ‘(atan -1.0)‘ returns ‘-0.785398‘.
- ‘(atan 2.0 3.0)‘ returns ‘0.588003‘.

- ‘(atan 2.0 -3.0)‘ returns ‘2.55359‘.
- ‘(atan 1.0 0.0)‘ returns ‘1.5708‘.

8.20.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.20.10 Source Notes

- [atan (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1A40A16E-8ABE-437D-9888-2F43CEA0B5CE.htm`)
- Status: documented (Phase 3 body fill).

8.21 Function Entry: COS

8.21.1 Name

‘cos‘

8.21.2 Class

Function

8.21.3 Syntax

```
(cos ang)
```

8.21.4 Arguments and Values

- ‘ang‘: integer or real, an angle expressed in radians.

8.21.5 Description

Returns the cosine of ‘ang‘.

8.21.6 Return Values

A real number.

8.21.7 Side Effects

None.

8.21.8 Examples

- ‘(cos 0.0)‘ returns ‘1.0‘.
- ‘(cos pi)‘ returns ‘-1.0‘.

8.21.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.21.10 Source Notes

- [cos (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5F39D690-A986-498D-929A-56BC891E08CD.htm`)
- Status: documented (Phase 3 body fill).

8.22 Function Entry: SIN

8.22.1 Name

‘sin‘

8.22.2 Class

Function

8.22.3 Syntax

`(sin ang)`

8.22.4 Arguments and Values

- ‘ang‘: integer or real, an angle expressed in radians.

8.22.5 Description

Returns the sine of ‘ang‘.

8.22.6 Return Values

A real number.

8.22.7 Side Effects

None.

8.22.8 Examples

- ‘(sin 1.0)‘ returns ‘0.841471‘.
- ‘(sin 0.0)‘ returns ‘0.0‘.

8.22.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.22.10 Source Notes

- [sin (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-83677851-6FA5-496E-A93C-AEA55ABCB527.htm`)
- Status: documented (Phase 3 body fill).

8.23 Function Entry: EXP

8.23.1 Name

‘exp‘

8.23.2 Class

Function

8.23.3 Syntax

`(exp num)`

8.23.4 Arguments and Values

- ‘num‘: integer or real, the exponent to which the natural constant ‘e‘ is raised.

8.23.5 Description

Returns ‘e‘ raised to the power ‘num‘ (the natural antilogarithm).

8.23.6 Return Values

A real number.

8.23.7 Side Effects

None.

8.23.8 Examples

- ‘(exp 1.0)‘ returns ‘2.71828‘.
- ‘(exp 2.2)‘ returns ‘9.02501‘.
- ‘(exp -0.4)‘ returns ‘0.67032‘.

8.23.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.23.10 Source Notes

- [exp (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-FD0C918B-A162-4939-9F4E-FFE8863C3FC8.htm`)
- Status: documented (Phase 3 body fill).

8.24 Function Entry: EXPT

8.24.1 Name

‘expt‘

8.24.2 Class

Function

8.24.3 Syntax

`(expt number power)`

8.24.4 Arguments and Values

- ‘number‘: integer or real, the base.
- ‘power‘: integer or real, the exponent.

8.24.5 Description

Returns ‘number‘ raised to the power ‘power‘.

8.24.6 Return Values

If both arguments are integers, the result is an integer; otherwise the result is a real.

8.24.7 Side Effects

None.

8.24.8 Examples

- ‘(expt 2 4)‘ returns ‘16‘.
- ‘(expt 3.0 2.0)‘ returns ‘9.0‘.

8.24.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: documented under the BricsCAD-only name ‘power‘ **and** presumed to support ‘expt‘. The BricsCAD analogue is recorded separately in chapter 24 — see Phase 4 cross-reference.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.24.10 See Also

- BricsCAD analogue: `power` (chapter 24). AutoCAD users portably write ‘(expt base exp)‘; BricsCAD users may use either.

8.24.11 Source Notes

- [expt (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-FCAC3150-7491-43DB-8042-37EDE3791A96.htm`)
- Status: documented (Phase 3 body fill).

8.25 Function Entry: LOG

8.25.1 Name

‘log‘

8.25.2 Class

Function

8.25.3 Syntax

`(log num)`

8.25.4 Arguments and Values

- ‘num‘: integer or real, a positive number.

8.25.5 Description

Returns the natural logarithm of ‘num‘ (base ‘e‘). The base is fixed; AutoLISP does not provide a multi-argument form.

8.25.6 Return Values

A real number.

8.25.7 Side Effects

None.

8.25.8 Examples

- ‘(log 4.5)‘ returns ‘1.50408‘.
- ‘(log 1.22)‘ returns ‘0.198851‘.

8.25.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible. The BricsCAD-only ‘log10‘ is documented separately in chapter 24.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.25.10 Source Notes

- [log (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4B1B6364-B1CB-4CE7-8C8F-706EA829780F.htm`)
- Status: documented (Phase 3 body fill).

8.26 Function Entry: SQRT

8.26.1 Name

‘sqrt‘

8.26.2 Class

Function

8.26.3 Syntax

`(sqrt num)`

8.26.4 Arguments and Values

- ‘num‘: integer or real, a non-negative number.

8.26.5 Description

Returns the square root of ‘num‘ as a real number.

8.26.6 Return Values

A real number.

8.26.7 Side Effects

None.

8.26.8 Examples

- ‘(sqrt 4)‘ returns ‘2.0‘.
- ‘(sqrt 2.0)‘ returns ‘1.41421‘.

8.26.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.26.10 Source Notes

- [sqrt (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B8D3BA69-95FE-4036-83BA-6AFEA45BADE5.htm`)
- Status: documented (Phase 3 body fill).

8.27 Function Entry: MAX

8.27.1 Name

‘max‘

8.27.2 Class

Function

8.27.3 Syntax

`(max [number number ...])`

8.27.4 Arguments and Values

- ‘number‘: zero or more integers or reals.

8.27.5 Description

Returns the largest of the supplied numbers. With no arguments, returns 0.

8.27.6 Return Values

If any argument is a real, the result is a real; otherwise the result is an integer.

8.27.7 Side Effects

None.

8.27.8 Examples

- ‘(max 4.07 -144)‘ returns ‘4.07‘.
- ‘(max -88 19 5 2)‘ returns ‘19‘.
- ‘(max 2.1 4 8)‘ returns ‘8.0‘.

8.27.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.27.10 Source Notes

- [max (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-057C56D3-8A15-406B-95CA-68DD859E64FB.htm`)
- Status: documented (Phase 3 body fill).

8.28 Function Entry: MIN

8.28.1 Name

‘min‘

8.28.2 Class

Function

8.28.3 Syntax

```
(min [number number ...])
```

8.28.4 Arguments and Values

- ‘number‘: zero or more integers or reals.

8.28.5 Description

Returns the smallest of the supplied numbers. With no arguments, returns 0.

8.28.6 Return Values

If any argument is a real, the result is a real; otherwise the result is an integer.

8.28.7 Side Effects

None.

8.28.8 Examples

- ‘(min 683 -10.0)‘ returns ‘-10.0‘.
- ‘(min 73 2 48 5)‘ returns ‘2‘.
- ‘(min 73.0 2 48 5)‘ returns ‘2.0‘.
- ‘(min 2 4 6.7)‘ returns ‘2.0‘.

8.28.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.28.10 Source Notes

- [min (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F401AD55-57B2-4B1A-9704-D7D8D45AE465.htm`)
- Status: documented (Phase 3 body fill).

8.29 Function Entry: GCD

8.29.1 Name

‘gcd‘

8.29.2 Class

Function

8.29.3 Syntax

`(gcd int1 int2)`

8.29.4 Arguments and Values

- ‘int1‘: integer greater than 0.
- ‘int2‘: integer greater than 0.

8.29.5 Description

Returns the greatest common divisor of ‘int1‘ and ‘int2‘.

8.29.6 Return Values

An integer.

8.29.7 Side Effects

None.

8.29.8 Examples

- ‘(gcd 81 57)‘ returns ‘3‘.
- ‘(gcd 12 20)‘ returns ‘4‘.

8.29.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.29.10 Source Notes

- [gcd (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3F748964-B6D7-4136-92D5-81A7DD02ED4A.htm`)
- Status: documented (Phase 3 body fill).

8.30 Function Entry: REM

8.30.1 Name

‘rem‘

8.30.2 Class

Function

8.30.3 Syntax

```
(rem [number number ...])
```

8.30.4 Arguments and Values

- ‘number‘: zero or more integers or reals.

8.30.5 Description

Divides the first number by the second, and returns the remainder. With more than two arguments, evaluates left to right: the remainder of ‘(rem n1 n2)‘ is divided by ‘n3‘, and so on. With no arguments, returns 0; with a single argument, returns that argument.

8.30.6 Return Values

If any argument is a real, the result is a real; otherwise the result is an integer.

8.30.7 Side Effects

None.

8.30.8 Examples

- ‘(rem 42 12)‘ returns ‘6‘.
- ‘(rem 12.0 16)‘ returns ‘12.0‘.
- ‘(rem 26 7 2)‘ returns ‘1‘ (‘(26 mod 7) = 5‘, then ‘(5 mod 2) = 1‘).

8.30.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: documented under the same name ‘rem‘ and additionally provides ‘mod‘. See Phase 4 cross-reference to BricsCAD ‘mod‘.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.30.10 Source Notes

- [rem (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CC43BEB-3215-41AE-839B-2DCE6354241A.htm`)
- Status: documented (Phase 3 body fill).

8.31 Function Entry: NUMBERP

8.31.1 Name

‘numberp‘

8.31.2 Class

Function

8.31.3 Syntax

`(numberp item)`

8.31.4 Arguments and Values

- ‘item‘: any AutoLISP value (integer, real, string, list, subroutine, ename, T, or nil).

8.31.5 Description

Verifies that ‘item‘ is a numeric value, that is, an integer or a real.

8.31.6 Return Values

Returns ‘T‘ if ‘item‘ is a real or integer; otherwise returns ‘nil‘.

8.31.7 Side Effects

None.

8.31.8 Examples

- ‘(numberp 4)‘ returns ‘T‘.
- ‘(numberp 3.8348)‘ returns ‘T‘.
- ‘(numberp ”Howdy”)‘ returns ‘nil‘.
- After ‘(setq a 123)‘, ‘(numberp a)‘ returns ‘T‘.

8.31.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.31.10 Source Notes

- [numberp (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1E5B591C-F60B-43CB-8CD8-E729D72B1 htm)
- Status: documented (Phase 3 body fill).

8.32 Function Entry: 1+

8.32.1 Name

‘1+‘

8.32.2 Class

Function

8.32.3 Syntax

`(1+ number)`

8.32.4 Arguments and Values

- ‘number‘: integer or real.

8.32.5 Description

Increments ‘number‘ by 1.

8.32.6 Return Values

The input number plus 1. Return type matches input type: integer for integer input, real for real input.

8.32.7 Side Effects

None.

8.32.8 Examples

- ‘(1+ 5)‘ returns ‘6‘.
- ‘(1+ -17.5)‘ returns ‘-16.5‘.

8.32.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.32.10 Source Notes

- [1+ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-EF06615A-5BE5-4A86-96CF-643B0712E43A.htm`)
- Status: documented (Phase 3 body fill).

8.33 Function Entry: 1-

8.33.1 Name

‘1-‘

8.33.2 Class

Function

8.33.3 Syntax

`(1- number)`

8.33.4 Arguments and Values

- ‘number‘: integer or real.

8.33.5 Description

Decrements ‘number‘ by 1.

8.33.6 Return Values

The input number minus 1. Return type matches input type.

8.33.7 Side Effects

None.

8.33.8 Examples

- ‘(1- 5)‘ returns ‘4‘.
- ‘(1- -17.5)‘ returns ‘-18.5‘.

8.33.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.33.10 Source Notes

- [1- (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4E3A882A-959D-4B46-BA5E-5A3C1BDB3079.htm)
- Status: documented (Phase 3 body fill).

8.34 Function Entry: ~

8.34.1 Name

‘~‘

8.34.2 Class

Function

8.34.3 Syntax

`(~ int)`

8.34.4 Arguments and Values

- ‘int‘: a positive or negative integer.

8.34.5 Description

Returns the bitwise NOT (one’s complement) of ‘int‘.

8.34.6 Return Values

An integer.

8.34.7 Side Effects

None.

8.34.8 Examples

- ‘(~ 3)‘ returns ‘-4‘.
- ‘(~ 100)‘ returns ‘-101‘.
- ‘(~ -4)‘ returns ‘3‘.

8.34.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.34.10 Source Notes

- [~ (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-98811E38-FCA8-4ADB-858C-60C3828D7DB4.htm)
- Status: documented (Phase 3 body fill).

8.35 Function Entry: BOOLE

8.35.1 Name

'boole'

8.35.2 Class

Function

8.35.3 Syntax

```
(boole operator int1 [int2 ...])
```

8.35.4 Arguments and Values

- 'operator': integer 0–15, the four-bit truth table identifying which Boolean function to apply (each bit of 'operator' selects one row of the two-input truth table).
- 'int1', 'int2', ...: integers to combine bitwise.

8.35.5 Description

Returns the bitwise Boolean combination of the integer arguments under the truth table identified by 'operator'. Common operator codes include:

- 1: AND (output 1 only when both input bits are 1).
- 6: XOR (output 1 only when exactly one input bit is 1).
- 7: OR (output 1 when at least one input bit is 1).
- 8: NOR (output 1 only when both input bits are 0).

With more than two integers, the operation is applied left-associatively: '(boole op a b c)' is '(boole op (boole op a b) c)'.

8.35.6 Return Values

An integer.

8.35.7 Side Effects

None.

8.35.8 Examples

- ‘(boole 1 12 5)‘ returns ‘3‘ (logical AND of ‘12‘ and ‘5‘).
- ‘(boole 4 3 14)‘ returns ‘12‘ (operator code 4: bits set in ‘int2‘ but not in ‘int1‘).

8.35.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.35.10 Source Notes

- [boole (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C8412B41-BED9-4182-94E6-9EDB3C176` `htm`)
- Status: documented (Phase 3 body fill).

8.36 Function Entry: LOGAND

8.36.1 Name

‘logand‘

8.36.2 Class

Function

8.36.3 Syntax

`(logand [int int ...])`

8.36.4 Arguments and Values

- ‘int‘: zero or more integers.

8.36.5 Description

Returns the bitwise AND of the supplied integers. With no arguments, returns 0.

8.36.6 Return Values

An integer.

8.36.7 Side Effects

None.

8.36.8 Examples

- ‘(logand 7 15 3)‘ returns ‘3‘.
- ‘(logand 2 3 15)‘ returns ‘2‘.
- ‘(logand 8 3 4)‘ returns ‘0‘.

8.36.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.36.10 Source Notes

- [logand (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4E042C63-8BD7-4FED-B9E1-B5CE12D18` `htm`)
- Status: documented (Phase 3 body fill).

8.37 Function Entry: LOGIOR

8.37.1 Name

‘logior‘

8.37.2 Class

Function

8.37.3 Syntax

```
(logior [int int ...])
```

8.37.4 Arguments and Values

- ‘int‘: zero or more integers.

8.37.5 Description

Returns the bitwise inclusive OR of the supplied integers. With no arguments, returns 0.

8.37.6 Return Values

An integer.

8.37.7 Side Effects

None.

8.37.8 Examples

- ‘(logior 1 2 4)‘ returns ‘7‘.
- ‘(logior 9 3)‘ returns ‘11‘.

8.37.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.37.10 Source Notes

- [logior (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-A4BF3B68-4988-42F0-AFE6-BFD06D0DB
htm)
- Status: documented (Phase 3 body fill).

8.38 Function Entry: LSH

8.38.1 Name

‘lsh‘

8.38.2 Class

Function

8.38.3 Syntax

`(lsh int numbits)`

8.38.4 Arguments and Values

- ‘int‘: integer to be shifted.
- ‘numbits‘: integer; positive shifts left, negative shifts right.

8.38.5 Description

Logical bitwise shift of ‘int‘ by ‘numbits‘ positions. Positive ‘numbits‘ shifts left; negative shifts right. Zero bits are shifted in, and bits shifted out are discarded.

8.38.6 Return Values

An integer. The result is positive if bit 31 (the sign bit) is 0 after the shift; otherwise the result is negative.

8.38.7 Side Effects

None.

8.38.8 Examples

- ‘(lsh 2 1)‘ returns ‘4‘.
- ‘(lsh 2 -1)‘ returns ‘1‘.
- ‘(lsh 40 2)‘ returns ‘160‘.

8.38.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

8.38.10 Source Notes

- [lsh (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3BC67533-3F4F-4C01-9053-CD82AA235AE8.htm`)
- Status: documented (Phase 3 body fill).

9 9 Characters and Strings

9.1 Overview

Characters and strings are visible both at reader level and through I/O and string-processing functions.

9.2 Normative Rules

- strings are quoted
- backslash escapes exist
- Unicode semantics changed significantly in AutoCAD 2021 and later

9.3 String Model

Autodesk documentation sampled here supports a semantic view of strings, but not a full implementation-level storage model.

Accordingly, this specification treats a string as:

- an ordered sequence of characters,
- addressable by string-processing functions,
- readable and writable through string and text-I/O facilities,
- subject to version-dependent character encoding behavior.

This specification does **not** currently claim a vendor-documented internal representation such as:

- byte vector,
- UTF-16 code-unit vector,
- Unicode scalar-value vector,
- implementation-specific opaque text object.

At the current evidence level, only the observable behavior is normative.

9.4 Pre-Unicode and Post-Unicode Semantics

9.4.1 Pre-2021 Autodesk Model

Before AutoCAD 2021, Autodesk documents string and character behavior in terms that were effectively tied to legacy text encodings.

In particular, Autodesk's current function notes say that earlier behavior for functions such as ‘ascii‘, ‘chr‘, ‘vl-string-*‘, and ‘read-char‘ reflected ASCII or MBCS-oriented handling rather than full Unicode semantics.

Observable consequences of the pre-2021 model include:

- character-oriented functions were effectively limited by legacy encoding assumptions,
- file text handling depended on older platform encoding behavior,
- string-position and character-code behavior for non-ASCII text was not Unicode-clean in the modern sense.

9.4.2 Post-2021 Autodesk Model

Beginning with AutoCAD 2021, Autodesk explicitly introduced Unicode-related changes controlled by ‘LISPSYS‘.

From the documentation sampled here, the post-2021 model includes at least:

- Unicode-aware string handling,
- revised behavior for string-search and character-code functions,
- revised ‘read-char‘ behavior,
- explicit file-opening encoding support,
- an implementation switch through ‘LISPSYS‘.

‘LISPSYS‘ values are documented as selecting different engine behavior:

- legacy ASCII engine behavior,
- Unicode engine behavior,
- Unicode engine behavior with compatibility adjustments, depending on Autodesk release details.

For the purposes of this specification, the important normative point is:

- string semantics after AutoCAD 2021 are versioned and mode-dependent,
- the language cannot be specified correctly without recording the active Unicode mode.

9.4.3 BricsCAD

BricsCAD documentation sampled here also indicates explicit Unicode-aware text-file handling through extended ‘open‘ modes.

This suggests that:

- BricsCAD should not be modeled purely as a pre-Unicode AutoLISP environment,
- text and encoding behavior must remain part of the dialect/environment profile.

9.5 Representation-Level Notes

At the observable language level, the following can be said:

- strings preserve character order,
- strings are delimited by double quotes in source representation,
- strings participate in search, comparison, concatenation, and I/O functions,
- the meaning of a ”character” is version-sensitive in Unicode-era Autodesk behavior.

At the implementation level, the following remain under-specified in the current source set:

- whether AutoCAD stores strings internally as Unicode code points, UTF-16 code units, or another representation,
- whether string indexing counts code units, Unicode scalar values, grapheme-like user-visible characters, or implementation-defined units in every function,
- whether normalization is performed or preserved,

- the exact interaction between source-file encoding, runtime string representation, and printer behavior in all releases.

Therefore, ‘clautolisp‘ should distinguish:

- external source encoding,
- external file I/O encoding,
- internal string representation,
- observable character/string API semantics.

Only the observable semantics should be treated as normative until stronger evidence is available.

9.6 Strict/Lax Considerations

- strict mode:
  - should avoid undocumented assumptions about internal encoding,
  - should reject or warn about string/character edge cases whose behavior is known to vary across Unicode modes or host versions,
  - should prefer explicitly encoded text I/O where supported.
- lax mode:
  - may accept broader compatibility behavior for legacy text handling,
  - may emulate legacy ASCII/MBCS behavior when targeting pre-2021 Autodesk semantics or equivalent host profiles.

9.7 Implementation Guidance

‘clautolisp‘ should model strings as abstract text objects with versioned observable behavior, not by prematurely committing the specification to a specific Common Lisp string implementation detail.

The implementation should provide at least:

- a legacy compatibility mode for pre-Unicode behavior,
- a Unicode-aware mode corresponding to modern Autodesk behavior,

- explicit control through dialect/environment/profile configuration,
- compatibility tests for character-code, indexing, file I/O, and search functions.

This is especially important because Common Lisp string behavior is implementation-dependent enough that naive reuse could leak incorrect semantics into AutoLISP compatibility.

9.8 Source Notes

- [About Strings (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-D7C33A49-9715-4E9B-9D8A-4C2E7BFC0FE5.htm)
- [LISPSYS](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-Core/files/GUID-1853092D-6E6D-4A06-8956-AD2C3DF203A3.htm)
- [ascii (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-03E1B586-72CB-4AB6-A151-B581AE530318.htm)
- [vl-string-search (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-4E7AE1DA-1ED5-4D96-A7D3-34241
htm)
- [read-char (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-AutoLISP-Reference/files/GUID-7E94BD14-F018-47D0-88DA-2B08DE32DB2C.htm)
- [open (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-089A323F-21FF-4337-99A9-375758E23BA4.htm)
- [read + write text files (BricsCAD)](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/readwritetextfiles.htm)
- Status: documented for major behavioral shift and observable effects; internal representation remains under-specified.

9.9 Function Entry: STRCAT

9.9.1 Name

‘strcat‘

9.9.2 Class

Function

9.9.3 Syntax

```
(strcat [string ...])
```

9.9.4 Arguments and Values

- zero or more strings

9.9.5 Description

Concatenates the argument strings.

9.9.6 Return Values

String.

Autodesk documents that if no arguments are supplied, the result is a zero-length string.

9.9.7 Side Effects

None.

9.9.8 Exceptional Situations

Erroneous if any argument is not acceptable as a string under host semantics.

9.9.9 Examples

- ‘(strcat ”A” ”B”) => ”AB”‘

9.9.10 Compatibility

String behavior is subject to the pre-Unicode / post-Unicode split discussed earlier in this chapter.

9.9.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.9.12 Source Notes

- [strcat (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-LT-AutoLISP-Reference/files/GUID-4430B1BF-DBB5-49D1-98F9-711B480976A htm`)

9.10 Function Entry: STRLEN

9.10.1 Name

‘strlen‘

9.10.2 Class

Function

9.10.3 Syntax

```
(strlen [str ...])
```

9.10.4 Arguments and Values

- ‘str‘ — zero or more strings

9.10.5 Description

Returns the length of the string.

9.10.6 Return Values

Integer.

Autodesk documents:

- the sum of the lengths when multiple arguments are supplied
- ‘0‘ if no arguments are supplied
- ‘0‘ for the empty string

9.10.7 Side Effects

None.

9.10.8 Compatibility

Autodesk explicitly documents the Unicode-era counting change.

9.10.9 Notes

This function is one of the places where internal representation and observable character counting must not be conflated.

Autodesk documents that AutoCAD 2021 changed the function so Unicode characters are counted correctly when ‘LISPSYS‘ is ‘1‘ or ‘2‘.

9.10.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.10.11 Source Notes

- [strlen (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F16AAC5F-5C87-4DC5-A7B9-BDCD25DC507A.htm`)

9.11 Function Entry: SUBSTR

9.11.1 Name

‘substr‘

9.11.2 Class

Function

9.11.3 Syntax

```
(substr string start [length])
```

9.11.4 Arguments and Values

- ‘string‘ — string
- ‘start‘ — positive integer starting position
- ‘length‘ — optional positive integer substring length

9.11.5 Description

Returns a substring of ‘string‘.

9.11.6 Return Values

String.

9.11.7 Side Effects

None.

9.11.8 Compatibility

String indexing and character-unit semantics are version-sensitive in Unicode-era environments.

9.11.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.11.10 Source Notes

- [substr (AutoLISP)](`https://help.autodesk.com/cloudhelp/2015/ESP/AutoCAD-AutoLISP/files/GUID-36F8701B-7AB7-47BE-AC31-8508A2DF46A2.htm`)

9.11.11 Notes

- Autodesk documents 1-based indexing: the first character of the string is position '1'.
- If 'length' is omitted, Autodesk documents that the substring extends to the end of the string.

9.12 Function Entry: STRCASE

9.12.1 Name

‘strcase‘

9.12.2 Class

Function

9.12.3 Syntax

```
(strcase string [which])
```

9.12.4 Arguments and Values

- ‘string‘ — string
- ‘which‘ — optional ‘T‘ or ‘nil‘

9.12.5 Description

Returns a string converted in case according to the supplied option.

9.12.6 Return Values

String.

9.12.7 Side Effects

None.

9.12.8 Compatibility

Case-conversion rules for non-ASCII characters are potentially version-sensitive in Unicode-aware modes.

9.12.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.12.10 Source Notes

- [strcase (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-108DCD2C-6597-4548-856D-937787AFE5E0.htm)

9.12.11 Notes

- Autodesk documents that ‘which‘ equal to ‘T‘ requests lowercase conversion; otherwise uppercase conversion is used.
- Autodesk documents that case mapping follows the currently configured character set.

9.13 Function Entry: ASCII

9.13.1 Name

‘ascii‘

9.13.2 Class

Function

9.13.3 Syntax

`(ascii string)`

9.13.4 Arguments and Values

- ‘string‘: string whose first character is examined

9.13.5 Description

Returns the code of the first character in the string.

9.13.6 Return Values

Integer character code.

9.13.7 Side Effects

None.

9.13.8 Compatibility

Autodesk explicitly documents behavior changes after AutoCAD 2021 due to Unicode support.

9.13.9 Notes

This function is one of the clearest probes into the observable character model of a given host/version.

9.13.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.13.11 Source Notes

- [ascii (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-03E1B586-72CB-4AB6-A151-B581AE530318.htm`)
- Status: documented.

9.14 Function Entry: CHR

9.14.1 Name

‘chr‘

9.14.2 Class

Function

9.14.3 Syntax

```
(chr integer)
```

9.14.4 Arguments and Values

- ‘integer‘: character code

Autodesk documents the accepted range as ‘1‘ to ‘65536‘ in Unicode-capable behavior.

9.14.5 Description

Returns a one-character string corresponding to the given character code.

9.14.6 Return Values

String of length one under ordinary observable semantics.

9.14.7 Side Effects

None.

9.14.8 Compatibility

Like ‘ascii‘, ‘chr‘ is affected by the pre-Unicode / post-Unicode split.

9.14.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.14.10 Source Notes

- [chr (AutoLISP)](https://help.autodesk.com/cloudhelp/2025/ESP/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5846BFDD-AE5F-4B2F-99E0-0B9DBA667739.htm)

9.14.11 Notes

- Autodesk documents that AutoCAD 2021 changed ‘chr‘ from legacy ASCII-oriented behavior to Unicode-code-point-oriented behavior when ‘LISPSYS‘ is ‘1‘ or ‘2‘.

9.15 Function Entry: ATOI

9.15.1 Name

‘atoi‘

9.15.2 Class

Conversion Function

9.15.3 Syntax

`(atoi string)`

9.15.4 Arguments and Values

- ‘string‘: a string representing any number.

9.15.5 Description

Converts ‘string‘ into an integer. Trailing non-numeric characters terminate parsing; the integer prefix is returned as the value. Decimal-fraction input is truncated toward zero.

9.15.6 Return Values

An integer; ‘0‘ if the string is not a valid number.

9.15.7 Side Effects

None.

9.15.8 Examples

- ‘(atoi ”12”)‘ returns ‘12‘.
- ‘(atoi ”12.34”)‘ returns ‘12‘ (truncating decimal input).
- ‘(atoi ”xyz”)‘ returns ‘0‘.

9.15.9 Implementation-defined Behaviour

The Phase-5 vendor citations below define the documented surface. Several lexical edges are not nailed down by either vendor's prose and remain implementation-defined unless and until product tests exist:

- Leading whitespace handling (‘(atoi " 17")‘).
- Sign acceptance (‘(atoi "+17")‘, ‘(atoi "-17")‘).
- Behaviour of leading-zero forms (‘(atoi "017")‘ — decimal vs. octal-like).
- Behaviour of ‘0x‘-prefixed forms (‘(atoi "0xbabe")‘).
- Trailing-junk strictness (‘(atoi "17x")‘).

For `clautolisp` the chosen lex model is: skip leading whitespace, accept optional sign, parse the longest run of decimal digits, truncate the fractional part if a ‘.‘ follows the digit run, treat all leading-zero forms as base 10, treat ‘0x‘-prefixed forms as ‘0‘ (no autodetection). The probe suite in `probe-atoi.lsp` records the AutoCAD / BricsCAD product behaviour when run via `scripts/run-probes.sh`.

9.15.10 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

The probe suite was run against BricsCAD V26 on macOS via the direct runner. Results from ‘results/bricscad/macos/20260426T122808Z/results.sexp‘:

Probe	Returned
‘(atoi "17")‘	‘17‘
‘(atoi "017")‘	‘17‘ (decimal interpretation; **not** octal)
‘(atoi "0xbabe")‘	‘0‘ (‘0x‘-prefix rejected)
‘(atoi "+17")‘	‘17‘ (leading ‘+‘ accepted)
‘(atoi "-17")‘	‘-17‘ (leading ‘-‘ accepted)
‘(atoi "17x")‘	‘17‘ (parsing stops at first non-digit)
‘(atoi " 17")‘	‘17‘ (leading whitespace accepted)
‘(atoi "3.9")‘	‘3‘ (decimal-fraction truncates toward zero)

Confirms a ‘strtol(..., 10)‘-style C-library model: skip leading whitespace, optional sign, longest decimal-digit prefix, return zero on no digits. The tested behaviour matches ‘clautolisp‘'s implementation choice, so the lex edges previously marked **implementation-defined** are now **tested** on BricsCAD V26.

9.15.11 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26).
- Status: documented in both vendors; lex-edge behaviour now tested against BricsCAD V26 / macOS. AutoCAD product test deferred until a local install becomes available.

9.15.12 See Also

- probe-atoi.lsp — executable Phase-5 probe suite.
- `itoa` — integer-to-string companion.

9.15.13 Source Notes

- [atoi (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-20EF237C-7079-4E29-860C-B8531D6C7F36.htm`)
- [atoi (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/atoi.htm`)
- Status: documented (Phase 5 closure via vendor citation; product-test coverage deferred until a CAD product is available locally).

9.16 Function Entry: ATOF

9.16.1 Name

‘atof‘

9.16.2 Class

Conversion Function

9.16.3 Syntax

`(atof string)`

9.16.4 Arguments and Values

- ‘string‘: a string representing any number.

9.16.5 Description

Converts ‘string‘ into a double-precision real. A trailing non-numeric tail terminates parsing; the numeric prefix is returned.

9.16.6 Return Values

A double-precision real; ‘0.0‘ if the string is not a valid number.

9.16.7 Side Effects

None.

9.16.8 Examples

- ‘(atof ”12”)‘ returns ‘12.0‘.
- ‘(atof ”12.34”)‘ returns ‘12.34‘.
- ‘(atof ”xyz”)‘ returns ‘0.0‘.

9.16.9 Implementation-defined Behaviour

The Phase-5 vendor citations below define the documented surface. Several lexical edges are not nailed down by either vendor’s prose and remain implementation-defined unless and until product tests exist:

- ‘.5‘, ‘1.‘ — leading/trailing decimal-point variants.
- Exponent notation (‘1e3‘, ‘1E+03‘).
- Leading whitespace handling.
- Locale sensitivity, in particular decimal-comma vs decimal-point.
- Acceptance of hexadecimal floating syntax (‘0x1p4‘).
- Trailing-junk strictness.

For `clautolisp` the chosen lex model is: skip leading whitespace, accept optional sign, accept the longest prefix matching ‘digits (’.’ digits?)? ((’e’|’E’) [+-]? digits)¿, no locale sensitivity (decimal-point only), no hexadecimal floating syntax. The probe suite in `probe-atof.lsp` records the AutoCAD / BricsCAD product behaviour when run via `scripts/run-probes.sh`.

9.16.10 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

Probe results from ‘results/bricscad/macos/20260426T122808Z/results.sexp‘:

Probe	Returned	Note
‘(atof ”3.5”)‘	‘3.5‘	baseline
‘(atof ”.5”)‘	‘0.5‘	leading-dot accepted
‘(atof ”1.”)‘	‘1.0‘	trailing-dot accepted
‘(atof ”1e3”)‘	‘1000.0‘	exponent accepted
‘(atof ”017”)‘	‘17.0‘	leading zero treated as decimal
‘(atof ”0x1p4”)‘	‘16.0‘	**hexadecimal floating syntax accepted** — BricsCAD V26 delegates
‘(atof ”3,5”)‘	‘3.0‘	decimal-comma rejected; parsing stops at the comma (no locale sensiti
‘(atof ” 3.5”)‘	‘3.5‘	leading whitespace accepted

The ‘0x1p4 → 16.0‘ result is the most significant finding: BricsCAD V26 **does** accept hex-float input. The conservative ‘clautolisp‘ lex model rejects it; conformant code that targets BricsCAD V26 may expand the lex model to include hex-floats. Locale sensitivity is **not** present on the macOS build; portable AutoLISP code should not rely on locale.

9.16.11 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26).
- Status: documented in both vendors; lex-edge behaviour now tested against BricsCAD V26 / macOS. AutoCAD product test deferred until a local install becomes available.

9.16.12 See Also

- probe-atof.lsp — executable Phase-5 probe suite.
- `rtos` — real-to-string companion.

9.16.13 Source Notes

- [atof (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8618A388-E4CF-40E1-813B-057367DD1840.htm`)
- [atof (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/atof.htm`)
- Status: documented (Phase 5 closure via vendor citation; product-test coverage deferred until a CAD product is available locally).

9.17 Function Entry: ITOA

9.17.1 Name

‘itoa‘

9.17.2 Class

Conversion Function

9.17.3 Syntax

`(itoa integer)`

9.17.4 Arguments and Values

- ‘integer‘ — integer

9.17.5 Description

Converts an integer to its string representation.

9.17.6 Return Values

String.

9.17.7 Side Effects

None.

9.17.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.17.9 Source Notes

- [itoa (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-7E247CA3-95D3-4497-BDE2-6EB0E727DD4D.htm`)

9.18 Function Entry: RTOS

9.18.1 Name

‘rtos‘

9.18.2 Class

Conversion Function

9.18.3 Syntax

```
(rtos number [mode [precision]])
```

9.18.4 Arguments and Values

- ‘number‘: number
- ‘mode‘: optional conversion mode
- ‘precision‘: optional precision

9.18.5 Description

Converts a real or number to a string according to AutoCAD formatting conventions.

9.18.6 Return Values

String.

9.18.7 Side Effects

None.

9.18.8 Affected By

- numeric formatting settings
- conversion mode and precision

9.18.9 Compatibility

This function is part of the language/host formatting boundary, not purely a generic printer primitive.

9.18.10 Notes

- Autodesk documents the accepted ‘mode‘ values as the linear-unit formats corresponding to ‘LUNITS‘:
  - ‘1‘ scientific
  - ‘2‘ decimal
  - ‘3‘ engineering
  - ‘4‘ architectural
  - ‘5‘ fractional
- If ‘mode‘ and ‘precision‘ are omitted, Autodesk documents that ‘rtos‘ uses the current ‘LUNITS‘ and ‘LUPREC‘ settings.
- Autodesk documents that the resulting string is affected by ‘UNITMODE‘ for engineering, architectural, and fractional output, and by ‘DIMZIN‘ for zero suppression.

9.18.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.18.12 Source Notes

- [rtos (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D03ABBC2-939A-44DB-8C93-FC63B64DE4A2.htm`)
- [About String Conversions (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-AutoLISP/files/GUID-A38E30D4-684E-44FA-A910-E08522876`
`htm`)

9.19 Function Entry: ANGTOF

9.19.1 Name

‘angtof‘

9.19.2 Class

Conversion Function

9.19.3 Syntax

```
(angtof string [unit])
```

9.19.4 Arguments and Values

- ‘string‘ — string
- ‘unit‘ — optional integer angular-unit selector

9.19.5 Description

Converts an angle string to a numeric angle value.

9.19.6 Return Values

Real angle value if successful; otherwise ‘nil‘.

9.19.7 Compatibility

Part of host-oriented numeric formatting and parsing semantics.

9.19.8 Notes

- Autodesk documents the accepted ‘unit‘ values as those of ‘AUNITS‘:
  - ‘0‘ degrees
  - ‘1‘ degrees/minutes/seconds
  - ‘2‘ grads
  - ‘3‘ radians
  - ‘4‘ surveyor’s units

- If ‘unit‘ is omitted, Autodesk documents that ‘angtof‘ uses the current ‘AUNITS‘ setting.
- Autodesk documents ‘angtof‘ and ‘angtos‘ as complementary functions when matching unit conventions are used.

9.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.19.10 Source Notes

- [angtof (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9DE9200B-F27E-4BB6-8AE6-BC5C8A64A527.htm`)
- [About Angular Conversion (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP/files/GUID-725BC3DE-3D4C-4365-A7F9-FA htm`)

9.20 Function Entry: ANGTOS

9.20.1 Name

‘angtos‘

9.20.2 Class

Conversion Function

9.20.3 Syntax

```
(angtos angle [unit [precision]])
```

9.20.4 Arguments and Values

- ‘angle‘ — integer or real, interpreted in radians
- ‘unit‘ — optional integer angular-unit selector
- ‘precision‘ — optional integer precision

9.20.5 Description

Converts a numeric angle value to a formatted string.

9.20.6 Return Values

A string if successful; otherwise ‘nil‘.

9.20.7 Affected By

- angle formatting conventions
- unit and precision parameters

9.20.8 Notes

- Autodesk documents the accepted ‘unit‘ values as those of ‘AUNITS‘:
  - ‘0‘ degrees
  - ‘1‘ degrees/minutes/seconds
  - ‘2‘ grads
  - ‘3‘ radians

  - ‘4‘ surveyor’s units
- If ‘unit‘ is omitted, Autodesk documents that ‘angtos‘ uses the current ‘AUNITS‘ value.
- If ‘precision‘ is omitted, Autodesk documents that ‘angtos‘ uses the current ‘AUPREC‘ value.
- Autodesk documents that the result is additionally affected by ‘UNITMODE‘ and ‘DIMZIN‘.
- Autodesk documents that negative angles are reduced to an equivalent positive angle between ‘0‘ and ‘2*pi‘ before formatting.

9.20.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.20.10 Source Notes

- [angtos (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-C2D277B4-2F59-423F-9C56-E5A1DEFE0CC htm`)
- [About Angular Conversion (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP/files/GUID-725BC3DE-3D4C-4365-A7F9-FA htm`)

9.21 Function Entry: DISTOF

9.21.1 Name

‘distof‘

9.21.2 Class

Conversion Function

9.21.3 Syntax

```
(distof string [mode])
```

9.21.4 Arguments and Values

- ‘string‘ — string
- ‘mode‘ — optional integer linear-unit selector

9.21.5 Description

Converts a distance string to a numeric distance value.

9.21.6 Return Values

Real number if successful; otherwise ‘nil‘.

9.21.7 Compatibility

Part of host-oriented unit parsing semantics.

9.21.8 Notes

- Autodesk documents the accepted ‘mode‘ values as those of ‘LUNITS‘:
  - ‘1‘ scientific
  - ‘2‘ decimal
  - ‘3‘ engineering
  - ‘4‘ architectural
  - ‘5‘ fractional

- Autodesk documents ‘distof‘ and ‘rtos‘ as complementary functions when the same mode conventions are used.
- Autodesk documents that modes ‘3‘ and ‘4‘ are parsed equivalently: engineering and architectural input forms are both accepted for either of those mode values.

9.21.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.21.10 Source Notes

- [distof (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-CC0490B9-29BC-44C4-A317-5BAA34B0168E.htm`)
- [About String Conversions (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-AutoLISP/files/GUID-A38E30D4-684E-44FA-A910-E08522876 htm`)

9.22 Dictionary Coverage Index: String and Character Functions

The Autodesk string-handling and related conversion families include at least:

- ‘read‘
- ‘strcase‘
- ‘strcat‘
- ‘strlen‘
- ‘substr‘
- ‘vl-prin1-to-string‘
- ‘vl-princ-to-string‘

- ‘vl-string->list‘
- ‘vl-string-elt‘
- ‘vl-string-left-trim‘
- ‘vl-string-mismatch‘
- ‘vl-string-position‘
- ‘vl-string-right-trim‘
- ‘vl-string-search‘
- ‘vl-string-subst‘
- ‘vl-string-translate‘
- ‘vl-string-trim‘
- ‘ascii‘
- ‘chr‘
- ‘itoa‘
- ‘rtos‘
- ‘angtof‘
- ‘angtos‘
- ‘atof‘
- ‘atoi‘
- ‘distof‘

Sources:

- [String-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8543549B-F0D7-43CD-` `htm`)
- [Conversion Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/KOR/AutoCAD-AutoLISP-Reference/files/GUID-5BDBD4BC-289E-4A49-` `htm`)

9.23 Function Entry: WCMATCH

9.23.1 Name

‘wcmatch‘

9.23.2 Class

Function

9.23.3 Syntax

```
(wcmatch str pattern)
```

9.23.4 Arguments and Values

- ‘str‘: string to test. Comparison is case-sensitive.
- ‘pattern‘: string containing wild-card pattern-match characters.

9.23.5 Description

Performs a wild-card pattern match. Only the first ~500 characters of both arguments are compared. The supported pattern syntax is:

Pattern char	Matches
‘#‘	a single numeric digit
‘@‘	a single alphabetic character
‘.‘	a single non-alphanumeric character
‘*‘	any character sequence, including empty
‘¿	a single character
‘~‘	negation (must be the first character of the pattern)
‘[...]‘	any one of the enclosed characters
‘[~ ...]‘	any character not in the enclosed set
‘-‘	character range, used inside ‘[ ]‘
‘,‘	separates multiple pattern alternatives
“ ‘ “	escape: take the next character literally

The ‘,‘ separator gives ‘wcmatch‘ an OR-of-patterns semantics: the result is ‘T‘ if any of the comma-separated patterns matches.

9.23.6 Return Values

Returns ‘T‘ when the pattern matches, otherwise ‘nil‘.

9.23.7 Side Effects

None.

9.23.8 Examples

- ‘(wcmatch ”Name” ”N*”)‘ returns ‘T‘.
- ‘(wcmatch ”Name” ”???,~*m*,N*”)‘ returns ‘T‘ — three alternatives joined by ‘,‘.

9.23.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

9.23.10 Source Notes

- [wcmatch (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-EC257AF7-72D4-4B38-99B6-9B09952A5` `htm`)
- Status: documented (Phase 3 body fill).

10 10 Lists, Dotted Pairs, and Association Lists

10.1 Overview

Lists are both code and data. Dotted pairs and association lists are especially important because they carry host entity and table data.

10.2 Normative Rules

- proper lists and dotted pairs are both first-class
- association lists are pervasive in CAD interaction
- points are lists, not a separate primitive vector type

10.3 Source Notes

- [About Lists (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-F8074AA9-F361-4960-B628-681465B74229.htm`)
- [About Dotted Pairs (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/PLK/AutoCAD-AutoLISP/files/GUID-877B87BA-6FA2-4894-9F94-184E7DF25B7D.htm`)

10.4 Function Entry: CAR

10.4.1 Name

‘car‘

10.4.2 Class

Function

10.4.3 Syntax

`(car lst)`

10.4.4 Arguments and Values

- ‘lst‘: list

10.4.5 Description

Returns the first element of a list.

10.4.6 Return Values

The first element of ‘lst‘.

If ‘lst‘ is the empty list, ‘car‘ returns ‘nil‘.

10.4.7 Side Effects

None.

10.4.8 Exceptional Situations

Erroneous if ‘lst‘ is not accepted as a list by the implementation.

10.4.9 Examples

- ‘(car ’(a b c)) => A‘
- ‘(car ’((a b) c)) => (A B)‘
- ‘(car ’()) => nil‘

10.4.10 Compatibility

Autodesk's dedicated page documents ‘car‘ for lists and explicitly documents empty-list behavior.

That page does not explicitly state dotted-pair behavior. Implementations may support ‘(car '(a . b)) => A‘ by Lisp tradition, but this specification treats dotted-pair acceptance by ‘car‘ as inferred or compatibility behavior until vendor evidence is collected.

10.4.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.4.12 Source Notes

- [car (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2DD1AF33-415C-4C1A-9631-DA958134C53A.htm)
- Status: documented for lists and empty-list behavior; dotted-pair behavior remains inferred.

10.5 Function Entry: CDR

10.5.1 Name

‘cdr‘

10.5.2 Class

Function

10.5.3 Syntax

`(cdr lst)`

10.5.4 Arguments and Values

- ‘lst‘: list, including the empty list; dotted pairs are also explicitly covered by Autodesk’s note

10.5.5 Description

Returns all but the first element of the specified list.

10.5.6 Return Values

If ‘lst‘ is a proper list with two or more elements, returns a list containing all elements of ‘lst‘ except the first.

If ‘lst‘ is the empty list, returns ‘nil‘.

If ‘lst‘ is a dotted pair, returns the second element without enclosing it in a list.

10.5.7 Side Effects

None.

10.5.8 Examples

- ‘(cdr ’(a b c)) => (B C)‘
- ‘(cdr ’((a b) c)) => (C)‘
- ‘(cdr ’()) => nil‘
- ‘(cdr ’(a . b)) => B‘

- ‘(cdr ’(1 . ”Text”)) => ”Text”‘

10.5.9 Compatibility

Unlike ‘car‘, Autodesk explicitly documents dotted-pair behavior for ‘cdr‘.

10.5.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.5.11 Source Notes

- [cdr (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-F9CD8FF3-022A-4323-BAE7-390174451537.htm`)
- Status: documented.

10.6 Function Entry: CONS

10.6.1 Name

‘cons‘

10.6.2 Class

Function

10.6.3 Syntax

`(cons new-first-element lst-or-atom)`

10.6.4 Arguments and Values

- ‘new-first-element‘: any value
- ‘lst-or-atom‘: list or atom

10.6.5 Description

Adds an element to the beginning of a list, or constructs a dotted pair.

10.6.6 Return Values

- a proper list if the second argument is a list
- a dotted pair if the second argument is an atom

10.6.7 Side Effects

None.

10.6.8 Examples

- ‘(cons ’a ’(b c)) => (A B C)‘
- ‘(cons ’(a) ’(b c d)) => ((A) B C D)‘
- ‘(cons ’a 2) => (A . 2)‘

10.6.9 Compatibility

This entry is central for understanding the list-versus-dotted-pair model of AutoLISP data.

10.6.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.6.11 Source Notes

- [cons (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-33B418E7-DB3D-4CBE-954E-F070F0A7CB2B.htm`)
- Status: documented.

10.7 Function Entry: LIST

10.7.1 Name

‘list‘

10.7.2 Class

Function

10.7.3 Syntax

`(list [expr ...])`

10.7.4 Arguments and Values

- zero or more expressions

10.7.5 Description

Evaluates its arguments and constructs a list containing their values.

10.7.6 Return Values

A list containing the argument values.

If no expressions are supplied, ‘list‘ returns ‘nil‘.

10.7.7 Side Effects

Side effects are those of evaluating the argument expressions.

10.7.8 Examples

- ‘(list ’a ’b ’c) => (A B C)‘
- ‘(list ’a ’(b c) ’d) => (A (B C) D)‘
- ‘(list 1 2 3) => (1 2 3)‘
- ‘(list) => nil‘

10.7.9 Notes

Autodesk explicitly notes that ‘list‘ is frequently used to construct 2D or 3D point values.

The dedicated page also notes that a quoted list literal may be used instead of ‘list‘ when there are no variables or undefined items in the list.

10.7.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.7.11 Source Notes

- [list (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2BA66308-DE64-4BA6-8F57-6E3EC1A93141.htm`)
- Status: documented.

10.8 Function Entry: APPEND

10.8.1 Name

‘append‘

10.8.2 Class

Function

10.8.3 Syntax

`(append list [list ...])`

10.8.4 Arguments and Values

- one or more lists

10.8.5 Description

Returns a list containing the elements of all argument lists in order.

10.8.6 Return Values

A list containing the concatenated elements.

If no arguments are supplied, Autodesk documents that ‘append‘ returns ‘nil‘.

10.8.7 Side Effects

None specified.

10.8.8 Exceptional Situations

Erroneous if arguments are not acceptable list values.

10.8.9 Examples

- ‘(append ’(a b c) ’(d e f)) => (A B C D E F)‘
- ‘(append ’((a) (b)) ’((c) (d))) => ((A) (B) (C) (D))‘

10.8.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.8.11 Source Notes

- [append (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-952B175A-D565-43A4-9208-0E6A27A5E742.htm`)
- Status: documented.

10.9 Function Entry: ASSOC

10.9.1 Name

‘assoc‘

10.9.2 Class

Function

10.9.3 Syntax

`(assoc item alist)`

10.9.4 Arguments and Values

- ‘item‘: key to search for
- ‘alist‘: association list

10.9.5 Description

Searches an association list for a dotted pair whose first element matches ‘item‘.

10.9.6 Return Values

Returns the matching dotted pair or ‘nil‘.

10.9.7 Side Effects

None.

10.9.8 Examples

- ‘(assoc ’size ’((name box) (width 3) (size 4.7263) (depth 5))) => (SIZE 4.7263)‘
- ‘(assoc ’weight ’((name box) (width 3) (size 4.7263) (depth 5))) => nil‘

10.9.9 Compatibility

This is a central function in CAD database idioms because entity and table data are commonly represented as alists.

The dedicated Autodesk page describes ‘item‘ as an integer, real, or string, but its own examples also use symbols. This inconsistency should be kept visible and eventually tested.

10.9.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.9.11 Source Notes

- [assoc (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-46309786-DAF6-4C28-8448-599FBC8A4F6A.htm`)
- Status: documented, with noted item-type inconsistency between prose and examples.

10.10 Function Entry: LAST

10.10.1 Name

‘last‘

10.10.2 Class

Function

10.10.3 Syntax

`(last lst)`

10.10.4 Description

Returns the last element or last cons structure of a list, according to AutoLISP list conventions.

10.10.5 Return Values

Returns the last element in the list.

Autodesk’s examples show that if the last element is itself a list, that list is returned as the last element.

10.10.6 Side Effects

None.

10.10.7 Examples

- ‘(last ’(a b c d e)) => E‘
- ‘(last ’(a b c (d e))) => (D E)‘

10.10.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.10.9 Source Notes

- [last (AutoLISP)](https://help.autodesk.com/cloudhelp/2025/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-463C97F2-B72F-44C6-B19B-659D231AA836.htm)
- Status: documented.

10.11 Function Entry: LENGTH

10.11.1 Name

‘length‘

10.11.2 Class

Function

10.11.3 Syntax

`(length item)`

10.11.4 Arguments and Values

- ‘item‘: list

10.11.5 Description

Returns the number of elements in a list.

10.11.6 Return Values

Integer length.

10.11.7 Side Effects

None.

10.11.8 Examples

- ‘(length ’(a b c d)) => 4‘
- ‘(length ’(a b (c d))) => 3‘
- ‘(length ’()) => 0‘

10.11.9 Compatibility

The dedicated Autodesk page specifies ‘length‘ for lists. Acceptance of strings or other sequence-like values should not be assumed without further evidence.

10.11.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.11.11 Source Notes

- [length (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ITA/AutoCAD-AutoLISP-Reference/files/GUID-DD227383-1895-46B6-A83B-43000FCC1018.htm`)
- Status: documented.

10.12 Function Entry: MAPCAR

10.12.1 Name

‘mapcar‘

10.12.2 Class

Function

10.12.3 Syntax

```
(mapcar function list [list ...])
```

10.12.4 Arguments and Values

- ‘function‘: function designator
- one or more lists

10.12.5 Description

Applies ‘function‘ element-wise across the supplied lists and returns a list of results.

10.12.6 Return Values

List of mapped results.

10.12.7 Side Effects

Side effects are those of applying the supplied function to the elements.

10.12.8 Exceptional Situations

Autodesk documents that the function must accept as many arguments as there are list arguments to ‘mapcar‘.

10.12.9 Examples

- ‘(mapcar ’1+ (list a b c)) => (11 21 31)‘ in Autodesk’s documented example context
- ‘(mapcar ’(lambda (x) (+ x 3)) ’(10 20 30)) => (13 23 33)‘

10.12.10 Compatibility

Closely tied to the semantics of ‘function‘, ‘lambda‘, and function-valued designators.

10.12.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.12.12 Source Notes

- [mapcar (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8802AE73-1A05-457E-8A51-09677C23E26E.htm`)
- Status: documented.

10.13 Function Entry: MEMBER

10.13.1 Name

‘member‘

10.13.2 Class

Function

10.13.3 Syntax

`(member item lst)`

10.13.4 Arguments and Values

- ‘item‘: search target
- ‘lst‘: list

10.13.5 Description

Searches a list for ‘item‘.

10.13.6 Return Values

Returns the remainder of the list, starting with the first occurrence of ‘item‘, or ‘nil‘ if no match is found.

10.13.7 Side Effects

None.

10.13.8 Examples

- ‘(member ’c ’(a b c d e)) => (C D E)‘
- ‘(member ’q ’(a b c d e)) => nil‘

10.13.9 Compatibility

Detailed matching predicate semantics still need tighter source coverage.

10.13.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.13.11 Source Notes

- [member (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-A2B08751-D966-44F5-9B02-1AAC4DA6AF59.htm`)
- Status: documented.

10.14 Function Entry: NTH

10.14.1 Name

‘nth‘

10.14.2 Class

Function

10.14.3 Syntax

`(nth n lst)`

10.14.4 Arguments and Values

- ‘n‘: index
- ‘lst‘: list

10.14.5 Description

Returns the element at index ‘n‘ in ‘lst‘.

10.14.6 Return Values

Element at the specified position, or ‘nil‘ if ‘n‘ is greater than the highest element number of the list.

10.14.7 Side Effects

None.

10.14.8 Examples

- ‘(nth 3 ’(a b c d e)) => D‘
- ‘(nth 0 ’(a b c d e)) => A‘
- ‘(nth 5 ’(a b c d e)) => nil‘

10.14.9 Compatibility

Autodesk explicitly documents zero-based indexing for ‘nth‘.

10.14.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.14.11 Source Notes

- [nth (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-0330FE17-6E15-4E34-BB50-E9040EABDADB.htm`)
- Status: documented.

10.15 Function Entry: REVERSE

10.15.1 Name

‘reverse‘

10.15.2 Class

Function

10.15.3 Syntax

`(reverse lst)`

10.15.4 Description

Returns a list containing the elements of ‘lst‘ in reverse order.

10.15.5 Return Values

Reversed list.

10.15.6 Side Effects

None.

10.15.7 Examples

- ‘(reverse ’((a) b c)) => (C B (A))‘

10.15.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.15.9 Source Notes

- [reverse (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-7669E8F1-2A4F-42C7-AAA8-74D1300F97 htm)
- Status: documented.

10.16 Function Entry: SUBST

10.16.1 Name

‘subst‘

10.16.2 Class

Function

10.16.3 Syntax

`(subst new old expr)`

10.16.4 Arguments and Values

- ‘new‘: replacement value
- ‘old‘: value to replace
- ‘expr‘: tree/list structure

10.16.5 Description

Substitutes ‘new‘ for occurrences of ‘old‘ within ‘expr‘.

10.16.6 Return Values

Returns a copy of the list with ‘new‘ substituted for every occurrence of ‘old‘.

If ‘old‘ is not found, Autodesk documents that the original list is returned unchanged.

10.16.7 Side Effects

None specified.

10.16.8 Examples

- ‘(subst 'qq 'b '(a b (c d) b)) => (A QQ (C D) QQ)‘
- ‘(subst '(qq rr) '(c d) '(a b (c d) b)) => (A B (QQ RR) B)‘

10.16.9 Compatibility

Exact structural sharing versus reconstruction behavior remains to be tightened from dedicated documentation or tests.

10.16.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.16.11 Source Notes

- [subst (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-25214E69-090A-45C3-8210-6D9801255E44.htm`)
- Status: documented.

10.17 Function Entry: LISTP

10.17.1 Name

'listp'

10.17.2 Class

Function

10.17.3 Syntax

```
(listp item)
```

10.17.4 Arguments and Values

- 'item': any AutoLISP value

10.17.5 Description

Determines whether 'item' is a list.

10.17.6 Return Values

Returns 'T' if 'item' is a list; otherwise returns 'nil'.

Autodesk explicitly documents that because 'nil' is both an atom and a list, '(listp nil)' returns 'T'.

10.17.7 Side Effects

None.

10.17.8 Examples

- '(listp '(a b c)) => T'
- '(listp 'a) => nil'
- '(listp 4.343) => nil'
- '(listp nil) => T'

10.17.9 Notes

This entry is one of the clearest vendor statements that ‘nil‘ denotes the empty list in AutoLISP value semantics.

10.17.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.17.11 Source Notes

- [listp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-8755F083-D3BA-45F0-BBAF-D60D3A73240C.htm`)
- Status: documented.

10.18 Function Entry: VL-CONSP

10.18.1 Name

‘vl-consp‘

10.18.2 Class

Visual LISP Predicate

10.18.3 Syntax

```
(vl-consp obj)
```

10.18.4 Arguments and Values

- ‘obj‘: any AutoLISP value

10.18.5 Description

Tests whether ‘obj‘ is a non-‘nil‘ list/cons-like structure under Visual LISP semantics.

10.18.6 Return Values

- ‘T‘ if ‘obj‘ is a cons-like list structure
- ‘nil‘ otherwise

10.18.7 Side Effects

None.

10.18.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.18.9 Source Notes

- [vl-consp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-36CFF983-0020-4300-845B-BA9768F1B4E8.htm`)
- Status: documented.

10.19 Function Entry: VL-EVERY

10.19.1 Name

‘vl-every‘

10.19.2 Class

Visual LISP List Function

10.19.3 Syntax

```
(vl-every predicate list [list ...])
```

10.19.4 Arguments and Values

- ‘predicate‘: subroutine or symbol
- ‘list‘: one or more lists

10.19.5 Description

Checks whether the predicate is true for every element combination drawn from the supplied lists.

10.19.6 Return Values

Returns ‘T‘ if the predicate is true for all tested combinations; otherwise returns ‘nil‘.

10.19.7 Side Effects

Side effects are those of calling ‘predicate‘.

10.19.8 Notes

- Autodesk documents that ‘predicate‘ may be supplied as:
  - a symbol naming a function,
  - a quoted lambda form,
  - a ‘(function (lambda ...))‘ form.
- Autodesk documents that elements are taken in lockstep from the supplied lists.

- Autodesk documents that evaluation stops as soon as one of the lists runs out.

10.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.19.10 Source Notes

- [vl-every (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/DEU/AutoCAD-AutoLISP-Reference/files/GUID-8BF61C9E-1382-4168-A778-FEA394A361CC.htm`)
- Status: documented.

10.20 Function Entry: VL-MEMBER-IF

10.20.1 Name

‘vl-member-if‘

10.20.2 Class

Visual LISP List Function

10.20.3 Syntax

```
(vl-member-if predicate list)
```

10.20.4 Arguments and Values

- ‘predicate‘: subroutine or symbol
- ‘list‘: list

10.20.5 Description

Determines whether the predicate is true for one of the list members and returns the corresponding tail of the list under member-style semantics.

10.20.6 Return Values

Returns a list tail beginning at the first matching element, or ‘nil‘.

10.20.7 Side Effects

Side effects are those of calling ‘predicate‘.

10.20.8 Notes

- Autodesk documents that ‘predicate‘ may be supplied as:
  - a symbol naming a function,
  - a quoted lambda form,
  - a ‘(function (lambda ...))‘ form.
- Autodesk documents that each element of ‘list‘ is passed to ‘predicate‘ in sequence until a non-‘nil‘ result is obtained or the list is exhausted.

10.20.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.20.10 Source Notes

- [vl-member-if (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-CFEA98F0-F6DE-4FF4-839A-0F6C8FBE1` `htm`)
- Status: documented.

10.21 Function Entry: VL-MEMBER-IF-NOT

10.21.1 Name

‘vl-member-if-not‘

10.21.2 Class

Visual LISP List Function

10.21.3 Syntax

```
(vl-member-if-not comparator lst)
```

10.21.4 Arguments and Values

- ‘comparator‘: a symbol, function, or lambda expression; must return ‘nil‘ or non-‘nil‘.
- ‘lst‘: any list of items.

10.21.5 Description

Returns the tail of ‘lst‘ starting with the first item for which ‘comparator‘ returns ‘nil‘. Validation of each list item stops at the first element whose ‘comparator‘ evaluation is ‘nil‘, and the remainder of the list from that element onward is returned. If ‘comparator‘ is non-‘nil‘ for every element, returns ‘nil‘.

10.21.6 Return Values

The list tail beginning at the first element where ‘comparator‘ returned ‘nil‘, or ‘nil‘ if every element matches the comparator.

10.21.7 Side Effects

Side effects are those of calling ‘comparator‘.

10.21.8 Examples

- ‘(vl-member-if-not ’(lambda (x) (< x 5)) ’(1 2 3 4 5 6 7 8 9 10))‘ returns ‘(5 6 7 8 9 10)‘ — the first element for which ‘(< x 5)‘ is false is ‘5‘, and the tail from ‘5‘ onward is returned.

10.21.9 Availability

- AutoCAD 2026: documented (Autodesk V Functions Reference; included in the family table).
- BricsCAD V26: documented (BricsCAD V26 dedicated per-symbol page).
- Status: documented in both vendors. Phase-5 closure achieved via direct citation of the BricsCAD V26 per-symbol page, which provides the per-element semantics that Autodesk's family-level prose only summarises.

10.21.10 See Also

- `vl-member-if` — companion that returns the tail from the first element where the comparator is non-'nil'.

10.21.11 Source Notes

- [V Functions Reference (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-htm`)
- [vl-member-if-not (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-member-if-not.htm`)
- Status: documented (Phase 5 closure; BricsCAD per-symbol page is the authoritative narrative source for the predicate-stop semantics).

10.22 Function Entry: VL-REMOVE

10.22.1 Name

‘vl-remove‘

10.22.2 Class

Visual LISP List Function

10.22.3 Syntax

```
(vl-remove item list)
```

10.22.4 Description

Removes elements equal to ‘item‘ from a list.

10.22.5 Return Values

List with the matching elements removed.

10.22.6 Side Effects

None.

10.22.7 Notes

- Autodesk documents that ‘item‘ may be of any Lisp data type accepted by the implementation.
- Autodesk documents that all matching elements are removed, not just the first.

10.22.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.22.9 See Also

- BricsCAD analogue: `remove` (chapter 24). Same role; different name.
- Pair documented in ‘vendor-inventory-2026.org‘ §10 item 7.

10.22.10 Source Notes

- [vl-remove (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/DEU/AutoCAD-AutoLISP/files/GUID-1BF33827-5C9C-49B2-A21B-656E0F429B21.htm`)
- Status: documented.

10.23 Function Entry: VL-REMOVE-IF

10.23.1 Name

‘vl-remove-if‘

10.23.2 Class

Visual LISP List Function

10.23.3 Syntax

```
(vl-remove-if predicate list)
```

10.23.4 Arguments and Values

- ‘predicate‘: subroutine or symbol
- ‘list‘: list

10.23.5 Description

Returns all elements of the supplied list that fail the test function.

10.23.6 Return Values

Filtered list.

10.23.7 Side Effects

Side effects are those of calling ‘predicate‘.

10.23.8 Notes

- Autodesk documents that ‘predicate‘ may be supplied as:
  - a symbol naming a function,
  - a quoted lambda form,
  - a ‘(function (lambda ...))‘ form.
- Autodesk documents that the result contains all elements for which ‘predicate‘ returns ‘nil‘.

10.23.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.23.10 Source Notes

- [vl-remove-if (AutoLISP)](`https://help.autodesk.com/cloudhelp/2019/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9934F630-4697-47BA-84E6-A0430A334` `htm`)
- Status: documented.

10.24 Function Entry: VL-REMOVE-IF-NOT

10.24.1 Name

‘vl-remove-if-not‘

10.24.2 Class

Visual LISP List Function

10.24.3 Syntax

```
(vl-remove-if-not predicate list)
```

10.24.4 Arguments and Values

- ‘predicate‘: subroutine or symbol
- ‘list‘: list

10.24.5 Description

Returns all elements of the supplied list that pass the test function.

10.24.6 Return Values

Filtered list.

10.24.7 Side Effects

Side effects are those of calling ‘predicate‘.

10.24.8 Notes

- Autodesk documents that ‘predicate‘ may be supplied as:
  - a symbol naming a function,
  - a quoted lambda form,
  - a ‘(function (lambda ...))‘ form.
- Autodesk documents that the result contains all elements for which ‘predicate‘ returns a non-‘nil‘ value.

10.24.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.24.10 Source Notes

- [vl-remove-if-not (AutoLISP)](`https://help.autodesk.com/cloudhelp/2015/DEU/AutoCAD-AutoLISP/files/GUID-53D12042-8DE3-4DAA-83BD-8ABB376ACA97.htm`)
- Status: documented.

10.25 Function Entry: VL-SOME

10.25.1 Name

‘vl-some‘

10.25.2 Class

Visual LISP List Function

10.25.3 Syntax

```
(vl-some predicate list [list ...])
```

10.25.4 Arguments and Values

- ‘predicate‘: subroutine or symbol
- ‘list‘: one or more lists

10.25.5 Description

Checks whether the predicate is non-‘nil‘ for one element combination.

10.25.6 Return Values

Returns the first non-‘nil‘ value produced by ‘predicate‘, or ‘nil‘ if none.

10.25.7 Side Effects

Side effects are those of calling ‘predicate‘.

10.25.8 Notes

- Autodesk documents that ‘predicate‘ may be supplied as:
  - a symbol naming a function,
  - a quoted lambda form,
  - a ‘(function (lambda ...))‘ form.
- Autodesk documents that elements are taken in lockstep from the supplied lists.
- Autodesk documents that evaluation stops as soon as ‘predicate‘ returns a non-‘nil‘ value, or when one of the lists runs out.

10.25.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.25.10 Source Notes

- [vl-some (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2840F793-DA88-4140-8A9D-EBAC47B62F9D.htm`)
- Status: documented.

10.26 Function Entry: VL-LIST*

10.26.1 Name

‘vl-list*‘

10.26.2 Class

Visual LISP List Constructor

10.26.3 Syntax

```
(vl-list* arg [arg ...])
```

10.26.4 Description

Constructs and returns a list.

10.26.5 Return Values

List or dotted-tail structure according to the final argument, under Lisp list* semantics.

10.26.6 Side Effects

None beyond evaluating the arguments.

10.26.7 Notes

- Autodesk documents that ‘vl-list*‘ is like ‘list‘, except that the last object is placed in the final ‘cdr‘.
- Autodesk documents these result cases:
  - a single atom argument yields that atom,
  - all-atom arguments of length two yield a dotted pair,
  - a final atom with more preceding arguments yields a dotted list,
  - a final list argument causes its elements to be appended as the tail.

10.26.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.26.9 Source Notes

- [vl-list* (AutoLISP)](`https://help.autodesk.com/cloudhelp/2021/ENU/AutoCAD-AutoLISP-Reference/files/GUID-959DE11B-95E8-4AC3-BFBD-02406977D343.htm`)
- Status: documented.

10.27 Function Entry: VL-LIST->STRING

10.27.1 Name

‘vl-list->string‘

10.27.2 Class

Visual LISP Conversion Function

10.27.3 Syntax

```
(vl-list->string list)
```

10.27.4 Description

Combines the characters associated with a list of integers into a string.

10.27.5 Return Values

String.

10.27.6 Side Effects

None.

10.27.7 Compatibility

Character-code interpretation is version-sensitive in Unicode-capable environments.

10.27.8 Notes

- Autodesk documents that the argument is a list of non-negative integers.
- Autodesk documents that in Unicode-capable behavior the accepted integer range is ‘1‘ to ‘65536‘.
- Autodesk documents that ‘(vl-list->string nil)‘ returns the empty string.
- Autodesk documents an AutoCAD 2021 behavior change controlled by ‘LISPSYS‘; earlier behavior accepted only legacy character-code ranges and produced different results for non-ASCII text.

10.27.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.27.10 Source Notes

- [vl-list->string (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PTB/AutoCAD-LT-AutoLISP-Reference/files/GUID-02C7058A-F648-48F5-BAF6-2A62EA` `htm`)
- [What's New or Changed with AutoLISP (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85CCEDF3E.htm`)
- Status: documented.

10.28 Dictionary Coverage Index: List Manipulation Functions

The Autodesk list manipulation family includes at least:

- ‘$\text{acad}_{\text{strlsort}}$‘
- ‘append‘
- ‘assoc‘
- ‘caddr‘
- ‘cadr‘
- ‘car‘
- ‘cdr‘
- ‘cons‘
- ‘foreach‘
- ‘last‘

- ‘length‘
- ‘list‘
- ‘listp‘
- ‘mapcar‘
- ‘member‘
- ‘nth‘
- ‘reverse‘
- ‘subst‘
- ‘vl-consp‘
- ‘vl-every‘
- ‘vl-member-if‘
- ‘vl-member-if-not‘
- ‘vl-remove‘
- ‘vl-remove-if‘
- ‘vl-remove-if-not‘
- ‘vl-some‘

Source:

- [List Manipulation Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-00A75AFB-5708-4D0C-B18C-F74DDC4D65FA.htm)

10.29 Function Entry: CADR

10.29.1 Name

‘cadr‘

10.29.2 Class

Function

10.29.3 Syntax

`(cadr list)`

10.29.4 Arguments and Values

- ‘list‘: a list. The function expects two or more elements; with fewer elements the result is ‘nil‘.

10.29.5 Description

Returns the second element of ‘list‘. Equivalent to ‘(car (cdr list))‘. Commonly used to extract the Y coordinate from a 2D or 3D point list.

10.29.6 Return Values

The second element of ‘list‘, or ‘nil‘ if ‘list‘ has fewer than two elements.

10.29.7 Side Effects

None.

10.29.8 Examples

- After ‘(setq pt2 ’(5.25 1.0))‘, ‘(cadr pt2)‘ returns ‘1.0‘.
- ‘(cadr ’(4.0))‘ returns ‘nil‘.
- ‘(cadr ’(5.25 1.0 3.0))‘ returns ‘1.0‘.

10.29.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.29.10 Source Notes

- [cadr (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F8E9E4F0-218D-4587-9D7E-922BE57C9F9B.htm`)
- Status: documented (Phase 3 body fill).

10.30 Function Entry: CADDR

10.30.1 Name

‘caddr‘

10.30.2 Class

Function

10.30.3 Syntax

```
(caddr list)
```

10.30.4 Arguments and Values

- ‘list‘: a list. The function expects three or more elements; with fewer elements the result is ‘nil‘.

10.30.5 Description

Returns the third element of ‘list‘. Equivalent to ‘(car (cdr (cdr list)))‘. Commonly used to extract the Z coordinate from a 3D point list.

10.30.6 Return Values

The third element of ‘list‘, or ‘nil‘ if ‘list‘ has fewer than three elements.

10.30.7 Side Effects

None.

10.30.8 Examples

- After ‘(setq pt3 ’(5.25 1.0 3.0))‘, ‘(caddr pt3)‘ returns ‘3.0‘.
- ‘(caddr ’(5.25 1.0))‘ returns ‘nil‘.

10.30.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.30.10 Source Notes

- [caddr (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0E81B9F0-7AAD-4BB4-98EB-378AEBE5C htm`)
- Status: documented (Phase 3 body fill).

10.31 Function Entry: VL-LIST-LENGTH

10.31.1 Name

‘vl-list-length‘

10.31.2 Class

Visual LISP List Function

10.31.3 Syntax

```
(vl-list-length list-or-cons-object)
```

10.31.4 Arguments and Values

- ‘list-or-cons-object‘: a true (proper) list, a dotted (improper) list, or ‘nil‘.

10.31.5 Description

Returns the length of a true list. Unlike Common Lisp’s ‘length‘, which signals an error on a dotted list, ‘vl-list-length‘ returns ‘nil‘ for a dotted list, providing graceful handling of malformed list structures.

10.31.6 Return Values

An integer for proper lists; ‘nil‘ for dotted lists.

10.31.7 Side Effects

None.

10.31.8 Examples

- ‘(vl-list-length nil)‘ returns ‘0‘.
- ‘(vl-list-length ’(1 2))‘ returns ‘2‘.
- ‘(vl-list-length ’(1 2 . 3))‘ returns ‘nil‘.

10.31.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

10.31.10 Source Notes

- [vl-list-length (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5BAF3505-B1E9-49BF-9202-AAD36043B` `htm`)
- Status: documented (Phase 3 body fill).

11 11 Filenames, Paths, and Files

11.1 Overview

File handling is part of the core AutoLISP runtime surface and is version-sensitive because of encoding support and trusted-path/security behavior.

11.2 Normative Rules: Source-File and File-Stream Encoding

11.2.1 Statement

AutoLISP source files and host file streams have two independent encoding axes:

1. **Source-file encoding** — the encoding the runtime uses when `load` (and any reader entry point that takes a filesystem path) reads bytes from disk. Historically AutoLISP ran on Windows with ANSI / MBCS source files, and that legacy default is observable in pre-2025 AutoCAD and pre-V25 BricsCAD. AutoCAD 2025 changed the default to UTF-8; BricsCAD V25/V26 followed suit. The value of the host system variable `LISPSYS` on Autodesk products selects between two engines: `0` keeps the legacy ANSI/MBCS reader, `1` and `2` enable Unicode and accept the optional `encoding` argument on `open`. `LISPSYS` requires an AutoCAD restart to take effect.

2. **File-stream encoding** — the encoding the runtime uses when `open` reads or writes the bytes of a host file the user is doing I/O on. This is the third (optional) argument to `open`. When the argument is omitted, the per-host default applies — ANSI/MBCS for the legacy engine, UTF-8 for the modern engine.

There is no documented per-string encoding API: AutoLISP strings are sequences of code units in the host engine's native domain (8-bit code page for the legacy engine, Unicode-capable host string for the modern engine). String **content** is exchanged with files through the encoding configured for that file's stream.

11.2.2 Encoding Names

The third argument to `open` accepts a small documented vocabulary. Implementations should additionally accept the listed aliases since deployed source uses them interchangeably:

Designator	Maps to (CL external-format)
`"utf8"`	`:utf-8`
`"utf8-bom"`	`:utf-8` with byte-order mark
`"utf16"`	`:utf-16`
`"ANSI"`	`:iso-8859-1` (the conservative 1-1 mapping that never fails on bytes)
`"ASCII"`	`:ascii`
`"latin1"` / `"iso-8859-1"`	`:iso-8859-1`
`"cp1252"` / `"windows-1252"`	`:cp1252`

Hosts derived from the BricsCAD codebase additionally accept the C-style `ccs=NAME` syntax embedded in the second (mode) argument, e.g. `(open path "r,ccs=UTF-8")`. The clautolisp `bricscad-v26` dialect honours this through the existing `open-ccs-mode-p` knob; the `strict` and `autocad-2026` dialects do not.

11.2.3 Per-Dialect Defaults

clautolisp resolves the absent third argument of `open` — and the absent `:external-format` keyword on `load` — using the active dialect descriptor:

Dialect	Default source encoding	Default file encoding
`:strict`	`:iso-8859-1`	`:iso-8859-1`
`:autocad-2026`	`:utf-8`	`:utf-8`
`:bricscad-v26`	`:utf-8`	`:utf-8`

The strict choice is **deliberately permissive at the byte level**: a 1-1 byte coding never raises a decoding error on any input, and existing AutoLISP corpora produced under the legacy ANSI/MBCS engine remain loadable. Source code that needs Unicode characters in literals can either (a) use the explicit `--dialect autocad-2026` / `--bricscad-v26` flag, or (b) pass `:external-format :utf-8` on the call site.

11.2.4 Vendor Evidence

- Autodesk reference for `open` (AutoCAD 2024+) documents `"utf8"` and `"utf8-bom"` as the only accepted encoding strings; "When a value isn't provided for the argument, the file is assumed to contain multi-byte character set (MBCS) which is the legacy behavior."
- Autodesk LISPSYS sysvar reference: 0 = ASCII / no encoding arg, 1 = Unicode, 2 = Unicode-with-extra-COM. Restart required.

- AutoCAD 2025 release-note quote: "In the previous versions of AutoCAD, the encoding of the file was 'ANSI', now it is 'UTF-8'." This applies to source files emitted by the Visual LISP IDE.

- BricsCAD V26 honours the C-runtime `ccs=NAME` embedded mode-string as documented in earlier Phase-6 probe evidence (`open-ccs-mode-p`).

11.2.5 Cross-references

- Special Form Entry: LOAD (chapter 11) — uses the dialect's source-encoding default.

- Function Entry: OPEN (chapter 11) — uses the dialect's file-encoding default when its third argument is omitted.

11.2.6 Source Notes

- [open (AutoLISP, AutoCAD 2024)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-089A323F-21FF-4337-99A9-375758E23` `htm`)

- [LISPSYS system variable (Autodesk)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-Customization/files/GUID-49E8AF16-D31E-4DB1-AB14-E0E9` `htm`)

- [AutoCAD 2025 — UTF-8 default for AutoLISP files (Autodesk Community)](`https://forums.autodesk.com/t5/net-forum/unicode-characters-problem-in-a` `td-p/12703850`)

- [BricsCAD Lisp documentation](`https://help.bricsys.com/en-us/document/bricscad/customization/lisp`)

11.3 Function Entry: OPEN

11.3.1 Name

‘open‘

11.3.2 Class

Function

11.3.3 Syntax

`(open filename mode [encoding])`

11.3.4 Arguments and Values

- ‘filename‘: string
- ‘mode‘: string designating read, write, or append
- ‘encoding‘: optional encoding designator in newer Autodesk versions

11.3.5 Description

Opens a file and returns a file descriptor.

11.3.6 Return Values

Returns a ‘FILE‘ descriptor or ‘nil‘ on failure, depending on host behavior and context.

11.3.7 Side Effects

Creates or opens an external file channel.

11.3.8 Affected By

- path-resolution rules
- trusted-path rules
- encoding support

11.3.9 Exceptional Situations

Errors are reported for invalid paths, modes, or inaccessible files.

11.3.10 Examples

- ‘(open ”foo.txt” ”r”)‘

11.3.11 Compatibility

- older Autodesk forms lacked explicit encoding
- BricsCAD documents richer text mode handling

11.3.12 Notes

‘clautolisp‘ should decouple path semantics from native OS via environment profiles.

11.3.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.3.14 See Also

- BricsCAD V26 extends ‘open‘ with encoding selectors of the form ‘”r,ccs=UNICODE”‘, ‘”r,ccs=UTF-8”‘, ‘”r,ccs=UTF-16LE”‘, ‘”w,ccs=UTF-8”‘, ‘”w,ccs=UTF-16LE”‘. AutoCAD does not accept these; AutoCAD’s Unicode story is governed by the ‘LISPSYS‘ system variable instead.
- Divergence documented in ‘vendor-inventory-2026.org‘ §10 item 2.

11.3.15 Source Notes

- [open (AutoLISP), current](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-089A323F-21FF-4337-99A9-375758htm`)

- [open (AutoLISP), earlier](`https://help.autodesk.com/cloudhelp/2015/ENU/AutoCAD-AutoLISP/files/GUID-089A323F-21FF-4337-99A9-375758E23BA4.htm`)
- [read + write text files (BricsCAD)](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/readwritetextfiles.htm`)
- Status: documented with version drift.

11.4 Function Entry: CLOSE

11.4.1 Name

‘close‘

11.4.2 Class

Function

11.4.3 Syntax

`(close file-desc)`

11.4.4 Arguments and Values

- ‘file-desc‘: open file descriptor

11.4.5 Description

Closes an open file.

11.4.6 Return Values

‘nil‘.

11.4.7 Side Effects

Terminates access through ‘file-desc‘ and flushes buffered output according to host behavior.

11.4.8 Exceptional Situations

Erroneous if ‘file-desc‘ is not an open file descriptor.

11.4.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.4.10 Source Notes

- [close (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5308ADB6-7C46-43BE-8B6C-B32FBA8982DB.htm)

11.5 Function Entry: FINDFILE

11.5.1 Name

‘findfile‘

11.5.2 Class

Function

11.5.3 Syntax

```
(findfile filename)
```

11.5.4 Arguments and Values

- ‘filename‘: string

11.5.5 Description

Searches the AutoCAD library path for the specified file.

11.5.6 Return Values

Returns a pathname string identifying the located file, or ‘nil‘.

11.5.7 Side Effects

None.

11.5.8 Affected By

- library-path configuration
- trusted-path configuration
- selected environment profile

11.5.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.5.10 Source Notes

- [findfile (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-LT-AutoLISP-Reference/files/GUID-D671F67D-F92B-41FF-B9FA-A48EF52CF6C` `htm`)

11.5.11 Notes

- Autodesk documents that ‘filename‘ must not contain a directory prefix.
- Autodesk documents that the support-file search path is searched.
- Autodesk documents that URL strings may also be returned when appropriate.

11.6 Function Entry: FINDTRUSTEDFILE

11.6.1 Name

‘findtrustedfile‘

11.6.2 Class

Function

11.6.3 Syntax

`(findtrustedfile filename)`

11.6.4 Arguments and Values

- ‘filename‘: string

11.6.5 Description

Searches the trusted file paths for the specified file.

11.6.6 Return Values

Returns a pathname string identifying the located file, or ‘nil‘.

11.6.7 Side Effects

None.

11.6.8 Affected By

- trusted-path configuration
- host path-search semantics

11.6.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.6.10 Source Notes

- [findtrustedfile (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2A654E6B-C489-4D70-97CB-CA7D94E65 htm)

11.6.11 Notes

- Autodesk documents that trusted paths are searched instead of the ordinary support-file search path.
- Autodesk documents that URL strings may also be returned when appropriate.

11.7 Function Entry: VL-DIRECTORY-FILES

11.7.1 Name

‘vl-directory-files‘

11.7.2 Class

Visual LISP File Function

11.7.3 Syntax

```
(vl-directory-files [directory [pattern [directories]]])
```

11.7.4 Arguments and Values

- ‘directory‘ — optional directory pathname; defaults to the current directory when omitted or ‘nil‘.
- ‘pattern‘ — optional wildcard pattern; defaults to ‘*.*‘.
- ‘directories‘ — optional selector:
  - ‘-1‘ directories only
  - ‘0‘ files and directories
  - ‘1‘ files only

11.7.5 Description

Lists files in a directory, optionally filtered by pattern and directory/file selection.

11.7.6 Return Values

List of strings or ‘nil‘.

Autodesk documents that only the matching names are returned, not full pathnames.

11.7.7 Side Effects

None.

11.7.8 Compatibility

Part of the Visual LISP extension layer, but documented as broadly available across Autodesk desktop and web platforms.

11.7.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.7.10 Source Notes

- [vl-directory-files (AutoLISP)](`https://help.autodesk.com/cloudhelp/2025/JPN/AutoCAD-AutoLISP-Reference/files/GUID-C28C0CB0-FBBE-4AA8-BAC4-2FF222772` `htm`)

11.7.11 Notes

- Autodesk documents a Unicode/LISPSYS behavior change in AutoCAD 2021.

11.8 Function Entry: VL-FILE-COPY

11.8.1 Name

‘vl-file-copy‘

11.8.2 Class

Visual LISP File Function

11.8.3 Syntax

```
(vl-file-copy source-filename destination-filename [append])
```

11.8.4 Arguments and Values

- ‘source-filename‘: source pathname string
- ‘destination-filename‘: destination pathname string
- ‘append‘: optional append flag

11.8.5 Description

Copies a file or appends the contents of one file to another.

11.8.6 Return Values

Integer byte count or ‘nil‘.

11.8.7 Side Effects

Creates or modifies the destination file.

11.8.8 Exceptional Situations

Errors are reported for missing source files, inaccessible destinations, or invalid pathnames.

11.8.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.8.10 Source Notes

- [vl-file-copy (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-12910252-217C-427D-8874-1323611FA74C.htm`)

11.8.11 Notes

- Autodesk documents that the function does not overwrite an existing destination file unless ‘append‘ is non-‘nil‘.
- Autodesk documents several ‘nil‘ cases including missing source, directory source, existing destination, invalid output name, or write-protected destination.

11.9 Function Entry: VL-FILE-DELETE

11.9.1 Name

‘vl-file-delete‘

11.9.2 Class

Visual LISP File Function

11.9.3 Syntax

```
(vl-file-delete filename)
```

11.9.4 Arguments and Values

- ‘filename‘: pathname string

11.9.5 Description

Deletes a file.

11.9.6 Return Values

‘T‘ or ‘nil‘.

11.9.7 Side Effects

Deletes the external file named by ‘filename‘.

11.9.8 Exceptional Situations

Errors are reported for missing files, permissions failures, or invalid pathnames.

11.9.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.9.10 Source Notes

- [vl-file-delete (AutoLISP)](https://help.autodesk.com/cloudhelp/2018/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CEDC718-EEBB-48A8-A9D3-387F216C1 htm)

11.10 Function Entry: VL-FILE-DIRECTORY-P

11.10.1 Name

‘vl-file-directory-p‘

11.10.2 Class

Visual LISP File Predicate

11.10.3 Syntax

```
(vl-file-directory-p filename)
```

11.10.4 Arguments and Values

- ‘filename‘: pathname string

11.10.5 Description

Determines whether a file name refers to a directory.

11.10.6 Return Values

- ‘T‘ if ‘filename‘ designates a directory
- ‘nil‘ otherwise

11.10.7 Side Effects

None.

11.10.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.10.9 Source Notes

- [vl-file-directory-p (AutoLISP)](https://help.autodesk.com/cloudhelp/2017/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7EEA5563-33A7-453C-9E96-860F75658 htm)

11.11 Function Entry: VL-FILE-RENAME

11.11.1 Name

‘vl-file-rename‘

11.11.2 Class

Visual LISP File Function

11.11.3 Syntax

```
(vl-file-rename old-filename new-filename)
```

11.11.4 Arguments and Values

- ‘old-filename‘: existing pathname string
- ‘new-filename‘: new pathname string

11.11.5 Description

Renames a file.

11.11.6 Return Values

‘T‘ or ‘nil‘.

11.11.7 Side Effects

Renames or moves the designated file under host file-system semantics.

11.11.8 Exceptional Situations

Errors are reported for missing files, conflicting destinations, or invalid pathnames.

11.11.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.11.10 Source Notes

- [vl-file-rename (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-188B957C-4CDF-41C7-99BE-8080FA htm)

11.11.11 Notes

- Autodesk documents that the function fails if the target file already exists.
- Autodesk documents a Unicode/LISPSYS behavior change in AutoCAD 2021.

11.12 Function Entry: VL-FILE-SIZE

11.12.1 Name

‘vl-file-size‘

11.12.2 Class

Visual LISP File Function

11.12.3 Syntax

```
(vl-file-size filename)
```

11.12.4 Arguments and Values

- ‘filename‘: pathname string

11.12.5 Description

Determines the size of a file in bytes.

11.12.6 Return Values

Integer or ‘nil‘.

Autodesk documents:

- ‘nil‘ if the file is not readable
- ‘0‘ if ‘filename‘ names a directory or an empty file

11.12.7 Side Effects

None.

11.12.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.12.9 Source Notes

- [vl-file-size (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-646032BF-CAC1-4166-9EBA-2C954DFF300 htm)

11.12.10 Notes

- Autodesk documents a Unicode/LISPSYS behavior change in AutoCAD 2021.

11.13 Function Entry: VL-FILE-SYSTIME

11.13.1 Name

‘vl-file-systime‘

11.13.2 Class

Visual LISP File Function

11.13.3 Syntax

```
(vl-file-systime filename)
```

11.13.4 Arguments and Values

- ‘filename‘: pathname string

11.13.5 Description

Returns the last modification time of the specified file.

11.13.6 Return Values

List or ‘nil‘.

Autodesk documents a list containing:

- year
- month
- day of week
- day of month
- hours
- minutes
- seconds

11.13.7 Side Effects

None.

11.13.8 Compatibility

Autodesk documents this on Windows and macOS, not Web.

11.13.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.13.10 Source Notes

- [vl-file-systime (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7F6A31B2-7D51-4C4D-B239-206F398D6` `htm`)

11.13.11 Notes

- Autodesk documents a Unicode/LISPSYS behavior change in AutoCAD 2021.

11.14 Function Entry: VL-FILENAME-BASE

11.14.1 Name

‘vl-filename-base‘

11.14.2 Class

Visual LISP Pathname Function

11.14.3 Syntax

`(vl-filename-base filename)`

11.14.4 Arguments and Values

- ‘filename‘: pathname string

11.14.5 Description

Returns the base name of a file after stripping the directory path and extension.

11.14.6 Return Values

String.

Autodesk documents the returned string as uppercase on Windows, with platform-native case behavior on macOS and Web examples.

11.14.7 Side Effects

None.

11.14.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.14.9 Source Notes

- [vl-filename-base (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CSY/AutoCAD-LT-AutoLISP-Reference/files/GUID-DA20B039-C252-4751-AA2F-EFC314` `htm`)

11.14.10 Notes

- Autodesk documents that the function does not check whether the specified file exists.

11.15 Function Entry: VL-FILENAME-DIRECTORY

11.15.1 Name

‘vl-filename-directory‘

11.15.2 Class

Visual LISP Pathname Function

11.15.3 Syntax

```
(vl-filename-directory filename)
```

11.15.4 Arguments and Values

- ‘filename‘: pathname string

11.15.5 Description

Returns the directory path of a file after stripping the name and extension.

11.15.6 Return Values

String.

Autodesk documents that the empty string is returned when no directory component is present.

11.15.7 Side Effects

None.

11.15.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.15.9 Source Notes

- [vl-filename-directory (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-AAB7ED2B-C119-4B30-A831-71A4DC htm)

11.15.10 Notes

- Autodesk documents that slashes and backslashes are both accepted as directory delimiters.
- Autodesk documents that the function does not check whether the specified file exists.

11.16 Function Entry: VL-FILENAME-EXTENSION

11.16.1 Name

‘vl-filename-extension‘

11.16.2 Class

Visual LISP Pathname Function

11.16.3 Syntax

```
(vl-filename-extension filename)
```

11.16.4 Arguments and Values

- ‘filename‘: pathname string

11.16.5 Description

Returns the extension from a file name after stripping out the rest of the name.

11.16.6 Return Values

String extension or ‘nil‘ if there is no extension under host semantics.

11.16.7 Side Effects

None.

11.16.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.16.9 Source Notes

- [vl-filename-extension (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-AutoLISP-Reference/files/GUID-6AF5AC95-2E38-4D2C-8A81-8D26F7E8C htm)

11.16.10 Notes

- Autodesk documents that the returned extension begins with a period.
- Autodesk documents that the function does not check whether the specified file exists.

11.17 Function Entry: VL-FILENAME-MKTEMP

11.17.1 Name

‘vl-filename-mktemp‘

11.17.2 Class

Visual LISP Pathname Function

11.17.3 Syntax

```
(vl-filename-mktemp [pattern [directory [extension]]])
```

11.17.4 Arguments and Values

- ‘pattern‘ — optional base pattern string.
- ‘directory‘ — optional directory string.
- ‘extension‘ — optional extension string.

11.17.5 Description

Calculates a unique file name to be used for a temporary file.

11.17.6 Return Values

String.

11.17.7 Side Effects

Allocates or reserves a temporary file name under host semantics; whether a file is also created should be treated as implementation-sensitive until confirmed.

11.17.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

11.17.9 Source Notes

- [vl-filename-mktemp (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F417A5EF-95BB-47EE-B60E-7C017 htm)

11.17.10 Notes

- Autodesk documents that the returned file name is unique and that no file is created by the function itself.
- Autodesk documents that the extension argument should not include the leading period.
- Autodesk documents a Unicode/LISPSYS behavior change in AutoCAD 2021.

11.18 Dictionary Coverage Index: File-Handling Functions

The Autodesk file-handling family includes at least:

- ‘close‘
- ‘findfile‘
- ‘findtrustedfile‘
- ‘open‘
- ‘read-char‘
- ‘read-line‘
- ‘vl-directory-files‘
- ‘vl-file-copy‘
- ‘vl-file-delete‘
- ‘vl-file-directory-p‘
- ‘vl-file-rename‘
- ‘vl-file-size‘
- ‘vl-file-systime‘

- ‘vl-filename-base‘
- ‘vl-filename-directory‘
- ‘vl-filename-extension‘
- ‘vl-filename-mktemp‘

Source:

- [File-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F70DECFC-DBE1-4F04-htm)

11.19 Source Notes

- current and historical ‘open‘ references above

12 12 Streams and Text Input

12.1 Overview

AutoLISP stream behavior is exposed mainly through file descriptors and text/character operations rather than through a large first-class stream protocol.

12.2 Function Entry: READ-CHAR

12.2.1 Name

‘read-char‘

12.2.2 Class

Function

12.2.3 Syntax

`(read-char [file-desc])`

12.2.4 Description

Reads a character from a file descriptor or the keyboard buffer.

12.2.5 Return Values

Returns a character code according to the host’s character model.

12.2.6 Compatibility

- Autodesk documents Unicode-related return behavior changes in 2021+
- newline normalization is documented

12.2.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

12.2.8 Source Notes

- [read-char (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-AutoLISP-Reference/files/GUID-7E94BD14-F018-47D0-88DA-2B08DE32DB2C.htm`)
- Status: documented.

12.3 Function Entry: READ-LINE

12.3.1 Name

‘read-line‘

12.3.2 Class

Function

12.3.3 Syntax

`(read-line file-desc)`

12.3.4 Description

Reads one line from a file.

12.3.5 Return Values

Returns the line without the end-of-line marker, or ‘nil‘ at end of file.

12.3.6 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

12.3.7 Source Notes

- [read-line (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-AC74D827-0969-4888-91C0-C3149DEC3659.htm`)
- Status: documented.

12.4 Function Entry: WRITE-CHAR

12.4.1 Name

‘write-char‘

12.4.2 Class

Function

12.4.3 Syntax

```
(write-char char [file-desc])
```

12.4.4 Arguments and Values

- ‘char‘: character value or code under host character semantics
- ‘file-desc‘: optional output file descriptor

12.4.5 Description

Writes one character to the command line or to an open file.

12.4.6 Return Values

Host-defined success value.

12.4.7 Side Effects

Produces output to the selected destination.

12.4.8 Compatibility

Character interpretation is version-sensitive in Unicode-capable environments.

12.4.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

12.4.10 Source Notes

- [About File Descriptors (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/CHT/AutoCAD-AutoLISP/files/GUID-EDADF856-4D64-4E19-A7BD-5017C4135 htm)
- [What's New or Changed with AutoLISP (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLisp/files/GUID-037BF4D4-755E-4A5C-8136-80E85CCEDF3E.htm)
- Status: documented.

12.5 Function Entry: WRITE-LINE

12.5.1 Name

‘write-line‘

12.5.2 Class

Function

12.5.3 Syntax

```
(write-line string [file-desc])
```

12.5.4 Arguments and Values

- ‘string‘: string
- ‘file-desc‘: optional output file descriptor

12.5.5 Description

Writes a line of text to the command line or to an open file.

12.5.6 Return Values

Host-defined success value.

12.5.7 Side Effects

Produces line-oriented output and appends an end-of-line marker under host file semantics.

12.5.8 Compatibility

End-of-line representation is an external-file concern and may differ from the implementation host defaults.

12.5.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

12.5.10 Source Notes

- [write-line (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-CB4F3ABC-F0F6-41DA-A911-75B90D9F974` `htm`)
- [About File Descriptors (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHT/AutoCAD-AutoLISP/files/GUID-EDADF856-4D64-4E19-A7BD-5017C4135` `htm`)
- Status: documented.

12.6 Function Entry: READ

12.6.1 Name

‘read‘

12.6.2 Class

Function

12.6.3 Syntax

`(read string)`

12.6.4 Description

Parses the first AutoLISP expression represented in a string.

12.6.5 Return Values

Returns the first object read from the string.

12.6.6 Notes

‘read‘ is the main vendor-documented entry point into surface syntax behavior.

12.6.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

12.6.8 Source Notes

- [read (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5B50BB3E-C244-46E8-85D8-6A2D48B1FE51.htm`)
- Status: documented.

13 13 Reader

13.1 Overview

The language has a de facto reader, but not a fully formal reader standard in the published vendor corpus.

13.2 Normative Rules

- chapter 2 defines the standardized reader subset
- chapter 12 'read' defines a documented string-to-object entry point
- all non-core reader syntax remains under-specified until documented or tested

13.3 Implementation Guidance

'clautolisp' should implement an explicit AutoLISP reader with:

- source locations,
- version/dialect switches,
- strict/lax acceptance policy,
- host-independent token semantics.

13.4 Source Notes

- [read (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5B50BB3E-C244-46E8-85D8-6A2D48B1FE51.htm`)
- 'autolisp-spec/documentation/specification-resolution-plan.org'

14 14 Printer and External Representations

14.1 Overview

Vendor documentation sampled so far is much richer on reading than on a formal printer specification.

14.2 Normative Rules

- examples print symbols in uppercase
- dotted pairs and lists print in conventional Lisp notation
- exact printer controls remain under-specified in this draft

14.3 Function Entry: PRIN1

14.3.1 Name

‘prin1‘

14.3.2 Class

Function

14.3.3 Syntax

```
(prin1 [expression [fileHandle]])
```

14.3.4 Arguments and Values

- ‘expression‘ (optional): any AutoLISP expression or value. If omitted, ‘nil‘ is used.
- ‘fileHandle‘ (optional): handle of a file as obtained from ‘(open ...)‘. If omitted or ‘nil‘, output goes to the command line.

14.3.5 Description

Prints ‘expression‘ to the command line, or writes it to the open file ‘fileHandle‘. The representation is **machine-readable**: control characters in a string expression (‘‘̑, “, ‘̈, the escaped double-quote, octal “) are written as their backslash escape sequences so that the output round-trips through the reader.

14.3.6 Return Values

Returns the evaluated value of ‘expression‘. With no argument, ‘(prin1)‘ returns the **void** symbol value, useful as the last form in a ‘defun‘ to emit ”nothing” to the command line.

14.3.7 Side Effects

Writes the printed representation to the command line or to ‘fileHandle‘.

14.3.8 Examples

With ‘(setq x 123)‘:

- ‘(prin1 x)‘ prints ‘123‘ to the command line and returns ‘123‘.
- ‘(prin1 x fh)‘ writes ‘123‘ to file handle ‘fh‘ and returns ‘123‘.

14.3.9 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

‘(prin1 ”hello” file-desc)‘ returns ‘”hello”‘ and writes the seven characters ‘”hello”‘ (the literal seven, including the surrounding double quotes — i.e. the round-trippable representation) to the file. The 2-argument form is fully accepted by BricsCAD V26.

14.3.10 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26). 2-argument form accepted; output is round-trippable.
- Status: documented in both vendors and tested against BricsCAD V26 / macOS.

14.3.11 See Also

- probe-output.lsp — executable Phase-5 probe suite for the printer surface.
- probe results, BricsCAD V26 / macOS, 2026-04-26.
- `princ` for display-oriented output (no escape rewriting).
- `print` for the bracketed prin1 + leading newline + trailing space form.

14.3.12 Source Notes

- [prin1 (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-BF4CEE45-BB6F-443B-A588-A40D9BCE3 htm`)
- [prin1 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/prin1.htm`)
- Status: documented and product-tested (Phase 5 closure).

14.4 Function Entry: PRINC

14.4.1 Name

‘princ‘

14.4.2 Class

Function

14.4.3 Syntax

`(princ [expression [fileHandle]])`

14.4.4 Arguments and Values

- ‘expression‘ (optional): any AutoLISP expression or value. If omitted, ‘nil‘ is used.
- ‘fileHandle‘ (optional): handle of a file as obtained from ‘(open ...)‘. If omitted or ‘nil‘, output goes to the command line.

14.4.5 Description

Prints ‘expression‘ to the command line, or writes it to the open file ‘fileHandle‘. The representation is **display-oriented**: unlike ‘prin1‘, control characters in a string expression are written verbatim (no “ / ‘‘̂ / escaped-double-quote rewriting). With no argument, ‘(princ)‘ returns the **void** symbol value and is the conventional way to terminate a ‘defun‘ without emitting anything.

14.4.6 Return Values

Returns the evaluated value of ‘expression‘, or **void** with no argument.

14.4.7 Side Effects

Writes the printed representation to the command line or to ‘fileHandle‘.

14.4.8 Examples

With ‘(setq x 123)‘:

- ‘(princ x)‘ prints ‘123‘ to the command line and returns ‘123‘.

- ‘(princ x fh)‘ writes ‘123‘ to file handle ‘fh‘ and returns ‘123‘.

14.4.9 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

‘(princ "hello" file-desc)‘ returns ‘"hello"‘ and writes the five characters ‘hello‘ to the file (no surrounding quotes, no trailing newline). Confirms the verbatim representation, distinct from ‘prin1‘’s round-trippable representation.

14.4.10 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26). 2-argument form accepted; output is verbatim (no escape rewriting).
- Status: documented in both vendors and tested against BricsCAD V26 / macOS.

14.4.11 See Also

- probe-output.lsp — executable Phase-5 probe suite.
- probe results, BricsCAD V26 / macOS, 2026-04-26.
- `prin1` for the round-trippable form.
- `vl-princ-to-string` for the string-returning companion.

14.4.12 Source Notes

- [princ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7DDA298B-A97C-49C9-94BA-46C77B50AE99.htm`)
- [princ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/princ.htm`)
- Status: documented and product-tested (Phase 5 closure).

14.5 Function Entry: PRINT

14.5.1 Name

‘print‘

14.5.2 Class

Function

14.5.3 Syntax

`(print [expression [fileHandle]])`

14.5.4 Arguments and Values

- ‘expression‘ (optional): any AutoLISP expression or value. If omitted, ‘nil‘ is used.
- ‘fileHandle‘ (optional): handle of a file as obtained from ‘(open ...)‘. If omitted or ‘nil‘, output goes to the command line.

14.5.5 Description

Same behaviour as ‘prin1‘ (control-character escapes; round-trippable representation), but with a leading newline before ‘expression‘ and an extra space afterwards.

14.5.6 Return Values

Returns the evaluated value of ‘expression‘.

14.5.7 Side Effects

Writes one newline, the printed representation of ‘expression‘, then a single trailing space.

14.5.8 Examples

With ‘(setq x 123)‘:

- ‘(print x)‘ prints ‘<newline>123 ‘ to the command line and returns ‘123‘.

- ‘(print x fh)‘ writes ‘<newline>123 ‘ to file handle ‘fh‘ and returns ‘123‘.

14.5.9 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

‘(print ”hello” file-desc)‘ returns ‘”hello”‘ and writes the nine characters ‘”hello” ‘ (literal newline, then the round-trippable seven characters of ‘prin1‘, then a single trailing space) to the file. Confirms the framing rule from the vendor page.

14.5.10 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26). Leading-newline / trailing-space framing confirmed empirically.
- Status: documented in both vendors and tested against BricsCAD V26 / macOS.

14.5.11 See Also

- probe-output.lsp — executable Phase-5 probe suite.
- probe results, BricsCAD V26 / macOS, 2026-04-26.
- `prin1` for the unframed round-trippable form.

14.5.12 Source Notes

- [print (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4EE74BA9-6ED4-4734-9C47-90A81E0B0971.htm`)
- [print (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/print.htm`)
- Status: documented (Phase 5 closure).

14.6 Function Entry: TERPRI

14.6.1 Name

‘terpri‘

14.6.2 Class

Function

14.6.3 Syntax

`(terpri)`

14.6.4 Arguments and Values

None.

14.6.5 Description

Prints a newline character to the command line. Has no file-handle parameter — the printer-surface family takes a file handle on the per-output-call functions (‘prin1‘, ‘princ‘, ‘print‘), but ‘terpri‘ is command-line-only.

14.6.6 Return Values

Always returns ‘nil‘.

14.6.7 Side Effects

Emits a newline to the command line.

14.6.8 Notes

Bricsys recommends ‘(princ)‘ (with no arguments) instead of ‘(terpri)‘ when ending a ‘defun‘ body, because ‘(princ)‘ returns **void** and does not print an extra newline.

14.6.9 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

The probe suite emits ‘(terpri stream)‘ against an open file handle. BricsCAD V26 reports:

Probe	Outcome
‘(terpri file-desc)‘	error: ‘too few / too many arguments at [TERPRI]‘
‘(princ ”A” file-desc) + (terpri file-desc)‘	first call writes ‘A‘; second call errors with the same mess

This empirically confirms BricsCAD V26’s documented zero-arity contract. AutoLISP code that wishes to write a newline to a file handle must use ‘(princ ”” stream)‘ (BricsCAD V26 confirmed acceptance: ‘princ-string.txt‘ test produced ‘hello‘). ‘terpri‘ is reserved for command-line output. The same code shape ported from Common Lisp via ‘(terpri stream)‘ will not portably work in AutoLISP.

14.6.10 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26). Arity is exactly zero; the 2-argument form ‘(terpri stream)‘ is rejected at runtime.
- Status: documented in both vendors and tested against BricsCAD V26 / macOS. Phase-5 closure: arity nailed down (zero arguments, command-line-only) by direct citation **and** product test.

14.6.11 See Also

- probe-output.lsp — executable Phase-5 probe suite.
- probe results, BricsCAD V26 / macOS, 2026-04-26.

14.6.12 Source Notes

- [terpri (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-16FB5411-7563-4FB9-AB59-A73457B09 htm`)
- [terpri (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/terpri.htm`)
- Status: documented and product-tested (Phase 5 closure).

14.7 Function Entry: PROMPT

14.7.1 Name

‘prompt‘

14.7.2 Class

Function

14.7.3 Syntax

`(prompt message)`

14.7.4 Arguments and Values

- ‘message‘: string; the message to display in the active output device(s).

14.7.5 Description

Sends ‘message‘ to the active output device(s). Bricsys explicitly recommends ‘prompt‘ over ‘prin1‘ / ‘princ‘ / ‘print‘ for command-line output — ‘prompt‘ is the dedicated end-user message channel, while the printer functions are general-purpose.

14.7.6 Return Values

Always returns ‘nil‘.

14.7.7 Side Effects

Writes the message to the active output channels.

14.7.8 Tested Behaviour (BricsCAD V26 / macOS, 2026-04-26)

‘(prompt "AUTOLISP-SPEC-PROMPT>")‘ returns ‘nil‘. The string was emitted to the BricsCAD command-line widget (visible to the user) but does not surface through file-handle capture, confirming the documented "active output device(s)" channel choice.

14.7.9 Availability

- AutoCAD 2026: documented (Autodesk reference page).
- BricsCAD V26: documented + product-tested on macOS (probe suite, 2026-04-26). Return value confirmed ‘nil‘; output goes to the active command-line channel.
- Status: documented in both vendors and tested against BricsCAD V26 / macOS.

14.7.10 See Also

- probe-output.lsp — executable Phase-5 probe suite.
- probe results, BricsCAD V26 / macOS, 2026-04-26.

14.7.11 Source Notes

- [prompt (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2F702ECF-FEC7-4DEE-8575-88CDEE05F htm`)
- [prompt (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/prompt.htm`)
- Status: documented and product-tested (Phase 5 closure).

14.8 Dictionary Coverage Index: Printer Functions

The core printer/output family includes at least:

- ‘prin1‘
- ‘princ‘
- ‘print‘
- ‘prompt‘
- ‘terpri‘
- ‘vl-prin1-to-string‘
- ‘vl-princ-to-string‘

Source:

- [P Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-088AD4D0-7088-4465-8CA5 htm)
- [T Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-424BAAF1-69C6-4DBF-9948 htm)

14.9 Source Notes

- inferred from vendor examples across the reference corpus
- Status: mostly inferred and under-specified.

15 15 Errors and Recovery

15.1 Overview

AutoLISP error handling is specified through ‘*error*‘, ‘ERRNO‘, and Visual LISP catch-all functions, not through a CL-style condition system.

15.2 Variable Entry: ERROR

15.2.1 Name

‘*error*‘

15.2.2 Class

Special variable / error hook

15.2.3 Value Type

A function of one string argument, or ‘nil‘.

15.2.4 Initial Value

Implementation-dependent.

15.2.5 Description

When defined, handles errors signaled to the AutoLISP environment.

15.2.6 Recovery and Handling

Typical uses include restoring system variables, cleanup, and filtering cancel messages.

15.2.7 Compatibility

Central across AutoCAD-compatible behavior; also documented by BricsCAD.

15.2.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

15.2.9 Source Notes

- [*error* (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-CF913180-17CC-43C7-B89F-3BC82FFBEFB9.htm`)
- [About Using the **error** Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-2AC47A3C-7E80-414D-9134-EABAC`
`htm`)
- [BricsCAD *error*](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/error.htm`)
- Status: documented.

15.3 Condition Entry: Catch-All Error Object

15.3.1 Name

Catch-All Error Object

15.3.2 Class

Visual LISP Error Facility

15.3.3 Triggering Situations

Produced by ‘vl-catch-all-apply‘ when an error is trapped.

15.3.4 Reported Form

Accessible via ‘vl-catch-all-error-p‘ and ‘vl-catch-all-error-message‘.

15.3.5 Recovery and Handling

Allows programmatic continuation after trapped errors.

15.3.6 Compatibility

Part of the Visual LISP extension layer.

15.3.7 Source Notes

- [About Catching Errors and Continuing Program Execution (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-9DDAD8C4-FB10-4C15-B59E-F6E05C1F9672.htm`)
- Status: documented.

15.4 Function Entry: ALERT

15.4.1 Name

‘alert‘

15.4.2 Class

Function

15.4.3 Syntax

`(alert message)`

15.4.4 Arguments and Values

- ‘message‘: string

15.4.5 Description

Displays an alert dialog containing an error or warning message.

15.4.6 Return Values

Host-defined dialog result.

15.4.7 Side Effects

Displays a user-interface alert; on some platforms Autodesk documents degraded behavior rather than a native alert box.

15.4.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

15.4.9 Source Notes

- [Error-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-MAC-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-8F3F-C43707B8212B.htm`)
- [About Using the **error** Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-2AC47A3C-7E80-414D-9134-EABAC` `htm`)
- Status: documented.

15.5 Function Entry: EXIT

15.5.1 Name

‘exit‘

15.5.2 Class

Function

15.5.3 Syntax

`(exit)`

15.5.4 Arguments and Values

None.

15.5.5 Description

Unconditionally exits the running LISP code and terminates LISP processing for the current invocation. It is a non-local exit from the current LISP evaluation, not a process-level termination of the host CAD application.

15.5.6 Return Values

Does not return normally. Bricsys describes the channel-level behaviour as printing the message ‘; error : quit / exit abort‘ to the prompt as the LISP top-level handles the abort.

15.5.7 Side Effects

Aborts the running LISP evaluation. Outstanding ‘*error*‘ handlers may run as the abort propagates.

15.5.8 Notes

‘exit‘ and ‘quit‘ are documented by Bricsys with identical signatures and identical channel-level effect (both raise the same ‘; error : quit / exit abort‘ message). Treat them as alternative spellings of the same primitive in the absence of further evidence; reserved separately so that AutoLISP code that names either one continues to read.

15.5.9 Availability

- AutoCAD 2026: documented (Autodesk Error-Handling Functions Reference).
- BricsCAD V26: documented (BricsCAD V26 dedicated per-symbol page).
- Status: documented in both vendors. Phase-5 closure: BricsCAD's per-symbol page makes the abort-message-channel behaviour explicit and confirms that ‘exit‘ and ‘quit‘ are documented identically.

15.5.10 See Also

- `quit` — companion form; identical signature and effect.
- `*error*` — error handler that runs as the abort propagates.
- `vl-exit-with-error`, `vl-exit-with-value` — namespace-scoped non-local exits for VLX.

15.5.11 Source Notes

- [Error-Handling Functions Reference (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-8F3F-C43707B8212B.htm`)
- [About Using the **error** Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-2AC47A3C-7E80-414D-9134-EABAC htm`)
- [exit (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/exit.htm`)
- Status: documented (Phase 5 closure).

15.6 Function Entry: QUIT

15.6.1 Name

‘quit‘

15.6.2 Class

Function

15.6.3 Syntax

`(quit)`

15.6.4 Arguments and Values

None.

15.6.5 Description

Unconditionally cancels the running LISP code and terminates LISP processing for the current invocation. Bricsys’s per-symbol pages distinguish the verb in prose only — ‘exit‘ ”exits” the LISP run, ‘quit‘ ”cancels” it — but the **signature, abort message, and effect on the host** are identical. Treat as an alias of ‘exit‘ in implementation.

15.6.6 Return Values

Does not return normally. Channel-level behaviour: prints ‘; error : quit / exit abort‘ to the prompt.

15.6.7 Side Effects

Aborts the running LISP evaluation; ‘*error*‘ handlers may run as the abort propagates.

15.6.8 Availability

- AutoCAD 2026: documented (Autodesk Error-Handling Functions Reference).
- BricsCAD V26: documented (BricsCAD V26 dedicated per-symbol page).

- Status: documented in both vendors. Phase-5 closure: the BricsCAD per-symbol page nails the previously open question down to ”no semantic distinction beyond the verb” — both forms are documented identically and produce the same abort-channel message. The historical hypothesis that some vendor or release distinguished ‘exit‘ and ‘quit‘ is not supported by current vendor evidence.

15.6.9 See Also

- `exit` — companion form; identical signature and effect.

15.6.10 Source Notes

- [Error-Handling Functions Reference (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-8F3F-C43707B8212B.htm`)
- [About Using the **error** Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP/files/GUID-2AC47A3C-7E80-414D-9134-EABAC` `htm`)
- [quit (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/quit.htm`)
- Status: documented (Phase 5 closure).

15.7 Function Entry: PUSH-ERROR-USING-COMMAND

15.7.1 Name

‘*push-error-using-command*‘

15.7.2 Class

Error-handling function

15.7.3 Syntax

`(*push-error-using-command*)`

15.7.4 Description

Marks that a custom ‘*error*‘ handler will use ‘command‘.

15.7.5 Return Values

Host-defined success value.

15.7.6 Side Effects

Changes the host error-handling mode.

15.7.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

15.7.8 Source Notes

- [Error-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-htm`)
- [*push-error-using-command* (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-620E034A-9151-427F-htm`)

- Status: documented.

15.8 Function Entry: PUSH-ERROR-USING-STACK

15.8.1 Name

‘*push-error-using-stack*‘

15.8.2 Class

Error-handling function

15.8.3 Syntax

`(*push-error-using-stack*)`

15.8.4 Description

Marks that a custom ‘*error*‘ handler will use AutoLISP stack variables.

15.8.5 Return Values

Host-defined success value.

15.8.6 Side Effects

Changes the host error-handling mode.

15.8.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

15.8.8 Source Notes

- [Error-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-` `htm`)
- [*push-error-using-stack* (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-C28420C9-2210-4EEC-A` `htm`)

- Status: documented.

15.9 Function Entry: POP-ERROR-MODE

15.9.1 Name

‘*pop-error-mode*‘

15.9.2 Class

Error-handling function

15.9.3 Syntax

`(*pop-error-mode*)`

15.9.4 Description

Ends the previous ‘*push-error-using-command*‘ or ‘*push-error-using-stack*‘ mode.

15.9.5 Return Values

Host-defined success value.

15.9.6 Side Effects

Restores the previous error-handling mode.

15.9.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

15.9.8 Source Notes

- [Error-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-` `htm`)
- [*pop-error-mode* (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLISP-Reference/files/GUID-60AFEFAD-1DF5-448F-A2E9-D726` `htm`)

- Status: documented.

15.10 Normative Rules

- CL errors must not leak directly to AutoLISP code in ‘clautolisp‘
- AutoLISP-visible errors must support ‘*error*‘, ‘vl-catch-all-*‘, and ‘ERRNO‘-style classification
- exact message identity is not always normative because Autodesk documents multiple valid reported errors in some cases

15.11 Source Notes

- [About Error Handling (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-AutoLISP/files/GUID-027AD2E0-5AC5-48DA-B451-112B7EECB htm)
- [Error Codes Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/HUN/AutoCAD-AutoLISP/files/GUID-97327347-2A13-4CBC-BDBF-979C7F1CA htm)

15.12 Dictionary Coverage Index: Error-Handling Functions

The Autodesk error-handling family includes at least:

- ‘*error*‘
- ‘*pop-error-mode*‘
- ‘*push-error-using-command*‘
- ‘*push-error-using-stack*‘
- ‘alert‘
- ‘exit‘
- ‘quit‘
- ‘vl-catch-all-apply‘
- ‘vl-catch-all-error-message‘
- ‘vl-catch-all-error-p‘

Source:

- [Error-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-htm)

16 16 Host Interaction

16.1 Overview

AutoLISP is a host-coupled language. Command execution, system variables, entity access, and selection sets are part of ordinary language practice.

16.2 Normative Rules

- host interaction is a language layer, not merely an external library
- product compatibility and operating-system compatibility must be tracked separately
- selected host-environment profiles may emulate Windows-like behavior on non-Windows hosts

16.3 Source Notes

- [Query and Command Functions Reference](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-F9ABE09D-9A71-4C00-8E5D-5BA7C` `htm`)
- project rule from ‘AGENTS.md‘

16.4 Function Entry: LOAD

16.4.1 Name

‘load‘

16.4.2 Class

Function

16.4.3 Syntax

```
(load filename [onfailure])
```

16.4.4 Arguments and Values

- ‘filename‘: source or compiled Lisp file designator
- ‘onfailure‘: optional failure-handling argument

16.4.5 Description

Loads an AutoLISP or compatible file into the current environment.

16.4.6 Return Values

If loading succeeds, Autodesk documents the return value as unspecified.

If loading fails:

- the value of ‘onfailure‘ is returned if that argument is supplied,
- if ‘onfailure‘ is a valid AutoLISP function, it is evaluated,
- if ‘onfailure‘ is omitted, an AutoLISP error is signaled.

16.4.7 Side Effects

- reads and evaluates forms from the target file
- may establish functions, variables, and other definitions

16.4.8 Affected By

- search path rules
- trusted-path rules
- environment profile
- file encoding rules
- filename extension search rules

16.4.9 Exceptional Situations

File and load errors are host- and version-sensitive.

16.4.10 Compatibility

Autodesk documents the extension search order used when ‘filename‘ has no extension:

1. ‘.vlx‘
2. ‘.fas‘
3. ‘.lsp‘

As soon as a match is found, loading stops and that file is loaded.

16.4.11 Notes

Autodesk also documents that VLX support is Windows-only.

16.4.12 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.4.13 Source Notes

- [load (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F3639BAA-FD70-487C-AEB5-9E6096EC0255.htm)
- Status: documented.

16.5 Function Entry: AUTOLOAD

16.5.1 Name

‘autoload‘

16.5.2 Class

Function

16.5.3 Syntax

```
(autoload filename function-list)
```

16.5.4 Arguments and Values

- ‘filename‘ — string naming the AutoLISP file to be loaded on demand.
- ‘function-list‘ — list of strings naming commands whose first use triggers the load.

16.5.5 Description

Registers deferred loading so that the named file is loaded when one of the named functions is invoked.

16.5.6 Return Values

Always returns ‘nil‘.

16.5.7 Side Effects

Installs autoload stubs or equivalent deferred-loading behavior.

16.5.8 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (AutoLISP-family function; no contradicting page found).
- Status: AutoCAD documented, BricsCAD presumed-both.

16.5.9 Compatibility

Part of the standard application-handling family.

16.5.10 Source Notes

- [autoload (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-421B36DE-38EA-4161-9768-01647B5492E8.htm)

16.5.11 Notes

- Autodesk documents that ‘filename‘ may omit both the path and the extension.
- Autodesk documents the extension search set as ‘.lsp‘, ‘.fas‘, or ‘.vlx‘ on Windows, and ‘.lsp‘ or ‘.fas‘ on macOS.
- Autodesk documents secure-mode restrictions beginning with AutoCAD 2014-based products through ‘SECURELOAD‘ and ‘TRUSTEDPATHS‘.

16.6 Function Entry: GETVAR

16.6.1 Name

‘getvar‘

16.6.2 Class

Host Interaction Function

16.6.3 Syntax

```
(getvar varname)
```

16.6.4 Arguments and Values

- ‘varname‘: system-variable name designator, usually a string

  Autodesk documents ‘varname‘ as a string naming a system variable.

16.6.5 Description

Retrieves the value of an AutoCAD system variable.

16.6.6 Return Values

Integer, real, string, list, or ‘nil‘.

Autodesk documents that ‘nil‘ is returned if ‘varname‘ is not a valid system variable.

16.6.7 Side Effects

None.

16.6.8 Affected By

- host drawing/session state
- selected environment profile

16.6.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.6.10 Source Notes

- [getvar (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-9C56BEA8-D473-4305-9E17-BEF7630C334 htm)
- [About System and Environment Variables (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/JPN/AutoCAD-MAC-AutoLisp/files/GUID-529819FD-9E9C-4789-BA18-B981BA32DD23.htm)

16.6.11 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

16.7 Function Entry: SETVAR

16.7.1 Name

‘setvar‘

16.7.2 Class

Host Interaction Function

16.7.3 Syntax

`(setvar varname value)`

16.7.4 Arguments and Values

- ‘varname‘: system-variable name designator
- ‘value‘: new value

Autodesk documents:

- ‘varname‘ as a string
- ‘value‘ as an integer, real, string, list, ‘T‘, or ‘nil‘

16.7.5 Description

Sets an AutoCAD system variable to a specified value.

16.7.6 Return Values

Returns ‘value‘ if successful.

16.7.7 Side Effects

Mutates host drawing/session state.

16.7.8 Exceptional Situations

Errors if the variable is read-only, absent, or rejects the supplied value.

16.7.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.7.10 Source Notes

- [setvar (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-B1CE1BFB-2448-47F1-95EC-10E9509F01FA.htm`)
- [About System and Environment Variables (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/JPN/AutoCAD-MAC-AutoLisp/files/GUID-529819FD-9E9C-4789-BA18-B981BA32DD23.htm`)

16.7.11 Notes

- Autodesk documents that for integer-valued system variables the supplied value must lie between ‘-32768‘ and ‘+32767‘.
- Autodesk documents that changes made with ‘setvar‘ during an active command may not take effect until the next command.
- Autodesk documents special angular semantics for ‘ANGBASE‘ and ‘SNAPANG‘: ‘setvar‘ interprets the values in radians, unlike the interactive ‘SETVAR‘ command.

16.7.12 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

16.8 Function Entry: COMMAND-S

16.8.1 Name

‘command-s‘

16.8.2 Class

Special command-interaction function

16.8.3 Syntax

```
(command-s [argument ...])
```

Autodesk documents the first argument as the command name:

```
(command-s [cmdname [arguments ...]])
```

16.8.4 Description

Executes an AutoCAD command and the supplied input.

16.8.5 Return Values

Returns ‘nil‘ when the command finishes successfully on the supplied arguments.

16.8.6 Side Effects

Invokes command processing and may modify the drawing, command stack, and UI state.

16.8.7 Compatibility

Closely related to ‘command‘, but documented separately by Autodesk. Precise sequencing differences should be treated as host-significant.

16.8.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.8.9 Source Notes

- [command-s (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5C9DC003-3DD2-4770-95E7-7E19A4EE1 htm)
- [About Using AutoCAD Commands (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/JPN/AutoCAD-AutoLISP/files/GUID-75F19C76-78D0-443B-BC39-EB9FE htm)

16.8.10 Notes

- Autodesk documents command arguments as strings, reals, integers, and lists/points matching the command prompt sequence.
- Autodesk documents that an empty string is equivalent to pressing Enter.
- Autodesk documents that failure to complete successfully is reported through '*error*'.

16.8.11 Compatibility

Supported on Windows and macOS in current Autodesk documentation.

16.9 Function Entry: VL-CMDF

16.9.1 Name

‘vl-cmdf‘

16.9.2 Class

Visual LISP Command Function

16.9.3 Syntax

```
(vl-cmdf [argument ...])
```

16.9.4 Arguments and Values

- ‘argument‘ — strings, reals, integers, lists/points, or entity names, as required by the target command prompt sequence.

16.9.5 Description

Executes an AutoCAD command.

16.9.6 Return Values

Always returns ‘T‘.

16.9.7 Side Effects

Invokes command processing.

16.9.8 Compatibility

Part of the Visual LISP command layer and related to ‘command‘ and ‘command-s‘.

16.9.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.9.10 Source Notes

- [vl-cmdf (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6F1AAB6B-D5B1-426A-A463-0CBE93E4D956.htm`)

16.9.11 Notes

- Autodesk documents that ‘vl-cmdf‘ evaluates all arguments before command execution begins.
- Autodesk contrasts this with ‘command‘, which may partially execute before an argument-evaluation error is detected.
- Autodesk documents that an empty string is equivalent to pressing Enter, and a no-argument call is equivalent to pressing Esc for most commands.

16.9.12 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

16.10 Function Entry: INITDIA

16.10.1 Name

‘initdia‘

16.10.2 Class

Application-handling function

16.10.3 Syntax

```
(initdia)
```

Autodesk documents the fuller syntax as:

```
(initdia [dialogflag])
```

16.10.4 Description

Forces the next compatible command to use a dialog box rather than command-line prompting.

16.10.5 Return Values

Always returns ‘nil‘.

16.10.6 Side Effects

Changes host command-invocation behavior for the next applicable command.

16.10.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.10.8 Source Notes

- [initdia (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-MAC-AutoLISP-Reference/files/GUID-D0F621C5-6D32-4D79-BC49-290CC4D0FI htm)

16.10.9 Notes

- If ‘dialogflag‘ is omitted or nonzero, Autodesk documents that the next supported command uses its dialog box.
- If ‘dialogflag‘ is ‘0‘, Autodesk documents that the effect is cleared.
- Autodesk documents a bounded set of commands that honor ‘initdia‘, such as ‘PLOT‘, ‘LAYER‘, ‘STYLE‘, and ‘INSERT‘.

16.10.10 Compatibility

Supported on Windows and macOS, not Web.

16.11 Function Entry: STARTAPP

16.11.1 Name

‘startapp‘

16.11.2 Class

Application-handling function

16.11.3 Syntax

```
(startapp appcmd [file])
```

16.11.4 Arguments and Values

- ‘appcmd‘: application command or executable designator
- ‘file‘: optional file or argument string

Autodesk documents both as strings.

16.11.5 Description

Starts an external application.

16.11.6 Return Values

Integer greater than ‘0‘, or ‘nil‘.

16.11.7 Side Effects

Launches an external application or process.

16.11.8 Compatibility

Highly host-environment-sensitive. ‘clautolisp‘ should model this through the selected environment profile rather than by exposing raw native-host behavior.

16.11.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.11.10 Source Notes

- [startapp (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-30EC493A-674B-4AAE-8C88-358E38CDAB21.htm`)

16.11.11 Notes

- Autodesk documents that if ‘appcmd‘ is not a full path, the host searches ‘PATH‘ or the platform equivalent.
- Autodesk documents that arguments containing spaces must be quoted explicitly inside the string.

16.11.12 Compatibility

Supported on Windows and macOS, not Web.

16.12 Function Entry: VL-LOAD-ALL

16.12.1 Name

‘vl-load-all‘

16.12.2 Class

Visual LISP application-loading function

16.12.3 Syntax

```
(vl-load-all filename)
```

16.12.4 Arguments and Values

- ‘filename‘ — string naming the file to load.

16.12.5 Description

Loads a Visual LISP application for all open documents under host semantics.

16.12.6 Return Values

Returns ‘nil‘ on success; otherwise signals an error.

16.12.7 Side Effects

Loads application code into multiple document contexts.

16.12.8 Compatibility

Part of the Visual LISP application-loading layer and more document-model-sensitive than ‘load‘.

16.12.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.12.10 Source Notes

- [vl-load-all (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-MAC-AutoLISP-Reference/files/GUID-116F4D02-4B1A-4298-B38C-5B38FD9828` `htm`)
- [To load functions across all document namespaces (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP/files/GUID-08F778AF-6A57-41AA-8AF5-3E56FEB1958B.htm`)
- [Namespace Communication Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2025/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F5DD5472-1695-4CA3-9110-E6008E87BB89.htm`)

16.12.11 Notes

- Autodesk documents that the filename extension is required; unlike ‘load‘, ‘vl-load-all‘ does not add one.
- Autodesk documents that the file is loaded into all open documents and into documents opened later in the same session.

16.12.12 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

16.13 Function Entry: ARXLOAD

16.13.1 Name

‘arxload‘

16.13.2 Class

Application-handling function

16.13.3 Syntax

```
(arxload appname)
```

16.13.4 Arguments and Values

- ‘appname‘: ObjectARX application designator

Autodesk documents the actual signature as:

```
(arxload application [onfailure])
```

with:

- ‘application‘ — string naming the ObjectARX executable or bundle
- ‘onfailure‘ — expression to evaluate if loading fails

16.13.5 Description

Loads an ObjectARX application.

16.13.6 Return Values

String naming the application if successful.

If loading fails and ‘onfailure‘ is supplied, Autodesk documents that the value of ‘onfailure‘ is returned; otherwise an error is reported.

16.13.7 Side Effects

Loads host extension code into the running CAD session.

16.13.8 Compatibility

Strongly host-specific and outside portable core AutoLISP semantics.

16.13.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.13.10 Source Notes

- [arxload (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-965A0D2A-CFD0-4D7C-9D2B-2D8188F0DAC8.htm`)

16.13.11 Notes

- Autodesk documents that the file extension may be omitted: '.arx' on Windows and '.bundle' on macOS.
- Autodesk documents that a full path is required unless the application is in the support-file search path.
- Autodesk documents that loading an already loaded application is an error, and suggests checking with 'arx' first.
- Autodesk documents secure-mode restrictions through 'SECURELOAD' and 'TRUSTEDPATHS'.

16.13.12 Compatibility

Supported on Windows and macOS, not Web.

16.14 Function Entry: ARXUNLOAD

16.14.1 Name

‘arxunload‘

16.14.2 Class

Application-handling function

16.14.3 Syntax

```
(arxunload appname)
```

16.14.4 Arguments and Values

- ‘appname‘: ObjectARX application designator

Autodesk documents the actual signature as:

```
(arxunload application [onfailure])
```

with:

- ‘application‘ — string naming an application previously loaded with ‘arxload‘
- ‘onfailure‘ — expression to evaluate if unloading fails

16.14.5 Description

Unloads an ObjectARX application.

16.14.6 Return Values

String naming the application if successful.

If unloading fails and ‘onfailure‘ is supplied, Autodesk documents that the value of ‘onfailure‘ is returned; otherwise an error is reported.

16.14.7 Side Effects

Removes host extension code from the running CAD session.

16.14.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.14.9 Source Notes

- [arxunload (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-29092087-D1F7-4DF5-863C-FAF4E4F1BC htm`)

16.14.10 Notes

- Autodesk documents that locked ObjectARX applications cannot be unloaded, and that they are locked by default.
- Autodesk documents that the extension may be omitted: '.arx' or '.crx' on Windows, '.bundle' on macOS.

16.14.11 Compatibility

Supported on Windows and macOS only.

16.15 Function Entry: AUTOARXLOAD

16.15.1 Name

‘autoarxload‘

16.15.2 Class

Application-handling function

16.15.3 Syntax

```
(autoarxload filename command-list)
```

16.15.4 Arguments and Values

- ‘filename‘ — string naming the ObjectARX file to be loaded on demand.
- ‘command-list‘ — list of strings naming commands that trigger the load.

16.15.5 Description

Predefines command names that trigger loading of an associated ObjectARX file.

16.15.6 Return Values

Always returns ‘nil‘.

16.15.7 Side Effects

Installs deferred-loading stubs for host extension commands.

16.15.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.15.9 Source Notes

- [autoarxload (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E1DDBABF-6F72-4747-B9E2-ACEEEA9FA htm)

16.15.10 Notes

- Autodesk documents that the path and extension may be omitted, in which case the support-file search path is used and '.arx' or '.bundle' is supplied by platform convention.
- Autodesk documents that on first use of a listed command, the application is loaded and the command then continues.
- Autodesk documents that secure-mode restrictions apply through 'SECURELOAD' and 'TRUSTEDPATHS'.

16.15.11 Compatibility

Supported on Windows and macOS, not Web.

16.16 Function Entry: SHOWHTMLMODALWINDOW

16.16.1 Name

‘showhtmlmodalwindow‘

16.16.2 Class

Application/UI function

16.16.3 Syntax

`(showhtmlmodalwindow uri)`

16.16.4 Arguments and Values

- ‘uri‘ — string URL or other URI naming the page to display.

16.16.5 Description

Displays a modal dialog box with a specified URI.

16.16.6 Return Values

Always returns ‘nil‘.

16.16.7 Side Effects

Displays modal user interface content based on a URI.

16.16.8 Compatibility

Highly host-UI-specific and potentially version-sensitive.

16.16.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.16.10 See Also

- AutoCAD-Windows-only. BricsCAD has no documented equivalent. Divergence documented in ‘vendor-inventory-2026.org‘ §10 item 10.

16.16.11 Source Notes

- [showhtmlmodalwindow (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-1330BB1E-866E-419A-8` `htm`)

16.16.12 Notes

- Autodesk documents that the dialog’s size and position are automatically persisted across sessions.
- Autodesk documents that the loaded page may use the AutoCAD JavaScript API.

16.16.13 Compatibility

Available on Windows only.

16.17 Function Entry: VL-VBALOAD

16.17.1 Name

‘vl-vbaload‘

16.17.2 Class

Visual LISP application-loading function

16.17.3 Syntax

```
(vl-vbaload project)
```

16.17.4 Arguments and Values

- ‘project‘ — string naming the VBA project file.

16.17.5 Description

Loads a VBA project.

16.17.6 Return Values

String containing the file name and path of the loaded project.

16.17.7 Side Effects

Loads VBA project state into the host environment.

16.17.8 Compatibility

Host- and platform-dependent legacy integration facility.

16.17.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.17.10 See Also

- AutoCAD-Windows-only. No BricsCAD analogue: BricsCAD does not embed a VBA module callable from LISP under this name. Divergence documented in ‘vendor-inventory-2026.org‘ §10 item 9.

16.17.11 Source Notes

- [vl-vbaload (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-54FD698E-8D94-4E27-9DF1-3F7E9D451E` `htm`)

16.17.12 Notes

- Autodesk documents that an error occurs if the file is not found.
- Autodesk documents a Unicode-related change in AutoCAD 2021 controlled by ‘LISPSYS‘.
- Autodesk documents secure-mode restrictions through ‘SECURELOAD‘ and ‘TRUSTEDPATHS‘.

16.17.13 Compatibility

Available on AutoCAD for Windows only; not available in AutoCAD LT for Windows, or on macOS or Web.

16.18 Function Entry: VL-VBARUN

16.18.1 Name

‘vl-vbarun‘

16.18.2 Class

Visual LISP application function

16.18.3 Syntax

```
(vl-vbarun macro-name)
```

16.18.4 Arguments and Values

- ‘macro-name‘ — string naming a macro in a loaded VBA project.

16.18.5 Description

Runs a VBA macro.

16.18.6 Return Values

Returns ‘macro-name‘ if successful.

16.18.7 Side Effects

Invokes host VBA automation.

16.18.8 Compatibility

Host- and platform-dependent legacy integration facility.

16.18.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.18.10 See Also

- AutoCAD-Windows-only. No BricsCAD analogue. Divergence documented in ‘vendor-inventory-2026.org‘ §10 item 9.

16.18.11 Source Notes

- [vl-vbarun (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-75387617-9144-49CB-97E4-03B4CD29973` `htm`)
- [To run a VBA macro](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-Customization/files/GUID-E50E8491-5BCB-4202-B3A9-BF6FD9CF68A3.htm`)

16.18.12 Notes

- Autodesk documents the macro naming syntax as ‘<project!module.macro>‘ when a fully qualified macro name is needed.

16.18.13 Compatibility

Available on AutoCAD for Windows only; not available in AutoCAD LT for Windows, or on macOS or Web.

16.19 Function Entry: VLAX-ADD-CMD

16.19.1 Name

‘vlax-add-cmd‘

16.19.2 Class

Visual LISP command-registration function

16.19.3 Syntax

```
(vlax-add-cmd global-name function-name [cmd-flags])
```

Autodesk documents the fuller syntax as:

```
(vlax-add-cmd global-name func-sym [local-name cmd-flags])
```

16.19.4 Arguments and Values

- ‘global-name‘ — string naming the global command.
- ‘func-sym‘ — symbol naming a no-argument AutoLISP function.
- ‘local-name‘ — optional string naming the local command alias; defaults to ‘global-name‘.
- ‘cmd-flags‘ — optional integer command flag word.

16.19.5 Description

Adds commands to the AutoCAD built-in command set.

16.19.6 Return Values

String naming the global command if successful.

16.19.7 Side Effects

Registers new commands in the host command system.

16.19.8 Compatibility

Part of the host integration layer rather than portable core language semantics.

16.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.19.10 Source Notes

- [vlax-add-cmd (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-42392DA4-4A2B-4D34-AA7B-A5ACAF727 htm)

16.19.11 Notes

- Autodesk documents modal and transparent command flags, plus additional flags such as pickset use.
- Autodesk documents that commands are assigned to command groups automatically.
- Autodesk documents that the function should not be used to expose functions that create reactor objects or serve as reactor callbacks.

16.19.12 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

16.20 Dictionary Coverage Index: Application-Handling Functions

The Autodesk application-handling family includes at least:

- ‘arx‘
- ‘arxload‘
- ‘arxunload‘
- ‘autoarxload‘
- ‘autoload‘

- ‘initdia‘
- ‘load‘
- ‘showhtmlmodalwindow‘
- ‘startapp‘
- ‘vl-load-all‘
- ‘vl-vbaload‘
- ‘vl-vbarun‘
- ‘vlax-add-cmd‘

Source:

- [Application-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-AutoLISP-Reference/files/GUID-E2B683FE-43FB-4F5D-A9AE-809772FE8D3B.htm`)

16.21 Function Entry: TRANS

16.21.1 Name

‘trans‘

16.21.2 Class

Function

16.21.3 Syntax

```
(trans pt from to [disp])
```

16.21.4 Arguments and Values

- ‘pt‘: a list of three reals, interpreted as a 3D point or as a 3D displacement.
- ‘from‘: an integer, list, or ename naming the source coordinate system. Designators are:
  - ‘0‘: WCS — World Coordinate System
  - ‘1‘: UCS — current User Coordinate System
  - ‘2‘: DCS — viewport Display Coordinate System
  - ‘3‘: PSDCS — paper-space DCS (only valid as the **to** designator and only from ‘2‘)
  - an ename: the Object Coordinate System of that entity
  - a list (extrusion vector): the OCS implied by that extrusion
- ‘to‘: same designator vocabulary as ‘from‘.
- ‘disp‘ (optional): when non-‘nil‘, treats ‘pt‘ as a displacement rather than a point (so translation components of the transform are skipped).

16.21.5 Description

Translates a point or a displacement from one coordinate system to another. When converting between systems whose Z-axes differ, this function applies the proper rotation.

16.21.6 Return Values

A 3D point (or displacement) expressed in the ‘to‘ coordinate system.

16.21.7 Side Effects

None.

16.21.8 Examples

- ‘(trans ’(1.0 2.0 3.0) 0 1)‘ returns ‘(2.0 -1.0 3.0)‘ when the UCS is rotated 90° counter-clockwise relative to the WCS.
- ‘(trans text-insert-point text-ename 1)‘ converts a text-entity insertion point from its OCS to the current UCS.
- ‘(trans ’(1 2 3) 1 circle-ename)‘ converts a UCS displacement to a circle’s OCS.

16.21.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.21.10 Source Notes

- [trans (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1A316343-0B68-4DBE-8F49-B4D601CB8FCC.htm`)
- Status: documented (Phase 3 body fill).

16.22 Function Entry: GETENV

16.22.1 Name

‘getenv‘

16.22.2 Class

Function

16.22.3 Syntax

```
(getenv variable-name)
```

16.22.4 Arguments and Values

- ‘variable-name‘: string; the environment-variable name. Names must be spelled and cased exactly as stored in the system registry.

16.22.5 Description

Returns the value bound to the named environment variable. Returns ‘nil‘ if the variable is not defined.

16.22.6 Return Values

A string, or ‘nil‘.

16.22.7 Side Effects

None.

16.22.8 Examples

- After ACAD is set to ‘/acad/support‘: ‘(getenv "ACAD")‘ returns ‘"/acad/support"‘.
- For an undefined variable: ‘(getenv "NOSUCH")‘ returns ‘nil‘.
- ‘(getenv "MaxArray")‘ returns ‘"10000"‘ if the variable is set to that value.

16.22.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.22.10 Source Notes

- [getenv (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0796D4FE-C07B-4DB6-9C01-9D772CAFA htm)
- Status: documented (Phase 3 body fill).

16.23 Function Entry: SETENV

16.23.1 Name

‘setenv‘

16.23.2 Class

Function

16.23.3 Syntax

```
(setenv varname value)
```

16.23.4 Arguments and Values

- ‘varname‘: string; the environment-variable name (must match registry casing exactly).
- ‘value‘: string; the value to assign.

16.23.5 Description

Sets the named environment variable to the given string. Some settings might not take effect until the next time AutoCAD is started.

16.23.6 Return Values

The ‘value‘ argument as a string.

16.23.7 Side Effects

Updates the environment-variable store.

16.23.8 Examples

- ‘(setenv ”MaxArray” ”10000”)‘ returns ‘”10000”‘.

16.23.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.23.10 Source Notes

- [setenv (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0C2E8222-62A8-4529-8A8A-58AAB2A5F` `htm`)
- Status: documented (Phase 3 body fill).

16.24 Function Entry: GETCFG

16.24.1 Name

‘getcfg‘

16.24.2 Class

Function (obsolete)

16.24.3 Syntax

`(getcfg cfgname)`

16.24.4 Arguments and Values

- ‘cfgname‘: string of the form ‘”AppData/application$_{\text{name}}$/section$_{\text{name}}$/.../param$_{\text{name}}$”‘, up to 496 characters.

16.24.5 Description

Obsolete. Retrieves application data from the ‘AppData‘ section of the ‘acad20xx.cfg‘ file. Autodesk recommends ‘vl-registry-read‘ for new code.

16.24.6 Return Values

The stored string on success, or ‘nil‘ if ‘cfgname‘ is invalid.

16.24.7 Side Effects

None.

16.24.8 Examples

- ‘(getcfg ”AppData/ArchStuff/WallThk”)‘ returns ‘”8”‘ when WallThk is set to 8.

16.24.9 Availability

- AutoCAD 2026: documented (obsolete; use ‘vl-registry-read‘).
- BricsCAD V26: presumed compatible (obsolete behaviour).
- Status: AutoCAD documented as obsolete; BricsCAD presumed-both pending Phase 4 verification.

16.24.10 Source Notes

- [getcfg (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D8D46A8C-82DF-4193-A1B7-3DC3CAC7C htm)
- Status: documented (Phase 3 body fill).

16.25 Function Entry: SETCFG

16.25.1 Name

‘setcfg‘

16.25.2 Class

Function (obsolete)

16.25.3 Syntax

```
(setcfg cfgname cfgval)
```

16.25.4 Arguments and Values

- ‘cfgname‘: string of the form ‘"AppData/application$_{name}$/section$_{name}$/.../param$_{name}$"‘, up to 496 characters.
- ‘cfgval‘: string up to 512 characters; longer values are written but cannot be retrieved.

16.25.5 Description

Obsolete. Writes application data to the ‘AppData‘ section of the ‘acad20xx.cfg‘ file. Autodesk recommends ‘vl-registry-write‘ for new code.

16.25.6 Return Values

The ‘cfgval‘ string on success, or ‘nil‘ if ‘cfgname‘ is invalid.

16.25.7 Side Effects

Writes to the AutoCAD configuration file.

16.25.8 Examples

- ‘(setcfg "AppData/ArchStuff/WallThk" "8")‘ returns ‘"8"‘.

16.25.9 Availability

- AutoCAD 2026: documented (obsolete; use ‘vl-registry-write‘).
- BricsCAD V26: presumed compatible (obsolete behaviour).
- Status: AutoCAD documented as obsolete; BricsCAD presumed-both pending Phase 4 verification.

16.25.10 Source Notes

- [setcfg (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8BABF3B0-C8E1-49C9-A367-9823211C1 htm)
- Status: documented (Phase 3 body fill).

16.26 Function Entry: GETCNAME

16.26.1 Name

‘getcname‘

16.26.2 Class

Function

16.26.3 Syntax

`(getcname cname)`

16.26.4 Arguments and Values

- ‘cname‘: string; a localized command name, or an underscored English command name. Up to 64 characters.

16.26.5 Description

Converts between localized and English command names. Without an underscore, returns the underscored English command name. With an underscore, returns the localized command name. Useful for writing language-agnostic AutoLISP routines.

16.26.6 Return Values

A string, or ‘nil‘ if ‘cname‘ is invalid.

16.26.7 Side Effects

None.

16.26.8 Examples

In French AutoCAD:

- ‘(getcname ”ETIRER”)‘ returns ‘”STRETCH”‘.
- ‘(getcname ”STRETCH”)‘ returns ‘”ETIRER”‘.

16.26.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (BricsCAD also localises command names).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.26.10 Source Notes

- [getcname (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1DCDFFA6-0372-479D-B00F-D165EC5B8 htm)
- Status: documented (Phase 3 body fill).

16.27 Function Entry: LAYOUTLIST

16.27.1 Name

‘layoutlist‘

16.27.2 Class

Function

16.27.3 Syntax

```
(layoutlist)
```

16.27.4 Arguments and Values

None.

16.27.5 Description

Returns a list naming all paper-space layouts in the current drawing.

16.27.6 Return Values

A list of strings.

16.27.7 Side Effects

None.

16.27.8 Examples

- ‘(layoutlist)‘ returns ‘(”Layout1” ”Layout2”)‘ for a typical drawing.

16.27.9 Availability

- AutoCAD 2026: documented (Windows + macOS).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.27.10 Source Notes

- [layoutlist (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-AE3F338F-0B11-45B0-AB12-2C119203A htm)
- Status: documented (Phase 3 body fill).

16.28 Function Entry: INITCOMMANDVERSION

16.28.1 Name

‘initcommandversion‘

16.28.2 Class

Function

16.28.3 Syntax

```
(initcommandversion [version])
```

16.28.4 Arguments and Values

- ‘version‘ (optional): integer specifying the command version to use. When omitted, the next supported command runs at its latest version.

16.28.5 Description

Forces the next AutoCAD command to run with the specified version. Only commands that have been updated to support versioning honour this call. Manual interactive use defaults to the latest version (typically 2); LISP / scripting contexts default to version 1, so ‘initcommandversion 2‘ is often used to invoke modern dialog-driven behaviour from a script, and ‘initcommandversion 1‘ is used to force command-prompt-only behaviour.

16.28.6 Return Values

Always returns ‘T‘.

16.28.7 Side Effects

Latches the version selector for the next command issued via ‘command‘ or ‘command-s‘.

16.28.8 Examples

- Running the FILLET command after ‘(initcommandversion 1)‘ shows the legacy prompt set.

- The COLOR command, normally dialog-driven from a script, can be forced to prompt at the command line by ‘(initcommandversion 1)‘ followed by the COLOR command.

16.28.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.28.10 Source Notes

- [initcommandversion (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6176FC98-DC5D-433E-8D76` `htm`)
- Status: documented (Phase 3 body fill).

16.29 Function Entry: SETFUNHELP

16.29.1 Name

‘setfunhelp‘

16.29.2 Class

Function

16.29.3 Syntax

```
(setfunhelp c:fname [helpfile [topic [command]]])
```

16.29.4 Arguments and Values

- ‘c:fname‘: string; the user-defined command name. Must include the ‘c:‘ prefix.
- ‘helpfile‘ (optional): string; help file name. Extension is optional; AutoCAD looks for ‘.chm‘ first, then ‘.hlp‘.
- ‘topic‘ (optional): string; help topic identifier. For CHM files, supply the topic file name without extension.
- ‘command‘ (optional): string; initial Help-window state expressed as an HtmlHelp() or WinHelp() command (e.g. ‘$\mathrm{HH}_{\mathrm{DISPLAYTOPIC}}$‘).

16.29.5 Description

Registers a user-defined command so the Help facility opens the supplied file and topic when the user requests help on that command. Calling ‘defun‘ to redefine the same command name removes its ‘setfunhelp‘ registration; therefore call ‘setfunhelp‘ after ‘defun‘.

16.29.6 Return Values

Returns the ‘c:fname‘ string on success, otherwise ‘nil‘. The function only validates the ‘c:‘ prefix; it does not verify that the function or topic exists.

16.29.7 Side Effects

Updates the Help-binding registry for the named command.

16.29.8 Examples

- After ‘(defun c:test () (getstring ”

TEST: ”) (princ))‘, ‘(setfunhelp ”c:test” ”acad$_{\text{acr.chm}}$” ”line”)‘ associates the LINE help topic with ‘c:test‘.

16.29.9 Availability

- AutoCAD 2026: documented (Windows-centric — CHM/HLP).
- BricsCAD V26: presumed compatible (BricsCAD has its own help system; binding semantics may differ).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.29.10 Source Notes

- [setfunhelp (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-61B2326A-8DE4-46C4-B9F9-22B8D163A htm`)
- Status: documented (Phase 3 body fill).

16.30 Function Entry: HELP

16.30.1 Name

‘help‘

16.30.2 Class

Function

16.30.3 Syntax

```
(help [helpfile [topic [command]]])
```

16.30.4 Arguments and Values

- ‘helpfile‘ (optional): string; help-file name. Extension is optional; AutoCAD looks for ‘.chm‘ first, then ‘.hlp‘. An empty string opens the online help.
- ‘topic‘ (optional): string; topic identifier. For CHM files, supply the topic file name without extension.
- ‘command‘ (optional, Windows only): string; initial Help-window state expressed as an HtmlHelp() / WinHelp() argument.

16.30.5 Description

Invokes the Help facility. Place help files in a folder whose path contains no Unicode characters; the Microsoft HTML Help executable is sensitive to that.

16.30.6 Return Values

Returns the ‘helpfile‘ string on success, otherwise ‘nil‘. With no arguments, returns the empty string ‘""‘ on success or ‘nil‘ on failure.

16.30.7 Side Effects

Opens a help window or browser tab.

16.30.8 Examples

- ‘(help ”achelp.chm” ”mycommand”)‘ opens the topic for MYCOMMAND in achelp.chm.

16.30.9 Availability

- AutoCAD 2026: documented (Windows, macOS, Web).
- BricsCAD V26: presumed compatible (BricsCAD has a HELP command).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.30.10 Source Notes

- [help (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CB4A951-1507-4F3D-9F7B-93FF3A2C9850.htm`)
- Status: documented (Phase 3 body fill).

16.31 Function Entry: MENUCMD

16.31.1 Name

'menucmd'

16.31.2 Class

Function

16.31.3 Syntax

`(menucmd str)`

16.31.4 Arguments and Values

- 'str': string of the form '"$menu_{area}$=value"' selecting a menu area and a value. Supported menu areas:
  - 'B1'–'B4': BUTTONS menus.
  - 'A1'–'A4': AUX (auxiliary device) menus.
  - 'P0'–'P16': pull-down menus.
  - 'I': image-tile menus.
  - 'T1'–'T4': TABLET menus.
  - 'M': DIESEL expressions.
  - 'Gmenugroup.nametag': a tagged menu inside a named menugroup.

16.31.5 Description

Issues menu commands. Switches between subpages of a menu, enables / disables menu and ribbon items, places marks, displays image-tile menus, and evaluates DIESEL expressions.

The BricsCAD interpretation of 'menucmd' is documented as supporting only the 'P0' cursor menu and 'P1'–'P16' pulldowns; auxiliary, button, icon, DIESEL, screen, and tablet forms are not supported there. See Phase 4 cross-reference.

16.31.6 Return Values

Always returns 'nil'.

16.31.7 Side Effects

Updates the menu state, evaluates DIESEL, or displays an image-tile menu.

16.31.8 Examples

- Display an image-tile menu: ‘(menucmd ”I=moreicons”)‘ then ‘(menucmd ”I=*”)‘.
- Check, then disable a pulldown item: ‘(setq s (menucmd ”P11.3=?”))‘ and ‘(if (= s ””) (menucmd ”P11.3=~”))‘.
- Evaluate DIESEL: ‘(menucmd ”M=$(edtime,$(getvar,date),DDDD\”,\” D MONTH YYYY)”)‘.

16.31.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: documented with reduced surface (only ‘P0‘/‘P1‘–‘P16‘).
- Status: cross-vendor divergence — see Phase 4 reconciliation in ‘vendor-inventory-2026.org‘ §10.

16.31.10 See Also

- BricsCAD documents only the ‘P0‘ cursor menu and ‘P1‘–‘P16‘ pull-down forms. AutoCAD additionally documents ‘B1‘–‘B4‘ (BUTTONS), ‘A1‘–‘A4‘ (AUX), ‘T1‘–‘T4‘ (TABLET), ‘I‘ (image-tile), ‘M‘ (DIESEL), and ‘Gmenugroup.nametag‘ forms — none are accepted by BricsCAD.
- Divergence documented in ‘vendor-inventory-2026.org‘ §10 item 4.

16.31.11 Source Notes

- [menucmd (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-20357966-71D4-4D55-881B-06AFDECD2 htm`)
- Status: documented (Phase 3 body fill).

16.32 Function Entry: MENUGROUP

16.32.1 Name

‘menugroup‘

16.32.2 Class

Function

16.32.3 Syntax

`(menugroup groupname)`

16.32.4 Arguments and Values

- ‘groupname‘: string; the menu group to look up.

16.32.5 Description

Verifies that the named menu group is currently loaded.

16.32.6 Return Values

Returns the ‘groupname‘ string when the group is loaded; otherwise ‘nil‘. On macOS / Web, the function always returns ‘groupname‘.

16.32.7 Side Effects

None.

16.32.8 Examples

Not documented on the vendor page.

16.32.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.32.10 Source Notes

- [menugroup (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-52FBC08B-F53F-4FC0-A5C5-ED3FA2BB8 htm)
- Status: documented (Phase 3 body fill).

16.33 Function Entry: SNVALID

16.33.1 Name

‘snvalid‘

16.33.2 Class

Function

16.33.3 Syntax

```
(snvalid sym_name [flag])
```

16.33.4 Arguments and Values

- ‘sym_{name}‘: string; a candidate symbol-table name.
- ‘flag‘ (optional): integer controlling vertical-bar handling.
  - ‘0‘ (default): vertical bars are disallowed.
  - ‘1‘: vertical bars are allowed except as the first or last character.

16.33.5 Description

Validates a symbol-table name against AutoCAD naming rules. The rules depend on the ‘EXTNAMES‘ system variable:

- ‘EXTNAMES = 1‘ (default): extended naming. Most characters are accepted, except ‘< > / \ ” : ? * | , = ‘ ;‘.
- ‘EXTNAMES = 0‘: restrictive naming. Only A–Z, 0–9, ‘$‘, ‘_‘, and ‘-‘ are accepted.

Control characters, graphic characters, the empty string, and vertical bars at name boundaries are always rejected.

16.33.6 Return Values

Returns ‘T‘ when the name is valid; otherwise ‘nil‘.

16.33.7 Side Effects

None.

16.33.8 Examples

With ‘EXTNAMES = 1‘:

- ‘(snvalid ”hocus-pocus”)‘ returns ‘T‘.
- ‘(snvalid ”hocus pocus”)‘ returns ‘T‘.
- ‘(snvalid ”hocus|pocus”)‘ returns ‘nil‘.
- ‘(snvalid ”hocus|pocus” 1)‘ returns ‘T‘.

With ‘EXTNAMES = 0‘:

- ‘(snvalid ”hocus-pocus”)‘ returns ‘T‘.
- ‘(snvalid ”hocus pocus”)‘ returns ‘nil‘.

16.33.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.33.10 Source Notes

- [snvalid (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2EFBE198-9860-456B-A090-206C743AC` `htm`)
- Status: documented (Phase 3 body fill).

16.34 Function Entry: ACDIMENABLEUPDATE

16.34.1 Name

‘acdimenableupdate‘

16.34.2 Class

Function

16.34.3 Syntax

`(acdimenableupdate flag)`

16.34.4 Arguments and Values

- ‘flag‘: ‘T‘ to enable automatic updating of associative dimensions, ‘nil‘ to suppress updates until DIMREGEN is invoked.

16.34.5 Description

Controls whether associative dimensions are automatically refreshed when their referenced geometry changes. Designed for batch-edit scenarios where deferring dimension recomputation until after several modifications avoids repeated work.

16.34.6 Return Values

Always returns ‘nil‘.

16.34.7 Side Effects

Updates the document-level associative-dimension auto-update flag.

16.34.8 Examples

- ‘(acdimenableupdate nil)‘ disables automatic updates.
- ‘(acdimenableupdate T)‘ re-enables them.

16.34.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (BricsCAD also supports associative dimensions).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.34.10 Source Notes

- [acdimenableupdate (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7E10D18A-6FD6-4B20-A981` `htm`)
- Status: documented (Phase 3 body fill).

16.35 Function Entry: ALLOC

16.35.1 Name

‘alloc‘

16.35.2 Class

Function

16.35.3 Syntax

```
(alloc n-alloc)
```

16.35.4 Arguments and Values

- ‘n-alloc‘: integer; the new segment-allocation size used by ‘expand‘. The number controls how many symbols, strings, usubrs, reals, and cons cells are allocated per future ‘expand‘ call.

16.35.5 Description

Sets the segment-allocation size used by the ‘expand‘ function.

16.35.6 Return Values

The previous ‘n-alloc‘ value (an integer).

16.35.7 Side Effects

Updates the AutoLISP allocator’s segment-size parameter.

16.35.8 Examples

- ‘(alloc 100)‘ returns ‘1000‘ (the previous segment size, on a fresh session with the documented default).

16.35.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (memory-knob behaviour may differ).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.35.10 Source Notes

- [alloc (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-DBC72668-153D-48F4-9C19-87053975A97C.htm`)
- Status: documented (Phase 3 body fill).

16.36 Function Entry: EXPAND

16.36.1 Name

‘expand‘

16.36.2 Class

Function

16.36.3 Syntax

`(expand n-expand)`

16.36.4 Arguments and Values

- ‘n-expand‘: integer; the number of additional segments to allocate. Each segment provides space for further symbols, strings, usubrs, reals, and cons cells based on the current ‘alloc‘ setting.

16.36.5 Description

Allocates additional memory for AutoLISP. The granularity of each segment is controlled by the prior ‘alloc‘ call.

16.36.6 Return Values

An integer: the number of free conses divided by ‘n-alloc‘.

16.36.7 Side Effects

Grows the AutoLISP heap by ‘n-expand‘ segments.

16.36.8 Examples

After ‘(alloc 100)‘ returns ‘1000‘:

- ‘(expand 2)‘ returns ‘82‘, providing 200+ additional symbols/strings/usubrs/reals plus 8200 available cons cells.

16.36.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (memory-knob behaviour may differ).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.36.10 Source Notes

- [expand (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-48C99C97-176D-4D9C-91E2-31B3C67D6` `htm`)
- Status: documented (Phase 3 body fill).

16.37 Function Entry: GC

16.37.1 Name

‘gc‘

16.37.2 Class

Function

16.37.3 Syntax

`(gc)`

16.37.4 Arguments and Values

None.

16.37.5 Description

Forces a garbage collection, freeing memory occupied by inaccessible objects.

16.37.6 Return Values

Always returns ‘nil‘.

16.37.7 Side Effects

Reclaims AutoLISP heap memory; may pause the runtime briefly.

16.37.8 Examples

Not documented on the vendor page.

16.37.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.37.10 Source Notes

- [gc (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6CABB3AF-9BED-4237-8A44-D36123246BF4.htm`)
- Status: documented (Phase 3 body fill).

16.38 Function Entry: MEM

16.38.1 Name

'mem'

16.38.2 Class

Function

16.38.3 Syntax

`(mem)`

16.38.4 Arguments and Values

None.

16.38.5 Description

Displays AutoLISP memory-usage statistics: garbage-collection counts and timings, dynamic-segment information (page size, used / free pages, largest contiguous free area, segment count), and per-type allocation (LISP stacks, bytecode area, CONS memory, strings, etc.). All heap memory is global; every AutoCAD document shares the same heap.

16.38.6 Return Values

Always returns 'nil'. The statistics are written to the command line as a side effect.

16.38.7 Side Effects

Writes a multi-line memory report to the AutoCAD command line.

16.38.8 Examples

- A typical '(mem)' invocation prints lines such as:

```
; GC calls: 23; GC run time: 298 ms
Dynamic memory segments statistic:
PgSz  Used  Free  FMCL  Segs  Type
 512    79    48    48     1  lisp stacks
```

```
 256  3706   423   142    16  bytecode area
4096   320    10    10    22  CONS memory
 ...
```

16.38.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (the report content is implementation-defined).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.38.10 Source Notes

- [mem (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F4AEB953-2117-4BF2-8056-EA1384AC3FFF.htm)
- Status: documented (Phase 3 body fill).

16.39 Function Entry: TABLET

16.39.1 Name

‘tablet‘

16.39.2 Class

Function

16.39.3 Syntax

```
(tablet code [row1 row2 row3 direction])
```

16.39.4 Arguments and Values

- ‘code‘: integer.
  - ‘0‘: retrieve the current digitiser calibration; the trailing arguments are omitted.
  - ‘1‘: set the calibration; the trailing arguments are required.
- ‘row1‘, ‘row2‘, ‘row3‘: lists of three reals forming the rows of a 3_3 transformation matrix. The Z component of ‘row3‘ is normalised to 1.
- ‘direction‘: list of three reals; the normal vector to the tablet surface in WCS. Automatically normalised.

16.39.5 Description

Retrieves or sets digitiser (tablet) calibrations. Calibration data lets AutoCAD transform raw digitiser coordinates into drawing-space coordinates.

16.39.6 Return Values

An integer on success. On failure, returns ‘nil‘ and sets the AutoCAD ‘ERRNO‘ system variable; typical failure cause is that the input device is not a tablet.

16.39.7 Side Effects

Updates the AutoCAD digitiser calibration state.

16.39.8 Examples

- Identity calibration: ‘(tablet 1 ’(1 0 0) ’(0 1 0) ’(0 0 1) ’(0 0 1))‘.

16.39.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible (Windows-only digitiser feature).
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

16.39.10 Source Notes

- [tablet (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5AD61389-1F8E-497B-8B0B-259488A5C` `htm`)
- Status: documented (Phase 3 body fill).

16.40 Function Entry: VPORTS

16.40.1 Name

‘vports‘

16.40.2 Class

Function

16.40.3 Syntax

`(vports)`

16.40.4 Arguments and Values

None.

16.40.5 Description

Returns a list of viewport descriptors for the current viewport configuration. The shape of the result depends on the ‘TILEMODE‘ system variable: tiled viewports (VPORTS command) when ‘TILEMODE‘ is 1, floating viewports (MVIEW command) when ‘TILEMODE‘ is 0. When ‘TILEMODE‘ is off, viewport number 1 is always paper space.

16.40.6 Return Values

A list of viewport descriptors. Each descriptor is ‘(viewport-id (lower-left) (upper-right))‘. With ‘TILEMODE‘ on, the corners are values in [0.0, 1.0] relative to the display screen. With ‘TILEMODE‘ off, the corners use paper-space coordinates. The current viewport's descriptor is always first in the list.

16.40.7 Side Effects

None.

16.40.8 Examples

- A single viewport with ‘TILEMODE‘ on: ‘((1 (0.0 0.0) (1.0 1.0)))‘.
- Four equal tiled viewports:

```
((5 (0.5 0.0) (1.0 0.5))
 (2 (0.5 0.5) (1.0 1.0))
 (3 (0.0 0.5) (0.5 1.0))
 (4 (0.0 0.0) (0.5 0.5)))
```

16.40.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.40.10 Source Notes

- [vports (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-929FC8B2-1235-4B2A-B330-FF2E219E0htm`)
- Status: documented (Phase 3 body fill).

16.41 Function Entry: SETVIEW

16.41.1 Name

‘setview‘

16.41.2 Class

Function

16.41.3 Syntax

```
(setview view_descriptor [vport_id])
```

16.41.4 Arguments and Values

- ‘$\text{view}_{\text{descriptor}}$‘: an entity-definition list similar to that returned by ‘tblsearch‘ for the ‘VIEW‘ symbol table.
- ‘vport_{id}‘ (optional): integer; the destination viewport. ‘0‘ selects the current viewport. The viewport identifier is held by the ‘CVPORT‘ system variable.

16.41.5 Description

Establishes a view for a specified viewport.

16.41.6 Return Values

The ‘$\text{view}_{\text{descriptor}}$‘ on success, or ‘nil‘ on failure.

16.41.7 Side Effects

Updates the named viewport’s view to the supplied descriptor.

16.41.8 Examples

Not documented on the vendor page.

16.41.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.41.10 Source Notes

- [setview (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-EEABEF66-423A-4D2B-BE16-AA6A56B08` `htm`)
- Status: documented (Phase 3 body fill).

16.42 Function Entry: ARX

16.42.1 Name

‘arx‘

16.42.2 Class

Function

16.42.3 Syntax

`(arx)`

16.42.4 Arguments and Values

None.

16.42.5 Description

Returns a list naming the ObjectARX applications currently loaded into the AutoCAD process.

16.42.6 Return Values

A list of strings. File extensions vary by platform: ‘.arx‘ on Windows / Web; ‘.bundle‘ or ‘.crx‘ on macOS.

16.42.7 Side Effects

None.

16.42.8 Examples

- Windows / Web: ‘(”acapp.arx” ”acmted.arx” …)‘.
- macOS: ‘(”acapp.bundle” ”acmted.crx” …)‘.

16.42.9 Availability

- AutoCAD 2026: documented (Windows, macOS, Web).
- BricsCAD V26: not applicable. BricsCAD uses the BRX runtime; querying the BRX module list is via a different interface.

- Status: AutoCAD documented; BricsCAD does not expose this name.

16.42.10 Source Notes

- [arx (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D74F3A2B-F506-415F-8F5A-258764C2E26D.htm`)
- Status: documented (Phase 3 body fill).

16.43 Function Entry: CVUNIT

16.43.1 Name

‘cvunit‘

16.43.2 Class

Function

16.43.3 Syntax

```
(cvunit value from-unit to-unit)
```

16.43.4 Arguments and Values

- ‘value‘: integer, real, or list (a 2D / 3D point).
- ‘from-unit‘: string naming the source unit.
- ‘to-unit‘: string naming the target unit.

16.43.5 Description

Converts ‘value‘ from one unit of measurement to another. Unit names must exist in ‘acad.unt‘ (or ‘acadlt.unt‘ for AutoCAD LT). Returns ‘nil‘ when a unit name is unknown or when the source and target units are incompatible (e.g. grams to years). For repeated conversions of the same unit pair, convert ‘1.0‘ once and use the result as a scale factor — except for temperature units, which involve an offset.

16.43.6 Return Values

The converted value as a real (or list of reals if ‘value‘ is a list); ‘nil‘ on failure.

16.43.7 Side Effects

None.

16.43.8 Examples

- ‘(cvunit 1 ”minute” ”second”)‘ returns ‘60.0‘.
- ‘(cvunit 1.0 ”inch” ”cm”)‘ returns ‘2.54‘.
- ‘(cvunit 1.0 ”acre” ”sq yard”)‘ returns ‘4840.0‘.
- ‘(cvunit ’(1.0 2.5) ”ft” ”in”)‘ returns ‘(12.0 30.0)‘.

16.43.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.43.10 Source Notes

- [cvunit (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-253E72F1-09B8-473E-A51B-B97BE2E29` `htm`)
- Status: documented (Phase 3 body fill).

16.44 Function Entry: REDRAW

16.44.1 Name

‘redraw‘

16.44.2 Class

Function

16.44.3 Syntax

```
(redraw [ename [mode]])
```

16.44.4 Arguments and Values

- ‘ename‘ (optional): an entity name; the entity to redraw.
- ‘mode‘ (optional): an integer:
  - 1: show entity
  - 2: hide entity
  - 3: highlight entity
  - 4: unhighlight entity

  Negative values affect only the header entity of a complex object (polyline, attributed block); positive values affect all subentities.

16.44.5 Description

Redraws the current viewport, or a specified entity in the current viewport. With no arguments, refreshes the entire viewport.

16.44.6 Return Values

Always returns ‘nil‘.

16.44.7 Side Effects

Refreshes the screen image of the affected viewport region.

16.44.8 Examples

Not documented on the vendor page.

16.44.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.44.10 Source Notes

- [redraw (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4A4DCECD-E85A-4860-A58F-56B48227
htm)
- Status: documented (Phase 3 body fill).

16.45 Function Entry: GRAPHSCR

16.45.1 Name

‘graphscr‘

16.45.2 Class

Function

16.45.3 Syntax

`(graphscr)`

16.45.4 Arguments and Values

None.

16.45.5 Description

Displays the AutoCAD graphics screen. Equivalent to the AutoCAD GRAPHSCR command. On macOS and Web platforms, the function is recognised but produces no visible effect; the complementary function ‘textscr‘ switches focus to the text screen on Windows.

16.45.6 Return Values

Always returns ‘nil‘.

16.45.7 Side Effects

Focuses the graphics window on Windows.

16.45.8 Examples

Not documented on the vendor page.

16.45.9 Availability

- AutoCAD 2026: documented (Windows; recognised but no-op on macOS / Web).
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.45.10 Source Notes

- [graphscr (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-FB9ABA5D-6DDA-4085-9E9F-18038EC31` `htm`)
- Status: documented (Phase 3 body fill).

16.46 Function Entry: TEXTPAGE

16.46.1 Name

‘textpage‘

16.46.2 Class

Function

16.46.3 Syntax

`(textpage)`

16.46.4 Arguments and Values

None.

16.46.5 Description

Switches focus from the drawing area to the text screen. Equivalent to ‘textscr‘. On macOS and Web platforms, the function is recognised but produces no visible effect.

16.46.6 Return Values

Always returns ‘nil‘.

16.46.7 Side Effects

Focuses the text-screen window on Windows.

16.46.8 Examples

Not documented on the vendor page.

16.46.9 Availability

- AutoCAD 2026: documented (Windows; recognised but no-op on macOS / Web).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.46.10 Source Notes

- [textpage (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C5F1E0AF-CAE8-4338-8C27-5A4235D37` `htm`)
- Status: documented (Phase 3 body fill).

16.47 Function Entry: TEXTSCR

16.47.1 Name

'textscr'

16.47.2 Class

Function

16.47.3 Syntax

`(textscr)`

16.47.4 Arguments and Values

None.

16.47.5 Description

Switches focus from the drawing area to the text screen. Equivalent to 'textpage'. On macOS and Web platforms, the function is recognised but produces no visible effect.

16.47.6 Return Values

Always returns 'nil'.

16.47.7 Side Effects

Focuses the text-screen window on Windows.

16.47.8 Examples

Not documented on the vendor page.

16.47.9 Availability

- AutoCAD 2026: documented (Windows; recognised but no-op on macOS / Web).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.47.10 Source Notes

- [textscr (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5F0A5635-0DCB-4A88-BD8E-60EC66534 htm)
- Status: documented (Phase 3 body fill).

16.48 Function Entry: GRCLEAR

16.48.1 Name

‘grclear‘

16.48.2 Class

Function (legacy / obsolete)

16.48.3 Syntax

`(grclear)`

16.48.4 Arguments and Values

None.

16.48.5 Description

Clears the current viewport. Documented as an obsolete legacy display function; modern code should use ‘redraw‘ instead.

16.48.6 Return Values

Returns ‘nil‘.

16.48.7 Side Effects

Clears the viewport on legacy display devices.

16.48.8 Examples

Not documented on the vendor page.

16.48.9 Availability

- AutoCAD 2026: documented as legacy / obsolete.
- BricsCAD V26: presumed compatible (legacy function, behaviour may be a no-op).
- Status: AutoCAD documented as obsolete; BricsCAD presumed-both pending Phase 4 verification.

16.48.10 Source Notes

- [G Functions Reference (AutoLISP 2026) — ‘grclear‘ summary](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-88320FBD-B49E-445A-A5C1-C2ABA98341BC.htm`)
- Status: legacy; per-symbol page not separately published.

16.49 Function Entry: GRDRAW

16.49.1 Name

‘grdraw‘

16.49.2 Class

Function

16.49.3 Syntax

```
(grdraw from to color [highlight])
```

16.49.4 Arguments and Values

- ‘from‘: list of two or three reals; one endpoint of the vector in the current UCS.
- ‘to‘: list of two or three reals; the other endpoint.
- ‘color‘: integer; AutoCAD color number 0–255. The value ‘-1‘ selects XOR ink, which complements anything it draws over and erases itself when overdrawn.
- ‘highlight‘ (optional): integer; if non-zero, the vector is drawn with the device’s default highlighting style (typically dashed).

16.49.5 Description

Draws a transient vector between two points in the current viewport. The drawing is automatically clipped to the screen.

16.49.6 Return Values

Always returns ‘nil‘.

16.49.7 Side Effects

Draws transient (non-persistent) graphics in the viewport.

16.49.8 Examples

Not documented on the vendor page.

16.49.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.49.10 Source Notes

- [grdraw (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C00544A7-AF63-403E-98EB-205248EB4` `htm`)
- Status: documented (Phase 3 body fill).

16.50 Function Entry: GRTEXT

16.50.1 Name

‘grtext‘

16.50.2 Class

Function

16.50.3 Syntax

```
(grtext [box text [highlight]])
```

16.50.4 Arguments and Values

- ‘box‘ (optional): integer; ‘0‘ or greater selects a screen-menu box, ‘-1‘ selects the mode status line, ‘-2‘ selects the coordinate status line.
- ‘text‘ (optional): string to display. Truncated by the device width.
- ‘highlight‘ (optional): integer; for screen-menu boxes (‘box >= 0‘) a positive value highlights and zero unhighlights. Ignored for status-line boxes; coordinate tracking can overwrite the ‘-2‘ area.

16.50.5 Description

Displays the supplied text in a screen-menu area or the status line; does not change the underlying menu item. With no arguments, restores all text areas to default values. Recognised but no-op on macOS and Web platforms.

16.50.6 Return Values

Returns the ‘text‘ argument when successful, otherwise ‘nil‘.

16.50.7 Side Effects

Updates the indicated screen-menu / status-line area.

16.50.8 Examples

Not documented on the vendor page.

16.50.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.50.10 Source Notes

- [grtext (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5B0B01CA-BDBA-4A77-B840-6EE64591A htm`)
- Status: documented (Phase 3 body fill).

16.51 Function Entry: GRVECS

16.51.1 Name

‘grvecs‘

16.51.2 Class

Function

16.51.3 Syntax

```
(grvecs vlist [trans])
```

16.51.4 Arguments and Values

- ‘vlist‘: a vector list of optional color integers paired with two point lists, in the form ‘([color1] from1 to1 [color2] from2 to2 ...)‘. Color values 0–255 are AutoCAD colors; values above 255 select XOR ink; negative values select the device’s highlight style.
- ‘trans‘ (optional): a 4_4 transformation matrix (a list of four 4-real lists) that repositions or scales every vector in ‘vlist‘.

16.51.5 Description

Draws multiple transient vectors in the drawing area. A specified color persists for all subsequent vectors until another color value is given.

16.51.6 Return Values

Always returns ‘nil‘.

16.51.7 Side Effects

Draws transient graphics; nothing is committed to the drawing database.

16.51.8 Examples

- Five vertical lines with five different colors:

```
(grvecs '(1 (1 2)(1 5)
          2 (2 2)(2 5)
```

```
3 (3 2)(3 5)
4 (4 2)(4 5)
5 (5 2)(5 5)))
```

- Identity transformation with translation ‘(5,5,0)‘:

```
'((1.0 0.0 0.0 5.0)
  (0.0 1.0 0.0 5.0)
  (0.0 0.0 1.0 0.0)
  (0.0 0.0 0.0 1.0))
```

16.51.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

16.51.10 Source Notes

- [grvecs (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-44B8C277-4140-4F4C-ACF2-3332CC463 htm)
- Status: documented (Phase 3 body fill).

17 17 Selection Sets and Entity Access

17.1 Overview

Selection sets and entity access are core host-facing language facilities.

17.2 Normative Rules

- entity data is represented as association lists of dotted pairs
- selection-set indices are zero-based
- some high indices require real-valued indices in Autodesk semantics

17.3 Source Notes

- [ssname (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-LT-AutoLISP-Reference/files/GUID-EFB83751-9AC5-4C52-AD3D-D971BC560C1` `htm`)
- [entmake (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-D47983BA-1E5D-417D-85B8-6F3DE5F506E` `htm`)

17.4 Function Entry: GETSTRING

17.4.1 Name

‘getstring‘

17.4.2 Class

Function

17.4.3 Syntax

`(getstring [cr] [msg])`

17.4.4 Arguments and Values

- ‘cr‘: optional flag controlling acceptance of spaces
- ‘msg‘: optional prompt string

17.4.5 Description

Prompts the user for a string response.

17.4.6 Return Values

Returns the entered string.

17.4.7 Side Effects

- prompts at the command line
- consumes user input

17.4.8 Exceptional Situations

Cancel and interactive-input errors are host-dependent.

17.4.9 Notes

Autodesk explicitly notes that the user cannot enter another AutoLISP expression as the response; this is user input, not reader input.

17.4.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.4.11 Source Notes

- [getstring (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/FRA/AutoCAD-LT-AutoLISP-Reference/files/GUID-B139EFBD-74B7-4276-B422-D2186F7D8DC htm)
- Status: documented.

17.5 Function Entry: GETPOINT

17.5.1 Name

‘getpoint‘

17.5.2 Class

Function

17.5.3 Syntax

```
(getpoint [pt] [msg])
```

17.5.4 Arguments and Values

- ‘pt‘: optional base point
- ‘msg‘: optional prompt string

Autodesk documents ‘pt‘ as a 2D or 3D base point in the current UCS. Autodesk also documents that a single integer or real may be supplied as ‘pt‘, invoking direct-distance entry from ‘LASTPOINT‘.

17.5.5 Description

Prompts the user to specify a point.

17.5.6 Return Values

Returns a 3D point list or ‘nil‘.

17.5.7 Side Effects

- prompts the user
- consumes interactive input

17.5.8 Compatibility

Host-interaction-specific; point representation remains ordinary list data.

17.5.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.5.10 Source Notes

- [getpoint (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-445F32F0-8A9D-4E1D-976F-DE87CC5267D htm`)

17.5.11 Notes

- Autodesk documents that if ‘pt‘ is present as a point, the host draws a rubber-band line from that point to the current crosshair position.
- Autodesk documents that the user cannot respond with another AutoLISP expression.

17.6 Function Entry: DISTANCE

17.6.1 Name

‘distance‘

17.6.2 Class

Function

17.6.3 Syntax

```
(distance pt1 pt2)
```

17.6.4 Arguments and Values

- ‘pt1‘, ‘pt2‘: points

17.6.5 Description

Returns the distance between two points.

17.6.6 Return Values

Real.

17.6.7 Side Effects

None.

17.6.8 Exceptional Situations

Erroneous for invalid point arguments.

17.6.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.6.10 Source Notes

- [distance (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-62F2AA98-1CE4-4E31-A2B7-E2C49A92E38` `htm`)

17.6.11 Notes

- Autodesk documents that if either argument is a 2D point, the function ignores any Z coordinates and returns the 2D distance projected into the current construction plane.

17.7 Function Entry: ANGLE

17.7.1 Name

‘angle‘

17.7.2 Class

Function

17.7.3 Syntax

`(angle pt1 pt2)`

17.7.4 Arguments and Values

- ‘pt1‘, ‘pt2‘ — 2D or 3D points.

17.7.5 Description

Returns the angle from ‘pt1‘ to ‘pt2‘.

17.7.6 Return Values

Real.

The returned angle is in radians.

17.7.7 Side Effects

None.

17.7.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.7.9 Source Notes

- [angle (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-AutoLISP-Reference/files/GUID-F28755D4-E89F-43EF-8E76-40518C7E2728.htm`)

17.7.10 Notes

- Autodesk documents that the angle is measured from the X axis of the current construction plane, increasing counterclockwise.
- Autodesk documents that if 3D points are supplied, they are projected onto the current construction plane.

17.8 Function Entry: NAMEDOBJDICT

17.8.1 Name

‘namedobjdict‘

17.8.2 Class

Host Function

17.8.3 Syntax

`(namedobjdict)`

17.8.4 Description

Returns the entity name of the named object dictionary.

17.8.5 Return Values

An ‘ENAME‘ designating the root named-object dictionary.

17.8.6 Side Effects

None.

17.8.7 Compatibility

This is a key dictionary-entry point into the drawing database.

17.8.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.8.9 Source Notes

- [namedobjdict (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-A1E43B30-EF8C-452E-97F5-8AC201 htm)
- Status: documented.

17.9 Function Entry: TBLNEXT

17.9.1 Name

‘tblnext‘

17.9.2 Class

Host Function

17.9.3 Syntax

```
(tblnext table-name [rewind])
```

17.9.4 Arguments and Values

- ‘table-name‘: symbol or string identifying a symbol table
- ‘rewind‘: optional flag

17.9.5 Description

Returns successive symbol-table entries as association-list data.

17.9.6 Return Values

An association list representing the next table entry, or ‘nil‘ when no further entry exists.

17.9.7 Side Effects

Advances table-iteration state.

17.9.8 Compatibility

Central to symbol-table iteration over host database structures.

17.9.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.9.10 Source Notes

- [tblnext (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PTB/AutoCAD-LT-AutoLISP-Reference/files/GUID-1720D8DC-5559-4AC1-8A75-3C932834D77` `htm`)
- Status: documented.

17.10 Function Entry: ENTGET

17.10.1 Name

‘entget‘

17.10.2 Class

Host Function

17.10.3 Syntax

```
(entget ename [applist])
```

17.10.4 Arguments and Values

- ‘ename‘: entity name
- ‘applist‘: optional list of registered application names

17.10.5 Description

Returns the definition data of an entity as an association list of dotted pairs.

17.10.6 Return Values

Association list representing the entity definition.

If ‘applist‘ is supplied, Autodesk documents that ‘entget‘ also retrieves extended data for the specified registered applications.

17.10.7 Side Effects

None.

17.10.8 Affected By

- validity of the entity name
- the optional application filter list

17.10.9 Exceptional Situations

Erroneous if the entity name is invalid or unacceptable to the host.

17.10.10 Examples

- ‘(entget ename) => ((-1 . <ename>) (0 . ”LINE”) …)‘

17.10.11 Compatibility

This is one of the central functions of the AutoLISP host database model.

17.10.12 Notes

‘entget‘ strongly reinforces the importance of alists and dotted pairs in the language specification.

Autodesk explicitly notes that AutoLISP DXF group codes differ slightly from those in DXF files.

17.10.13 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.10.14 Source Notes

- [entget (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-12540DAE-C84B-4BDB-AEEC-DDFE5BE3C42A.htm`)
- Status: documented.

17.11 Function Entry: ENTMOD

17.11.1 Name

‘entmod‘

17.11.2 Class

Host Function

17.11.3 Syntax

`(entmod elist)`

17.11.4 Arguments and Values

- ‘elist‘: entity definition association list

Autodesk documents that the entity name to modify is taken from group ‘-1‘ in ‘elist‘.

17.11.5 Description

Modifies an entity in the drawing database according to the supplied entity definition list.

17.11.6 Return Values

List or ‘nil‘.

If successful, Autodesk documents that ‘entmod‘ returns the supplied ‘elist‘; otherwise it returns ‘nil‘.

17.11.7 Side Effects

Modifies the drawing database.

17.11.8 Exceptional Situations

Erroneous if the entity definition list is invalid or requests an illegal modification.

17.11.9 Compatibility

Semantically paired with ‘entget‘.

17.11.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.11.11 Source Notes

- [entmod (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-LT-AutoLISP-Reference/files/GUID-C7D27797-247E-49B9-937C-0D8C58F4C83` `htm`)
- [About Modifying an Entity without the Command Function (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-MAC-AutoLisp/files/GUID-125DC058-BBAA-4CA9-A203-53F0A27B87D0.htm`)

17.11.12 Notes

- Autodesk documents that an entity’s type and handle cannot be changed.
- Autodesk documents that certain internal fields, such as the ‘-2‘ group of a ‘SEQEND‘, are ignored if modified.
- Autodesk documents that viewport entities cannot be modified with ‘entmod‘.
- Autodesk documents that changing entities inside a block definition affects all inserts of that block.
- Autodesk documents special color behavior in AutoCAD 2004 and later: if true-color group ‘420‘ conflicts with ACI group ‘62‘, the true-color value takes precedence.

17.12 Function Entry: ENTMAKE

17.12.1 Name

‘entmake‘

17.12.2 Class

Host Function

17.12.3 Syntax

`(entmake elist)`

17.12.4 Arguments and Values

- ‘elist‘: entity definition association list

17.12.5 Description

Creates a new entity from a list of dotted pairs.

17.12.6 Return Values

Entity definition list for the new entity, or ‘nil‘ on failure according to Autodesk documentation.

17.12.7 Side Effects

Creates an entity in the drawing database.

17.12.8 Affected By

- validity of entity type and group codes
- current database context

17.12.9 Exceptional Situations

Invalid definitions may fail or be erroneous depending on host behavior.

17.12.10 Compatibility

One of the most characteristic AutoLISP functions because it exposes CAD entities through list structure.

17.12.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.12.12 Source Notes

- [entmake (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-D47983BA-1E5D-417D-85B8-6F3DE5F506` `htm`)
- Status: documented.

17.13 Function Entry: ENTLAST

17.13.1 Name

‘entlast‘

17.13.2 Class

Host Function

17.13.3 Syntax

`(entlast)`

17.13.4 Description

Returns the entity name of the last non-deleted main object in the drawing.

17.13.5 Return Values

‘ENAME‘ or ‘nil‘.

17.13.6 Side Effects

None.

17.13.7 Compatibility

Frequently used in create-and-query workflows together with ‘entmake‘ and ‘entget‘.

17.13.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.13.9 Source Notes

- [entlast (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-75DBA9B2-034B-4377-A4E2-21D37B298D8` `htm`)

17.13.10 Notes

- Autodesk documents that the entity need not be visible on screen or on a thawed layer.
- Autodesk documents that ‘entlast‘ returns the last nondeleted main entity only; if the last created object has trailing subentities, a loop using ‘entnext‘ is needed to find the last nondeleted subentity.

17.14 Function Entry: SSGET

17.14.1 Name

‘ssget‘

17.14.2 Class

Host Function

17.14.3 Syntax

```
(ssget [sel-method] [pt1 [pt2]] [pt-list] [filter-list])
```

17.14.4 Arguments and Values

- optional selection method
- optional points
- optional point list
- optional filter list

17.14.5 Description

Constructs a selection set by interactive or programmatic selection.

17.14.6 Return Values

Selection set (‘PICKSET‘) or ‘nil‘.

17.14.7 Side Effects

- may prompt the user
- may consume interactive input
- depends on drawing state and selection semantics

17.14.8 Affected By

- selection mode
- filter list syntax
- current host selection settings
- vendor-specific extensions

17.14.9 Exceptional Situations

Autodesk documents many ‘ERRNO‘ cases related to invalid filter lists and selection usage.

17.14.10 Compatibility

This is one of the most host-specific functions in the core language.

BricsCAD has historically documented additional ‘ssget‘ modes and differences.

17.14.11 Notes

Autodesk explicitly documents selection methods including at least:

- ‘A‘
- ‘C‘
- ‘CP‘
- ‘F‘
- ‘I‘
- ‘L‘
- ‘P‘
- ‘W‘
- ‘WP‘
- ‘X‘

The selection-method sublanguage should later be given its own dedicated specification subsection.

17.14.12 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.14.13 See Also

- BricsCAD V9 documented ‘ssget‘ as compatible-with-enhancements. The current BricsCAD V26 surface is presumed compatible with AutoCAD; specific BricsCAD enhancements should be verified in Phase 5 against a live product.

17.14.14 Source Notes

- [About Selection Set Filter Lists (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-AutoLISP/files/GUID-7BE77062-C359-4D01-915B-69CF6` `htm`)
- [ssget (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-0F37CC5E-1559-4011-B8CF-A3BA0973B2C3.htm`)
- [LISP Compatibility, BricsCAD V9](`https://developer.bricsys.com/bricscad/help/en_US/V9/DevRef/source/IDR_LISP_B100.htm`)
- [Error Codes Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/HUN/AutoCAD-AutoLISP/files/GUID-97327347-2A13-4CBC-BDBF-979C7F1CA` `htm`)
- Status: documented in role and partially in detail; exhaustive selection-mode semantics still to be expanded.

17.15 Function Entry: SSNAME

17.15.1 Name

‘ssname‘

17.15.2 Class

Host Function

17.15.3 Syntax

```
(ssname ss index)
```

17.15.4 Arguments and Values

- ‘ss‘: selection set
- ‘index‘: selection index

17.15.5 Description

Returns the entity name at the specified index in the selection set.

17.15.6 Return Values

‘ENAME‘.

17.15.7 Side Effects

None.

17.15.8 Exceptional Situations

Erroneous for invalid sets or indices.

17.15.9 Compatibility

Autodesk documents zero-based indexing and special handling for indices above 32,767.

17.15.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.15.11 Source Notes

- [ssname (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-LT-AutoLISP-Reference/files/GUID-EFB83751-9AC5-4C52-AD3D-D971BC560C1`
`htm`)
- Status: documented.

17.16 Function Entry: SSLENGTH

17.16.1 Name

‘sslength‘

17.16.2 Class

Host Function

17.16.3 Syntax

`(sslength ss)`

17.16.4 Arguments and Values

- ‘ss‘: selection set

17.16.5 Description

Returns the number of entities in the selection set.

17.16.6 Return Values

Integer count.

17.16.7 Side Effects

None.

17.16.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.16.9 Source Notes

- [sslength (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-034B9EC8-0945-48A0-802A-9725DDBA0EF2.htm`)

17.16.10 Compatibility

Supported on Windows, macOS, and Web in current Autodesk documentation.

17.17 Function Entry: ENTSEL

17.17.1 Name

'entsel'

17.17.2 Class

Host Function

17.17.3 Syntax

```
(entsel [msg])
```

17.17.4 Arguments and Values

- 'msg': optional prompt string

17.17.5 Description

Prompts the user to select an entity.

17.17.6 Return Values

List or 'nil'.

If successful, Autodesk documents a two-element list:

- the selected entity name
- the pick point in the current UCS

17.17.7 Side Effects

- prompts the user
- consumes interactive input

17.17.8 Compatibility

This is a standard bridge between interactive selection and Lisp data.

17.17.9 Notes

- Autodesk documents that the returned pick point is the crosshair position at selection time and need not lie on the selected object.
- Autodesk documents that the current Osnap setting is ignored unless explicitly requested while in the function.
- Autodesk documents that ‘entsel‘ honors keywords established by a preceding ‘initget‘.

17.17.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.17.11 Source Notes

- [entsel (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9D4CF74D-8B8B-4D66-A952-564AFBA254E7.htm`)

17.18 Function Entry: DICTSEARCH

17.18.1 Name

'dictsearch'

17.18.2 Class

Host Function

17.18.3 Syntax

`(dictsearch ename symbol [setnext])`

17.18.4 Arguments and Values

- 'ename': dictionary entity name
- 'symbol': key name
- 'setnext': optional iteration-control flag

Autodesk documents 'symbol' as a string.

17.18.5 Description

Searches a dictionary for a named entry.

17.18.6 Return Values

Association list for the matching entry, or 'nil'.

17.18.7 Side Effects

May affect dictionary-iteration state when the optional flag is used.

17.18.8 Compatibility

Part of the drawing-database dictionary API.

17.18.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.18.10 Source Notes

- [dictsearch (AutoLISP)](`https://help.autodesk.com/cloudhelp/2018/JPN/AutoCAD-AutoLISP-Reference/files/GUID-FCB62959-7C51-41B6-8A0E-1350580F8364.htm`)

17.18.11 Notes

- If ‘setnext‘ is supplied and non-‘nil‘, Autodesk documents that the next ‘dictnext‘ call returns the entry following the one found by ‘dictsearch‘.

17.19 Function Entry: TBLSEARCH

17.19.1 Name

‘tblsearch‘

17.19.2 Class

Host Function

17.19.3 Syntax

`(tblsearch table-name symbol [setnext])`

17.19.4 Arguments and Values

- ‘table-name‘: symbol table designator
- ‘symbol‘: entry name
- ‘setnext‘: optional iteration-control flag

Autodesk documents both ‘table-name‘ and ‘symbol‘ as strings and case-insensitive.

17.19.5 Description

Searches a symbol table for a named entry.

17.19.6 Return Values

Association list for the matching entry, or ‘nil‘.

17.19.7 Side Effects

May affect table-iteration state when the optional flag is used.

17.19.8 Compatibility

Part of the core CAD database querying model.

17.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.19.10 Source Notes

- [tblsearch (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-MAC-AutoLISP-Reference/files/GUID-2AEB84A6-E3D0-4DD9-A29C-54D4099ED9 htm)

17.19.11 Notes

- If ‘setnext‘ is supplied and non-‘nil‘, Autodesk documents that the next ‘tblnext‘ call returns the entry following the one found by ‘tblsearch‘.

17.20 Function Entry: DICTADD

17.20.1 Name

‘dictadd‘

17.20.2 Class

Host Function

17.20.3 Syntax

```
(dictadd ename symbol newobj)
```

17.20.4 Arguments and Values

- ‘ename‘: dictionary entity name
- ‘symbol‘: entry name
- ‘newobj‘: entity name of the object to add

Autodesk documents ‘symbol‘ as a string naming a unique key in the dictionary.

17.20.5 Description

Adds an object to a dictionary under the given name.

17.20.6 Return Values

Entity name of the object added to the dictionary.

17.20.7 Side Effects

Modifies the dictionary object in the drawing database.

17.20.8 Exceptional Situations

Erroneous for invalid dictionaries, invalid object references, or illegal naming collisions according to host rules.

17.20.9 Compatibility

Part of the named-object and extension-dictionary manipulation model.

17.20.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.20.11 Source Notes

- [dictadd (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-5931D6D8-7F6E-4773-B08C-DEC5F9C4A22` `htm`)

17.20.12 Notes

- Autodesk documents a general uniqueness rule for objects within a dictionary.

17.21 Function Entry: DICTNEXT

17.21.1 Name

‘dictnext‘

17.21.2 Class

Host Function

17.21.3 Syntax

```
(dictnext ename [rewind])
```

17.21.4 Arguments and Values

- ‘ename‘: dictionary entity name
- ‘rewind‘: optional iteration reset flag

17.21.5 Description

Returns successive entries from a dictionary.

17.21.6 Return Values

Association list representing the next dictionary entry, or ‘nil‘.

Autodesk documents that deleted dictionary entries are not returned.

17.21.7 Side Effects

Advances dictionary-iteration state.

17.21.8 Compatibility

Closely parallel to ‘tblnext‘.

17.21.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.21.10 Source Notes

- [dictnext (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-AutoLISP-Reference/files/GUID-8596E2E9-FEB5-4995-ADE4-31287864C0CB.htm`)

17.21.11 Notes

- If ‘rewind‘ is non-‘nil‘, Autodesk documents that iteration restarts at the beginning.
- Autodesk documents that switching dictionaries during an iteration sequence makes the previous current position indeterminate.

17.22 Function Entry: DICTREMOVE

17.22.1 Name

‘dictremove‘

17.22.2 Class

Host Function

17.22.3 Syntax

`(dictremove ename symbol)`

17.22.4 Arguments and Values

- ‘ename‘: dictionary entity name
- ‘symbol‘: entry name

Autodesk documents ‘symbol‘ as a string.

17.22.5 Description

Removes a named entry from a dictionary.

17.22.6 Return Values

Entity name or ‘nil‘.

If successful, Autodesk documents that the entity name of the removed entry is returned; otherwise ‘nil‘ is returned.

17.22.7 Side Effects

Modifies the dictionary in the drawing database.

17.22.8 Compatibility

Part of the dictionary mutation API.

17.22.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.22.10 Source Notes

- [dictremove (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/ENU/AutoCAD-AutoLISP/files/GUID-0A888F13-7C2A-4CB6-B849-56B72EAC972A.htm`)

17.22.11 Notes

- Autodesk documents that removing an entry from a dictionary does not normally delete the object from the database; ‘entdel‘ must be used separately when deletion is desired.
- Autodesk documents exceptions for groups and multiline styles, where the implementing code may automatically delete the removed entity.
- Autodesk documents that an actively referenced multiline style cannot be removed from the multiline style dictionary.

17.23 Function Entry: DICTRENAME

17.23.1 Name

'dictrename'

17.23.2 Class

Host Function

17.23.3 Syntax

`(dictrename ename old-name new-name)`

17.23.4 Arguments and Values

- 'ename': dictionary entity name
- 'old-name': old entry name
- 'new-name': new entry name

Autodesk documents 'old-name' and 'new-name' as strings.

17.23.5 Description

Renames an entry in a dictionary.

17.23.6 Return Values

String or 'nil'.
If successful, returns 'new-name'.

17.23.7 Side Effects

Modifies dictionary naming state.

17.23.8 Compatibility

Part of the dictionary mutation API.

17.23.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.23.10 Source Notes

- [dictrename (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/ITA/AutoCAD-AutoLISP-Reference/files/GUID-F99779A7-2A1A-4B04-84B5-C80CC8406 htm)

17.23.11 Notes

- Autodesk documents that ‘nil‘ is returned if the old name is not present, if the dictionary is invalid, if the new name is invalid, or if the new name already exists.

17.24 Function Entry: TBLOBJNAME

17.24.1 Name

‘tblobjname‘

17.24.2 Class

Host Function

17.24.3 Syntax

```
(tblobjname table-name symbol)
```

17.24.4 Arguments and Values

- ‘table-name‘: symbol table designator
- ‘symbol‘: entry name

Autodesk documents both as strings and case-insensitive.

17.24.5 Description

Returns the entity name of the specified symbol-table entry.

17.24.6 Return Values

‘ENAME‘ or ‘nil‘.

17.24.7 Side Effects

None.

17.24.8 Compatibility

Companion to ‘tblsearch‘ and ‘tblnext‘.

17.24.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.24.10 Source Notes

- [tblobjname (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-MAC-AutoLISP-Reference/files/GUID-B75D58C9-8ED9-4025-906A-76B24 htm)

17.24.11 Notes

- Autodesk documents that the returned entity name can be passed to ‘entget‘ and ‘entmod‘.

17.25 Function Entry: ENTDEL

17.25.1 Name

‘entdel‘

17.25.2 Class

Host Function

17.25.3 Syntax

```
(entdel ename)
```

17.25.4 Arguments and Values

- ‘ename‘: entity name

17.25.5 Description

Deletes or undeletes an entity, according to AutoLISP host semantics.

17.25.6 Return Values

Entity name.

17.25.7 Side Effects

Changes the deletion state of an entity in the drawing database.

17.25.8 Compatibility

Part of the core object-handling family.

17.25.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.25.10 Source Notes

- [entdel (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-AF320BD5-C83C-4EA0-983E-1EC885F5FC7 htm)

17.25.11 Notes

- Autodesk documents toggle behavior within the current editing session: a currently present entity is deleted, and a previously deleted entity is restored.
- Autodesk documents that deleted entities are purged when the drawing is exited.
- Autodesk documents that the function operates only on main entities, not attributes or polyline vertices independently.
- Autodesk documents that entities within a block definition cannot be deleted with ‘entdel‘.

17.26 Function Entry: ENTNEXT

17.26.1 Name

‘entnext‘

17.26.2 Class

Host Function

17.26.3 Syntax

`(entnext [ename])`

17.26.4 Arguments and Values

- ‘ename‘: optional entity name

17.26.5 Description

Returns the next entity following the supplied entity, or the first non-deleted entity if no argument is given.

17.26.6 Return Values

‘ENAME‘ or ‘nil‘.

17.26.7 Side Effects

None.

17.26.8 Compatibility

Important for object traversal and subentity walking.

17.26.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.26.10 Source Notes

- [entnext (AutoLISP)](`https://help.autodesk.com/cloudhelp/2021/ENU/AutoCAD-AutoLISP-Reference/files/GUID-65924CF5-0C51-4E36-8B38-7A5513951A04.htm`)

17.26.11 Notes

- Autodesk documents that ‘entnext‘ returns both main entities and subentities.
- Autodesk documents that ‘ssget‘ returns only main entities; ‘entnext‘ is used to walk subentity structure.

17.27 Function Entry: HANDENT

17.27.1 Name

‘handent‘

17.27.2 Class

Host Function

17.27.3 Syntax

`(handent handle)`

17.27.4 Arguments and Values

- ‘handle‘: entity handle string

17.27.5 Description

Returns the entity name associated with the supplied handle.

17.27.6 Return Values

‘ENAME‘ or ‘nil‘.

17.27.7 Side Effects

None.

17.27.8 Compatibility

Provides a bridge between persistent handle identity and transient entity-name identity.

17.27.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.27.10 Source Notes

- [handent (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-AF6DD533-1A24-4687-96EB-F03F26050C07.htm`)

17.27.11 Notes

- Autodesk documents that ‘handent‘ can return entities deleted during the current editing session, allowing them to be restored with ‘entdel‘.
- Autodesk documents that an entity name may change between editing sessions even though the handle remains constant.
- Autodesk documents that both graphical and nongraphical entities are supported.

17.28 Function Entry: SSADD

17.28.1 Name

‘ssadd‘

17.28.2 Class

Host Function

17.28.3 Syntax

```
(ssadd [ename [ss]])
```

17.28.4 Arguments and Values

- ‘ename‘: optional entity name
- ‘ss‘: optional selection set

17.28.5 Description

Creates a new selection set or adds an entity to an existing one.

17.28.6 Return Values

Selection set or ‘nil‘.

17.28.7 Side Effects

Creates or mutates a selection set object.

17.28.8 Compatibility

Part of the programmatic selection-set manipulation API.

17.28.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.28.10 See Also

- BricsCAD V9 documented ‘ssadd‘ as compatible-with-enhancements. The current BricsCAD V26 surface is presumed compatible; verify in Phase 5.

17.28.11 Source Notes

- [ssadd (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-22337977-82F2-4394-B209-39DE2B4B9E86.htm`)
- [About Modifying Selection Sets (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-AutoLISP/files/GUID-BF30AA6B-AB31-4085-BEC8-ABA51`
`htm`)

17.28.12 Notes

- With no arguments, Autodesk documents that a new empty selection set is created.
- If the entity is already in the set, Autodesk documents that the operation is ignored and no error is reported.
- If a selection set is shared through multiple variables, Autodesk documents that mutation is visible through all of them because the same set object is returned.

17.29 Function Entry: SSDEL

17.29.1 Name

‘ssdel‘

17.29.2 Class

Host Function

17.29.3 Syntax

```
(ssdel ename ss)
```

17.29.4 Arguments and Values

- ‘ename‘: entity name
- ‘ss‘: selection set

17.29.5 Description

Removes an entity from a selection set.

17.29.6 Return Values

Selection set or ‘nil‘.

Autodesk documents that the existing selection set object is returned when deletion succeeds, and ‘nil‘ is returned if the specified entity is not a member of the set.

17.29.7 Side Effects

Mutates the selection set object.

17.29.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.29.9 Source Notes

- [ssdel (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F84DB8CE-B535-453D-977D-AE9D7AD2D7AA.htm)

17.29.10 Notes

- Autodesk explicitly states that ‘ssdel‘ mutates the existing selection set rather than returning a fresh set.

17.30 Function Entry: GETANGLE

17.30.1 Name

‘getangle‘

17.30.2 Class

Function

17.30.3 Syntax

```
(getangle [pt] [msg])
```

17.30.4 Arguments and Values

- ‘pt‘: optional base point
- ‘msg‘: optional prompt string

Autodesk documents ‘pt‘ as a 2D base point in the current UCS; a 3D point may be supplied, but measurement remains in the current construction plane.

17.30.5 Description

Prompts the user to specify an angle.

17.30.6 Return Values

Real or ‘nil‘.

The returned angle is in radians.

17.30.7 Side Effects

- prompts the user
- consumes interactive input

17.30.8 Affected By

- current angle conventions
- current unit settings
- optional base point

17.30.9 Exceptional Situations

Cancel and interactive-input errors are host-dependent.

17.30.10 Compatibility

Part of the standard user-input family and closely related to angle conversion functions such as ‘angtof‘ and ‘angtos‘.

17.30.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.30.12 Source Notes

- [getangle (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-947F34FA-5E58-4C7C-A169-556D4B8E2208.htm`)

17.30.13 Notes

- Autodesk documents that the returned value is measured in radians with respect to the current construction plane.
- Autodesk documents that zero direction is ANGBASE, with positive angles increasing counterclockwise.
- Autodesk documents that the user cannot respond with another AutoLISP expression.

17.31 Function Entry: GETDIST

17.31.1 Name

‘getdist‘

17.31.2 Class

Function

17.31.3 Syntax

```
(getdist [pt] [msg])
```

17.31.4 Arguments and Values

- ‘pt‘: optional base point
- ‘msg‘: optional prompt string

Autodesk documents ‘pt‘ as a 2D or 3D base point in the current UCS.

17.31.5 Description

Prompts the user to specify a distance.

17.31.6 Return Values

Real or ‘nil‘.

17.31.7 Side Effects

- prompts the user
- consumes interactive input

17.31.8 Affected By

- current distance/unit conventions
- optional base point

17.31.9 Compatibility

Part of the standard user-input family.

17.31.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.31.11 Source Notes

- [getdist (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-7D381B2A-79AA-4133-9C96-3DA63F8D863` `htm`)

17.31.12 Notes

- Autodesk documents that if a 3D point is provided, the result is a 3D distance unless ‘initget‘ bit 64 is set, in which case Z is ignored and a 2D distance is returned.
- Autodesk documents that the user cannot respond with another AutoLISP expression.
- Autodesk documents that the result is always a real, even when the user enters values in architectural units.

17.32 Function Entry: GETINT

17.32.1 Name

‘getint‘

17.32.2 Class

Function

17.32.3 Syntax

`(getint [msg])`

17.32.4 Arguments and Values

- ‘msg‘: optional prompt string

17.32.5 Description

Prompts the user for an integer value.

17.32.6 Return Values

Integer or ‘nil‘.

17.32.7 Side Effects

- prompts the user
- consumes interactive input

17.32.8 Exceptional Situations

Invalid input, cancel, or interactive state errors are host-dependent.

17.32.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.32.10 Source Notes

- [getint (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-LT-AutoLISP-Reference/files/GUID-18D6FF0F-B1DF-447D-BC6C-A46C933EF78 htm)

17.32.11 Notes

- Autodesk documents an accepted range of ‘-32768‘ to ‘+32767‘.
- Autodesk documents that invalid non-integer input causes a retry prompt with a diagnostic message.
- Autodesk documents that the user cannot respond with another AutoLISP expression.

17.33 Function Entry: GETREAL

17.33.1 Name

‘getreal‘

17.33.2 Class

Function

17.33.3 Syntax

`(getreal [msg])`

17.33.4 Arguments and Values

- ‘msg‘: optional prompt string

17.33.5 Description

Prompts the user for a real-number value.

17.33.6 Return Values

Real or ‘nil‘.

17.33.7 Side Effects

- prompts the user
- consumes interactive input

17.33.8 Compatibility

This function is an important bridge between textual/numeric input and the host’s numeric conventions.

17.33.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.33.10 Source Notes

- [getreal (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-503BBE86-9B22-456A-8883-1B5F5832B7 htm)

17.33.11 Notes

- Autodesk documents that the user cannot respond with another AutoLISP expression.

17.34 Function Entry: GETKWORD

17.34.1 Name

‘getkword‘

17.34.2 Class

Function

17.34.3 Syntax

`(getkword [msg])`

17.34.4 Arguments and Values

- ‘msg‘: optional prompt string

17.34.5 Description

Prompts the user for a keyword response.

17.34.6 Return Values

String or ‘nil‘.

17.34.7 Side Effects

- prompts the user
- consumes interactive input

17.34.8 Affected By

- ‘initget‘ keyword configuration

17.34.9 Compatibility

Closely tied to ‘initget‘.

17.34.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.34.11 Source Notes

- [getkword (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-9F940144-0D7B-4DA1-BF50-BBF8FB8DFF2` `htm`)

17.34.12 Notes

- Autodesk documents that ‘getkword‘ returns ‘nil‘ if the user presses Enter without typing a keyword.
- Autodesk documents that ‘getkword‘ also returns ‘nil‘ if it is called without a preceding ‘initget‘ that establishes keywords.
- Autodesk documents that invalid keyword input causes a warning and a re-prompt.
- Autodesk documents that the user cannot respond with another AutoLISP expression.

17.35 Function Entry: INITGET

17.35.1 Name

‘initget‘

17.35.2 Class

Function

17.35.3 Syntax

`(initget [bits] [keywords])`

17.35.4 Arguments and Values

- ‘bits‘: optional control bit mask
- ‘keywords‘: optional keyword specification

17.35.5 Description

Initializes constraints and keyword handling for the next user-input function call.

17.35.6 Return Values

Always returns ‘nil‘.

17.35.7 Side Effects

Changes the interpretation of the immediately following user-input request.

17.35.8 Affected By

- bit-mask semantics
- keyword list syntax

17.35.9 Compatibility

This function is foundational for structured user-input workflows.

17.35.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.35.11 Source Notes

- [initget (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-LT-AutoLISP-Reference/files/GUID-9ED8841B-5C1D-4B3F-9F3B-84A4408A6B htm`)
- [About Controlling User-Input Function Conditions (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-AutoLISP/files/GUID-44553A7D-EFFF-4C26-8CF9-DF163ECACCB9.htm`)

17.35.12 Notes

- Autodesk documents that the constraints and keywords apply only to the next user-input call and are then discarded.
- Autodesk documents that keywords are honored by ‘getint‘, ‘getreal‘, ‘getdist‘, ‘getangle‘, ‘getorient‘, ‘getpoint‘, ‘getcorner‘, ‘getkword‘, ‘entsel‘, ‘nentsel‘, and ‘nentselp‘, but not by ‘getstring‘.
- Autodesk documents that bits irrelevant to the next input function are ignored.

17.36 Function Entry: GETCORNER

17.36.1 Name

‘getcorner‘

17.36.2 Class

Function

17.36.3 Syntax

`(getcorner pt [msg])`

17.36.4 Arguments and Values

- ‘pt‘: first corner point
- ‘msg‘: optional prompt string

Autodesk documents ‘pt‘ as a 2D or 3D point.

17.36.5 Description

Prompts the user for the opposite corner of a rectangular area.

17.36.6 Return Values

Point list or ‘nil‘.

17.36.7 Side Effects

- prompts the user
- consumes interactive input

17.36.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.36.9 Source Notes

- [About The Getxxx Functions (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-MAC-AutoLisp/files/GUID-5D9F55C4-97FE-4FF1-81D2-C` `htm`)
- [User Input Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-49DBE7DD-D1D2-4B3A-9718-E3F4E2FD5E81.htm`)

17.36.10 Notes

- Autodesk documents that 'getcorner' obtains the opposite corner of a box.
- Autodesk documents that it honors keywords established by 'initget'.

17.37 Function Entry: GETFILED

17.37.1 Name

‘getfiled‘

17.37.2 Class

Function

17.37.3 Syntax

```
(getfiled title default ext flags)
```

17.37.4 Arguments and Values

- ‘title‘: dialog title
- ‘default‘: default file/path
- ‘ext‘: default extension
- ‘flags‘: file-dialog flags

Autodesk documents ‘flags‘ as a bit-coded integer.

17.37.5 Description

Displays a file-selection dialog and returns a selected file name/path.

17.37.6 Return Values

String pathname or ‘nil‘.

17.37.7 Side Effects

- opens a host dialog
- consumes user interaction

17.37.8 Compatibility

This function is highly host-UI-specific and therefore environment-profile-sensitive.

17.37.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.37.10 Source Notes

- [getfiled (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP-Reference/files/GUID-AD65DF88-5218-4655-B877-B4D33B9FB6D1.htm`)

17.37.11 Notes

- Autodesk documents that ‘ext‘ defaults to ‘*‘ when the empty string is supplied.
- Autodesk documents specific ‘flags‘ bits including new-file mode, path-only mode, overwrite suppression, and URL restrictions.
- Autodesk documents that on Web the function is supported but does not display the standard dialog and always returns ‘1‘.

17.38 Function Entry: GETORIENT

17.38.1 Name

‘getorient‘

17.38.2 Class

Function

17.38.3 Syntax

```
(getorient [pt] [msg])
```

17.38.4 Arguments and Values

- ‘pt‘: optional base point
- ‘msg‘: optional prompt

Autodesk documents ‘pt‘ as a 2D base point in the current UCS; a 3D point may be supplied, but measurement remains in the current construction plane.

17.38.5 Description

Prompts the user for an orientation angle.

17.38.6 Return Values

Real or ‘nil‘.

17.38.7 Side Effects

- prompts the user
- consumes interactive input

17.38.8 Compatibility

Closely related to ‘getangle‘, but documented separately.

17.38.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.38.10 Source Notes

- [getorient (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4F8606DD-6453-4D01-84C6-EC8EEBBE2D1A.htm`)

17.38.11 Notes

- Autodesk documents that ‘getorient‘ returns radians measured counterclockwise from the right (east), ignoring ‘ANGBASE‘ and ‘ANGDIR‘ in the returned value.
- Autodesk documents that the user cannot respond with another AutoLISP expression.
- Autodesk explicitly distinguishes ‘getorient‘ from ‘getangle‘: ‘getorient‘ is for an absolute orientation, ‘getangle‘ for a rotation amount.

17.39 Function Entry: NENTSEL

17.39.1 Name

‘nentsel‘

17.39.2 Class

Host Function

17.39.3 Syntax

```
(nentsel [msg])
```

17.39.4 Arguments and Values

- ‘msg‘: optional prompt string

17.39.5 Description

Prompts the user to select a non-main or nested subentity and returns selection information.

17.39.6 Return Values

List or ‘nil‘.

Autodesk documents:

- for non-complex objects, the same information as ‘entsel‘
- for a 3D polyline selection, a list containing the selected vertex name and the pick point
- for nested block selections, additional context information including a transformation matrix and containing block names

17.39.7 Side Effects

- prompts the user
- consumes interactive input

17.39.8 Compatibility

Nested-selection semantics are host-specific and more complex than ‘entsel‘.

17.39.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.39.10 Source Notes

- [nentsel (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-LT-AutoLISP-Reference/files/GUID-A7AC0917-66CE-4BAA-BBAF-D49F8ADB26F htm`)
- [About Accessing an Object's Entity Name (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-AutoLISP/files/GUID-3EEA3D4B-36B6-4E3B-B9E8-62B6265FA68C.htm`)

17.39.11 Notes

- Autodesk documents that 'nentsel' ignores the current object-snap mode unless the user explicitly requests it.
- Autodesk documents that 'nentsel' honors keywords established by 'initget'.
- Autodesk documents that for 3D polylines, the returned subentity is the starting vertex of the picked segment, never the 'SEQEND' subentity.

17.40 Function Entry: NENTSELP

17.40.1 Name

‘nentselp‘

17.40.2 Class

Host Function

17.40.3 Syntax

```
(nentselp [msg] [pt])
```

17.40.4 Arguments and Values

- ‘msg‘: optional prompt string
- ‘pt‘: optional point at which to perform the selection without further user input

17.40.5 Description

Prompts for a nested entity selection and returns extended nested-selection information.

17.40.6 Return Values

List or ‘nil‘.

Autodesk documents that ‘nentselp‘ returns information similar to ‘nentsel‘, but with a transformation matrix in the format used by other AutoLISP and ObjectARX functions.

17.40.7 Side Effects

- prompts the user
- consumes interactive input

17.40.8 Compatibility

Nested-selection and subentity semantics vary across host capabilities.

17.40.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.40.10 Source Notes

- [nentselp (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/JPN/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5CE182FE-6455-4C62-B953-B1CA441455 htm)
- [About Entity Context and Coordinate Transform Data (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-LT-AutoLISP/files/GUID-4F71D4FB-FBF4-4EB3-B546-24B8858FBB94.htm)

17.40.11 Notes

- Autodesk documents that ‘nentselp‘ can operate without user input when ‘pt‘ is supplied.
- Autodesk documents that ‘nentselp‘ and ‘nentsel‘ return subentity names rather than complex-entity headers when a complex object is selected.
- Current Autodesk documentation marks ‘nentselp‘ as supported on Windows and macOS.

17.41 Function Entry: GRREAD

17.41.1 Name

‘grread‘

17.41.2 Class

Host Function

17.41.3 Syntax

```
(grread [track] [allkeys [curtype]])
```

17.41.4 Arguments and Values

- ‘track‘: optional tracking flag
- ‘allkeys‘: optional key/gesture mode
- ‘curtype‘: optional cursor-type control

Autodesk documents ‘allkeys‘ as a bit-coded integer.

17.41.5 Description

Reads low-level graphical or keyboard input events.

17.41.6 Return Values

List describing the input event.

Autodesk documents that the first list element is an event type code, with the second element depending on that type.

17.41.7 Side Effects

- enters low-level input mode
- consumes input events

17.41.8 Compatibility

One of the most host- and UI-specific user-input functions.

17.41.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.41.10 Source Notes

- [grread (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2484FFE3-95B3-4C2B-AF79-BE7772B07419.htm`)
- [About Accessing and Requesting User Input (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-7F024C1B-EFB8-4F6C-9EFE-A6491210A4CD.htm`)

17.41.11 Notes

- Autodesk documents ‘allkeys‘ bit values including drag-mode coordinates, all-key reporting, and cursor control.
- Autodesk explicitly warns that applications using ‘grread‘ are unlikely to be upward compatible across hardware and platform configurations.
- Current Autodesk reference marks ‘grread‘ as supported on Windows and macOS, not Web.

17.42 Function Entry: INTERS

17.42.1 Name

‘inters‘

17.42.2 Class

Geometric Function

17.42.3 Syntax

`(inters pt1 pt2 pt3 pt4 [onseg])`

17.42.4 Arguments and Values

- ‘pt1‘, ‘pt2‘, ‘pt3‘, ‘pt4‘: points defining two lines
- ‘onseg‘: optional segment-only flag

17.42.5 Description

Returns the intersection point of two lines or segments.

17.42.6 Return Values

Point list or ‘nil‘.

17.42.7 Side Effects

None.

17.42.8 Exceptional Situations

Autodesk explicitly uses ‘inters‘ as an example in error-handling documentation to show that more than one valid error may be reported depending on evaluation/checking order.

17.42.9 Compatibility

This function is useful as a test vehicle for argument evaluation and error-reporting semantics.

17.42.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.42.11 Source Notes

- [Geometric Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-8C29D7A3-9CBB-47A8-A936-87118782303B.htm)
- [About Error Handling (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-AutoLISP/files/GUID-027AD2E0-5AC5-48DA-B451-112B7EECB htm)
- Status: documented.

17.43 Function Entry: OSNAP

17.43.1 Name

‘osnap‘

17.43.2 Class

Geometric / Host Function

17.43.3 Syntax

`(osnap pt mode)`

17.43.4 Arguments and Values

- ‘pt‘: point
- ‘mode‘: object-snap mode designator

17.43.5 Description

Returns the object-snap point for the given point and snap mode.

17.43.6 Return Values

Point list or ‘nil‘.

17.43.7 Side Effects

None specified.

17.43.8 Compatibility

- heavily host-dependent
- Autodesk documents version changes
- BricsCAD historically documented additional snap modes

17.43.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.43.10 Source Notes

- [Geometric Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-8C29D7A3-9CBB-47A8-A936-87118782303B.htm`)
- [What's New or Changed with AutoLISP](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85 htm`)
- [LISP Compatibility, BricsCAD V9](`https://developer.bricsys.com/bricscad/help/en_US/V9/DevRef/source/IDR_LISP_B100.htm`)
- Status: documented with version/vendor differences.

17.44 Function Entry: POLAR

17.44.1 Name

‘polar‘

17.44.2 Class

Geometric Function

17.44.3 Syntax

```
(polar pt angle distance)
```

17.44.4 Arguments and Values

- ‘pt‘: base point
- ‘angle‘: angle
- ‘distance‘: distance

17.44.5 Description

Returns a point located at the given angle and distance from ‘pt‘.

17.44.6 Return Values

Point list.

Autodesk documents a 3D point result.

17.44.7 Side Effects

None.

17.44.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.44.9 Source Notes

- [About Geometric Utilities (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-AutoLISP/files/GUID-56A7AF2E-F3CC-45B3-915C-A34290A59 htm`)
- [Geometric Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-8C29D7A3-9CBB-47A8-A936-87118782303B.htm`)

17.44.10 Notes

- Autodesk documents that the angle is expressed in radians relative to the world X axis and independent of the current construction plane.

17.45 Function Entry: TEXTBOX

17.45.1 Name

‘textbox‘

17.45.2 Class

Geometric / Host Function

17.45.3 Syntax

```
(textbox elist)
```

17.45.4 Arguments and Values

- ‘elist‘: entity definition list for a text object

17.45.5 Description

Returns two corner points that describe the text bounding box.

17.45.6 Return Values

List of two points, or ‘nil‘.

17.45.7 Side Effects

None.

17.45.8 Compatibility

Depends on host text and font metrics behavior.

17.45.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.45.10 Source Notes

- [textbox (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PTB/AutoCAD-AutoLISP-Reference/files/GUID-D6C31C79-ECBF-4C7D-9285-1CFA95AC0918.htm)

17.45.11 Notes

- Autodesk documents that omitted text parameters in ‘elist‘ default to the current or default settings.
- Autodesk documents that the minimum acceptable list is one containing only the text string.
- Autodesk documents that the returned points are measured as if the text insertion point were ‘(0,0,0)‘ and rotation were ‘0‘, regardless of the actual insertion point and orientation.

17.46 Dictionary Coverage Index: User Input, Geometric, and Dictionary Functions

Important standardized families in this area include:

User input:

- ‘entsel‘
- ‘getangle‘
- ‘getcorner‘
- ‘getdist‘
- ‘getfiled‘
- ‘getint‘
- ‘getkword‘
- ‘getorient‘
- ‘getpoint‘
- ‘getreal‘
- ‘getstring‘

- ‘initget‘
- ‘nentsel‘
- ‘nentselp‘
- ‘grread‘

Geometric:

- ‘angle‘
- ‘distance‘
- ‘inters‘
- ‘osnap‘
- ‘polar‘
- ‘textbox‘

Symbol table and dictionary handling:

- ‘dictadd‘
- ‘dictnext‘
- ‘dictremove‘
- ‘dictrename‘
- ‘dictsearch‘
- ‘layoutlist‘
- ‘namedobjdict‘
- ‘tblnext‘
- ‘tblobjname‘
- ‘tblsearch‘

Sources:

- [User Input Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-49DBE7DD-D1D2-4B3A-htm`)

- [About Accessing and Requesting User Input (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/CSY/AutoCAD-AutoLISP/files/GUID-7F024C1B-EFB8-4F6C-9EFE-A6491210A4CD.htm)
- [Geometric Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-8C29D7A3-9CBB-47A8-A936-87118782303B.htm)
- [Symbol Table and Dictionary-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/CSY/AutoCAD-AutoLISP-Reference/files/GUID-7AACB386-1AA1-4BDF-B5D7-A1E23A51CA15.htm)

17.47 Function Entry: SSGETFIRST

17.47.1 Name

‘ssgetfirst‘

17.47.2 Class

Function

17.47.3 Syntax

`(ssgetfirst)`

17.47.4 Arguments and Values

None.

17.47.5 Description

Returns the currently selected and gripped objects. Operates only on entities in model space and paper space; nongraphical objects and entities inside block definitions are excluded.

17.47.6 Return Values

A two-element list ‘(grip-set selection-set)‘ or ‘nil‘. The first element is always ‘nil‘ because AutoCAD no longer supports grips on unselected objects. The second element is a selection set of selected-and-gripped entities, or ‘nil‘ if there are none.

17.47.7 Side Effects

None.

17.47.8 Examples

Not documented on the vendor page.

17.47.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.47.10 Source Notes

- [ssgetfirst (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F18CB64C-1B18-46F3-BAD4-24D83214C htm`)
- Status: documented (Phase 3 body fill).

17.48 Function Entry: SSSETFIRST

17.48.1 Name

‘sssetfirst‘

17.48.2 Class

Function

17.48.3 Syntax

```
(sssetfirst gripset [pickset])
```

17.48.4 Arguments and Values

- ‘gripset‘: ignored. Pass ‘nil‘. With no ‘pickset‘ and ‘gripset = nil‘, ‘sssetfirst‘ clears any previous grip and selection state set by an earlier call.
- ‘pickset‘ (optional): a selection set; objects to be selected and gripped.

17.48.5 Description

Programmatically selects objects and shows their grip handles. The gripset parameter is no longer honoured because AutoCAD discontinued grips on unselected objects. Do not call this function while AutoCAD is in the middle of a command — the call is for use between commands.

17.48.6 Return Values

A list naming the selection-set arguments.

17.48.7 Side Effects

Replaces the current selection set and grip state with ‘pickset‘.

17.48.8 Examples

Given ‘pickset1‘ and ‘pickset2‘ from prior ‘entmake‘ / ‘ssadd‘ calls:

- ‘(sssetfirst nil pickset1)‘ returns ‘(nil <Selection set: a5>)‘.
- ‘(sssetfirst nil pickset2)‘ returns ‘(nil <Selection set: ab>)‘.

17.48.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.48.10 Source Notes

- [sssetfirst (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4076CD9D-1C66-4C73-ACC0-3A6CD0B0D
htm)
- Status: documented (Phase 3 body fill).

17.49 Function Entry: SSMEMB

17.49.1 Name

‘ssmemb‘

17.49.2 Class

Function

17.49.3 Syntax

```
(ssmemb ename ss)
```

17.49.4 Arguments and Values

- ‘ename‘: an entity name to test.
- ‘ss‘: a selection set (pickset).

17.49.5 Description

Tests whether ‘ename‘ is a member of ‘ss‘.

17.49.6 Return Values

Returns ‘ename‘ if it is in ‘ss‘; otherwise ‘nil‘.

17.49.7 Side Effects

None.

17.49.8 Examples

- When ‘e2‘ is in ‘ss‘: ‘(ssmemb e2 ss)‘ returns ‘<Entity name: 1d62d68>‘.
- When ‘e1‘ is not in ‘ss‘: ‘(ssmemb e1 ss)‘ returns ‘nil‘.

17.49.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.49.10 Source Notes

- [ssmemb (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8F291146-6A1A-4A31-9380-C800590BF` `htm`)
- Status: documented (Phase 3 body fill).

17.50 Function Entry: SSNAMEX

17.50.1 Name

‘ssnamex‘

17.50.2 Class

Function

17.50.3 Syntax

```
(ssnamex ss [index])
```

17.50.4 Arguments and Values

- ‘ss‘: a selection set.
- ‘index‘ (optional): integer or real (zero-based); the element position. With no ‘index‘, every element is described.

17.50.5 Description

Returns the entity name at ‘index‘, plus data describing how the entity was selected. Without ‘index‘, returns the same information for every element in the selection set. Operates only on entities from the current drawing's model and paper spaces; nongraphical objects and entities in block definitions are excluded.

17.50.6 Return Values

A list of descriptors, each of the form ‘(selection-id ename data)‘ where ‘selection-id‘ is one of:

- ‘0‘: nonspecific (Last, All).
- ‘1‘: pick.
- ‘2‘: window or WPolygon.
- ‘3‘: crossing or CPolygon.
- ‘4‘: fence.

Returns ‘nil‘ if ‘index‘ is invalid.

17.50.7 Side Effects

None.

17.50.8 Examples

- Pick selection: ‘((1 <Entity name: 1d62da0> 0 (0 (1.0 1.0 0.0))))‘.
- Crossing selection: ‘((3 <Entity name: 1d62d60> 0 -1) (-1 (0 (-1.80879 8.85536 0.0)) ...))‘.
- Fence selection lists fence-edge intersections.

17.50.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.50.10 Source Notes

- [ssnamex (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0A0E41A1-8F97-4B2E-A6A6-FF6223C6` `htm`)
- Status: documented (Phase 3 body fill).

17.51 Function Entry: ENTMAKEX

17.51.1 Name

‘entmakex‘

17.51.2 Class

Function

17.51.3 Syntax

`(entmakex [elist])`

17.51.4 Arguments and Values

- ‘elist‘ (optional): an entity-definition list in the format returned by ‘entget‘. Required group codes must be present; optional codes default if omitted.

17.51.5 Description

Creates a new entity, gives it a handle and entity name, and returns the entity name. Unlike ‘entmake‘, the new entity has no owner; it must be assigned ownership before being saved (e.g. by adding it to a dictionary via ‘dictadd‘). Supports both graphical and nongraphical entities.

17.51.6 Return Values

The new entity name on success; ‘nil‘ if the entity cannot be created.

17.51.7 Side Effects

Allocates a new entity in the current drawing.

17.51.8 Examples

- ‘(entmakex ’((0 . ”CIRCLE”) (62 . 1) (10 4.0 3.0 0.0) (40 . 1.0)))‘ returns ‘<Entity name: 1d45558>‘.

17.51.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.51.10 Source Notes

- [entmakex (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3F9A2EB2-082D-49DD-8EFD-DAD8F6E9A htm)
- Status: documented (Phase 3 body fill).

17.52 Function Entry: ENTUPD

17.52.1 Name

‘entupd‘

17.52.2 Class

Function

17.52.3 Syntax

```
(entupd ename)
```

17.52.4 Arguments and Values

- ‘ename‘: an entity name.

17.52.5 Description

Updates the screen image of ‘ename‘. Most useful after editing the vertices of a polyline or the attributes of a block with ‘entmod‘, since modifying a sub-entity does not automatically refresh the parent. May be called on any sub-entity of a complex entity, not only on the head; AutoCAD regenerates the whole complex entity. For nested-block contexts, some sub-entities may not regenerate; use the REGEN command for full coverage in those cases.

17.52.6 Return Values

The entity name on success; ‘nil‘ if no update occurred.

17.52.7 Side Effects

Forces a screen-display refresh for the affected entity.

17.52.8 Examples

```
(setq e1 (entnext))
(setq e2 (entnext e1))
(setq ed (entget e2))
(setq ed (subst '(10 1.0 2.0) (assoc 10 ed) ed))
(entmod ed)
(entupd e1)
```

17.52.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.52.10 Source Notes

- [entupd (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-78313F48-676D-49D7-842A-288DD347A` `htm`)
- Status: documented (Phase 3 body fill).

17.53 Function Entry: REGAPP

17.53.1 Name

‘regapp‘

17.53.2 Class

Function

17.53.3 Syntax

`(regapp application)`

17.53.4 Arguments and Values

- ‘application‘: string; an application name. Must be a valid symbol-table name (see ‘snvalid‘).

17.53.5 Description

Registers an application name with the current AutoCAD drawing in preparation for attaching extended (xdata) object data. The name is entered into the ‘APPID‘ symbol table, which lists every application using xdata in the drawing. A naming scheme that uses the company or product name combined with a unique suffix (telephone number, date/time) is recommended.

17.53.6 Return Values

Returns the application name on success; ‘nil‘ if an application of the same name is already registered.

17.53.7 Side Effects

Adds an entry to the drawing’s ‘APPID‘ symbol table.

17.53.8 Examples

- ‘(regapp ”ADESK$_{4153322344}$”)‘ returns ‘”ADESK$_{4153322344}$”‘.
- ‘(regapp ”DESIGNER-v2.1-124753”)‘ returns ‘”DESIGNER-v2.1-124753”‘.

17.53.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.53.10 Source Notes

- [regapp (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D331A88A-9B6E-49C9-B745-D0A063F38` `htm`)
- Status: documented (Phase 3 body fill).

17.54 Function Entry: XDROOM

17.54.1 Name

‘xdroom‘

17.54.2 Class

Function

17.54.3 Syntax

```
(xdroom ename)
```

17.54.4 Arguments and Values

- ‘ename‘: an entity name.

17.54.5 Description

Returns the amount of extended-data (xdata) space available for the named entity. AutoCAD limits xdata to about 16 KB per entity, shared across all applications that have attached xdata to it. Use this function to verify that enough room exists before appending more xdata.

17.54.6 Return Values

An integer (free bytes), or ‘nil‘ on failure.

17.54.7 Side Effects

None.

17.54.8 Examples

- ‘(xdroom vpname)‘ returns ‘16162‘, indicating 221 bytes of the 16 383-byte budget are already in use.

17.54.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.54.10 Source Notes

- [xdroom (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B7AD1D4B-FAFD-40F2-8A7C-7E35EC23F` `htm`)

- Status: documented (Phase 3 body fill).

17.55 Function Entry: XDSIZE

17.55.1 Name

‘xdsize‘

17.55.2 Class

Function

17.55.3 Syntax

```
(xdsize lst)
```

17.55.4 Arguments and Values

- ‘lst‘: a valid extended-data list whose application name has been registered by ‘regapp‘. The list may optionally start with a ‘-3‘ group-code sentinel; outer parentheses are required either way. Brace fields (group code 1002) must be balanced.

17.55.5 Description

Returns the number of bytes the list will occupy when linked to an entity as xdata. If the application name is not registered, an error is signalled and ‘ERRNO‘ is set.

17.55.6 Return Values

An integer (byte size), or ‘nil‘ on failure.

17.55.7 Side Effects

None.

17.55.8 Examples

Both of these forms are accepted:

```
(-3 ("MYAPP" (1000 . "SUITOFARMOR")
             (1002 . "{")
             (1040 . 0.0)
             (1040 . 1.0)
```

```
            (1002 . "}")
))
( ("MYAPP" (1000 . "SUITOFARMOR")
          (1002 . "{")
          (1040 . 0.0)
          (1040 . 1.0)
          (1002 . "}")
))
```

17.55.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

17.55.10 Source Notes

- [xdsize (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F183738E-43BC-4820-AC51-5D0C1A3DA htm`)
- Status: documented (Phase 3 body fill).

17.56 Function Entry: LAYERSTATE-ADDLAYERS

17.56.1 Name

'layerstate-addlayers'

17.56.2 Class

Layer-State Function

17.56.3 Syntax

```
(layerstate-addlayers layerstatename (list layername state color linetype lineweight
```

17.56.4 Arguments and Values

- 'layerstatename': string; the layer state to update.
- One or more lists, each describing a layer:
  - 'layername': string.
  - 'state': integer; sum of '1' (off), '2' (freeze), '4' (lock), '8' (no plot), '16' (frozen in new viewports). 'nil' uses defaults.
  - 'color': dotted pair '(62 . ColorIndex)' (or 420/430 codes for true-color or color-book).
  - 'linetype': string; layer linetype name. 'nil' defaults to '"Continuous"'.
  - 'lineweight': integer; valid lineweight. 'nil' defaults to '"Default"'.
  - 'plotstyle': string; plot-style name. 'nil' defaults to '"Normal"'.

17.56.5 Description

Adds or updates layers within an existing layer state.

17.56.6 Return Values

'T' on success; otherwise 'nil'.

17.56.7 Side Effects

Updates the named layer state in the current drawing.

17.56.8 Examples

```
(layerstate-addlayers
  "myLayerState"
  (list "Walls" 4 '(62 . 45) "Divide" 35 "10% Screen")
  (list "Floors" 6 '(420 . 16235019) "Continuous" 40 "60% Screen")
  (list "Ceiling" 0 '(430 . "RAL CLASSIC$RAL 1003") "DOT" nil nil))
```

17.56.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. The BricsCAD parallel family is ‘vl-layerstates-*‘; see chapter 24 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.56.10 See Also

- No direct BricsCAD analogue. BricsCAD’s parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.56.11 Source Notes

- [layerstate-addlayers (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F392F4F2-B2C1-4C8E-AEBE htm`)
- Status: documented (Phase 3 body fill).

17.57 Function Entry: LAYERSTATE-COMPARE

17.57.1 Name

‘layerstate-compare‘

17.57.2 Class

Layer-State Function

17.57.3 Syntax

```
(layerstate-compare layerstatename)
```

17.57.4 Arguments and Values

- ‘layerstatename‘: string; the layer state to compare against the current drawing’s layer settings.

17.57.5 Description

Compares a saved layer state with the current state of layers in the drawing and reports which layers have differences. Useful for previewing the effect of ‘layerstate-restore‘ before applying it.

17.57.6 Return Values

A list of layer names that differ from the saved state, or ‘nil‘ when there are no differences.

17.57.7 Side Effects

None.

17.57.8 Examples

Not documented on the vendor page.

17.57.9 Availability

- AutoCAD 2026: documented (Windows, macOS).

- BricsCAD V26: not under this name (parallel BricsCAD family ‘vl-layerstates-*‘ does not currently document a comparator). See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.57.10 See Also

- No direct BricsCAD analogue. BricsCAD’s parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.57.11 Source Notes

- [layerstate-compare (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C2555813-0DD3-46AA-B22F htm`)
- Status: documented (Phase 3 body fill).

17.58 Function Entry: LAYERSTATE-DELETE

17.58.1 Name

‘layerstate-delete‘

17.58.2 Class

Layer-State Function

17.58.3 Syntax

```
(layerstate-delete layerstatename)
```

17.58.4 Arguments and Values

- ‘layerstatename‘: string; the layer state to delete.

17.58.5 Description

Removes a saved layer state from the current drawing.

17.58.6 Return Values

‘T‘ if the layer state existed and was deleted; ‘nil‘ otherwise.

17.58.7 Side Effects

Removes the named layer state from the drawing.

17.58.8 Examples

Not documented on the vendor page.

17.58.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: parallel function is ‘vl-layerstates-delete‘ (chapter 24).
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.58.10 See Also

- BricsCAD analogue: `vl-layerstates-delete` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.58.11 Source Notes

- [layerstate-delete (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-828C4B51-7C59-4FF0-AA69-907C53F31` `htm`)
- Status: documented (Phase 3 body fill).

17.59 Function Entry: LAYERSTATE-EXPORT

17.59.1 Name

‘layerstate-export‘

17.59.2 Class

Layer-State Function

17.59.3 Syntax

```
(layerstate-export layerstatename filename)
```

17.59.4 Arguments and Values

- ‘layerstatename‘: string; the layer state to export.
- ‘filename‘: string; the destination LAS file path.

17.59.5 Description

Exports a saved layer state to an external LAS (Layer State) file for sharing across drawings.

17.59.6 Return Values

‘T‘ on success; ‘nil‘ on failure.

17.59.7 Side Effects

Writes a file to the file system.

17.59.8 Examples

Not documented on the vendor page.

17.59.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.59.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.

- Family pair documented in 'vendor-inventory-2026.org' §10 item 6.

17.59.11 Source Notes

- [layerstate-export (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E14DB236-6DD9-48A8-91F3-8A44A6E8E htm`)

- Status: documented (Phase 3 body fill).

17.60 Function Entry: LAYERSTATE-GETLASTRESTORED

17.60.1 Name

‘layerstate-getlastrestored‘

17.60.2 Class

Layer-State Function

17.60.3 Syntax

```
(layerstate-getlastrestored)
```

17.60.4 Arguments and Values

None.

17.60.5 Description

Returns the name of the layer state most recently restored in the current drawing.

17.60.6 Return Values

A string naming the last-restored layer state, or ‘nil‘ if no restore has happened in this session.

17.60.7 Side Effects

None.

17.60.8 Examples

Not documented on the vendor page.

17.60.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.60.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.

- Family pair documented in 'vendor-inventory-2026.org' §10 item 6.

17.60.11 Source Notes

- [layerstate-getlastrestored (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-404D79AB-B213-4EF4-` `htm`)

- Status: documented (Phase 3 body fill).

17.61 Function Entry: LAYERSTATE-GETLAYERS

17.61.1 Name

‘layerstate-getlayers‘

17.61.2 Class

Layer-State Function

17.61.3 Syntax

`(layerstate-getlayers layerstatename)`

17.61.4 Arguments and Values

- ‘layerstatename‘: string.

17.61.5 Description

Returns the layers stored in a layer state.

17.61.6 Return Values

A list of layer names, or ‘nil‘ if the layer state does not exist or contains no layers.

17.61.7 Side Effects

None.

17.61.8 Examples

Not documented on the vendor page.

17.61.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.61.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.61.11 Source Notes

- [layerstate-getlayers (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B0B12E74-D5EF-47E2-9B1C` `htm`)
- Status: documented (Phase 3 body fill).

17.62 Function Entry: LAYERSTATE-GETNAMES

17.62.1 Name

‘layerstate-getnames‘

17.62.2 Class

Layer-State Function

17.62.3 Syntax

`(layerstate-getnames [includehidden] [includexref])`

17.62.4 Arguments and Values

- ‘includehidden‘ (optional): ‘T‘ or ‘nil‘. With ‘T‘, the result includes hidden layer states.
- ‘includexref‘ (optional): ‘T‘ or ‘nil‘. With ‘nil‘, xref layer states are excluded.

17.62.5 Description

Returns the names of layer states defined in the drawing, optionally filtered to exclude hidden or xref-borrowed states.

17.62.6 Return Values

A list of strings, or ‘nil‘ if no layer states exist.

17.62.7 Side Effects

None.

17.62.8 Examples

- ‘(layerstate-getnames)‘ returns ‘(”First Floor” ”Second Floor” ”Foundation”)‘.

17.62.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: parallel function is ‘vl-layerstates-list‘ (chapter 24).
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.62.10 See Also

- BricsCAD analogue: `vl-layerstates-list` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.62.11 Source Notes

- [layerstate-getnames (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B0EF0543-F792-4391-B54F`
`htm`)
- Status: documented (Phase 3 body fill).

17.63 Function Entry: LAYERSTATE-HAS

17.63.1 Name

‘layerstate-has‘

17.63.2 Class

Layer-State Function

17.63.3 Syntax

```
(layerstate-has layerstatename)
```

17.63.4 Arguments and Values

- ‘layerstatename‘: string.

17.63.5 Description

Tests whether the named layer state exists in the current drawing.

17.63.6 Return Values

‘T‘ if the layer state exists; otherwise ‘nil‘.

17.63.7 Side Effects

None.

17.63.8 Examples

Not documented on the vendor page.

17.63.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: parallel function is ‘vl-layerstates-has‘ (chapter 24).
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.63.10 See Also

- BricsCAD analogue: `vl-layerstates-has` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.63.11 Source Notes

- [layerstate-has (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0936B91C-7DE0-49DD-9BCD-FD097417F` `htm`)
- Status: documented (Phase 3 body fill).

17.64 Function Entry: LAYERSTATE-IMPORT

17.64.1 Name

‘layerstate-import‘

17.64.2 Class

Layer-State Function

17.64.3 Syntax

`(layerstate-import filename)`

17.64.4 Arguments and Values

- ‘filename‘: string; path to a LAS (Layer State) file.

17.64.5 Description

Imports one or more layer states from a LAS file into the current drawing.

17.64.6 Return Values

‘T‘ on success; ‘nil‘ on failure.

17.64.7 Side Effects

Creates layer-state entries in the current drawing.

17.64.8 Examples

Not documented on the vendor page.

17.64.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.64.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.64.11 Source Notes

- [layerstate-import (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E8B76E26-4E16-4AC6-B5E1-2CEA0C21` `htm`)
- Status: documented (Phase 3 body fill).

17.65 Function Entry: LAYERSTATE-IMPORTFROMDB

17.65.1 Name

‘layerstate-importfromdb‘

17.65.2 Class

Layer-State Function

17.65.3 Syntax

```
(layerstate-importfromdb filename layerstatename)
```

17.65.4 Arguments and Values

- ‘filename‘: string; path to a DWG or DWT drawing file.
- ‘layerstatename‘: string; the layer state to import from the source drawing.

17.65.5 Description

Imports a specific layer state from another drawing file into the current drawing.

17.65.6 Return Values

‘T‘ on success; ‘nil‘ on failure.

17.65.7 Side Effects

Adds the imported layer state to the current drawing.

17.65.8 Examples

Not documented on the vendor page.

17.65.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.

- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.65.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.

- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.65.11 Source Notes

- [layerstate-importfromdb (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-A41D1A72-5DD2-469B-htm`)

- Status: documented (Phase 3 body fill).

17.66 Function Entry: LAYERSTATE-REMOVELAYERS

17.66.1 Name

'layerstate-removelayers'

17.66.2 Class

Layer-State Function

17.66.3 Syntax

```
(layerstate-removelayers layerstatename layername [layername ...])
```

17.66.4 Arguments and Values

- 'layerstatename': string.
- 'layername': one or more strings naming layers to drop from the layer state.

17.66.5 Description

Removes the named layers from a layer state.

17.66.6 Return Values

'T' on success; 'nil' on failure.

17.66.7 Side Effects

Updates the layer-state record in the drawing.

17.66.8 Examples

Not documented on the vendor page.

17.66.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: not under this name. See Phase 4 cross-reference.
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.66.10 See Also

- No direct BricsCAD analogue. BricsCAD's parallel `vl-layerstates-*` family does not currently document this operation; consider implementing in user code or treating as AutoCAD-only.

- Family pair documented in 'vendor-inventory-2026.org' §10 item 6.

17.66.11 Source Notes

- [layerstate-removelayers (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5F9FFCEB-D7D6-4233-` `htm`)

- Status: documented (Phase 3 body fill).

17.67 Function Entry: LAYERSTATE-RENAME

17.67.1 Name

‘layerstate-rename‘

17.67.2 Class

Layer-State Function

17.67.3 Syntax

```
(layerstate-rename oldname newname)
```

17.67.4 Arguments and Values

- ‘oldname‘: string; the existing layer-state name.
- ‘newname‘: string; the replacement name.

17.67.5 Description

Renames a layer state.

17.67.6 Return Values

‘T‘ on success; ‘nil‘ if ‘oldname‘ does not exist or ‘newname‘ is already in use.

17.67.7 Side Effects

Updates the layer-state record in the drawing.

17.67.8 Examples

Not documented on the vendor page.

17.67.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: parallel function is ‘vl-layerstates-rename‘ (chapter 24).

- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.67.10 See Also

- BricsCAD analogue: `vl-layerstates-rename` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.67.11 Source Notes

- [layerstate-rename (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-843B8FC9-A1F5-48CE-B83D` `htm`)
- Status: documented (Phase 3 body fill).

17.68 Function Entry: LAYERSTATE-RESTORE

17.68.1 Name

‘layerstate-restore‘

17.68.2 Class

Layer-State Function

17.68.3 Syntax

```
(layerstate-restore layerstatename viewport [restoreflags])
```

17.68.4 Arguments and Values

- ‘layerstatename‘: string.
- ‘viewport‘: ename of a viewport, or ‘nil‘ for model space.
- ‘restoreflags‘ (optional): integer; sum of:
  - ‘1‘: turn off all layers not in the restored state.
  - ‘2‘: freeze all layers not in the restored state.
  - ‘4‘: apply the layer state as viewport overrides (requires non-‘nil‘ ‘viewport‘).

17.68.5 Description

Applies a saved layer state to the current drawing.

17.68.6 Return Values

A list of layer names if successful, or ‘nil‘ if the layer state does not exist or contains no layers.

17.68.7 Side Effects

Mutates the per-layer visibility / property settings, or applies viewport overrides.

17.68.8 Examples

- ‘(layerstate-restore ”myLayerState” viewportId 5)‘.

17.68.9 Availability

- AutoCAD 2026: documented (Windows, macOS only).
- BricsCAD V26: parallel function is ‘vl-layerstates-restore‘ (chapter 24).
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.68.10 See Also

- BricsCAD analogue: `vl-layerstates-restore` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.68.11 Source Notes

- [layerstate-restore (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-84C7195D-12C1-4282-B5EE-8B9D6DDB7` `htm`)
- Status: documented (Phase 3 body fill).

17.69 Function Entry: LAYERSTATE-SAVE

17.69.1 Name

‘layerstate-save‘

17.69.2 Class

Layer-State Function

17.69.3 Syntax

```
(layerstate-save layerstatename mask viewport)
```

17.69.4 Arguments and Values

- ‘layerstatename‘: string.
- ‘mask‘: integer; sum of property bits to capture:
  - ‘1‘: on/off
  - ‘2‘: frozen/thawed
  - ‘4‘: lock
  - ‘8‘: plot
  - ‘16‘: VPVSDFLT
  - ‘32‘: color
  - ‘64‘: linetype
  - ‘128‘: lineweight
- ‘viewport‘: ename of a viewport (to capture VPLAYER settings), or ‘nil‘ to exclude viewport-specific settings.

17.69.5 Description

Saves a layer state in the current drawing under the given name with the selected property bits.

17.69.6 Return Values

‘T‘ on success; ‘nil‘ on failure.

17.69.7 Side Effects

Creates or updates the named layer-state record in the drawing.

17.69.8 Examples

- ‘(layerstate-save ”myLayerState” 21 viewportId)‘ returns ‘T‘.
- ‘(layerstate-save ”myLayerState” nil nil)‘ returns ‘nil‘.

17.69.9 Availability

- AutoCAD 2026: documented (Windows, macOS).
- BricsCAD V26: parallel function is ‘vl-layerstates-save‘ (chapter 24).
- Status: AutoCAD-named family with BricsCAD divergence — see Phase 4 reconciliation.

17.69.10 See Also

- BricsCAD analogue: `vl-layerstates-save` (chapter 24).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

17.69.11 Source Notes

- [layerstate-save (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05B5EA5C-2E7F-47A5-9C6F-27902DB6C htm`)
- Status: documented (Phase 3 body fill).

18 18 DCL

18.1 Overview

DCL is the standard dialog-description companion technology used by AutoLISP applications.

18.2 Normative Rules

- DCL is not part of the core expression language
- DCL is nevertheless part of the practical standard application model
- platform and product support vary by version

18.3 Function Entry: LOAD$_{\text{DIALOG}}$

18.3.1 Name

‘load$_{\text{dialog}}$‘

18.3.2 Class

DCL Function

18.3.3 Syntax

`(load_dialog dclfile)`

18.3.4 Arguments and Values

- ‘dclfile‘: DCL pathname string

18.3.5 Description

Loads a DCL file and returns a dialog-resource identifier.

18.3.6 Return Values

Dialog-resource identifier, or failure indicator under host semantics.

18.3.7 Side Effects

Loads external dialog definitions into the host dialog subsystem.

18.3.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.3.9 Source Notes

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- Status: documented in the DCL workflow documentation.

18.4 Function Entry: NEW$_{\mathbf{DIALOG}}$

18.4.1 Name

‘new$_{\text{dialog}}$‘

18.4.2 Class

DCL Function

18.4.3 Syntax

`(new_dialog dialog-name dcl-id [action [screen-pt]])`

18.4.4 Description

Creates or activates a dialog from a previously loaded DCL resource.

18.4.5 Return Values

Success indicator under host semantics.

18.4.6 Side Effects

Initializes the active dialog state and dialog tiles.

18.4.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.4.8 Source Notes

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- [Displaying Nested Dialog Boxes](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-DA0D5D75-C954-408A-AC43-FFD09E6D2517.htm`)

- Status: documented in the DCL workflow documentation.

18.5 Function Entry: START$_{\text{DIALOG}}$

18.5.1 Name

‘start$_{\text{dialog}}$‘

18.5.2 Class

DCL Function

18.5.3 Syntax

`(start_dialog)`

18.5.4 Description

Displays the current dialog and enters modal dialog processing.

18.5.5 Return Values

Dialog status code when the dialog terminates.

18.5.6 Side Effects

Transfers control to the dialog event loop.

18.5.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.5.8 Source Notes

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- Status: documented in the DCL workflow documentation.

18.6 Function Entry: DONE$_{\text{DIALOG}}$

18.6.1 Name

'done$_{\text{dialog}}$'

18.6.2 Class

DCL Function

18.6.3 Syntax

`(done_dialog [status])`

18.6.4 Description

Terminates the current dialog and returns control to 'start$_{\text{dialog}}$'.

18.6.5 Return Values

Dialog termination code or host-defined success value.

18.6.6 Side Effects

Ends the active dialog session.

18.6.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.6.8 Source Notes

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- Status: documented in the DCL workflow documentation.

18.7 Function Entry: UNLOAD$_{\mathbf{DIALOG}}$

18.7.1 Name

‘unload$_{\text{dialog}}$‘

18.7.2 Class

DCL Function

18.7.3 Syntax

`(unload_dialog dcl-id)`

18.7.4 Description

Unloads a previously loaded DCL resource.

18.7.5 Return Values

Host-defined success value.

18.7.6 Side Effects

Releases dialog-resource state.

18.7.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.7.8 Source Notes

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- Status: documented in the DCL workflow documentation.

18.8 Function Entry: ACTION$_{\text{TILE}}$

18.8.1 Name

‘action$_{\text{tile}}$‘

18.8.2 Class

DCL Function

18.8.3 Syntax

```
(action_tile key action-expression)
```

18.8.4 Description

Associates an action expression or callback with a tile.

18.8.5 Return Values

Host-defined success value.

18.8.6 Side Effects

Changes dialog callback behavior.

18.8.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.8.8 Source Notes

- [Action Expressions and Callbacks](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-78FDD902-C159-4654-8E3B-228E344A78E8.htm`)
- Status: documented in the DCL workflow documentation.

18.9 Function Entry: GET$_{\text{TILE}}$

18.9.1 Name

‘get$_{\text{tile}}$‘

18.9.2 Class

DCL Function

18.9.3 Syntax

`(get_tile key)`

18.9.4 Description

Retrieves the current value of a dialog tile.

18.9.5 Return Values

String tile value under DCL semantics.

18.9.6 Side Effects

None.

18.9.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.9.8 Source Notes

- [Initializing Modes and Values](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-EBCB6710-0290-44CB-94B3-6C1064E82BB4.htm`)
- Status: documented in the DCL workflow documentation.

18.10 Function Entry: SET_{TILE}

18.10.1 Name

‘set_{tile}‘

18.10.2 Class

DCL Function

18.10.3 Syntax

`(set_tile key value)`

18.10.4 Description

Sets the current value of a dialog tile.

18.10.5 Return Values

Host-defined success value.

18.10.6 Side Effects

Mutates dialog tile state.

18.10.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.10.8 Source Notes

- [Initializing Modes and Values](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-EBCB6710-0290-44CB-94B3-6C1064E82BB4.htm`)
- Status: documented in the DCL workflow documentation.

18.11 Function Entry: MODE$_{\text{TILE}}$

18.11.1 Name

‘mode$_{\text{tile}}$‘

18.11.2 Class

DCL Function

18.11.3 Syntax

`(mode_tile key mode)`

18.11.4 Description

Changes the mode of a dialog tile, such as enabling, disabling, or focusing it, according to DCL semantics.

18.11.5 Return Values

Host-defined success value.

18.11.6 Side Effects

Mutates dialog UI state.

18.11.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.11.8 Source Notes

- [Initializing Modes and Values](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-EBCB6710-0290-44CB-94B3-6C1064E82BB4.htm`)
- Status: documented in the DCL workflow documentation.

18.12 Dictionary Coverage Index: DCL Functions

The standard DCL workflow includes at least:

- ‘action$_{\text{tile}}$‘
- ‘done$_{\text{dialog}}$‘
- ‘get$_{\text{tile}}$‘
- ‘load$_{\text{dialog}}$‘
- ‘mode$_{\text{tile}}$‘
- ‘new$_{\text{dialog}}$‘
- ‘set$_{\text{tile}}$‘
- ‘start$_{\text{dialog}}$‘
- ‘unload$_{\text{dialog}}$‘

Sources:

- [Function Sequence to Display and Work with a Dialog](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-FD32ADB3-F16E-42AA-BD37-53259B824AAB.htm`)
- [Initializing Modes and Values](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-EBCB6710-0290-44CB-94B3-6C1064E82BB4.htm`)
- [Action Expressions and Callbacks](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-78FDD902-C159-4654-8E3B-228E344A78E8.htm`)

18.13 Source Notes

- [AutoLISP Developer’s Guide](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-265AADB3-FB89-4D34-AA9D-6ADF70FF7D4B.htm`)
- [AutoLISP Improvements (AutoCAD 2021)](`https://help.autodesk.com/cloudhelp/2021/ENU/AutoCAD-WhatsNew/files/GUID-80F564DA-A5C4-4242-91DB-2331E`
`htm`)

18.14 Function Entry: TERM$_{\textbf{DIALOG}}$

18.14.1 Name

'term$_{\text{dialog}}$'

18.14.2 Class

DCL Function

18.14.3 Syntax

```
(term_dialog)
```

18.14.4 Arguments and Values

None.

18.14.5 Description

Terminates all currently active dialog boxes as if the user had cancelled each of them. Mainly used to abort nested dialog boxes. AutoCAD invokes this automatically when an application closes with open DCL files.

18.14.6 Return Values

Always returns 'nil'.

18.14.7 Side Effects

Closes every active dialog box.

18.14.8 Examples

Not documented on the vendor page.

18.14.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.14.10 Source Notes

- [$\text{term}_{\text{dialog}}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7276E779-2FBA-47A5-AF3D-36927B738` `htm`)
- Status: documented (Phase 3 body fill).

18.15 Function Entry: ADD$_{\text{LIST}}$

18.15.1 Name

‘add$_{\text{list}}$‘

18.15.2 Class

DCL Function

18.15.3 Syntax

```
(add_list str)
```

18.15.4 Arguments and Values

- ‘str‘: string; the value to add or substitute in the active list.

18.15.5 Description

Adds or modifies a string in the currently active dialog-box list. Must be called between ‘start$_{\text{list}}$‘ and ‘end$_{\text{list}}$‘. The semantics depend on the operation specified by ‘start$_{\text{list}}$‘: append (operation ‘2‘), change (operation ‘1‘), or replace-after-clear (operation ‘3‘).

18.15.6 Return Values

The added string on success; ‘nil‘ on failure.

18.15.7 Side Effects

Updates the contents of the active list-box or pop-up-list tile.

18.15.8 Examples

- Initialise:

  ```
  (setq llist '("first line" "second line" "third line"))
  (start_list "longlist")
  (mapcar 'add_list llist)
  (end_list)
  ```

- Modify the second list item:

```
(start_list "longlist" 1 0)
(add_list "2nd line")
(end_list)
```

18.15.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.15.10 Source Notes

- [add$_{\text{list}}$ (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9BF538FC-BB15-4843-9796-DDCDB6D38 htm)
- Status: documented (Phase 3 body fill).

18.16 Function Entry: $\mathrm{START}_{\mathrm{LIST}}$

18.16.1 Name

‘$\mathrm{start}_{\mathrm{list}}$‘

18.16.2 Class

DCL Function

18.16.3 Syntax

```
(start_list key [operation [index]])
```

18.16.4 Arguments and Values

- ‘key‘: case-sensitive string; the list-box or pop-up-list tile.
- ‘operation‘ (optional): integer.
  - ‘1‘: change the contents of a selected list item.
  - ‘2‘: append a new entry to the existing list.
  - ‘3‘ (default): delete the existing list and create a new one.
- ‘index‘ (optional): integer; the zero-based item position to modify. Defaults to ‘0‘. Ignored unless ‘operation = 1‘.

18.16.5 Description

Starts processing of a list in a list-box or pop-up-list tile. Subsequent ‘$\mathrm{add}_{\mathrm{list}}$‘ calls modify the active list until ‘$\mathrm{end}_{\mathrm{list}}$‘ is invoked. Do not call ‘$\mathrm{set}_{\mathrm{tile}}$‘ between ‘$\mathrm{start}_{\mathrm{list}}$‘ and ‘$\mathrm{end}_{\mathrm{list}}$‘.

18.16.6 Return Values

The ‘key‘ string on success; ‘nil‘ on failure.

18.16.7 Side Effects

Latches the named tile as the active list target.

18.16.8 Examples

Not documented on the vendor page.

18.16.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.16.10 Source Notes

- [$start_{list}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-84FAC8F6-59B1-4FBD-9446-0CB323B3D htm`)
- Status: documented (Phase 3 body fill).

18.17 Function Entry: $\textbf{END}_{\textbf{LIST}}$

18.17.1 Name

‘end_{list}‘

18.17.2 Class

DCL Function

18.17.3 Syntax

```
(end_list)
```

18.17.4 Arguments and Values

None.

18.17.5 Description

Ends processing of the currently active list-box or pop-up-list tile. Counterpart to ‘$\text{start}_{\text{list}}$‘.

18.17.6 Return Values

Always returns ‘nil‘.

18.17.7 Side Effects

Commits pending list updates.

18.17.8 Examples

See ‘$\text{start}_{\text{list}}$‘ and ‘$\text{add}_{\text{list}}$‘.

18.17.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.17.10 Source Notes

- [end$_{list}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-389C46C7-3E8A-41CA-96B6-8BD63577E` `htm`)
- Status: documented (Phase 3 body fill).

18.18 Function Entry: START$_{\mathrm{IMAGE}}$

18.18.1 Name

‘start$_{\mathrm{image}}$‘

18.18.2 Class

DCL Function

18.18.3 Syntax

```
(start_image key)
```

18.18.4 Arguments and Values

- ‘key‘: case-sensitive string; the image tile.

18.18.5 Description

Starts the creation of an image in a dialog-box image tile. Subsequent ‘fill$_{\mathrm{image}}$‘, ‘slide$_{\mathrm{image}}$‘, and ‘vector$_{\mathrm{image}}$‘ calls draw into the active image until ‘end$_{\mathrm{image}}$‘ is invoked. Do not call ‘set$_{\mathrm{tile}}$‘ between ‘start$_{\mathrm{image}}$‘ and ‘end$_{\mathrm{image}}$‘.

18.18.6 Return Values

The ‘key‘ string on success; ‘nil‘ on failure.

18.18.7 Side Effects

Latches the named tile as the active image target.

18.18.8 Examples

See ‘fill$_{\mathrm{image}}$‘ and ‘slide$_{\mathrm{image}}$‘.

18.18.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.18.10 Source Notes

- [start$_{\text{image}}$ (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CB5C93A-6357-4853-B918-0F3EE3752 htm)
- Status: documented (Phase 3 body fill).

18.19 Function Entry: END$_{\text{IMAGE}}$

18.19.1 Name

'end$_{\text{image}}$'

18.19.2 Class

DCL Function

18.19.3 Syntax

```
(end_image)
```

18.19.4 Arguments and Values

None.

18.19.5 Description

Ends creation of the currently active dialog-box image. Counterpart to 'start$_{\text{image}}$'.

18.19.6 Return Values

Always returns 'nil'.

18.19.7 Side Effects

Commits pending image-tile updates.

18.19.8 Examples

See 'fill$_{\text{image}}$' and 'slide$_{\text{image}}$'.

18.19.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.19.10 Source Notes

- [$\text{end}_{\text{image}}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C5DDD271-AA70-4546-9346-09C540979`
`htm`)
- Status: documented (Phase 3 body fill).

18.20 Function Entry: $\text{FILL}_{\text{IMAGE}}$

18.20.1 Name

‘$\text{fill}_{\text{image}}$‘

18.20.2 Class

DCL Function

18.20.3 Syntax

```
(fill_image x1 y1 width height color)
```

18.20.4 Arguments and Values

- ‘x1‘: positive integer; X coordinate of the upper-left corner of the filled rectangle.
- ‘y1‘: positive integer; Y coordinate of the upper-left corner.
- ‘width‘: integer; width of the fill area in pixels relative to ‘x1‘.
- ‘height‘: integer; height of the fill area in pixels relative to ‘y1‘.
- ‘color‘: integer; an AutoCAD color number, or a logical-color value in ‘[-2, -18]‘.

18.20.5 Description

Draws a filled rectangle in the currently active dialog-box image tile. The image-tile origin ‘(0, 0)‘ is its upper-left corner. Must be called between ‘$\text{start}_{\text{image}}$‘ and ‘$\text{end}_{\text{image}}$‘.

18.20.6 Return Values

The fill color (an integer).

18.20.7 Side Effects

Updates the active image tile.

18.20.8 Examples

```
(setq color -2)
(fill_image 0 0 (dimx_tile "slide_tile") (dimy_tile "slide_tile") color)
(end_image)
```

18.20.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.20.10 Source Notes

- [fill$_{image}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1C95F5E1-C063-4334-A296-40E7E123`…`htm`)
- Status: documented (Phase 3 body fill).

18.21 Function Entry: SLIDE$_{\text{IMAGE}}$

18.21.1 Name

‘slide$_{\text{image}}$‘

18.21.2 Class

DCL Function

18.21.3 Syntax

```
(slide_image x1 y1 width height sldname)
```

18.21.4 Arguments and Values

- ‘x1‘: positive integer; X-offset of the upper-left corner of the slide within the image tile, in pixels.
- ‘y1‘: positive integer; Y-offset of the upper-left corner.
- ‘width‘: integer; slide width in pixels.
- ‘height‘: integer; slide height in pixels.
- ‘sldname‘: string; either a slide file (‘.sld‘) or ‘libname(sldname)‘ to select a slide inside a ‘.slb‘ library.

18.21.5 Description

Displays an AutoCAD slide in the currently active image tile. The insertion point ‘(x1, y1)‘ is the upper-left corner of the slide; the lower-right corner is ‘(x1 + width, y1 + height)‘. Must be called between ‘start$_{\text{image}}$‘ and ‘end$_{\text{image}}$‘.

18.21.6 Return Values

The ‘sldname‘ string on success.

18.21.7 Side Effects

Draws the slide into the active image tile.

18.21.8 Examples

```
(slide_image 0 0 (dimx_tile "slide_tile") (dimy_tile "slide_tile") "myslide")
(end_image)
```

18.21.9 Availability

- AutoCAD 2026: documented (primarily Windows; slide rendering is platform-dependent).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.21.10 Source Notes

- [slide$_{\text{image}}$ (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D5B5A0AF-82BE-4A00-832D-04D496D80 htm)
- Status: documented (Phase 3 body fill).

18.22 Function Entry: DIMX$_{\text{TILE}}$

18.22.1 Name

‘dimx$_{\text{tile}}$‘

18.22.2 Class

DCL Function

18.22.3 Syntax

`(dimx_tile key)`

18.22.4 Arguments and Values

- ‘key‘: case-sensitive string; the tile to measure.

18.22.5 Description

Returns the X dimension of a tile, expressed in absolute tile coordinates. Designed for use with image-tile functions (‘vector$_{\text{image}}$‘, ‘fill$_{\text{image}}$‘, ‘slide$_{\text{image}}$‘) that need absolute pixel coordinates. The returned value is one less than the total X dimension because coordinates are zero-based; it is therefore the maximum allowed X coordinate.

18.22.6 Return Values

An integer.

18.22.7 Side Effects

None.

18.22.8 Examples

- ‘(setq tile$_{\text{width}}$ (dimx$_{\text{tile}}$ ”my$_{\text{tile}}$”))‘.

18.22.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.22.10 Source Notes

- [dimx$_{\text{tile}}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-AE8CC93E-1F87-4023-BC87-80584229E htm`)
- Status: documented (Phase 3 body fill).

18.23 Function Entry: DIMY$_{\mathbf{TILE}}$

18.23.1 Name

‘dimy$_{\text{tile}}$‘

18.23.2 Class

DCL Function

18.23.3 Syntax

`(dimy_tile key)`

18.23.4 Arguments and Values

- ‘key‘: case-sensitive string; the tile to measure.

18.23.5 Description

Returns the Y dimension of a tile, expressed in absolute tile coordinates. The returned value is one less than the total Y dimension because coordinates are zero-based; it is therefore the maximum allowed Y coordinate.

18.23.6 Return Values

An integer.

18.23.7 Side Effects

None.

18.23.8 Examples

- ‘(setq tile$_{\text{height}}$ (dimy$_{\text{tile}}$ ”my$_{\text{tile}}$”))‘.

18.23.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.23.10 Source Notes

- [dimy$_{\text{tile}}$ (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-FAA2950E-5374-48BF-8BCF-8B1A7C101 htm)
- Status: documented (Phase 3 body fill).

18.24 Function Entry: $\mathbf{GET_{ATTR}}$

18.24.1 Name

'get$_{attr}$'

18.24.2 Class

DCL Function

18.24.3 Syntax

```
(get_attr key attribute)
```

18.24.4 Arguments and Values

- 'key': case-sensitive string; the tile.
- 'attribute': string; the attribute name as it appears in the tile's DCL description.

18.24.5 Description

Returns the DCL value of a dialog-box attribute as defined in the DCL file. This is the **initial** value, not the current run-time value (use 'get$_{tile}$' for the latter).

18.24.6 Return Values

A string.

18.24.7 Side Effects

None.

18.24.8 Examples

Not documented on the vendor page.

18.24.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.24.10 Source Notes

- [get$_{attr}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-AB9100B0-F19E-4C29-B82A-28E8A607E htm`)
- Status: documented (Phase 3 body fill).

18.25 Function Entry: $\textbf{CLIENT}_{\textbf{DATA TILE}}$

18.25.1 Name

'client$_{datatile}$'

18.25.2 Class

DCL Function

18.25.3 Syntax

```
(client_data_tile key clientdata)
```

18.25.4 Arguments and Values

- 'key': case-sensitive string; the tile.
- 'clientdata': string; the value to associate with 'key'. Action expressions and callback functions can reference it as '$data'.

18.25.5 Description

Attaches application-managed data to a dialog-box tile so that callbacks can later read it via '$data'.

18.25.6 Return Values

Always returns 'nil'.

18.25.7 Side Effects

Updates the per-tile client-data binding.

18.25.8 Examples

Not documented on the vendor page.

18.25.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

18.25.10 Source Notes

- [client$_{\text{datatile}}$ (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9790A095-68BA-40FB-B89E-036F276CC` `htm`)
- Status: documented (Phase 3 body fill).

19 19 Visual LISP Core Extensions

19.1 Overview

Visual LISP extends AutoLISP with helper families, runtime services, IDE-related features, and error-handling facilities.

19.2 Normative Rules

- ‘vl-*‘ functions belong to the Visual LISP layer
- they are not universally available on every Autodesk platform
- BricsCAD implements overlapping but not identical functionality

19.3 Source Notes

- [AutoLISP Extensions Reference](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-07C3599E-EC87-4EC0-8F0B-8DE4BD htm)
- [What’s New and What’s More in BricsCAD Lisp](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/WhatsNew.htm)

19.4 Function Entry: VL-LOAD-COM

19.4.1 Name

‘vl-load-com‘

19.4.2 Class

Visual LISP Function

19.4.3 Syntax

`(vl-load-com)`

19.4.4 Description

Initializes ActiveX support for AutoLISP and loads the relevant extension environment.

19.4.5 Return Values

Host-defined success value.

19.4.6 Side Effects

Enables the ActiveX/COM extension layer in environments that support it.

19.4.7 Compatibility

- Windows-only in Autodesk documentation
- may map to broader compatibility/emulation behavior in ‘clautolisp‘

19.4.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.4.9 Source Notes

- [AutoLISP Extensions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-07C3599E-EC87-4EC0-8F0B-8DE4BD150A64.htm`)
- Status: documented.

19.5 Function Entry: VL-CATCH-ALL-APPLY

19.5.1 Name

‘vl-catch-all-apply‘

19.5.2 Class

Visual LISP Function

19.5.3 Syntax

```
(vl-catch-all-apply function arg-list)
```

19.5.4 Arguments and Values

- ‘function‘: function designator
- ‘arg-list‘: list of arguments

19.5.5 Description

Calls ‘function‘ with ‘arg-list‘, trapping errors and returning a catch-all error object instead of unwinding directly into the caller.

19.5.6 Return Values

- normal function result on success
- catch-all error object on failure

19.5.7 Side Effects

Side effects are those of the invoked function up to the point of failure or success.

19.5.8 Compatibility

Part of the Visual LISP error-handling layer.

19.5.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.5.10 Source Notes

- [About Catching Errors and Continuing Program Execution (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-9DDAD8C4-FB10-4C15-B59E-F6E05C1F9672.htm`)
- [Error-Handling Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-D114F1C4-DE8E-422D-` `htm`)
- Status: documented.

19.6 Function Entry: VL-CATCH-ALL-ERROR-P

19.6.1 Name

‘vl-catch-all-error-p‘

19.6.2 Class

Visual LISP Function

19.6.3 Syntax

```
(vl-catch-all-error-p obj)
```

19.6.4 Description

Tests whether ‘obj‘ is a catch-all error object.

19.6.5 Return Values

- ‘T‘ if ‘obj‘ is a catch-all error object
- ‘nil‘ otherwise

19.6.6 Side Effects

None.

19.6.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.6.8 Source Notes

- [vl-catch-all-error-p (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-14CADE53-54C6-437D-8F3C-14845E085 htm)

- [About Catching Errors and Continuing Program Execution (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP/files/GUID-9DDAD8C4-FB10-4C15-B59E-F6E05C1F9672.htm`)
- Status: documented.

19.7 Function Entry: VL-CATCH-ALL-ERROR-MESSAGE

19.7.1 Name

‘vl-catch-all-error-message‘

19.7.2 Class

Visual LISP Function

19.7.3 Syntax

`(vl-catch-all-error-message err)`

19.7.4 Description

Returns the message string associated with a catch-all error object.

19.7.5 Return Values

String error message.

19.7.6 Side Effects

None.

19.7.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.7.8 Source Notes

- [vl-catch-all-error-message (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-C703867D-7B64-480D-BE34` `htm`)
- Status: documented.

19.8 Function Entry: VL-PRIN1-TO-STRING

19.8.1 Name

‘vl-prin1-to-string‘

19.8.2 Class

Visual LISP Function

19.8.3 Syntax

```
(vl-prin1-to-string obj)
```

19.8.4 Description

Returns a printed string representation of ‘obj‘ in a ‘prin1‘-like style.

19.8.5 Return Values

String.

19.8.6 Side Effects

None.

19.8.7 Arguments and Values

- ‘obj‘: any AutoLISP value documented as printable by Autodesk, including numbers, strings, lists, entity names, ‘T‘, and ‘nil‘

19.8.8 Notes

- Autodesk documents that the result is the same printed representation that ‘prin1‘ would produce.
- Autodesk examples show that strings are represented with surrounding quotes and necessary escape characters.

19.8.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.8.10 Source Notes

- [vl-prin1-to-string (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-4AA25012-28D5-4723-8AD3-B8BF9A` `htm`)
- Status: documented.

19.9 Function Entry: VL-PRINC-TO-STRING

19.9.1 Name

‘vl-princ-to-string‘

19.9.2 Class

Visual LISP Function

19.9.3 Syntax

`(vl-princ-to-string obj)`

19.9.4 Description

Returns a printed string representation of ‘obj‘ in a ‘princ‘-like style.

19.9.5 Return Values

String.

19.9.6 Side Effects

None.

19.9.7 Arguments and Values

- ‘obj‘: any AutoLISP value documented as printable by Autodesk, including numbers, strings, lists, entity names, ‘T‘, and ‘nil‘

19.9.8 Notes

- Autodesk documents that the result is the same printed representation that ‘princ‘ would produce.
- Autodesk examples show platform-sensitive pathname formatting in printed strings, while preserving ‘princ‘-style omission of string quotes.

19.9.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.9.10 Source Notes

- [vl-princ-to-string (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-33B0CA43-84F2-46DE-88BE-8FFF22C83 htm)
- Status: documented.

19.10 Function Entry: VL-STRING-SEARCH

19.10.1 Name

‘vl-string-search‘

19.10.2 Class

Visual LISP Function

19.10.3 Syntax

```
(vl-string-search pattern string [start-pos])
```

19.10.4 Arguments and Values

- ‘pattern‘: search string
- ‘string‘: target string
- ‘start-pos‘: optional starting position

19.10.5 Description

Searches for ‘pattern‘ within ‘string‘.

19.10.6 Return Values

Returns the index of the first match at or after ‘start-pos‘, or ‘nil‘ if no match is found.

19.10.7 Side Effects

None.

19.10.8 Compatibility

Autodesk explicitly documents Unicode-related behavioral changes beginning in AutoCAD 2021.

19.10.9 Notes

This function is a useful probe for the host’s character-indexing semantics in Unicode-era environments.

19.10.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.10.11 Source Notes

- [vl-string-search (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-4E7AE1DA-1ED5-4D96-A7D3-34241A` `htm`)
- Status: documented.

19.11 Function Entry: VL-STRING-POSITION

19.11.1 Name

‘vl-string-position‘

19.11.2 Class

Visual LISP Function

19.11.3 Syntax

```
(vl-string-position char string [start-pos [from-end]])
```

19.11.4 Arguments and Values

- ‘char‘: integer character code
- ‘string‘: target string
- ‘start-pos‘: optional integer starting position; ‘0‘ if omitted
- ‘from-end‘: optional reverse-search flag

19.11.5 Description

Searches for a character within a string and returns its position.

19.11.6 Return Values

Index of the matching character, or ‘nil‘.

19.11.7 Side Effects

None.

19.11.8 Compatibility

Character/index semantics are potentially version-sensitive in Unicode-aware hosts.

19.11.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.11.10 Source Notes

- [vl-string-position (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2B0BF48B-FA26-431B-8EC3-DD0BD9F14 htm)
- Status: documented.

19.11.11 Notes

- Autodesk documents zero-based indexing: the first character is at position ‘0‘.
- Autodesk documents that when ‘from-end‘ is ‘T‘, the search proceeds backward from the end of the string toward ‘start-pos‘.
- Autodesk documents an AutoCAD 2021 change from ASCII character codes to Unicode character codes when ‘LISPSYS‘ selects the Unicode engine.

19.12 Function Entry: VL-STRING-MISMATCH

19.12.1 Name

‘vl-string-mismatch‘

19.12.2 Class

Visual LISP Function

19.12.3 Syntax

```
(vl-string-mismatch str1 str2 [start1 [start2 [ignore-case]]])
```

19.12.4 Arguments and Values

- ‘str1‘, ‘str2‘: strings
- ‘start1‘: optional integer starting position in ‘str1‘; ‘0‘ if omitted
- ‘start2‘: optional integer starting position in ‘str2‘; ‘0‘ if omitted
- ‘ignore-case‘: optional ‘T‘ or ‘nil‘

19.12.5 Description

Compares two strings and returns the position of the first mismatch.

19.12.6 Return Values

Index of mismatch, or ‘nil‘ if the compared regions match.

19.12.7 Side Effects

None.

19.12.8 Compatibility

Case and character-unit behavior should be treated as Unicode-sensitive.

19.12.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.12.10 Source Notes

- [vl-string-mismatch (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-D1EA6F47-146E-44BC-BBCA-40B7A` `htm`)
- Status: documented.

19.12.11 Notes

- Autodesk documents that the function returns the length of the longest common prefix of the compared suffixes.
- Autodesk documents zero-based positions for ‘start1‘ and ‘start2‘.
- Autodesk documents that AutoCAD 2021 changed the accepted text model from ASCII-oriented to Unicode-aware behavior.

19.13 Function Entry: VL-STRING-TRIM

19.13.1 Name

‘vl-string-trim‘

19.13.2 Class

Visual LISP Function

19.13.3 Syntax

`(vl-string-trim char-set string)`

19.13.4 Arguments and Values

- ‘char-set‘: trimming character set
- ‘string‘: source string

19.13.5 Description

Removes matching leading and trailing characters from the string.

19.13.6 Return Values

Trimmed string.

19.13.7 Side Effects

None.

19.13.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.13.9 Source Notes

- [vl-string-trim (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PTB/AutoCAD-AutoLISP-Reference/files/GUID-FF1DA441-C9D8-4C99-BC6E-B2A72B562 htm)
- Status: documented.

19.14 Function Entry: VL-STRING-LEFT-TRIM

19.14.1 Name

‘vl-string-left-trim‘

19.14.2 Class

Visual LISP Function

19.14.3 Syntax

```
(vl-string-left-trim char-set string)
```

19.14.4 Arguments and Values

- ‘char-set‘: string listing characters to remove
- ‘string‘: source string

19.14.5 Description

Removes matching leading characters from the string.

19.14.6 Return Values

Trimmed string.

19.14.7 Side Effects

None.

19.14.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.14.9 Source Notes

- [vl-string-left-trim (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-24CB6E8B-9F4F-46DD-AB28-6481B86BA htm)
- Status: documented.

19.15 Function Entry: VL-STRING-RIGHT-TRIM

19.15.1 Name

‘vl-string-right-trim‘

19.15.2 Class

Visual LISP Function

19.15.3 Syntax

`(vl-string-right-trim char-set string)`

19.15.4 Arguments and Values

- ‘char-set‘: string listing characters to remove
- ‘string‘: source string

19.15.5 Description

Removes matching trailing characters from the string.

19.15.6 Return Values

Trimmed string.

19.15.7 Side Effects

None.

19.15.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.15.9 Source Notes

- [vl-string-right-trim (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-MAC-AutoLISP-Reference/files/GUID-532CDF83-7A7F-4B54-8362-F5B7F htm)
- Status: documented.

19.16 Function Entry: VL-STRING-TRANSLATE

19.16.1 Name

‘vl-string-translate‘

19.16.2 Class

Visual LISP Function

19.16.3 Syntax

```
(vl-string-translate from-set to-set string)
```

19.16.4 Arguments and Values

- ‘from-set‘: source character set
- ‘to-set‘: replacement character set
- ‘string‘: source string

19.16.5 Description

Translates characters in the source string according to the mapping from ‘from-set‘ to ‘to-set‘.

19.16.6 Return Values

Translated string.

19.16.7 Side Effects

None.

19.16.8 Compatibility

Character mapping semantics should be treated as Unicode-sensitive in modern hosts.

19.16.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.16.10 Source Notes

- [vl-string-translate (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-57060085-C79D-4613-B438-506AC443B htm)
- Status: documented.

19.16.11 Notes

- Autodesk documents that characters in ‘string‘ matching characters in ‘from-set‘ are replaced by corresponding characters from ‘to-set‘.
- Autodesk documents an AutoCAD 2021 change from ASCII-oriented to Unicode-aware behavior under ‘LISPSYS‘.

19.17 Function Entry: VL-STRING-SUBST

19.17.1 Name

‘vl-string-subst‘

19.17.2 Class

Visual LISP Function

19.17.3 Syntax

```
(vl-string-subst new old string [start-pos])
```

19.17.4 Arguments and Values

- ‘new‘: replacement string
- ‘old‘: substring to replace
- ‘string‘: source string
- ‘start-pos‘: optional starting position

19.17.5 Description

Replaces the first matching occurrence of ‘old‘ with ‘new‘ in the source string.

19.17.6 Return Values

Modified string.

19.17.7 Side Effects

None.

19.17.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.17.9 Source Notes

- [vl-string-subst (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-D8EE91DC-D4DB-43E0-9AFE-5FA166C08 htm`)
- Status: documented.

19.17.10 Notes

- Autodesk documents that the search is case-sensitive.
- Autodesk documents that only the first matching occurrence is replaced.
- Autodesk documents zero-based indexing for ‘start-pos‘.
- Autodesk documents an AutoCAD 2021 change from ASCII-oriented to Unicode-aware behavior under ‘LISPSYS‘.

19.18 Function Entry: VL-STRING-ELT

19.18.1 Name

‘vl-string-elt‘

19.18.2 Class

Visual LISP Function

19.18.3 Syntax

`(vl-string-elt string index)`

19.18.4 Arguments and Values

- ‘string‘: source string
- ‘index‘: integer character position

19.18.5 Description

Returns the character code or character-designating value at the given position in the string.

19.18.6 Return Values

Integer character code.

19.18.7 Side Effects

None.

19.18.8 Compatibility

This function is especially sensitive to the underlying character-indexing model and therefore to Unicode-era changes.

19.18.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.18.10 Source Notes

- [vl-string-elt (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-5A68B31E-1EA7-434A-93FF-C05D9514A htm`)

- Status: documented.

19.18.11 Notes

- Autodesk documents zero-based indexing: the first character is at position ‘0‘.

- Autodesk documents that an error occurs if ‘index‘ is outside the bounds of ‘string‘.

- Autodesk documents an AutoCAD 2021 change from legacy ASCII-range behavior to Unicode character codes under ‘LISPSYS‘.

19.19 Function Entry: VL-STRING->LIST

19.19.1 Name

‘vl-string->list‘

19.19.2 Class

Visual LISP Function

19.19.3 Syntax

```
(vl-string->list string)
```

19.19.4 Arguments and Values

- ‘string‘: source string

19.19.5 Description

Returns a list representation of the string in terms of its character values.

19.19.6 Return Values

List of integer character codes, or ‘nil‘ for the empty string.

19.19.7 Side Effects

None.

19.19.8 Compatibility

Like ‘ascii‘, ‘chr‘, and ‘vl-string-elt‘, this function exposes the host’s observable character model.

19.19.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.19.10 Source Notes

- [vl-string->list (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-3C1B15F7-CF97-4783-A725-231C3A htm)
- Status: documented.

19.19.11 Notes

- Autodesk documents that each list element is an integer character code corresponding to a character of ‘string‘.
- Autodesk documents that ‘(vl-string->list "")‘ returns ‘nil‘.
- Autodesk documents an AutoCAD 2021 change from ASCII-oriented to Unicode-aware behavior under ‘LISPSYS‘.

19.20 Function Entry: VER

19.20.1 Name

‘ver‘

19.20.2 Class

Visual LISP environment inquiry function

19.20.3 Syntax

```
(ver)
```

19.20.4 Description

Returns a string that identifies the current AutoLISP environment version.

19.20.5 Return Values

A string of the documented form:

```
"environment version (nn)"
```

where ‘environment‘ is an environment name and ‘nn‘ is a two-letter language code.

19.20.6 Side Effects

None.

19.20.7 Notes

- Autodesk documents example outputs such as ‘”Visual LISP <release> (en)”‘ on Windows/Web and ‘”MacOS AutoLISP <release> (en)”‘ on macOS.
- This function is useful for compatibility checks, but specification code should prefer feature/version probes where possible.

19.20.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.20.9 Source Notes

- [ver (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6E5DF4E1-5386-47A2-AEF3-488AA068FA15.htm`)

19.21 Function Entry: VL-ACAD-DEFUN

19.21.1 Name

‘vl-acad-defun‘

19.21.2 Class

Visual LISP external-exposure function

19.21.3 Syntax

```
(vl-acad-defun 'symbol)
```

19.21.4 Arguments and Values

- ‘symbol‘ — symbol naming a function.

19.21.5 Description

Defines an AutoLISP function symbol as an external subroutine callable from ObjectARX.

19.21.6 Return Values

Returns an integer success value.

19.21.7 Side Effects

Exposes the designated function to external ObjectARX callers.

19.21.8 Notes

- Autodesk documents this as useful for non-‘c:‘ functions that should be callable from external ObjectARX applications.
- This is part of the host integration boundary rather than portable core language semantics.

19.21.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.21.10 Source Notes

- [vl-acad-defun (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-D15CB87F-5B69-4BC3-B8BB-E45D57` `htm`)

19.22 Function Entry: VL-ACAD-UNDEFUN

19.22.1 Name

‘vl-acad-undefun‘

19.22.2 Class

Visual LISP external-exposure function

19.22.3 Syntax

```
(vl-acad-undefun 'symbol)
```

19.22.4 Arguments and Values

- ‘symbol‘ — subroutine or symbol naming a function.

19.22.5 Description

Undefines an AutoLISP function symbol so it is no longer available to ObjectARX applications.

19.22.6 Return Values

Returns an integer if successful; returns ‘nil‘ if unsuccessful.

19.22.7 Side Effects

Removes external visibility of the designated function.

19.22.8 Notes

- Autodesk documents use on ‘c:‘ functions and functions previously exposed with ‘vl-acad-defun‘.

19.22.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.22.10 Source Notes

- [vl-acad-undefun (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-81CEB01F-8A57-4465-A4F9-CE2E3
htm)

19.23 Function Entry: VL-GET-RESOURCE

19.23.1 Name

‘vl-get-resource‘

19.23.2 Class

Visual LISP VLX resource function

19.23.3 Syntax

```
(vl-get-resource text-file)
```

19.23.4 Arguments and Values

- ‘text-file‘ — string naming a ‘.txt‘ resource packaged in a VLX, without the ‘.txt‘ extension.

19.23.5 Description

Returns the text stored in a packaged VLX text resource.

19.23.6 Return Values

Returns a string containing the resource contents.

19.23.7 Exceptional Situations

Signals an AutoLISP-visible error if the named resource cannot be retrieved.

19.23.8 Compatibility

- VLX files are Windows-only in Autodesk’s execution model.
- Autodesk also documents the function as supported on macOS/Web but having no effect there in some recent references.

19.23.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.23.10 Source Notes

- [vl-get-resource (AutoLISP/Visual LISP IDE)](`https://help.autodesk.com/cloudhelp/2025/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-947B44CE-EAC4-4B17-83E1-8521AD5F5341.htm`)
- [vl-get-resource (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-947B44CE-EAC4-4B17-83E1-8521AD5F5341.htm`)

19.24 Function Entry: VL-LIST-EXPORTED-FUNCTIONS

19.24.1 Name

‘vl-list-exported-functions‘

19.24.2 Class

Visual LISP VLX namespace inquiry function

19.24.3 Syntax

```
(vl-list-exported-functions [appname])
```

19.24.4 Arguments and Values

- ‘appname‘ — optional string naming a loaded VLX application, without the ‘.vlx‘ extension.

19.24.5 Description

Lists functions exported from a separate-namespace VLX application.

19.24.6 Return Values

Returns a list of strings naming exported functions, or ‘nil‘ if none are exported from the specified application.

19.24.7 Notes

- If ‘appname‘ is omitted or ‘nil‘, Autodesk documents that the function returns all exported functions except those exported from VLX namespaces.
- Autodesk’s newer references say the function is supported on macOS and Web but has no effect there because VLX files are Windows-only.

19.24.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.24.9 Source Notes

- [vl-list-exported-functions (AutoLISP/Visual LISP IDE)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-3D87AA2E-DB4B-4B28-B5A8-04A8183BF8AD.htm`)
- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)

19.25 Function Entry: VL-LIST-LOADED-VLX

19.25.1 Name

‘vl-list-loaded-vlx‘

19.25.2 Class

Visual LISP VLX namespace inquiry function

19.25.3 Syntax

```
(vl-list-loaded-vlx)
```

19.25.4 Description

Returns the set of separate-namespace VLX applications associated with the current document.

19.25.5 Return Values

Returns a list of symbols naming separate-namespace VLX applications, or ‘nil‘ if none are associated with the current document.

19.25.6 Notes

- Autodesk documents that document-namespace VLX applications are not reported by this function.
- Recent Autodesk references say the function is supported on macOS and Web but has no effect there because VLX files are Windows-only.

19.25.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.25.8 Source Notes

- [vl-list-loaded-vlx (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-0F0C7A45-41AA-410F-AEF0-EECAE8 htm)
- [VLX Namespace Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm)

19.26 Function Entry: VL-LOAD-REACTORS

19.26.1 Name

‘vl-load-reactors‘

19.26.2 Class

Visual LISP compatibility loader

19.26.3 Syntax

`(vl-load-reactors)`

19.26.4 Description

Loads reactor support functions.

19.26.5 Return Values

Always returns ‘nil‘.

19.26.6 Notes

- Autodesk documents this as equivalent to ‘vl-load-com‘ and retained for backward compatibility.
- Autodesk currently documents it as unavailable on macOS and Web.

19.26.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.26.8 Source Notes

- [vl-load-reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-05F7E56E-51C7-43E2-` `htm`)

19.27 Function Entry: VL-MKDIR

19.27.1 Name

‘vl-mkdir‘

19.27.2 Class

Visual LISP file-system function

19.27.3 Syntax

`(vl-mkdir directoryname)`

19.27.4 Arguments and Values

- ‘directoryname‘ — string naming the directory to create.

19.27.5 Description

Creates a directory.

19.27.6 Return Values

Returns ‘T‘ if successful; returns ‘nil‘ if the directory already exists or creation fails.

19.27.7 Side Effects

Creates a directory in the host file system.

19.27.8 Compatibility

- Autodesk documents a Unicode-related behavior change in AutoCAD 2021 controlled by ‘LISPSYS‘.

19.27.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.27.10 Source Notes

- [vl-mkdir (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-02D5CC9E-9394-4630-AE14-133F5F03D7B1.htm`)

19.28 Function Entry: VL-POSITION

19.28.1 Name

‘vl-position‘

19.28.2 Class

Visual LISP list function

19.28.3 Syntax

```
(vl-position item list)
```

19.28.4 Arguments and Values

- ‘item‘ — any supported AutoLISP datum.
- ‘list‘ — a true list.

19.28.5 Description

Returns the index of the specified list item.

19.28.6 Return Values

Returns a zero-based integer index if ‘item‘ occurs in ‘list‘; otherwise returns ‘nil‘.

19.28.7 Notes

- Autodesk explicitly documents zero-based indexing.
- The exact comparison predicate is not stated in the currently harvested source and therefore remains under-specified in this draft.

19.28.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.28.9 See Also

- BricsCAD analogue: `position` (chapter 24). Same role; different name.
- Pair documented in ‘vendor-inventory-2026.org‘ §10 item 7.

19.28.10 Source Notes

- [vl-position (AutoLISP)](`https://help.autodesk.com/cloudhelp/2022/ESP/AutoCAD-AutoLISP-Reference/files/GUID-E2C60C39-B46E-4779-82CE-CFFCCCE93AF0.htm`)

19.29 Function Entry: VL-PROPAGATE

19.29.1 Name

‘vl-propagate‘

19.29.2 Class

Namespace communication function

19.29.3 Syntax

```
(vl-propagate 'symbol)
```

19.29.4 Arguments and Values

- ‘symbol‘ — symbol naming an AutoLISP variable.

19.29.5 Description

Copies the value of a variable into all open document namespaces and causes that value to be established in subsequently opened drawings during the current session.

19.29.6 Return Values

Always returns ‘nil‘.

19.29.7 Side Effects

Propagates variable state across multiple document namespaces.

19.29.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.29.9 Source Notes

- [vl-propagate (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ESP/AutoCAD-AutoLISP-Reference/files/GUID-9B30560A-5037-4C2D-8175-429E11027 htm)
- [Namespace Communication Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2025/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F5DD5472-1695-4CA3-9110-E6008E87BB89.htm)

19.30 Function Entry: VL-SORT

19.30.1 Name

‘vl-sort‘

19.30.2 Class

Visual LISP list function

19.30.3 Syntax

```
(vl-sort lst comparison-function)
```

19.30.4 Arguments and Values

- ‘lst‘ — list to sort.
- ‘comparison-function‘ — function designator of two arguments returning non-‘nil‘ when the first argument should precede the second.

19.30.5 Description

Sorts a list according to the specified comparison relation.

19.30.6 Return Values

Returns a list containing the elements of ‘lst‘ in sorted order.

19.30.7 Notes

- Autodesk documents that duplicate elements may be eliminated from the result.
- The comparison function may be supplied as a symbol, quoted lambda expression, or ‘function‘ form.

19.30.8 Side Effects

None specified by Autodesk at the language level.

19.30.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.30.10 Source Notes

- [vl-sort (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F3B27BD2-27FA-4185-B22C-85509175C171.htm`)

19.31 Function Entry: VL-SORT-I

19.31.1 Name

‘vl-sort-i‘

19.31.2 Class

Visual LISP list function

19.31.3 Syntax

```
(vl-sort-i lst comparison-function)
```

19.31.4 Arguments and Values

- ‘lst‘ — list to sort indirectly.
- ‘comparison-function‘ — function designator of two arguments returning non-‘nil‘ when the first argument should precede the second.

19.31.5 Description

Sorts a list indirectly and returns the indices of the original elements in sorted order.

19.31.6 Return Values

Returns a list of zero-based indices into ‘lst‘.

19.31.7 Notes

- Autodesk documents that duplicate elements are retained in the result.
- This contrasts with ‘vl-sort‘, which may eliminate duplicates.

19.31.8 Side Effects

None.

19.31.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.31.10 Source Notes

- [vl-sort-i (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B21C91DD-D359-4370-84ED-317F8BBAF414.htm`)

19.32 Function Entry: VECTOR$_{\text{IMAGE}}$

19.32.1 Name

‘vector$_{\text{image}}$‘

19.32.2 Class

DCL image-drawing function

19.32.3 Syntax

```
(vector_image x1 y1 x2 y2 color)
```

19.32.4 Arguments and Values

- ‘x1‘, ‘y1‘, ‘x2‘, ‘y2‘ — integer image coordinates.
- ‘color‘ — integer color selector, including documented logical color numbers.

19.32.5 Description

Draws a vector in the currently active dialog box image from ‘(x1,y1)‘ to ‘(x2,y2)‘.

19.32.6 Return Values

Returns an integer representing the vector color.

19.32.7 Side Effects

Draws into the active DCL image context.

19.32.8 Notes

- Autodesk documents the origin as the upper-left corner of the image.
- This function operates between ‘start$_{\text{image}}$‘ and ‘end$_{\text{image}}$‘.

19.32.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.32.10 Source Notes

- [vector$_{\text{image}}$ (AutoLISP/DCL)](https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-LT-AutoLISP-Reference/files/GUID-28CC07ED-CBD2-47EF-9866-1AB31D
htm)

19.33 Dictionary Coverage Index: Core Visual LISP Names

Important Visual LISP names covered or referenced in this specification include:

- ‘ver‘
- ‘vector$_{\text{image}}$‘
- ‘vl-acad-defun‘
- ‘vl-acad-undefun‘
- ‘vl-load-all‘
- ‘vl-load-com‘
- ‘vl-load-reactors‘
- ‘vl-get-resource‘
- ‘vl-list-exported-functions‘
- ‘vl-list-loaded-vlx‘
- ‘vl-mkdir‘
- ‘vl-position‘
- ‘vl-propagate‘
- ‘vl-sort‘

- ‘vl-sort-i‘
- ‘vl-catch-all-apply‘
- ‘vl-catch-all-error-message‘
- ‘vl-catch-all-error-p‘
- ‘vl-prin1-to-string‘
- ‘vl-princ-to-string‘
- ‘vl-string-*‘
- ‘vl-symbol-name‘
- ‘vl-symbol-value‘
- ‘vl-symbolp‘

Sources:

- [AutoLISP Extensions Reference](https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-07C3599E-EC87-4EC0-8F0B-8DE4B htm)
- [String-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8543549B-F0D7-43CD- htm)
- [Symbol-Handling Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-796EFF94-E904-4F5B-B120-C6D2DCD8646A.htm)

19.34 Function Entry: VLAX-ENAME->VLA-OBJECT

19.34.1 Name

‘vlax-ename->vla-object‘

19.34.2 Class

Visual LISP Object Function

19.34.3 Syntax

```
(vlax-ename->vla-object ename)
```

19.34.4 Arguments and Values

- ‘ename‘: entity name

19.34.5 Description

Converts an AutoLISP entity name to a corresponding ActiveX/VLA object.

19.34.6 Return Values

‘VLA-object‘.

19.34.7 Side Effects

None.

19.34.8 Affected By

- availability of the ActiveX/automation layer
- validity of the entity name

19.34.9 Exceptional Situations

Erroneous if the entity name is invalid or the automation layer is unavailable.

19.34.10 Compatibility

- part of the Visual LISP automation bridge
- Autodesk documents this in the ActiveX/Windows-oriented layer
- ‘clautolisp‘ may emulate this through a compatibility layer on non-Windows hosts

19.34.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.34.12 Source Notes

- [vlax-ename->vla-object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-BD451263-76D5-4A93-B50A-3C27E89A0AD4.htm`)
- Status: documented.

19.35 Function Entry: VLAX-VLA-OBJECT->ENAME

19.35.1 Name

'vlax-vla-object->ename'

19.35.2 Class

Visual LISP Object Function

19.35.3 Syntax

```
(vlax-vla-object->ename obj)
```

19.35.4 Arguments and Values

- 'obj': 'VLA-object'

19.35.5 Description

Converts a VLA object back to an AutoLISP entity name.

19.35.6 Return Values

'ENAME'.

19.35.7 Side Effects

None.

19.35.8 Compatibility

Inverse companion of 'vlax-ename->vla-object' within the automation bridge.

19.35.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.35.10 Source Notes

- [vlax-vla-object->ename (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-9127B839-D275-473F-` `htm`)
- Status: documented.

19.36 Function Entry: VLAX-GET-PROPERTY

19.36.1 Name

‘vlax-get-property‘

19.36.2 Class

Visual LISP Property Function

19.36.3 Syntax

```
(vlax-get-property obj property [index ...])
```

19.36.4 Arguments and Values

- ‘obj‘: automation object
- ‘property‘: property designator
- optional index arguments for indexed properties

19.36.5 Description

Returns the value of a property on an automation object.

19.36.6 Return Values

Property value in the appropriate AutoLISP/Visual LISP representation, possibly including ‘VARIANT‘, ‘SAFEARRAY‘, scalar, string, or object values.

19.36.7 Side Effects

None beyond object-model access.

19.36.8 Notes

Autodesk documents that this function was formerly named ‘vlax-get‘.

19.36.9 Exceptional Situations

Erroneous if the property does not exist, is inaccessible, or the automation layer is unavailable.

19.36.10 Compatibility

Core part of the ActiveX/VLA property access model.

19.36.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.36.12 Source Notes

- [vlax-get-property (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-B3F22E35-4666-452F-89C8 htm)
- Status: documented.

19.37 Function Entry: VLAX-PUT-PROPERTY

19.37.1 Name

‘vlax-put-property‘

19.37.2 Class

Visual LISP Property Function

19.37.3 Syntax

```
(vlax-put-property obj property value [index ...])
```

19.37.4 Arguments and Values

- ‘obj‘: automation object
- ‘property‘: property designator
- ‘value‘: new property value
- optional index arguments

19.37.5 Description

Sets the value of a property on an automation object.

19.37.6 Return Values

Host-defined success value.

19.37.7 Side Effects

Mutates the state of the target automation object.

19.37.8 Notes

Autodesk’s property-access overview treats this as the property-setting counterpart of ‘vlax-get-property‘ when wrapper functions are not used.

19.37.9 Exceptional Situations

Erroneous if the property is absent, read-only, incompatible with the supplied value, or the automation layer is unavailable.

19.37.10 Compatibility

Core part of the ActiveX/VLA property-setting model.

19.37.11 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.37.12 Source Notes

- [About Obtaining and Setting an Object's Property (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-3221619A-85E4-470E-AF2E-34048BB3DED5.htm`)
- Status: documented by official overview; dedicated per-function page still desirable.

19.38 Function Entry: GETPROPERTYVALUE

19.38.1 Name

‘getpropertyvalue‘

19.38.2 Class

Property Function

19.38.3 Syntax

```
(getpropertyvalue ename property-name)
```

19.38.4 Arguments and Values

- ‘ename‘: entity name
- ‘property-name‘: property identifier

19.38.5 Description

Returns the value of a named property of an object.

19.38.6 Return Values

Property value.

19.38.7 Side Effects

None.

19.38.8 Compatibility

- added by Autodesk in AutoCAD 2011
- belongs to the generic property-function family rather than the low-level ‘vlax-*‘ automation layer
- BricsCAD later added a corresponding family with broader property coverage

19.38.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.38.10 Source Notes

- [What's New or Changed with AutoLISP](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85 htm`)
- [Generic Properties Functions (BricsCAD)](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/GenericPropertiesFunctions.htm`)
- Status: documented by release history and family documentation.

19.39 Function Entry: SETPROPERTYVALUE

19.39.1 Name

‘setpropertyvalue‘

19.39.2 Class

Property Function

19.39.3 Syntax

```
(setpropertyvalue ename property-name value)
```

19.39.4 Arguments and Values

- ‘ename‘: entity name
- ‘property-name‘: property identifier
- ‘value‘: new property value

19.39.5 Description

Sets the value of a named property of an object.

19.39.6 Return Values

Host-defined success value.

19.39.7 Side Effects

Mutates the target object’s property state.

19.39.8 Compatibility

- added by Autodesk in AutoCAD 2011
- BricsCAD later documents a broader related family

19.39.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.39.10 Source Notes

- [What's New or Changed with AutoLISP](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85` `htm`)
- [Generic Properties Functions (BricsCAD)](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/GenericPropertiesFunctions.htm`)
- Status: documented by release history and family documentation.

19.40 Function Entry: ISPROPERTYREADONLY

19.40.1 Name

‘ispropertyreadonly‘

19.40.2 Class

Property Function

19.40.3 Syntax

`(ispropertyreadonly ename property-name)`

19.40.4 Arguments and Values

- ‘ename‘: entity name
- ‘property-name‘: property identifier

19.40.5 Description

Tests whether a named property is read-only.

19.40.6 Return Values

- ‘T‘ if the property is read-only
- ‘nil‘ otherwise

19.40.7 Side Effects

None.

19.40.8 Compatibility

Part of the Autodesk generic property-function additions from AutoCAD 2011.

19.40.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.40.10 Source Notes

- [What's New or Changed with AutoLISP](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85 htm)
- [Generic Properties Functions (BricsCAD)](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/GenericPropertiesFunctions.htm)
- Status: documented by release history and family documentation.

19.41 Function Entry: DUMPALLPROPERTIES

19.41.1 Name

‘dumpallproperties‘

19.41.2 Class

Property Function

19.41.3 Syntax

```
(dumpallproperties ename [show-readonly])
```

19.41.4 Arguments and Values

- ‘ename‘: entity name
- ‘show-readonly‘: optional flag

19.41.5 Description

Displays or reports all available properties of an object.

19.41.6 Return Values

Host-defined diagnostic/reporting value.

19.41.7 Side Effects

Produces diagnostic output or property listing side effects in the host environment.

19.41.8 Compatibility

Added by Autodesk in AutoCAD 2011 as part of the generic property family.

19.41.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.41.10 Source Notes

- [What's New or Changed with AutoLISP](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85htm)
- [Generic Properties Functions (BricsCAD)](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/GenericPropertiesFunctions.htm)
- Status: documented by release history and family documentation.

19.42 Function Entry: VLAX-INVOKE-METHOD

19.42.1 Name

‘vlax-invoke-method‘

19.42.2 Class

Visual LISP Object Function

19.42.3 Syntax

```
(vlax-invoke-method obj method [arg ...])
```

19.42.4 Arguments and Values

- ‘obj‘: automation object
- ‘method‘: method designator
- ‘arg‘: zero or more method arguments

19.42.5 Description

Invokes a method on an automation object.

19.42.6 Return Values

Method result in the appropriate AutoLISP/Visual LISP representation.

19.42.7 Side Effects

Method-dependent object or host side effects.

19.42.8 Exceptional Situations

Erroneous if the method is invalid, the arguments are invalid, or the automation layer is unavailable.

19.42.9 Compatibility

Core part of the ‘vlax-*‘ automation bridge.

19.42.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.42.11 Source Notes

- [vlax-invoke-method (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-A8B097A2-CE86-4B38-B2A9-D6F53EACA8ED.htm`)
- Status: documented.

19.43 Function Entry: VLAX-METHOD-APPLICABLE-P

19.43.1 Name

‘vlax-method-applicable-p‘

19.43.2 Class

Visual LISP Object Function

19.43.3 Syntax

```
(vlax-method-applicable-p obj method [arg ...])
```

19.43.4 Description

Tests whether the specified method is applicable to the object and arguments.

19.43.5 Return Values

- ‘T‘ if the method is applicable
- ‘nil‘ otherwise

19.43.6 Side Effects

None beyond automation-model introspection.

19.43.7 Compatibility

Part of the introspection layer for automation objects.

19.43.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.43.9 Source Notes

- [vlax-method-applicable-p (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-63B81424-11BA-4CB3-` `htm`)
- [About Determining If an ActiveX Method or Property Applies to an Object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2022/PTB/AutoCAD-AutoLISP/files/GUID-F6607E76-FD9B-4B8B-B714-99CB34C988A5.htm`)
- Status: documented.

19.44 Function Entry: VLAX-3D-POINT

19.44.1 Name

‘vlax-3D-point‘

19.44.2 Class

Visual LISP automation utility

19.44.3 Syntax

```
(vlax-3D-point point)
```

Autodesk also documents these equivalent calling patterns:

```
(vlax-3D-point x y)
(vlax-3D-point x y z)
```

19.44.4 Arguments and Values

- ‘point‘: point-like AutoLISP value

When numeric arguments are supplied directly:

- ‘x‘, ‘y‘, ‘z‘ — numbers

19.44.5 Description

Creates an ActiveX-compatible variant 3D point structure.

19.44.6 Return Values

Variant-like automation value representing a 3D point.

19.44.7 Side Effects

None.

19.44.8 Compatibility

Part of the Windows-oriented automation bridge.

19.44.9 Notes

- Autodesk documents that when only ‘x‘ and ‘y‘ are supplied, ‘z‘ defaults to ‘0.0‘.
- Autodesk documents that the resulting object is suitable for methods requiring a variant containing a three-element array of doubles.

19.44.10 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.44.11 Source Notes

- [About Safearrays with Variants (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP/files/GUID-FF4F04CC-9DEF-41C7-8` `htm`)

19.44.12 Compatibility

Available only on Windows; not available on macOS or Web.

19.45 Function Entry: VLAX-DUMP-OBJECT

19.45.1 Name

‘vlax-dump-object‘

19.45.2 Class

Visual LISP automation utility

19.45.3 Syntax

```
(vlax-dump-object obj [detail])
```

19.45.4 Arguments and Values

- ‘obj‘ — VLA-object.
- ‘detail‘ — optional generalized boolean; if non-‘nil‘, methods are listed in addition to properties.

19.45.5 Description

Lists an object’s properties and optionally the methods that apply to the object.

19.45.6 Return Values

Returns ‘T‘.

19.45.7 Side Effects

Produces diagnostic output about the object.

19.45.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.45.9 Source Notes

- [About Listing an Object's Properties and Methods (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-4805C525-0E7F-4407-A473-D83ECAF1A940.htm`)
- [Object-Handling Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-6FB61071-0F0D-40EF-A873-7C3561F03C29.htm`)

19.45.10 Notes

- Autodesk documents this as requiring 'vl-load-com' in the classic extension model.

19.45.11 Compatibility

Available on AutoCAD and AutoCAD LT for Windows; not available on macOS or Web.

19.46 Function Entry: VLAX-GET-ACAD-OBJECT

19.46.1 Name

‘vlax-get-acad-object‘

19.46.2 Class

Visual LISP automation utility

19.46.3 Syntax

`(vlax-get-acad-object)`

19.46.4 Description

Retrieves the top-level AutoCAD application object for the current session.

19.46.5 Return Values

VLA-object representing the AutoCAD Application object.

19.46.6 Side Effects

None beyond automation-layer access.

19.46.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.46.8 Source Notes

- [About Accessing the AutoCAD Application Object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-AutoLISP/files/GUID-62F07541-9663-4D83-B5EE-24D562351FCE.htm`)
- [Object-Handling Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-6FB61071-0F0D-40EF-A873-7C3561F03C29.htm`)

19.46.9 Notes

- Autodesk documents this as requiring ‘vl-load-com‘ in the classic extension model.
- The returned object is the root of the AutoCAD object model hierarchy.

19.46.10 Compatibility

Available on AutoCAD and AutoCAD LT for Windows; not available on macOS or Web.

19.47 Function Entry: VLAX-GET-OR-CREATE-OBJECT

19.47.1 Name

‘vlax-get-or-create-object‘

19.47.2 Class

Visual LISP automation utility

19.47.3 Syntax

```
(vlax-get-or-create-object progid)
```

19.47.4 Arguments and Values

- ‘progid‘ — string naming the desired ActiveX object.

  Autodesk documents the format as:

```
<Vendor>.<Component>.<Version>
```

19.47.5 Description

Returns a running instance of an application object, or creates a new instance if the application is not currently running.

19.47.6 Return Values

VLA-object.

19.47.7 Side Effects

May create an external automation server process.

19.47.8 Compatibility

Available only on Windows; not available on macOS or Web.

19.47.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.47.10 Source Notes

- [vlax-get-or-create-object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/PTB/AutoCAD-LT-AutoLISP-Reference/files/GUID-7AFF597F-D630-4489-8349-94EF67CCE1F0.htm`)
- [About Establishing a Connection to an External Application with ActiveX (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHT/AutoCAD-AutoLISP/files/GUID-4A45548C-2556-4C92-9801-2996A56448DC.htm`)

19.48 Function Entry: VLAX-IMPORT-TYPE-LIBRARY

19.48.1 Name

‘vlax-import-type-library‘

19.48.2 Class

Visual LISP automation utility

19.48.3 Syntax

```
(vlax-import-type-library [options ...])
```

Autodesk documents the concrete keyword syntax as:

```
(vlax-import-type-library
  :tlb-filename filename
  [:methods-prefix mprefix
   :properties-prefix pprefix
   :constants-prefix cprefix])
```

Keywords are required for the arguments.

19.48.4 Description

Imports information from a type library.

19.48.5 Return Values

‘T‘ if successful; otherwise ‘nil‘.

19.48.6 Side Effects

Creates or registers bindings based on external type-library metadata.

19.48.7 Notes

- Autodesk documents that imported wrapper functions are available only in the context of the document from which the import was issued.
- Autodesk recommends calling this at top level, before any code that uses the generated wrappers or constants.
- Autodesk recommends avoiding absolute paths for portability.

19.48.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.48.9 Source Notes

- [vlax-import-type-library (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-LT-AutoLISP-Reference/files/GUID-1699B6A0-C4A6-4BC4-8 htm)
- [About Importing a Type Library (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2023/HUN/AutoCAD-AutoLISP/files/GUID-B140BF54-AD94-47EE-B htm)

19.48.10 Compatibility

Available on AutoCAD for Windows only; not available in AutoCAD LT for Windows, or on macOS or Web.

19.49 Function Entry: VLAX-MAKE-SAFEARRAY

19.49.1 Name

‘vlax-make-safearray‘

19.49.2 Class

Visual LISP automation utility

19.49.3 Syntax

`(vlax-make-safearray type bounds)`

19.49.4 Arguments and Values

- ‘type‘ — integer constant naming the element type.
- ‘bounds‘ — one or more dotted pairs ‘(l-bound . u-bound)‘ defining dimensions.

Autodesk documents the element-type constants:

- ‘vlax-vbInteger‘
- ‘vlax-vbLong‘
- ‘vlax-vbSingle‘
- ‘vlax-vbDouble‘
- ‘vlax-vbString‘
- ‘vlax-vbObject‘
- ‘vlax-vbBoolean‘
- ‘vlax-vbVariant‘

19.49.5 Description

Creates a safearray.

19.49.6 Return Values

Safearray object.

19.49.7 Side Effects

Allocates automation-array storage.

19.49.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.49.9 Source Notes

- [vlax-make-safearray (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-E0D40331-096B-4A52-A0D7-10204829C4A6.htm`)
- [About Safearrays (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/PTB/AutoCAD-AutoLISP/files/GUID-986313E1-0AEB-41DA-BC8B-093437F42 htm`)

19.49.10 Notes

- Autodesk documents a maximum of 16 dimensions.

19.49.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.50 Function Entry: VLAX-MAKE-VARIANT

19.50.1 Name

‘vlax-make-variant‘

19.50.2 Class

Visual LISP automation utility

19.50.3 Syntax

```
(vlax-make-variant value [type])
```

19.50.4 Arguments and Values

- ‘value‘ — integer, real, string, VLA-object, safearray, ‘T‘, ‘nil‘, or existing variant.
- ‘type‘ — optional integer type constant.

19.50.5 Description

Creates a variant data type.

19.50.6 Return Values

Variant object.

19.50.7 Side Effects

None.

19.50.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.50.9 Source Notes

- [vlax-make-variant (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-MAC-AutoLISP-Reference/files/GUID-A1B964E3-89B4-4655-` `htm`)

- [About Working with Variants (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-AutoLISP/files/GUID-5AEC3752-6612-4289-8` `htm`)

19.50.10 Notes

- If ‘type‘ is omitted, Autodesk documents a default mapping from the supplied AutoLISP value to a variant type.

- In particular, an integer defaults to ‘vlax-vbLong‘, not ‘vlax-vbInteger‘.

19.50.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.51 Function Entry: VLAX-OBJECT-RELEASED-P

19.51.1 Name

‘vlax-object-released-p‘

19.51.2 Class

Visual LISP automation predicate

19.51.3 Syntax

```
(vlax-object-released-p obj)
```

19.51.4 Arguments and Values

- ‘obj‘ — VLA-object.

19.51.5 Description

Determines whether an object has been released.

19.51.6 Return Values

- ‘T‘ if the object has been released
- ‘nil‘ otherwise

19.51.7 Notes

- Autodesk documents that erasing an object does not release it.
- Autodesk documents that an object becomes released when ‘vlax-release-object‘ is called, when AutoLISP garbage collection reclaims it, or when the drawing database is destroyed at session end.

19.51.8 Side Effects

None.

19.51.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.51.10 Source Notes

- [vlax-object-released-p (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/KOR/AutoCAD-LT-AutoLISP-Reference/files/GUID-23DA7B30-FBB5-40DC-A93E-E1FC2A5D8B6F.htm`)
- [About Releasing Objects and Freeing Memory (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-4A2C849B-C0E0-4991-905D-A916B2CD3F25.htm`)

19.51.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.52 Function Entry: VLAX-PROPERTY-AVAILABLE-P

19.52.1 Name

‘vlax-property-available-p‘

19.52.2 Class

Visual LISP automation predicate

19.52.3 Syntax

`(vlax-property-available-p obj property [check-modify])`

19.52.4 Arguments and Values

- ‘obj‘ — VLA-object.
- ‘property‘ — symbol or string naming the property to test.
- ‘check-modify‘ — optional ‘T‘; if supplied, also requires that the property be modifiable.

19.52.5 Description

Determines whether an object has a specified property.

19.52.6 Return Values

- ‘T‘ if the property is available
- ‘nil‘ otherwise

If ‘check-modify‘ is ‘T‘, returns ‘nil‘ when the property exists but is read-only.

19.52.7 Side Effects

None.

19.52.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.52.9 Source Notes

- [vlax-property-available-p (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2025/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-E8A5B009-46D6-4BA7-9655-88104F8BE792.htm`)
- [About Determining If an ActiveX Method or Property Applies to an Object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP/files/GUID-F6607E76-FD9B-4B8B-B714-99CB34C988A5.htm`)

19.52.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.53 Function Entry: VLAX-READ-ENABLED-P

19.53.1 Name

‘vlax-read-enabled-p‘

19.53.2 Class

Visual LISP automation predicate

19.53.3 Syntax

```
(vlax-read-enabled-p obj)
```

19.53.4 Arguments and Values

- ‘obj‘ — VLA-object.

19.53.5 Description

Determines whether an object can be read.

19.53.6 Return Values

- ‘T‘ if the object is readable
- ‘nil‘ otherwise

19.53.7 Side Effects

None.

19.53.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.53.9 Source Notes

- [vlax-read-enabled-p (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-8BBBF683-BA93-4942-9B8C-AD43D490E52A.htm)

19.53.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.54 Function Entry: VLAX-RELEASE-OBJECT

19.54.1 Name

‘vlax-release-object‘

19.54.2 Class

Visual LISP automation utility

19.54.3 Syntax

```
(vlax-release-object obj)
```

19.54.4 Arguments and Values

- ‘obj‘ — VLA-object.

19.54.5 Description

Releases a drawing or automation object.

19.54.6 Return Values

Integer on success.

19.54.7 Side Effects

Releases underlying automation resources.

19.54.8 Exceptional Situations

Signals an AutoLISP-visible error if the object has already been released.

19.54.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.54.10 Source Notes

- [vlax-release-object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-85FA59E7-9EC9-4690-8CEE` `htm`)
- [About Releasing Objects and Freeing Memory (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-4A2C849B-C0E0-4991-905D-A916B2CD3F25.htm`)

19.54.11 Notes

- Autodesk documents that the actual COM release can be deferred until garbage collection.
- Autodesk documents that ‘gc‘ can be used to force collection, though excessive use may hurt performance.

19.54.12 Compatibility

Available only on Windows; not available on macOS or Web.

19.55 Function Entry: VLAX-SAFEARRAY->LIST

19.55.1 Name

‘vlax-safearray->list‘

19.55.2 Class

Visual LISP automation conversion

19.55.3 Syntax

```
(vlax-safearray->list array)
```

19.55.4 Arguments and Values

- ‘array‘ — safearray.

19.55.5 Description

Returns the elements of a safearray in list form.

19.55.6 Return Values

List or ‘nil‘.

For multidimensional arrays, Autodesk documents a nested list structure.

19.55.7 Side Effects

None.

19.55.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.55.9 Source Notes

- [vlax-safearray->list (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-AutoLISP-Reference/files/GUID-A38DDA9E-1E84-4529-htm)

19.55.10 Compatibility

Available only on Windows.

19.56 Function Entry: VLAX-SAFEARRAY-FILL

19.56.1 Name

‘vlax-safearray-fill‘

19.56.2 Class

Visual LISP automation utility

19.56.3 Syntax

```
(vlax-safearray-fill array data)
```

19.56.4 Arguments and Values

- ‘array‘ — safearray.
- ‘data‘ — list of element values; for multidimensional arrays, nested lists.

19.56.5 Description

Stores data in the elements of a safearray.

19.56.6 Return Values

The safearray supplied in ‘array‘, or ‘nil‘.

19.56.7 Side Effects

Mutates safearray contents.

19.56.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.56.9 Source Notes

- [vlax-safearray-fill (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP-Reference/files/GUID-7C2331B4-1A4D-40FF-B59A htm)
- [About Safearrays (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/PTB/AutoCAD-AutoLISP/files/GUID-986313E1-0AEB-41DA-BC8B-093437F42 htm)

19.56.10 Notes

- If fewer values are supplied than there are array elements, Autodesk documents that remaining elements retain their current values.

19.56.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.57 Function Entry: VLAX-SAFEARRAY-GET-DIM

19.57.1 Name

‘vlax-safearray-get-dim‘

19.57.2 Class

Visual LISP automation accessor

19.57.3 Syntax

`(vlax-safearray-get-dim array)`

19.57.4 Arguments and Values

- ‘array‘ — safearray.

19.57.5 Description

Returns the number of dimensions in a safearray object.

19.57.6 Return Values

Integer.

19.57.7 Side Effects

None.

19.57.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.57.9 Source Notes

- [About Getting an Element of a Safearray (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-AutoLISP/files/GUID-4BEC64A1-05AB-4202-A873-55C8C7EFA67A.htm`)

19.57.10 Exceptional Situations

Signals an AutoLISP-visible error if ‘array‘ is not a safearray.

19.57.11 Compatibility

Available only on Windows.

19.58 Function Entry: VLAX-SAFEARRAY-GET-ELEMENT

19.58.1 Name

‘vlax-safearray-get-element‘

19.58.2 Class

Visual LISP automation accessor

19.58.3 Syntax

```
(vlax-safearray-get-element array index [index ...])
```

19.58.4 Arguments and Values

- ‘array‘ — safearray.
- ‘index‘ — one integer per dimension, selecting the element to retrieve.

19.58.5 Description

Returns an element from an array.

19.58.6 Return Values

Integer, real, string, VLA-object, safearray, variant, ‘T‘, or ‘nil‘.

19.58.7 Side Effects

None.

19.58.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.58.9 Source Notes

- [vlax-safearray-get-element (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-AutoLISP-Reference/files/GUID-09386E77-DBBF-4E4D-`
`htm`)

19.58.10 Exceptional Situations

Signals an AutoLISP-visible error if the argument is not a safearray or if the supplied indices are invalid.

19.58.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.59 Function Entry: VLAX-SAFEARRAY-GET-L-BOUND

19.59.1 Name

‘vlax-safearray-get-l-bound‘

19.59.2 Class

Visual LISP automation accessor

19.59.3 Syntax

```
(vlax-safearray-get-l-bound array dimension)
```

19.59.4 Arguments and Values

- ‘array‘ — safearray.
- ‘dimension‘ — integer dimension number, where the first dimension is ‘1‘.

19.59.5 Description

Returns the lower boundary of a safearray dimension.

19.59.6 Return Values

Integer lower bound.

19.59.7 Side Effects

None.

19.59.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.59.9 Source Notes

- [vlax-safearray-get-l-bound (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2025/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-8ABBA8A8-D421-48D2-BCBC-34DA1C728A78.htm`)

19.59.10 Exceptional Situations

Signals an AutoLISP-visible error if ‘array‘ is not a safearray or if ‘dimension‘ is invalid.

19.59.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.60 Function Entry: VLAX-SAFEARRAY-GET-U-BOUND

19.60.1 Name

‘vlax-safearray-get-u-bound‘

19.60.2 Class

Visual LISP automation accessor

19.60.3 Syntax

```
(vlax-safearray-get-u-bound array dimension)
```

19.60.4 Arguments and Values

- ‘array‘ — safearray.
- ‘dimension‘ — integer dimension number, where the first dimension is ‘1‘.

19.60.5 Description

Returns the upper boundary of a safearray dimension.

19.60.6 Return Values

Integer upper bound.

19.60.7 Side Effects

None.

19.60.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.60.9 Source Notes

- [vlax-safearray-get-u-bound (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-07070F34-A919-4BF5-` `htm`)

19.60.10 Exceptional Situations

Signals an AutoLISP-visible error if ‘array‘ is not a safearray or if ‘dimension‘ is invalid.

19.60.11 Compatibility

Available only on Windows.

19.61 Function Entry: VLAX-SAFEARRAY-PUT-ELEMENT

19.61.1 Name

‘vlax-safearray-put-element‘

19.61.2 Class

Visual LISP automation mutator

19.61.3 Syntax

`(vlax-safearray-put-element array index value)`

19.61.4 Arguments and Values

- ‘array‘ — safearray.
- ‘index‘ — one integer per dimension selecting the destination element.
- ‘value‘ — integer, real, string, VLA-object, safearray, variant, ‘T‘, or ‘nil‘.

19.61.5 Description

Adds or stores an element in an array.

19.61.6 Return Values

The value assigned to the element.

19.61.7 Side Effects

Mutates safearray contents.

19.61.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.61.9 Source Notes

- [vlax-safearray-put-element (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2015/ENU/AutoCAD-AutoLISP/files/GUID-D4D51BF6-E954-4EA8-9A8D-47FE6
htm)
- [About Creating Array Elements (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-AutoLISP/files/GUID-04690F92-A1E6-42DC-9
htm)

19.61.10 Exceptional Situations

Signals an AutoLISP-visible error if ‘array‘ is not a safearray or the indices are invalid.

19.61.11 Compatibility

Available only on Windows.

19.62 Function Entry: VLAX-SAFEARRAY-TYPE

19.62.1 Name

‘vlax-safearray-type‘

19.62.2 Class

Visual LISP automation accessor

19.62.3 Syntax

`(vlax-safearray-type array)`

19.62.4 Arguments and Values

- ‘array‘ — safearray.

19.62.5 Description

Returns the data type of a safearray.

19.62.6 Return Values

Integer type code.

Autodesk documents values such as ‘2‘, ‘3‘, ‘4‘, ‘5‘, ‘8‘, ‘9‘, ‘11‘, and ‘12‘, corresponding to the ‘vlax-vb*‘ element-type constants.

19.62.7 Side Effects

None.

19.62.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.62.9 Source Notes

- [vlax-safearray-type (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2025/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-64051CE8-BDDF-4AFF-A440-B93F1E24AF78.htm`)
- [About Getting an Element of a Safearray (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-AutoLISP/files/GUID-4BEC64A1-05AB-4202-A873-55C8C7EFA67A.htm`)

19.62.10 Exceptional Situations

Signals an AutoLISP-visible error if ‘array‘ is not a safearray.

19.62.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.63 Function Entry: VLAX-VARIANT-CHANGE-TYPE

19.63.1 Name

‘vlax-variant-change-type‘

19.63.2 Class

Visual LISP automation conversion

19.63.3 Syntax

```
(vlax-variant-change-type variant type)
```

19.63.4 Arguments and Values

- ‘variant‘ — variant.
- ‘type‘ — integer variant-type constant.

19.63.5 Description

Returns the value of a variant after changing it from one data type to another.

19.63.6 Return Values

Integer, real, string, VLA-object, safearray, ‘T‘, or ‘nil‘.

Autodesk documents that the original ‘variant‘ value is unchanged.

19.63.7 Side Effects

None.

19.63.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.63.9 Source Notes

- [vlax-variant-change-type (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP-Reference/files/GUID-64714B92-0662-496E-htm)

19.63.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.64 Function Entry: VLAX-VARIANT-TYPE

19.64.1 Name

‘vlax-variant-type‘

19.64.2 Class

Visual LISP automation accessor

19.64.3 Syntax

```
(vlax-variant-type variant)
```

19.64.4 Arguments and Values

- ‘variant‘ — variant.

19.64.5 Description

Determines the data type of a variant.

19.64.6 Return Values

Integer type code.

Autodesk documents values such as ‘0‘, ‘1‘, ‘2‘, ‘3‘, ‘4‘, ‘5‘, ‘8‘, ‘9‘, ‘11‘, and ‘8192+n‘ for safearray-bearing variants.

19.64.7 Side Effects

None.

19.64.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.64.9 Source Notes

- [vlax-variant-type (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-14029CC8-2839-432B-` `htm`)
- [About Working with Variants (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-AutoLISP/files/GUID-5AEC3752-6612-4289-8` `htm`)

19.64.10 Exceptional Situations

Signals an AutoLISP-visible error if the argument is not a variant.

19.64.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.65 Function Entry: VLAX-VARIANT-VALUE

19.65.1 Name

‘vlax-variant-value‘

19.65.2 Class

Visual LISP automation accessor

19.65.3 Syntax

```
(vlax-variant-value variant)
```

19.65.4 Arguments and Values

- ‘variant‘ — variant.

19.65.5 Description

Returns the value of a variant.

19.65.6 Return Values

Integer, real, string, VLA-object, safearray, ‘T‘, or ‘nil‘.

19.65.7 Side Effects

None.

19.65.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.65.9 Source Notes

- [vlax-variant-value (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-D26E25ED-6D9A-4E0F-` `htm`)

19.65.10 Exceptional Situations

Signals an AutoLISP-visible error if the argument is not a variant.

19.65.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.66 Function Entry: VLAX-WRITE-ENABLED-P

19.66.1 Name

‘vlax-write-enabled-p‘

19.66.2 Class

Visual LISP automation predicate

19.66.3 Syntax

```
(vlax-write-enabled-p obj)
```

19.66.4 Arguments and Values

- ‘obj‘ — VLA-object or entity name.

19.66.5 Description

Determines whether an AutoCAD drawing object can be modified.

19.66.6 Return Values

- ‘T‘ if the object is writable
- ‘nil‘ otherwise

19.66.7 Side Effects

None.

19.66.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.66.9 Source Notes

- [vlax-write-enabled-p (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-67DAD30A-B359-4566-9B11-C6203EB71247.htm)

19.66.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.67 Function Entry: VLAX-CURVE-GETAREA

19.67.1 Name

‘vlax-curve-getArea‘

19.67.2 Class

Visual LISP curve function

19.67.3 Syntax

```
(vlax-curve-getArea curve)
```

19.67.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.67.5 Description

Returns the area inside the curve.

19.67.6 Return Values

- real or ‘nil‘

19.67.7 Side Effects

None.

19.67.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.67.9 Source Notes

- [vlax-curve-getArea (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ITA/AutoCAD-AutoLISP-Reference/files/GUID-93C0A361-465A-400F-htm)

19.67.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.68 Function Entry: VLAX-CURVE-GETCLOSESTPOINTTO

19.68.1 Name

‘vlax-curve-getClosestPointTo‘

19.68.2 Class

Visual LISP curve function

19.68.3 Syntax

```
(vlax-curve-getClosestPointTo curve point [extend])
```

19.68.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘point‘ — 3D point list in WCS.
- ‘extend‘ — optional generalized boolean requesting search on the extended curve.

19.68.5 Description

Returns the point in WCS on a curve that is nearest to the specified point.

19.68.6 Return Values

- 3D point list or ‘nil‘

19.68.7 Side Effects

None.

19.68.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.68.9 Source Notes

- [vlax-curve-getClosestPointTo (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/JPN/AutoCAD-AutoLISP-Reference/files/GUID-90F585FE-D5C8-4C08-htm`)

19.68.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.69 Function Entry: VLAX-CURVE-GETCLOSESTPOINTTOPROJECTION

19.69.1 Name

‘vlax-curve-getClosestPointToProjection‘

19.69.2 Class

Visual LISP curve function

19.69.3 Syntax

```
(vlax-curve-getClosestPointToProjection curve point normal [extend])
```

19.69.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘point‘ — 3D point list in WCS.
- ‘normal‘ — 3D normal vector in WCS.
- ‘extend‘ — optional generalized boolean requesting search on the extended curve.

19.69.5 Description

Returns the closest point in WCS on a curve after projecting the curve onto a plane.

19.69.6 Return Values

- 3D point list or ‘nil‘

19.69.7 Notes

- Autodesk documents that the curve is projected onto the plane through ‘point‘ with normal ‘normal‘, then the nearest point is found and mapped back to the original curve.

19.69.8 Side Effects

None.

19.69.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.69.10 Source Notes

- [vlax-curve-getClosestPointToProjection (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-23A1B8CE-9FF5-482C-AF32-3EE48784B905.htm`)

19.69.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.70 Function Entry: VLAX-CURVE-GETDISTATPARAM

19.70.1 Name

‘vlax-curve-getDistAtParam‘

19.70.2 Class

Visual LISP curve function

19.70.3 Syntax

```
(vlax-curve-getDistAtParam curve param)
```

19.70.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘param‘ — number specifying a parameter on the curve.

19.70.5 Description

Returns the length of the curve’s segment from the curve’s beginning to the specified parameter.

19.70.6 Return Values

- real or ‘nil‘

19.70.7 Side Effects

None.

19.70.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.70.9 Source Notes

- [vlax-curve-getDistAtParam (AutoLISP/ActiveX)](https://help.autodesk.com/view/CIV3D/2025/ENU/?guid=GUID-8FF6D3D5-7EA5-4E9A-8A61-295C0562E7AE)
- Status: documented.

19.70.10 Compatibility

ActiveX curve measurement is Windows-only in Autodesk's platform model.

19.71 Function Entry: VLAX-CURVE-GETDISTATPOINT

19.71.1 Name

‘vlax-curve-getDistAtPoint‘

19.71.2 Class

Visual LISP curve function

19.71.3 Syntax

```
(vlax-curve-getDistAtPoint curve point)
```

19.71.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘point‘ — 3D point list in WCS on the curve.

19.71.5 Description

Returns the length of the curve’s segment between the curve’s start point and the specified point.

19.71.6 Return Values

- real or ‘nil‘

19.71.7 Side Effects

None.

19.71.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.71.9 Source Notes

- [vlax-curve-getDistAtPoint (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-627E64B0-B34C-4E7A-htm`)

19.71.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.72 Function Entry: VLAX-CURVE-GETENDPARAM

19.72.1 Name

‘vlax-curve-getEndParam‘

19.72.2 Class

Visual LISP curve function

19.72.3 Syntax

`(vlax-curve-getEndParam curve)`

19.72.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.72.5 Description

Returns the parameter of the endpoint of the curve.

19.72.6 Return Values

- real or ‘nil‘

19.72.7 Side Effects

None.

19.72.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.72.9 Source Notes

- [vlax-curve-getEndParam (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-B939A11A-DF89-4450-B520-11D59BD2A126.htm`)

19.72.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.73 Function Entry: VLAX-CURVE-GETENDPOINT

19.73.1 Name

‘vlax-curve-getEndPoint‘

19.73.2 Class

Visual LISP curve function

19.73.3 Syntax

```
(vlax-curve-getEndPoint curve)
```

19.73.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.73.5 Description

Returns the endpoint in WCS of the curve.

19.73.6 Return Values

- 3D point list or ‘nil‘

19.73.7 Side Effects

None.

19.73.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.73.9 Source Notes

- [vlax-curve-getEndPoint (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-AutoLISP-Reference/files/GUID-11309952-C617-4467-htm`)
- Status: documented.

19.74 Function Entry: VLAX-CURVE-GETFIRSTDERIV

19.74.1 Name

‘vlax-curve-getFirstDeriv‘

19.74.2 Class

Visual LISP curve function

19.74.3 Syntax

```
(vlax-curve-getFirstDeriv curve param)
```

19.74.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘param‘ — number specifying a parameter on the curve.

19.74.5 Description

Returns the first derivative in WCS of a curve at the specified location.

19.74.6 Return Values

- 3D vector list or ‘nil‘

19.74.7 Side Effects

None.

19.74.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.74.9 Source Notes

- [vlax-curve-getFirstDeriv (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-37282EB1-D47C-4039-B1AF-744EB5B53`
`htm`)

19.75 Function Entry: VLAX-CURVE-GETPARAMATDIST

19.75.1 Name

‘vlax-curve-getParamAtDist‘

19.75.2 Class

Visual LISP curve function

19.75.3 Syntax

```
(vlax-curve-getParamAtDist curve dist)
```

19.75.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘dist‘ — integer or real distance from the curve’s beginning.

19.75.5 Description

Returns the parameter of a curve at the specified distance from the beginning of the curve.

19.75.6 Return Values

- real or ‘nil‘

19.75.7 Side Effects

None.

19.75.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.75.9 Source Notes

- [vlax-curve-getParamAtDist (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/KOR/AutoCAD-MAC-AutoLISP-Reference/files/GUID-4F02FF68-0D68-453D-A63E-2CB96DB62B05.htm)

19.75.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.76 Function Entry: VLAX-CURVE-GETPARAMATPOINT

19.76.1 Name

‘vlax-curve-getParamAtPoint‘

19.76.2 Class

Visual LISP curve function

19.76.3 Syntax

```
(vlax-curve-getParamAtPoint curve point)
```

19.76.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘point‘ — 3D point list in WCS on the curve.

19.76.5 Description

Returns the parameter of the curve at the point.

19.76.6 Return Values

- real or ‘nil‘

19.76.7 Side Effects

None.

19.76.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.76.9 Source Notes

- [vlax-curve-getParamAtPoint (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-1827742B-21DE-4454-90A0-19551DE79B5A.htm`)

19.76.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.77 Function Entry: VLAX-CURVE-GETPOINTATDIST

19.77.1 Name

‘vlax-curve-getPointAtDist‘

19.77.2 Class

Visual LISP curve function

19.77.3 Syntax

```
(vlax-curve-getPointAtDist curve dist)
```

19.77.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘dist‘ — real distance from the curve’s start.

19.77.5 Description

Returns the point in WCS along a curve at the specified distance.

19.77.6 Return Values

- 3D point list or ‘nil‘

19.77.7 Side Effects

None.

19.77.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.77.9 Source Notes

- [vlax-curve-getPointAtDist (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2017/JPN/AutoCAD-AutoLISP/files/GUID-F41FB58A-4645-404E-98B7-D4978` `htm`)

19.77.10 Compatibility

Available only on Windows.

19.78 Function Entry: VLAX-CURVE-GETPOINTATPARAM

19.78.1 Name

‘vlax-curve-getPointAtParam‘

19.78.2 Class

Visual LISP curve function

19.78.3 Syntax

```
(vlax-curve-getPointAtParam curve param)
```

19.78.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘param‘ — integer or real parameter on the curve.

19.78.5 Description

Returns the point at the specified parameter value along a curve.

19.78.6 Return Values

- 3D point list or ‘nil‘

19.78.7 Side Effects

None.

19.78.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.78.9 Source Notes

- [vlax-curve-getPointAtParam (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-B506C696-DBDA-466B-9EF8-1AF3B1A485F8.htm`)

19.78.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.79 Function Entry: VLAX-CURVE-GETSECONDDERIV

19.79.1 Name

‘vlax-curve-getSecondDeriv‘

19.79.2 Class

Visual LISP curve function

19.79.3 Syntax

```
(vlax-curve-getSecondDeriv curve param)
```

19.79.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.
- ‘param‘ — number specifying a parameter on the curve.

19.79.5 Description

Returns the second derivative in WCS of a curve at the specified location.

19.79.6 Return Values

- 3D vector list or ‘nil‘

19.79.7 Side Effects

None.

19.79.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.79.9 Source Notes

- [vlax-curve-getSecondDeriv (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-E1C78CD1-2347-4A2B-B054-62149738B` `htm`)

19.80 Function Entry: VLAX-CURVE-GETSTARTPARAM

19.80.1 Name

‘vlax-curve-getStartParam‘

19.80.2 Class

Visual LISP curve function

19.80.3 Syntax

`(vlax-curve-getStartParam curve)`

19.80.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.80.5 Description

Returns the start parameter on the curve.

19.80.6 Return Values

- real or ‘nil‘

19.80.7 Side Effects

None.

19.80.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.80.9 Source Notes

- [vlax-curve-getStartParam (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-854C3C4D-1305-4920-htm)

19.80.10 Compatibility

Available only on Windows.

19.81 Function Entry: VLAX-CURVE-GETSTARTPOINT

19.81.1 Name

‘vlax-curve-getStartPoint‘

19.81.2 Class

Visual LISP curve function

19.81.3 Syntax

`(vlax-curve-getStartPoint curve)`

19.81.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.81.5 Description

Returns the start point in WCS of the curve.

19.81.6 Return Values

- 3D point list or ‘nil‘

19.81.7 Side Effects

None.

19.81.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.81.9 Source Notes

- [vlax-curve-getStartPoint (AutoLISP)](`https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-AA1C450F-AA6E-4B05-AFB3-989AD91AD` `htm`)

19.82 Function Entry: VLAX-CURVE-ISCLOSED

19.82.1 Name

‘vlax-curve-isClosed‘

19.82.2 Class

Visual LISP curve predicate

19.82.3 Syntax

```
(vlax-curve-isClosed curve)
```

19.82.4 Arguments and Values

- ‘curve‘ — VLA-object designating a curve.

19.82.5 Description

Determines whether the specified curve is closed.

19.82.6 Return Values

- ‘T‘ if the curve is closed
- ‘nil‘ otherwise

19.82.7 Side Effects

None.

19.82.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.82.9 Source Notes

- [vlax-curve-isClosed (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/JPN/AutoCAD-AutoLISP-Reference/files/GUID-DC3C19D3-D0EA-4638-` `htm`)

19.82.10 Notes

- Autodesk glosses "closed" here as having the same start point and end point.

19.82.11 Compatibility

Available only on Windows.

19.83 Function Entry: VLAX-CURVE-ISPERIODIC

19.83.1 Name

‘vlax-curve-isPeriodic‘

19.83.2 Class

Visual LISP curve predicate

19.83.3 Syntax

```
(vlax-curve-isPeriodic curve)
```

19.83.4 Arguments and Values

- ‘curve‘ — VLA-object to test.

19.83.5 Description

Determines whether the specified curve has an infinite periodic parameterization.

19.83.6 Return Values

- ‘T‘ if the curve is periodic
- ‘nil‘ otherwise

19.83.7 Notes

- Autodesk defines periodicity here as the existence of a period value ‘dT‘ such that, for any parameter ‘u‘, the point at ‘(u + dT)‘ equals the point at ‘u‘.

19.83.8 Side Effects

None.

19.83.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.83.10 Source Notes

- [vlax-curve-isPeriodic (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-0A082EE2-ACE0-49CE-981D-2665738C7DC2.htm`)

19.83.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.84 Function Entry: VLAX-CURVE-ISPLANAR

19.84.1 Name

‘vlax-curve-isPlanar‘

19.84.2 Class

Visual LISP curve predicate

19.84.3 Syntax

```
(vlax-curve-isPlanar curve)
```

19.84.4 Arguments and Values

- ‘curve‘ — VLA-object to test.

19.84.5 Description

Determines whether there is a plane that contains the curve.

19.84.6 Return Values

- ‘T‘ if the curve is planar
- ‘nil‘ otherwise

19.84.7 Side Effects

None.

19.84.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.84.9 Source Notes

- [vlax-curve-isPlanar (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-798B0177-0B90-4393-AF9D-C1D4607A0E7F.htm`)

19.84.10 Compatibility

Available only on Windows; not available on macOS or Web.

19.85 Dictionary Coverage Index: VLAX Curve Functions

The Autodesk ‘vlax-curve-*‘ family includes at least:

- ‘vlax-curve-getArea‘
- ‘vlax-curve-getClosestPointTo‘
- ‘vlax-curve-getClosestPointToProjection‘
- ‘vlax-curve-getDistAtParam‘
- ‘vlax-curve-getDistAtPoint‘
- ‘vlax-curve-getEndParam‘
- ‘vlax-curve-getEndPoint‘
- ‘vlax-curve-getFirstDeriv‘
- ‘vlax-curve-getParamAtDist‘
- ‘vlax-curve-getParamAtPoint‘
- ‘vlax-curve-getPointAtDist‘
- ‘vlax-curve-getPointAtParam‘
- ‘vlax-curve-getSecondDeriv‘
- ‘vlax-curve-getStartParam‘
- ‘vlax-curve-getStartPoint‘
- ‘vlax-curve-isClosed‘
- ‘vlax-curve-isPeriodic‘

- ‘vlax-curve-isPlanar‘

Source:

- [V Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A617 htm)

19.86 Function Entry: VLAX-CREATE-OBJECT

19.86.1 Name

‘vlax-create-object‘

19.86.2 Class

Visual LISP automation constructor

19.86.3 Syntax

```
(vlax-create-object prog-id)
```

19.86.4 Arguments and Values

- ‘prog-id‘ — string naming an automation class.

  The documented format is:

```
<Vendor>.<Component>.<Version>
```

  For example, ‘AutoCAD.AcCmColor.25‘.

19.86.5 Description

Creates a new instance of an application automation object.

19.86.6 Return Values

Returns a VLA-object on success.

19.86.7 Exceptional Situations

Signals an AutoLISP-visible error if automation support is unavailable or object creation fails.

19.86.8 Notes

- Autodesk documents this as available only in AutoCAD for Windows, not in AutoCAD LT for Windows, nor on macOS or Web.
- This differs from ‘vlax-get-object‘, which retrieves an already-running object.

- If the target application registers itself as a single-instance object, repeated calls may still yield only one instance.

19.86.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.86.10 Source Notes

- [vlax-create-object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-D5A6811A-F110-49DD-A` `htm`)

19.87 Function Entry: VLAX-GET-OBJECT

19.87.1 Name

‘vlax-get-object‘

19.87.2 Class

Visual LISP automation locator

19.87.3 Syntax

```
(vlax-get-object prog-id)
```

19.87.4 Arguments and Values

- ‘prog-id‘ — string naming an automation class.

  The documented format is:

```
appname.objecttype[.version]
```

19.87.5 Description

Returns a running instance of an application automation object.

19.87.6 Return Values

Returns a VLA-object if a matching running object exists; otherwise returns ‘nil‘.

19.87.7 Exceptional Situations

Signals an AutoLISP-visible error if the underlying automation layer fails.

19.87.8 Notes

- Autodesk documents this as available only in AutoCAD for Windows, not in AutoCAD LT for Windows, nor on macOS or Web.
- This is the non-creating counterpart of ‘vlax-create-object‘.

19.87.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.87.10 Source Notes

- [vlax-get-object (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/HUN/AutoCAD-AutoLISP-Reference/files/GUID-A408BF2E-6C7D-453A-A71A` `htm`)

19.88 Function Entry: VLAX-ERASED-P

19.88.1 Name

‘vlax-erased-p‘

19.88.2 Class

Visual LISP automation predicate

19.88.3 Syntax

```
(vlax-erased-p obj)
```

19.88.4 Arguments and Values

- ‘obj‘ — VLA-object.

19.88.5 Description

Determines whether a drawing object represented through the automation layer has been erased.

19.88.6 Return Values

Returns non-‘nil‘ if the object is erased; otherwise returns ‘nil‘.

19.88.7 Exceptional Situations

Signals an AutoLISP-visible error if ‘obj‘ is not a suitable automation object.

19.88.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.88.9 Source Notes

- [vlax-erased-p (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-423F047A-F8B8-4DD0-ABEC-52E3C7
htm)

19.88.10 Compatibility

Available on AutoCAD for Windows only; not available on macOS or Web.

19.89 Function Entry: VLAX-MAP-COLLECTION

19.89.1 Name

‘vlax-map-collection‘

19.89.2 Class

Visual LISP automation iterator

19.89.3 Syntax

```
(vlax-map-collection collection function)
```

19.89.4 Arguments and Values

- ‘collection‘ — automation collection object.
- ‘function‘ — function designator applied to each member.

19.89.5 Description

Applies a function to each element of an automation collection.

19.89.6 Return Values

Autodesk’s documentation specifies traversal semantics but does not state a precise aggregate return-value contract. This specification therefore treats the return value as under-specified.

19.89.7 Exceptional Situations

Signals an AutoLISP-visible error if ‘collection‘ is not an automation collection or if applying ‘function‘ fails.

19.89.8 Notes

- ‘vlax-for‘ is the corresponding special-form iteration facility.
- Autodesk documents this as applying ‘function‘ to every object in the collection.
- The primary observable contract is side-effecting traversal.

19.89.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.89.10 Source Notes

- [About Working With Collection Objects (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-AutoLISP/files/GUID-100F8BDB-2852-4986-BC09-53F77AFCDB96.htm`)
- [V Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A617` `htm`)

19.89.11 Compatibility

Available only on Windows; not available on macOS or Web.

19.90 Function Entry: VLAX-LDATA-GET

19.90.1 Name

‘vlax-ldata-get‘

19.90.2 Class

Visual LISP persistent-data accessor

19.90.3 Syntax

```
(vlax-ldata-get dictionary key [default [private]])
```

19.90.4 Arguments and Values

- ‘dictionary‘ — dictionary or object designator supporting ldata.
- ‘key‘ — string naming the ldata key.
- ‘default‘ — default value returned if the key is absent.
- ‘private‘ — privacy selector in Autodesk’s ldata protocol.

  Autodesk documents:

- ‘dictionary‘ as either a VLA-object or a string naming a global dictionary
- ‘default‘ as one of integer, real, string, list, ename, VLA-object, variant, safearray, ‘T‘, or ‘nil‘
- ‘private‘ as ‘T‘ or ‘nil‘

19.90.5 Description

Retrieves persistent AutoLISP data associated with a dictionary or object.

19.90.6 Return Values

Returns the stored value if present; otherwise returns ‘default‘ if supplied, or ‘nil‘.

19.90.7 Exceptional Situations

Signals an AutoLISP-visible error if the target does not support ldata access.

19.90.8 Notes

- If ‘private‘ is non-‘nil‘ and the call originates from a separate-namespace VLX, Autodesk documents that private ldata is read instead of public ldata.
- If ‘private‘ is supplied, Autodesk requires that ‘default‘ also be supplied, though ‘default‘ may itself be ‘nil‘.
- Autodesk explicitly notes that the same ‘dict‘ and ‘key‘ can hold both private and non-private data.

19.90.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.90.10 Source Notes

- [vlax-ldata-get (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/JPN/AutoCAD-AutoLISP-Reference/files/GUID-C279216F-9568-4176-A9E7` `htm`)

19.90.11 Compatibility

Available on Windows only; not available on macOS or Web.

19.91 Function Entry: VLAX-LDATA-DELETE

19.91.1 Name

‘vlax-ldata-delete‘

19.91.2 Class

Visual LISP persistent-data deletion function

19.91.3 Syntax

```
(vlax-ldata-delete dictionary key [private])
```

19.91.4 Arguments and Values

- ‘dictionary‘ — VLA-object or string naming a global dictionary.
- ‘key‘ — string naming the ldata key.
- ‘private‘ — ‘T‘ or ‘nil‘.

19.91.5 Description

Deletes persistent AutoLISP data associated with a dictionary or object.

19.91.6 Return Values

Returns ‘T‘ if deletion succeeds; otherwise returns ‘nil‘, for example when the data does not exist.

19.91.7 Notes

- If ‘private‘ is non-‘nil‘ and the call originates from a separate-namespace VLX, Autodesk documents that private ldata is deleted.

19.91.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.91.9 Source Notes

- [vlax-ldata-delete (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/JPN/AutoCAD-AutoLISP-Reference/files/GUID-06F6C8B7-A55C-42DE-BD51
htm)

19.91.10 Compatibility

Available on Windows only; not available on macOS or Web.

19.92 Function Entry: VLAX-LDATA-LIST

19.92.1 Name

‘vlax-ldata-list‘

19.92.2 Class

Visual LISP persistent-data enumerator

19.92.3 Syntax

```
(vlax-ldata-list dictionary [private])
```

19.92.4 Arguments and Values

- ‘dictionary‘ — dictionary or object designator supporting ldata.
- ‘private‘ — privacy selector in Autodesk’s ldata protocol.

Autodesk documents:

- ‘dictionary‘ as either a VLA-object or a string naming a global dictionary
- ‘private‘ as ‘T‘ or ‘nil‘

19.92.5 Description

Lists ldata entries associated with a dictionary or object.

19.92.6 Return Values

Returns an association list of ‘(key . value)‘ pairs.

19.92.7 Exceptional Situations

Signals an AutoLISP-visible error if ldata enumeration is not supported for the target.

19.92.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.92.9 Source Notes

- [vlax-ldata-list (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ENU/AutoCAD-AutoLISP-Reference/files/GUID-A124867A-FBB8-4B36-9BBC` `htm`)

19.92.10 Compatibility

Available on Windows only; not available on macOS or Web.

19.93 Function Entry: VLAX-LDATA-PUT

19.93.1 Name

‘vlax-ldata-put‘

19.93.2 Class

Visual LISP persistent-data writer

19.93.3 Syntax

```
(vlax-ldata-put dictionary key value [private])
```

19.93.4 Arguments and Values

- ‘dictionary‘ — dictionary or object designator supporting ldata.
- ‘key‘ — string naming the ldata key.
- ‘value‘ — value to persist.
- ‘private‘ — privacy selector in Autodesk’s ldata protocol.

Autodesk documents ‘value‘ as one of integer, real, string, list, ename, VLA-object, variant, safearray, ‘T‘, or ‘nil‘.

19.93.5 Description

Stores persistent AutoLISP data associated with a dictionary or object.

19.93.6 Return Values

Returns ‘value‘.

19.93.7 Exceptional Situations

Signals an AutoLISP-visible error if the target does not support the requested storage operation.

19.93.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.93.9 Source Notes

- [vlax-ldata-put (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2022/KOR/AutoCAD-AutoLISP-Reference/files/GUID-D2459C22-223C-4C8E-A69C` `htm`)

19.93.10 Notes

- If ‘private‘ is non-‘nil‘ and the call originates from a separate-namespace VLX, Autodesk documents that the data is stored as private to that VLX.

19.93.11 Compatibility

Available on Windows only; not available on macOS or Web.

19.94 Function Entry: VLAX-LDATA-TEST

19.94.1 Name

‘vlax-ldata-test‘

19.94.2 Class

Visual LISP persistent-data predicate

19.94.3 Syntax

```
(vlax-ldata-test value)
```

19.94.4 Arguments and Values

- ‘value‘ — any AutoLISP value.

Autodesk specifically documents the storable categories as integer, real, string, list, ename, VLA-object, variant, safearray, ‘T‘, or ‘nil‘.

19.94.5 Description

Determines whether a value can be stored as ldata.

19.94.6 Return Values

Returns non-‘nil‘ if the value is admissible for ldata persistence; otherwise returns ‘nil‘.

19.94.7 Exceptional Situations

Signals an AutoLISP-visible error if the implementation cannot determine whether the value is storable.

19.94.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.94.9 Source Notes

- [vlax-ldata-test (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2025/ENU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-CD72D160-25BF-43F5-htm`)

19.94.10 Compatibility

Available on Windows only; not available on macOS or Web.

19.95 Function Entry: VLAX-MACHINE-PRODUCT-KEY

19.95.1 Name

‘vlax-machine-product-key‘

19.95.2 Class

Visual LISP Windows registry helper

19.95.3 Syntax

```
(vlax-machine-product-key)
```

19.95.4 Description

Returns the Windows registry path for the AutoCAD product in the machine-wide registry hive.

19.95.5 Return Values

Returns a string.

19.95.6 Notes

- Autodesk documents this as returning the HKLM registry path.
- Autodesk documents that this function replaces ‘vlax-product-key‘, which was used in AutoCAD 2012 and earlier.
- Autodesk also documents that use requires a prior call to ‘vl-load-com‘.

19.95.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.95.8 Source Notes

- [vlax-machine-product-key (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-2EB68815-FDF9-4235-918B-025FB9A42220.htm`)

- [Windows Registry and Property List File Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E3D3DA29-AF63-41D3-865C-39E50FF31596.htm`)

19.95.9 Compatibility

Available on AutoCAD for Windows only; not available in AutoCAD LT for Windows.

19.96 Function Entry: VLAX-PRODUCT-KEY

19.96.1 Name

‘vlax-product-key‘

19.96.2 Class

Visual LISP Windows registry helper

19.96.3 Syntax

```
(vlax-product-key)
```

19.96.4 Description

Returns the Windows registry path for the current AutoCAD product.

19.96.5 Return Values

Returns a string.

19.96.6 Notes

- Autodesk documents this function as obsolete and recommends ‘vlax-machine-product-key‘ instead.
- Autodesk documents that the returned path can be used to register an application for demand loading.

19.96.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.96.8 Source Notes

- [vlax-product-key (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/JPN/AutoCAD-AutoLISP-Reference/files/GUID-237CEDFA-A02E-443E-BB21
htm)
- [Windows Registry and Property List File Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-E3D3DA29-AF63-41D3-865C-39E50FF31596.htm)

19.96.9 Compatibility

Available on Windows only; not available on macOS or Web.

19.97 Function Entry: VLAX-USER-PRODUCT-KEY

19.97.1 Name

‘vlax-user-product-key‘

19.97.2 Class

Visual LISP Windows registry helper

19.97.3 Syntax

```
(vlax-user-product-key)
```

19.97.4 Description

Returns the Windows registry path for the current AutoCAD product in ‘HKCU‘.

19.97.5 Return Values

Returns a string.

19.97.6 Notes

- This is the user-hive counterpart of ‘vlax-machine-product-key‘.
- Autodesk documents the returned path as usable for demand-loading registration.

19.97.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.97.8 Source Notes

- [Windows Registry and Property List File Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/ files/GUID-E3D3DA29-AF63-41D3-865C-39E50FF31596.htm)
- [V Functions Reference (AutoLISP)](https://help.autodesk.com/ cloudhelp/2025/JPN/AutoCAD-LT-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A htm)

19.97.9 Compatibility

Available on Windows only; not available on macOS or Web.

19.98 Function Entry: VLAX-REMOVE-CMD

19.98.1 Name

‘vlax-remove-cmd‘

19.98.2 Class

Visual LISP command-registration function

19.98.3 Syntax

```
(vlax-remove-cmd global-name)
```

19.98.4 Arguments and Values

- ‘global-name‘ — string naming a command, or ‘T‘.

19.98.5 Description

Removes a command previously registered through the ‘vlax‘ command-registration interface.

19.98.6 Return Values

Returns ‘T‘ if successful; otherwise returns ‘nil‘.

19.98.7 Exceptional Situations

No error is documented for an undefined command; Autodesk documents ‘nil‘ in that case.

19.98.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

19.98.9 Source Notes

- [vlax-remove-cmd (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-LT-AutoLISP-Reference/files/GUID-EA671EF1-4EF9-49FE-9` `htm`)

19.98.10 Notes

- If 'global-name' is 'T', Autodesk documents that the whole command group 'VLC-AppName' is removed.
- Removal applies only to the current AutoCAD session.

19.99 Dictionary Coverage Index: Additional VLAX Utility Functions

This specification currently covers the following additional low-level 'vlax' names:

- 'vlax-create-object'
- 'vlax-erased-p'
- 'vlax-get-object'
- 'vlax-ldata-delete'
- 'vlax-ldata-get'
- 'vlax-ldata-list'
- 'vlax-ldata-put'
- 'vlax-ldata-test'
- 'vlax-machine-product-key'
- 'vlax-map-collection'
- 'vlax-product-key'
- 'vlax-remove-cmd'
- 'vlax-user-product-key'

Source:

- [V Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A617
htm)
- [Drawing Object Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2016/FRA/AutoCAD-MAC-Core/files/GUID-DF1CB039-265D-4D17-A86E-EF966
htm)

20 20 ActiveX, COM, and Automation

20.1 Overview

This chapter covers ‘vlax-*‘, ‘vla-*‘, and related automation facilities.

20.2 Normative Rules

- Autodesk treats full ActiveX support as Windows-oriented
- BricsCAD extends COM-like compatibility beyond Windows in some versions
- ‘clautolisp‘ may emulate Windows-like COM behavior on non-Windows hosts through an explicit compatibility layer
- for ‘vla-*‘ wrappers, the AutoCAD ActiveX object model documentation is normative in addition to the AutoLISP wrapper documentation
- for ‘vlax-*‘ property, method, variant, safearray, and COM-object functions, COM / OLE Automation semantics are normative insofar as Autodesk exposes them through AutoLISP
- where the AutoLISP wrapper constrains or renames the underlying API, the AutoLISP-layer contract takes precedence for call shape and error surface

20.3 Source Notes

- [vlax-import-type-library (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHS/AutoCAD-LT-AutoLISP-Reference/files/GUID-1699B6A0-C4A6-4BC4-8` `htm`)
- [BricsCAD What’s New](`https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/WhatsNew.htm`)

20.4 Function Entry: VLAX-TMATRIX

20.4.1 Name

‘vlax-tmatrix‘

20.4.2 Class

ActiveX Function

20.4.3 Syntax

```
(vlax-tmatrix lst)
```

20.4.4 Arguments and Values

- ‘lst‘: a list of four sublists, each containing four numbers; the rows of a 4_4 transformation matrix.

20.4.5 Description

Converts a nested list into a variant suitable for VLA methods that take a 4_4 transformation matrix (e.g. ‘vla-transformby‘).

20.4.6 Return Values

A variant of type safearray representing the 4_4 matrix.

20.4.7 Side Effects

None.

20.4.8 Examples

- Build a matrix:

```
(setq tmatrix (vlax-tmatrix '((1 1 1 0) (1 2 3 0) (2 3 4 5) (2 9 8 3))))
```

- Inspect the result:

```
(vlax-safearray->list (vlax-variant-value tmatrix))
```

20.4.9 Availability

- AutoCAD 2026: documented (Windows only — ActiveX surface).
- BricsCAD V26: presumed compatible (BricsCAD ActiveX layer is Windows-centric).
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

20.4.10 Source Notes

- [vlax-tmatrix (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-7B1BBCDD-69DD-40E6-81CB-370A05725` `htm`)
- Status: documented (Phase 3 body fill).

20.5 Function Entry: VLAX-TYPEINFO-AVAILABLE-P

20.5.1 Name

‘vlax-typeinfo-available-p‘

20.5.2 Class

ActiveX Predicate

20.5.3 Syntax

```
(vlax-typeinfo-available-p obj)
```

20.5.4 Arguments and Values

- ‘obj‘: a VLA-object.

20.5.5 Description

Determines whether TypeLib information is present for ‘obj‘. AutoLISP requires TypeLib information to determine whether a method or property is available on an object; some objects (notably ‘AcadDocument‘) may lack it.

20.5.6 Return Values

Returns ‘T‘ if TypeLib information is available; otherwise ‘nil‘.

20.5.7 Side Effects

None.

20.5.8 Examples

Not documented on the vendor page.

20.5.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: documented (confirmed via the live TOC walk on 2026-04-26; entry ‘vlax-typeinfo-available-p‘ appears in the BricsCAD V26 Standard Functions section).

- Status: documented in both vendors.

20.5.10 See Also

- BricsCAD V26 confirmation came from the live-DOM TOC walk recorded in ‘vendor-inventory-2026.org‘ §3.2.

20.5.11 Source Notes

- [vlax-typeinfo-available-p (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1DAB4433-A0FA-43EA-htm`)

- Status: documented (Phase 3 body fill).

21 21 Reactors and Event Integration

21.1 Overview

Reactors are part of the Visual LISP event model.

A reactor is a host-side event-callback subscription object. The host calls the registered callback automatically whenever a documented event occurs (entity modified, document opened, command executed, sysvar changed, mouse-down, ...). Reactors are the AutoLISP equivalent of ”event listeners” in GUI frameworks and the ”observer pattern” in OOP, layered over a fixed reactor **type** taxonomy that the vendor reference ties to specific event classes.

21.2 Normative Rules

- Reactor support belongs to the Visual LISP extension layer.
- Autodesk documents reactor support as requiring ‘(vl-load-com)‘ before reactor functions are used.
- Autodesk documents reactor support as available only on Windows.
- Reactor constructors accept application data and callback bindings; object reactors additionally accept an owner list.
- Reactors are host-event-driven objects; callback execution order is therefore determined by host events rather than ordinary expression sequencing.

21.3 Normative Rules: Host Object Ontology and Lifecycles

21.3.1 Statement

The reactor catalogue is interpretable only against the host's **object ontology**. Each reactor type subscribes to lifecycle transitions of one specific class of host object; the published reaction names are systematic transition labels in those classes' finite-state machines.

This section pins down: the host object graph, the lifecycle of each class, the canonical event vocabulary, and the scope rule (per-document vs per-application).

21.3.2 Host Object Graph

The graph is small and stable; it has been the same shape since AutoCAD R14's Visual LISP merger.

```
Application                        (one per process)
+--__ DocumentManager                (the documents collection)
_    ___ Document  (_N)            (one per open drawing)
_         +--__ ModelSpace, PaperSpace, Blocks, Layers, Linetypes,
_         _   TextStyles, DimStyles, Views, UCSs, Viewports,
_         _   RegisteredApplications, NamedObjectsDictionary,
_         _   SystemVariables (per-document), ActiveSelectionSet,
_         _   Picksets, Entities, PendingCommand, UndoStack
+--__ Preferences, ARXModules, LispEnvironment,
_   ApplicationReactors,
___ PromptHistory / CommandLine
```

21.3.3 Per-Class Lifecycles

Class	States
`application`	`:running` → `:quitting` → `:quit`
`document-manager`	always `:running`
`document`	`:loading` → `:loaded` _ `:active` _ `:inactive` → `:closing` → `:closed` (tran
`command`	`:pending` → `:started` → `:running` → (`:ended` _ `:cancelled` _ `:failed`)
`entity`	`:created` → `:committed` _ `:modified` → (`:erased` _ `:unerased`) → `:purgec`
`selection-set`	`:created` → `:populated` _ `:mutated` → `:released`
`dictionary-entry`	`:added` → `:present` → (`:renamed` _ `:mutated`) → `:removed`
`sysvar`	always `:bound`; transitions `value-will-change` / `value-changed`
`xref`	`:unresolved` → `:resolved` _ `:unloaded` _ `:reloaded` → `:detached`
`wblock-op`	`:started` → (`:completed` _ `:cancelled`)
`undo-op`	`:begin` → (`:commit` _ `:rollback`)
`mouse-event`	`:down` → (`:drag*`) → `:up`
`vlx-app`	`:loading` → `:loaded` → `:unloaded`
`lisp-eval`	`:loading-source` → `:running` → `:idle`
`reactor`	`:armed` _ `:disabled` → `:removed`

21.3.4 Event-Naming Convention

Reactor reactions are systematic names for transitions in the table above. The convention has two halves:

- A `*WillChange` / `*ToBeFooed` reaction fires **before** the transition; the world is still in the old state. The callback may sometimes veto the transition.
- A `*Changed` / `*BecameFoo` / `*Fooed` reaction fires **after** the transition; the world is in the new state.

Most documented transitions emit a **pair** of reactions; this is why the vendor reactor reactions look so verbose.

21.3.5 Event Scope Rule

Every event has a single canonical **scope** — application or document — and a single canonical **reactor type**:

Event	Scope	Reactor type
`:vlr-documentCreated`	application	`vlr-docmanager-reactor`
`:vlr-documentToBeDestroyed`	application	`vlr-docmanager-reactor`
`:vlr-beginDocumentClose`	document	`vlr-document-reactor`
`:vlr-objectModified`	document	`vlr-acdb-reactor` / `vlr-object-reactor`
`:vlr-sysVarChanged`	document	`vlr-sysvar-reactor`
`:vlr-commandWillStart`	document	`vlr-command-reactor`
`:vlr-beginQuit`	application	`vlr-editor-reactor`
`:vlr-beginARXLoad`	application	`vlr-linker-reactor`

Some events have **both** an application-scope and a document-scope reactor type subscribed to the same transition. Document-activation is the canonical case: `vlr-docmanager-reactor` subscribers see every document's `:vlr-documentBecameCurrent`; `vlr-document-reactor` subscribers on document X see only X's. The runtime fires both.

21.3.6 Vetoing

A few `*WillChange` events let the callback veto the transition. clautolisp models this through the callback's **return value**:

- A non-nil return permits the transition.
- The keyword `:veto` rejects it; the runtime aborts the transition and signals `:transition-vetoed` to the caller that initiated it.

Only events documented as vetoable by the vendor honour this: `:vlr-documentLockModeWillChang` `:vlr-aboutToBeDeactivated`, and the `*WillStart` family of command reactions.

21.3.7 Persistence

A persistent reactor (registered through `vlr-pers`) is stored in the **document's** named-object dictionary, traditionally under the key `ACAD_REACTORS`. Its callback is referenced by **symbol name**, not by closure: closures cannot survive serialisation, but symbol names re-resolve when the document is opened.

On document open the host walks `ACAD_REACTORS` and reconstructs each persistent reactor **before** user code executes. If the callback's symbol is undefined at fire time, dispatch records the failure and continues; the reactor is not removed.

21.3.8 Vendor Evidence

- Autodesk reference for the Visual LISP reactor family — the per-class reaction tables in `AutoCAD-AutoLISP-Reference/files/GUID-...` pages — define the reactor types, the reaction names per type, and the per-event argument shape.
- Bricsys `Lisp - BricsCAD` help page documents the same taxonomy with no functional divergence on per-document vs per-application scope.
- The reactor data model (callback by symbol name, persistence in `ACAD_REACTORS`) is documented in the AutoCAD ObjectARX guide and surfaced through Visual LISP's `vlr-pers` family.

21.3.9 Strict / Lax Policy

The ontology is **strict** in every supported dialect; the only behaviour that varies between hosts is the set of reactor types recognised. The clautolisp `strict`, `autocad-2026`, and `bricscad-v26` dialects accept the union of reactor types documented by either vendor.

21.3.10 Cross-references

- Reactor type entries (this chapter) — each lists the reactions it subscribes to.
- Function Entry: `vlr-pers`, `vlr-pers-release`, `vlr-pers-list` — the persistence interface.
- Function Entry: `vlr-set-notification` — the notification-mode toggle (`:all-documents` / `:current-document-only`).

21.4 Type Entry: Reactor Object

21.4.1 Name

Reactor Object

21.4.2 Class

Runtime class / Visual LISP extension object

21.4.3 Description

Object representing a registered event source and its associated callback set.

21.4.4 Valid Operations

- construction by ‘vlr-*reactor‘ constructor functions
- enabling and disabling with ‘vlr-add‘ and ‘vlr-remove‘
- interrogation via ‘vlr-type‘, ‘vlr-data‘, ‘vlr-reactions‘, and related accessors
- persistence-related operations in implementations that expose persistent reactors

21.4.5 Compatibility

Part of the Visual LISP extension layer and available only on Windows in Autodesk’s platform model.

21.4.6 Notes

- Autodesk distinguishes transient reactors and persistent reactors.
- Autodesk documents transient reactors as not surviving drawing closure.
- Autodesk documents persistent reactors as being stored with the drawing.

21.4.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.4.8 Source Notes

- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)
- [About Transient and Persistent Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/ESP/AutoCAD-AutoLISP/files/GUID-0A632CD0-99B8-4E7E-A958-A0723E50ABB5.htm`)
- Status: documented.

21.5 Function Entry: VLR-OBJECT-REACTOR

21.5.1 Name

‘vlr-object-reactor‘

21.5.2 Class

Reactor constructor function

21.5.3 Syntax

```
(vlr-object-reactor owners data callbacks)
```

21.5.4 Description

Constructs a drawing object reactor.

21.5.5 Return Values

Reactor object.

21.5.6 Side Effects

Registers reactor state with the host event system.

21.5.7 Arguments and Values

- ‘owners‘: list of VLA-objects monitored by the reactor
- ‘data‘: application-specific AutoLISP data
- ‘callbacks‘: list of ‘(event-name . callback-function)‘ pairs

21.5.8 Notes

- Autodesk documents object reactors as associated with specific owner objects.
- Autodesk documents that callback specifications are event/callback pairs.
- Autodesk cautions that callbacks should not modify the objects that own the object reactor.

21.5.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.5.10 Source Notes

- [About Using Object Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/CHT/AutoCAD-AutoLISP/files/GUID-373FCF92-4E98-448A-9` `htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)
- Status: documented.

21.6 Function Entry: VLR-COMMAND-REACTOR

21.6.1 Name

‘vlr-command-reactor‘

21.6.2 Class

Reactor constructor function

21.6.3 Syntax

`(vlr-command-reactor data callbacks)`

21.6.4 Description

Constructs an editor reactor that notifies of command events.

21.6.5 Return Values

Reactor object.

21.6.6 Side Effects

Registers command-event callbacks with the host event system.

21.6.7 Arguments and Values

- ‘data‘: application-specific AutoLISP data
- ‘callbacks‘: list of ‘(event-name . callback-function)‘ pairs

21.6.8 Notes

- Autodesk documents non-object reactor constructors as taking application data plus event/callback bindings.
- The accepted event names depend on the reactor class and are part of Autodesk’s reactor-type tables.

21.6.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.6.10 Source Notes

- [About Creating Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ITA/AutoCAD-AutoLISP/files/GUID-396A2538-4C9E-4A12-8174-289A1` `htm`)
- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- Status: documented.

21.7 Function Entry: VLR-ADD

21.7.1 Name

‘vlr-add‘

21.7.2 Class

Reactor control function

21.7.3 Syntax

`(vlr-add reactor)`

21.7.4 Description

Enables a disabled reactor object.

21.7.5 Return Values

Reactor object or host-defined success value.

21.7.6 Side Effects

Enables event delivery for the reactor.

21.7.7 Notes

- Autodesk documents ‘vlr-add‘ as re-enabling a previously removed reactor.

21.7.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.7.9 Source Notes

- [About Removing Reactors (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2018/CHS/AutoCAD-AutoLISP/files/GUID-DC1713AC-D79F-456B-87E8-16561 htm)
- [Reactor Functions Reference (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm)
- Status: documented.

21.8 Function Entry: VLR-REMOVE

21.8.1 Name

‘vlr-remove‘

21.8.2 Class

Reactor control function

21.8.3 Syntax

```
(vlr-remove reactor)
```

21.8.4 Description

Disables a reactor.

21.8.5 Return Values

Reactor object or host-defined success value.

21.8.6 Side Effects

Disables event delivery for the reactor.

21.8.7 Notes

- Autodesk documents that removed reactors are disabled, not destroyed.
- Autodesk documents that a removed reactor may later be re-enabled with ‘vlr-add‘.

21.8.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.8.9 Source Notes

- [vlr-remove (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2025/ENU/AutoCAD-AutoLISP-Reference/files/GUID-2580A278-39B4-4BBA-99B4-20DA9565D htm)
- [About Removing Reactors (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2018/CHS/AutoCAD-AutoLISP/files/GUID-DC1713AC-D79F-456B-87E8-16561 htm)
- Status: documented.

21.9 Function Entry: VLR-REMOVE-ALL

21.9.1 Name

‘vlr-remove-all‘

21.9.2 Class

Reactor control function

21.9.3 Syntax

```
(vlr-remove-all reactor-type)
```

21.9.4 Description

Disables all reactors of the specified type.

21.9.5 Return Values

List of affected reactors or other host-defined result.

21.9.6 Side Effects

Bulk reactor deactivation.

21.9.7 Arguments and Values

- ‘reactor-type‘: optional reactor type symbol; if omitted, all active reactor types are targeted

21.9.8 Notes

- Autodesk documents the return value as a list of reactor lists.
- Autodesk documents that each reactor list begins with the reactor type symbol and is followed by the disabled reactor objects.
- Autodesk documents that the result is ‘nil‘ when no active reactors match.

21.9.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.9.10 Source Notes

- [vlr-remove-all (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-3F8E47F1-4B6C-4C59-99C1-860195 htm`)
- [About Removing Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2018/CHS/AutoCAD-AutoLISP/files/GUID-DC1713AC-D79F-456B-87E8-16561 htm`)
- Status: documented.

21.10 Function Entry: VLR-ACDB-REACTOR

21.10.1 Name

‘vlr-acdb-reactor‘

21.10.2 Class

Reactor constructor function

21.10.3 Syntax

`(vlr-acdb-reactor data callbacks)`

21.10.4 Description

Constructs a reactor object that notifies when an object is added to, modified in, or erased from a drawing database.

21.10.5 Return Values

Reactor object.

21.10.6 Side Effects

Registers database-event callbacks with the host event system.

21.10.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.10.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.11 Function Entry: VLR-DOCMANAGER-REACTOR

21.11.1 Name

‘vlr-docmanager-reactor‘

21.11.2 Class

Reactor constructor function

21.11.3 Syntax

`(vlr-docmanager-reactor data callbacks)`

21.11.4 Description

Constructs a reactor object that notifies of events relating to drawing documents.

21.11.5 Return Values

Reactor object.

21.11.6 Side Effects

Registers document-management callbacks with the host event system.

21.11.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.11.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.12 Function Entry: VLR-DEEPCLONE-REACTOR

21.12.1 Name

‘vlr-deepclone-reactor‘

21.12.2 Class

Reactor constructor function

21.12.3 Syntax

`(vlr-deepclone-reactor data callbacks)`

21.12.4 Description

Constructs an editor reactor object that notifies of a deep clone event.

21.12.5 Return Values

Reactor object.

21.12.6 Side Effects

Registers deep-clone callbacks with the host event system.

21.12.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.12.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.13 Function Entry: VLR-DWG-REACTOR

21.13.1 Name

‘vlr-dwg-reactor‘

21.13.2 Class

Reactor constructor function

21.13.3 Syntax

```
(vlr-dwg-reactor data callbacks)
```

21.13.4 Description

Constructs an editor reactor object that notifies of drawing events, such as opening or closing a drawing file.

21.13.5 Return Values

Reactor object.

21.13.6 Side Effects

Registers drawing-event callbacks with the host event system.

21.13.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.13.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.14 Function Entry: VLR-DXF-REACTOR

21.14.1 Name

‘vlr-dxf-reactor‘

21.14.2 Class

Reactor constructor function

21.14.3 Syntax

```
(vlr-dxf-reactor data callbacks)
```

21.14.4 Description

Constructs an editor reactor object that notifies of events related to reading or writing a DXF file.

21.14.5 Return Values

Reactor object.

21.14.6 Side Effects

Registers DXF I/O callbacks with the host event system.

21.14.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.14.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.15 Function Entry: VLR-EDITOR-REACTOR

21.15.1 Name

‘vlr-editor-reactor‘

21.15.2 Class

Reactor constructor function

21.15.3 Syntax

`(vlr-editor-reactor data callbacks)`

21.15.4 Description

Constructs a generic editor reactor object.

21.15.5 Return Values

Reactor object.

21.15.6 Side Effects

Registers editor callbacks with the host event system.

21.15.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.15.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.16 Function Entry: VLR-INSERT-REACTOR

21.16.1 Name

‘vlr-insert-reactor‘

21.16.2 Class

Reactor constructor function

21.16.3 Syntax

`(vlr-insert-reactor data callbacks)`

21.16.4 Description

Constructs an editor reactor object that notifies of events related to block insertion.

21.16.5 Return Values

Reactor object.

21.16.6 Side Effects

Registers insert-event callbacks with the host event system.

21.16.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.16.8 Source Notes

- [About Querying and Modifying Reactor Data (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.17 Function Entry: VLR-LINKER-REACTOR

21.17.1 Name

‘vlr-linker-reactor‘

21.17.2 Class

Reactor constructor function

21.17.3 Syntax

```
(vlr-linker-reactor data callbacks)
```

21.17.4 Description

Constructs a reactor object that notifies when an ObjectARX application is loaded or unloaded.

21.17.5 Return Values

Reactor object.

21.17.6 Side Effects

Registers linker/application-load callbacks with the host event system.

21.17.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.17.8 Source Notes

- [About Querying and Modifying Reactor Data (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- [vlr-data-set (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-84D67D6D-4F95-48CF-AF59-795C386CD htm`)

- Status: documented.

21.18 Function Entry: VLR-LISP-REACTOR

21.18.1 Name

‘vlr-lisp-reactor‘

21.18.2 Class

Reactor constructor function

21.18.3 Syntax

```
(vlr-lisp-reactor data callbacks)
```

21.18.4 Description

Constructs an editor reactor object that notifies of a Lisp event.

21.18.5 Return Values

Reactor object.

21.18.6 Side Effects

Registers Lisp-event callbacks with the host event system.

21.18.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.18.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm)
- [Reactor Functions Reference (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm)

- Status: documented.

21.19 Function Entry: VLR-MISCELLANEOUS-REACTOR

21.19.1 Name

'vlr-miscellaneous-reactor'

21.19.2 Class

Reactor constructor function

21.19.3 Syntax

`(vlr-miscellaneous-reactor data callbacks)`

21.19.4 Description

Constructs an editor reactor object for events that do not fall under the other editor-reactor categories.

21.19.5 Return Values

Reactor object.

21.19.6 Side Effects

Registers miscellaneous editor callbacks with the host event system.

21.19.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.19.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.20 Function Entry: VLR-MOUSE-REACTOR

21.20.1 Name

‘vlr-mouse-reactor‘

21.20.2 Class

Reactor constructor function

21.20.3 Syntax

```
(vlr-mouse-reactor data callbacks)
```

21.20.4 Description

Constructs an editor reactor object that notifies of mouse events, such as a double-click.

21.20.5 Return Values

Reactor object.

21.20.6 Side Effects

Registers mouse-event callbacks with the host event system.

21.20.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.20.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.21 Function Entry: VLR-SYSVAR-REACTOR

21.21.1 Name

‘vlr-sysvar-reactor‘

21.21.2 Class

Reactor constructor function

21.21.3 Syntax

```
(vlr-sysvar-reactor data callbacks)
```

21.21.4 Description

Constructs an editor reactor object that notifies of a change to a system variable.

21.21.5 Return Values

Reactor object.

21.21.6 Side Effects

Registers system-variable change callbacks with the host event system.

21.21.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.21.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.22 Function Entry: VLR-UNDO-REACTOR

21.22.1 Name

‘vlr-undo-reactor‘

21.22.2 Class

Reactor constructor function

21.22.3 Syntax

`(vlr-undo-reactor data callbacks)`

21.22.4 Description

Constructs an editor reactor object that notifies of an undo event.

21.22.5 Return Values

Reactor object.

21.22.6 Side Effects

Registers undo callbacks with the host event system.

21.22.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.22.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.23 Function Entry: VLR-WBLOCK-REACTOR

21.23.1 Name

‘vlr-wblock-reactor‘

21.23.2 Class

Reactor constructor function

21.23.3 Syntax

```
(vlr-wblock-reactor data callbacks)
```

21.23.4 Description

Constructs an editor reactor object that notifies of events related to writing a block.

21.23.5 Return Values

Reactor object.

21.23.6 Side Effects

Registers block-write callbacks with the host event system.

21.23.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.23.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.24 Function Entry: VLR-WINDOW-REACTOR

21.24.1 Name

‘vlr-window-reactor‘

21.24.2 Class

Reactor constructor function

21.24.3 Syntax

```
(vlr-window-reactor data callbacks)
```

21.24.4 Description

Constructs an editor reactor object that notifies of events related to moving or sizing an AutoCAD window.

21.24.5 Return Values

Reactor object.

21.24.6 Side Effects

Registers window-event callbacks with the host event system.

21.24.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.24.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.25 Function Entry: VLR-XREF-REACTOR

21.25.1 Name

‘vlr-xref-reactor‘

21.25.2 Class

Reactor constructor function

21.25.3 Syntax

`(vlr-xref-reactor data callbacks)`

21.25.4 Description

Constructs an editor reactor object that notifies of events related to attaching or modifying XREFs.

21.25.5 Return Values

Reactor object.

21.25.6 Side Effects

Registers XREF-related callbacks with the host event system.

21.25.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.25.8 Source Notes

- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.26 Function Entry: VLR-ADDED-P

21.26.1 Name

‘vlr-added-p‘

21.26.2 Class

Reactor predicate

21.26.3 Syntax

```
(vlr-added-p reactor)
```

21.26.4 Description

Tests whether a reactor object is enabled.

21.26.5 Return Values

- ‘T‘ if the reactor is enabled
- ‘nil‘ otherwise

21.26.6 Side Effects

None.

21.26.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.26.8 Source Notes

- [vlr-current-reaction-name (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-462F86A0-1452-4F17-htm)

- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- Status: documented.

21.27 Function Entry: VLR-CURRENT-REACTION-NAME

21.27.1 Name

‘vlr-current-reaction-name‘

21.27.2 Class

Reactor introspection function

21.27.3 Syntax

```
(vlr-current-reaction-name)
```

21.27.4 Description

Returns the name of the current event when called from within a reactor callback.

21.27.5 Return Values

Symbol naming the current reaction, or ‘nil‘ outside a callback context.

21.27.6 Side Effects

None.

21.27.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.27.8 Source Notes

- [vlr-data (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2023/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3DEDE2D3-7A27-4C51-A47D-C2A6661C1` `htm`)
- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)

- Status: documented.

21.28 Function Entry: VLR-DATA

21.28.1 Name

‘vlr-data‘

21.28.2 Class

Reactor accessor function

21.28.3 Syntax

`(vlr-data reactor)`

21.28.4 Description

Returns the application-specific data associated with a reactor.

21.28.5 Return Values

Application-defined data object.

21.28.6 Side Effects

None.

21.28.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.28.8 Source Notes

- [vlr-reaction-name (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-42AFF892-3DB5-4924-B2EC htm`)
- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)

- Status: documented.

21.29 Function Entry: VLR-DATA-SET

21.29.1 Name

‘vlr-data-set‘

21.29.2 Class

Reactor mutator function

21.29.3 Syntax

`(vlr-data-set reactor data)`

21.29.4 Description

Overwrites the application-specific data associated with a reactor.

21.29.5 Return Values

Assigned data or another host-defined success value.

21.29.6 Side Effects

Mutates reactor-associated application data.

21.29.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.29.8 Source Notes

- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.30 Function Entry: VLR-REACTION-NAME

21.30.1 Name

‘vlr-reaction-name‘

21.30.2 Class

Reactor introspection function

21.30.3 Syntax

```
(vlr-reaction-name reactor)
```

21.30.4 Description

Returns a list of all possible callback conditions for the reactor type.

21.30.5 Return Values

List of reaction/event names.

21.30.6 Side Effects

None.

21.30.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.30.8 Source Notes

- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.31 Function Entry: VLR-REACTION-SET

21.31.1 Name

‘vlr-reaction-set‘

21.31.2 Class

Reactor mutator function

21.31.3 Syntax

```
(vlr-reaction-set reactor event-name callback)
```

21.31.4 Description

Adds or replaces a callback function in a reactor.

21.31.5 Return Values

Host-defined success value.

21.31.6 Side Effects

Mutates the callback set associated with the reactor.

21.31.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.31.8 Source Notes

- [vlr-reactors (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-10C29F2A-8DF9-4511-8A5C-7DA92A4DA htm)
- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm)

- Status: documented.

21.32 Function Entry: VLR-REACTIONS

21.32.1 Name

‘vlr-reactions‘

21.32.2 Class

Reactor accessor function

21.32.3 Syntax

```
(vlr-reactions reactor)
```

21.32.4 Description

Returns a list of ‘(event-name . callback-function)‘ pairs for the reactor.

21.32.5 Return Values

Association list of event/callback pairs.

21.32.6 Side Effects

None.

21.32.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.32.8 Source Notes

- [vlr-type (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6CFDF8C3-0C66-41F1-9254-6D112AFEE htm)
- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm)

- Status: documented.

21.33 Function Entry: VLR-REACTORS

21.33.1 Name

‘vlr-reactors‘

21.33.2 Class

Reactor accessor function

21.33.3 Syntax

```
(vlr-reactors [reactor-type])
```

21.33.4 Description

Returns a list of existing reactors, optionally filtered by type.

21.33.5 Return Values

List of reactor objects.

21.33.6 Side Effects

None.

21.33.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.33.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.34 Function Entry: VLR-TYPE

21.34.1 Name

‘vlr-type‘

21.34.2 Class

Reactor accessor function

21.34.3 Syntax

`(vlr-type reactor)`

21.34.4 Description

Returns a symbol representing the reactor type.

21.34.5 Return Values

Type symbol.

21.34.6 Side Effects

None.

21.34.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.34.8 Source Notes

- [vlr-type (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6CFDF8C3-0C66-41F1-9254-6D112AFEE` `htm`)
- [About Querying and Modifying Reactors (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/FRA/AutoCAD-AutoLISP/files/GUID-8A2E5A06-9479-4E9D-B99A-9D4DC79BD77E.htm`)

- Status: documented.

21.35 Function Entry: VLR-TYPES

21.35.1 Name

‘vlr-types‘

21.35.2 Class

Reactor accessor function

21.35.3 Syntax

`(vlr-types)`

21.35.4 Description

Returns a list of all reactor types.

21.35.5 Return Values

List of type symbols.

21.35.6 Side Effects

None.

21.35.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.35.8 Source Notes

- [About Reactor Types and Events (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2024/CHS/AutoCAD-AutoLISP/files/GUID-EC61BEC2-39B9-4CD1-BE3B-7781DF1E3530.htm`)
- [Reactor Functions Reference (AutoLISP/ActiveX)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-LT-AutoLISP-Reference/files/GUID-B83B512E-CEF6-43C5-9099-398999E254AF.htm`)

- Status: documented.

21.36 Dictionary Coverage Index: Reactor Functions

The Autodesk reactor family includes at least:

- ‘vlr-acdb-reactor‘
- ‘vlr-add‘
- ‘vlr-added-p‘
- ‘vlr-command-reactor‘
- ‘vlr-current-reaction-name‘
- ‘vlr-data‘
- ‘vlr-data-set‘
- ‘vlr-docmanager-reactor‘
- ‘vlr-dwg-reactor‘
- ‘vlr-editor-reactor‘
- ‘vlr-lisp-reactor‘
- ‘vlr-object-reactor‘
- ‘vlr-reaction-name‘
- ‘vlr-reaction-set‘
- ‘vlr-reactions‘
- ‘vlr-reactors‘
- ‘vlr-remove‘
- ‘vlr-remove-all‘
- ‘vlr-sysvar-reactor‘
- ‘vlr-type‘
- ‘vlr-types‘

Source:

- [V Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A617`
`htm`)

21.37 Source Notes

- Visual LISP function-family documentation and product overviews
- Status: acknowledged, not yet fully specified.

21.38 Function Entry: VLR-BEEP-REACTION

21.38.1 Name

‘vlr-beep-reaction‘

21.38.2 Class

Reactor Callback Function

21.38.3 Syntax

```
(vlr-beep-reaction [args])
```

21.38.4 Arguments and Values

- ‘args‘: ignored. The function accepts whatever arguments the reactor framework passes to a callback.

21.38.5 Description

A predefined callback function that produces a beep sound. Useful as a placeholder during reactor development to confirm that a callback is being invoked.

21.38.6 Return Values

Returns ‘nil‘.

21.38.7 Side Effects

Produces a system beep.

21.38.8 Examples

- ‘(vlr-mouse-reactor nil ’((:vlr-beginDoubleClick . vlr-beep-reaction)))‘ installs an audible beep on the begin-double-click event.

21.38.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.38.10 Source Notes

- [vlr-beep-reaction (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-47765811-714A-455C-87FF-806B21483` `htm`)
- Status: documented (Phase 3 body fill).

21.39 Function Entry: VLR-TRACE-REACTION

21.39.1 Name

‘vlr-trace-reaction‘

21.39.2 Class

Reactor Callback Function

21.39.3 Syntax

`(vlr-trace-reaction [args])`

21.39.4 Arguments and Values

- ‘args‘: ignored.

21.39.5 Description

A predefined callback function that prints one or more callback arguments to the Trace window. Used during reactor development to inspect callback parameters.

21.39.6 Return Values

Returns ‘nil‘.

21.39.7 Side Effects

Writes to the Trace window.

21.39.8 Examples

Not documented on the vendor page.

21.39.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.39.10 Source Notes

- [vlr-trace-reaction (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-489E0189-FCB3-4B6C-95F7 htm)
- Status: documented (Phase 3 body fill).

21.40 Function Entry: VLR-NOTIFICATION

21.40.1 Name

‘vlr-notification‘

21.40.2 Class

Reactor Function

21.40.3 Syntax

```
(vlr-notification reactor)
```

21.40.4 Arguments and Values

- ‘reactor‘: a VLR object.

21.40.5 Description

Returns the reactor’s notification scope: whether the callback fires only when the associated document is active, or for events in any document.

21.40.6 Return Values

Returns the symbol ‘'active-document-only‘ if the reactor fires only when its document is current, or ‘'all-documents‘ if it fires regardless.

21.40.7 Side Effects

None.

21.40.8 Examples

Not documented on the vendor page.

21.40.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.40.10 Source Notes

- [vlr-notification (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0A10A687-C72A-407C-AD71-C9DD76084 htm)
- Status: documented (Phase 3 body fill).

21.41 Function Entry: VLR-SET-NOTIFICATION

21.41.1 Name

‘vlr-set-notification‘

21.41.2 Class

Reactor Function

21.41.3 Syntax

```
(vlr-set-notification reactor scope)
```

21.41.4 Arguments and Values

- ‘reactor‘: a VLR object.
- ‘scope‘: the symbol ‘’active-document-only‘ or ‘’all-documents‘.

21.41.5 Description

Defines whether a reactor’s callback function will execute when its associated namespace is not active.

21.41.6 Return Values

Returns the new scope symbol on success, or ‘nil‘ on failure.

21.41.7 Side Effects

Mutates the reactor object.

21.41.8 Examples

- ‘(vlr-set-notification myReactor ’all-documents)‘ makes the reactor fire even when its source document is not current.

21.41.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.41.10 Source Notes

- [vlr-set-notification (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-1EBB6848-E863-4067-8A61 htm)
- Status: documented (Phase 3 body fill).

21.42 Function Entry: VLR-TOOLBAR-REACTOR

21.42.1 Name

‘vlr-toolbar-reactor‘

21.42.2 Class

Reactor Constructor Function

21.42.3 Syntax

```
(vlr-toolbar-reactor data callbacks)
```

21.42.4 Arguments and Values

- ‘data‘: integer, real, string, list, VLA-object, safearray, variant, ‘T‘, or ‘nil‘. Application-specific data attached to the reactor.
- ‘callbacks‘: a list of ‘(event-name . callback-function)‘ pairs. Supported events:
  - ‘:vlr-toolbarBitmapSizeWillChange‘
  - ‘:vlr-toolbarBitmapSizeChanged‘

  Each callback receives two arguments: the reactor object and a list containing event data (notably whether large or small icons are being applied).

21.42.5 Description

Constructs an editor reactor object that fires when toolbar bitmap sizes change.

21.42.6 Return Values

A new VLR reactor object.

21.42.7 Side Effects

Registers a reactor with the host editor.

21.42.8 Examples

Not documented on the vendor page.

21.42.9 Availability

- AutoCAD 2026: documented.
- BricsCAD V26: presumed compatible (toolbar bitmap-change semantics may differ).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

21.42.10 Source Notes

- [vlr-toolbar-reactor (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-ABCA865A-DFBE-4057-BE9E` `htm`)
- Status: documented (Phase 3 body fill).

21.43 Function Entry: VLR-PERS

21.43.1 Name

‘vlr-pers‘

21.43.2 Class

Reactor Function

21.43.3 Syntax

```
(vlr-pers reactor)
```

21.43.4 Arguments and Values

- ‘reactor‘: a VLR object to make persistent.

21.43.5 Description

Makes a reactor persistent — the reactor is saved with the drawing and remains active across AutoCAD sessions. By default reactors are transient and live only for the current session.

21.43.6 Return Values

Returns the reactor object on success; ‘nil‘ on failure.

21.43.7 Side Effects

Flags the reactor as persistent in the drawing database.

21.43.8 Examples

```
(setq circleReactor (vlr-object-reactor (list myCircle) "Radius size"
                    '((:vlr-modified . print-radius))))
(vlr-pers circleReactor)
```

21.43.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.43.10 Source Notes

- [vlr-pers (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0F0F7B01-2CE7-4F3C-B686-B255866C9 htm`)
- Status: documented (Phase 3 body fill).

21.44 Function Entry: VLR-PERS-LIST

21.44.1 Name

‘vlr-pers-list‘

21.44.2 Class

Reactor Function

21.44.3 Syntax

```
(vlr-pers-list)
```

21.44.4 Arguments and Values

None.

21.44.5 Description

Returns the persistent reactors stored in the current drawing document.

21.44.6 Return Values

A list of VLR reactor objects, or ‘nil‘ if there are none.

21.44.7 Side Effects

None.

21.44.8 Examples

Not documented on the vendor page.

21.44.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.44.10 Source Notes

- [vlr-pers-list (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0C7564C7-1C8B-4F17-A5F8-374C2860E htm)
- Status: documented (Phase 3 body fill).

21.45 Function Entry: VLR-PERS-P

21.45.1 Name

‘vlr-pers-p‘

21.45.2 Class

Reactor Predicate

21.45.3 Syntax

```
(vlr-pers-p reactor)
```

21.45.4 Arguments and Values

- ‘reactor‘: a VLR object.

21.45.5 Description

Tests whether ‘reactor‘ is persistent.

21.45.6 Return Values

Returns the reactor object if it is persistent; otherwise ‘nil‘.

21.45.7 Side Effects

None.

21.45.8 Examples

Not documented on the vendor page.

21.45.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.45.10 Source Notes

- [vlr-pers-p (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4C03EA1B-E274-49B2-8DEE-EBCCCEE2E htm)
- Status: documented (Phase 3 body fill).

21.46 Function Entry: VLR-PERS-RELEASE

21.46.1 Name

‘vlr-pers-release‘

21.46.2 Class

Reactor Function

21.46.3 Syntax

```
(vlr-pers-release reactor)
```

21.46.4 Arguments and Values

- ‘reactor‘: a VLR object.

21.46.5 Description

Makes a previously persistent reactor transient — removes it from the drawing’s persistent-reactor set so it no longer survives a session boundary.

21.46.6 Return Values

Returns the reactor object on success; ‘nil‘ on failure.

21.46.7 Side Effects

Clears the persistent flag on the reactor.

21.46.8 Examples

Not documented on the vendor page.

21.46.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.46.10 Source Notes

- [vlr-pers-release (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-79BE8AC1-4C54-49AD-B13D-CC12E06EE htm)
- Status: documented (Phase 3 body fill).

21.47 Function Entry: VLR-OWNERS

21.47.1 Name

‘vlr-owners‘

21.47.2 Class

Reactor Function

21.47.3 Syntax

```
(vlr-owners reactor)
```

21.47.4 Arguments and Values

- ‘reactor‘: an object reactor (VLR object).

21.47.5 Description

Returns the list of owners of an object reactor — the entities to which the reactor is attached.

21.47.6 Return Values

A list of VLA-objects (or entity names), or ‘nil‘ if the reactor has no owners.

21.47.7 Side Effects

None.

21.47.8 Examples

Not documented on the vendor page.

21.47.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.47.10 Source Notes

- [vlr-owners (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-AD8525AF-411C-4002-A025-CD36CD59D htm`)
- Status: documented (Phase 3 body fill).

21.48 Function Entry: VLR-OWNER-ADD

21.48.1 Name

‘vlr-owner-add‘

21.48.2 Class

Reactor Function

21.48.3 Syntax

```
(vlr-owner-add reactor owner)
```

21.48.4 Arguments and Values

- ‘reactor‘: an object reactor.
- ‘owner‘: a VLA-object or entity name to attach.

21.48.5 Description

Adds an object to the list of owners of an object reactor — extending the set of entities whose events will fire the reactor.

21.48.6 Return Values

The owner reference on success; ‘nil‘ on failure.

21.48.7 Side Effects

Mutates the reactor’s owner list.

21.48.8 Examples

Not documented on the vendor page.

21.48.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.48.10 Source Notes

- [vlr-owner-add (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8DF13914-E66F-410D-BFDF-8A9A1860A htm)
- Status: documented (Phase 3 body fill).

21.49 Function Entry: VLR-OWNER-REMOVE

21.49.1 Name

‘vlr-owner-remove‘

21.49.2 Class

Reactor Function

21.49.3 Syntax

```
(vlr-owner-remove reactor owner)
```

21.49.4 Arguments and Values

- ‘reactor‘: an object reactor.
- ‘owner‘: a VLA-object or entity name previously added with ‘vlr-owner-add‘.

21.49.5 Description

Removes an object from the list of owners of an object reactor.

21.49.6 Return Values

The owner reference on success; ‘nil‘ on failure.

21.49.7 Side Effects

Mutates the reactor’s owner list.

21.49.8 Examples

Not documented on the vendor page.

21.49.9 Availability

- AutoCAD 2026: documented (Windows only).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD-Windows documented; BricsCAD parity TBD.

21.49.10 Source Notes

- [vlr-owner-remove (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-EE5ABE92-4589-482A-95B3` `htm`)
- Status: documented (Phase 3 body fill).

22 22 Packaging, Compilation, and Namespaces

22.1 Overview

This chapter covers '.fas', '.vlx', namespace isolation, and related packaging/compilation interfaces.

22.2 Normative Rules

- these features affect more than delivery; they influence name visibility and execution environment
- BricsCAD supports related but different packaging mechanisms such as DEScoder

22.3 Function Entry: VL-DOC-EXPORT

22.3.1 Name

‘vl-doc-export‘

22.3.2 Class

Namespace function

22.3.3 Syntax

```
(vl-doc-export 'function)
```

22.3.4 Description

Makes a function loaded in a VLX namespace available to the current document namespace.

22.3.5 Return Values

Host-defined success value.

22.3.6 Side Effects

Exports a symbol/function binding across namespace boundaries.

22.3.7 Compatibility

Applies to separate-namespace VLX applications. Autodesk documents behavior differences on non-Windows platforms.

22.3.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.3.9 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm)
- Status: documented by family page.

22.4 Function Entry: VL-DOC-IMPORT

22.4.1 Name

‘vl-doc-import‘

22.4.2 Class

Namespace function

22.4.3 Syntax

```
(vl-doc-import ['function | application])
```

22.4.4 Description

Imports a function previously exported from another separate-namespace VLX.

22.4.5 Return Values

Host-defined success value.

22.4.6 Side Effects

Imports bindings across namespace boundaries.

22.4.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.4.8 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)
- Status: documented by family page.

22.5 Function Entry: VL-DOC-REF

22.5.1 Name

‘vl-doc-ref‘

22.5.2 Class

Namespace function

22.5.3 Syntax

`(vl-doc-ref symbol)`

22.5.4 Description

Retrieves the value of a variable from the associated document namespace.

22.5.5 Return Values

Current value of ‘symbol‘ in the document namespace.

22.5.6 Side Effects

None.

22.5.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.5.8 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)
- Status: documented by family page.

22.6 Function Entry: VL-DOC-SET

22.6.1 Name

‘vl-doc-set‘

22.6.2 Class

Namespace function

22.6.3 Syntax

`(vl-doc-set symbol value)`

22.6.4 Description

Sets the value of a variable in the associated document namespace.

22.6.5 Return Values

The value assigned or another host-defined success value.

22.6.6 Side Effects

Mutates document-namespace state.

22.6.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.6.8 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)
- Status: documented by family page.

22.7 Function Entry: VL-BB-REF

22.7.1 Name

‘vl-bb-ref‘

22.7.2 Class

Namespace function

22.7.3 Syntax

`(vl-bb-ref symbol)`

22.7.4 Description

Retrieves the value of a variable from the blackboard namespace.

22.7.5 Return Values

Current value of ‘symbol‘, or a host-specific absence result.

22.7.6 Side Effects

None.

22.7.7 Arguments and Values

- ‘symbol‘: quoted symbol naming the blackboard variable

22.7.8 Notes

- Autodesk documents blackboard addressing as namespace communication across documents.
- Autodesk documents ‘vl-bb-ref‘ as paired with ‘vl-bb-set‘.

22.7.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.7.10 Source Notes

- [Namespace Communication Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2025/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F5DD5472-1695-4CA3-9110-E6008E87BB89.htm)
- [vl-bb-set (AutoLISP)](https://help.autodesk.com/cloudhelp/2024/CSY/AutoCAD-AutoLISP-Reference/files/GUID-2B1484FC-BBD1-408B-A8A0-DB931363630F.htm)
- Status: documented.

22.8 Function Entry: VL-BB-SET

22.8.1 Name

‘vl-bb-set‘

22.8.2 Class

Namespace function

22.8.3 Syntax

`(vl-bb-set symbol value)`

22.8.4 Description

Sets a variable in the blackboard namespace.

22.8.5 Return Values

The value assigned or another host-defined success value.

22.8.6 Side Effects

Mutates blackboard-namespace state.

22.8.7 Arguments and Values

- ‘symbol‘: quoted symbol naming the blackboard variable
- ‘value‘: any AutoLISP value except a function

22.8.8 Notes

- Autodesk documents the return value as the assigned value.
- Autodesk documents blackboard addressing as namespace communication across documents.

22.8.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.8.10 Source Notes

- [vl-bb-set (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/CSY/AutoCAD-AutoLISP-Reference/files/GUID-2B1484FC-BBD1-408B-A8A0-DB931363630F.htm`)
- [Namespace Communication Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2025/ITA/AutoCAD-MAC-AutoLISP-Reference/files/GUID-F5DD5472-1695-4CA3-9110-E6008E87BB89.htm`)
- Status: documented.

22.9 Function Entry: VL-EXIT-WITH-ERROR

22.9.1 Name

‘vl-exit-with-error‘

22.9.2 Class

Namespace/error-bridge function

22.9.3 Syntax

`(vl-exit-with-error msg)`

22.9.4 Description

Passes control from a VLX error handler to the ‘*error*‘ function of the calling namespace.

22.9.5 Return Values

Does not return normally to the local caller.

22.9.6 Side Effects

Transfers control across namespace/error boundaries.

22.9.7 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.9.8 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)
- Status: documented by family page.

22.10 Function Entry: VL-EXIT-WITH-VALUE

22.10.1 Name

‘vl-exit-with-value‘

22.10.2 Class

Namespace/error-bridge function

22.10.3 Syntax

```
(vl-exit-with-value value)
```

22.10.4 Description

Returns a value to the function that invoked the ‘*error*‘ handler from another namespace.

22.10.5 Return Values

Transfers ‘value‘ to the calling namespace.

22.10.6 Side Effects

Transfers control across namespace boundaries.

22.10.7 Arguments and Values

- ‘value‘: any AutoLISP value accepted by Autodesk for namespace return transfer, including integer, real, string, list, file, entity name, ‘T‘, or ‘nil‘

22.10.8 Notes

- Autodesk documents this function as valid only in the context of a VLX application’s ‘*error*‘ handler.
- Autodesk documents that pending instructions are flushed when control transfers back to the calling namespace.
- Autodesk documents the function as part of separate-namespace VLX error handling.

22.10.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.10.10 Source Notes

- [vl-exit-with-value (AutoLISP)](`https://help.autodesk.com/cloudhelp/2024/DEU/AutoCAD-MAC-AutoLISP-Reference/files/GUID-80622A39-A5E8-4E68-824E-E66BD htm`)
- [About Handling Errors in a VLX Application Running in Its Own Namespace (Visual LISP IDE)](`https://help.autodesk.com/cloudhelp/2026/ESP/AutoCAD-AutoLISP/files/GUID-B682DF97-DFD5-48B4-9FFF-8A0DF7CAC05F.htm`)
- Status: documented.

22.11 Function Entry: VL-ARX-IMPORT

22.11.1 Name

‘vl-arx-import‘

22.11.2 Class

Namespace / extension-bridge function

22.11.3 Syntax

```
(vl-arx-import [function | application])
```

22.11.4 Description

Imports ObjectARX or ADSR/X-defined functions into a separate-namespace VLX environment.

22.11.5 Return Values

Host-defined success value.

22.11.6 Side Effects

Imports externally defined callable interfaces into the current namespace.

22.11.7 Compatibility

Applies to separate-namespace VLX applications and has documented behavior differences on non-Windows platforms.

22.11.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.11.9 Source Notes

- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)
- Status: documented by family page.

22.12 Function Entry: VL-VLX-LOADED-P

22.12.1 Name

‘vl-vlx-loaded-p‘

22.12.2 Class

VLX namespace inquiry function

22.12.3 Syntax

```
(vl-vlx-loaded-p appname)
```

22.12.4 Arguments and Values

- ‘appname‘ — string naming a VLX application.

22.12.5 Description

Determines whether a separate-namespace VLX application is loaded.

22.12.6 Return Values

Returns ‘T‘ if the specified separate-namespace VLX application is loaded; otherwise returns ‘nil‘.

22.12.7 Notes

- Autodesk documents this for separate-namespace VLX applications, not document-namespace loads.
- Recent Autodesk references say the function is supported on macOS and Web but has no effect there because VLX files are Windows-only.

22.12.8 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.12.9 Source Notes

- [vl-vlx-loaded-p (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/HUN/AutoCAD-AutoLISP-Reference/files/GUID-2322688B-F1FA-416E-AEF0-E1425671
htm)

- [VLX Namespace Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm)

22.13 Function Entry: VL-UNLOAD-VLX

22.13.1 Name

‘vl-unload-vlx‘

22.13.2 Class

VLX namespace control function

22.13.3 Syntax

```
(vl-unload-vlx appname)
```

22.13.4 Arguments and Values

- ‘appname‘ — string naming a VLX application loaded in its own namespace, without the ‘.vlx‘ extension.

22.13.5 Description

Unloads a VLX application that was loaded in a separate namespace.

22.13.6 Return Values

Returns ‘T‘ if successful.

22.13.7 Exceptional Situations

Signals an AutoLISP-visible error if the named application cannot be unloaded.

22.13.8 Notes

- Autodesk documents that this does not apply to VLX applications loaded into the current document namespace.
- Recent Autodesk references say the function is supported on macOS and Web but has no effect there because VLX files are Windows-only.

22.13.9 Availability

- AutoCAD 2026: documented (per Autodesk reference page).
- BricsCAD V26: presumed compatible (no contradicting page found).
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.13.10 Source Notes

- [vl-unload-vlx (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/PLK/AutoCAD-LT-AutoLISP-Reference/files/GUID-7AF6AC48-3435-4625-B210-C8FFF` `htm`)
- [VLX Namespace Functions Reference (AutoLISP)](`https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm`)

22.14 Dictionary Coverage Index: Namespace and Packaging Functions

The Autodesk VLX/namespace family includes at least:

- ‘vl-arx-import‘
- ‘vl-bb-ref‘
- ‘vl-bb-set‘
- ‘vl-doc-export‘
- ‘vl-doc-import‘
- ‘vl-doc-ref‘
- ‘vl-doc-set‘
- ‘vl-exit-with-error‘
- ‘vl-exit-with-value‘
- ‘vl-list-exported-functions‘
- ‘vl-list-loaded-vlx‘

- ‘vl-unload-vlx‘
- ‘vl-vlx-loaded-p‘

Sources:

- [VLX Namespace Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1.htm)
- [V Functions Reference (AutoLISP)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-26FF2BE2-B5BF-4DD1-A617 htm)

22.15 Source Notes

- [VLX Namespace Functions Reference](https://help.autodesk.com/cloudhelp/2026/CHT/AutoCAD-AutoLISP-Reference/files/GUID-5784FC6F-82DD-4459-879B htm)
- [VLX NameSpace Support (BricsCAD)](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/VLXNameSpaceSupport.htm)
- [DEScoder LISP encryption (BricsCAD)](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/DEScoderLISPencryption.htm)

22.16 Function Entry: VLISP-COMPILE

22.16.1 Name

‘vlisp-compile‘

22.16.2 Class

Visual LISP Function (vendor-divergent)

22.16.3 Syntax (AutoCAD 2026)

```
(vlisp-compile 'mode 'filename [output-filename])
```

22.16.4 Syntax (BricsCAD V26)

```
(vlisp-compile 'mode sourcefile [outfile])
```

22.16.5 Arguments and Values

- ‘mode‘ (AutoCAD): symbol; one of
  - ‘’st‘ — standard build mode; smallest output, suitable for single-file programs.
  - ‘’lsm‘ — optimise + link indirectly; optimised but no direct cross-file references.
  - ‘’lsa‘ — optimise + link directly; optimised with direct cross-file references.
- ‘mode‘ (BricsCAD): symbol; the same three names are accepted but the value is documented as ignored in BricsCAD V26.
- ‘filename‘ / ‘sourcefile‘: string; AutoLISP source file. AutoCAD: path may be omitted if the file is on the AutoCAD support search path; the extension defaults to ‘.lsp‘.
- ‘output-filename‘ / ‘outfile‘ (optional): string; output path.
  - AutoCAD: when omitted, the output file shares the source name with extension ‘.fas‘. Starting with AutoCAD 2021, FAS files can be produced in either Unicode or MBCS form.
  - BricsCAD: when omitted, the output file shares the source name with extension ‘.des‘ (DEScoder-encrypted).

22.16.6 Description

Compiles AutoLISP source code into a vendor-defined compiled form. The cross-vendor divergence is significant: AutoCAD produces '.fas' (and, via larger build options, '.vlx' archives), while BricsCAD V26 produces '.des' and ignores the 'mode' argument. Phase 4 reconciliation in 'vendor-inventory-2026.org' §10.

22.16.7 Return Values

- AutoCAD 2026: returns 'T' on success, otherwise 'nil'.
- BricsCAD V26: returns 'T' if the encrypted output file is created, otherwise 'nil'.

22.16.8 Side Effects

Writes the compiled output file to disk.

22.16.9 Examples

- AutoCAD: '(vlisp-compile 'st "yinyang.lsp")' produces 'yinyang.fas' and returns 'T'.
- BricsCAD: '(vlisp-compile 'st "yinyang.lsp")' produces 'yinyang.des' and returns 'T'.

22.16.10 Availability

- AutoCAD 2026: documented; produces '.fas' (and '.vlx' archives via auxiliary tooling).
- BricsCAD V26: documented; produces '.des'; the 'mode' argument is parsed but ignored.
- Status: cross-vendor divergence — see Phase 4 reconciliation in 'vendor-inventory-2026.org' §10.

22.16.11 See Also

- Cross-vendor divergence: AutoCAD produces '.fas' (and '.vlx' archives via auxiliary tooling); BricsCAD produces '.des' and ignores the 'mode' argument. See body Description above and 'vendor-inventory-2026.org' §10 item 1.

22.16.12 Source Notes

- [vlisp-compile (AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-8184531C-EDE5-4949-9EAE-25CE98BFF htm)
- [vlisp-compile (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlisp-compile.htm)
- Status: documented (Phase 3 body fill).

22.17 Function Entry: VL-REGISTRY-DELETE

22.17.1 Name

‘vl-registry-delete‘

22.17.2 Class

Visual LISP Function

22.17.3 Syntax

`(vl-registry-delete reg-key [val-name])`

22.17.4 Arguments and Values

- ‘reg-key‘: string; a Windows-registry key path (or a property-list-file key path on macOS). May not start with ‘$\text{HKEY}_{\text{USERS}}$‘ or ‘$\text{HKEY}_{\text{LOCALMACHINE}}$‘.
- ‘val-name‘ (optional): string; the value name to delete. With no ‘val-name‘, the entire key is deleted; with ‘val-name‘, only that named value is deleted.

22.17.5 Description

Deletes the specified key, or the named value within a key, from the Windows registry, or the equivalent property-list file on macOS.

22.17.6 Return Values

Returns the deleted value-name (or key) on success; ‘nil‘ on failure.

22.17.7 Side Effects

Mutates the user-level persistent settings store.

22.17.8 Examples

Not documented on the vendor page.

22.17.9 Availability

- AutoCAD 2026: documented (Windows + macOS).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.17.10 Source Notes

- [vl-registry-delete (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-BF5FB9C5-A015-464C-A475-B320C5B5E htm`)
- Status: documented (Phase 3 body fill).

22.18 Function Entry: VL-REGISTRY-DESCENDENTS

22.18.1 Name

‘vl-registry-descendents‘

22.18.2 Class

Visual LISP Function

22.18.3 Syntax

```
(vl-registry-descendents reg-key [val-names])
```

22.18.4 Arguments and Values

- ‘reg-key‘: string; a registry / plist key path.
- ‘val-names‘ (optional): when supplied as a non-‘nil‘ value, the function returns the value names under ‘reg-key‘ rather than its sub-keys.

22.18.5 Description

Returns the immediate children of ‘reg-key‘ — sub-key names by default, or value names when ‘val-names‘ is non-‘nil‘.

22.18.6 Return Values

A list of strings, or ‘nil‘ if the key does not exist or has no children.

22.18.7 Side Effects

None.

22.18.8 Examples

Not documented on the vendor page.

22.18.9 Availability

- AutoCAD 2026: documented (Windows + macOS).
- BricsCAD V26: presumed compatible.

- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.18.10 Source Notes

- [vl-registry-descendents (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-3A2A4AA0-8799-4B0A-` `htm`)
- Status: documented (Phase 3 body fill).

22.19 Function Entry: VL-REGISTRY-READ

22.19.1 Name

‘vl-registry-read‘

22.19.2 Class

Visual LISP Function

22.19.3 Syntax

```
(vl-registry-read reg-key [val-name])
```

22.19.4 Arguments and Values

- ‘reg-key‘: string; the registry / plist key path.
- ‘val-name‘ (optional): string; the value name. With no ‘val-name‘, the key’s default value is read.

22.19.5 Description

Returns the data stored under the named value (or the default value) of a registry key, or the equivalent in a macOS property-list file.

22.19.6 Return Values

The stored data, typed as the registry value’s data type, or ‘nil‘ if the key/value does not exist.

22.19.7 Side Effects

None.

22.19.8 Examples

- After ‘(vl-registry-write ”HKEY$_{\text{CURRENTUSER}}$” ”” ”test data”)‘, ‘(vl-registry-read ”HKEY$_{\text{CURRENTUSER}}$”)‘ returns ‘”test data”‘.

22.19.9 Availability

- AutoCAD 2026: documented (Windows + macOS).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.19.10 Source Notes

- [vl-registry-read (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-0B9E26CF-2410-482E-B568-7FE05B9F0 htm`)
- Status: documented (Phase 3 body fill).

22.20 Function Entry: VL-REGISTRY-WRITE

22.20.1 Name

‘vl-registry-write‘

22.20.2 Class

Visual LISP Function

22.20.3 Syntax

`(vl-registry-write reg-key [val-name val-data])`

22.20.4 Arguments and Values

- ‘reg-key‘: string; the key path. May not start with ‘$HKEY_{USERS}$‘ or ‘$HKEY_{LOCALMACHINE}$‘.
- ‘val-name‘ (optional): string; the value name. With no ‘val-name‘, the default value is written.
- ‘val-data‘ (optional): string; the data. With ‘val-name‘ but no ‘val-data‘, an empty string is written.

22.20.5 Description

Creates a key, or sets a named value within a key, in the Windows registry, or the equivalent in a macOS property-list file.

22.20.6 Return Values

The stored data string on success; ‘nil‘ on failure.

22.20.7 Side Effects

Mutates the user-level persistent settings store.

22.20.8 Examples

- ‘(vl-registry-write "$HKEY_{CURRENTUSER}$" "" "test data")‘ returns ‘"test data"‘.

22.20.9 Availability

- AutoCAD 2026: documented (Windows + macOS).
- BricsCAD V26: presumed compatible.
- Status: AutoCAD documented; BricsCAD presumed-both pending Phase 4 verification.

22.20.10 Source Notes

- [vl-registry-write (AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9DDE1962-43C5-40D2-85E3-B7A729CAB` `htm`)
- Status: documented (Phase 3 body fill).

23 23 Express Tools

23.1 Overview

The Express Tools are an AutoCAD-specific add-on shipped with full AutoCAD on Windows. They are not part of AutoCAD LT and are not part of BricsCAD. The functions enumerated below are available only after `(acet-load-expresstools)` has succeeded.

This chapter exists so the specification carries every documented AutoLISP-visible Express Tools function as a dictionary entry, but the entries are scoped to the AutoCAD-Windows profile only.

23.2 Normative Rules

- Express Tools functions are AutoCAD-only.
- A conforming BricsCAD profile must not provide them.
- A conforming AutoCAD profile must load Express Tools before any of these functions are callable.

23.3 Function Entry: ACAD$_{\text{COLORDLG}}$

23.3.1 Name

'acad$_{\text{colordlg}}$'

23.3.2 Class

Express Tools Function

23.3.3 Syntax

```
(acad_colordlg color [flag])
```

23.3.4 Description

Displays the standard AutoCAD color-selection dialog. The flag suppresses the BYLAYER and BYBLOCK choices.

23.3.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.3.6 Side Effects

Vendor-defined.

23.3.7 Examples

Not documented in the current category page.

23.3.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires '(acet-load-expresstools)').
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.3.9 Source Notes

- [acad$_{\text{colordlg}}$ (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.4 Function Entry: ACAD$_{\text{HELPDLG}}$

23.4.1 Name

‘acad$_{\text{helpdlg}}$‘

23.4.2 Class

Express Tools Function

23.4.3 Syntax

```
(acad_helpdlg helpfile topic)
```

23.4.4 Description

Invokes the Help facility (obsolete; use ‘help‘ instead).

23.4.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.4.6 Side Effects

Vendor-defined.

23.4.7 Examples

Not documented in the current category page.

23.4.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.4.9 Source Notes

- [acad$_{\text{helpdlg}}$ (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.5 Function Entry: ACAD$_{\text{STRLSORT}}$

23.5.1 Name

‘acad$_{\text{strlsort}}$‘

23.5.2 Class

Express Tools Function

23.5.3 Syntax

`(acad_strlsort lst)`

23.5.4 Description

Sorts a list of strings in alphabetical order.

23.5.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.5.6 Side Effects

Vendor-defined.

23.5.7 Examples

Not documented in the current category page.

23.5.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.5.9 Source Notes

- [acad$_{\text{strlsort}}$ (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.6 Function Entry: ACAD$_{\text{TRUECOLORCLI}}$

23.6.1 Name

‘acad$_{\text{truecolorcli}}$‘

23.6.2 Class

Express Tools Function

23.6.3 Syntax

```
(acad_truecolorcli flag [allowcolorbook])
```

23.6.4 Description

Prompts for colors at the Command prompt with truecolor support.

23.6.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.6.6 Side Effects

Vendor-defined.

23.6.7 Examples

Not documented in the current category page.

23.6.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.6.9 Source Notes

- [acad$_{\text{truecolorcli}}$ (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.7 Function Entry: ACAD$_{\text{TRUECOLORDLG}}$

23.7.1 Name

‘acad$_{\text{truecolordlg}}$‘

23.7.2 Class

Express Tools Function

23.7.3 Syntax

```
(acad_truecolordlg color [flag [allowcolorbook]])
```

23.7.4 Description

Displays the color-selection dialog with index, true color, and color-book tabs.

23.7.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.7.6 Side Effects

Vendor-defined.

23.7.7 Examples

Not documented in the current category page.

23.7.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.7.9 Source Notes

- [acad$_{\text{truecolordlg}}$ (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.8 Function Entry: ACAD-POP-DBMOD

23.8.1 Name

‘acad-pop-dbmod‘

23.8.2 Class

Express Tools Function

23.8.3 Syntax

`(acad-pop-dbmod)`

23.8.4 Description

Restores the DBMOD system variable to a previously stored value.

23.8.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.8.6 Side Effects

Vendor-defined.

23.8.7 Examples

Not documented in the current category page.

23.8.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.8.9 Source Notes

- [acad-pop-dbmod (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.9 Function Entry: ACAD-PUSH-DBMOD

23.9.1 Name

‘acad-push-dbmod‘

23.9.2 Class

Express Tools Function

23.9.3 Syntax

```
(acad-push-dbmod)
```

23.9.4 Description

Stores the current value of the DBMOD system variable on a private stack.

23.9.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.9.6 Side Effects

Vendor-defined.

23.9.7 Examples

Not documented in the current category page.

23.9.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.9.9 Source Notes

- [acad-push-dbmod (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.10 Function Entry: ACET-ENT-GEOMEXTENTS

23.10.1 Name

‘acet-ent-geomextents‘

23.10.2 Class

Express Tools Function

23.10.3 Syntax

```
(acet-ent-geomextents ename)
```

23.10.4 Description

Returns the geometric extents of the given object.

23.10.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.10.6 Side Effects

Vendor-defined.

23.10.7 Examples

Not documented in the current category page.

23.10.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.10.9 Source Notes

- [acet-ent-geomextents (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-676F8B1B-7C23-4480-9DD4-19D0836EAE08.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.11 Function Entry: ACET-FILE-ATTR

23.11.1 Name

‘acet-file-attr‘

23.11.2 Class

Express Tools Function

23.11.3 Syntax

```
(acet-file-attr filename [attr])
```

23.11.4 Description

Gets or sets the protection attributes of a file or directory.

23.11.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.11.6 Side Effects

Vendor-defined.

23.11.7 Examples

Not documented in the current category page.

23.11.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.11.9 Source Notes

- [acet-file-attr (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.12 Function Entry: ACET-FILE-CHDIR

23.12.1 Name

‘acet-file-chdir‘

23.12.2 Class

Express Tools Function

23.12.3 Syntax

```
(acet-file-chdir directory)
```

23.12.4 Description

Changes the current working directory.

23.12.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.12.6 Side Effects

Vendor-defined.

23.12.7 Examples

Not documented in the current category page.

23.12.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.12.9 Source Notes

- [acet-file-chdir (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.13 Function Entry: ACET-FILE-COPY

23.13.1 Name

‘acet-file-copy‘

23.13.2 Class

Express Tools Function

23.13.3 Syntax

```
(acet-file-copy from to [force])
```

23.13.4 Description

Copies a file.

23.13.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.13.6 Side Effects

Vendor-defined.

23.13.7 Examples

Not documented in the current category page.

23.13.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.13.9 Source Notes

- [acet-file-copy (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.14 Function Entry: ACET-FILE-CWD

23.14.1 Name

‘acet-file-cwd‘

23.14.2 Class

Express Tools Function

23.14.3 Syntax

`(acet-file-cwd)`

23.14.4 Description

Returns the current working directory.

23.14.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.14.6 Side Effects

Vendor-defined.

23.14.7 Examples

Not documented in the current category page.

23.14.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.14.9 Source Notes

- [acet-file-cwd (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.15 Function Entry: ACET-FILE-DIR

23.15.1 Name

‘acet-file-dir‘

23.15.2 Class

Express Tools Function

23.15.3 Syntax

`(acet-file-dir pattern [attributes] [starting-directory])`

23.15.4 Description

Locates files matching a pattern.

23.15.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.15.6 Side Effects

Vendor-defined.

23.15.7 Examples

Not documented in the current category page.

23.15.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.15.9 Source Notes

- [acet-file-dir (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.16 Function Entry: ACET-FILE-MKDIR

23.16.1 Name

‘acet-file-mkdir‘

23.16.2 Class

Express Tools Function

23.16.3 Syntax

```
(acet-file-mkdir directory)
```

23.16.4 Description

Creates a new directory.

23.16.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.16.6 Side Effects

Vendor-defined.

23.16.7 Examples

Not documented in the current category page.

23.16.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.16.9 Source Notes

- [acet-file-mkdir (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.17 Function Entry: ACET-FILE-MOVE

23.17.1 Name

‘acet-file-move‘

23.17.2 Class

Express Tools Function

23.17.3 Syntax

```
(acet-file-move from to)
```

23.17.4 Description

Moves or renames files.

23.17.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.17.6 Side Effects

Vendor-defined.

23.17.7 Examples

Not documented in the current category page.

23.17.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.17.9 Source Notes

- [acet-file-move (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.18 Function Entry: ACET-FILE-REMOVE

23.18.1 Name

‘acet-file-remove‘

23.18.2 Class

Express Tools Function

23.18.3 Syntax

```
(acet-file-remove filespec [force])
```

23.18.4 Description

Deletes one or more files.

23.18.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.18.6 Side Effects

Vendor-defined.

23.18.7 Examples

Not documented in the current category page.

23.18.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.18.9 Source Notes

- [acet-file-remove (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.19 Function Entry: ACET-FILE-RMDIR

23.19.1 Name

‘acet-file-rmdir‘

23.19.2 Class

Express Tools Function

23.19.3 Syntax

```
(acet-file-rmdir directory)
```

23.19.4 Description

Removes a directory.

23.19.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.19.6 Side Effects

Vendor-defined.

23.19.7 Examples

Not documented in the current category page.

23.19.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.19.9 Source Notes

- [acet-file-rmdir (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-05EB15E4-F0ED-4AC7-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.20 Function Entry: ACET-HELP

23.20.1 Name

‘acet-help‘

23.20.2 Class

Express Tools Function

23.20.3 Syntax

`(acet-help [topic])`

23.20.4 Description

Displays a help page in the Express Tools help system.

23.20.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.20.6 Side Effects

Vendor-defined.

23.20.7 Examples

Not documented in the current category page.

23.20.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.20.9 Source Notes

- [acet-help (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.21 Function Entry: ACET-HELP-TRAP

23.21.1 Name

‘acet-help-trap‘

23.21.2 Class

Express Tools Function

23.21.3 Syntax

```
(acet-help-trap command [topic [file]])
```

23.21.4 Description

Installs an F1 key monitor for context-sensitive help.

23.21.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.21.6 Side Effects

Vendor-defined.

23.21.7 Examples

Not documented in the current category page.

23.21.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.21.9 Source Notes

- [acet-help-trap (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.22 Function Entry: ACET-INI-GET

23.22.1 Name

‘acet-ini-get‘

23.22.2 Class

Express Tools Function

23.22.3 Syntax

```
(acet-ini-get inifile [section [key [default]]])
```

23.22.4 Description

Retrieves data from an INI file.

23.22.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.22.6 Side Effects

Vendor-defined.

23.22.7 Examples

Not documented in the current category page.

23.22.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.22.9 Source Notes

- [acet-ini-get (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.23 Function Entry: ACET-INI-SET

23.23.1 Name

‘acet-ini-set‘

23.23.2 Class

Express Tools Function

23.23.3 Syntax

```
(acet-ini-set inifile section [key [value]])
```

23.23.4 Description

Writes data to an INI file.

23.23.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.23.6 Side Effects

Vendor-defined.

23.23.7 Examples

Not documented in the current category page.

23.23.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.23.9 Source Notes

- [acet-ini-set (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.24 Function Entry: ACET-LAYERP-MARK

23.24.1 Name

‘acet-layerp-mark‘

23.24.2 Class

Express Tools Function

23.24.3 Syntax

`(acet-layerp-mark)`

23.24.4 Description

Places a mark for Layer Previous (LAYERP) recording.

23.24.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.24.6 Side Effects

Vendor-defined.

23.24.7 Examples

Not documented in the current category page.

23.24.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.24.9 Source Notes

- [acet-layerp-mark (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.25 Function Entry: ACET-LAYERP-MODE

23.25.1 Name

‘acet-layerp-mode‘

23.25.2 Class

Express Tools Function

23.25.3 Syntax

```
(acet-layerp-mode [flag])
```

23.25.4 Description

Queries or sets the LAYERPMODE setting.

23.25.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.25.6 Side Effects

Vendor-defined.

23.25.7 Examples

Not documented in the current category page.

23.25.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.25.9 Source Notes

- [acet-layerp-mode (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.26 Function Entry: ACET-LAYTRANS

23.26.1 Name

‘acet-laytrans‘

23.26.2 Class

Express Tools Function

23.26.3 Syntax

```
(acet-laytrans [filename [flag]])
```

23.26.4 Description

Translates drawing layers to standards.

23.26.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.26.6 Side Effects

Vendor-defined.

23.26.7 Examples

Not documented in the current category page.

23.26.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.26.9 Source Notes

- [acet-laytrans (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.27 Function Entry: ACET-LOAD-EXPRESSTOOLS

23.27.1 Name

‘acet-load-expresstools‘

23.27.2 Class

Express Tools Function

23.27.3 Syntax

`(acet-load-expresstools)`

23.27.4 Description

Initializes the Express Tools utility functions; must be called before other acet-* functions.

23.27.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.27.6 Side Effects

Vendor-defined.

23.27.7 Examples

Not documented in the current category page.

23.27.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.27.9 Source Notes

- [acet-load-expresstools (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.28 Function Entry: ACET-MS-TO-PS

23.28.1 Name

‘acet-ms-to-ps‘

23.28.2 Class

Express Tools Function

23.28.3 Syntax

```
(acet-ms-to-ps value [vport])
```

23.28.4 Description

Converts a value from model space to paper space units.

23.28.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.28.6 Side Effects

Vendor-defined.

23.28.7 Examples

Not documented in the current category page.

23.28.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.28.9 Source Notes

- [acet-ms-to-ps (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.29 Function Entry: ACET-PS-TO-MS

23.29.1 Name

‘acet-ps-to-ms‘

23.29.2 Class

Express Tools Function

23.29.3 Syntax

```
(acet-ps-to-ms value [vport])
```

23.29.4 Description

Converts a value from paper space to model space units.

23.29.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.29.6 Side Effects

Vendor-defined.

23.29.7 Examples

Not documented in the current category page.

23.29.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.29.9 Source Notes

- [acet-ps-to-ms (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-985B33C7-0A4B-4FBE-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.30 Function Entry: ACET-REG-DEL

23.30.1 Name

‘acet-reg-del‘

23.30.2 Class

Express Tools Function

23.30.3 Syntax

```
(acet-reg-del key [name])
```

23.30.4 Description

Deletes a key or value from the Windows registry.

23.30.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.30.6 Side Effects

Vendor-defined.

23.30.7 Examples

Not documented in the current category page.

23.30.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.30.9 Source Notes

- [acet-reg-del (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.31 Function Entry: ACET-REG-GET

23.31.1 Name

‘acet-reg-get‘

23.31.2 Class

Express Tools Function

23.31.3 Syntax

`(acet-reg-get key [name])`

23.31.4 Description

Returns a value from the Windows registry.

23.31.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.31.6 Side Effects

Vendor-defined.

23.31.7 Examples

Not documented in the current category page.

23.31.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.31.9 Source Notes

- [acet-reg-get (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.32 Function Entry: ACET-REG-MACHINE-PRODKEY

23.32.1 Name

‘acet-reg-machine-prodkey‘

23.32.2 Class

Express Tools Function

23.32.3 Syntax

```
(acet-reg-machine-prodkey)
```

23.32.4 Description

Returns the AutoCAD registry path under $\text{HKEY}_{\text{LOCALMACHINE}}$.

23.32.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.32.6 Side Effects

Vendor-defined.

23.32.7 Examples

Not documented in the current category page.

23.32.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.32.9 Source Notes

- [acet-reg-machine-prodkey (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-87C3-906B908AF5B8.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.33 Function Entry: ACET-REG-PRODKEY

23.33.1 Name

‘acet-reg-prodkey‘

23.33.2 Class

Express Tools Function

23.33.3 Syntax

```
(acet-reg-prodkey)
```

23.33.4 Description

Returns the current product key (obsolete; use the machine/user variants).

23.33.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.33.6 Side Effects

Vendor-defined.

23.33.7 Examples

Not documented in the current category page.

23.33.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.33.9 Source Notes

- [acet-reg-prodkey (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-87C3-906B908AF5B8.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.34 Function Entry: ACET-REG-PUT

23.34.1 Name

‘acet-reg-put‘

23.34.2 Class

Express Tools Function

23.34.3 Syntax

```
(acet-reg-put key [name value])
```

23.34.4 Description

Writes a Windows registry key or value.

23.34.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.34.6 Side Effects

Vendor-defined.

23.34.7 Examples

Not documented in the current category page.

23.34.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.34.9 Source Notes

- [acet-reg-put (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-
htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.35 Function Entry: ACET-REG-USER-PRODKEY

23.35.1 Name

‘acet-reg-user-prodkey‘

23.35.2 Class

Express Tools Function

23.35.3 Syntax

`(acet-reg-user-prodkey)`

23.35.4 Description

Returns the AutoCAD registry path under $\text{HKEY}_{\text{CURRENTUSER}}$.

23.35.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.35.6 Side Effects

Vendor-defined.

23.35.7 Examples

Not documented in the current category page.

23.35.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.35.9 Source Notes

- [acet-reg-user-prodkey (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-9F789A1F-D7E1-4655-87C3-906B908AF5B8.htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.36 Function Entry: ACET-SS-DRAG-MOVE

23.36.1 Name

‘acet-ss-drag-move‘

23.36.2 Class

Express Tools Function

23.36.3 Syntax

```
(acet-ss-drag-move ss pt [prompt] [highlight [cursor]])
```

23.36.4 Description

Drags a selection set to change location.

23.36.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.36.6 Side Effects

Vendor-defined.

23.36.7 Examples

Not documented in the current category page.

23.36.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.36.9 Source Notes

- [acet-ss-drag-move (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-676F8B1B-7C23-4480-9DD4-19D0836EAE08.htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.37 Function Entry: ACET-SS-DRAG-ROTATE

23.37.1 Name

‘acet-ss-drag-rotate‘

23.37.2 Class

Express Tools Function

23.37.3 Syntax

```
(acet-ss-drag-rotate ss pt [prompt] [highlight [cursor]])
```

23.37.4 Description

Drags a selection set to change rotation.

23.37.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.37.6 Side Effects

Vendor-defined.

23.37.7 Examples

Not documented in the current category page.

23.37.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.37.9 Source Notes

- [acet-ss-drag-rotate (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-676F8B1B-7C23-4480-9DD4-19D0836EAE08.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.38 Function Entry: ACET-SS-DRAG-SCALE

23.38.1 Name

‘acet-ss-drag-scale‘

23.38.2 Class

Express Tools Function

23.38.3 Syntax

```
(acet-ss-drag-scale ss pt [prompt] [highlight [cursor]])
```

23.38.4 Description

Drags a selection set to change scale.

23.38.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.38.6 Side Effects

Vendor-defined.

23.38.7 Examples

Not documented in the current category page.

23.38.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.38.9 Source Notes

- [acet-ss-drag-scale (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-676F8B1B-7C23-4480-9DD4-19D0836EAE08.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.39 Function Entry: ACET-STR-COLLATE

23.39.1 Name

‘acet-str-collate‘

23.39.2 Class

Express Tools Function

23.39.3 Syntax

```
(acet-str-collate left right [matchCase])
```

23.39.4 Description

Compares two strings for sort order using locale rules.

23.39.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.39.6 Side Effects

Vendor-defined.

23.39.7 Examples

Not documented in the current category page.

23.39.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.39.9 Source Notes

- [acet-str-collate (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.40 Function Entry: ACET-STR-EQUAL

23.40.1 Name

‘acet-str-equal‘

23.40.2 Class

Express Tools Function

23.40.3 Syntax

```
(acet-str-equal left right [matchCase])
```

23.40.4 Description

Compares two strings for equality, with optional case sensitivity.

23.40.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.40.6 Side Effects

Vendor-defined.

23.40.7 Examples

Not documented in the current category page.

23.40.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.40.9 Source Notes

- [acet-str-equal (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.41 Function Entry: ACET-STR-FIND

23.41.1 Name

'acet-str-find'

23.41.2 Class

Express Tools Function

23.41.3 Syntax

```
(acet-str-find find string [ignoreCase [useRegExp]])
```

23.41.4 Description

Finds a substring within a string. With useRegExp, find is interpreted as a regular expression.

23.41.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.41.6 Side Effects

Vendor-defined.

23.41.7 Examples

Not documented in the current category page.

23.41.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires '(acet-load-expresstools)').
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.41.9 Source Notes

- [acet-str-find (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.42 Function Entry: ACET-STR-FORMAT

23.42.1 Name

‘acet-str-format‘

23.42.2 Class

Express Tools Function

23.42.3 Syntax

```
(acet-str-format format [args ...])
```

23.42.4 Description

Formats a message with embedded arguments (printf-style).

23.42.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.42.6 Side Effects

Vendor-defined.

23.42.7 Examples

Not documented in the current category page.

23.42.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.42.9 Source Notes

- [acet-str-format (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.43 Function Entry: ACET-STR-REPLACE

23.43.1 Name

‘acet-str-replace‘

23.43.2 Class

Express Tools Function

23.43.3 Syntax

```
(acet-str-replace find replace string [ignoreCase [useRegExp [count]]])
```

23.43.4 Description

Replaces one substring with another within a string. With useRegExp, the find/replace pattern is treated as a regular expression.

23.43.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.43.6 Side Effects

Vendor-defined.

23.43.7 Examples

Not documented in the current category page.

23.43.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.43.9 Source Notes

- [acet-str-replace (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.44 Function Entry: ACET-STR-WCMATCH

23.44.1 Name

‘acet-str-wcmatch‘

23.44.2 Class

Express Tools Function

23.44.3 Syntax

`(acet-str-wcmatch string pattern [matchCase])`

23.44.4 Description

Performs wild-card matching with control over case sensitivity, similar to ‘wcmatch‘.

23.44.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.44.6 Side Effects

Vendor-defined.

23.44.7 Examples

Not documented in the current category page.

23.44.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.44.9 Source Notes

- [acet-str-wcmatch (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CE7D81F-BFE6-4221-B9FE-EC7D7680546E.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.45 Function Entry: ACET-SYS-BEEP

23.45.1 Name

‘acet-sys-beep‘

23.45.2 Class

Express Tools Function

23.45.3 Syntax

```
(acet-sys-beep sound)
```

23.45.4 Description

Produces a system beep.

23.45.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.45.6 Side Effects

Vendor-defined.

23.45.7 Examples

Not documented in the current category page.

23.45.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.45.9 Source Notes

- [acet-sys-beep (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.46 Function Entry: ACET-SYS-COMMAND

23.46.1 Name

‘acet-sys-command‘

23.46.2 Class

Express Tools Function

23.46.3 Syntax

`(acet-sys-command shell-command)`

23.46.4 Description

Runs a shell command and waits for completion before returning.

23.46.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.46.6 Side Effects

Vendor-defined.

23.46.7 Examples

Not documented in the current category page.

23.46.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.46.9 Source Notes

- [acet-sys-command (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-88F8-9F6FBBE7427A.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.47 Function Entry: ACET-SYS-FOREGROUND

23.47.1 Name

‘acet-sys-foreground‘

23.47.2 Class

Express Tools Function

23.47.3 Syntax

`(acet-sys-foreground)`

23.47.4 Description

Forces the AutoCAD window to be topmost and receive focus.

23.47.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.47.6 Side Effects

Vendor-defined.

23.47.7 Examples

Not documented in the current category page.

23.47.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.47.9 Source Notes

- [acet-sys-foreground (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-88F8-9F6FBBE7427A.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.48 Function Entry: ACET-SYS-KEYSTATE

23.48.1 Name

‘acet-sys-keystate‘

23.48.2 Class

Express Tools Function

23.48.3 Syntax

```
(acet-sys-keystate keycode)
```

23.48.4 Description

Checks the state of a virtual key.

23.48.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.48.6 Side Effects

Vendor-defined.

23.48.7 Examples

Not documented in the current category page.

23.48.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.48.9 Source Notes

- [acet-sys-keystate (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-88F8-9F6FBBE7427A.htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.49 Function Entry: ACET-SYS-LASTERR

23.49.1 Name

‘acet-sys-lasterr‘

23.49.2 Class

Express Tools Function

23.49.3 Syntax

`(acet-sys-lasterr)`

23.49.4 Description

Reports the last system error.

23.49.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.49.6 Side Effects

Vendor-defined.

23.49.7 Examples

Not documented in the current category page.

23.49.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.49.9 Source Notes

- [acet-sys-lasterr (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.50 Function Entry: ACET-SYS-PROCID

23.50.1 Name

‘acet-sys-procid‘

23.50.2 Class

Express Tools Function

23.50.3 Syntax

```
(acet-sys-procid)
```

23.50.4 Description

Returns the current process identifier.

23.50.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.50.6 Side Effects

Vendor-defined.

23.50.7 Examples

Not documented in the current category page.

23.50.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.50.9 Source Notes

- [acet-sys-procid (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF- htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.51 Function Entry: ACET-SYS-SLEEP

23.51.1 Name

‘acet-sys-sleep‘

23.51.2 Class

Express Tools Function

23.51.3 Syntax

```
(acet-sys-sleep msecs)
```

23.51.4 Description

Delays execution for the given number of milliseconds.

23.51.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.51.6 Side Effects

Vendor-defined.

23.51.7 Examples

Not documented in the current category page.

23.51.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.51.9 Source Notes

- [acet-sys-sleep (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.52 Function Entry: ACET-SYS-SPAWN

23.52.1 Name

‘acet-sys-spawn‘

23.52.2 Class

Express Tools Function

23.52.3 Syntax

```
(acet-sys-spawn flags command [argument ...])
```

23.52.4 Description

Executes an external program asynchronously.

23.52.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.52.6 Side Effects

Vendor-defined.

23.52.7 Examples

Not documented in the current category page.

23.52.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.52.9 Source Notes

- [acet-sys-spawn (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.53 Function Entry: ACET-SYS-TERM

23.53.1 Name

‘acet-sys-term‘

23.53.2 Class

Express Tools Function

23.53.3 Syntax

```
(acet-sys-term processID)
```

23.53.4 Description

Terminates the given process.

23.53.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.53.6 Side Effects

Vendor-defined.

23.53.7 Examples

Not documented in the current category page.

23.53.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.53.9 Source Notes

- [acet-sys-term (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.54 Function Entry: ACET-SYS-WAIT

23.54.1 Name

‘acet-sys-wait‘

23.54.2 Class

Express Tools Function

23.54.3 Syntax

```
(acet-sys-wait processID [time])
```

23.54.4 Description

Waits for a spawned process to complete.

23.54.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.54.6 Side Effects

Vendor-defined.

23.54.7 Examples

Not documented in the current category page.

23.54.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.54.9 Source Notes

- [acet-sys-wait (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-6BCE925F-D717-4BCF-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.55 Function Entry: ACET-UI-MESSAGE

23.55.1 Name

‘acet-ui-message‘

23.55.2 Class

Express Tools Function

23.55.3 Syntax

```
(acet-ui-message message [caption [type]])
```

23.55.4 Description

Displays a message box.

23.55.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.55.6 Side Effects

Vendor-defined.

23.55.7 Examples

Not documented in the current category page.

23.55.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.55.9 Source Notes

- [acet-ui-message (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.56 Function Entry: ACET-UI-PICKDIR

23.56.1 Name

‘acet-ui-pickdir‘

23.56.2 Class

Express Tools Function

23.56.3 Syntax

```
(acet-ui-pickdir [message [startDir [caption]]])
```

23.56.4 Description

Displays a directory-selection dialog.

23.56.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.56.6 Side Effects

Vendor-defined.

23.56.7 Examples

Not documented in the current category page.

23.56.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.56.9 Source Notes

- [acet-ui-pickdir (Express Tools, AutoLISP 2026)](https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-htm)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.57 Function Entry: ACET-UI-PROGRESS

23.57.1 Name

‘acet-ui-progress‘

23.57.2 Class

Express Tools Function

23.57.3 Syntax

```
(acet-ui-progress [label [max]])
```

23.57.4 Description

Displays a progress meter dialog. Calling ‘(acet-ui-progress current)‘ updates the meter.

23.57.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.57.6 Side Effects

Vendor-defined.

23.57.7 Examples

Not documented in the current category page.

23.57.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.57.9 Source Notes

- [acet-ui-progress (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.58 Function Entry: ACET-UI-STATUS

23.58.1 Name

‘acet-ui-status‘

23.58.2 Class

Express Tools Function

23.58.3 Syntax

```
(acet-ui-status [message [caption]])
```

23.58.4 Description

Displays a modeless status dialog.

23.58.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.58.6 Side Effects

Vendor-defined.

23.58.7 Examples

Not documented in the current category page.

23.58.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.58.9 Source Notes

- [acet-ui-status (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

23.59 Function Entry: ACET-UI-TXTED

23.59.1 Name

‘acet-ui-txted‘

23.59.2 Class

Express Tools Function

23.59.3 Syntax

```
(acet-ui-txted [text [caption [note]]])
```

23.59.4 Description

Displays a multiline text-editing dialog.

23.59.5 Return Values

Vendor-defined; see the per-category page cited under Source Notes.

23.59.6 Side Effects

Vendor-defined.

23.59.7 Examples

Not documented in the current category page.

23.59.8 Availability

- AutoCAD 2026: documented (Windows-only Express Tools; requires ‘(acet-load-expresstools)‘).
- BricsCAD V26: not documented.
- Status: AutoCAD-only.

23.59.9 Source Notes

- [acet-ui-txted (Express Tools, AutoLISP 2026)](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-F2967662-438E-425A-` `htm`)
- Status: documented (Phase 3 body fill from the Express Tools category page).

24 24 BricsCAD Extensions

24.1 Overview

This chapter enumerates the LISP functions documented by Bricsys as BricsCAD-specific extensions to AutoLISP. They have no Autodesk counterpart and a conforming AutoCAD profile must not provide them.

These functions appear on the BricsCAD-Specific LISP Functions page of the BricsCAD V26 Developer Reference and on the related Generic Properties, VLE library, VLX namespace, and Lisp Optimiser pages.

24.2 Normative Rules

- BricsCAD Extensions are BricsCAD-only.
- A conforming AutoCAD profile must not provide them.
- Where a BricsCAD-only function has an AutoCAD analogue with a different name (for example BricsCAD `position` vs AutoCAD `vl-position`, BricsCAD `remove` vs AutoCAD `vl-remove`, BricsCAD `mod` vs AutoCAD `rem`, BricsCAD `power` vs AutoCAD `expt`, the BricsCAD `vl-layerstates-*` family vs the AutoCAD `layerstate-*` family), Phase 4 cross-references must connect the two entries.

24.3 Function Entry: BCAD$LICENSELEVELS

24.3.1 Name

‘bcad$licenselevels‘

24.3.2 Class

BricsCAD Extension Function

24.3.3 Syntax

`(bcad$licenselevels)`

24.3.4 Description

Returns BricsCAD license-level information for the current installation.

24.3.5 Return Values

Vendor-defined; see Source Notes.

24.3.6 Side Effects

Vendor-defined.

24.3.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.3.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.3.9 Source Notes

- [bcad$licenselevels (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.4 Function Entry: INSPECTOR

24.4.1 Name

‘inspector‘

24.4.2 Class

BricsCAD Extension Function

24.4.3 Syntax

`(inspector [obj])`

24.4.4 Description

Opens the BricsCAD object inspector on ‘obj‘.

24.4.5 Return Values

Vendor-defined; see Source Notes.

24.4.6 Side Effects

Vendor-defined.

24.4.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.4.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V18.2).
- Status: BricsCAD-only.

24.4.9 Source Notes

- [inspector (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.5 Function Entry: ACOS

24.5.1 Name

‘acos‘

24.5.2 Class

BricsCAD Extension Function

24.5.3 Syntax

`(acos num)`

24.5.4 Description

Returns the arc cosine of a number, in radians.

24.5.5 Return Values

Vendor-defined; see Source Notes.

24.5.6 Side Effects

Vendor-defined.

24.5.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.5.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.5.9 See Also

- AutoCAD has no ‘acos‘. Portable code uses ‘(atan (sqrt (- 1.0 (* x x))) x)‘.

24.5.10 Source Notes

- [acos (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.6 Function Entry: ASIN

24.6.1 Name

‘asin‘

24.6.2 Class

BricsCAD Extension Function

24.6.3 Syntax

`(asin num)`

24.6.4 Description

Returns the arc sine of a number, in radians.

24.6.5 Return Values

Vendor-defined; see Source Notes.

24.6.6 Side Effects

Vendor-defined.

24.6.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.6.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.6.9 See Also

- AutoCAD has no ‘asin‘. Portable code uses ‘(atan x (sqrt (- 1.0 (* x x))))‘.

24.6.10 Source Notes

- [asin (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.7 Function Entry: ATANH

24.7.1 Name

‘atanh‘

24.7.2 Class

BricsCAD Extension Function

24.7.3 Syntax

`(atanh num)`

24.7.4 Description

Returns the inverse hyperbolic tangent of a number.

24.7.5 Return Values

Vendor-defined; see Source Notes.

24.7.6 Side Effects

Vendor-defined.

24.7.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.7.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.7.9 Source Notes

- [atanh (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.8 Function Entry: CEILING

24.8.1 Name

‘ceiling‘

24.8.2 Class

BricsCAD Extension Function

24.8.3 Syntax

`(ceiling num)`

24.8.4 Description

Returns the smallest integer greater than or equal to ‘num‘.

24.8.5 Return Values

Vendor-defined; see Source Notes.

24.8.6 Side Effects

Vendor-defined.

24.8.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.8.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.8.9 See Also

- AutoCAD has no ‘ceiling‘. AutoCAD’s ‘fix‘ truncates toward zero rather than rounding upward; portable code emulates ‘ceiling‘ as ‘(if (= n (fix n)) (fix n) (1+ (fix n)))‘ for non-negative ‘n‘.

24.8.10 Source Notes

- [ceiling (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.9 Function Entry: COSH

24.9.1 Name

‘cosh‘

24.9.2 Class

BricsCAD Extension Function

24.9.3 Syntax

`(cosh num)`

24.9.4 Description

Returns the hyperbolic cosine of a number.

24.9.5 Return Values

Vendor-defined; see Source Notes.

24.9.6 Side Effects

Vendor-defined.

24.9.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.9.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.9.9 Source Notes

- [cosh (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.10 Function Entry: FLOOR

24.10.1 Name

‘floor‘

24.10.2 Class

BricsCAD Extension Function

24.10.3 Syntax

```
(floor num)
```

24.10.4 Description

Returns the largest integer less than or equal to ‘num‘.

24.10.5 Return Values

Vendor-defined; see Source Notes.

24.10.6 Side Effects

Vendor-defined.

24.10.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.10.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.10.9 See Also

- AutoCAD has no ‘floor‘. For non-negative numbers, ‘fix‘ is equivalent; for negative numbers portable code uses ‘(if (= n (fix n)) (fix n) (1- (fix n)))‘.

24.10.10 Source Notes

- [floor (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.11 Function Entry: GET$_{\text{DISKSERIALID}}$

24.11.1 Name

‘get$_{\text{diskserialid}}$‘

24.11.2 Class

BricsCAD Extension Function

24.11.3 Syntax

`(get_diskserialid)`

24.11.4 Description

Returns the disk serial identifier of the current volume.

24.11.5 Return Values

Vendor-defined; see Source Notes.

24.11.6 Side Effects

Vendor-defined.

24.11.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.11.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.11.9 Source Notes

- [get$_{\text{diskserialid}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.12 Function Entry: GETPID

24.12.1 Name

‘getpid‘

24.12.2 Class

BricsCAD Extension Function

24.12.3 Syntax

`(getpid)`

24.12.4 Description

Returns the BricsCAD process identifier.

24.12.5 Return Values

Vendor-defined; see Source Notes.

24.12.6 Side Effects

Vendor-defined.

24.12.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.12.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.12.9 Source Notes

- [getpid (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.13 Function Entry: GRARC

24.13.1 Name

‘grarc‘

24.13.2 Class

BricsCAD Extension Function

24.13.3 Syntax

`(grarc center radius start-angle end-angle [color])`

24.13.4 Description

Draws a transient graphical arc in the current viewport.

24.13.5 Return Values

Vendor-defined; see Source Notes.

24.13.6 Side Effects

Vendor-defined.

24.13.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.13.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.13.9 Source Notes

- [grarc (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.14 Function Entry: GRFILL

24.14.1 Name

‘grfill‘

24.14.2 Class

BricsCAD Extension Function

24.14.3 Syntax

`(grfill points color)`

24.14.4 Description

Draws a filled polygon (transient) defined by ‘points‘.

24.14.5 Return Values

Vendor-defined; see Source Notes.

24.14.6 Side Effects

Vendor-defined.

24.14.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.14.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.14.9 Source Notes

- [grfill (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.15 Function Entry: INIT$_{\text{DIALOG}}$

24.15.1 Name

‘init$_{\text{dialog}}$‘

24.15.2 Class

BricsCAD Extension Function

24.15.3 Syntax

`(init_dialog)`

24.15.4 Description

Initializes a DCL dialog box (BricsCAD-specific helper).

24.15.5 Return Values

Vendor-defined; see Source Notes.

24.15.6 Side Effects

Vendor-defined.

24.15.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.15.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.15.9 Source Notes

- [init$_{\text{dialog}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.16 Function Entry: LISP$INSTALL

24.16.1 Name

‘lisp$install‘

24.16.2 Class

BricsCAD Extension Function

24.16.3 Syntax

`(lisp$install [flag])`

24.16.4 Description

Manages installation state of the BricsCAD LISP runtime.

24.16.5 Return Values

Vendor-defined; see Source Notes.

24.16.6 Side Effects

Vendor-defined.

24.16.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.16.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.16.9 Source Notes

- [lisp$install (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.17 Function Entry: LISP$ENABLEFASTCOM

24.17.1 Name

'lisp$enablefastcom'

24.17.2 Class

BricsCAD Extension Function

24.17.3 Syntax

`(lisp$enablefastcom [flag])`

24.17.4 Description

Enables or disables fast COM operations in BricsCAD LISP.

24.17.5 Return Values

Vendor-defined; see Source Notes.

24.17.6 Side Effects

Vendor-defined.

24.17.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.17.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.17.9 Source Notes

- [lisp$enablefastcom (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.18 Function Entry: LISP$VERSION

24.18.1 Name

‘lisp$version‘

24.18.2 Class

BricsCAD Extension Function

24.18.3 Syntax

`(lisp$version)`

24.18.4 Description

Returns the BricsCAD LISP engine version as a string or numeric tuple.

24.18.5 Return Values

Vendor-defined; see Source Notes.

24.18.6 Side Effects

Vendor-defined.

24.18.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.18.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.18.9 Source Notes

- [lisp$version (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.19 Function Entry: LOG10

24.19.1 Name

‘log10‘

24.19.2 Class

BricsCAD Extension Function

24.19.3 Syntax

```
(log10 num)
```

24.19.4 Description

Returns the base-10 logarithm of a positive number.

24.19.5 Return Values

Vendor-defined; see Source Notes.

24.19.6 Side Effects

Vendor-defined.

24.19.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.19.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.19.9 See Also

- AutoCAD has no direct base-10 logarithm; portable AutoLISP code uses ‘(/ (log x) (log 10))‘.

24.19.10 Source Notes

- [log10 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.20 Function Entry: MOD

24.20.1 Name

‘mod‘

24.20.2 Class

BricsCAD Extension Function

24.20.3 Syntax

`(mod n divisor)`

24.20.4 Description

Returns the modulo (remainder) of ‘n / divisor‘. AutoCAD analogue: ‘rem‘.

24.20.5 Return Values

Vendor-defined; see Source Notes.

24.20.6 Side Effects

Vendor-defined.

24.20.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.20.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.20.9 See Also

- AutoCAD analogue: `rem` (chapter 8). BricsCAD ships both ‘mod‘ and ‘rem‘; AutoCAD ships only ‘rem‘.

24.20.10 Source Notes

- [mod (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.21 Function Entry: POSITION

24.21.1 Name

‘position‘

24.21.2 Class

BricsCAD Extension Function

24.21.3 Syntax

`(position item lst)`

24.21.4 Description

Returns the zero-based position (or ‘nil‘) of ‘item‘ in ‘lst‘. AutoCAD analogue: ‘vl-position‘.

24.21.5 Return Values

Vendor-defined; see Source Notes.

24.21.6 Side Effects

Vendor-defined.

24.21.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.21.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.21.9 See Also

- AutoCAD analogue: `vl-position` (chapter 19).
- Pair documented in ‘vendor-inventory-2026.org‘ §10 item 7.

24.21.10 Source Notes

- [position (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.22 Function Entry: POWER

24.22.1 Name

‘power‘

24.22.2 Class

BricsCAD Extension Function

24.22.3 Syntax

`(power base exponent)`

24.22.4 Description

Raises ‘base‘ to ‘exponent‘. AutoCAD analogue: ‘expt‘.

24.22.5 Return Values

Vendor-defined; see Source Notes.

24.22.6 Side Effects

Vendor-defined.

24.22.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.22.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.22.9 See Also

- AutoCAD analogue: `expt` (chapter 8).

24.22.10 Source Notes

- [power (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.23 Function Entry: RANDOM

24.23.1 Name

‘random‘

24.23.2 Class

BricsCAD Extension Function

24.23.3 Syntax

`(random [n])`

24.23.4 Description

Returns a random integer (in ‘[0, n)‘ if ‘n‘ is supplied; otherwise from the full integer range).

24.23.5 Return Values

Vendor-defined; see Source Notes.

24.23.6 Side Effects

Vendor-defined.

24.23.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.23.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.23.9 Source Notes

- [random (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.24 Function Entry: REDRAW$_{\text{DIALOG}}$

24.24.1 Name

'redraw$_{\text{dialog}}$'

24.24.2 Class

BricsCAD Extension Function

24.24.3 Syntax

`(redraw_dialog)`

24.24.4 Description

Refreshes the currently active DCL dialog display.

24.24.5 Return Values

Vendor-defined; see Source Notes.

24.24.6 Side Effects

Vendor-defined.

24.24.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.24.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.24.9 Source Notes

- [redraw$_{\text{dialog}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.25 Function Entry: REMOVE

24.25.1 Name

‘remove‘

24.25.2 Class

BricsCAD Extension Function

24.25.3 Syntax

```
(remove item lst)
```

24.25.4 Description

Returns ‘lst‘ with all occurrences of ‘item‘ removed. AutoCAD analogue: ‘vl-remove‘.

24.25.5 Return Values

Vendor-defined; see Source Notes.

24.25.6 Side Effects

Vendor-defined.

24.25.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.25.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.25.9 See Also

- AutoCAD analogue: `vl-remove` (chapter 19).
- Pair documented in ‘vendor-inventory-2026.org‘ §10 item 7.

24.25.10 Source Notes

- [remove (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.26 Function Entry: ROUND

24.26.1 Name

‘round‘

24.26.2 Class

BricsCAD Extension Function

24.26.3 Syntax

`(round num)`

24.26.4 Description

Rounds ‘num‘ to the nearest integer.

24.26.5 Return Values

Vendor-defined; see Source Notes.

24.26.6 Side Effects

Vendor-defined.

24.26.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.26.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.26.9 See Also

- AutoCAD has no ‘round‘. Portable code uses ‘(fix (+ n 0.5))‘ for non-negative numbers (asymmetric rounding).

24.26.10 Source Notes

- [round (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.27 Function Entry: SINH

24.27.1 Name

‘sinh‘

24.27.2 Class

BricsCAD Extension Function

24.27.3 Syntax

`(sinh num)`

24.27.4 Description

Returns the hyperbolic sine of a number.

24.27.5 Return Values

Vendor-defined; see Source Notes.

24.27.6 Side Effects

Vendor-defined.

24.27.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.27.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.27.9 Source Notes

- [sinh (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.28 Function Entry: SLEEP

24.28.1 Name

‘sleep‘

24.28.2 Class

BricsCAD Extension Function

24.28.3 Syntax

`(sleep ms)`

24.28.4 Description

Pauses execution for ‘ms‘ milliseconds.

24.28.5 Return Values

Vendor-defined; see Source Notes.

24.28.6 Side Effects

Vendor-defined.

24.28.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.28.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.28.9 Source Notes

- [sleep (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.29 Function Entry: STRING-SPLIT

24.29.1 Name

‘string-split‘

24.29.2 Class

BricsCAD Extension Function

24.29.3 Syntax

`(string-split string delimiter)`

24.29.4 Description

Splits ‘string‘ on ‘delimiter‘ and returns a list of substrings.

24.29.5 Return Values

Vendor-defined; see Source Notes.

24.29.6 Side Effects

Vendor-defined.

24.29.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.29.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.29.9 Source Notes

- [string-split (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.30 Function Entry: TAN

24.30.1 Name

‘tan‘

24.30.2 Class

BricsCAD Extension Function

24.30.3 Syntax

`(tan ang)`

24.30.4 Description

Returns the tangent of an angle in radians. AutoCAD users compute as ‘(/ (sin x) (cos x))‘.

24.30.5 Return Values

Vendor-defined; see Source Notes.

24.30.6 Side Effects

Vendor-defined.

24.30.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.30.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.30.9 See Also

- AutoCAD has no ‘tan‘. Portable AutoLISP code uses ‘(/ (sin x) (cos x))‘.

24.30.10 Source Notes

- [tan (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.31 Function Entry: TANH

24.31.1 Name

‘tanh‘

24.31.2 Class

BricsCAD Extension Function

24.31.3 Syntax

`(tanh num)`

24.31.4 Description

Returns the hyperbolic tangent of a number.

24.31.5 Return Values

Vendor-defined; see Source Notes.

24.31.6 Side Effects

Vendor-defined.

24.31.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.31.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.31.9 Source Notes

- [tanh (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.32 Function Entry: UNTIL

24.32.1 Name

‘until‘

24.32.2 Class

BricsCAD Special Form

24.32.3 Syntax

```
(until test-form body...)
```

24.32.4 Description

Repeatedly evaluates ‘body‘ until ‘test-form‘ evaluates to non-nil; the last evaluation of the last form in ‘body‘ is returned. AutoCAD has no direct equivalent; idiomatic AutoLISP uses ‘(while (not test) ...)‘.

24.32.5 Return Values

Vendor-defined; see Source Notes.

24.32.6 Side Effects

Vendor-defined.

24.32.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.32.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.32.9 Source Notes

- [until (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.33 Function Entry: VL-ENABLE-USER-CANCEL

24.33.1 Name

‘vl-enable-user-cancel‘

24.33.2 Class

BricsCAD Extension Function

24.33.3 Syntax

`(vl-enable-user-cancel [flag])`

24.33.4 Description

Enables or disables user cancellation in long-running LISP routines.

24.33.5 Return Values

Vendor-defined; see Source Notes.

24.33.6 Side Effects

Vendor-defined.

24.33.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.33.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.33.9 Source Notes

- [vl-enable-user-cancel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.34 Function Entry: VL-GETCURRENTDIR

24.34.1 Name

‘vl-getcurrentdir‘

24.34.2 Class

BricsCAD Extension Function

24.34.3 Syntax

`(vl-getcurrentdir)`

24.34.4 Description

Returns the active working directory.

24.34.5 Return Values

Vendor-defined; see Source Notes.

24.34.6 Side Effects

Vendor-defined.

24.34.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.34.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.34.9 Source Notes

- [vl-getcurrentdir (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.35 Function Entry: VL-GETGEOMEXTENTS

24.35.1 Name

'vl-getgeomextents'

24.35.2 Class

BricsCAD Extension Function

24.35.3 Syntax

`(vl-getgeomextents ename)`

24.35.4 Description

Returns the geometric extents of an entity.

24.35.5 Return Values

Vendor-defined; see Source Notes.

24.35.6 Side Effects

Vendor-defined.

24.35.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.35.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.35.9 Source Notes

- [vl-getgeomextents (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.36 Function Entry: VL-GETSTARTUPDIR

24.36.1 Name

‘vl-getstartupdir‘

24.36.2 Class

BricsCAD Extension Function

24.36.3 Syntax

`(vl-getstartupdir)`

24.36.4 Description

Returns the BricsCAD startup directory.

24.36.5 Return Values

Vendor-defined; see Source Notes.

24.36.6 Side Effects

Vendor-defined.

24.36.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.36.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.36.9 Source Notes

- [vl-getstartupdir (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.37 Function Entry: VL-HIDEPROMPTMENU

24.37.1 Name

‘vl-hidepromptmenu‘

24.37.2 Class

BricsCAD Extension Function

24.37.3 Syntax

`(vl-hidepromptmenu)`

24.37.4 Description

Hides the BricsCAD prompt menu display.

24.37.5 Return Values

Vendor-defined; see Source Notes.

24.37.6 Side Effects

Vendor-defined.

24.37.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.37.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.37.9 Source Notes

- [vl-hidepromptmenu (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.38 Function Entry: VL-LAYERSTATES-DELETE

24.38.1 Name

‘vl-layerstates-delete‘

24.38.2 Class

BricsCAD Extension Function

24.38.3 Syntax

`(vl-layerstates-delete name)`

24.38.4 Description

Removes a layer state. Parallel to AutoCAD ‘layerstate-delete‘.

24.38.5 Return Values

Vendor-defined; see Source Notes.

24.38.6 Side Effects

Vendor-defined.

24.38.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.38.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.38.9 See Also

- AutoCAD analogue: `layerstate-delete` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.38.10 Source Notes

- [vl-layerstates-delete (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.39 Function Entry: VL-LAYERSTATES-LIST

24.39.1 Name

‘vl-layerstates-list‘

24.39.2 Class

BricsCAD Extension Function

24.39.3 Syntax

`(vl-layerstates-list)`

24.39.4 Description

Returns the list of layer-state names. Parallel to AutoCAD ‘layerstate-getnames‘.

24.39.5 Return Values

Vendor-defined; see Source Notes.

24.39.6 Side Effects

Vendor-defined.

24.39.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.39.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.39.9 See Also

- AutoCAD analogue: `layerstate-getnames` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.39.10 Source Notes

- [vl-layerstates-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.40 Function Entry: VL-LAYERSTATES-SETPROPERTYMASK

24.40.1 Name

‘vl-layerstates-setpropertymask‘

24.40.2 Class

BricsCAD Extension Function

24.40.3 Syntax

`(vl-layerstates-setpropertymask name mask)`

24.40.4 Description

Configures a layer state’s property mask.

24.40.5 Return Values

Vendor-defined; see Source Notes.

24.40.6 Side Effects

Vendor-defined.

24.40.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.40.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.40.9 See Also

- No direct AutoCAD analogue. AutoCAD’s parallel `layerstate-*` family does not document property-mask or description accessors as separate functions; in AutoCAD they are folded into the ‘mask‘ / state arguments of `layerstate-save` and `layerstate-getlayers`.

- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.40.10 Source Notes

- [vl-layerstates-setpropertymask (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.41 Function Entry: VL-LAYERSTATES-SETDESCRIPTION

24.41.1 Name

‘vl-layerstates-setdescription‘

24.41.2 Class

BricsCAD Extension Function

24.41.3 Syntax

```
(vl-layerstates-setdescription name description)
```

24.41.4 Description

Assigns a description to a layer state.

24.41.5 Return Values

Vendor-defined; see Source Notes.

24.41.6 Side Effects

Vendor-defined.

24.41.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.41.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.41.9 See Also

- No direct AutoCAD analogue. AutoCAD’s parallel `layerstate-*` family does not document property-mask or description accessors as separate functions; in AutoCAD they are folded into the ‘mask‘ / state arguments of `layerstate-save` and `layerstate-getlayers`.

- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.41.10 Source Notes

- [vl-layerstates-setdescription (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.42 Function Entry: VL-LAYERSTATES-RESTORE

24.42.1 Name

‘vl-layerstates-restore‘

24.42.2 Class

BricsCAD Extension Function

24.42.3 Syntax

`(vl-layerstates-restore name [vport [flags]])`

24.42.4 Description

Activates a saved layer state. Parallel to AutoCAD ‘layerstate-restore‘.

24.42.5 Return Values

Vendor-defined; see Source Notes.

24.42.6 Side Effects

Vendor-defined.

24.42.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.42.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.42.9 See Also

- AutoCAD analogue: `layerstate-restore` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.42.10 Source Notes

- [vl-layerstates-restore (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.43 Function Entry: VL-LAYERSTATES-HAS

24.43.1 Name

‘vl-layerstates-has‘

24.43.2 Class

BricsCAD Extension Function

24.43.3 Syntax

`(vl-layerstates-has name)`

24.43.4 Description

Tests whether a named layer state exists. Parallel to AutoCAD ‘layerstate-has‘.

24.43.5 Return Values

Vendor-defined; see Source Notes.

24.43.6 Side Effects

Vendor-defined.

24.43.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.43.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.43.9 See Also

- AutoCAD analogue: `layerstate-has` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.43.10 Source Notes

- [vl-layerstates-has (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.44 Function Entry: VL-LAYERSTATES-GETPROPERTYMASK

24.44.1 Name

‘vl-layerstates-getpropertymask‘

24.44.2 Class

BricsCAD Extension Function

24.44.3 Syntax

`(vl-layerstates-getpropertymask name)`

24.44.4 Description

Retrieves a layer state’s property mask.

24.44.5 Return Values

Vendor-defined; see Source Notes.

24.44.6 Side Effects

Vendor-defined.

24.44.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.44.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.44.9 See Also

- No direct AutoCAD analogue. AutoCAD’s parallel `layerstate-*` family does not document property-mask or description accessors as separate functions; in AutoCAD they are folded into the ‘mask‘ / state arguments of `layerstate-save` and `layerstate-getlayers`.

- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.44.10 Source Notes

- [vl-layerstates-getpropertymask (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.45 Function Entry: VL-LAYERSTATES-GETDESCRIPTION

24.45.1 Name

‘vl-layerstates-getdescription‘

24.45.2 Class

BricsCAD Extension Function

24.45.3 Syntax

`(vl-layerstates-getdescription name)`

24.45.4 Description

Returns a layer state’s description.

24.45.5 Return Values

Vendor-defined; see Source Notes.

24.45.6 Side Effects

Vendor-defined.

24.45.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.45.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.45.9 See Also

- No direct AutoCAD analogue. AutoCAD’s parallel `layerstate-*` family does not document property-mask or description accessors as separate functions; in AutoCAD they are folded into the ‘mask‘ / state arguments of `layerstate-save` and `layerstate-getlayers`.

- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.45.10 Source Notes

- [vl-layerstates-getdescription (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.46 Function Entry: VL-LAYERSTATES-RENAME

24.46.1 Name

‘vl-layerstates-rename‘

24.46.2 Class

BricsCAD Extension Function

24.46.3 Syntax

`(vl-layerstates-rename oldname newname)`

24.46.4 Description

Renames a layer state. Parallel to AutoCAD ‘layerstate-rename‘.

24.46.5 Return Values

Vendor-defined; see Source Notes.

24.46.6 Side Effects

Vendor-defined.

24.46.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.46.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.46.9 See Also

- AutoCAD analogue: `layerstate-rename` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.46.10 Source Notes

- [vl-layerstates-rename (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.47 Function Entry: VL-LAYERSTATES-SAVE

24.47.1 Name

‘vl-layerstates-save‘

24.47.2 Class

BricsCAD Extension Function

24.47.3 Syntax

```
(vl-layerstates-save name [mask [vport]])
```

24.47.4 Description

Saves a layer state. Parallel to AutoCAD ‘layerstate-save‘.

24.47.5 Return Values

Vendor-defined; see Source Notes.

24.47.6 Side Effects

Vendor-defined.

24.47.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.47.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.47.9 See Also

- AutoCAD analogue: `layerstate-save` (chapter 17).
- Family pair documented in ‘vendor-inventory-2026.org‘ §10 item 6.

24.47.10 Source Notes

- [vl-layerstates-save (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.48 Function Entry: VL-LIST-LOADED-LISP

24.48.1 Name

‘vl-list-loaded-lisp‘

24.48.2 Class

BricsCAD Extension Function

24.48.3 Syntax

`(vl-list-loaded-lisp)`

24.48.4 Description

Returns a list of currently loaded LISP files.

24.48.5 Return Values

Vendor-defined; see Source Notes.

24.48.6 Side Effects

Vendor-defined.

24.48.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.48.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.48.9 Source Notes

- [vl-list-loaded-lisp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.49 Function Entry: VL-SETCURRENTDIR

24.49.1 Name

‘vl-setcurrentdir‘

24.49.2 Class

BricsCAD Extension Function

24.49.3 Syntax

```
(vl-setcurrentdir directory)
```

24.49.4 Description

Changes the active working directory.

24.49.5 Return Values

Vendor-defined; see Source Notes.

24.49.6 Side Effects

Vendor-defined.

24.49.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.49.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.49.9 Source Notes

- [vl-setcurrentdir (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.50 Function Entry: VL-SHOWPROMPTMENU

24.50.1 Name

'vl-showpromptmenu'

24.50.2 Class

BricsCAD Extension Function

24.50.3 Syntax

`(vl-showpromptmenu)`

24.50.4 Description

Shows the BricsCAD prompt menu display.

24.50.5 Return Values

Vendor-defined; see Source Notes.

24.50.6 Side Effects

Vendor-defined.

24.50.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.50.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.50.9 Source Notes

- [vl-showpromptmenu (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.51 Function Entry: VL-RMDIR

24.51.1 Name

‘vl-rmdir‘

24.51.2 Class

BricsCAD Extension Function

24.51.3 Syntax

`(vl-rmdir directory)`

24.51.4 Description

Removes a directory.

24.51.5 Return Values

Vendor-defined; see Source Notes.

24.51.6 Side Effects

Vendor-defined.

24.51.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.51.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.51.9 Source Notes

- [vl-rmdir (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.52 Function Entry: VL-STRING-SPLIT

24.52.1 Name

‘vl-string-split‘

24.52.2 Class

BricsCAD Extension Function

24.52.3 Syntax

```
(vl-string-split string delimiter)
```

24.52.4 Description

Splits ‘string‘ on ‘delimiter‘, returning a list of substrings.

24.52.5 Return Values

Vendor-defined; see Source Notes.

24.52.6 Side Effects

Vendor-defined.

24.52.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.52.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.52.9 Source Notes

- [vl-string-split (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.53 Function Entry: VLA-COLLECTION->LIST

24.53.1 Name

'vla-collection->list'

24.53.2 Class

BricsCAD Extension Function

24.53.3 Syntax

```
(vla-collection->list collection)
```

24.53.4 Description

Converts an ActiveX collection into an AutoLISP list.

24.53.5 Return Values

Vendor-defined; see Source Notes.

24.53.6 Side Effects

Vendor-defined.

24.53.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.53.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.53.9 Source Notes

- [vla-collection->list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.54 Function Entry: VLA-POSTCOMMAND

24.54.1 Name

‘vla-postcommand‘

24.54.2 Class

BricsCAD Extension Function

24.54.3 Syntax

`(vla-postcommand acaddoc command-string)`

24.54.4 Description

Posts a command to the BricsCAD execution queue, to run after the current call returns.

24.54.5 Return Values

Vendor-defined; see Source Notes.

24.54.6 Side Effects

Vendor-defined.

24.54.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.54.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V22).
- Status: BricsCAD-only.

24.54.9 Source Notes

- [vla-postcommand (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.55 Function Entry: DLG-SYSVARS

24.55.1 Name

‘dlg-sysvars‘

24.55.2 Class

BricsCAD Extension Function

24.55.3 Syntax

`(dlg-sysvars)`

24.55.4 Description

Opens or manages the BricsCAD system-variables dialog.

24.55.5 Return Values

Vendor-defined; see Source Notes.

24.55.6 Side Effects

Vendor-defined.

24.55.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.55.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V20).
- Status: BricsCAD-only.

24.55.9 Source Notes

- [dlg-sysvars (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.56 Function Entry: VL-VECTOR-PROJECT-POINTTOENTITY

24.56.1 Name

‘vl-vector-project-pointtoentity‘

24.56.2 Class

BricsCAD Extension Function

24.56.3 Syntax

`(vl-vector-project-pointtoentity pt vec ename)`

24.56.4 Description

Projects a point onto an entity along a vector.

24.56.5 Return Values

Vendor-defined; see Source Notes.

24.56.6 Side Effects

Vendor-defined.

24.56.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.56.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V20).
- Status: BricsCAD-only.

24.56.9 Source Notes

- [vl-vector-project-pointtoentity (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.57 Function Entry: _VLAX-SAFEARRAY-MODE

24.57.1 Name

‘$_{\text{vlax}}$-safearray-mode‘

24.57.2 Class

BricsCAD Extension Function

24.57.3 Syntax

`(_vlax-safearray-mode [mode])`

24.57.4 Description

Configures the safe-array implementation mode for vlax operations.

24.57.5 Return Values

Vendor-defined; see Source Notes.

24.57.6 Side Effects

Vendor-defined.

24.57.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.57.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.57.9 Source Notes

- [$_{\text{vlax}}$-safearray-mode (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.58 Function Entry: VLISP-OPTIMIZER

24.58.1 Name

‘vlisp-optimizer‘

24.58.2 Class

BricsCAD Extension Function

24.58.3 Syntax

`(vlisp-optimizer [flag])`

24.58.4 Description

Toggles or queries the BricsCAD LISP optimiser; the flag is ‘T‘ to enable, ‘nil‘ to disable.

24.58.5 Return Values

Vendor-defined; see Source Notes.

24.58.6 Side Effects

Vendor-defined.

24.58.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.58.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.58.9 Source Notes

- [vlisp-optimizer (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.59 Function Entry: DICTOBJNAME

24.59.1 Name

‘dictobjname‘

24.59.2 Class

BricsCAD Extension Function

24.59.3 Syntax

`(dictobjname dictname keyname)`

24.59.4 Description

Converts a dictionary entry reference to an entity name.

24.59.5 Return Values

Vendor-defined; see Source Notes.

24.59.6 Side Effects

Vendor-defined.

24.59.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.59.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.59.9 Source Notes

- [dictobjname (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.60 Function Entry: VL-ANNOTATIVE-GET

24.60.1 Name

‘vl-annotative-get‘

24.60.2 Class

BricsCAD Extension Function

24.60.3 Syntax

`(vl-annotative-get ename)`

24.60.4 Description

Returns the annotative status of an entity.

24.60.5 Return Values

Vendor-defined; see Source Notes.

24.60.6 Side Effects

Vendor-defined.

24.60.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.60.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.60.9 Source Notes

- [vl-annotative-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.61 Function Entry: VL-ANNOTATIVE-GETSCALES

24.61.1 Name

‘vl-annotative-getscales‘

24.61.2 Class

BricsCAD Extension Function

24.61.3 Syntax

`(vl-annotative-getscales ename)`

24.61.4 Description

Returns the annotative-scales list of an entity.

24.61.5 Return Values

Vendor-defined; see Source Notes.

24.61.6 Side Effects

Vendor-defined.

24.61.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.61.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.61.9 Source Notes

- [vl-annotative-getscales (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.62 Function Entry: VL-ANNOTATIVE-SET

24.62.1 Name

‘vl-annotative-set‘

24.62.2 Class

BricsCAD Extension Function

24.62.3 Syntax

`(vl-annotative-set ename flag)`

24.62.4 Description

Sets the annotative status of an entity.

24.62.5 Return Values

Vendor-defined; see Source Notes.

24.62.6 Side Effects

Vendor-defined.

24.62.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.62.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.62.9 Source Notes

- [vl-annotative-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.63 Function Entry: VL-ANNOTATIVE-SETSCALES

24.63.1 Name

‘vl-annotative-setscales‘

24.63.2 Class

BricsCAD Extension Function

24.63.3 Syntax

```
(vl-annotative-setscales ename scales)
```

24.63.4 Description

Establishes the annotative-scales list of an entity.

24.63.5 Return Values

Vendor-defined; see Source Notes.

24.63.6 Side Effects

Vendor-defined.

24.63.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.63.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.63.9 Source Notes

- [vl-annotative-setscales (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.64 Function Entry: VL-ANNOTATIVE-SCALELIST

24.64.1 Name

‘vl-annotative-scalelist‘

24.64.2 Class

BricsCAD Extension Function

24.64.3 Syntax

`(vl-annotative-scalelist)`

24.64.4 Description

Returns the available annotative-scale list.

24.64.5 Return Values

Vendor-defined; see Source Notes.

24.64.6 Side Effects

Vendor-defined.

24.64.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.64.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.64.9 Source Notes

- [vl-annotative-scalelist (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.65 Function Entry: VL-ANNOTATIVE-SUPPORTED

24.65.1 Name

‘vl-annotative-supported‘

24.65.2 Class

BricsCAD Extension Function

24.65.3 Syntax

`(vl-annotative-supported ename)`

24.65.4 Description

Tests whether annotative behaviour is supported on an entity.

24.65.5 Return Values

Vendor-defined; see Source Notes.

24.65.6 Side Effects

Vendor-defined.

24.65.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.65.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.65.9 Source Notes

- [vl-annotative-supported (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.66 Function Entry: VL-ANNOTATIVE-REMOVE

24.66.1 Name

‘vl-annotative-remove‘

24.66.2 Class

BricsCAD Extension Function

24.66.3 Syntax

`(vl-annotative-remove ename)`

24.66.4 Description

Removes annotative behaviour from an entity.

24.66.5 Return Values

Vendor-defined; see Source Notes.

24.66.6 Side Effects

Vendor-defined.

24.66.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.66.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.66.9 Source Notes

- [vl-annotative-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.67 Function Entry: VL-ANNOTATIVE-RESET

24.67.1 Name

‘vl-annotative-reset‘

24.67.2 Class

BricsCAD Extension Function

24.67.3 Syntax

`(vl-annotative-reset ename)`

24.67.4 Description

Resets the annotative defaults for an entity.

24.67.5 Return Values

Vendor-defined; see Source Notes.

24.67.6 Side Effects

Vendor-defined.

24.67.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.67.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.67.9 Source Notes

- [vl-annotative-reset (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.68 Function Entry: VL-ANNOTATIVE-ADDSCALE

24.68.1 Name

‘vl-annotative-addscale‘

24.68.2 Class

BricsCAD Extension Function

24.68.3 Syntax

`(vl-annotative-addscale ename scale)`

24.68.4 Description

Adds a scale to an annotative entity.

24.68.5 Return Values

Vendor-defined; see Source Notes.

24.68.6 Side Effects

Vendor-defined.

24.68.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.68.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.68.9 Source Notes

- [vl-annotative-addscale (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.69 Function Entry: VL-ANNOTATIVE-REMOVESCALE

24.69.1 Name

‘vl-annotative-removescale‘

24.69.2 Class

BricsCAD Extension Function

24.69.3 Syntax

```
(vl-annotative-removescale ename scale)
```

24.69.4 Description

Removes a scale from an annotative entity.

24.69.5 Return Values

Vendor-defined; see Source Notes.

24.69.6 Side Effects

Vendor-defined.

24.69.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.69.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.69.9 Source Notes

- [vl-annotative-removescale (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.70 Function Entry: VL-SUBENT-ATPOINT

24.70.1 Name

‘vl-subent-atpoint‘

24.70.2 Class

BricsCAD Extension Function

24.70.3 Syntax

```
(vl-subent-atpoint ename pt)
```

24.70.4 Description

Returns the sub-entity of ‘ename‘ at the given point.

24.70.5 Return Values

Vendor-defined; see Source Notes.

24.70.6 Side Effects

Vendor-defined.

24.70.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.70.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V23.1).
- Status: BricsCAD-only.

24.70.9 Source Notes

- [vl-subent-atpoint (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.71 Function Entry: VL-SUBENT-SELECT

24.71.1 Name

‘vl-subent-select‘

24.71.2 Class

BricsCAD Extension Function

24.71.3 Syntax

`(vl-subent-select [prompt])`

24.71.4 Description

Enables sub-entity selection mode.

24.71.5 Return Values

Vendor-defined; see Source Notes.

24.71.6 Side Effects

Vendor-defined.

24.71.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.71.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V23.1).
- Status: BricsCAD-only.

24.71.9 Source Notes

- [vl-subent-select (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.72 Function Entry: VL-SUBENT-SSADD

24.72.1 Name

‘vl-subent-ssadd‘

24.72.2 Class

BricsCAD Extension Function

24.72.3 Syntax

```
(vl-subent-ssadd subent ss)
```

24.72.4 Description

Adds a sub-entity to a selection set.

24.72.5 Return Values

Vendor-defined; see Source Notes.

24.72.6 Side Effects

Vendor-defined.

24.72.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.72.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V23.1).
- Status: BricsCAD-only.

24.72.9 Source Notes

- [vl-subent-ssadd (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.73 Function Entry: VL-SUBENT-SSDEL

24.73.1 Name

‘vl-subent-ssdel‘

24.73.2 Class

BricsCAD Extension Function

24.73.3 Syntax

`(vl-subent-ssdel subent ss)`

24.73.4 Description

Removes a sub-entity from a selection set.

24.73.5 Return Values

Vendor-defined; see Source Notes.

24.73.6 Side Effects

Vendor-defined.

24.73.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.73.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V23.1).
- Status: BricsCAD-only.

24.73.9 Source Notes

- [vl-subent-ssdel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.74 Function Entry: VL-SUBENT-SSMEMB

24.74.1 Name

‘vl-subent-ssmemb‘

24.74.2 Class

BricsCAD Extension Function

24.74.3 Syntax

`(vl-subent-ssmemb subent ss)`

24.74.4 Description

Tests sub-entity membership in a selection set.

24.74.5 Return Values

Vendor-defined; see Source Notes.

24.74.6 Side Effects

Vendor-defined.

24.74.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.74.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V23.1).
- Status: BricsCAD-only.

24.74.9 Source Notes

- [vl-subent-ssmemb (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.75 Function Entry: VL-LOCAL-UNDO-PUSH

24.75.1 Name

‘vl-local-undo-push‘

24.75.2 Class

BricsCAD Extension Function

24.75.3 Syntax

`(vl-local-undo-push)`

24.75.4 Description

Saves the current local undo state.

24.75.5 Return Values

Vendor-defined; see Source Notes.

24.75.6 Side Effects

Vendor-defined.

24.75.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.75.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V24.1).
- Status: BricsCAD-only.

24.75.9 Source Notes

- [vl-local-undo-push (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.76 Function Entry: VL-LOCAL-UNDO-POP

24.76.1 Name

‘vl-local-undo-pop‘

24.76.2 Class

BricsCAD Extension Function

24.76.3 Syntax

`(vl-local-undo-pop)`

24.76.4 Description

Restores the previous local undo state.

24.76.5 Return Values

Vendor-defined; see Source Notes.

24.76.6 Side Effects

Vendor-defined.

24.76.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.76.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V24.1).
- Status: BricsCAD-only.

24.76.9 Source Notes

- [vl-local-undo-pop (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.77 Function Entry: VL-LOCAL-UNDO-STEPS

24.77.1 Name

‘vl-local-undo-steps‘

24.77.2 Class

BricsCAD Extension Function

24.77.3 Syntax

`(vl-local-undo-steps)`

24.77.4 Description

Returns the local undo step count.

24.77.5 Return Values

Vendor-defined; see Source Notes.

24.77.6 Side Effects

Vendor-defined.

24.77.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.77.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V24.1).
- Status: BricsCAD-only.

24.77.9 Source Notes

- [vl-local-undo-steps (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.78 Function Entry: VL-LOCAL-UNDO-RESET

24.78.1 Name

‘vl-local-undo-reset‘

24.78.2 Class

BricsCAD Extension Function

24.78.3 Syntax

`(vl-local-undo-reset)`

24.78.4 Description

Clears the local undo history.

24.78.5 Return Values

Vendor-defined; see Source Notes.

24.78.6 Side Effects

Vendor-defined.

24.78.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.78.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V24.1).
- Status: BricsCAD-only.

24.78.9 Source Notes

- [vl-local-undo-reset (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.79 Function Entry: VL-LOCAL-UNDO-CLEAR

24.79.1 Name

‘vl-local-undo-clear‘

24.79.2 Class

BricsCAD Extension Function

24.79.3 Syntax

`(vl-local-undo-clear)`

24.79.4 Description

Removes all local undo records.

24.79.5 Return Values

Vendor-defined; see Source Notes.

24.79.6 Side Effects

Vendor-defined.

24.79.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.79.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented (since V24.1).
- Status: BricsCAD-only.

24.79.9 Source Notes

- [vl-local-undo-clear (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/BricsCADLISPFunctions.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.80 Function Entry: VLE-OPTIMISER

24.80.1 Name

‘vle-optimiser‘

24.80.2 Class

BricsCAD Extension Function

24.80.3 Syntax

```
(vle-optimiser flag)
```

24.80.4 Description

Toggles the BricsCAD LISP optimiser within a ‘defun‘ block; ‘nil‘ disables the optimiser, ‘T‘ re-enables it. Used inside source files for troubleshooting.

24.80.5 Return Values

Vendor-defined; see Source Notes.

24.80.6 Side Effects

Vendor-defined.

24.80.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.80.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.80.9 Source Notes

- [vle-optimiser (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/Howitworks.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.81 Function Entry: VLE-NTH0

24.81.1 Name

‘vle-nth0‘

24.81.2 Class

BricsCAD Extension Function

24.81.3 Syntax

`(vle-nth0 lst)`

24.81.4 Description

Optimised positional accessor: returns the 0th element. Equivalent to ‘car‘.

24.81.5 Return Values

Vendor-defined; see Source Notes.

24.81.6 Side Effects

Vendor-defined.

24.81.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.81.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.81.9 Source Notes

- [vle-nth0 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.82 Function Entry: VLE-NTH1

24.82.1 Name

‘vle-nth1‘

24.82.2 Class

BricsCAD Extension Function

24.82.3 Syntax

`(vle-nth1 lst)`

24.82.4 Description

Optimised positional accessor: returns the 1st element. Equivalent to ‘cadr‘.

24.82.5 Return Values

Vendor-defined; see Source Notes.

24.82.6 Side Effects

Vendor-defined.

24.82.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.82.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.82.9 Source Notes

- [vle-nth1 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.83 Function Entry: VLE-NTH2

24.83.1 Name

‘vle-nth2‘

24.83.2 Class

BricsCAD Extension Function

24.83.3 Syntax

`(vle-nth2 lst)`

24.83.4 Description

Optimised positional accessor: returns the 2nd element. Equivalent to ‘caddr‘.

24.83.5 Return Values

Vendor-defined; see Source Notes.

24.83.6 Side Effects

Vendor-defined.

24.83.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.83.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.83.9 Source Notes

- [vle-nth2 (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.84 Function Entry: VLE-NTH3

24.84.1 Name

‘vle-nth3‘

24.84.2 Class

BricsCAD Extension Function

24.84.3 Syntax

`(vle-nth3 lst)`

24.84.4 Description

Optimised positional accessor: returns the 3rd element.

24.84.5 Return Values

Vendor-defined; see Source Notes.

24.84.6 Side Effects

Vendor-defined.

24.84.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.84.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.84.9 Source Notes

- [vle-nth3 (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.85 Function Entry: VLE-NTH4

24.85.1 Name

‘vle-nth4‘

24.85.2 Class

BricsCAD Extension Function

24.85.3 Syntax

`(vle-nth4 lst)`

24.85.4 Description

Optimised positional accessor: returns the 4th element.

24.85.5 Return Values

Vendor-defined; see Source Notes.

24.85.6 Side Effects

Vendor-defined.

24.85.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.85.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.85.9 Source Notes

- [vle-nth4 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.86 Function Entry: VLE-NTH5

24.86.1 Name

‘vle-nth5‘

24.86.2 Class

BricsCAD Extension Function

24.86.3 Syntax

`(vle-nth5 lst)`

24.86.4 Description

Optimised positional accessor: returns the 5th element.

24.86.5 Return Values

Vendor-defined; see Source Notes.

24.86.6 Side Effects

Vendor-defined.

24.86.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.86.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.86.9 Source Notes

- [vle-nth5 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.87 Function Entry: VLE-NTH6

24.87.1 Name

‘vle-nth6‘

24.87.2 Class

BricsCAD Extension Function

24.87.3 Syntax

`(vle-nth6 lst)`

24.87.4 Description

Optimised positional accessor: returns the 6th element.

24.87.5 Return Values

Vendor-defined; see Source Notes.

24.87.6 Side Effects

Vendor-defined.

24.87.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.87.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.87.9 Source Notes

- [vle-nth6 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.88 Function Entry: VLE-NTH7

24.88.1 Name

'vle-nth7'

24.88.2 Class

BricsCAD Extension Function

24.88.3 Syntax

`(vle-nth7 lst)`

24.88.4 Description

Optimised positional accessor: returns the 7th element.

24.88.5 Return Values

Vendor-defined; see Source Notes.

24.88.6 Side Effects

Vendor-defined.

24.88.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.88.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.88.9 Source Notes

- [vle-nth7 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.89 Function Entry: VLE-NTH8

24.89.1 Name

‘vle-nth8‘

24.89.2 Class

BricsCAD Extension Function

24.89.3 Syntax

```
(vle-nth8 lst)
```

24.89.4 Description

Optimised positional accessor: returns the 8th element.

24.89.5 Return Values

Vendor-defined; see Source Notes.

24.89.6 Side Effects

Vendor-defined.

24.89.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.89.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.89.9 Source Notes

- [vle-nth8 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.90 Function Entry: VLE-NTH9

24.90.1 Name

‘vle-nth9‘

24.90.2 Class

BricsCAD Extension Function

24.90.3 Syntax

`(vle-nth9 lst)`

24.90.4 Description

Optimised positional accessor: returns the 9th element.

24.90.5 Return Values

Vendor-defined; see Source Notes.

24.90.6 Side Effects

Vendor-defined.

24.90.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.90.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.90.9 Source Notes

- [vle-nth9 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.91 Function Entry: VLE-CDRASSOC

24.91.1 Name

'vle-cdrassoc'

24.91.2 Class

BricsCAD Extension Function

24.91.3 Syntax

`(vle-cdrassoc key alist)`

24.91.4 Description

Optimised idiom for '(cdr (assoc key alist))'.

24.91.5 Return Values

Vendor-defined; see Source Notes.

24.91.6 Side Effects

Vendor-defined.

24.91.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.91.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.91.9 Source Notes

- [vle-cdrassoc (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.92 Function Entry: VLE-REMOVE-LAST

24.92.1 Name

‘vle-remove-last‘

24.92.2 Class

BricsCAD Extension Function

24.92.3 Syntax

`(vle-remove-last lst)`

24.92.4 Description

Optimised drop-the-last-element on a list.

24.92.5 Return Values

Vendor-defined; see Source Notes.

24.92.6 Side Effects

Vendor-defined.

24.92.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.92.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.92.9 Source Notes

- [vle-remove-last (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.93 Function Entry: VLE-ENTGET

24.93.1 Name

‘vle-entget‘

24.93.2 Class

BricsCAD Extension Function

24.93.3 Syntax

`(vle-entget ename)`

24.93.4 Description

Optimised ‘entget‘ for known entity names.

24.93.5 Return Values

Vendor-defined; see Source Notes.

24.93.6 Side Effects

Vendor-defined.

24.93.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.93.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.93.9 Source Notes

- [vle-entget (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.94 Function Entry: VLE-TBLSEARCH

24.94.1 Name

‘vle-tblsearch‘

24.94.2 Class

BricsCAD Extension Function

24.94.3 Syntax

```
(vle-tblsearch table-name name)
```

24.94.4 Description

Optimised ‘tblsearch‘ for symbol-table lookups.

24.94.5 Return Values

Vendor-defined; see Source Notes.

24.94.6 Side Effects

Vendor-defined.

24.94.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.94.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.94.9 Source Notes

- [vle-tblsearch (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.95 Function Entry: VLE-DICTSEARCH

24.95.1 Name

‘vle-dictsearch‘

24.95.2 Class

BricsCAD Extension Function

24.95.3 Syntax

`(vle-dictsearch dict key)`

24.95.4 Description

Optimised ‘dictsearch‘ for dictionary lookups.

24.95.5 Return Values

Vendor-defined; see Source Notes.

24.95.6 Side Effects

Vendor-defined.

24.95.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.95.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.95.9 Source Notes

- [vle-dictsearch (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.96 Function Entry: VLE-DICTOBJNAME

24.96.1 Name

‘vle-dictobjname‘

24.96.2 Class

BricsCAD Extension Function

24.96.3 Syntax

`(vle-dictobjname dictname keyname)`

24.96.4 Description

Optimised dictionary-to-entity-name access.

24.96.5 Return Values

Vendor-defined; see Source Notes.

24.96.6 Side Effects

Vendor-defined.

24.96.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.96.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.96.9 Source Notes

- [vle-dictobjname (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.97 Function Entry: VLE-ENAME-VALID

24.97.1 Name

‘vle-ename-valid‘

24.97.2 Class

BricsCAD Extension Function

24.97.3 Syntax

`(vle-ename-valid ename)`

24.97.4 Description

Optimised entity-name validity check.

24.97.5 Return Values

Vendor-defined; see Source Notes.

24.97.6 Side Effects

Vendor-defined.

24.97.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.97.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.97.9 Source Notes

- [vle-ename-valid (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/OptimisedCodePatterns.htm`)

- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.98 Function Entry: VLE-FILE-ENCODING

24.98.1 Name

‘vle-file-encoding‘

24.98.2 Class

BricsCAD Extension Function

24.98.3 Syntax

`(vle-file-encoding handle [encoding])`

24.98.4 Description

Encoding declarator for BricsCAD text-file I/O. Encodings include ‘”UTF-8”‘, ‘”UTF-16LE”‘, and ‘”UNICODE”‘.

24.98.5 Return Values

Vendor-defined; see Source Notes.

24.98.6 Side Effects

Vendor-defined.

24.98.7 Examples

Not documented in the BricsCAD-Specific LISP Functions catalogue page.

24.98.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.98.9 Source Notes

- [vle-file-encoding (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/readwritetextfiles.htm)
- Status: documented (Phase 3 body fill from the BricsCAD V26 DevRef catalogue page).

24.99 Phase-2 Follow-up Catalogue (2026-04-26)

The following 377 entries were added in the Phase-2 follow-up after a live walk of the BricsCAD V26 LISP TOC. See ‘vendor-inventory-2026.org‘ §8.13 and §3.2 for the source data.

24.99.1 Standard / Miscellaneous (BricsCAD-documented) (5 entries)

24.100 Function Entry: ADS

24.100.1 Name

'ads'

24.100.2 Class

BricsCAD Standard / Misc Function

24.100.3 Syntax

`(ads)`

24.100.4 Arguments and Values

- none

24.100.5 Description

This function returns the list of loaded BRX, TX and NET application modules (obsolete, use (arx) function instead).

24.100.6 Return Values

the list of loaded BRX, TX and NET applications

24.100.7 Examples

```
(ads)
```

```
returns
```

```
("td_dynblocks_19.5_15.tx"
"plotstyleservices_19.5_15.tx" "td_sm_19.5_15.tx" ...)
```

24.100.8 Notes

obsolete, please use (arx) function instead

24.100.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.100.10 Source Notes

- [ads (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/ads.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.101 Function Entry: BPOLY

24.101.1 Name

‘bpoly‘

24.101.2 Class

BricsCAD Standard / Misc Function

24.101.3 Syntax

`(bpoly point)`

24.101.4 Arguments and Values

- point (2d/3d point list) the point to create the boundary
- around

24.101.5 Description

This function returns a polyline from the boundary found around point.

24.101.6 Return Values

ENAME of the boundary polyline (if created), NIL otherwise

24.101.7 Notes

In AutoLISP, there is only the first argument point allowed, the original syntax (bpoly point selectionset) is no longer supported;

alternative : (command ”$_{\text{-boundary}}$” ....)

24.101.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.101.9 Source Notes

- [bpoly (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/bpoly.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.102 Function Entry: FNSPLITL

24.102.1 Name

‘fnsplitl‘

24.102.2 Class

BricsCAD Standard / Misc Function

24.102.3 Syntax

```
(fnsplitl pathname)
```

24.102.4 Arguments and Values

- pathname string, any file or folder name to
- splitted

24.102.5 Description

This function returns the file or folder name pathname splitted into components as listof strings (path, name, extension).

24.102.6 Return Values

list of strings (path name extension)

24.102.7 Examples

```
(fnsplitl
"C:\MyAppFolder\BootStrap.lsp")

("C:\MyAppFolder\" "BootStrap"
".lsp")

(fnsplitl
"/MyAppFolder/BootStrap.lsp")

("/MyAppFolder/" "BootStrap"
".lsp")
```

```
(fnsplitl
"MyAppFolder\BootStrap.lsp")

("MyAppFolder\" "BootStrap"
".lsp")

(fnsplitl
"\MyAppFolder\BootStrap.lsp")

("\MyAppFolder\" "BootStrap" ".lsp")
```

24.102.8 Notes

the specified pathname must not exist on disk

24.102.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.102.10 Source Notes

- [fnsplitl (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/fnsplitl.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.103 Function Entry: VMON

24.103.1 Name

‘vmon‘

24.103.2 Class

BricsCAD Standard / Misc Function

24.103.3 Syntax

`(vmon)`

24.103.4 Arguments and Values

- none

24.103.5 Description

This function enables ”virtual memory”, obsolete, has no effect.

24.103.6 Return Values

always returns NIL

24.103.7 Notes

included in BricsCAD Lisp for backward compatibility with old AutoLISP code

24.103.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.103.9 Source Notes

- [vmon (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vmon.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.104 Function Entry: XSTRCASE

24.104.1 Name

‘xstrcase‘

24.104.2 Class

BricsCAD Standard / Misc Function

24.104.3 Syntax

```
(xstrcase "aAbBcC")
```

24.104.4 Arguments and Values

- string (string) the string to be converted to
- uppercase

24.104.5 Description

xstrcase string) This function returns the string in uppercase.

24.104.6 Return Values

string, always in uppercase

24.104.7 Examples

```
(xstrcase "aAbBcC")
 "AABBCC"
```

24.104.8 Notes

see (strcase) function

24.104.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.104.10 Source Notes

- [xstrcase (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/xstrcase.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.104.11 BricsCAD System Functions (1 entries)

24.105 Function Entry: BCAD$DISABLE-EXTENDED-ERROR

24.105.1 Name

‘bcad$disable-extended-error‘

24.105.2 Class

BricsCAD System Function

24.105.3 Syntax

```
(bcad$disable-extended-error)
```

24.105.4 Description

this variable defines, whether extended error information (i.e. error location) is shown on commandline : non-NIL extended error details are not shown (disabled) NIL other value will enable the extended error informations

24.105.5 Return Values

Vendor-defined; see Source Notes.

24.105.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.105.7 Source Notes

- [bcad$disable-extended-error (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/bcaddisableextendederror.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.105.8 Generic-Properties Helpers (2 entries)

24.106 Function Entry: ISPROPERTYVALID

24.106.1 Name

‘ispropertyvalid‘

24.106.2 Class

BricsCAD Generic-Properties Function

24.106.3 Syntax

```
(ispropertyvalid ename property)
```

24.106.4 Arguments and Values

- ename (classic entity name) object or entity ename to
- query for the given property
- property (string) the name of the property to test; can be
- specified as ”Name~NameSpace”; property name is always
- case-insensitive

24.106.5 Description

isPropertyValid (ispropertyvalid ename property) This function verifies whether the specified property is applicable for the object or entity ename.

24.106.6 Return Values

T if property is applicable for the given object or entity

24.106.7 Examples

```
(ispropertyvalid
 (entlast)  "LayerId")  T

(ispropertyvalid  (entlast)
 "LayerId~Native")  T
```

24.106.8 Notes

this is a BricsCAD specific function !

24.106.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.106.10 Source Notes

- [ispropertyvalid (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/isPropertyValid.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.107 Function Entry: LISTALLPROPERTIES

24.107.1 Name

‘listallproperties‘

24.107.2 Class

BricsCAD Generic-Properties Function

24.107.3 Syntax

`(listallproperties)`

24.107.4 Description

BricsCAD-only function. See vendor page cited under Source Notes.

24.107.5 Return Values

Vendor-defined; see Source Notes.

24.107.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.107.7 Source Notes

- [listallproperties (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/listAllProperties.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.107.8 Error-Control Toggles (3 entries)

24.108 Function Entry: VL-BT

24.108.1 Name

‘vl-bt‘

24.108.2 Class

BricsCAD Error-Control Function

24.108.3 Syntax

`(vl-bt [depth ])`

24.108.4 Arguments and Values

- depth (optional) number, specifying the depth of the
- backtrace;
- if depth
- is omitted, the default of
- -1 is
- applied

24.108.5 Description

This function returns the actual error back trace information as a list of strings.

24.108.6 Return Values

list of error backtrace information strings, as shown on commandline, optionally restricted to specified depth

24.108.7 Examples

```
(vl-bt) full error output as
list of strings

(vl-bt 5) error output as list of strings,
maximum length of the list (depth of backtrace) is 5
```

24.108.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.108.9 Source Notes

- [vl-bt (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-bt.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.109 Function Entry: VL-BT-OFF

24.109.1 Name

'vl-bt-off'

24.109.2 Class

BricsCAD Error-Control Function

24.109.3 Syntax

`(vl-bt-off)`

24.109.4 Arguments and Values

- none

24.109.5 Description

Disables the output of the Lisp function call stack (backtrace) in case of a Lisp error

24.109.6 Return Values

always T

24.109.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.109.8 Source Notes

- [vl-bt-off (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlbtoff.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.110 Function Entry: VL-BT-ON

24.110.1 Name

‘vl-bt-on‘

24.110.2 Class

BricsCAD Error-Control Function

24.110.3 Syntax

`(vl-bt-on)`

24.110.4 Arguments and Values

- none

24.110.5 Description

Enables the output of the Lisp function call stack (backtrace) in case of a Lisp error

24.110.6 Return Values

always T

24.110.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.110.8 Source Notes

- [vl-bt-on (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlbton.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.110.9 Visual LISP Extensions (BricsCAD-only) (3 entries)

24.111 Function Entry: VL-INFP

24.111.1 Name

‘vl-infp‘

24.111.2 Class

BricsCAD Visual LISP Extension Function

24.111.3 Syntax

`(vl-infp number)`

24.111.4 Arguments and Values

- number (integer or double) number to verify

24.111.5 Description

This function verifies whether the specified number is an infinite value.

24.111.6 Return Values

T if the value is infinite, NIL otherwise

24.111.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.111.8 Source Notes

- [vl-infp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-infp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.112 Function Entry: VL-INIT

24.112.1 Name

‘vl-init‘

24.112.2 Class

BricsCAD Visual LISP Extension Function

24.112.3 Syntax

`(vl-init)`

24.112.4 Arguments and Values

- none

24.112.5 Description

This function is obsolete and provided only for backward compatibility with old AutoLISP code.

24.112.6 Return Values

”void” (like (princ) function)

24.112.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.112.8 Source Notes

- [vl-init (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-init.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.113 Function Entry: VL-NANP

24.113.1 Name

'vl-nanp'

24.113.2 Class

BricsCAD Visual LISP Extension Function

24.113.3 Syntax

`(vl-nanp number)`

24.113.4 Arguments and Values

- number (integer or double) number to verify

24.113.5 Description

This function verifies whether the specified number is a valid number (or a deformed, invalid [double] number).

24.113.6 Return Values

T if the value is a valid number, NIL otherwise

24.113.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.113.8 Source Notes

- [vl-nanp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-nanp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.113.9 Viewport-Layer Property Accessors (9 entries)

24.114 Function Entry: VL-VPLAYER-GET-COLOR

24.114.1 Name

‘vl-vplayer-get-color‘

24.114.2 Class

BricsCAD Viewport-Layer Function

24.114.3 Syntax

`(vl-vplayer-get-color layer viewport)`

24.114.4 Arguments and Values

- layer specifies the layer to retrieve the
- viewport-specific color override
- can be :
- 1. name (string) the name of the
- layer
- 2. ename (ename) the entity name
- of the layer
- viewport specifies the viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

24.114.5 Description

This function returns the viewport-specific color override value for the layer.

24.114.6 Return Values

list (dxf color isOverride)

dxf 62 indicates an ACI color index, 420 indicates a truecolor value

color (integer) for dxf = 62 it is the ACI color index, for dxf = 420 it is the trucolor value

isOverride T/NIL indicates whether the returned color value is the override color, or the normal color

24.114.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.114.8 Source Notes

- [vl-vplayer-get-color (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayergetcolor.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.115 Function Entry: VL-VPLAYER-GET-LINETYPE

24.115.1 Name

‘vl-vplayer-get-linetype‘

24.115.2 Class

BricsCAD Viewport-Layer Function

24.115.3 Syntax

`(vl-vplayer-get-linetype layer viewport)`

24.115.4 Arguments and Values

- layer specifies the layer to retrieve the
- viewport-specific linetype override
- can be :
- 1. name (string) the name of the
- layer
- 2. ename (ename) the entity name
- of the layer
- viewport specifies the viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

24.115.5 Description

This function returns the viewport-specific linetype override value for the layer.

24.115.6 Return Values

list (linetype isOverride)
linetype (string) the linetype name
isOverride T indicates that the value is a true override, NIL indicates a normal property value

24.115.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.115.8 Source Notes

- [vl-vplayer-get-linetype (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayergetlinetype.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.116 Function Entry: VL-VPLAYER-GET-LINEWEIGHT

24.116.1 Name

‘vl-vplayer-get-lineweight‘

24.116.2 Class

BricsCAD Viewport-Layer Function

24.116.3 Syntax

`(vl-vplayer-get-lineweight layer viewport)`

24.116.4 Arguments and Values

- layer specifies the layer to retrieve the
- viewport-specific lineweight override
- can be :
- 1. name (string) the name of the
- layer
- 2. ename (ename) the entity name
- of the layer
- viewport specifies the viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

24.116.5 Description

This function returns the viewport-specific lineweight override value for the layer.

24.116.6 Return Values

list (lineweight isOverride)

isOverride T indicates that the value is a true override, NIL indicates a normal property value

lineweight (integer) can be (symbolic name = integer value)

acLnWt000 = 0
acLnWt005 = 5
acLnWt009 = 9
acLnWt013 = 13
acLnWt015 = 15
acLnWt018 = 18
acLnWt020 = 20
acLnWt025 = 25
acLnWt030 = 30
acLnWt035 = 35
acLnWt040 = 40
acLnWt050 = 50
acLnWt053 = 53
acLnWt060 = 60
acLnWt070 = 70
acLnWt080 = 80
acLnWt090 = 90
acLnWt100 = 100
acLnWt106 = 106
acLnWt120 = 120
acLnWt140 = 140
acLnWt158 = 158
acLnWt200 = 200
acLnWt211 = 211
acLnWtByLayer = -1
acLnWtByBlock = -2
acLnWtByLwDefault = -3

24.116.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.116.8 Source Notes

- [vl-vplayer-get-lineweight (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayergetlineweight.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.117 Function Entry: VL-VPLAYER-GET-TRANSPARENCY

24.117.1 Name

‘vl-vplayer-get-transparency‘

24.117.2 Class

BricsCAD Viewport-Layer Function

24.117.3 Syntax

`(vl-vplayer-get-transparency layer viewport)`

24.117.4 Arguments and Values

- layer specifies the layer to retrieve the
- viewport-specific transparency override
- can be :
- 1. name (string) the name of the
- layer
- 2. ename (ename) the entity name
- of the layer
- viewport specifies the viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

24.117.5 Description

This function returns the viewport-specific transparency override value for the layer.

24.117.6 Return Values

list (transparency isOverride)

transparency (integer) 0…255

isOverride T indicates that the value is a true override, NIL indicates a normal property value

24.117.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.117.8 Source Notes

- [vl-vplayer-get-transparency (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayergettransparency.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.118 Function Entry: VL-VPLAYER-SET-COLOR

24.118.1 Name

‘vl-vplayer-set-color‘

24.118.2 Class

BricsCAD Viewport-Layer Function

24.118.3 Syntax

`(vl-vplayer-set-color layer viewport color)`

24.118.4 Arguments and Values

- layer specifies one or more layer(s) to apply the
- viewport-specific color
- can be :
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer
- 3. enameList (list of name | ename)
- list of 1 or more layer entity string names and/or
- enames
- viewport specifies one or more viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

- 3. (list ename cvport ...) list ename(s)
- and/or cvport(s)
- color (integer) the index color value to be
- assigned; if NIL, any color override is removed

24.118.5 Description

This function sets the viewport-specific color override for the layer; (layer can specify one or more layers)

24.118.6 Return Values

T if the color value is successfully applied resp. removed, NIL otherwise

24.118.7 Notes

this is a BricsCAD-specific function !

24.118.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.118.9 Source Notes

- [vl-vplayer-set-color (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vl-vplayer-set-color1.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.119 Function Entry: VL-VPLAYER-SET-LINETYPE

24.119.1 Name

‘vl-vplayer-set-linetype‘

24.119.2 Class

BricsCAD Viewport-Layer Function

24.119.3 Syntax

`(vl-vplayer-set-linetype layer viewport linetype)`

24.119.4 Arguments and Values

- layer specifies one or more layer(s) to apply the
- viewport-specific linetype
- can be :
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer
- 3. enameList (list of name | ename)
- list of 1 or more layer entity string names and/or
- enames
- viewport specifies one or more viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

- 3. (list ename cvport ...) list ename(s)
- and/or cvport(s)
- linetype the
- linetype value to be assigned; if NIL, any linetype override is
- removed
- can be
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer

24.119.5 Description

This function sets the viewport-specific linetype override for the layer. (layer can specify one or more layers)

24.119.6 Return Values

T if the linetype value is successfully applied resp. removed, NIL otherwise

24.119.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.119.8 Source Notes

- [vl-vplayer-set-linetype (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayersetlinetype.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.120 Function Entry: VL-VPLAYER-SET-LINEWEIGHT

24.120.1 Name

‘vl-vplayer-set-lineweight‘

24.120.2 Class

BricsCAD Viewport-Layer Function

24.120.3 Syntax

`(vl-vplayer-set-lineweight layer viewport lineweight)`

24.120.4 Arguments and Values

- layer specifies one or more layer(s) to apply the
- viewport-specific lineweight
- can be :
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer
- 3. enameList (list of name | ename)
- list of 1 or more layer entity string names and/or
- enames
- viewport specifies one or more viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

- 3. (list ename cvport …) list ename(s)
- and/or cvport(s)
- lineweight (integer) the lineweight value to be assigned;
- if NIL, any lineweight override is removed
- can be (symbolic name = integer
- value)
- acLnWt000 = 0
- acLnWt005 = 5
- acLnWt009 = 9
- acLnWt013 = 13
- acLnWt015 = 15
- acLnWt018 = 18
- acLnWt020 = 20
- acLnWt025 = 25
- acLnWt030 = 30
- acLnWt035 = 35
- acLnWt040 = 40
- acLnWt050 = 50
- acLnWt053 = 53
- acLnWt060 = 60
- acLnWt070 = 70
- acLnWt080 = 80
- acLnWt090 = 90
- acLnWt100 = 100

- acLnWt106 = 106
- acLnWt120 = 120
- acLnWt140 = 140
- acLnWt158 = 158
- acLnWt200 = 200
- acLnWt211 = 211
- acLnWtByLayer = -1
- acLnWtByBlock = -2
- acLnWtByLwDefault = -3

24.120.5 Description

This function sets the viewport-specific lineweight override for the layer. (layer can specify one or more layers)

24.120.6 Return Values

T if the lineweight value is successfully applied, NIL otherwise

24.120.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.120.8 Source Notes

- [vl-vplayer-set-lineweight (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayersetlineweight.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.121 Function Entry: VL-VPLAYER-SET-TRANSPARENCY

24.121.1 Name

‘vl-vplayer-set-transparency‘

24.121.2 Class

BricsCAD Viewport-Layer Function

24.121.3 Syntax

`(vl-vplayer-set-transparency layer viewport transparency)`

24.121.4 Arguments and Values

- layer specifies one or more layer(s) to apply the
- viewport-specific transparency
- can be :
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer
- 3. enameList (list of name | ename)
- list of 1 or more layer entity string names and/or
- enames
- viewport specifies one or more viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

- 3. (list ename cvport …) list ename(s)
- and/or cvport(s)
- transparency (integer or double) the transparency value to
- be assigned; if NIL, any transparency override is
- removed
- can be
- 1. integer 0…255
- 2. double 0…1.0

24.121.5 Description

This function sets the viewport-specific transparency override for the layer. (layer can specify one or more layers)

24.121.6 Return Values

T if the transparency value is successfully applied, NIL otherwise

24.121.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.121.8 Source Notes

- [vl-vplayer-set-transparency (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayersettransparency.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.122 Function Entry: VL-VPLAYER-SET-TRUECOLOR

24.122.1 Name

‘vl-vplayer-set-truecolor‘

24.122.2 Class

BricsCAD Viewport-Layer Function

24.122.3 Syntax

`(vl-vplayer-set-truecolor layer viewport rgbColor)`

24.122.4 Arguments and Values

- layer specifies one or more layer(s) to apply the
- viewport-specific truecolor
- can be :
- 1. name (string) the name of 1
- layer
- 2. ename (ename) the entity name
- of the layer
- 3. enameList (list of name | ename)
- list of 1 or more layer entity string names and/or
- enames
- viewport specifies one or more viewport
- can be
- 1. ename (ename) the viewport’s entity
- name
- 2. cvport (integer) the viewport’s
- CVPORT viewport number

- 3. (list ename cvport …) list ename(s)
- and/or cvport(s)
- rgbColor (integer) the truecolor value to be assigned;
- if NIL, any truecolor override is removed

24.122.5 Description

This function sets the viewport-specific rgbColor truecolor override for the layer; (layer can specify one or more layers)

24.122.6 Return Values

T if the truecolor value is successfully applied resp. removed, NIL otherwise

24.122.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.122.8 Source Notes

- [vl-vplayer-set-truecolor (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlvplayersettruecolor.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.122.9 ActiveX Extensions (BricsCAD-only vlax-*) (4 entries)

24.123 Function Entry: VLAX-2D-POINT

24.123.1 Name

‘vlax-2d-point‘

24.123.2 Class

BricsCAD ActiveX Extension Function

24.123.3 Syntax

`(vlax-2d-point lst)`

24.123.4 Arguments and Values

- lst (list) list of 2 double values for X and Y
- coordinate
- or
- x (double)
- the X coordinate value
- y (double)
- the Y coordinate value
- z (double)
- the Z coordinate value

24.123.5 Description

These functions creates a COM VARIANT object for the 2D point.

24.123.6 Return Values

VARIANT; returns a VARIANT, containing a SafeArray with 2 double values

24.123.7 Examples

```
(vlax-2d-point 10.5 10.6)
 returns #<variant 8197 ...>

(vlax-2d-point '(10.6 10.7))  returns
#<variant 8197 ...>
```

24.123.8 Notes

this function is specific for BricsCAD, not available in AutoLISP !

24.123.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.123.10 Source Notes

- [vlax-2d-point (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlax-2d-point.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.124 Function Entry: VLAX-CURVE-GETPERIMETER

24.124.1 Name

‘vlax-curve-getperimeter‘

24.124.2 Class

BricsCAD ActiveX Extension Function

24.124.3 Syntax

```
(vlax-curve-getperimeter curveEnt)
```

24.124.4 Arguments and Values

- curveEnt (ename or vla-object) the COM object of any ”curve”
- entity

24.124.5 Description

This function returns the perimeter of the curveEnt curve entity.

24.124.6 Return Values

double; the perimeter of the curve or NIL

24.124.7 Examples

```
last entity is a CIRCLE with
radius = 1.0 :

(setq ent (vlax-ename->vla-object
(entlast))) #<VLA-OBJECT IAcadCircle
000000004E8E3920>

(vlax-curve-getperimeter ent)
 6.28318530717959
```

24.124.8 Notes

this function is specific for BricsCAD, not available in AutoLISP !

24.124.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.124.10 Source Notes

- [vlax-curve-getperimeter (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlax-curve-getperimeter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.125 Function Entry: VLAX-QUEUEEXPR

24.125.1 Name

'vlax-queueexpr'

24.125.2 Class

BricsCAD ActiveX Extension Function

24.125.3 Syntax

```
(vlax-queueexpr commandOrLispString)
```

24.125.4 Arguments and Values

- commandOrLispString
- (string) command name or Lisp expression
- to be sent to the commands' queue

24.125.5 Description

This function sends the commandOrLispString to commands' queue for asynchronous execution.

24.125.6 Return Values

always NIL

24.125.7 Examples

```
(vlax-queueexpr "(alert \"*
DONE *\")
")

puts the (alert "* DONE *)
statement
```

24.125.8 Notes

note : usually, as with interactive input, a trailing space or <return> is required to start the execution at command line;
note : the commandOrLispString is not echoed to command line;
note : this is a BricsCAD specific function, please use with care !

24.125.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.125.10 Source Notes

- [vlax-queueexpr (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlax-queueexpr.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.126 Function Entry: VLAX-SAFEARRAY-VALUE

24.126.1 Name

‘vlax-safearray-value‘

24.126.2 Class

BricsCAD ActiveX Extension Function

24.126.3 Syntax

```
(vlax-safearray-value saVar)
```

24.126.4 Arguments and Values

- saVar (SafeArray) the SafeArray to query for its
- value

24.126.5 Description

This function returns the data contained in saVar SafeArray as a Lisp list.

24.126.6 Return Values

list; for multi-dimensional SafeArray, it returns a list of sub-lists

24.126.7 Examples

```
(setq sa (vlax-make-safearray
vlax-vbDouble '(0 . 1) '(0 . 2)))
 #<safearray...>

(vlax-safearray-fill sa '((1.0 2.0 3.0) (10.0
11.0 12.0)))  #<safearray...>

(vlax-safearray-value sa)  ((1.0 2.0 3.0)
(10.0 11.0 12.0))
```

24.126.8 Notes

this function is a BricsCAD specific function, not available in AutoCAD AutoLISP - please use with case !

same as (vlax-safearray->list), provided for consistency + similarity with (vlax-variant-value)

24.126.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.126.10 Source Notes

- [vlax-safearray-value (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlax-safearray-value.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.126.11 Reactor Extensions (BricsCAD-only vlr-*) (3 entries)

24.127 Function Entry: VLR-DOCUMENT

24.127.1 Name

‘vlr-document‘

24.127.2 Class

BricsCAD Reactor Extension Function

24.127.3 Syntax

```
(vlr-document reactor)
```

24.127.4 Arguments and Values

- reactor (any reactor object) the reactor to query for
- attached custom data

24.127.5 Description

This function returns the document associated with the specified reactor object.

24.127.6 Return Values

IAcadDocument object, the associated document

24.127.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.127.8 Source Notes

- [vlr-document (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlr-document.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.128 Function Entry: VLR-PERS-DICTNAME

24.128.1 Name

‘vlr-pers-dictname‘

24.128.2 Class

BricsCAD Reactor Extension Function

24.128.3 Syntax

```
(vlr-pers-dictname)
```

24.128.4 Arguments and Values

- none

24.128.5 Description

This function returns the name of the dictionary used to store persistent Lisp reactors.

24.128.6 Return Values

string; the name of the dictionary used to store persistent reactors

24.128.7 Notes

see also (vlr-pers) and (vlr-pers-list) function

24.128.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.128.9 Source Notes

- [vlr-pers-dictname (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlr-pers-dictname.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.129 Function Entry: VLR-REACTION-NAMES

24.129.1 Name

‘vlr-reaction-names‘

24.129.2 Class

BricsCAD Reactor Extension Function

24.129.3 Syntax

```
(vlr-reaction-names reactorType)
```

24.129.4 Arguments and Values

- reactorType
- (symbol) specifies the reactor
- type to be queried;
- can be one of these symbols :
- :VLR-AcDb-Reactor
- database reactor

24.129.5 Description

This function returns a list of all available events for reactors of reactorType.

24.129.6 Return Values

list of reactor type specific events

24.129.7 Examples

```
(vlr-reaction-names
':vlr-object-reactor)  returns :

(:VLR-SUBOBJMODIFIED :VLR-COPIED
:VLR-OBJECTCLOSED :VLR-REAPPENDED :VLR-UNAPPENDED
:VLR-MODIFIEDXDATA :VLR-MODIFYUNDONE :VLR-MODIFIED
```

```
:VLR-OPENEDFORMODIFY :VLR-GOODBYE :VLR-UNERASED :VLR-ERASED
:VLR-CANCELLED)
```

24.129.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.129.9 Source Notes

- [vlr-reaction-names (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlr-reaction-names.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.129.10 Obsolete VLX-Namespace Functions (3 entries)

24.130 Function Entry: VLISP-EXPORT-SYMBOL

24.130.1 Name

‘vlisp-export-symbol‘

24.130.2 Class

BricsCAD VLX Namespace Function (obsolete)

24.130.3 Syntax

```
(vlisp-export-symbol symbolOrList)
```

24.130.4 Arguments and Values

- symbolOrList
- a single Lisp symbol (string or symbol)
- or a list of Lisp symbols

24.130.5 Description

This function exports the specified symbol / variable (or the list of symbols / variables) to the global document namespace (where normal Lisp code is operating); then those symbols / variables are accessible from any other normal Lisp code and function. It exports the actual value of the symbol / variable, not a ”reference” to the symbol / variable ! If the value is changed within the VLX NameSpace code, the symbol / variable need to be re-exported again. see (vl-doc-set)

24.130.6 Return Values

the value of the last evaluated symbol / variable

24.130.7 Examples

```
(vlisp-export-symbol 'MyVar1)
   returns the value of "MyVar1"

(vlisp-export-symbol (list 'MyVar1 'MyVar2
```

```
'MyVar3))    returns the value of "MyVar3"
```

24.130.8 Notes

if (vlisp-export-symbol) is called from the global document namespace, then it does nothing and returns the value of the last evaluated symbol / variable;

obsolete function ! please use (vl-doc-set)

24.130.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.130.10 Source Notes

- [vlisp-export-symbol (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlispexportsymbol.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.131 Function Entry: VLISP-IMPORT-EXSUBRS

24.131.1 Name

‘vlisp-import-exsubrs‘

24.131.2 Class

BricsCAD VLX Namespace Function (obsolete)

24.131.3 Syntax

`(vlisp-import-exsubrs)`

24.131.4 Arguments and Values

- filename a single Lisp symbol (string or symbol) or a list of
- Lisp symbols
- function1 function2 ...
- string names of functions to be
- imported

24.131.5 Description

(vlisp-import-exsubrs (list filename ”function1” ”function2” ...)) This function loads the specified ARX/BRX module filename and imports the specified functions into current VLX namespace; those functions function1, function2, ... should be exported by the ARX/BRX module. see (vl-arx-import)

24.131.6 Return Values

the filename if loaded successfully, or NIL

24.131.7 Notes

In BricsCAD LISP, all functions exported by ARX/BRX modules are always exported to the the Lisp core namespace, while AutoCAD exports those functions to the document’s namespace.

obsolete function ! please use (vl-arx-import)

24.131.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.131.9 Source Notes

- [vlisp-import-exsubrs (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlispimportexsubrs.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.132 Function Entry: VLISP-IMPORT-SYMBOL

24.132.1 Name

'vlisp-import-symbol'

24.132.2 Class

BricsCAD VLX Namespace Function (obsolete)

24.132.3 Syntax

```
(vlisp-import-symbol symbolOrList)
```

24.132.4 Arguments and Values

- symbolOrList
- a single Lisp symbol (string or symbol)
- or a list of Lisp symbols

24.132.5 Description

This function imports the specified symbol / variable (or the list of symbols / variables) from the global document namespace (where normal Lisp code is operating) to the current VLX namespace. It imports the actual value of the symbol / variable, not a "reference" to the symbol / variable ! If the value is changed in global document namespace, the symbol / variable need to be re-imported again. see (vl-doc-ref)

24.132.6 Return Values

the value of the last evaluated symbol / variable

24.132.7 Examples

```
(vlisp-import-symbol 'MyVar1)
   returns the value of "MyVar1"

(vlisp-import-symbol (list 'MyVar1 'MyVar2
'MyVar3))    returns the value of "MyVar3"
```

24.132.8 Notes

if (vlisp-import-symbol) is called from the global document namespace, then it does nothing and returns the value of the last evaluated symbol / variable;

obsolete function ! please use (vl-doc-ref)

24.132.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.132.10 Source Notes

- [vlisp-import-symbol (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlispimportsymbol.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.132.11 VLE LISP Function Library (105 entries)

24.133 Function Entry: VLE-ACI2RGB

24.133.1 Name

‘vle-aci2rgb‘

24.133.2 Class

BricsCAD VLE Library Function

24.133.3 Syntax

```
(vle-aci2rgb color)
```

24.133.4 Arguments and Values

- color an integer number specifying the ACI color index
- value (1… 256)

24.133.5 Description

Returns the Red/Green/Blue true color values for specified ACI color index as list (Red Green Blue).

24.133.6 Return Values

returns a list of format /Red Green Blue)

24.133.7 Examples

```
(vle-aci2rgb 1)  returns
-(255 0 0)

(vle-aci2rgb 5)  returns  (0 0
255)

(vle-aci2rgb 128)  returns  (0 76
57)
```

24.133.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.133.9 Source Notes

- [vle-aci2rgb (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleaci2rgb.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.134 Function Entry: VLE-ALERT

24.134.1 Name

‘vle-alert‘

24.134.2 Class

BricsCAD VLE Library Function

24.134.3 Syntax

`(vle-alert title msg flags)`

24.134.4 Arguments and Values

- title the title for the message box (the caption
- string)
- msg the
- message to be displayed; will be word-wrapped
- flags combination (addition) of integers specifying
- behaviour
- flags for button :
- 0 MB_{OK}
- 1 $MB_{OKCANCEL}$
- 2 $MB_{ABORTRETRYIGNORE}$
- 3 $MB_{YESNOCANCEL}$
- 4 MB_{YESNO}
- 5 $MB_{RETRYCANCEL}$
- 6 $MB_{CANCELTRYCONTINUE}$
- flags for icons :
- 16 $MB_{ICONHAND}$ / $MB_{ICONSTOP}$

- 32 $\mathrm{MB}_{\text{ICONQUESTION}}$
- 48 $\mathrm{MB}_{\text{ICONEXCLAMATION}}$
- 64 $\mathrm{MB}_{\text{ICONASTERISK}}$ /
- $\mathrm{MB}_{\text{ICONINFORMATION}}$
- flags for default button (for <return>
- input) :
- 0
- $\mathrm{MB}_{\text{DEFBUTTON1}}$
- 256 $\mathrm{MB}_{\text{DEFBUTTON2}}$
- 512 $\mathrm{MB}_{\text{DEFBUTTON3}}$
- 768 $\mathrm{MB}_{\text{DEFBUTTON4}}$
- flags for behaviour :
- 0
- $\mathrm{MB}_{\text{APPLMODAL}}$

24.134.5 Description

shows a message box, which can be customised in wide range (based on Windows' ::MessageBox() function) Author: Lee Mac, Copyright © 2012 - www.lee-mac.com, original code: `http://lee-mac.com/popup.html`

24.134.6 Return Values

number of button which was used to finish the dialogue
1 OK button
2 Cancel button
3 Abort button
4 Retry button
5 Ignore button
6 Yes button
7 No button
10 Try Again button
11 Continue button

24.134.7 Examples

```
(vle-alert "My CAD App" "Dear
Customer ...." (+ 4 32 4096))

will show Yes+No button, the question mark
icon, as system modal dialogue
```

24.134.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.134.9 Source Notes

- [vle-alert (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlealert.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.135 Function Entry: VLE-APPEND

24.135.1 Name

‘vle-append‘

24.135.2 Class

BricsCAD VLE Library Function

24.135.3 Syntax

```
(vle-append lst item)
```

24.135.4 Arguments and Values

- lst any valid list
- item any
- valid item to be appended

24.135.5 Description

append ’item’ to the list ’lst’; ’item’ can be any expression; fast replacement for (append lst (list item))

24.135.6 Return Values

the list lst appended by item

24.135.7 Examples

```
(vle-append '(1 2 3) '(4 5
6)) returns (1 2 3 (4 5 6))
```

24.135.8 Notes

other than (append) function, the item to be appended does not need to be wrapped into extra (list …) around !

24.135.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.135.10 Source Notes

- [vle-append (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleappend.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.136 Function Entry: VLE-ATOI32

24.136.1 Name

‘vle-atoi32‘

24.136.2 Class

BricsCAD VLE Library Function

24.136.3 Syntax

```
(vle-atoi32 numstr)
```

24.136.4 Arguments and Values

- numstr an integer number string

24.136.5 Description

returns the 32 bit integer value of input string; if input value is above 32 bit range (a 64 bit number string) then this is truncated to a 32 bit value, to match the behaviour of other CAD systems using 32 bit integers even in x64 builds

24.136.6 Return Values

always returns a 32 bit integer value

24.136.7 Examples

```
(vle-atoi32
"2684354560")

returns -1610612736 instead of
2684354560
```

24.136.8 Notes

this is effective only with Bricscad x64, otherwise it does nothing;

as BricsCAD x64 allows 64 bit integers, there is a minor conflict in compatibility with other CAD systems here;

if a Lisp application needs that incorrect behaviour of other CAD systems, then this function can help as workaround

24.136.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.136.10 Source Notes

- [vle-atoi32 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleatoi32.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.137 Function Entry: VLE-CADRASSOC

24.137.1 Name

‘vle-cadrassoc‘

24.137.2 Class

BricsCAD VLE Library Function

24.137.3 Syntax

```
(vle-cadrassoc key lst)
```

24.137.4 Arguments and Values

- key key value for the dotted pair value to search
- for
- lst list of
- dotted pairs

24.137.5 Description

searches ’lst’ list for a dotted-pair with key ’key’ and returns the (cadr) of its value;

24.137.6 Return Values

the (cadr) of the value of a dotted pair (key . value), if a dotted pair with key key, if present in the list lst;
NIL if not present

24.137.7 Examples

```
(vle-cadrassoc 11 '((1 (0 0
0)) (2 (1 1 1)) (11 (1 2 3))))

returns (1 2 3)
```

24.137.8 Notes

replacement for (cadr (assoc key lst))

24.137.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.137.10 Source Notes

- [vle-cadrassoc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-cadrassoc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.138 Function Entry: VLE-CEILING

24.138.1 Name

‘vle-ceiling‘

24.138.2 Class

BricsCAD VLE Library Function

24.138.3 Syntax

```
(vle-ceiling x)
```

24.138.4 Arguments and Values

- x any number (integer or double)

24.138.5 Description

returns the smallest integer that is not less than x

24.138.6 Return Values

the smallest integer that is not less than x

24.138.7 Examples

```
(vle-ceiling  3.4) =>
 4

(vle-ceiling -3.4) => -3
```

24.138.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.138.9 Source Notes

- [vle-ceiling (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleceiling.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.139 Function Entry: VLE-COLLECTION->LIST

24.139.1 Name

‘vle-collection->list‘

24.139.2 Class

BricsCAD VLE Library Function

24.139.3 Syntax

```
(vle-collection->list collection)
```

24.139.4 Arguments and Values

- collection
- the COM collection object to get the
- items as normal list from

24.139.5 Description

creates a normal list of items from given COM collection collection

24.139.6 Return Values

list of items, or NIL if collection is empty

24.139.7 Examples

```
(vle-collection->list
(vla-get-layers (vlax-get-acad-object)))

returns (#<VLA-OBJECT IAcadLayer>
#<VLA-OBJECT IAcadLayer> ...)
```

24.139.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.139.9 Source Notes

- [vle-collection->list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlecollectionlist.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.140 Function Entry: VLE-COMPILE-SHAPE

24.140.1 Name

‘vle-compile-shape‘

24.140.2 Class

BricsCAD VLE Library Function

24.140.3 Syntax

```
(vle-compile-shape shapefile)
```

24.140.4 Arguments and Values

- shapefile (string) the .shp file to be compiled into .shx
- format

24.140.5 Description

Compiles the specified symbol font file ’shapefile’ from .shp into .shx format.

24.140.6 Return Values

(string) name of the compiled .shx file, or NIL of compilation fails

24.140.7 Examples

```
(vle-file->list
"MyData.dat" ";")

loads all file lines into returned list, and
ignores all lines starting with ";" character (omitted)
```

24.140.8 Notes

if does not contain a path specification, the file is searched in SupportPaths sequence the normal way like (findfile);

the .shx file is always generated in same folder as the input .shp file

24.140.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.140.10 Source Notes

- [vle-compile-shape (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-compile-shape.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.141 Function Entry: VLE-CURVE-GETPERIMETER

24.141.1 Name

‘vle-curve-getperimeter‘

24.141.2 Class

BricsCAD VLE Library Function

24.141.3 Syntax

```
(vle-curve-getperimeter entity)
```

24.141.4 Arguments and Values

- entity can be an ENAME or VLA-OBJECT

24.141.5 Description

returns the curve length (perimeter) for the specified entity

24.141.6 Return Values

’length’ (perimeter) value of any ’curve’ entity, or NIL in case of an (unexpected) error

24.141.7 Examples

```
(vle-curve-getperimeter
(entlast))

returns 2.0 for a LINE with (0,0,0) -
(2,0,0)
```

24.141.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.141.9 Source Notes

- [vle-curve-getperimeter (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlecurvegetperimeter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.142 Function Entry: VLE-DICTIONARY-LIST

24.142.1 Name

‘vle-dictionary-list‘

24.142.2 Class

BricsCAD VLE Library Function

24.142.3 Syntax

```
(vle-dictionary-list dict subDict asNames)
```

24.142.4 Arguments and Values

- dict entity name or string of dictionary to be
- scanned
- subDict if
- not NIL, the name of the sub-dictionary to be scanned; if NIL, the
- ’dict’ is scanned
- asNames specifies whether item keys are listed or the entity
- names of items; ’T’ specifies item keys

24.142.5 Description

scans the specified dictionary or sub-dictionary and lists all entries

24.142.6 Return Values

list of dictionary entries, as list of key strings or entity names

24.142.7 Examples

```
(vle-dictionary-list
(namedobjdict) nil T) => ("ACAD_COLOR" "ACAD_GROUP"
"ACAD_LAYOUT" ...)

(vle-dictionary-list (namedobjdict)
```

```
"ACAD_TABLESTYLE" T) => ("Standard")

(vle-dictionary-list (namedobjdict)
"ACAD_TABLESTYLE" nil) => (<Entity name: nnnnnnnn>)
```

24.142.8 Notes

depending on 'subDict', the function scans either the given 'dict' or the 'subDict' dictionary for its entries;

supported dictionary name strings (case-insensitive) : $ACAD_{COLOR}$

24.142.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.142.10 Source Notes

- [vle-dictionary-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vledictionarylist.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.143 Function Entry: VLE-DISPLAYPAUSE

24.143.1 Name

‘vle-displaypause‘

24.143.2 Class

BricsCAD VLE Library Function

24.143.3 Syntax

```
(vle-displaypause pauseDisplay)
```

24.143.4 Arguments and Values

- pauseDisplay
- T disables display updates; NIL
- enables display updates

24.143.5 Description

enables or disables display updates - when disabled, any update of the screen is suppressed until enabled again

24.143.6 Return Values

T when display updates are disabled

NIL when display updates are enabled

24.143.7 Notes

this function can be nested - but each call with ’T’ should be paired by a call with ’NIL’;

when Lisp execution is finished, the display will be automatically enabled for safety

24.143.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.143.9 Source Notes

- [vle-displaypause (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vledisplaypause.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.144 Function Entry: VLE-DISPLAYUPDATE

24.144.1 Name

‘vle-displayupdate‘

24.144.2 Class

BricsCAD VLE Library Function

24.144.3 Syntax

`(vle-displayupdate)`

24.144.4 Arguments and Values

- none

24.144.5 Description

triggers an immediate display update/refresh; temporary graphics is not erased, other pending updates are processed

24.144.6 Return Values

T always

24.144.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.144.8 Source Notes

- [vle-displayupdate (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vledisplayupdate.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.145 Function Entry: VLE-EDITTEXTINPLACE

24.145.1 Name

‘vle-edittextinplace‘

24.145.2 Class

BricsCAD VLE Library Function

24.145.3 Syntax

```
(vle-edittextinplace ename)
```

24.145.4 Arguments and Values

- ename entity name to be edited; if NIL is provided, the
- function asks the user to select a normal or nested
- entity

24.145.5 Description

Starts inplace-editing for several entity types containing text values; works in 'modal' way, (vle-edittextinplace) returns **after** editing is finished.

24.145.6 Return Values

(string) the edited text value
NIL if editing failed or was cancelled

24.145.7 Examples

```
(vle-edittextinplace-en)
=> returns "edited text value"
```

24.145.8 Notes

supported entity types

Text MText Dimension MLeader Attribute AttributeDefinition

24.145.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.145.10 Source Notes

- [vle-edittextinplace (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-edittextinplace.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.146 Function Entry: VLE-ENABLESERVERBUSY

24.146.1 Name

‘vle-enableserverbusy‘

24.146.2 Class

BricsCAD VLE Library Function

24.146.3 Syntax

`(vle-enableserverbusy enableFlag)`

24.146.4 Arguments and Values

- enableFlag
- T enables, NIL disables the ”Server Busy
- …” message box

24.146.5 Description

enables or disables the ”Server Busy …” message box; There are 2 similar ”Server Busy …” and ”Server does not respond …” messages, triggered and shown from Microsoft OLE/COM, when an out-of-process ”server application” does not respond within some seconds, and is usually that is a false-positive, but annoying always; this function allows to disable both message boxes.

24.146.6 Return Values

always T

24.146.7 Examples

```
(vle-enableserverbusy nil)
disables "Server Busy ..." message box

(vle-enableserverbusy t)    enables
"Server Busy ..." message box
```

24.146.8 Notes

this is effective only with BricsCAD, otherwise it is a No-Operation;
(on Linux and Mac platform, this function does nothing)

24.146.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.146.10 Source Notes

- [vle-enableserverbusy (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-enableserverbusy.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.147 Function Entry: VLE-ENAMEP

24.147.1 Name

‘vle-enamep‘

24.147.2 Class

BricsCAD VLE Library Function

24.147.3 Syntax

`(vle-enamep obj)`

24.147.4 Arguments and Values

- obj any Lisp item to be verified

24.147.5 Description

predicator function to determine whether ’obj’ is of ENAME type

24.147.6 Return Values

T - if ’obj’ is a ENAME object
NIL - if ’obj’ is not a ENAME object

24.147.7 Notes

replacement for : (= (type obj) ’ENAME) and similar

24.147.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.147.9 Source Notes

- [vle-enamep (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleenamep.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.148 Function Entry: VLE-END-TRANSACTION

24.148.1 Name

‘vle-end-transaction‘

24.148.2 Class

BricsCAD VLE Library Function

24.148.3 Syntax

```
(vle-end-transaction)
```

24.148.4 Arguments and Values

- none

24.148.5 Description

finishes the active database transaction for current drawing

24.148.6 Return Values

T when transaction is active, NIL otherwise

24.148.7 Examples

```
(vle-end-transaction)

T : transaction is active

NIL : transaction is not active
```

24.148.8 Notes

see (vle-start-transaction)

when outermost Lisp execution finishes, a pending / open transaction is automatically closed (also when Lisp code errors)

calling (vle-end-transaction) when no transaction is active does not harm at all; it is a No-Operation then

24.148.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.148.10 Source Notes

- [vle-end-transaction (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleendtransaction.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.149 Function Entry: VLE-ENTGET-M

24.149.1 Name

‘vle-entget-m‘

24.149.2 Class

BricsCAD VLE Library Function

24.149.3 Syntax

```
(vle-entget-m dxfCodes ename)
```

24.149.4 Arguments and Values

- dxfCodes list of DXF group codes to be retrieved
- ename the
- entity’s ename

24.149.5 Description

retrieves the values for specified DXF group codes ’dxfCodes’ for given entity ’ename’, returns a list of (dxf . value) assoc items; DXF group codes are specified as a simple list of DXF codes;

24.149.6 Return Values

list of assoc items ’((dxf1 . value1)(dxf2 . value2) ...)

24.149.7 Examples

```
(vle-entget-m '(8 62 39)
entity)

or

(setq dxf6 6 dxf52 52 dxf39 39)

(vle-entget-m (list dxf8 dxf62 dxf39) entity)
- if variables are used
```

```
returns '((8 . "LayerX")(62 . 4)(39 .
0.0))
```

24.149.8 Notes

(vle-entget-m) does always return non-NIL values for omitted ”default” values (ByLayer etc.);

if the entity does not support particular dxf code, that dxf code is not part of returned list

24.149.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.149.10 Source Notes

- [vle-entget-m (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleentgetm.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.150 Function Entry: VLE-ENTGET-MASSOC

24.150.1 Name

‘vle-entget-massoc‘

24.150.2 Class

BricsCAD VLE Library Function

24.150.3 Syntax

```
(vle-entget-massoc dxfCode ename)
```

24.150.4 Arguments and Values

- dxfCode key to be used to parse the (entget) list of
- ’ename’
- ename entity
- name

24.150.5 Description

returns a list of all values using same key ’dxfCode’ in the (entget) list of an entity ’ename’

24.150.6 Return Values

a list of all cdr values of assoc items with key ’dxfCode’;
all values are in their original sequence

24.150.7 Examples

```
assume 'pline' to be a
LWPOLYLINE with 4 vertices

(vle-entget-massoc 10 pline)

=>  ((11.08 76.272)(14.54 14.2)(23.45
16.71)(26.18 11.62)(19.84 5.064))
```

24.150.8 Notes

functional identical with : (vle-list-massoc dxfCode (entget ename)), but with higher performance

24.150.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.150.10 Source Notes

- [vle-entget-massoc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleentgetmassoc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.151 Function Entry: VLE-ENTMOD

24.151.1 Name

‘vle-entmod‘

24.151.2 Class

BricsCAD VLE Library Function

24.151.3 Syntax

```
(vle-entmod dxf ename value)
```

24.151.4 Arguments and Values

- dxf the DXF group code to set the new value
- for
- ename the
- entity’s ename
- value new
- value to be set for the entity’s property

24.151.5 Description

modifies the value for specified DXF group code ’dxf’ for specified entity ’ename’

24.151.6 Return Values

T if the entity’s value has been updated (or is identical)
NIL if updating the entity’s DXF value failed (dxf code not supported)

24.151.7 Examples

```
(vle-entmod 62 entity 112)
sets the color index as 112
```

24.151.8 Notes

this is a high-performance replacement for usual code like (entmod (subst (cons dxf value) olditem lst))

24.151.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.151.10 Source Notes

- [vle-entmod (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleentmod.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.152 Function Entry: VLE-ENTMOD-M

24.152.1 Name

'vle-entmod-m'

24.152.2 Class

BricsCAD VLE Library Function

24.152.3 Syntax

```
(vle-entmod-m dxfData ename)
```

24.152.4 Arguments and Values

- dxfData list of (dxf . value) assoc items to be
- set
- ename
- the entity's ename

24.152.5 Description

modifies the DXF values for DXF group codes as specified with 'dxfData' for given entity 'ename'

24.152.6 Return Values

T if the entity's values have been updated (or is identical)
NIL if updating the entity's DXF values failed

24.152.7 Examples

```
(vle-entmod-m '((8 .
"LayerX") (62 . 112) (39 . 2.0)) entity)

or

(setq layer "LayerX" color 112 elev
2.0)
```

```
(vle-entmod-m (list (cons 8  layer) (cons
62 color) (cons 39 elev)) entity) - if variables are
used

sets the layer to "LayerX"

sets the colour index as 112

sets the elevation as 2.0
```

24.152.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.152.9 Source Notes

- [vle-entmod-m (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleentmodm.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.153 Function Entry: VLE-EXTENSIONS-ACTIVE

24.153.1 Name

‘vle-extensions-active‘

24.153.2 Class

BricsCAD VLE Library Function

24.153.3 Syntax

`(vle-extensions-active)`

24.153.4 Description

this variable indicates whether VLE extensions are present and active : T VLE extensions are active NIL VLE extensions are not active

24.153.5 Return Values

Vendor-defined; see Source Notes.

24.153.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.153.7 Source Notes

- [vle-extensions-active (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleextensionsactive.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.154 Function Entry: VLE-FASTCOM

24.154.1 Name

‘vle-fastcom‘

24.154.2 Class

BricsCAD VLE Library Function

24.154.3 Syntax

```
(vle-fastcom enableFlag)
```

24.154.4 Arguments and Values

- enableFlag
- T enables, NIL disables the ”Fast-COM”
- feature

24.154.5 Description

enables or disables ”Fast-COM” feature; Fast-COM bypasses the Windows COM interface for CAD-built-in COM objects, and therefore gives a significant performance advantage for (vla) and (vlax) functions

24.154.6 Return Values

currently active status, after function call

24.154.7 Examples

```
(vle-fastcom t)
 enables "Fast-COM" mode

(vle-fastcom nil) disables "Fast-COM" mode -
i.e. for troubleshooting
```

24.154.8 Notes

this is effective only with BricsCAD, otherwise it is a No-Operation; can be used for trouble-shooting

24.154.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.154.10 Source Notes

- [vle-fastcom (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlefastcom.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.155 Function Entry: VLE-FILE->LIST

24.155.1 Name

‘vle-file->list‘

24.155.2 Class

BricsCAD VLE Library Function

24.155.3 Syntax

```
(vle-file->list filename commentchar)
```

24.155.4 Arguments and Values

- filename name of the file to be loaded
- commentchar the ”comment” character to ignore comment lines;
- specify NIL for ”no comment character”

24.155.5 Description

loads the text lines of specified file ’filename’ as a list of strings, ignoring all comment lines as defined by ’commentchar’

24.155.6 Return Values

list of file lines;
 or NIL, if the file is not found or empty

24.155.7 Examples

```
(vle-file->list
"MyData.dat" ";")

loads all file lines into returned list, and
ignores all lines starting with ";" character (omitted)
```

24.155.8 Notes

’filename’ is searched in all support pathes, if it does not contain a path;
 ’commentchar’ only used if not NIL; if NIL, all file lines are loaded

24.155.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.155.10 Source Notes

- [vle-file->list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlefilelist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.156 Function Entry: VLE-FILEP

24.156.1 Name

‘vle-filep‘

24.156.2 Class

BricsCAD VLE Library Function

24.156.3 Syntax

`(vle-filep obj)`

24.156.4 Arguments and Values

- obj any Lisp item to be verified

24.156.5 Description

predicator function to determine whether ’obj’ is of FILE type

24.156.6 Return Values

T - if ’obj’ is a FILE object
NIL - if ’obj’ is not a FILE object

24.156.7 Notes

replacement for : (= (type obj) ’FILE) and similar

24.156.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.156.9 Source Notes

- [vle-filep (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlefilep.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.157 Function Entry: VLE-FLOOR

24.157.1 Name

‘vle-floor‘

24.157.2 Class

BricsCAD VLE Library Function

24.157.3 Syntax

```
(vle-floor x)
```

24.157.4 Arguments and Values

- x any number (integer or double)

24.157.5 Description

returns the largest integer that is not greater than x

24.157.6 Return Values

the largest integer that is not greater than x

24.157.7 Examples

```
(vle-floor  3.4) =>
 3

(vle-floor -3.4) => -4
```

24.157.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.157.9 Source Notes

- [vle-floor (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlefloor.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.158 Function Entry: VLE-GETGEOMEXTENTS

24.158.1 Name

‘vle-getgeomextents‘

24.158.2 Class

BricsCAD VLE Library Function

24.158.3 Syntax

```
(vle-getgeomextents ename | selectionset | list-of-enames)
```

24.158.4 Arguments and Values

- ename single entity name
- or
- selectionset set of entities
- or
- list-of-enames list of entity names

24.158.5 Description

returns the bounding box (geometric extents) for given entity or entities

24.158.6 Return Values

’(minpoint maxpoint)
 list of minimum (lower-left) and maximum (upper-right) 3d WCS points;
 NIL in case of error

24.158.7 Examples

```
(vle-getgeomextents
(entlast))

returns  '((-1.0 2.0 0.0) (10.0 12.0
2.0))
```

24.158.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.158.9 Source Notes

- [vle-getgeomextents (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlegetgeomextents.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.159 Function Entry: VLE-HIDEPROMPTMENU

24.159.1 Name

‘vle-hidepromptmenu‘

24.159.2 Class

BricsCAD VLE Library Function

24.159.3 Syntax

`(vle-hidepromptmenu)`

24.159.4 Arguments and Values

- none

24.159.5 Description

closes the PromptMenu, if displayed; otherwise this function does nothing

24.159.6 Return Values

NIL always

24.159.7 Notes

this function can always be used, if a PromptMenu was (potentially) shown before; therefore, Lisp programs do not need to track whether the PromptMenu was show before;

under other CAD systems (and older BricsCAD versions), this is a No-Operation and does nothing

24.159.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.159.9 Source Notes

- [vle-hidepromptmenu (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-hidepromptmenu.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.160 Function Entry: VLE-INT64TO32

24.160.1 Name

‘vle-int64to32‘

24.160.2 Class

BricsCAD VLE Library Function

24.160.3 Syntax

```
(vle-int64to32 intVal)
```

24.160.4 Arguments and Values

- intVal any integer value

24.160.5 Description

truncates a given integer value to be in 32 bit range if input value is above 32 bit range (a 64 bit value) then this is truncated to a 32 bit value, to match the behaviour of other CAD systems using 32 bit integers even in x64 builds

24.160.6 Return Values

always returns a 32 bit integer value

24.160.7 Examples

```
(vle-int64to32 (* 10
268435456))

returns -1610612736 instead of
2684354560
```

24.160.8 Notes

this is effective only with Bricscad x64, otherwise it does nothing;

as BricsCAD x64 allows 64 bit integers, there is a minor conflict in compatibility with other CAD systems here;

if a Lisp application needs that incorrect behaviour of other CAD systems, then this function can help as workaround

24.160.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.160.10 Source Notes

- [vle-int64to32 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleint64to32.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.161 Function Entry: VLE-INTEGERP

24.161.1 Name

‘vle-integerp‘

24.161.2 Class

BricsCAD VLE Library Function

24.161.3 Syntax

`(vle-integerp obj)`

24.161.4 Arguments and Values

- obj any Lisp item to be verified

24.161.5 Description

predicator function to determine whether ’obj’ is of INT type

24.161.6 Return Values

T - if ’obj’ is a INT object
NIL - if ’obj’ is not a INT object

24.161.7 Notes

replacement for : (= (type obj) ’INT) and similar

24.161.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.161.9 Source Notes

- [vle-integerp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleintegerp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.162 Function Entry: VLE-IS-CURVE

24.162.1 Name

‘vle-is-curve‘

24.162.2 Class

BricsCAD VLE Library Function

24.162.3 Syntax

```
(vle-is-curve en)
```

24.162.4 Arguments and Values

- entity entity name to be verified (ENAME or
- VLA-OBJECT

24.162.5 Description

verifies whether ’entity’ is a valid curve-based entity name or not.

24.162.6 Return Values

T if ’entity’ is a valid curve-based entity
NIL if ’entity’ is not a valid curve-based entity

24.162.7 Examples

```
(vle-is-curve en) =>
return T for a LINE entity

(vle-is-curve en) => return NIL for a
3DFACE entity
```

24.162.8 Notes

curve-based entities

AcDbLine
AcDbPolyline
AcDb2dPolyline
AcDb3dPolyline
AcDbArc
AcDbCircle
AcDbEllipse
AcDbSpline
AcDbHelix
AcDbLeader
AcDbRay
AcDbXline

24.162.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.162.10 Source Notes

- [vle-is-curve (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-is-curve.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.163 Function Entry: VLE-ITOA32

24.163.1 Name

‘vle-itoa32‘

24.163.2 Class

BricsCAD VLE Library Function

24.163.3 Syntax

```
(vle-itoa32 intVal)
```

24.163.4 Arguments and Values

- intVal
- an integer number

24.163.5 Description

returns the string representation of the 32 bit value of input number; if input value is above 32 bit range (a 64 bit number string) then this is truncated to a 32 bit value, to match the behaviour of other CAD systems using 32 bit integers even in x64 builds

24.163.6 Return Values

returns the string of the 32 bit value of input number

24.163.7 Examples

```
(vle-itoa32
2684354560)

returns "-1610612736" instead of
"2684354560"
```

24.163.8 Notes

this is effective only with Bricscad x64, otherwise it does nothing;

as BricsCAD x64 allows 64 bit integers, there is a minor conflict in compatibility with other CAD systems here;

if a Lisp application needs that incorrect behaviour of other CAD systems, then this function can help as workaround

24.163.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.163.10 Source Notes

- [vle-itoa32 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-itoa32.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.164 Function Entry: VLE-LICENSELEVEL

24.164.1 Name

‘vle-licenselevel‘

24.164.2 Class

BricsCAD VLE Library Function

24.164.3 Syntax

```
(vle-licenselevel)
```

24.164.4 Arguments and Values

- none

24.164.5 Description

returns the actual BricsCAD LicenseLevel including the optionally licensed components (Communicator, SheetMetal, Bim); ”RunAsLevel” setting is respected in result list;

24.164.6 Return Values

a list like (”CLASSIC” ”LITE” ”PROFESSIONAL” ”PLATINUM” ”COMMUNICATOR” ”SHEETMETAL” ”MECHANICAL” ”BIM”)

24.164.7 Examples

```
(vle-licenselevel) =>
("CLASSIC" "PROFESSIONAL" "PLATINUM"  "MECHANICAL"
"SHEETMETAL" "BIM")

under emulation using vle-extension.lsp (i.e.
with AutoCAD) :

("CLASSIC" "PROFESSIONAL"))
```

24.164.8 Notes

items are always uppercase, ”CLASSIC” is always present (”CLASSIC” and ”PROFESSIONAL” always on non-BricsCAD systems);

since V21, there is also the ”LITE” entry present, both ”LITE” and ”CLASSIC” are identical, and maintained for backward compatibility

24.164.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.164.10 Source Notes

- [vle-licenselevel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelicenselevel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.165 Function Entry: VLE-LISPINSTALL

24.165.1 Name

‘vle-lispinstall‘

24.165.2 Class

BricsCAD VLE Library Function

24.165.3 Syntax

```
(vle-lispinstall)
```

24.165.4 Arguments and Values

- none

24.165.5 Description

returns the path (folder) where the Lisp engine is running from

24.165.6 Return Values

the path (folder) where the Lisp engine is running from

24.165.7 Examples

```
(vle-lispinstall) =>
"D:\BricsCAD V13"
```

24.165.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.165.9 Source Notes

- [vle-lispinstall (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelispinstall.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.166 Function Entry: VLE-LISPVERSION

24.166.1 Name

‘vle-lispversion‘

24.166.2 Class

BricsCAD VLE Library Function

24.166.3 Syntax

```
(vle-lispversion)
```

24.166.4 Arguments and Values

- none

24.166.5 Description

returns the version of the Lisp engine as string of a number

24.166.6 Return Values

the version of the Lisp engine as string of a version number and platform like ”3.0 (x86)”

24.166.7 Examples

```
(vle-lispversion) => "5.0
(x86)" for x86 BricsCAD or "5.0 (x64)" for x64 BricsCAD;

(vle-lispversion) => "2013
(x86)" for AutoCAD 2013 x86, "2013 (x64)" for AutoCAD 2013
x64
```

24.166.8 Notes

using (atof (vle-lispversion)) will return the (double) number instead

24.166.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.166.10 Source Notes

- [vle-lispversion (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelispversion.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.167 Function Entry: VLE-LIST-DIFF

24.167.1 Name

‘vle-list-diff‘

24.167.2 Class

BricsCAD VLE Library Function

24.167.3 Syntax

```
(vle-list-diff lst1 lst2)
```

24.167.4 Arguments and Values

- lst1 list to be compared against ’lst2’
- lst2 list to
- be compared against ’lst1’

24.167.5 Description

returns the ”list difference” of 2 lists, containing all items, which are member of either ’lst1’ or ’lst2’, but not member of both lists

24.167.6 Return Values

a list containing all items which are unique for their owner lists; in other words, which are not member of both lists

24.167.7 Examples

```
(vle-list-diff '(1 2 3 3 4)
'(3 4 5 6))  => '(1 2 5 6)

(vle-list-diff '(1 2 3 3 4) '(5 6 7 8))
 => '(1 2 3 3 4 5 6 7 8)

(vle-list-diff '(1 2 3 3 4) nil)  =>
'(1 2 3 3 4)
```

24.167.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.167.9 Source Notes

- [vle-list-diff (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistdiff.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.168 Function Entry: VLE-LIST-INTERSECT

24.168.1 Name

‘vle-list-intersect‘

24.168.2 Class

BricsCAD VLE Library Function

24.168.3 Syntax

```
(vle-list-intersect lst1 lst2)
```

24.168.4 Arguments and Values

- lst1 list to be verified against ’lst2’
- lst2 list to
- be verified against ’lst1’

24.168.5 Description

creates a list which contains all those items contained in both input lists

24.168.6 Return Values

a list containing all unique items, which are member in both input lists

24.168.7 Examples

```
(vle-list-intersect '(1 2 3 4
5) '(3 4 5 6 7))  => '(3 4 5)

(vle-list-intersect '(1 2 3 4) '(4 5 6))
 => '(4)

(vle-list-intersect '(1 2 3 4) '(5 6 7))
 => nil
```

24.168.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.168.9 Source Notes

- [vle-list-intersect (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistintersect.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.169 Function Entry: VLE-LIST-MASSOC

24.169.1 Name

‘vle-list-massoc‘

24.169.2 Class

BricsCAD VLE Library Function

24.169.3 Syntax

```
(vle-list-massoc key lst)
```

24.169.4 Arguments and Values

- key key to be used to parse the assoc list
- lst assoc
- list, may contain multiple items with same key

24.169.5 Description

returns a list of all values using same ’key’ in assoc-list ’lst’

24.169.6 Return Values

a list containing all cdr values of assoc items with key ’key’;
the list may contain multiple assoc items with key ’key’;
returned list contains all values in original sequence

24.169.7 Examples

```
(setq lst '((1 . "a")(2 .
"b")(1 . 11)(2 . 22)))

(vle-list-massoc 1 lst) => ("a"
11)

(vle-list-massoc 2 lst) => ("b" 22)
```

24.169.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.169.9 Source Notes

- [vle-list-massoc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistmassoc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.170 Function Entry: VLE-LIST-SPLIT

24.170.1 Name

‘vle-list-split‘

24.170.2 Class

BricsCAD VLE Library Function

24.170.3 Syntax

```
(vle-list-split lst item)
```

24.170.4 Arguments and Values

- lst list to be splitted at ’item’
- item the
- data where ’lst’ is to be splitted

24.170.5 Description

splits a list at a given item and returns both list parts

24.170.6 Return Values

a list containing the ”left” and the ”right” sub-lists of input list lst; item is added to the ”right” list, if present;

if item is not present in lst, the ”right” list part is NIL

24.170.7 Examples

```
(vle-list-split '(1 2 3 4 5)
3)  => '((1 2) (3 4 5))

(vle-list-split '(1 2 3 4 5) 8)  =>
'((1 2 3 4 5) nil)

(vle-list-split '(1 2 3 4 5) 1)  =>
'(NIL (1 2 3 4 5))
```

24.170.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.170.9 Source Notes

- [vle-list-split (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistsplit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.171 Function Entry: VLE-LIST-SUBTRACT

24.171.1 Name

‘vle-list-subtract‘

24.171.2 Class

BricsCAD VLE Library Function

24.171.3 Syntax

```
(vle-list-subtract lst1 lst2)
```

24.171.4 Arguments and Values

- lst1 list ”subtract” the items of ’lst2’ from
- lst2 list to
- be ”subtracted” from ’lst1’

24.171.5 Description

creates a list which contains all items of ’lst1’ list without items of list ’lst2’

24.171.6 Return Values

a list containing all items of ’lst1’ without items of ’lst2’

24.171.7 Examples

```
(vle-list-subtract '(1 2 3 4
5) '(7 3 6))  => '(1 2 4 5)

(vle-list-subtract '(1 2 3 4 5) '(7 8 9))
 => '(1 2 3 4 5)

(vle-list-subtract '(1 2 3 3 4) '(7 3 6))
 => '(1 2 4)
```

24.171.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.171.9 Source Notes

- [vle-list-subtract (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistsubtract.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.172 Function Entry: VLE-LIST-UNION

24.172.1 Name

‘vle-list-union‘

24.172.2 Class

BricsCAD VLE Library Function

24.172.3 Syntax

```
(vle-list-union lst1 lst2)
```

24.172.4 Arguments and Values

- lst1 first list to merge into result list
- lst2 second
- list to merge into result list

24.172.5 Description

merges 2 lists into 1 result list; all duplicates are removed

24.172.6 Return Values

a list containing all unique items of ’lst1’ and ’lst2’;
result list does not contain any duplicates

24.172.7 Examples

```
(vle-list-union '(1 2 3 4)
'(7 8 9))  => '(1 2 3 4 7 8 9)

(vle-list-union '(1 2 3 4) '(4 5 6))
 => '(1 2 3 4 5 6)

(vle-list-union '(1 2 3 3) '(4 4 5))
 => '(1 2 3 4 5)
```

24.172.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.172.9 Source Notes

- [vle-list-union (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlelistunion.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.173 Function Entry: VLE-MEMBER

24.173.1 Name

‘vle-member‘

24.173.2 Class

BricsCAD VLE Library Function

24.173.3 Syntax

`(vle-member item lst)`

24.173.4 Arguments and Values

- item expression to search for in ’lst’
- lst normal
- Lisp list

24.173.5 Description

verifies whether list lst contains the expression item.

24.173.6 Return Values

T if the expression ’item’ is found in list ’lst’;
NIL if not found

24.173.7 Examples

```
(vle-member 123 '(1 2 3 123 4
5))  returns T

(vle-member 456 '(1 2 3 123 4 5))
 returns NIL
```

24.173.8 Notes

replacement for (member item lst)

24.173.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.173.10 Source Notes

- [vle-member (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlemember.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.174 Function Entry: VLE-NTH<X>

24.174.1 Name

‘vle-nth<x>‘

24.174.2 Class

BricsCAD VLE Library Function

24.174.3 Syntax

```
(vle-nth<x> lst)
```

24.174.4 Arguments and Values

- lst list of items

24.174.5 Description

vle-nth0 vle-nth1 vle-nth2 vle-nth3 vle-nth4 vle-nth5 vle-nth6 vle-nth7 vle-nth8 vle-nth9 returns the item at index <x> of the list ’lst’ for placeholder <x> the functions are implemented for range 0...9

24.174.6 Return Values

item at index <x> of the input list ’lst’

24.174.7 Examples

```
(vle-nth4 '(0 11 22 33 44 55
66 77 88 99) returns 44

(vle-nth8 '(0 11 22 33 44 55 66 77 88 99)
returns 88
```

24.174.8 Notes

replacement for (nth <x> lst); index is 0-based

24.174.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.174.10 Source Notes

- [vle-nth<x> (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlenthx.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.175 Function Entry: VLE-NUMBERP

24.175.1 Name

‘vle-numberp‘

24.175.2 Class

BricsCAD VLE Library Function

24.175.3 Syntax

`(vle-numberp obj)`

24.175.4 Arguments and Values

- obj any Lisp item to be verified

24.175.5 Description

predicator function to determine whether 'obj' is of any number type (INT or REAL)

24.175.6 Return Values

T - if 'obj' is a INT or REAL object
NIL - if 'obj' is not a INT or REAL object

24.175.7 Notes

identical to (numberp), implemented for completeness

24.175.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.175.9 Source Notes

- [vle-numberp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlenumberp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.176 Function Entry: VLE-OPTIMIZER

24.176.1 Name

‘vle-optimizer‘

24.176.2 Class

BricsCAD VLE Library Function

24.176.3 Syntax

```
(vle-optimizer enableFlag)
```

24.176.4 Arguments and Values

- enableFlag
- T enables, NIL disables the
- Optimiser

24.176.5 Description

enables or disables ”Lisp Optimiser” features, which analyses (defun) code blocks at load-time to optimise inefficient code to be more efficient (i.e. by combining statements into efficient built-in VLE functions)

24.176.6 Return Values

previous activation status

24.176.7 Examples

```
(vle-optimizer t)
 enables automatic Lisp Optimiser

(vle-optimizer nil) disables automatic Lisp
Optimiser
```

24.176.8 Notes

the Optimiser feature operates on files only, not on Lisp input via commandline;

to disable Lisp Optimiser for a particular Lisp file (i.e. for troubleshooting), add :

(vle-optimizer nil)

at beginning of the Lisp file;

after loading each file, the Lisp Optimiser is automatically reset to be active, therefore using (vle-optimizer nil) only affects the particular file

24.176.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.176.10 Source Notes

- [vle-optimizer (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleoptimizer.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.177 Function Entry: VLE-PICKSETP

24.177.1 Name

‘vle-picksetp‘

24.177.2 Class

BricsCAD VLE Library Function

24.177.3 Syntax

`(vle-picksetp obj)`

24.177.4 Arguments and Values

- obj any Lisp item to be verified

24.177.5 Description

predicator function to determine whether ’obj’ is of PICKSET type

24.177.6 Return Values

T - if ’obj’ is a PICKSET object
NIL - if ’obj’ is not a PICKSET object

24.177.7 Notes

replacement for : (= (type obj) ’PICKSET) and similar

24.177.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.177.9 Source Notes

- [vle-picksetp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlepicksetp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.178 Function Entry: VLE-PING-ALIVE

24.178.1 Name

‘vle-ping-alive‘

24.178.2 Class

BricsCAD VLE Library Function

24.178.3 Syntax

`(vle-ping-alive)`

24.178.4 Arguments and Values

- none

24.178.5 Description

sends a signal to Windows, prevents Windows’ false ”Application does not respond …” screen

24.178.6 Return Values

t always

24.178.7 Notes

this function can be used during heavy computation (loops etc.) to prevent that Windows sets the BricsCAD application window to the irritating (and in fact, wrong !) ”not responding” state

24.178.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.178.9 Source Notes

- [vle-ping-alive (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-ping-alive.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.179 Function Entry: VLE-POINTP

24.179.1 Name

‘vle-pointp‘

24.179.2 Class

BricsCAD VLE Library Function

24.179.3 Syntax

```
(vle-pointp obj)
```

24.179.4 Arguments and Values

- obj any Lisp item to be verified

24.179.5 Description

This function returns whether obj represents a 2D or 3D point or vector.

24.179.6 Return Values

T if obj represents a 2D or 3D point/vector, NIL if obj does not represent a 2D or 3D point or vector

24.179.7 Examples

```
(vle-pointp "abc")
 NIL

(vle-pointp '(1.1 2.2)  T
```

24.179.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.179.9 Source Notes

- [vle-pointp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-pointp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.180 Function Entry: VLE-PUT-NTH

24.180.1 Name

‘vle-put-nth‘

24.180.2 Class

BricsCAD VLE Library Function

24.180.3 Syntax

`(vle-put-nth lst idx val)`

24.180.4 Arguments and Values

- ls’ input list where the item at position
- ’idx’ is to be
- set
- idx the
- 0-based index of the item to be set by new ’val’
- val the new
- value to be set for the item at index ’idx’

24.180.5 Description

sets / replaces the item at index ’idx’ in list ’lst’ with specified new value ’val’; if ’idx’ is larger then the list length, the list is filled with NIL items before ’val’

24.180.6 Return Values

list ’lst’ with item at position ’idx’ set to ’val’;
if ’idx’ is < 0, the unchanged list is returned
if ’idx’ is greater than the list length, the list is filled with NIL values before added item ’val’

24.180.7 Notes

index 'idx' is 0-based

24.180.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.180.9 Source Notes

- [vle-put-nth (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleputnth.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.181 Function Entry: VLE-REALP

24.181.1 Name

‘vle-realp‘

24.181.2 Class

BricsCAD VLE Library Function

24.181.3 Syntax

`(vle-realp obj)`

24.181.4 Arguments and Values

- obj any Lisp item to be verified

24.181.5 Description

predicator function to determine whether ’obj’ is of REAL (double) type

24.181.6 Return Values

T - if ’obj’ is a REAL object
NIL - if ’obj’ is not a REAL object

24.181.7 Notes

replacement for : (= (type obj) ’REAL) and similar

24.181.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.181.9 Source Notes

- [vle-realp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlerealp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.182 Function Entry: VLE-REMOVE-ALL

24.182.1 Name

‘vle-remove-all‘

24.182.2 Class

BricsCAD VLE Library Function

24.182.3 Syntax

```
(vle-remove-all item lst)
```

24.182.4 Arguments and Values

- item object to remove from lst
- lst normal
- Lisp list

24.182.5 Description

removes all occurrences of ’item’ from list ’lst’ (like vl-remove)

24.182.6 Return Values

list ’lst’ without all occurrences of ’item’, if found;
unchanged list ’lst’ if not found

24.182.7 Examples

```
(vle-remove 123 '(1 2 123 4 5
123 6 7 8))

returns '(1 2 4 5 6 7 8)
```

24.182.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.182.9 Source Notes

- [vle-remove-all (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleremoveall.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.183 Function Entry: VLE-REMOVE-FIRST

24.183.1 Name

‘vle-remove-first‘

24.183.2 Class

BricsCAD VLE Library Function

24.183.3 Syntax

```
(vle-remove-first item lst)
```

24.183.4 Arguments and Values

- item object to remove from lst
- lst normal
- Lisp list

24.183.5 Description

removes the first occurrence of ’item’ from list ’lst’

24.183.6 Return Values

list ’lst’ without first occurrence of ’item’, if found;
unchanged list ’lst’ if not found

24.183.7 Examples

```
(vle-remove-first 123 '(1 2
123 4 5 123 6 7 8))

returns '(1 2 4 5 123 6 7 8)
```

24.183.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.183.9 Source Notes

- [vle-remove-first (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleremovefirst.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.184 Function Entry: VLE-REMOVE-NTH

24.184.1 Name

‘vle-remove-nth‘

24.184.2 Class

BricsCAD VLE Library Function

24.184.3 Syntax

```
(vle-remove-nth idx lst)
```

24.184.4 Arguments and Values

- idx index of the item to be removed; if negative,
- or lst contains less items, nothing is removed
- lst’ normal
- Lisp list

24.184.5 Description

removes the object at index idx from list lst

24.184.6 Return Values

list lst without item at position idx;
if idx is not in the list, the unchanged list lst is returned

24.184.7 Examples

```
(vle-remove-nth 3 '(1 2 123 4
5 123 6 7 8))

returns '(1 2 123 5 123 6 7 8)
```

24.184.8 Notes

index is 0-based

24.184.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.184.10 Source Notes

- [vle-remove-nth (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleremoventh.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.185 Function Entry: VLE-RGB2ACI

24.185.1 Name

‘vle-rgb2aci‘

24.185.2 Class

BricsCAD VLE Library Function

24.185.3 Syntax

`(vle-rgb2aci rgb)`

24.185.4 Arguments and Values

- rgb an integer specifying the RGB value
- red an
- integer specifying the red value part (0…255)
- green an
- integer specifying the green value part (0…255)
- blue an
- integer specifying the blue value part (0…255)

24.185.5 Description

(vle-rgb2aci red green blue) Returns the closest ACI color index for specified RGB color value.

24.185.6 Return Values

always returns an integer value

24.185.7 Examples

```
(vle-rgb2aci 0 0 255)
 returns  5

(vle-rgb2aci 0 76 57)  returns
128

(vle-rgb2aci 16711680)  returns 5
```

24.185.8 Notes

the vle-extension.lsp emulation for non-BricsCAD systems only supports the (vle-rgb2aci red green blue) syntax !

24.185.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.185.10 Source Notes

- [vle-rgb2aci (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlergb2aci.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.186 Function Entry: VLE-ROUND

24.186.1 Name

‘vle-round‘

24.186.2 Class

BricsCAD VLE Library Function

24.186.3 Syntax

```
(vle-round x)
```

24.186.4 Arguments and Values

- x any integer or double number

24.186.5 Description

returns the integer nearest to x

24.186.6 Return Values

the nearest integer to x

24.186.7 Examples

```
(vle-round  13.2) =>
 13

(vle-round -13.2) => -13

(vle-round  13.8) =>
 14

(vle-round -13.8) => -14
```

24.186.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.

- Status: BricsCAD-only.

24.186.9 Source Notes

- [vle-round (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleround.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.187 Function Entry: VLE-ROUNDTO

24.187.1 Name

‘vle-roundto‘

24.187.2 Class

BricsCAD VLE Library Function

24.187.3 Syntax

```
(vle-roundto x digits)
```

24.187.4 Arguments and Values

- x any integer or double number
- digits integer
- number of digital places to round to

24.187.5 Description

returns the input value ’x’ rounded to ’digits’ number of decimal places (precision)

24.187.6 Return Values

the input value ’x’ rounded to ’digits’ number of decimal places (precision), as double

24.187.7 Examples

```
(vle-roundto  1.23456789
 1) => 1.2

(vle-roundto  1.23456789
 2) => 1.23

(vle-roundto  1.23456789
 3) => 1.235
```

```
(vle-roundto  1.23456789
 4) => 1.2346

(vle-roundto  1.23456789
 5) => 1.23457

(vle-roundto  1.23456789
 6) => 1.234568

(vle-roundto  1.23456789
 7) => 1.2345679

(vle-roundto  1.23456789
 8) => 1.23456789
```

24.187.8 Notes

if input value x is an integer, it is automatically converted into double

24.187.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.187.10 Source Notes

- [vle-roundto (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleroundto.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.188 Function Entry: VLE-SAFEARRAY->LIST

24.188.1 Name

‘vle-safearray->list‘

24.188.2 Class

BricsCAD VLE Library Function

24.188.3 Syntax

```
(vle-safearray->list safearray)
```

24.188.4 Arguments and Values

- safearray the SAFEARRAY to be returned as list

24.188.5 Description

creates a normal list of items from given SafeArray safearray

24.188.6 Return Values

list of items, or NIL if safearray is empty

24.188.7 Examples

```
(vle-safearray->list
sa)

returns  (<#IDispatch object>
<#IDispatch object> ...) or NIL
```

24.188.8 Notes

other than the normal (vlax-safearray->list), this function does not trigger an error if the SafeArray is empty (which is normal + valid for any SafeArray)

24.188.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.188.10 Source Notes

- [vle-safearray->list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-safearray-list.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.189 Function Entry: VLE-SAFEARRAYP

24.189.1 Name

‘vle-safearrayp‘

24.189.2 Class

BricsCAD VLE Library Function

24.189.3 Syntax

```
(vle-safearrayp obj)
```

24.189.4 Arguments and Values

- obj any Lisp item to be verified

24.189.5 Description

predicator function to determine whether ’obj’ is of SAFEARRAY type

24.189.6 Return Values

T - if ’obj’ is a SAFEARRAY object
NIL - if ’obj’ is not a SAFEARRAY object

24.189.7 Notes

replacement for : (= (type obj) ’SAFEARRAY) and similar

24.189.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.189.9 Source Notes

- [vle-safearrayp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlesafearrayp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.190 Function Entry: VLE-SEARCH

24.190.1 Name

‘vle-search‘

24.190.2 Class

BricsCAD VLE Library Function

24.190.3 Syntax

```
(vle-search item lst asIdx)
```

24.190.4 Arguments and Values

- item object to search for in lst
- lst list of
- items to search in
- asIdx flag
- defining the type of result;:if NIL, the list starting with
- item is returned;
- if non-NIL, the index of item
- in lst
- is returned

24.190.5 Description

searches for ’item’ in list ’lst’ and returns either the index of ’item’ in ’lst’ (if ’asIdx’ is non-NIL), or the list starting with ’item’ (if ’asIdx’ is NIL), or NIL if ’item’ is not contained in ’lst’

24.190.6 Return Values

the index of item in lst, or the sub-list starting with item, depending on asIdx flag;

if item is not contained in ’lst’, NIL is returned

24.190.7 Examples

```
(vle-search 11 '(1 2 3 11 22
33 ) nil)   returns   '(11 22 33)

(vle-search 11 '(1 2 3 11 22 33 ) t)
   returns   3
```

24.190.8 Notes

returned index is 0-based

24.190.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.190.10 Source Notes

- [vle-search (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlesearch.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.191 Function Entry: VLE-SELECTIONSET->LIST

24.191.1 Name

‘vle-selectionset->list‘

24.191.2 Class

BricsCAD VLE Library Function

24.191.3 Syntax

```
(vle-selectionset->list selectionset)
```

24.191.4 Arguments and Values

- selectionset
- the selection set to be returned as
- entity list

24.191.5 Description

creates a normal list of entities from given selection set ’selectionset’

24.191.6 Return Values

list of plain entity names, or NIL if selectionset is empty

24.191.7 Examples

```
(vle-selectionset->list
ss)

returns  (<#Entity name>
<#Entity name>)
```

24.191.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.191.9 Source Notes

- [vle-selectionset->list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleselectionsetlist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.192 Function Entry: VLE-SET-CDRASSOC

24.192.1 Name

‘vle-set-cdrassoc‘

24.192.2 Class

BricsCAD VLE Library Function

24.192.3 Syntax

```
(vle-set-cdrassoc key lst val)
```

24.192.4 Arguments and Values

- key key value for the dotted pair value to search
- for
- lst list of
- dotted pairs
- val new
- value for dotted pairs with key ’key’

24.192.5 Description

searches ’lst’ list for dotted-pairs with key ’key’ and sets the value of the dotted pairs (if found) to ’val’;

24.192.6 Return Values

the input list with changed values for dotted pairs with ’key’;
if there is no dotted-pair with ’key’, list ’lst’ returns unchanged

24.192.7 Examples

```
(setq lst '((1 . "a")(2 .
"b")(1 . "a")(2 . "b")))

(setq lst (vle-set-cdrassoc 1 lst
```

```
"A"))

returns '((1 . "A")(2 . "b")(1 . "A")(2 .
"b"))
```

24.192.8 Notes

NOT a general replacement for (subst new old lst)
but useful replacement for : (subst new (assoc key lst) lst)

24.192.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.192.10 Source Notes

- [vle-set-cdrassoc (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlesetcdrassoc.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.193 Function Entry: VLE-SHOWPROMPTMENU

24.193.1 Name

‘vle-showpromptmenu‘

24.193.2 Class

BricsCAD VLE Library Function

24.193.3 Syntax

```
(vle-showpromptmenu keywords)
```

24.193.4 Arguments and Values

- keywords list of keywords to be shown in the PromptMenu

24.193.5 Description

shows the PromptMenu, providing the options as in ’keywords’; the options keywords should follow the rules for (initget) keywords string; especially (but not only) useful in combination with (grread)

24.193.6 Return Values

NIL always

24.193.7 Examples

```
(vle-ShowPromptMenu "Continue
Finish Abort")
```

24.193.8 Notes

the PromptMenu is shown ”modeless”, Lisp does not wait for user interaction !

under other CAD systems (and older BricsCAD versions), this is a No-Operation and does nothing

24.193.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.193.10 Source Notes

- [vle-showpromptmenu (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vleshowpromptmenu.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.194 Function Entry: VLE-START-TRANSACTION

24.194.1 Name

‘vle-start-transaction‘

24.194.2 Class

BricsCAD VLE Library Function

24.194.3 Syntax

```
(vle-start-transaction)
```

24.194.4 Arguments and Values

- none

24.194.5 Description

starts a database transaction for current drawing

24.194.6 Return Values

T when transaction is active, NIL otherwise

24.194.7 Examples

```
(vle-start-transaction)

T : transaction is active

NIL : transaction is not active
```

24.194.8 Notes

with an active transaction, database and display updates are delayed until the transaction is finished - this can significantly improve performance when many entity and database modifications are done;

multiple / nested transactions are not supported - but it is gracefully tolerated, Lisp keeps a single (global) transaction only;

when outermost Lisp execution finishes, a pending / open transaction is automatically closed (also when Lisp code errors)

24.194.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.194.10 Source Notes

- [vle-start-transaction (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlestarttransaction.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.195 Function Entry: VLE-STARTAPP

24.195.1 Name

‘vle-startapp‘

24.195.2 Class

BricsCAD VLE Library Function

24.195.3 Syntax

`(vle-startapp cmd args mode)`

24.195.4 Arguments and Values

- cmd the command / program to run (*.exe *.com
- *.bat)
- args optional arguments as string, or NIL if no
- arguments
- mode T
- specifies to wait for the process, NIL specifies to run
- parallel

24.195.5 Description

runs the specified command, optionally with arguments, and optionally waits for the process to end (synchronous execution)

24.195.6 Return Values

process ID if started parallel, or the program’s return status if started synchronously

24.195.7 Examples

```
(vle-startapp "notepad.exe"
"test.dat" t)

starts notepad.exe with "test.dat" file,
running synchronously
```

24.195.8 Notes

replaces (startapp), which does not wait for the process (asynchronous)

24.195.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.195.10 Source Notes

- [vle-startapp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlestartapp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.196 Function Entry: VLE-STRING-REPLACE

24.196.1 Name

‘vle-string-replace‘

24.196.2 Class

BricsCAD VLE Library Function

24.196.3 Syntax

```
(vle-string-replace newStr oldStr inString)
```

24.196.4 Arguments and Values

- newStr new string replacing oldStr
- oldStr string to be replaced
- inStr target
- string for replacement

24.196.5 Description

replaces all occurrences of oldStr with newStr in string inStr

24.196.6 Return Values

new string with all oldStr replaced by newStr

24.196.7 Examples

```
(vle-string-replace "." ","
"12,345,678")

will return  "12.345.678"
```

24.196.8 Notes

both ’newStr’ and ’oldStr’ can have any length, and do not need to be of same length

24.196.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.196.10 Source Notes

- [vle-string-replace (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlestringreplace.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.197 Function Entry: VLE-STRING-SPLIT

24.197.1 Name

'vle-string-split'

24.197.2 Class

BricsCAD VLE Library Function

24.197.3 Syntax

```
(vle-string-split keys string)
```

24.197.4 Arguments and Values

- keys string containing all delimiter characters to be
- used
- string the
- string to be splitted into tokens

24.197.5 Description

splits the given string string into string-tokens, based on delimiter characters as specified by keys

24.197.6 Return Values

list of string tokens;

if none of the keys characters is present in string, the list contains only the original string (1 token)

24.197.7 Examples

```
(vle-string-split ",;"
"Sample, using different keys; voila")

returns   '("Sample" " using different
keys" " voila")
```

24.197.8 Notes

if no <delimiter> character is contained, the returned list will contain v the original string

24.197.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.197.10 Source Notes

- [vle-string-split (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlestringsplit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.198 Function Entry: VLE-STRINGP

24.198.1 Name

‘vle-stringp‘

24.198.2 Class

BricsCAD VLE Library Function

24.198.3 Syntax

`(vle-stringp obj)`

24.198.4 Arguments and Values

- obj any Lisp item to be verified

24.198.5 Description

predicator function to determine whether ’obj’ is of STRING type

24.198.6 Return Values

T - if ’obj’ is a STRING object
NIL - if ’obj’ is not a STRING object

24.198.7 Notes

replacement for : (= (type obj) ’STR) and similar

24.198.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.198.9 Source Notes

- [vle-stringp (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlestringp.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.199 Function Entry: VLE-SUBLIST

24.199.1 Name

‘vle-sublist‘

24.199.2 Class

BricsCAD VLE Library Function

24.199.3 Syntax

`(vle-sublist lst startidx nritems)`

24.199.4 Arguments and Values

- lst list to extract a sublist from
- startidx index of first item for sublist (0-based); if
- startidx is $<$
- 0, 0 is used
- nritems
- number of items to be copied from lst list; if
- nritems is < 0,
- all items after startidx
- are copied to result list

24.199.5 Description

creates a sublist from given input list lst, starting with item at index startidx and of maximum nritems number of list items from lst

24.199.6 Return Values

a sublist of lst starting with index startidx and containing nritems of input list lst;

if lst has not enough items, only the items of lst are copied to result list, no NIL is appended

24.199.7 Examples

```
(vle-sublist '(1 2 3 4 5 6 7)
2 3)  => '(3 4 5)

(vle-sublist '(1 2 3 4 5 6 7) 2 -1) => '(3
4 5 6 7)

(vle-sublist '(1 2 3 4 5 6 7) 2 10) => '(3
4 5 6 7)
```

24.199.8 Notes

index is 0-based

24.199.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.199.10 Source Notes

- [vle-sublist (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlesublist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.200 Function Entry: VLE-SUBST-NTH

24.200.1 Name

‘vle-subst-nth‘

24.200.2 Class

BricsCAD VLE Library Function

24.200.3 Syntax

`(vle-subst-nth lst idx val)`

24.200.4 Arguments and Values

- lst input list where the item at position
- idx is to be
- set
- idx the
- index of the item to be replaced by new val
- val the new
- value to be set for the item

24.200.5 Description

sets / replaces the item at index idx in list ’lst’ with specified new value val

24.200.6 Return Values

list lst without item at position idx;
if there is no item idx in lst the list remains unchanged;
if idx is < 0 or greater than the list length, the unchanged list is returned

24.200.7 Examples

```
(vle-subst-nth '(1 2 3 4 5) 2
99)    => '(1 2 99 4 5)

(vle-subst-nth '(1 2 3 4 5) 10 99)  =>
'(1 2 3 4 5)
```

24.200.8 Notes

index idx is 0-based

24.200.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.200.10 Source Notes

- [vle-subst-nth (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlesubstnth.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.201 Function Entry: VLE-SUNID

24.201.1 Name

‘vle-sunid‘

24.201.2 Class

BricsCAD VLE Library Function

24.201.3 Syntax

```
(vle-sunid)
```

24.201.4 Arguments and Values

- none

24.201.5 Description

This function returns.the ENAME of the actual (ModelSpace or PaperSpace) viewport.

24.201.6 Return Values

ENAME of the sun object, or NIL

(if NIL is returned, the sun needs to be enabled : (setvar ”SUNSTATUS” 1)

24.201.7 Notes

each ModelSpace viewport (AcDbViewportTableRecord) and each PaperSpace viewport (AcDbViewport) uses an own sun object

24.201.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.201.9 Source Notes

- [vle-sunid (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-sunid.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.202 Function Entry: VLE-TABLE-LIST

24.202.1 Name

‘vle-table-list‘

24.202.2 Class

BricsCAD VLE Library Function

24.202.3 Syntax

```
(vle-table-list tablename asNames)
```

24.202.4 Arguments and Values

- tablename table name (string) to be scanned
- asNames specifies whether item names are listed or the
- entity names of items; ’T’ specifies item names

24.202.5 Description

scans the specified table and returns a list of all entries, excluding anonymous entries.

24.202.6 Return Values

list of table entries, as list of strings or entity names

24.202.7 Examples

```
(vle-table-list "LAYER" T)
  => ("0" "1" "Lay_A" ... "Lay_X")

(vle-table-list "LAYER" nil) => (<Entity
name> <Entity name> ... )
```

24.202.8 Notes

supported tables :
  BLOCK LAYER LTYPE VIEW STYLE UCS APPID DIMSTYLE VPORT
  see function (vle-table-list-all-all)

24.202.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.202.10 Source Notes

- [vle-table-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-table-list.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.203 Function Entry: VLE-TABLE-LIST-ALL

24.203.1 Name

‘vle-table-list-all‘

24.203.2 Class

BricsCAD VLE Library Function

24.203.3 Syntax

```
(vle-table-list-all tablename asNames)
```

24.203.4 Arguments and Values

- tablename table name (string) to be scanned
- asNames specifies whether item names are listed or the
- entity names of items; ’T’ specifies item names

24.203.5 Description

scans the specified table and returns a list of all entries, including anonymous entries.

24.203.6 Return Values

list of table entries, as list of strings or entity names

24.203.7 Examples

```
(vle-table-list-all "BLOCK"
T)  returns ("*Model_Space" "*Paper_Space" ...)
```

24.203.8 Notes

supported tables :
BLOCK LAYER LTYPE VIEW STYLE UCS APPID DIMSTYLE VPORT
see function (vle-table-list-all-all)

24.203.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.203.10 Source Notes

- [vle-table-list-all (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-table-list-all.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.204 Function Entry: VLE-TAN

24.204.1 Name

‘vle-tan‘

24.204.2 Class

BricsCAD VLE Library Function

24.204.3 Syntax

```
(vle-tan x)
```

24.204.4 Arguments and Values

- x any integer or double number (except PI*0.5 and multiples,
- where (cos x) is 0.0)

24.204.5 Description

returns the tangent of x

24.204.6 Return Values

the tangent of x

24.204.7 Examples

```
(vle-tan 0.5) =>
0.546302
```

24.204.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.204.9 Source Notes

- [vle-tan (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vletan.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.205 Function Entry: VLE-VARIANTP

24.205.1 Name

‘vle-variantp‘

24.205.2 Class

BricsCAD VLE Library Function

24.205.3 Syntax

`(vle-variantp obj)`

24.205.4 Arguments and Values

- obj any Lisp item to be verified

24.205.5 Description

predicator function to determine whether ’obj’ is of VARIANT type

24.205.6 Return Values

T - if ’obj’ is a VARIANT object
NIL - if ’obj’ is not a VARIANT object

24.205.7 Notes

replacement for : (= (type obj) ’VARIANT) and similar

24.205.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.205.9 Source Notes

- [vle-variantp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlevariantp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.206 Function Entry: VLE-VECTOR-ADD

24.206.1 Name

‘vle-vector-add‘

24.206.2 Class

BricsCAD VLE Library Function

24.206.3 Syntax

```
(vle-vector-add vec1 vec2)
```

24.206.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.206.5 Description

This function returns the addition of vector vec2 to vector vec1.

24.206.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.206.7 Examples

```
(vle-vector-add '(1 1 1) '(2
3 4))  returns '(3 4 5)
```

24.206.8 Notes

’vec1’ can also be interpreted as ’point1’, adding vector ’vec2’ to point ’vec1’ then

24.206.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.206.10 Source Notes

- [vle-vector-add (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-add.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.207 Function Entry: VLE-VECTOR-ANGLETO

24.207.1 Name

‘vle-vector-angleto‘

24.207.2 Class

BricsCAD VLE Library Function

24.207.3 Syntax

```
(vle-vector-angleto vec1 vec2)
```

24.207.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.207.5 Description

This function returns the angle between vectors vec1 and vec2.

24.207.6 Return Values

double, angle in radians, in range 0 … PI

24.207.7 Examples

```
(setq v1 '(2 0 0) v2 '(0 3
0))

(vle-vector-angleto v1 v2)

1.5707963267949

(setq v1 '(2 0 0) v2 '(-1 -1 0))

(vle-vector-angleto v1 v2)

2.35619449019234
```

24.207.8 Notes

input vectors must not be unit-length (non-normalised vectors are allowed)

24.207.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.207.10 Source Notes

- [vle-vector-angleto (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-angleto.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.208 Function Entry: VLE-VECTOR-ANGLETOREF

24.208.1 Name

‘vle-vector-angletoref‘

24.208.2 Class

BricsCAD VLE Library Function

24.208.3 Syntax

```
(vle-vector-angletoref vec1 vec2 normal)
```

24.208.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list
- normal a 2d
- or 3d vector list, specifies the plane normal

24.208.5 Description

This function returns the angle between vectors vec1 and vec2, using normal as reference vector, which defines the vector plane.

24.208.6 Return Values

double, angle in radians, in range 0 … 2PI

24.208.7 Examples

```
(setq v1 '(2 0 0) v2 '(0 3 0)
normal '(0 0 1))

(vle-vector-angletoref v1 v2
normal)

1.5707963267949
```

```
(setq v1 '(2 0 0) v2 '(-1 -1 0) normal '(0 0
1))

(vle-vector-angletoref v1 v2
normal)

3.92699081698724
```

24.208.8 Notes

input vectors must not be unit-length (non-normalised vectors are allowed)

24.208.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.208.10 Source Notes

- [vle-vector-angletoref (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-angletoref.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.209 Function Entry: VLE-VECTOR-CROSSPRODUCT

24.209.1 Name

‘vle-vector-crossproduct‘

24.209.2 Class

BricsCAD VLE Library Function

24.209.3 Syntax

```
(vle-vector-crossproduct vec1 vec2)
```

24.209.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.209.5 Description

This function returns the cross-product vector of vector vec1 and vector vec2.

24.209.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.209.7 Examples

```
(setq v1 '(2 0 0) v2 '(0 3
0))

(vle-vector-crossproduct v1 v2)

(0.0 0.0 6.0)
```

24.209.8 Notes

the cross-product vector is orthogonal to both argument vectors;
the returned crossproduct vector is not normalised !

24.209.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.209.10 Source Notes

- [vle-vector-crossproduct (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-crossproduct.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.210 Function Entry: VLE-VECTOR-DOTPRODUCT

24.210.1 Name

‘vle-vector-dotproduct‘

24.210.2 Class

BricsCAD VLE Library Function

24.210.3 Syntax

```
(vle-vector-dotproduct vec1 vec2)
```

24.210.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.210.5 Description

This function returns the dot-product vector of vector vec1 and vector vec2.

24.210.6 Return Values

double

24.210.7 Examples

```
(setq v1 '(2 0 0) v2 '(0 3
0))

(vle-vector-dotproduct v1 v2)

0.0

(setq v1 '(2 0 0) v2 '(-2 0 0))

(vle-vector-dotproduct v1 v2)
```

```
-4.0
```

```
(setq v1 '(2 1 0) v2 '(-2 2 1))
```

```
(vle-vector-dotproduct v1 v2)
```

```
-2.0
```

24.210.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.210.9 Source Notes

- [vle-vector-dotproduct (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-dotproduct.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.211 Function Entry: VLE-VECTOR-GET

24.211.1 Name

‘vle-vector-get‘

24.211.2 Class

BricsCAD VLE Library Function

24.211.3 Syntax

```
(vle-vector-get fromPt toPt)
```

24.211.4 Arguments and Values

- fromPt a 2d or 3d point list
- toPt a 2d or
- 3d point list

24.211.5 Description

This function returns the vector fromPt -> toPt as 3d list.

24.211.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.211.7 Examples

```
(vle-vector-get  '(10 10
10)  '(21 22 23))  returns (11 12 13)
```

24.211.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.211.9 Source Notes

- [vle-vector-get (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-get.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.212 Function Entry: VLE-VECTOR-GETPERPVECTOR

24.212.1 Name

‘vle-vector-getperpvector‘

24.212.2 Class

BricsCAD VLE Library Function

24.212.3 Syntax

```
(vle-vector-getperpvector vec)
```

24.212.4 Arguments and Values

- vec a 2d or 3d vector list

24.212.5 Description

This function returns a vector which is perpendicular to vec.

24.212.6 Return Values

(vecX vecY vecZ) always a 3d vector list, perpendicular (orthogonal) to input vector vec

24.212.7 Examples

```
(setq vec '(2 0
0))

(vle-vector-getperpvector vec)

(0.0 1.0 0.0)

(setq vec '(2 0 3))

(vle-vector-getperpvector vec)

(0.0 1.0 0.0)

(setq vec '(2 -1 1))
```

```
(vle-vector-getperpvector vec)
```

```
(0.447213595499958 0.894427190999916
0.0)
```

24.212.8 Notes

input vector must not be unit-length (non-normalised vectors are allowed)

24.212.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.212.10 Source Notes

- [vle-vector-getperpvector (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-getperpvector.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.213 Function Entry: VLE-VECTOR-GETTOLERANCE

24.213.1 Name

‘vle-vector-gettolerance‘

24.213.2 Class

BricsCAD VLE Library Function

24.213.3 Syntax

`(vle-vector-gettolerance)`

24.213.4 Arguments and Values

- none

24.213.5 Description

This function returns the current tolerance used in vector functions.

24.213.6 Return Values

double the actual tolerance as used in vector functions

24.213.7 Notes

please do not modify ’$\mathrm{vle}_{\mathrm{gvecTol}}$’ directly, use this function

24.213.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.213.9 Source Notes

- [vle-vector-gettolerance (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-gettolerance.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.214 Function Entry: VLE-VECTOR-GETUCS

24.214.1 Name

‘vle-vector-getucs‘

24.214.2 Class

BricsCAD VLE Library Function

24.214.3 Syntax

```
(vle-vector-getucs normal)
```

24.214.4 Arguments and Values

- normal a 2d or 3d vector list

24.214.5 Description

This function returns the list of u-axis and v-axis for the given normal vector.

24.214.6 Return Values

’((uX uY uZ) (vX vY vZ)) always a list of 2 3d vectors, representing the x-axis and y-axis

24.214.7 Examples

```
(setq vec '(2 0
0))

(vle-vector-getucs vec)

((0.0 1.0 0.0) (0.0 0.0 1.0)

(setq vec '(2 2 1))

(vle-vector-getucs vec)

((-0.707106781186547 0.707106781186547 0.0)
(-0.235702260395516 -0.235702260395516 0.942809041582063))
```

24.214.8 Notes

input vector must not be unit-length (non-normalised vectors are allowed);
returned X and Y axis vectors are always normalised (1-unit length)

24.214.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.214.10 Source Notes

- [vle-vector-getucs (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-getucs.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.215 Function Entry: VLE-VECTOR-ISCODIRECTIONAL

24.215.1 Name

‘vle-vector-iscodirectional‘

24.215.2 Class

BricsCAD VLE Library Function

24.215.3 Syntax

```
(vle-vector-iscodirectional vec1 vec2)
```

24.215.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.215.5 Description

This function returns whether the vectors vec1 and vec2 are codirectional or not.

24.215.6 Return Values

T or NIL

24.215.7 Examples

```
(setq v1 '(2 0 0) v2'(-2 0
0))

(vle-vector-iscodirectional v1 v2)

NIL

(setq v1 '(2 0 0) v2'(3 0 0))

(vle-vector-iscodirectional v1 v2)
```

`T`

24.215.8 Notes

input vectors must not be unit-length (non-normalised vectors are allowed)

24.215.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.215.10 Source Notes

- [vle-vector-iscodirectional (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-iscodirectional.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.216 Function Entry: VLE-VECTOR-ISEQUAL

24.216.1 Name

‘vle-vector-isequal‘

24.216.2 Class

BricsCAD VLE Library Function

24.216.3 Syntax

```
(vle-vector-isequal vec1 vec2)
```

24.216.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.216.5 Description

This function returns whether vector vec1 is equal to vector vec2.

24.216.6 Return Values

T or NIL

24.216.7 Examples

```
(setq v1 '(2 0 0) v2'(-2 0
0))

(vle-vector-isequal v1 v2)

NIL

(setq v1 '(2 0 0) v2'(2 0 0))

(vle-vector-isequal v1 v2)

T
```

24.216.8 Notes

both arguments can as well be interpreted as 'points'; same as (equal vec1 vec2 {tolerance])

24.216.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.216.10 Source Notes

- [vle-vector-isequal (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isequal.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.217 Function Entry: VLE-VECTOR-ISPARALLEL

24.217.1 Name

‘vle-vector-isparallel‘

24.217.2 Class

BricsCAD VLE Library Function

24.217.3 Syntax

```
(vle-vector-isparallel vec1 vec2)
```

24.217.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.217.5 Description

This function returns whether the vectors vec1 and vec2 are parallel or not.

24.217.6 Return Values

T or NIL

24.217.7 Examples

```
(setq v1 '(2 0 0) v2'(-2 0
0))

(vle-vector-isparallel v1 v2)

T

(setq v1 '(2 0 0) v2'(3 0 0))

(vle-vector-isparallel v1 v2)
```

```
T
```

```
(setq v1 '(2 0 0) v2'(3 0 1))
```

```
(vle-vector-isparallel v1 v2)
```

```
NIL
```

24.217.8 Notes

input vectors must not be unit-length (non-normalised vectors are allowed)

24.217.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.217.10 Source Notes

- [vle-vector-isparallel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isparallel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.218 Function Entry: VLE-VECTOR-ISPERPENDICULAR

24.218.1 Name

‘vle-vector-isperpendicular‘

24.218.2 Class

BricsCAD VLE Library Function

24.218.3 Syntax

```
(vle-vector-isperpendicular vec1 vec2)
```

24.218.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.218.5 Description

This function returns whether the vectors vec1 and vec2 are orthogonal to each other.

24.218.6 Return Values

T or NIL

24.218.7 Examples

```
(setq v1 '(2 0 0) v2'(0 3
0))

(vle-vector-isperpendicular v1 v2)

T

(setq v1 '(2 1 0) v2'(-1 2 0))

(vle-vector-isperpendicular v1 v2)
```

```
T

(setq v1 '(2 1 0) v2'(1 2 0))

(vle-vector-isperpendicular v1 v2)

NIL
```

24.218.8 Notes

input vectors must not be unit-length (non-normalised vectors are allowed)

24.218.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.218.10 Source Notes

- [vle-vector-isperpendicular (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isperpendicular.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.219 Function Entry: VLE-VECTOR-ISUNITLENGTH

24.219.1 Name

‘vle-vector-isunitlength‘

24.219.2 Class

BricsCAD VLE Library Function

24.219.3 Syntax

```
(vle-vector-isunitlength vec)
```

24.219.4 Arguments and Values

- vec a 2d or 3d vector list

24.219.5 Description

This function returns whether vec has a length of 1.0 (1 unit length).

24.219.6 Return Values

T or NIL

24.219.7 Examples

```
(setq vec '(1 1
0))

(vle-vector-isunitlength v1 v2)

NIL

(setq vec '(0 1 0))

(vle-vector-isunitlength v1 v2)

T
```

24.219.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.219.9 Source Notes

- [vle-vector-isunitlength (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isunitlength.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.220 Function Entry: VLE-VECTOR-ISXAXIS

24.220.1 Name

‘vle-vector-isxaxis‘

24.220.2 Class

BricsCAD VLE Library Function

24.220.3 Syntax

```
(vle-vector-isxaxis vec)
```

24.220.4 Arguments and Values

- vec a 2d or 3d vector list

24.220.5 Description

This function returns whether vec is equal to X-axis vector (1 0 0).

24.220.6 Return Values

T or NIL

24.220.7 Examples

```
(setq vec '(1 1
0))

(vle-vector-isxaxis vec)

NIL

(setq vec '(1 0 0))

(vle-vector-isxaxis vec)

T
```

24.220.8 Notes

the vec must be in 1-unit length (normalised)

24.220.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.220.10 Source Notes

- [vle-vector-isxaxis (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isxaxis.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.221 Function Entry: VLE-VECTOR-ISYAXIS

24.221.1 Name

‘vle-vector-isyaxis‘

24.221.2 Class

BricsCAD VLE Library Function

24.221.3 Syntax

```
(vle-vector-isyaxis vec)
```

24.221.4 Arguments and Values

- vec a 2d or 3d vector list

24.221.5 Description

This function returns whether vec is equal to Y-axis vector (0 1 0).

24.221.6 Return Values

T or NIL

24.221.7 Examples

```
(setq vec '(1 1
0))

(vle-vector-isyaxis vec)

NIL

(setq vec '(0 1 0))

(vle-vector-isyaxis vec)

T
```

24.221.8 Notes

the vec must be in 1-unit length (normalised)

24.221.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.221.10 Source Notes

- [vle-vector-isyaxis (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-isyaxis.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.222 Function Entry: VLE-VECTOR-ISZAXIS

24.222.1 Name

‘vle-vector-iszaxis‘

24.222.2 Class

BricsCAD VLE Library Function

24.222.3 Syntax

```
(vle-vector-iszaxis vec)
```

24.222.4 Arguments and Values

- vec a 2d or 3d vector list

24.222.5 Description

This function returns whether vec is equal to Z-axis vector (0 0 1).

24.222.6 Return Values

T or NIL

24.222.7 Examples

```
(setq vec '(1 1
0))

(vle-vector-iszaxis vec)

NIL

(setq vec '(0 0 1))

(vle-vector-iszaxis vec)

T
```

24.222.8 Notes

the vec must be in 1-unit length (normalised)

24.222.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.222.10 Source Notes

- [vle-vector-iszaxis (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-iszaxis.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.223 Function Entry: VLE-VECTOR-ISZEROLENGTH

24.223.1 Name

‘vle-vector-iszerolength‘

24.223.2 Class

BricsCAD VLE Library Function

24.223.3 Syntax

```
(vle-vector-iszerolength vec)
```

24.223.4 Arguments and Values

- vec a 2d or 3d vector list

24.223.5 Description

This function returns whether vec has a length of 0.0 (zero length).

24.223.6 Return Values

T or NIL

24.223.7 Examples

```
(setq vec '(1 1
0))

(vle-vector-iszerolength v1 v2)

NIL

(setq vec '(0 0 0))

(vle-vector-iszerolength v1 v2)

T
```

24.223.8 Notes

if vec is interpreted as point, then it compares as "equal to point (0 0 0)"

24.223.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.223.10 Source Notes

- [vle-vector-iszerolength (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-iszerolength.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.224 Function Entry: VLE-VECTOR-LENGTH

24.224.1 Name

‘vle-vector-length‘

24.224.2 Class

BricsCAD VLE Library Function

24.224.3 Syntax

```
(vle-vector-length vec)
```

24.224.4 Arguments and Values

- vec a 2d or 3d vector list

24.224.5 Description

This function returns the 3D length of the vector vec.

24.224.6 Return Values

double, the 3D length of the vector in XYZ space

24.224.7 Examples

```
(setq vec '(1 1
3))

(vle-vector-length vec)

3.3166247903554
```

24.224.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.224.9 Source Notes

- [vle-vector-length (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-length.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.225 Function Entry: VLE-VECTOR-LENGTH2D

24.225.1 Name

‘vle-vector-length2d‘

24.225.2 Class

BricsCAD VLE Library Function

24.225.3 Syntax

```
(vle-vector-length2d vec)
```

24.225.4 Arguments and Values

- vec a 2d or 3d vector list

24.225.5 Description

This function returns the 2D length of the vector vec in XY plane.

24.225.6 Return Values

double, the 2D length of the vector in XY plane

24.225.7 Examples

```
(setq vec '(1 1
3))

(vle-vector-length2d vec)

1.4142135623731
```

24.225.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.225.9 Source Notes

- [vle-vector-length2d (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-length2d.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.226 Function Entry: VLE-VECTOR-LENGTH2DXZ

24.226.1 Name

‘vle-vector-length2dxz‘

24.226.2 Class

BricsCAD VLE Library Function

24.226.3 Syntax

```
(vle-vector-length2dxz vec)
```

24.226.4 Arguments and Values

- vec a 2d or 3d vector list

24.226.5 Description

This function returns the 2D length of the vector vec in XZ plane.

24.226.6 Return Values

double, the 2D length of the vector in XZ plane

24.226.7 Examples

```
(setq vec '(1 1
3))

(vle-vector-length2dxz vec)

3.16227766016838
```

24.226.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.226.9 Source Notes

- [vle-vector-length2dxz (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-length2dxz.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.227 Function Entry: VLE-VECTOR-LENGTH2DYZ

24.227.1 Name

‘vle-vector-length2dyz‘

24.227.2 Class

BricsCAD VLE Library Function

24.227.3 Syntax

```
(vle-vector-length2dyz vec)
```

24.227.4 Arguments and Values

- vec a 2d or 3d vector list

24.227.5 Description

This function returns the 2D length of the vector vec in YZ plane.

24.227.6 Return Values

double, the 2D length of the vector in YZ plane

24.227.7 Examples

```
(setq vec '(1 1
3))
```

```
(vle-vector-length2dyz vec)
```

```
3.16227766016838
```

24.227.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.227.9 Source Notes

- [vle-vector-length2dyz (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-length2dyz.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.228 Function Entry: VLE-VECTOR-MIDPOINT

24.228.1 Name

‘vle-vector-midpoint‘

24.228.2 Class

BricsCAD VLE Library Function

24.228.3 Syntax

```
(vle-vector-midpoint pnt1 pnt2)
```

24.228.4 Arguments and Values

- pnt1 a 2d or 3d point list
- pnt2 a 2d or
- 3d point list

24.228.5 Description

This function returns the midpoint between point pnt1 and point pnt2.

24.228.6 Return Values

list (midX midY midZ) always a 3d point list

24.228.7 Examples

```
(vle-vector-midpoint
 '(1 2 3)  '(10 20 30))  returns '(5.5  11
 16.5)
```

24.228.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.228.9 Source Notes

- [vle-vector-midpoint (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-midpoint.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.229 Function Entry: VLE-VECTOR-NEGATE

24.229.1 Name

‘vle-vector-negate‘

24.229.2 Class

BricsCAD VLE Library Function

24.229.3 Syntax

```
(vle-vector-negate vec)
```

24.229.4 Arguments and Values

- vec a 2d or 3d vector list

24.229.5 Description

This function returns the negated (inverted) vector of vec.

24.229.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.229.7 Examples

```
(setq vec '(1 1
3))
```

```
(vle-vector-negate vec)
```

```
(-1.0 -1.0 -3.0)
```

24.229.8 Notes

vle-vector-negate is identical to (but faster as) (vle-vector-scale vec -1.0)

24.229.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.229.10 Source Notes

- [vle-vector-negate (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-negate.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.230 Function Entry: VLE-VECTOR-NORMALISE

24.230.1 Name

‘vle-vector-normalise‘

24.230.2 Class

BricsCAD VLE Library Function

24.230.3 Syntax

```
(vle-vector-normalise vec)
```

24.230.4 Arguments and Values

- vec a 2d or 3d vector list

24.230.5 Description

This function returns the normalised vector of vector vec (length is 1.0).

24.230.6 Return Values

(vecX vecY vecZ) always a 3d vector list, always in unit-length (1.0)

24.230.7 Examples

```
(setq vec '(1 1
3))

(vle-vector-normalise vec)

(0.301511344577764 0.301511344577764
0.904534033733291)

(setq vec '(2 2 2))

(vle-vector-normalise vec)

(0.577350269189626 0.577350269189626
0.577350269189626)
```

24.230.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.230.9 Source Notes

- [vle-vector-normalise (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-normalise.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.231 Function Entry: VLE-VECTOR-SCALE

24.231.1 Name

‘vle-vector-scale‘

24.231.2 Class

BricsCAD VLE Library Function

24.231.3 Syntax

```
(vle-vector-scale vec factor)
```

24.231.4 Arguments and Values

- vec a 2d or 3d vector list
- factor integer or double, the scaling factor

24.231.5 Description

This function returns the vector vec scaled/multiplied by factor.

24.231.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.231.7 Examples

```
(setq vec '(1 1
3))

(vle-vector-scale vec 2.5)

(2.5 2.5 7.5)
```

24.231.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.231.9 Source Notes

- [vle-vector-scale (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-scale.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.232 Function Entry: VLE-VECTOR-SETTOLERANCE

24.232.1 Name

‘vle-vector-settolerance‘

24.232.2 Class

BricsCAD VLE Library Function

24.232.3 Syntax

```
(vle-vector-settolerance tol)
```

24.232.4 Arguments and Values

- tol double, the new tolerance (usually 1e-06 …
- 1e-12)

24.232.5 Description

This function sets the current tolerance used in vector functions.

24.232.6 Return Values

always T

24.232.7 Examples

```
(vle-vector-settolerance
0.000001)  T

sets default tolerance to 1.0e-06
```

24.232.8 Notes

please do not modify ’$\mathrm{vle}_{\mathrm{gvecTol}}$’ directly, use this function

24.232.9 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.232.10 Source Notes

- [vle-vector-settolerance (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-settolerance.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.233 Function Entry: VLE-VECTOR-SUB

24.233.1 Name

‘vle-vector-sub‘

24.233.2 Class

BricsCAD VLE Library Function

24.233.3 Syntax

```
(vle-vector-sub vec1 vec2)
```

24.233.4 Arguments and Values

- vec1 a 2d or 3d vector list
- vec2 a 2d or
- 3d vector list

24.233.5 Description

This function returns the subtraction of vector vec2 from vec1.

24.233.6 Return Values

(vecX vecY vecZ) always a 3d vector list

24.233.7 Examples

```
(vle-vector-sub  '(10 10
10)  '(1 2 3))  returns '(9 8 7)
```

24.233.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.233.9 Source Notes

- [vle-vector-sub (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-sub.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.234 Function Entry: VLE-VECTOR-TO2D

24.234.1 Name

‘vle-vector-to2d‘

24.234.2 Class

BricsCAD VLE Library Function

24.234.3 Syntax

```
(vle-vector-to2d vec)
```

24.234.4 Arguments and Values

- vec (list) a 2d or 3d point/vector list of 2 or 3
- integer or double values

24.234.5 Description

This function returns a 2-element list representing a 2D point/vector for a given vec 2D or 3D list.

24.234.6 Return Values

a 2-element list representing a 2D point/vector for a given 2D or 3D list, or NIL

24.234.7 Examples

```
(vle-vector-to3d  '(10
11 12))  returns '(10 11)
```

24.234.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.234.9 Source Notes

- [vle-vector-to2d (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-to2d.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.235 Function Entry: VLE-VECTOR-TO3D

24.235.1 Name

‘vle-vector-to3d‘

24.235.2 Class

BricsCAD VLE Library Function

24.235.3 Syntax

```
(vle-vector-to3d vec)
```

24.235.4 Arguments and Values

- vec (list) a 2d or 3d point/vector list of 2 or 3
- integer or double values

24.235.5 Description

This function returns a 3-element list representing a 3D point/vector for a given vec 2D or 3D list.

24.235.6 Return Values

a 3-element list representing a 3D point/vector for a given 2D or 3D list, or NIL

24.235.7 Examples

```
(vle-vector-to3d  '(10
11))  returns '(10 11 0)
```

24.235.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.235.9 Source Notes

- [vle-vector-to3d (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle-vector-to3d.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.236 Function Entry: VLE-VLAOBJECTP

24.236.1 Name

‘vle-vlaobjectp‘

24.236.2 Class

BricsCAD VLE Library Function

24.236.3 Syntax

`(vle-vlaobjectp obj)`

24.236.4 Arguments and Values

- obj any Lisp item to be verified

24.236.5 Description

predicator function to determine whether ’obj’ is of VLA-OBJECT type

24.236.6 Return Values

T - if ’obj’ is a VLA-OBJECT object
NIL - if ’obj’ is not a VLA-OBJECT object

24.236.7 Notes

replacement for : (= (type obj) ’VLA-OBJECT) and similar

24.236.8 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.236.9 Source Notes

- [vle-vlaobjectp (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vlevlaobjectp.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.237 Function Entry: $VLE_{GVECTOL}$

24.237.1 Name

'$vle_{gvectol}$'

24.237.2 Class

BricsCAD VLE Library Function

24.237.3 Syntax

`(vle_g_vectol)`

24.237.4 Description

$vle_{gvecTol}$ $vle_{gvecTol}$ double 1.0e-10 Preset tolerance for (vle-vector-xxx) functions. Lisp code can retrieve and adjust to a different value if needed : (vle-vector-setTolerance 0.00000001) ;; preset as 1.0e-08 Remarks : Please do not access "$vle_{gvecTol}$" directly, always use (vle-vector-getTolerance) and (vle-vector-setTolerance) !

24.237.5 Return Values

Vendor-defined; see Source Notes.

24.237.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.237.7 Source Notes

- [$vle_{gvectol}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/vle_g_vecTol.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.237.8 ExpressTools API Support (208 entries)

24.238 Function Entry: ACET-ACAD-REFRESH

24.238.1 Name

‘acet-acad-refresh‘

24.238.2 Class

BricsCAD ExpressTools API Function

24.238.3 Syntax

`(acet-acad-refresh)`

24.238.4 Description

Causes a refresh of the drawing window.

24.238.5 Return Values

Vendor-defined; see Source Notes.

24.238.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.238.7 Source Notes

- [acet-acad-refresh (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetacadrefresh.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.239 Function Entry: ACET-ALERT

24.239.1 Name

‘acet-alert‘

24.239.2 Class

BricsCAD ExpressTools API Function

24.239.3 Syntax

`(acet-alert msg)`

24.239.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.239.5 Description

Shows a message box displaying the msg information.

24.239.6 Return Values

Vendor-defined; see Source Notes.

24.239.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.239.8 Source Notes

- [acet-alert (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetalert.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.240 Function Entry: ACET-ANGLE-EQUAL

24.240.1 Name

‘acet-angle-equal‘

24.240.2 Class

BricsCAD ExpressTools API Function

24.240.3 Syntax

`(acet-angle-equal a b fuz)`

24.240.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.240.5 Description

Compares two angle a and b values based on tolerance fuz; both angles are normalised to 0...2Pi range. Returns T or NIL.

24.240.6 Return Values

Vendor-defined; see Source Notes.

24.240.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.240.8 Source Notes

- [acet-angle-equal (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetangleequal.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.241 Function Entry: ACET-ANGLE-FORMAT

24.241.1 Name

‘acet-angle-format‘

24.241.2 Class

BricsCAD ExpressTools API Function

24.241.3 Syntax

`(acet-angle-format angle)`

24.241.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.241.5 Description

Returns the input angle value angle as normalised angle in range 0…2Pi.

24.241.6 Return Values

Vendor-defined; see Source Notes.

24.241.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.241.8 Source Notes

- [acet-angle-format (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetangleformat.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.242 Function Entry: ACET-ANNOTATION-SS

24.242.1 Name

‘acet-annotation-ss‘

24.242.2 Class

BricsCAD ExpressTools API Function

24.242.3 Syntax

`(acet-annotation-ss ss)`

24.242.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.242.5 Description

This function is identical to (acet-ss-annotation-filter).

24.242.6 Return Values

Vendor-defined; see Source Notes.

24.242.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.242.8 Source Notes

- [acet-annotation-ss (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetannotationss.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.243 Function Entry: ACET-APPID-DELETE

24.243.1 Name

‘acet-appid-delete‘

24.243.2 Class

BricsCAD ExpressTools API Function

24.243.3 Syntax

`(acet-appid-delete appId)`

24.243.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.243.5 Description

This function removes XData from all entities using such APPID based XData, and finally removes the appid from the APPID table. appid can contain wildcard characters (like ”*” and ”?”).

24.243.6 Return Values

Vendor-defined; see Source Notes.

24.243.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.243.8 Source Notes

- [acet-appid-delete (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetappiddelete.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.244 Function Entry: ACET-ARXLOAD-OR-BUST

24.244.1 Name

‘acet-arxload-or-bust‘

24.244.2 Class

BricsCAD ExpressTools API Function

24.244.3 Syntax

`(acet-arxload-or-bust filename)`

24.244.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.244.5 Description

Identical to (arxload filename).

24.244.6 Return Values

Vendor-defined; see Source Notes.

24.244.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.244.8 Source Notes

- [acet-arxload-or-bust (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetarxloadorbust.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.245 Function Entry: ACET-ATT-SUBSCRIPT-DUPLICATES

24.245.1 Name

‘acet-att-subscript-duplicates‘

24.245.2 Class

BricsCAD ExpressTools API Function

24.245.3 Syntax

```
(acet-att-subscript-duplicates inplist)
```

24.245.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.245.5 Description

Verifies the input list inplist for duplicate tag names - any duplicate tag names are subscripted by "(<idx>)". inplist must be a list of (tag value) pairs - verifying the tags is case-insensitive.

24.245.6 Return Values

Vendor-defined; see Source Notes.

24.245.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.245.8 Source Notes

- [acet-att-subscript-duplicates (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetattsubscriptduplicates.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.246 Function Entry: ACET-AUTOLOAD

24.246.1 Name

‘acet-autoload‘

24.246.2 Class

BricsCAD ExpressTools API Function

24.246.3 Syntax

`(acet-autoload loadlist)`

24.246.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.246.5 Description

Defines AutoLoader commands and functions, based on definitions in ; loadlist has the 2 entries : first entry is the application file to be loaded (.lsp, .arx, .brx), second item is the Example : (autoload '("MyTool.lsp" "(myTool arg1 arg2"))

24.246.6 Return Values

Vendor-defined; see Source Notes.

24.246.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.246.8 Source Notes

- [acet-autoload (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetautoload.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.247 Function Entry: ACET-AUTOLOAD2

24.247.1 Name

‘acet-autoload2‘

24.247.2 Class

BricsCAD ExpressTools API Function

24.247.3 Syntax

```
(acet-autoload2 loadlist)
```

24.247.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.247.5 Description

Similar to (acet-autoload), loadlist can have 1 or more arguments; the second argument can be a string or a Lisp function definition. NOTE : only the first argument (application file) is supported This function searches for a matching application file, and if found, loads the Lisp/BRX module immediately. If the application file is specified without extension, .lsp is automatically appended to the filename. supported file types : .lsp, .arx, .brx example : (acet-autoload2 '("MyTool" "C:MyTool" nil/t "MyTool" 2)) searches for MyTool.lsp in all support pathes, and loads it; other arguments are ignored (acet-autoload2 '("MyTool.brx" "MyTool")) searches for MyTool.brx in all support pathes, and loads it; other arguments are ignored

24.247.6 Return Values

Vendor-defined; see Source Notes.

24.247.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.247.8 Source Notes

- [acet-autoload2 (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetautoload2.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.248 Function Entry: ACET-AUTOLOADARX

24.248.1 Name

'acet-autoloadarx'

24.248.2 Class

BricsCAD ExpressTools API Function

24.248.3 Syntax

```
(acet-autoloadarx loadlist)
```

24.248.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.248.5 Description

Similar to (acet-autoload2) function, but primarily intended for loading ARX/BRX modules; the loadlist argument can have a 3rd argument, which is a dedicated command function call which is executed after the ARX/BRX module is loaded. example : (acet-autoloadarx '("MyTool.brx" "(C:MyToolCmd)" "(command \"MyToolCmd\")")) The "MyTool.brx" module will be loaded, when triggered from a "(C:MyToolCmd)" call, and after successful module loading, it will automatically call the Lisp expression "(command \"MyToolCmd\")"

24.248.6 Return Values

Vendor-defined; see Source Notes.

24.248.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.248.8 Source Notes

- [acet-autoloadarx (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetautoloadarx.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.249 Function Entry: ACET-BLINK-AND-SHOW-OBJECT

24.249.1 Name

‘acet-blink-and-show-object‘

24.249.2 Class

BricsCAD ExpressTools API Function

24.249.3 Syntax

`(acet-blink-and-show-object inplist)`

24.249.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.249.5 Description

This function draws a rectangle around the entity specified in inplist, using specified color, and draws + undraws the rectangle several times, to show a kind of ”blinking” effect; the inplist argument must be of format (ename color).

24.249.6 Return Values

Vendor-defined; see Source Notes.

24.249.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.249.8 Source Notes

- [acet-blink-and-show-object (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetblinkandshowobject.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.250 Function Entry: ACET-BLK-MATCH

24.250.1 Name

‘acet-blk-match‘

24.250.2 Class

BricsCAD ExpressTools API Function

24.250.3 Syntax

`(acet-blk-match filter)`

24.250.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.250.5 Description

Returns a list of entities in BlockTable, due to specified filter filter.

24.250.6 Return Values

Vendor-defined; see Source Notes.

24.250.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.250.8 Source Notes

- [acet-blk-match (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetblkmatch.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.251 Function Entry: ACET-BLKTBL-MATCH

24.251.1 Name

‘acet-blktbl-match‘

24.251.2 Class

BricsCAD ExpressTools API Function

24.251.3 Syntax

```
(acet-blktbl-match blkname filter lst flag)
```

24.251.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.251.5 Description

Returns a list of entities in specified blocks, due to specified filter blkname, filter, lst and flag.

24.251.6 Return Values

Vendor-defined; see Source Notes.

24.251.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.251.8 Source Notes

- [acet-blktbl-match (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetblktblmatch.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.252 Function Entry: ACET-BLOCK-MAKE-ANON

24.252.1 Name

‘acet-block-make-anon‘

24.252.2 Class

BricsCAD ExpressTools API Function

24.252.3 Syntax

`(acet-block-make-anon ss blockname)`

24.252.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.252.5 Description

Create a new anonymous block from selectionset ss, using specified block name blockname. If blockname is NIL, a new block name is automatically determined.

24.252.6 Return Values

Vendor-defined; see Source Notes.

24.252.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.252.8 Source Notes

- [acet-block-make-anon (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetblockmakeanon.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.253 Function Entry: ACET-BLOCK-PURGE

24.253.1 Name

‘acet-block-purge‘

24.253.2 Class

BricsCAD ExpressTools API Function

24.253.3 Syntax

`(acet-block-purge blockname)`

24.253.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.253.5 Description

Purges the specified block blockname from the drawing.

24.253.6 Return Values

Vendor-defined; see Source Notes.

24.253.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.253.8 Source Notes

- [acet-block-purge (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetblockpurge.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.254 Function Entry: ACET-BS-STRIP

24.254.1 Name

‘acet-bs-strip‘

24.254.2 Class

BricsCAD ExpressTools API Function

24.254.3 Syntax

```
(acet-bs-strip string)
```

24.254.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.254.5 Description

Removes any backspace characters (0x08) from input string string.

24.254.6 Return Values

Vendor-defined; see Source Notes.

24.254.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.254.8 Source Notes

- [acet-bs-strip (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetbsstrip.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.255 Function Entry: ACET-CALC-BITLIST

24.255.1 Name

‘acet-calc-bitlist‘

24.255.2 Class

BricsCAD ExpressTools API Function

24.255.3 Syntax

```
(acet-calc-bitlist num)
```

24.255.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.255.5 Description

Returns num as a list of bit values (splitting num into bit values). Example : (acet-calc-bitlist 7) returns ’(1 2 4)

24.255.6 Return Values

Vendor-defined; see Source Notes.

24.255.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.255.8 Source Notes

- [acet-calc-bitlist (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcalcbitlist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.256 Function Entry: ACET-CALC-ROUND

24.256.1 Name

‘acet-calc-round‘

24.256.2 Class

BricsCAD ExpressTools API Function

24.256.3 Syntax

`(acet-calc-round num prec)`

24.256.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.256.5 Description

Returns the number num rounded to numerical prec; similar to (but not identical with !) (vle-roundto). Examples : (acet-calc-round 1.2345 0.1) results in 1.2 (acet-calc-round 1.2345 0.01) results in 1.23 (acet-calc-round 1.2345 0.001) results in 1.234 (acet-calc-round 1.2345 0.0001) results in 1.2345

24.256.6 Return Values

Vendor-defined; see Source Notes.

24.256.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.256.8 Source Notes

- [acet-calc-round (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcalcround.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.257 Function Entry: ACET-CALC-TAN

24.257.1 Name

‘acet-calc-tan‘

24.257.2 Class

BricsCAD ExpressTools API Function

24.257.3 Syntax

`(acet-calc-tan num)`

24.257.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.257.5 Description

Returns the tangent value for num; same as (tan) and (vle-tan).

24.257.6 Return Values

Vendor-defined; see Source Notes.

24.257.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.257.8 Source Notes

- [acet-calc-tan (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcalctan.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.258 Function Entry: ACET-CLEAR-TEMP-GRAPHICS

24.258.1 Name

‘acet-clear-temp-graphics‘

24.258.2 Class

BricsCAD ExpressTools API Function

24.258.3 Syntax

`(acet-clear-temp-graphics)`

24.258.4 Description

This functions removes any temporary graphics from screen, which were generated by dragging + jigger operations, (grdraw), (grvecs) and similar functions, as well as removing visual artefacts.

24.258.5 Return Values

Vendor-defined; see Source Notes.

24.258.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.258.7 Source Notes

- [acet-clear-temp-graphics (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcleartempgraphics.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.259 Function Entry: ACET-CMD-EXIT

24.259.1 Name

‘acet-cmd-exit‘

24.259.2 Class

BricsCAD ExpressTools API Function

24.259.3 Syntax

```
(acet-cmd-exit)
```

24.259.4 Description

Cancels all running commands (by using (command) calls).

24.259.5 Return Values

Vendor-defined; see Source Notes.

24.259.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.259.7 Source Notes

- [acet-cmd-exit (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcmdexit.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.260 Function Entry: ACET-COPYM

24.260.1 Name

‘acet-copym‘

24.260.2 Class

BricsCAD ExpressTools API Function

24.260.3 Syntax

```
(acet-copym sset refPt)
```

24.260.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.260.5 Description

This function provides the user to copy the given sset (PickSet)SelectionSet one or multiple times; refPt (2D or 3D point list) is the initial reference point for the copy operation.

24.260.6 Return Values

Vendor-defined; see Source Notes.

24.260.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.260.8 Source Notes

- [acet-copym (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-copym.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.261 Function Entry: ACET-COPYTOLAYER

24.261.1 Name

‘acet-copytolayer‘

24.261.2 Class

BricsCAD ExpressTools API Function

24.261.3 Syntax

```
(acet-copytolayer sset layer)
```

24.261.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.261.5 Description

This function copies the entities of the given sset (PickSet) SelectionSet to the specified layer (string).

24.261.6 Return Values

Vendor-defined; see Source Notes.

24.261.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.261.8 Source Notes

- [acet-copytolayer (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-copytolayer.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.262 Function Entry: ACET-CURRENTVIEWPORT-ENAME

24.262.1 Name

'acet-currentviewport-ename'

24.262.2 Class

BricsCAD ExpressTools API Function

24.262.3 Syntax

`(acet-currentviewport-ename)`

24.262.4 Description

Returns the entity name of the active viewport, when actually in a PaperSpace layout, NIL when in modelspace layout ("Model"), or when no active viewport is present.

24.262.5 Return Values

Vendor-defined; see Source Notes.

24.262.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.262.7 Source Notes

- [acet-currentviewport-ename (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetcurrentviewportename.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.263 Function Entry: ACET-DCL-LIST-MAKE

24.263.1 Name

‘acet-dcl-list-make‘

24.263.2 Class

BricsCAD ExpressTools API Function

24.263.3 Syntax

```
(acet-dcl-list-make dcltile valuelist)
```

24.263.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.263.5 Description

Fills a DCL list box dcltile with the values from valuelist.

24.263.6 Return Values

Vendor-defined; see Source Notes.

24.263.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.263.8 Source Notes

- [acet-dcl-list-make (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetdcllistmake.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.264 Function Entry: ACET-DICT-ENAME

24.264.1 Name

‘acet-dict-ename‘

24.264.2 Class

BricsCAD ExpressTools API Function

24.264.3 Syntax

`(acet-dict-ename dictname entry)`

24.264.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.264.5 Description

Returns the entity name of item entry in dictionary dictname (STRING or ENAME), or NIL if not found.

24.264.6 Return Values

Vendor-defined; see Source Notes.

24.264.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.264.8 Source Notes

- [acet-dict-ename (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetdictename.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.265 Function Entry: ACET-DICT-NAME-LIST

24.265.1 Name

‘acet-dict-name-list‘

24.265.2 Class

BricsCAD ExpressTools API Function

24.265.3 Syntax

`(acet-dict-name-list dictname)`

24.265.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.265.5 Description

Returns the list of entries in dictionary dictname (STRING or ENAME) as a list of strings.

24.265.6 Return Values

Vendor-defined; see Source Notes.

24.265.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.265.8 Source Notes

- [acet-dict-name-list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetdictnamelist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.266 Function Entry: ACET-DTOR

24.266.1 Name

‘acet-dtor‘

24.266.2 Class

BricsCAD ExpressTools API Function

24.266.3 Syntax

`(acet-dtor ang)`

24.266.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.266.5 Description

Returns the angle value of ang in radians.

24.266.6 Return Values

Vendor-defined; see Source Notes.

24.266.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.266.8 Source Notes

- [acet-dtor (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetdtor.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.267 Function Entry: ACET-DXF

24.267.1 Name

‘acet-dxf‘

24.267.2 Class

BricsCAD ExpressTools API Function

24.267.3 Syntax

`(acet-dxf key assoclist)`

24.267.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.267.5 Description

Returns the (cdr (assoc key assoclist)) value; assoclist is usually an (entget) list, but can be any assoc list.

24.267.6 Return Values

Vendor-defined; see Source Notes.

24.267.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.267.8 Source Notes

- [acet-dxf (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetdxf.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.268 Function Entry: ACET-ELIST-ADD-DEFAULTS

24.268.1 Name

‘acet-elist-add-defaults‘

24.268.2 Class

BricsCAD ExpressTools API Function

24.268.3 Syntax

`(acet-elist-add-defaults elist)`

24.268.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.268.5 Description

This function updates and completes the given entity definition list with default entries for layer, color, lineweight and linetype.

24.268.6 Return Values

Vendor-defined; see Source Notes.

24.268.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.268.8 Source Notes

- [acet-elist-add-defaults (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetelistadddefaults.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.269 Function Entry: ACET-ENT-CURVEPOINTS

24.269.1 Name

‘acet-ent-curvepoints‘

24.269.2 Class

BricsCAD ExpressTools API Function

24.269.3 Syntax

`(acet-ent-curvepoints ename resolution)`

24.269.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.269.5 Description

Returns a list of points along the curve entity ename (line, arc, circle, polyline, spline, ellipse, leader, ray, xline); the number resolution must be in range 0.0 < resolution < 1.0 and controls the smoothness of curve scan points.

24.269.6 Return Values

Vendor-defined; see Source Notes.

24.269.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.269.8 Source Notes

- [acet-ent-curvepoints (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetentcurvepoints.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.270 Function Entry: ACET-ERROR-INIT

24.270.1 Name

‘acet-error-init‘

24.270.2 Class

BricsCAD ExpressTools API Function

24.270.3 Syntax

```
(acet-error-init initdata)
```

24.270.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.270.5 Description

Initialises an error handler due to arguments contained in initdata list; initdata is a list of up to three items or NIL : item 1 : list of system variables and value to set - ’(”cmdecho” 0 ”osmode” 0 ...) or NIL the actual values of the specified system variables are saved for later (acet-error-restore) call, and will also be restored in case of an error item 2 : flag indicating UNDO usage : NIL means not using UNDO at all 0 means Undo/Begin is placed, but no UNDO is done in case of an error 1 means means Undo/Begin is placed, and UNDO operation is done in case of an error item 3 : optional a custom function or expression to be called in case of an error (can be a quoted expression, function, lambda, defun etc.); this optional error expression will only be used if the item 2 is provided as 1 When using this function, the Lisp code should also call the matching (acet-error-restore) at end of execution. Note : functions can be used in nested function calls. In such case, the UNDO operation handling is initialised resp. executed only in the first, outermost calls to (acet-error-init) and (acet-error-restore), while system variables are properly saved and restored with each nesting level.

24.270.6 Return Values

Vendor-defined; see Source Notes.

24.270.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.270.8 Source Notes

- [acet-error-init (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/aceterrorinit.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.271 Function Entry: ACET-ERROR-RESTORE

24.271.1 Name

‘acet-error-restore‘

24.271.2 Class

BricsCAD ExpressTools API Function

24.271.3 Syntax

`(acet-error-restore)`

24.271.4 Description

Resets the error handler to original error handler which was active at preceding call of (acet-error-init) and also restores saved system variables (if any were stored). Note : functions can be used in nested function calls. In such case, the UNDO operation handling is initialised resp. executed only in the first, outermost calls to (acet-error-init) and (acet-error-restore), while system variables are properly saved and restored with each nesting level.

24.271.5 Return Values

Vendor-defined; see Source Notes.

24.271.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.271.7 Source Notes

- [acet-error-restore (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/aceterrorrestore.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.272 Function Entry: ACET-EXPLODE

24.272.1 Name

‘acet-explode‘

24.272.2 Class

BricsCAD ExpressTools API Function

24.272.3 Syntax

`(acet-explode EnameOrSelectionset)`

24.272.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.272.5 Description

Returns a selectionset, containing all the entities created by exploding EnameOrSelectionset, which can be a single entity or a selectionset.

24.272.6 Return Values

Vendor-defined; see Source Notes.

24.272.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.272.8 Source Notes

- [acet-explode (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetexplode.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.273 Function Entry: ACET-FILE-BACKUP

24.273.1 Name

‘acet-file-backup‘

24.273.2 Class

BricsCAD ExpressTools API Function

24.273.3 Syntax

```
(acet-file-backup filename)
```

24.273.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.273.5 Description

Creates a backup copy of given file filename; the name of the backup file copy will use the additional "$_{bak}$" postfix in its filename. All backup files created by this functions will internally be collected, and can be deleted by (acet-file-backup-delete). Return : NIL always

24.273.6 Return Values

Vendor-defined; see Source Notes.

24.273.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.273.8 Source Notes

- [acet-file-backup (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilebackup.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.274 Function Entry: ACET-FILE-BACKUP-DELETE

24.274.1 Name

‘acet-file-backup-delete‘

24.274.2 Class

BricsCAD ExpressTools API Function

24.274.3 Syntax

`(acet-file-backup-delete)`

24.274.4 Description

This functions deletes all backup files created by (acet-file-backup). Return : T always

24.274.5 Return Values

Vendor-defined; see Source Notes.

24.274.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.274.7 Source Notes

- [acet-file-backup-delete (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilebackupdelete.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.275 Function Entry: ACET-FILE-FIND

24.275.1 Name

‘acet-file-find‘

24.275.2 Class

BricsCAD ExpressTools API Function

24.275.3 Syntax

```
(acet-file-find filename)
```

24.275.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.275.5 Description

Searches for the file given by filename on all support pathes; this is similar to normal (findfile) so far - but if the file is not found, the path is removed from filename, and the search is repeated again. Returns : the name of the file found, or NIL

24.275.6 Return Values

Vendor-defined; see Source Notes.

24.275.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.275.8 Source Notes

- [acet-file-find (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilefind.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.276 Function Entry: ACET-FILE-FIND-FONT

24.276.1 Name

‘acet-file-find-font‘

24.276.2 Class

BricsCAD ExpressTools API Function

24.276.3 Syntax

`(acet-file-find-font)`

24.276.4 Description

(acet-file-find-image fontfile) Searches for an image file specified as fontfile, using (acet-file-find); if no file extension is specified, it searches for .shx and .ttf files (in that sequence); additionally, the ./Fonts folder of the operating system is also searched. Returns : the name of the file found, or NIL

24.276.5 Return Values

Vendor-defined; see Source Notes.

24.276.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.276.7 Source Notes

- [acet-file-find-font (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilefindfont.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.277 Function Entry: ACET-FILE-FIND-IMAGE

24.277.1 Name

‘acet-file-find-image‘

24.277.2 Class

BricsCAD ExpressTools API Function

24.277.3 Syntax

```
(acet-file-find-image imagefile)
```

24.277.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.277.5 Description

Searches for an image file specified as imagefile, using (acet-file-find); if no file extension is specified, the functions uses .bmp, .gif, .jpg, .png, .tif - in this sequence. Returns : the name of the file found, or NIL

24.277.6 Return Values

Vendor-defined; see Source Notes.

24.277.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.277.8 Source Notes

- [acet-file-find-image (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilefindimage.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.278 Function Entry: ACET-FILE-FIND-ON-PATH

24.278.1 Name

‘acet-file-find-on-path‘

24.278.2 Class

BricsCAD ExpressTools API Function

24.278.3 Syntax

`(acet-file-find-on-path filename pathes)`

24.278.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.278.5 Description

Searches the file filename by using on all folders as specified by pathes; pathes is a string, with one of more folder names, separated by ”;” - see

24.278.6 Return Values

Vendor-defined; see Source Notes.

24.278.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.278.8 Source Notes

- [acet-file-find-on-path (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilefindonpath.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.279 Function Entry: ACET-FILE-OPEN

24.279.1 Name

‘acet-file-open‘

24.279.2 Class

BricsCAD ExpressTools API Function

24.279.3 Syntax

`(acet-file-open filename mode)`

24.279.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.279.5 Description

Opens the specified file filename with specified mode; mode can be ”r” for read-mode, ”w” for write mode, ”a” for append mode; additionally the ”b” suffix indicates ”binary” mode, without newline character translation. Returns the file handle, or NIL in case of an error

24.279.6 Return Values

Vendor-defined; see Source Notes.

24.279.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.279.8 Source Notes

- [acet-file-open (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfileopen.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.280 Function Entry: ACET-FILE-READDIALOG

24.280.1 Name

‘acet-file-readdialog‘

24.280.2 Class

BricsCAD ExpressTools API Function

24.280.3 Syntax

`(acet-file-readdialog title folder extension helptopic flags)`

24.280.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.280.5 Description

Opens the file selection dialog with specified arguments to select an existing file, and returns selected filename or NIL. All arguments can be provided as NIL which uses default arguments then.

24.280.6 Return Values

Vendor-defined; see Source Notes.

24.280.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.280.8 Source Notes

- [acet-file-readdialog (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-file-readdialog.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.281 Function Entry: ACET-FILE-WRITEDIALOG

24.281.1 Name

'acet-file-writedialog'

24.281.2 Class

BricsCAD ExpressTools API Function

24.281.3 Syntax

`(acet-file-writedialog title folder extension helptopic flags)`

24.281.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.281.5 Description

Opens the file selection dialog with specified arguments to create a new file (or overwrite an existing file), and returns selected filename or NIL. All arguments can be provided as NIL which uses default arguments then.

24.281.6 Return Values

Vendor-defined; see Source Notes.

24.281.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.281.8 Source Notes

- [acet-file-writedialog (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilewritedialog.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.282 Function Entry: ACET-FILENAME-ASSOCIATED-APP

24.282.1 Name

‘acet-filename-associated-app‘

24.282.2 Class

BricsCAD ExpressTools API Function

24.282.3 Syntax

`(acet-filename-associated-app filename)`

24.282.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.282.5 Description

Returns the name of the application, associated with given file filename; based on Windows Registry settings, so this will not work correctly on Linux/Mac platforms.

24.282.6 Return Values

Vendor-defined; see Source Notes.

24.282.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.282.8 Source Notes

- [acet-filename-associated-app (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenameassociatedapp.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.283 Function Entry: ACET-FILENAME-DIRECTORY

24.283.1 Name

‘acet-filename-directory‘

24.283.2 Class

BricsCAD ExpressTools API Function

24.283.3 Syntax

`(acet-filename-directory filename)`

24.283.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.283.5 Description

Returns the folder part of given filename.

24.283.6 Return Values

Vendor-defined; see Source Notes.

24.283.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.283.8 Source Notes

- [acet-filename-directory (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-filename-directory.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.284 Function Entry: ACET-FILENAME-EXT-REMOVE

24.284.1 Name

‘acet-filename-ext-remove‘

24.284.2 Class

BricsCAD ExpressTools API Function

24.284.3 Syntax

`(acet-filename-ext-remove filename)`

24.284.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.284.5 Description

Returns the filename without file extension.

24.284.6 Return Values

Vendor-defined; see Source Notes.

24.284.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.284.8 Source Notes

- [acet-filename-ext-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenameextremove.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.285 Function Entry: ACET-FILENAME-EXTENSION

24.285.1 Name

‘acet-filename-extension‘

24.285.2 Class

BricsCAD ExpressTools API Function

24.285.3 Syntax

`(acet-filename-extension filename)`

24.285.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.285.5 Description

Returns the extension part of given filename.

24.285.6 Return Values

Vendor-defined; see Source Notes.

24.285.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.285.8 Source Notes

- [acet-filename-extension (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenameextension.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.286 Function Entry: ACET-FILENAME-PATH-REMOVE

24.286.1 Name

‘acet-filename-path-remove‘

24.286.2 Class

BricsCAD ExpressTools API Function

24.286.3 Syntax

```
(acet-filename-path-remove filename)
```

24.286.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.286.5 Description

Returns the filename without path.

24.286.6 Return Values

Vendor-defined; see Source Notes.

24.286.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.286.8 Source Notes

- [acet-filename-path-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenamepathremove.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.287 Function Entry: ACET-FILENAME-SUPPORTPATH-REMOVE

24.287.1 Name

‘acet-filename-supportpath-remove‘

24.287.2 Class

BricsCAD ExpressTools API Function

24.287.3 Syntax

```
(acet-filename-supportpath-remove filename)
```

24.287.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.287.5 Description

Returns the filename without path, if the file is located in support pathes; (this means, the filename without path can be found by (findfile) operation).

24.287.6 Return Values

Vendor-defined; see Source Notes.

24.287.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.287.8 Source Notes

- [acet-filename-supportpath-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenamesupportpathremove.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.288 Function Entry: ACET-FILENAME-VALID

24.288.1 Name

‘acet-filename-valid‘

24.288.2 Class

BricsCAD ExpressTools API Function

24.288.3 Syntax

```
(acet-filename-valid filename)
```

24.288.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.288.5 Description

Returns T if the file specified by filename is valid, NIL otherwise.

24.288.6 Return Values

Vendor-defined; see Source Notes.

24.288.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.288.8 Source Notes

- [acet-filename-valid (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfilenamevalid.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.289 Function Entry: ACET-FSCREEN-TOGGLE

24.289.1 Name

‘acet-fscreen-toggle‘

24.289.2 Class

BricsCAD ExpressTools API Function

24.289.3 Syntax

`(acet-fscreen-toggle)`

24.289.4 Description

Provided for backward compatibility, implementation does nothing (no operation).

24.289.5 Return Values

Vendor-defined; see Source Notes.

24.289.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.289.7 Source Notes

- [acet-fscreen-toggle (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfscreentoggle.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.290 Function Entry: ACET-GENERAL-PROPS-GET

24.290.1 Name

‘acet-general-props-get‘

24.290.2 Class

BricsCAD ExpressTools API Function

24.290.3 Syntax

`(acet-general-props-get object)`

24.290.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.290.5 Description

Returns a list with entity properties, as sequence of ("$_{\text{layer}}$" <layername> "$_{\text{color}}$" ...); this sequence can directly be used by (command); object can be an entity name (ENAME) or an (entget) list

24.290.6 Return Values

Vendor-defined; see Source Notes.

24.290.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.290.8 Source Notes

- [acet-general-props-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeneralpropsget.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.291 Function Entry: ACET-GENERAL-PROPS-GET-PAIRS

24.291.1 Name

‘acet-general-props-get-pairs‘

24.291.2 Class

BricsCAD ExpressTools API Function

24.291.3 Syntax

`(acet-general-props-get-pairs)`

24.291.4 Description

Returns a list with entity properties, each property is given as (dxf . value) item in that list; object can be an entity name (ENAME) or an (entget) list

24.291.5 Return Values

Vendor-defined; see Source Notes.

24.291.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.291.7 Source Notes

- [acet-general-props-get-pairs (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeneralpropsgetpairs.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.292 Function Entry: ACET-GENERAL-PROPS-SET

24.292.1 Name

‘acet-general-props-set‘

24.292.2 Class

BricsCAD ExpressTools API Function

24.292.3 Syntax

`(acet-general-props-set ss properties)`

24.292.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.292.5 Description

Sets all entities in selectionset ss to the properties as given by properties; the properties list must have the format as returned by (acet-general-props-get).

24.292.6 Return Values

Vendor-defined; see Source Notes.

24.292.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.292.8 Source Notes

- [acet-general-props-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeneralpropsset.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.293 Function Entry: ACET-GENERAL-PROPS-SET-PAIRS

24.293.1 Name

‘acet-general-props-set-pairs‘

24.293.2 Class

BricsCAD ExpressTools API Function

24.293.3 Syntax

`(acet-general-props-set-pairs object properties)`

24.293.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.293.5 Description

Sets the entity object to the properties as given by properties; the properties list must have the assoc format as returned by (acet-general-props-get-pairs). object can be an entity name (ENAME) or an (entget) list

24.293.6 Return Values

Vendor-defined; see Source Notes.

24.293.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.293.8 Source Notes

- [acet-general-props-set-pairs (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeneralpropssetpairs.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.294 Function Entry: ACET-GEOM-ANGLE-TRANS

24.294.1 Name

‘acet-geom-angle-trans‘

24.294.2 Class

BricsCAD ExpressTools API Function

24.294.3 Syntax

`(acet-geom-angle-trans angle fromCS toCS)`

24.294.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.294.5 Description

Returns the angle transformed from fromCS to toCS coordinate system. angle is in radians.

24.294.6 Return Values

Vendor-defined; see Source Notes.

24.294.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.294.8 Source Notes

- [acet-geom-angle-trans (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomangletrans.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.295 Function Entry: ACET-GEOM-ARBITRARY-X

24.295.1 Name

‘acet-geom-arbitrary-x‘

24.295.2 Class

BricsCAD ExpressTools API Function

24.295.3 Syntax

`(acet-geom-arbitrary-x zvec)`

24.295.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.295.5 Description

Returns the x-vector for a given normal vector, calculated by the arbitrary axis algorithm, as used by the CAD system to express an UCS. zvec is a normal vector (3 real values), i.e. as obtained from 210 dxf group in an (entget) list.

24.295.6 Return Values

Vendor-defined; see Source Notes.

24.295.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.295.8 Source Notes

- [acet-geom-arbitrary-x (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomarbitraryx.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.296 Function Entry: ACET-GEOM-ARC-3P-D-ANGLE

24.296.1 Name

‘acet-geom-arc-3p-d-angle‘

24.296.2 Class

BricsCAD ExpressTools API Function

24.296.3 Syntax

```
(acet-geom-arc-3p-d-angle p1 p2 p3)
```

24.296.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.296.5 Description

Returns the difference angle for a bulge between points p1 -> p2 -> p3 (this is, the included angle center-p1 and center-p3, with center defined by the 3 arcus/center points); the sign of the result specifies the direction. angle is in radians.

24.296.6 Return Values

Vendor-defined; see Source Notes.

24.296.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.296.8 Source Notes

- [acet-geom-arc-3p-d-angle (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomarc3pdangle.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.297 Function Entry: ACET-GEOM-ARC-BULGE

24.297.1 Name

'acet-geom-arc-bulge'

24.297.2 Class

BricsCAD ExpressTools API Function

24.297.3 Syntax

`(acet-geom-arc-bulge center p1 deltaang)`

24.297.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.297.5 Description

Returns the bulge value for specified difference angle deltaang. arguments center and p1 are unused The bulge is the tangent of 1/4 of the included angle for an arc segment, made negative if the arc goes clockwise from the start point to the endpoint. A bulge of 0 indicates a straight segment, and a bulge of 1 is a semicircle.

24.297.6 Return Values

Vendor-defined; see Source Notes.

24.297.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.297.8 Source Notes

- [acet-geom-arc-bulge (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomarcbulge.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.298 Function Entry: ACET-GEOM-ARC-CENTER

24.298.1 Name

‘acet-geom-arc-center‘

24.298.2 Class

BricsCAD ExpressTools API Function

24.298.3 Syntax

```
(acet-geom-arc-center p1 p2 p3)
```

24.298.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.298.5 Description

Returns the center point for for an arc or circle specified by 3 points p1, p2, p3.

24.298.6 Return Values

Vendor-defined; see Source Notes.

24.298.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.298.8 Source Notes

- [acet-geom-arc-center (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomarccenter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.299 Function Entry: ACET-GEOM-ARC-D-ANGLE

24.299.1 Name

‘acet-geom-arc-d-angle‘

24.299.2 Class

BricsCAD ExpressTools API Function

24.299.3 Syntax

```
(acet-geom-arc-d-angle center p1 p2)
```

24.299.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.299.5 Description

Returns the angle included between center-p1 and center-p2. angle is in radians.

24.299.6 Return Values

Vendor-defined; see Source Notes.

24.299.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.299.8 Source Notes

- [acet-geom-arc-d-angle (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomarcdangle.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.300 Function Entry: ACET-GEOM-CALC-ARC-ERROR

24.300.1 Name

‘acet-geom-calc-arc-error‘

24.300.2 Class

BricsCAD ExpressTools API Function

24.300.3 Syntax

`(acet-geom-calc-arc-error param)`

24.300.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.300.5 Description

Calculates an error factor based on input parameter param; for internal calculations; do not use in application code.

24.300.6 Return Values

Vendor-defined; see Source Notes.

24.300.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.300.8 Source Notes

- [acet-geom-calc-arc-error (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomcalcarcerror.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.301 Function Entry: ACET-GEOM-CROSS-PRODUCT

24.301.1 Name

‘acet-geom-cross-product‘

24.301.2 Class

BricsCAD ExpressTools API Function

24.301.3 Syntax

```
(acet-geom-cross-product v1 v2)
```

24.301.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.301.5 Description

Returns the cross-product vector of vectors v1 and v2 (normal vector of plane defined by v1 and v2, as list of 3 doubles).

24.301.6 Return Values

Vendor-defined; see Source Notes.

24.301.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.301.8 Source Notes

- [acet-geom-cross-product (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomcrossproduct.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.302 Function Entry: ACET-GEOM-DELTA-VECTOR

24.302.1 Name

'acet-geom-delta-vector'

24.302.2 Class

BricsCAD ExpressTools API Function

24.302.3 Syntax

`(acet-geom-delta-vector p1 p2)`

24.302.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.302.5 Description

Returns the vector from p1 to p2, as list of 3 doubles.

24.302.6 Return Values

Vendor-defined; see Source Notes.

24.302.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.302.8 Source Notes

- [acet-geom-delta-vector (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomdeltavector.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.303 Function Entry: ACET-GEOM-IMAGE-BOUNDS

24.303.1 Name

‘acet-geom-image-bounds‘

24.303.2 Class

BricsCAD ExpressTools API Function

24.303.3 Syntax

`(acet-geom-image-bounds ename)`

24.303.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.303.5 Description

Returns the boundary points of an image entity, as list (pLL pLR pUR pUL pLL).

24.303.6 Return Values

Vendor-defined; see Source Notes.

24.303.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.303.8 Source Notes

- [acet-geom-image-bounds (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomimagebounds.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.304 Function Entry: ACET-GEOM-INTERSECTWITH

24.304.1 Name

‘acet-geom-intersectwith‘

24.304.2 Class

BricsCAD ExpressTools API Function

24.304.3 Syntax

`(acet-geom-intersectwith ename1 ename2 flags)`

24.304.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.304.5 Description

Returns all intersection points between entities ename1 and entity2; flags is an bit-coded integer, bits can be combined : bit 0 (1) : virtually extend entity1 bit 1 (2) : virtually extend entity2

24.304.6 Return Values

Vendor-defined; see Source Notes.

24.304.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.304.8 Source Notes

- [acet-geom-intersectwith (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomintersectwith.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.305 Function Entry: ACET-GEOM-IS-ARC

24.305.1 Name

‘acet-geom-is-arc‘

24.305.2 Class

BricsCAD ExpressTools API Function

24.305.3 Syntax

`(acet-geom-is-arc p1 p2 p3 p4 fuz)`

24.305.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.305.5 Description

Verifies, whether given 4 points p1 p2 p3 p4 define a valid arc/circle; fuz is a numerical tolerance - if NIL, tolerance 0.0 is used. Returns T if the 4 points belong to an arc/circle, NIL otherwise

24.305.6 Return Values

Vendor-defined; see Source Notes.

24.305.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.305.8 Source Notes

- [acet-geom-is-arc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomisarc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.306 Function Entry: ACET-GEOM-LIST-EXTENTS

24.306.1 Name

‘acet-geom-list-extents‘

24.306.2 Class

BricsCAD ExpressTools API Function

24.306.3 Syntax

```
(acet-geom-list-extents pointlist)
```

24.306.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.306.5 Description

Determines the extents of all points given in pointlist; pointlist must be a list of 2d or 3d points (mixed format is allowed). Returns a list of format (pLowerLeft pUpperRight); if all points of pointlist are 3d points, the returned pLowerLeft and pUpperRight are 3d points.

24.306.6 Return Values

Vendor-defined; see Source Notes.

24.306.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.306.8 Source Notes

- [acet-geom-list-extents (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomlistextents.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.307 Function Entry: ACET-GEOM-M-TRANS

24.307.1 Name

‘acet-geom-m-trans‘

24.307.2 Class

BricsCAD ExpressTools API Function

24.307.3 Syntax

`(acet-geom-m-trans pointlst from to)`

24.307.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.307.5 Description

Returns the list of transformed points from pointlst; applies (trans point from to) to each point of pointlst.

24.307.6 Return Values

Vendor-defined; see Source Notes.

24.307.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.307.8 Source Notes

- [acet-geom-m-trans (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeommtrans.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.308 Function Entry: ACET-GEOM-MID-POINT

24.308.1 Name

‘acet-geom-mid-point‘

24.308.2 Class

BricsCAD ExpressTools API Function

24.308.3 Syntax

```
(acet-geom-mid-point p1 p2)
```

24.308.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.308.5 Description

Returns the mid-point of given points p1 and p2, as 3D point if both p1 and p2 are 3D points, or as 2D points if either p1 or p2 is only in 2D.

24.308.6 Return Values

Vendor-defined; see Source Notes.

24.308.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.308.8 Source Notes

- [acet-geom-mid-point (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeommidpoint.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.309 Function Entry: ACET-GEOM-MIDPOINT

24.309.1 Name

‘acet-geom-midpoint‘

24.309.2 Class

BricsCAD ExpressTools API Function

24.309.3 Syntax

`(acet-geom-midpoint p1 p2)`

24.309.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.309.5 Description

Please see (acet-geom-mid-point) function; both are identical in behaviour.

24.309.6 Return Values

Vendor-defined; see Source Notes.

24.309.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.309.8 Source Notes

- [acet-geom-midpoint (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeommidpoint1.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.310 Function Entry: ACET-GEOM-OBJECT-END-POINTS

24.310.1 Name

‘acet-geom-object-end-points‘

24.310.2 Class

BricsCAD ExpressTools API Function

24.310.3 Syntax

`(acet-geom-object-end-points ename)`

24.310.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.310.5 Description

Returns a list of defining end points for entity ename, using format (ptA ptB); both points can be 2d/3d points, depending on the entity.

24.310.6 Return Values

Vendor-defined; see Source Notes.

24.310.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.310.8 Source Notes

- [acet-geom-object-end-points (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomobjectendpoints.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.311 Function Entry: ACET-GEOM-OBJECT-FUZ

24.311.1 Name

‘acet-geom-object-fuz‘

24.311.2 Class

BricsCAD ExpressTools API Function

24.311.3 Syntax

```
(acet-geom-object-fuz ename)
```

24.311.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.311.5 Description

Returns a tolerance value for the given entity (depending on object dimension + location).

24.311.6 Return Values

Vendor-defined; see Source Notes.

24.311.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.311.8 Source Notes

- [acet-geom-object-fuz (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomobjectfuz.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.312 Function Entry: ACET-GEOM-OBJECT-NORMAL-VECTOR

24.312.1 Name

‘acet-geom-object-normal-vector‘

24.312.2 Class

BricsCAD ExpressTools API Function

24.312.3 Syntax

```
(acet-geom-object-normal-vector ename)
```

24.312.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.312.5 Description

Returns the normal vector for entity ename, if ename is a planar entity; otherwise it returns (0 0 1) vector.

24.312.6 Return Values

Vendor-defined; see Source Notes.

24.312.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.312.8 Source Notes

- [acet-geom-object-normal-vector (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomobjectnormalvector.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.313 Function Entry: ACET-GEOM-OBJECT-POINT-LIST

24.313.1 Name

‘acet-geom-object-point-list‘

24.313.2 Class

BricsCAD ExpressTools API Function

24.313.3 Syntax

```
(acet-geom-object-point-list ename smoothness)
```

24.313.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.313.5 Description

Returns a list of defining points for entity ename, using format (pt1 pt2 … ptN); all points can be 2d/3d points, depending on the entity; smoothness is an optional double value, defining the smoothness when rasterizing the entity; if NIL, an internal default is used.

24.313.6 Return Values

Vendor-defined; see Source Notes.

24.313.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.313.8 Source Notes

- [acet-geom-object-point-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomobjectpointlist.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.314 Function Entry: ACET-GEOM-OBJECT-Z-AXIS

24.314.1 Name

‘acet-geom-object-z-axis‘

24.314.2 Class

BricsCAD ExpressTools API Function

24.314.3 Syntax

`(acet-geom-object-z-axis elist)`

24.314.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.314.5 Description

Returns the normal vector for entity given by its entity data elist, if entity is a planar entity; otherwise it returns (0 0 1) vector.

24.314.6 Return Values

Vendor-defined; see Source Notes.

24.314.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.314.8 Source Notes

- [acet-geom-object-z-axis (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomobjectzaxis.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.315 Function Entry: ACET-GEOM-PIXEL-UNIT

24.315.1 Name

‘acet-geom-pixel-unit‘

24.315.2 Class

BricsCAD ExpressTools API Function

24.315.3 Syntax

`(acet-geom-pixel-unit)`

24.315.4 Description

Returns the pixel size in dwg units.

24.315.5 Return Values

Vendor-defined; see Source Notes.

24.315.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.315.7 Source Notes

- [acet-geom-pixel-unit (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeompixelunit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.316 Function Entry: ACET-GEOM-PLINE-ARC-INFO

24.316.1 Name

‘acet-geom-pline-arc-info‘

24.316.2 Class

BricsCAD ExpressTools API Function

24.316.3 Syntax

`(acet-geom-pline-arc-info p1 p2 bulge)`

24.316.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.316.5 Description

Returns the definition data of a polyline arc, given by p1, p2, and bulge values, using following format : (center radius deltaang start end).

24.316.6 Return Values

Vendor-defined; see Source Notes.

24.316.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.316.8 Source Notes

- [acet-geom-pline-arc-info (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomplinearcinfo.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.317 Function Entry: ACET-GEOM-POINT-INSIDE

24.317.1 Name

‘acet-geom-point-inside‘

24.317.2 Class

BricsCAD ExpressTools API Function

24.317.3 Syntax

`(acet-geom-point-inside pt ptList halfwidth)`

24.317.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.317.5 Description

Determines whether given point pt is contained in a band defined by ptList, using halfwidth. Returns T or NIL.

24.317.6 Return Values

Vendor-defined; see Source Notes.

24.317.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.317.8 Source Notes

- [acet-geom-point-inside (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeompointinside.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.318 Function Entry: ACET-GEOM-POINT-ROTATE

24.318.1 Name

‘acet-geom-point-rotate‘

24.318.2 Class

BricsCAD ExpressTools API Function

24.318.3 Syntax

`(acet-geom-point-rotate pt refPoint rotAngle)`

24.318.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.318.5 Description

Returns a point created by rotating point pt around refPoint using angle rotAngle.

24.318.6 Return Values

Vendor-defined; see Source Notes.

24.318.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.318.8 Source Notes

- [acet-geom-point-rotate (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeompointrotate.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.319 Function Entry: ACET-GEOM-POINT-SCALE

24.319.1 Name

‘acet-geom-point-scale‘

24.319.2 Class

BricsCAD ExpressTools API Function

24.319.3 Syntax

```
(acet-geom-point-scale toPoint fromPoint scale)
```

24.319.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.319.5 Description

Returns a point created by scaling the vector from fromPoint to toPoint with scale, and applied to fromPoint; if scale is NIL, 1.0 is used.

24.319.6 Return Values

Vendor-defined; see Source Notes.

24.319.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.319.8 Source Notes

- [acet-geom-point-scale (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeompointscale.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.320 Function Entry: ACET-GEOM-RECT-POINTS

24.320.1 Name

‘acet-geom-rect-points‘

24.320.2 Class

BricsCAD ExpressTools API Function

24.320.3 Syntax

`(acet-geom-rect-points p1 p2)`

24.320.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.320.5 Description

Creates a list of ”rectangle corner points” for the rectangle defined by p1 and p2; returns a list of (ptLL ptLR ptUR ptUL ptLL).

24.320.6 Return Values

Vendor-defined; see Source Notes.

24.320.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.320.8 Source Notes

- [acet-geom-rect-points (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomrectpoints.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.321 Function Entry: ACET-GEOM-SELF-INTERSECT

24.321.1 Name

‘acet-geom-self-intersect‘

24.321.2 Class

BricsCAD ExpressTools API Function

24.321.3 Syntax

```
(acet-geom-self-intersect pointlist flag)
```

24.321.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.321.5 Description

Checks a polygon point sequence defined by pointlist for self-intersection; flag is actually ignored. Returns the first intersection point found, or NIL if the polygon does not self-intersect.

24.321.6 Return Values

Vendor-defined; see Source Notes.

24.321.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.321.8 Source Notes

- [acet-geom-self-intersect (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomselfintersect.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.322 Function Entry: ACET-GEOM-SS-EXTENTS

24.322.1 Name

‘acet-geom-ss-extents‘

24.322.2 Class

BricsCAD ExpressTools API Function

24.322.3 Syntax

`(acet-geom-ss-extents ss scale)`

24.322.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.322.5 Description

Returns the geometrical extents of selectionset ss as list of 2 points (argument scale is actually ignored).

24.322.6 Return Values

Vendor-defined; see Source Notes.

24.322.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.322.8 Source Notes

- [acet-geom-ss-extents (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomssextents.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.323 Function Entry: ACET-GEOM-TEXTBOX

24.323.1 Name

‘acet-geom-textbox‘

24.323.2 Class

BricsCAD ExpressTools API Function

24.323.3 Syntax

```
(acet-geom-textbox textobject offset)
```

24.323.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.323.5 Description

Returns a list of 4 boundary rectangle points for a Text or AttributeDefinition or Attribute or MText, given by textobject; textobject can be an ename or an (entget) list; offset specifies an offset applied to expand the boundary rectangle.

24.323.6 Return Values

Vendor-defined; see Source Notes.

24.323.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.323.8 Source Notes

- [acet-geom-textbox (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomtextbox.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.324 Function Entry: ACET-GEOM-UNIT-VECTOR

24.324.1 Name

‘acet-geom-unit-vector‘

24.324.2 Class

BricsCAD ExpressTools API Function

24.324.3 Syntax

```
(acet-geom-unit-vector p1 p2)
```

24.324.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.324.5 Description

Returns the unit vector (length is 1.0) for the vector from p1 to p2.

24.324.6 Return Values

Vendor-defined; see Source Notes.

24.324.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.324.8 Source Notes

- [acet-geom-unit-vector (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomunitvector.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.325 Function Entry: ACET-GEOM-VECTOR-ADD

24.325.1 Name

‘acet-geom-vector-add‘

24.325.2 Class

BricsCAD ExpressTools API Function

24.325.3 Syntax

`(acet-geom-vector-add vec addVec)`

24.325.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.325.5 Description

Adds vector addVec to vector vec and returns the new vector.

24.325.6 Return Values

Vendor-defined; see Source Notes.

24.325.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.325.8 Source Notes

- [acet-geom-vector-add (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvectoradd.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.326 Function Entry: ACET-GEOM-VECTOR-D-ANGLE

24.326.1 Name

‘acet-geom-vector-d-angle‘

24.326.2 Class

BricsCAD ExpressTools API Function

24.326.3 Syntax

`(acet-geom-vector-d-angle v1 v2)`

24.326.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.326.5 Description

Returns the angle between 2 vectors v1 and v2; can be 3d vectors.

24.326.6 Return Values

Vendor-defined; see Source Notes.

24.326.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.326.8 Source Notes

- [acet-geom-vector-d-angle (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvectordangle.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.327 Function Entry: ACET-GEOM-VECTOR-PARALLEL

24.327.1 Name

‘acet-geom-vector-parallel‘

24.327.2 Class

BricsCAD ExpressTools API Function

24.327.3 Syntax

`(acet-geom-vector-parallel v1 v2)`

24.327.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.327.5 Description

Returns T if the vectors v1 and v2 are parallel, otherwise returns NIL.

24.327.6 Return Values

Vendor-defined; see Source Notes.

24.327.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.327.8 Source Notes

- [acet-geom-vector-parallel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvectorparallel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.328 Function Entry: ACET-GEOM-VECTOR-SCALE

24.328.1 Name

‘acet-geom-vector-scale‘

24.328.2 Class

BricsCAD ExpressTools API Function

24.328.3 Syntax

`(acet-geom-vector-scale vec scale)`

24.328.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.328.5 Description

Returns a vector from scaling the vector vec by scale; vec can be a 2d or 3d vector; result vector is always 3d.

24.328.6 Return Values

Vendor-defined; see Source Notes.

24.328.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.328.8 Source Notes

- [acet-geom-vector-scale (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvectorscale.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.329 Function Entry: ACET-GEOM-VECTOR-SIDE

24.329.1 Name

‘acet-geom-vector-side‘

24.329.2 Class

BricsCAD ExpressTools API Function

24.329.3 Syntax

```
(acet-geom-vector-side pt pA pB)
```

24.329.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.329.5 Description

Determines the side of point pt, relative to the direction vector pA->pB. Returns 1.0, if pt is ”left” of direction vector (mathematically positive), or -1.0 if pt is ”right” of direction vector (mathematically negative), or 0.0 if pt is located on the direction vector.

24.329.6 Return Values

Vendor-defined; see Source Notes.

24.329.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.329.8 Source Notes

- [acet-geom-vector-side (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvectorside.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.330 Function Entry: ACET-GEOM-VERTEX-LIST

24.330.1 Name

‘acet-geom-vertex-list‘

24.330.2 Class

BricsCAD ExpressTools API Function

24.330.3 Syntax

```
(acet-geom-vertex-list entity)
```

24.330.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.330.5 Description

Returns the list of definition points for a POLYLINE or LWPOLYLINE entity given by entity; entity can be : a) an ename : result points are transformed into current UCS b) an (entsel) lst : result points are transformed into current UCS c) a list (ename toCS) : result points are transformed into UCS as defined by toCS, see (trans) function

24.330.6 Return Values

Vendor-defined; see Source Notes.

24.330.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.330.8 Source Notes

- [acet-geom-vertex-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomvertexlist.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.331 Function Entry: ACET-GEOM-VIEW-POINTS

24.331.1 Name

‘acet-geom-view-points‘

24.331.2 Class

BricsCAD ExpressTools API Function

24.331.3 Syntax

`(acet-geom-view-points)`

24.331.4 Description

Returns a list of the 2 extent points for current screen area, transformed into current UCS.

24.331.5 Return Values

Vendor-defined; see Source Notes.

24.331.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.331.7 Source Notes

- [acet-geom-view-points (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomviewpoints.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.332 Function Entry: ACET-GEOM-Z-AXIS

24.332.1 Name

‘acet-geom-z-axis‘

24.332.2 Class

BricsCAD ExpressTools API Function

24.332.3 Syntax

`(acet-geom-z-axis)`

24.332.4 Description

Returns the Z vector of the current UCS (cross-product of (getvar ”UCSXDIR”) and (getvar ”UCSXDIR”)).

24.332.5 Return Values

Vendor-defined; see Source Notes.

24.332.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.332.7 Source Notes

- [acet-geom-z-axis (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomzaxis.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.333 Function Entry: ACET-GEOM-ZOOM-FOR-SELECT

24.333.1 Name

‘acet-geom-zoom-for-select‘

24.333.2 Class

BricsCAD ExpressTools API Function

24.333.3 Syntax

```
(acet-geom-zoom-for-select pntlist)
```

24.333.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.333.5 Description

Verifies whether all points in pntlist are visible on screen - in this case, the function returns NIL; otherwise it returns a list of 2 point, suitable for Zoom/Window to ensure all points in pntlist are visible.

24.333.6 Return Values

Vendor-defined; see Source Notes.

24.333.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.333.8 Source Notes

- [acet-geom-zoom-for-select (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgeomzoomforselect.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.334 Function Entry: ACET-GET-ATT

24.334.1 Name

‘acet-get-att‘

24.334.2 Class

BricsCAD ExpressTools API Function

24.334.3 Syntax

`(acet-get-att inplist)`

24.334.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.334.5 Description

This function is an alias for (acet-insert-attrib-get inplist).

24.334.6 Return Values

Vendor-defined; see Source Notes.

24.334.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.334.8 Source Notes

- [acet-get-att (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgetatt.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.335 Function Entry: ACET-GET-WINFONT-PATH

24.335.1 Name

‘acet-get-winfont-path‘

24.335.2 Class

BricsCAD ExpressTools API Function

24.335.3 Syntax

`(acet-get-winfont-path)`

24.335.4 Description

Returns the path of the operating system’s special font folder (also on Linux and OSX platforms).

24.335.5 Return Values

Vendor-defined; see Source Notes.

24.335.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.335.7 Source Notes

- [acet-get-winfont-path (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgetwinfontpath.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.336 Function Entry: ACET-GETVAR

24.336.1 Name

‘acet-getvar‘

24.336.2 Class

BricsCAD ExpressTools API Function

24.336.3 Syntax

`(acet-getvar varlist)`

24.336.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.336.5 Description

Loads custom variables from several storage places. varlist is a list of ’(var1 saveMode var2 saveMode …); saveMode can be a combination (bitflags) of : 1 = dictionary 2 = registry 4 = environment; if saveMode is NIL, the variable with its value is loaded from dictionary and registry

24.336.6 Return Values

Vendor-defined; see Source Notes.

24.336.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.336.8 Source Notes

- [acet-getvar (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgetvar.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.337 Function Entry: ACET-GROUP-MAKE-ANON

24.337.1 Name

‘acet-group-make-anon‘

24.337.2 Class

BricsCAD ExpressTools API Function

24.337.3 Syntax

`(acet-group-make-anon enamelist description)`

24.337.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.337.5 Description

Creates a new anonymous group (*Ann named), using the entity list enamelist with optional description text, if description is not NIL.

24.337.6 Return Values

Vendor-defined; see Source Notes.

24.337.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.337.8 Source Notes

- [acet-group-make-anon (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgroupmakeanon.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.338 Function Entry: ACET-GROUPS-SEL

24.338.1 Name

‘acet-groups-sel‘

24.338.2 Class

BricsCAD ExpressTools API Function

24.338.3 Syntax

`(acet-groups-sel groupenames)`

24.338.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.338.5 Description

Sets the groups specified in groupenames as ”selectable”.

24.338.6 Return Values

Vendor-defined; see Source Notes.

24.338.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.338.8 Source Notes

- [acet-groups-sel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgroupssel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.339 Function Entry: ACET-GROUPS-UNSEL

24.339.1 Name

‘acet-groups-unsel‘

24.339.2 Class

BricsCAD ExpressTools API Function

24.339.3 Syntax

`(acet-groups-unsel)`

24.339.4 Description

Sets all groups to be ”unselectable”.

24.339.5 Return Values

Vendor-defined; see Source Notes.

24.339.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.339.7 Source Notes

- [acet-groups-unsel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetgroupsunsel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.340 Function Entry: ACET-INIT-FAS-LIB

24.340.1 Name

‘acet-init-fas-lib‘

24.340.2 Class

BricsCAD ExpressTools API Function

24.340.3 Syntax

`(acet-init-fas-lib flag1 flag2)`

24.340.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.340.5 Description

Provided for backward compatibility, implementation does nothing (no operation); arguments flags and flag2 are ignored.

24.340.6 Return Values

Vendor-defined; see Source Notes.

24.340.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.340.8 Source Notes

- [acet-init-fas-lib (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetinitfaslib.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.341 Function Entry: ACET-INSERT-ATTRIB-GET

24.341.1 Name

‘acet-insert-attrib-get‘

24.341.2 Class

BricsCAD ExpressTools API Function

24.341.3 Syntax

```
(acet-insert-attrib-get inplist)
```

24.341.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.341.5 Description

Returns a list of attributes (tags and/or values) for the block reference as specified by inplist : a) if inplist is an entity name pairs b) if inplist is a list of (ename mode), the mode argument defines the structure of returned list : mode = 1 returns a list of (tag value) pairs mode = 2 returns a list of tags only mode = 3 returns a list of values only mode = 4 returns a list of (tag ename) pairs, where ename is the entity name of the attribute

24.341.6 Return Values

Vendor-defined; see Source Notes.

24.341.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.341.8 Source Notes

- [acet-insert-attrib-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetinsertattribget.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.342 Function Entry: ACET-INSERT-ATTRIB-MOD

24.342.1 Name

‘acet-insert-attrib-mod‘

24.342.2 Class

BricsCAD ExpressTools API Function

24.342.3 Syntax

```
(acet-insert-attrib-mod ename inplist suppress)
```

24.342.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.342.5 Description

Modifies the attributes values for a BlockReference entity ename, based on the inplist arguments; inplist must be a list of (tag value) pairs - verifying the tags is case-insensitive. suppress argument controls whether a hint message is shown, if the BlockReference does not contain an attribute with tag key tag.

24.342.6 Return Values

Vendor-defined; see Source Notes.

24.342.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.342.8 Source Notes

- [acet-insert-attrib-mod (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetinsertattribmod.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.343 Function Entry: ACET-INSERT-ATTRIB-SET

24.343.1 Name

‘acet-insert-attrib-set‘

24.343.2 Class

BricsCAD ExpressTools API Function

24.343.3 Syntax

`(acet-insert-attrib-set ename inplist suppress)`

24.343.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.343.5 Description

This function behaves identical as (acet-insert-attrib-mod).

24.343.6 Return Values

Vendor-defined; see Source Notes.

24.343.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.343.8 Source Notes

- [acet-insert-attrib-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetinsertattribset.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.344 Function Entry: ACET-LAYER-LOCKED

24.344.1 Name

‘acet-layer-locked‘

24.344.2 Class

BricsCAD ExpressTools API Function

24.344.3 Syntax

`(acet-layer-locked layerName)`

24.344.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.344.5 Description

Returns T if specified layer layerName is locked, NIL otherwise.

24.344.6 Return Values

Vendor-defined; see Source Notes.

24.344.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.344.8 Source Notes

- [acet-layer-locked (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlayerlocked.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.345 Function Entry: ACET-LAYER-OFF

24.345.1 Name

‘acet-layer-off‘

24.345.2 Class

BricsCAD ExpressTools API Function

24.345.3 Syntax

`(acet-layer-off layerName)`

24.345.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.345.5 Description

Returns T if specified layer layerName is OFF, NIL otherwise.

24.345.6 Return Values

Vendor-defined; see Source Notes.

24.345.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.345.8 Source Notes

- [acet-layer-off (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-layer-off.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.346 Function Entry: ACET-LAYER-UNLOCK-ALL

24.346.1 Name

‘acet-layer-unlock-all‘

24.346.2 Class

BricsCAD ExpressTools API Function

24.346.3 Syntax

`(acet-layer-unlock-all)`

24.346.4 Description

This function unlocks all layers in the drawing. Returns a list of those layers (strings), which were locked before.

24.346.5 Return Values

Vendor-defined; see Source Notes.

24.346.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.346.7 Source Notes

- [acet-layer-unlock-all (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlayerunlockall.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.347 Function Entry: ACET-LIST-ASSOC-APPEND

24.347.1 Name

‘acet-list-assoc-append‘

24.347.2 Class

BricsCAD ExpressTools API Function

24.347.3 Syntax

`(acet-list-assoc-append assoclst lst)`

24.347.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.347.5 Description

Returns lst as updated or appended by assoclst.

24.347.6 Return Values

Vendor-defined; see Source Notes.

24.347.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.347.8 Source Notes

- [acet-list-assoc-append (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistassocappend.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.348 Function Entry: ACET-LIST-ASSOC-PUT

24.348.1 Name

‘acet-list-assoc-put‘

24.348.2 Class

BricsCAD ExpressTools API Function

24.348.3 Syntax

`(acet-list-assoc-put dpValue assoclst)`

24.348.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.348.5 Description

dpValue a single dotted pair value assoclst a dotted pair list to be updated by dpValue; dpValue is appended to assoclst if not present yet. Returns assoclst as updated by dpValue.

24.348.6 Return Values

Vendor-defined; see Source Notes.

24.348.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.348.8 Source Notes

- [acet-list-assoc-put (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistassocput.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.349 Function Entry: ACET-LIST-ASSOC-REMOVE

24.349.1 Name

‘acet-list-assoc-remove‘

24.349.2 Class

BricsCAD ExpressTools API Function

24.349.3 Syntax

`(acet-list-assoc-remove key assoclst)`

24.349.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.349.5 Description

Removes the item with key key from assoclst.

24.349.6 Return Values

Vendor-defined; see Source Notes.

24.349.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.349.8 Source Notes

- [acet-list-assoc-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistassocremove.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.350 Function Entry: ACET-LIST-GROUP-BY-ASSOC

24.350.1 Name

‘acet-list-group-by-assoc‘

24.350.2 Class

BricsCAD ExpressTools API Function

24.350.3 Syntax

`(acet-list-group-by-assoc lst)`

24.350.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.350.5 Description

Returns the list lst as sorted by the assoc keys.

24.350.6 Return Values

Vendor-defined; see Source Notes.

24.350.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.350.8 Source Notes

- [acet-list-group-by-assoc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistgroupbyassoc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.351 Function Entry: ACET-LIST-IS-DOTTED-PAIR

24.351.1 Name

‘acet-list-is-dotted-pair‘

24.351.2 Class

BricsCAD ExpressTools API Function

24.351.3 Syntax

`(acet-list-is-dotted-pair item)`

24.351.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.351.5 Description

Returns T if item is a true DottedPair, NIL otherwise.

24.351.6 Return Values

Vendor-defined; see Source Notes.

24.351.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.351.8 Source Notes

- [acet-list-is-dotted-pair (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistisdottedpair.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.352 Function Entry: ACET-LIST-ISORT

24.352.1 Name

‘acet-list-isort‘

24.352.2 Class

BricsCAD ExpressTools API Function

24.352.3 Syntax

`(acet-list-isort lst index)`

24.352.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.352.5 Description

Returns the list lst as sorted by index. lst input list is a ”list of sub-lists”, index specifies the entry in each sub-list, which is used as sorting value. Examples : (setq lst (list ’(1 2 5) ’(2 1 4) ’(3 4 3) ’(4 3 2) ’(5 5 1))) (acet-list-isort lst 0) returns : ((1 2 5) (2 1 4) (3 4 3) (4 3 2) (5 5 1)) (acet-list-isort lst 1) returns : ((2 1 4) (1 2 5) (4 3 2) (3 4 3) (5 5 1)) (acet-list-isort lst 2) returns : ((5 5 1) (4 3 2) (3 4 3) (2 1 4) (1 2 5))

24.352.6 Return Values

Vendor-defined; see Source Notes.

24.352.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.352.8 Source Notes

- [acet-list-isort (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistisort.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.353 Function Entry: ACET-LIST-M-ASSOC

24.353.1 Name

‘acet-list-m-assoc‘

24.353.2 Class

BricsCAD ExpressTools API Function

24.353.3 Syntax

`(acet-list-m-assoc key lst)`

24.353.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.353.5 Description

Returns a list with all items in lst using the assoc key key.

24.353.6 Return Values

Vendor-defined; see Source Notes.

24.353.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.353.8 Source Notes

- [acet-list-m-assoc (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistmassoc.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.354 Function Entry: ACET-LIST-PUT-NTH

24.354.1 Name

‘acet-list-put-nth‘

24.354.2 Class

BricsCAD ExpressTools API Function

24.354.3 Syntax

```
(acet-list-put-nth newval datalist atidx)
```

24.354.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.354.5 Description

Returns the list datalist, with updated item newval at index atidx; index is 0-based.

24.354.6 Return Values

Vendor-defined; see Source Notes.

24.354.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.354.8 Source Notes

- [acet-list-put-nth (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistputnth.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.355 Function Entry: ACET-LIST-REMOVE-ADJACENT-DUPS

24.355.1 Name

‘acet-list-remove-adjacent-dups‘

24.355.2 Class

BricsCAD ExpressTools API Function

24.355.3 Syntax

```
(acet-list-remove-adjacent-dups datalist)
```

24.355.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.355.5 Description

Returns a list with any duplicates removed from datalist.

24.355.6 Return Values

Vendor-defined; see Source Notes.

24.355.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.355.8 Source Notes

- [acet-list-remove-adjacent-dups (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistremoveadjacentdups.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.356 Function Entry: ACET-LIST-REMOVE-DUPLICATE-POINTS

24.356.1 Name

‘acet-list-remove-duplicate-points‘

24.356.2 Class

BricsCAD ExpressTools API Function

24.356.3 Syntax

```
(acet-list-remove-duplicate-points datalist fuz)
```

24.356.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.356.5 Description

Returns a list by removing all duplicate entries from datalist, using optional tolerance fuz; a NIL fuz value means 0.0 tolerance. 0.0; datalist mus be a list of points !

24.356.6 Return Values

Vendor-defined; see Source Notes.

24.356.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.356.8 Source Notes

- [acet-list-remove-duplicate-points (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistremoveduplicatepoints.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.357 Function Entry: ACET-LIST-REMOVE-DUPLICATES

24.357.1 Name

‘acet-list-remove-duplicates‘

24.357.2 Class

BricsCAD ExpressTools API Function

24.357.3 Syntax

`(acet-list-remove-duplicates datalist fuz)`

24.357.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.357.5 Description

Returns a list by removing all duplicate entries from datalist, using optional tolerance fuz; a NIL fuz value means 0.0 tolerance. 0.0

24.357.6 Return Values

Vendor-defined; see Source Notes.

24.357.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.357.8 Source Notes

- [acet-list-remove-duplicates (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistremoveduplicates.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.358 Function Entry: ACET-LIST-REMOVE-NTH

24.358.1 Name

'acet-list-remove-nth'

24.358.2 Class

BricsCAD ExpressTools API Function

24.358.3 Syntax

`(acet-list-remove-nth atidx datalist)`

24.358.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.358.5 Description

Returns the list datalist, with removed item at index atidx; index is 0-based.

24.358.6 Return Values

Vendor-defined; see Source Notes.

24.358.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.358.8 Source Notes

- [acet-list-remove-nth (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistremoventh.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.359 Function Entry: ACET-LIST-SPLIT

24.359.1 Name

‘acet-list-split‘

24.359.2 Class

BricsCAD ExpressTools API Function

24.359.3 Syntax

```
(acet-list-split lst item)
```

24.359.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.359.5 Description

Splits the list lst at item item into 2 sub-lists (left-side and right-side part) and returns a list of both sub-lists; the split-item item is first entry in the right-side list part (if present).

24.359.6 Return Values

Vendor-defined; see Source Notes.

24.359.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.359.8 Source Notes

- [acet-list-split (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlistsplit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.360 Function Entry: ACET-LIST-TO-SS

24.360.1 Name

‘acet-list-to-ss‘

24.360.2 Class

BricsCAD ExpressTools API Function

24.360.3 Syntax

`(acet-list-to-ss lst)`

24.360.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.360.5 Description

Returns a selectionset containing all entities in list lst, or NIL if list is NIL,

24.360.6 Return Values

Vendor-defined; see Source Notes.

24.360.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.360.8 Source Notes

- [acet-list-to-ss (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlisttoss.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.361 Function Entry: ACET-LWPLINE-MAKE

24.361.1 Name

'acet-lwpline-make'

24.361.2 Class

BricsCAD ExpressTools API Function

24.361.3 Syntax

```
(acet-lwpline-make inplist)
```

24.361.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.361.5 Description

Creates a LWPOYLINE entity based in inplist data; inplist must be a list of format (defaults points). defaults : is a list of (dxf . value) pairs, specifying desired standard properties like color, layer etc.; can be NIL points : is a list of 2d or 3d points Returns nothing (like (princ) function). Example : (setq points Create a closed LWPOYLINE with color cyan and global width 2, using 3 points.

24.361.6 Return Values

Vendor-defined; see Source Notes.

24.361.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.361.8 Source Notes

- [acet-lwpline-make (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetlwplinemake.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.362 Function Entry: ACET-MOD-ATT

24.362.1 Name

‘acet-mod-att‘

24.362.2 Class

BricsCAD ExpressTools API Function

24.362.3 Syntax

`(acet-mod-att ename inplist)`

24.362.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.362.5 Description

This function is a simplified wrapper for (acet-insert-attrib-mod ename inplist T).

24.362.6 Return Values

Vendor-defined; see Source Notes.

24.362.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.362.8 Source Notes

- [acet-mod-att (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetmodatt.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.363 Function Entry: ACET-PLINE-IS-2D

24.363.1 Name

‘acet-pline-is-2d‘

24.363.2 Class

BricsCAD ExpressTools API Function

24.363.3 Syntax

`(acet-pline-is-2d entdata)`

24.363.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.363.5 Description

Verifies whether the entity specified by entdata is a 2d polyline (POLYLINE, LWPOLYLINE); returns T for a 2d polyline, NIL otherwise

24.363.6 Return Values

Vendor-defined; see Source Notes.

24.363.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.363.8 Source Notes

- [acet-pline-is-2d (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetplineis2d.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.364 Function Entry: ACET-PLINE-SEGMENT-LIST

24.364.1 Name

‘acet-pline-segment-list‘

24.364.2 Class

BricsCAD ExpressTools API Function

24.364.3 Syntax

`(acet-pline-segment-list entdata)`

24.364.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.364.5 Description

Returns the definition data of polyline specified by entdata as a list of type (vertex-list startwidth-list endwidth-list bulge-list).

24.364.6 Return Values

Vendor-defined; see Source Notes.

24.364.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.364.8 Source Notes

- [acet-pline-segment-list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetplinesegmentlist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.365 Function Entry: ACET-PLINE-SEGMENT-LIST-APPLY

24.365.1 Name

‘acet-pline-segment-list-apply‘

24.365.2 Class

BricsCAD ExpressTools API Function

24.365.3 Syntax

```
(acet-pline-segment-list-apply entity plinedata)
```

24.365.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.365.5 Description

Applies polyline definition data plinedata to the polyline specified by entity; entity can either by an ENAME or an (entget) list including the -1 DXF group, and must specify a polyline entity; the definition data plinedata to be applied to the polyline must be a list of type (vertex-list startwidth-list endwidth-list bulge-list) as returned by (acet-pline-segment-list).

24.365.6 Return Values

Vendor-defined; see Source Notes.

24.365.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.365.8 Source Notes

- [acet-pline-segment-list-apply (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetplinesegmentlistapply.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.366 Function Entry: ACET-PLINES-EXPLODE

24.366.1 Name

‘acet-plines-explode‘

24.366.2 Class

BricsCAD ExpressTools API Function

24.366.3 Syntax

```
(acet-plines-explode ss)
```

24.366.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.366.5 Description

Explodes all polylines contained in selectionset ss, and returns a a list of definition data, 1 entry for each exploded polyline : (ExplodedData ExplodeData …). ExplodedData : is a list of format (ltypegen enames widthlist ucs startpt flag1 flag2) ltypegen : string specifying whether linetype generation is on or off for the source polyline enames : list of entity names created by the explode operation widthlist : list of (start-width end-width) entries of the source polyline ucs : entity coordinate system of the source polyline startpt : start point of the source polyline flag1 : actually always NIL flag2 : actually always NIL

24.366.6 Return Values

Vendor-defined; see Source Notes.

24.366.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.366.8 Source Notes

- [acet-plines-explode (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetplinesexplode.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.367 Function Entry: ACET-PLINES-REBUILD

24.367.1 Name

‘acet-plines-rebuild‘

24.367.2 Class

BricsCAD ExpressTools API Function

24.367.3 Syntax

```
(acet-plines-rebuild plinedata)
```

24.367.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.367.5 Description

Recreates new polylines based on plinedata definition data. plinedata : is a list of polyline definition data as returned by (acet-plines-explode) Returns a selectionset containing the recreated polylines, or NIL if no polyline was recreated.

24.367.6 Return Values

Vendor-defined; see Source Notes.

24.367.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.367.8 Source Notes

- [acet-plines-rebuild (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetplinesrebuild.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.368 Function Entry: ACET-POINT-FLAT

24.368.1 Name

‘acet-point-flat‘

24.368.2 Class

BricsCAD ExpressTools API Function

24.368.3 Syntax

`(acet-point-flat)`

24.368.4 Description

Transforms the input point pt from the fromCS.coordinate system to toCS system, and returns the resulting point with Z value 0

24.368.5 Return Values

Vendor-defined; see Source Notes.

24.368.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.368.7 Source Notes

- [acet-point-flat (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetpointflat.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.369 Function Entry: ACET-PREF-SUPPORTPATH-LIST

24.369.1 Name

‘acet-pref-supportpath-list‘

24.369.2 Class

BricsCAD ExpressTools API Function

24.369.3 Syntax

`(acet-pref-supportpath-list)`

24.369.4 Description

Returns the list of support pathes defined.

24.369.5 Return Values

Vendor-defined; see Source Notes.

24.369.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.369.7 Source Notes

- [acet-pref-supportpath-list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetprefsupportpathlist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.370 Function Entry: ACET-ROTATE-TEXT

24.370.1 Name

‘acet-rotate-text‘

24.370.2 Class

BricsCAD ExpressTools API Function

24.370.3 Syntax

`(acet-rotate-text)`

24.370.4 Description

This function rotate the specified text entity (ename) by the given angle (double, in radians); returns T or NIL.

24.370.5 Return Values

Vendor-defined; see Source Notes.

24.370.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.370.7 Source Notes

- [acet-rotate-text (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-rotate-text.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.371 Function Entry: ACET-RTOD

24.371.1 Name

‘acet-rtod‘

24.371.2 Class

BricsCAD ExpressTools API Function

24.371.3 Syntax

`(acet-rtod rad)`

24.371.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.371.5 Description

Returns the angle value of rad in degrees.

24.371.6 Return Values

Vendor-defined; see Source Notes.

24.371.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.371.8 Source Notes

- [acet-rtod (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetrtod.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.372 Function Entry: ACET-SAFE-COMMAND

24.372.1 Name

‘acet-safe-command‘

24.372.2 Class

BricsCAD ExpressTools API Function

24.372.3 Syntax

`(acet-safe-command)`

24.372.4 Description

(acet-safe-command- bStart bAutoEnd cmdList) Executes the command list cmdList in safe way. If bStart is T, all items of cmdList are processed, and if bAutoEnd is T, the command sequence will be finished by an extra (command). If bStart is NIL, then an already running command is continued by all items of cmdList, bAutoEnd is ignored.

24.372.5 Return Values

Vendor-defined; see Source Notes.

24.372.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.372.7 Source Notes

- [acet-safe-command (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsafecommand.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.373 Function Entry: ACET-SET-CMDECHO

24.373.1 Name

‘acet-set-cmdecho‘

24.373.2 Class

BricsCAD ExpressTools API Function

24.373.3 Syntax

`(acet-set-cmdecho)`

24.373.4 Description

Sets the CMDECHO system variable : if val is NIL : 1 is used for any other value, 0 is used

24.373.5 Return Values

Vendor-defined; see Source Notes.

24.373.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.373.7 Source Notes

- [acet-set-cmdecho (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsetcmdecho.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.374 Function Entry: ACET-SETVAR

24.374.1 Name

'acet-setvar'

24.374.2 Class

BricsCAD ExpressTools API Function

24.374.3 Syntax

`(acet-setvar varlist)`

24.374.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.374.5 Description

Saves custom variables with their values to several storage places. varlist is a list of '(var1 var2 saveMode var2 val2 saveMode ...); saveMode can be a combination (bitflags) of : 1 = dictionary 2 = registry 4 = environment; if saveMode is NIL, the variable with its value is stored in dictionary and registry

24.374.6 Return Values

Vendor-defined; see Source Notes.

24.374.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.374.8 Source Notes

- [acet-setvar (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsetvar.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.375 Function Entry: ACET-SHX-LOADED

24.375.1 Name

‘acet-shx-loaded‘

24.375.2 Class

BricsCAD ExpressTools API Function

24.375.3 Syntax

`(acet-shx-loaded)`

24.375.4 Description

This function verifies whether the shxFile (string) SHX file is loaded: returns T or NIL.

24.375.5 Return Values

Vendor-defined; see Source Notes.

24.375.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.375.7 Source Notes

- [acet-shx-loaded (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-shx-loaded.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.376 Function Entry: ACET-SHX-RELOAD

24.376.1 Name

'acet-shx-reload'

24.376.2 Class

BricsCAD ExpressTools API Function

24.376.3 Syntax

`(acet-shx-reload)`

24.376.4 Description

This function reloads the specified shxFile (string) SHX file; an internal (findfile) operation is included; returns T or NIL.

24.376.5 Return Values

Vendor-defined; see Source Notes.

24.376.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.376.7 Source Notes

- [acet-shx-reload (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-shx-reload.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.377 Function Entry: ACET-SPINNER

24.377.1 Name

‘acet-spinner‘

24.377.2 Class

BricsCAD ExpressTools API Function

24.377.3 Syntax

`(acet-spinner)`

24.377.4 Description

Shows a ”spinning wheel” on the commandline; each call of (acet-spinner) turns the spinning wheel by 90 degrees. Applications can show this during long-running routines to give some visual feedback to user.

24.377.5 Return Values

Vendor-defined; see Source Notes.

24.377.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.377.7 Source Notes

- [acet-spinner (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetspinner.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.378 Function Entry: ACET-SS-ANNOTATION-FILTER

24.378.1 Name

‘acet-ss-annotation-filter‘

24.378.2 Class

BricsCAD ExpressTools API Function

24.378.3 Syntax

`(acet-ss-annotation-filter ss)`

24.378.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.378.5 Description

This function takes a selectionset ss, and removes all INSERT entities from it; all attributes contained in the INSERT entities are in turn added to resulting selectionset. Returns a new selectionset always (can be empty).

24.378.6 Return Values

Vendor-defined; see Source Notes.

24.378.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.378.8 Source Notes

- [acet-ss-annotation-filter (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssannotationfilter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.379 Function Entry: ACET-SS-CLEAR-PREV

24.379.1 Name

‘acet-ss-clear-prev‘

24.379.2 Class

BricsCAD ExpressTools API Function

24.379.3 Syntax

`(acet-ss-clear-prev)`

24.379.4 Description

This function clears the ”previous selectionset” in the current drawing document. Returns NIL always.

24.379.5 Return Values

Vendor-defined; see Source Notes.

24.379.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.379.7 Source Notes

- [acet-ss-clear-prev (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssclearprev.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.380 Function Entry: ACET-SS-ENTDEL

24.380.1 Name

‘acet-ss-entdel‘

24.380.2 Class

BricsCAD ExpressTools API Function

24.380.3 Syntax

`(acet-ss-entdel ss)`

24.380.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.380.5 Description

Erases all entities contained in selectionset ss; already erased entities are unerased instead. Returns the number of processed entities.

24.380.6 Return Values

Vendor-defined; see Source Notes.

24.380.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.380.8 Source Notes

- [acet-ss-entdel (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssentdel.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.381 Function Entry: ACET-SS-FILTER

24.381.1 Name

‘acet-ss-filter‘

24.381.2 Class

BricsCAD ExpressTools API Function

24.381.3 Syntax

`(acet-ss-filter ss filter)`

24.381.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.381.5 Description

Creates a new selectionset based on filter input. ss selectionset to be filtered by filter filterdata must be a list Returns a new selectionset, containing the entities from ss which passed the filter

24.381.6 Return Values

Vendor-defined; see Source Notes.

24.381.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.381.8 Source Notes

- [acet-ss-filter (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssfilter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.382 Function Entry: ACET-SS-FILTER-CURRENT-UCS

24.382.1 Name

‘acet-ss-filter-current-ucs‘

24.382.2 Class

BricsCAD ExpressTools API Function

24.382.3 Syntax

```
(acet-ss-filter-current-ucs ss printError)
```

24.382.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.382.5 Description

Creates a new selectionset with all entities not being in current UCS removed from selectionset ss. Returns the new selectionset, or NIL if all entities were removed by filtering.

24.382.6 Return Values

Vendor-defined; see Source Notes.

24.382.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.382.8 Source Notes

- [acet-ss-filter-current-ucs (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssfiltercurrentucs.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.383 Function Entry: ACET-SS-FLT-CSPACE

24.383.1 Name

‘acet-ss-flt-cspace‘

24.383.2 Class

BricsCAD ExpressTools API Function

24.383.3 Syntax

`(acet-ss-flt-cspace)`

24.383.4 Description

Creates a filter list, specifying the actual ”space”, containing the model/paperspace flag and the actual layout name. Returns a list of format ((410 . ”<layout>”) (67 . 0/1)) Example : ((410 . ”Layout2”) (67 . 1)) or ((410 . ”Model”) (67 . 0))

24.383.5 Return Values

Vendor-defined; see Source Notes.

24.383.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.383.7 Source Notes

- [acet-ss-flt-cspace (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssfltcspace.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.384 Function Entry: ACET-SS-INTERSECTION

24.384.1 Name

‘acet-ss-intersection‘

24.384.2 Class

BricsCAD ExpressTools API Function

24.384.3 Syntax

```
(acet-ss-intersection ss ssMaster)
```

24.384.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.384.5 Description

Creates and returns a new selectionset, as ”intersection” of selectionset ss with selectionset ssMaster, containing the entities contained in ss and ssMaster. Returns the intersection selectionset or NIL, if either input argument is NIL.

24.384.6 Return Values

Vendor-defined; see Source Notes.

24.384.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.384.8 Source Notes

- [acet-ss-intersection (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssintersection.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.385 Function Entry: ACET-SS-MOD

24.385.1 Name

'acet-ss-mod'

24.385.2 Class

BricsCAD ExpressTools API Function

24.385.3 Syntax

`(acet-ss-mod ss flags printit)`

24.385.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.385.5 Description

Creates a new selectionset from input selectionset ss by filtering based on flags; flags is a combination of bit values as follows : Bit 0 (1) : remove entities from locked layers Bit 1 (2) : remove entities from other layouts Bit 2 (4) : remove mesh entities printit is currently ignored Returns a list of format (selectionset 0 0 0); selectionset can be NIL.

24.385.6 Return Values

Vendor-defined; see Source Notes.

24.385.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.385.8 Source Notes

- [acet-ss-mod (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssmod.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.386 Function Entry: ACET-SS-NEW

24.386.1 Name

‘acet-ss-new‘

24.386.2 Class

BricsCAD ExpressTools API Function

24.386.3 Syntax

`(acet-ss-new ename)`

24.386.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.386.5 Description

Returns a new selectionset with all entities following the entity ename; can be empty. Vertex and Sequend entities are not added to resulting entities. If ename argument is NIL, the entity iteration starts with the last entity (entlast).

24.386.6 Return Values

Vendor-defined; see Source Notes.

24.386.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.386.8 Source Notes

- [acet-ss-new (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssnew.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.387 Function Entry: ACET-SS-REDRAW

24.387.1 Name

‘acet-ss-redraw‘

24.387.2 Class

BricsCAD ExpressTools API Function

24.387.3 Syntax

```
(acet-ss-redraw ss mode)
```

24.387.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.387.5 Description

Redraws all entities contained in selectionset ss, using mode; mode argument is the same as used with (redraw) function : 1 show entity 2 hide entity 3 highlight entity 4 unhighlight entity If mode argument is NIL, mode 1 is used.

24.387.6 Return Values

Vendor-defined; see Source Notes.

24.387.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.387.8 Source Notes

- [acet-ss-redraw (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssredraw.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.388 Function Entry: ACET-SS-REMOVE

24.388.1 Name

‘acet-ss-remove‘

24.388.2 Class

BricsCAD ExpressTools API Function

24.388.3 Syntax

`(acet-ss-remove ss ssmaster)`

24.388.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.388.5 Description

Removes any entity contained in selectionset ss from ’master’ selectionset ssmaster. Returns ssmaster without the entities from ss.

24.388.6 Return Values

Vendor-defined; see Source Notes.

24.388.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.388.8 Source Notes

- [acet-ss-remove (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssremove.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.389 Function Entry: ACET-SS-REMOVE-DUPS

24.389.1 Name

‘acet-ss-remove-dups‘

24.389.2 Class

BricsCAD ExpressTools API Function

24.389.3 Syntax

```
(acet-ss-remove-dups ss fuz dxfToIgnore)
```

24.389.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.389.5 Description

Filters the input selectionset ss and removes any geometrically duplicate entity. The argument fuz specifies tolerance for geometrical comparison (like 1e08); dxfToIgnore is a list of DXF group codes to be ignored in geometrical comparison, like (40 41 42). These DXF codes are always ignored : 6 8 39 62 370 390. Returns list of modified selectionset ss and selectionset, containing the removed entities : (list ss ssRemoved).

24.389.6 Return Values

Vendor-defined; see Source Notes.

24.389.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.389.8 Source Notes

- [acet-ss-remove-dups (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssremovedups.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.390 Function Entry: ACET-SS-SCALE-TO-FIT

24.390.1 Name

'acet-ss-scale-to-fit'

24.390.2 Class

BricsCAD ExpressTools API Function

24.390.3 Syntax

```
(acet-ss-scale-to-fit ss p1 p2 scale)
```

24.390.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.390.5 Description

Returns a list (ptRefPoint effectiveScale).

24.390.6 Return Values

Vendor-defined; see Source Notes.

24.390.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.390.8 Source Notes

- [acet-ss-scale-to-fit (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssscaletofit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.391 Function Entry: ACET-SS-SORT

24.391.1 Name

‘acet-ss-sort‘

24.391.2 Class

BricsCAD ExpressTools API Function

24.391.3 Syntax

```
(acet-ss-sort ss filter)
```

24.391.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.391.5 Description

Creates a new, sorted selectionset from input selectionset ss, based on filter input. filter must be an expression, which can be evaluated by (lambda (x) (cons x ((eval filter) (entget x)))) Returns the sorted selectionset.

24.391.6 Return Values

Vendor-defined; see Source Notes.

24.391.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.391.8 Source Notes

- [acet-ss-sort (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsssort.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.392 Function Entry: ACET-SS-SSGET-FILTER

24.392.1 Name

‘acet-ss-ssget-filter‘

24.392.2 Class

BricsCAD ExpressTools API Function

24.392.3 Syntax

`(acet-ss-ssget-filter ss filterdata)`

24.392.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.392.5 Description

Creates a new selectionset by filtering the input selectionset ss with filterdata. The filterdata is same as used by (acet-filter-match). Returns the filtered selectionset, or NIL if not any entity matches the provided filter.

24.392.6 Return Values

Vendor-defined; see Source Notes.

24.392.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.392.8 Source Notes

- [acet-ss-ssget-filter (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssssgetfilter.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.393 Function Entry: ACET-SS-TO-LIST

24.393.1 Name

‘acet-ss-to-list‘

24.393.2 Class

BricsCAD ExpressTools API Function

24.393.3 Syntax

`(acet-ss-to-list ss)`

24.393.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.393.5 Description

Returns a list of entities contained in selectionset ss; can be NIL.

24.393.6 Return Values

Vendor-defined; see Source Notes.

24.393.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.393.8 Source Notes

- [acet-ss-to-list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsstolist1.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.394 Function Entry: ACET-SS-UNION

24.394.1 Name

'acet-ss-union'

24.394.2 Class

BricsCAD ExpressTools API Function

24.394.3 Syntax

`(acet-ss-union ssList)`

24.394.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.394.5 Description

Creates and returns a new selectionset, as "union" of all selectionsets contained in ssList, which is a list of selectionsets. Returns the summary of all selectionsets, can be empty.

24.394.6 Return Values

Vendor-defined; see Source Notes.

24.394.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.394.8 Source Notes

- [acet-ss-union (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssunion.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.395 Function Entry: ACET-SS-VISIBLE

24.395.1 Name

'acet-ss-visible'

24.395.2 Class

BricsCAD ExpressTools API Function

24.395.3 Syntax

`(acet-ss-visible ss flag)`

24.395.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.395.5 Description

Sets all entities contained in selectionset ss to visible (flag=0) or invisible (flag=1); the flag must be an integer 0 or 1 (as used by DXF code 60). Returns the number of entities switched.

24.395.6 Return Values

Vendor-defined; see Source Notes.

24.395.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.395.8 Source Notes

- [acet-ss-visible (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetssvisible.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.396 Function Entry: ACET-SS-ZOOM-EXTENTS

24.396.1 Name

‘acet-ss-zoom-extents‘

24.396.2 Class

BricsCAD ExpressTools API Function

24.396.3 Syntax

`(acet-ss-zoom-extents ss)`

24.396.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.396.5 Description

This function zooms the display to the extents of given selectionset ss (scaled by factor 1.05). Returns a list (ptMin ptMax) specifying the extents used for zooming.

24.396.6 Return Values

Vendor-defined; see Source Notes.

24.396.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.396.8 Source Notes

- [acet-ss-zoom-extents (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsszoomextents.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.397 Function Entry: ACET-STR-ESC-WILDCARDS

24.397.1 Name

‘acet-str-esc-wildcards‘

24.397.2 Class

BricsCAD ExpressTools API Function

24.397.3 Syntax

```
(acet-str-esc-wildcards string)
```

24.397.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.397.5 Description

Returns a string, with all wildcard characters contained in string escaped by ‘ character.

24.397.6 Return Values

Vendor-defined; see Source Notes.

24.397.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.397.8 Source Notes

- [acet-str-esc-wildcards (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetstrescwildcards.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.398 Function Entry: ACET-STR-LR-TRIM

24.398.1 Name

‘acet-str-lr-trim‘

24.398.2 Class

BricsCAD ExpressTools API Function

24.398.3 Syntax

`(acet-str-lr-trim trimchars string)`

24.398.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.398.5 Description

Returns the string string which is left- and right-trimmed for any character in trimchars.

24.398.6 Return Values

Vendor-defined; see Source Notes.

24.398.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.398.8 Source Notes

- [acet-str-lr-trim (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetstrlrtrim.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.399 Function Entry: ACET-STR-M-FIND

24.399.1 Name

‘acet-str-m-find‘

24.399.2 Class

BricsCAD ExpressTools API Function

24.399.3 Syntax

```
(acet-str-m-find what string)
```

24.399.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.399.5 Description

Returns the string string for occurrences of what.and returns a list of found positions.

24.399.6 Return Values

Vendor-defined; see Source Notes.

24.399.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.399.8 Source Notes

- [acet-str-m-find (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetstrmfind.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.400 Function Entry: ACET-STR-SPACE-TRIM

24.400.1 Name

‘acet-str-space-trim‘

24.400.2 Class

BricsCAD ExpressTools API Function

24.400.3 Syntax

`(acet-str-space-trim string)`

24.400.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.400.5 Description

Returns the string string which is trimmed for any preceding and trailing spaces.

24.400.6 Return Values

Vendor-defined; see Source Notes.

24.400.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.400.8 Source Notes

- [acet-str-space-trim (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetstrspacetrim.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.401 Function Entry: ACET-STR-TO-LIST

24.401.1 Name

'acet-str-to-list'

24.401.2 Class

BricsCAD ExpressTools API Function

24.401.3 Syntax

`(acet-str-to-list deli line)`

24.401.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.401.5 Description

Returns a list of string tokens from splitting line at delimiter deli.

24.401.6 Return Values

Vendor-defined; see Source Notes.

24.401.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.401.8 Source Notes

- [acet-str-to-list (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetstrtolist.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.402 Function Entry: ACET-SYS-CONTROL-DOWN

24.402.1 Name

‘acet-sys-control-down‘

24.402.2 Class

BricsCAD ExpressTools API Function

24.402.3 Syntax

`(acet-sys-control-down)`

24.402.4 Description

Returns T if the ”Ctrl” key is just pressed, NIL otherwise.

24.402.5 Return Values

Vendor-defined; see Source Notes.

24.402.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.402.7 Source Notes

- [acet-sys-control-down (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsyscontroldown.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.403 Function Entry: ACET-SYS-LMOUSE-DOWN

24.403.1 Name

‘acet-sys-lmouse-down‘

24.403.2 Class

BricsCAD ExpressTools API Function

24.403.3 Syntax

`(acet-sys-lmouse-down)`

24.403.4 Description

Returns T if the left mouse button key is just pressed, NIL otherwise.

24.403.5 Return Values

Vendor-defined; see Source Notes.

24.403.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.403.7 Source Notes

- [acet-sys-lmouse-down (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsyslmousedown.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.404 Function Entry: ACET-SYS-MMOUSE-DOWN

24.404.1 Name

‘acet-sys-mmouse-down‘

24.404.2 Class

BricsCAD ExpressTools API Function

24.404.3 Syntax

`(acet-sys-mmouse-down)`

24.404.4 Description

Returns T if the middle mouse button key is just pressed, NIL otherwise.

24.404.5 Return Values

Vendor-defined; see Source Notes.

24.404.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.404.7 Source Notes

- [acet-sys-mmouse-down (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsysmmousedown.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.405 Function Entry: ACET-SYS-RMOUSE-DOWN

24.405.1 Name

‘acet-sys-rmouse-down‘

24.405.2 Class

BricsCAD ExpressTools API Function

24.405.3 Syntax

`(acet-sys-rmouse-down)`

24.405.4 Description

Returns T if the right mouse button key is just pressed, NIL otherwise.

24.405.5 Return Values

Vendor-defined; see Source Notes.

24.405.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.405.7 Source Notes

- [acet-sys-rmouse-down (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsysrmousedown.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.406 Function Entry: ACET-SYS-SHIFT-DOWN

24.406.1 Name

‘acet-sys-shift-down‘

24.406.2 Class

BricsCAD ExpressTools API Function

24.406.3 Syntax

`(acet-sys-shift-down)`

24.406.4 Description

Returns T if the shift key is just pressed, NIL otherwise.

24.406.5 Return Values

Vendor-defined; see Source Notes.

24.406.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.406.7 Source Notes

- [acet-sys-shift-down (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsysshiftdown.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.407 Function Entry: ACET-SYSVAR-RESTORE

24.407.1 Name

‘acet-sysvar-restore‘

24.407.2 Class

BricsCAD ExpressTools API Function

24.407.3 Syntax

`(acet-sysvar-restore)`

24.407.4 Description

Restores the system variables as saved by a preceding call to (acet-sysvar-set) to their previous values. Returns void (princ) always.

24.407.5 Return Values

Vendor-defined; see Source Notes.

24.407.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.407.7 Source Notes

- [acet-sysvar-restore (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsysvarrestore.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.408 Function Entry: ACET-SYSVAR-SET

24.408.1 Name

‘acet-sysvar-set‘

24.408.2 Class

BricsCAD ExpressTools API Function

24.408.3 Syntax

```
(acet-sysvar-set varlist)
```

24.408.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.408.5 Description

Sets system variables from given list varlist and saves the actual values of those system variables to an internal stack list; varlist is a list containing the system variable names and there new values as a flat sequence. Returns the list of saved system variables with their values. Example : (acet-sysvar-set '(”cmdecho” 0 ”osmode” 0 ...))

24.408.6 Return Values

Vendor-defined; see Source Notes.

24.408.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.408.8 Source Notes

- [acet-sysvar-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetsysvarset.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.409 Function Entry: ACET-TABLE-NAME-LIST

24.409.1 Name

‘acet-table-name-list‘

24.409.2 Class

BricsCAD ExpressTools API Function

24.409.3 Syntax

```
(acet-table-name-list tblORlist)
```

24.409.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.409.5 Description

Returns a list of table entry names; the table is specified by tblORlist; if tblORlist is a list, it must contain the table name as first item, optionally followed by integer values, used to filter for entries NOT having one of the values in their flag 70. Example : (acet-table-name-list (list ”block” 1 4 8 16)) returns a list ob block names, which are : - not anonymous (1) - not external references (4) - not an xref overlay (8) - not depending on external resources / references (16)

24.409.6 Return Values

Vendor-defined; see Source Notes.

24.409.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.409.8 Source Notes

- [acet-table-name-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettablenamelist.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.410 Function Entry: ACET-TABLE-PURGE

24.410.1 Name

‘acet-table-purge‘

24.410.2 Class

BricsCAD ExpressTools API Function

24.410.3 Syntax

```
(acet-table-purge table entry flag)
```

24.410.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.410.5 Description

Purges the entry object from specified symbol table table, flagis actually ignored.

24.410.6 Return Values

Vendor-defined; see Source Notes.

24.410.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.410.8 Source Notes

- [acet-table-purge (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettablepurge.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.411 Function Entry: ACET-TBL-MATCH

24.411.1 Name

‘acet-tbl-match‘

24.411.2 Class

BricsCAD ExpressTools API Function

24.411.3 Syntax

`(acet-tbl-match tablename filter)`

24.411.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.411.5 Description

Returns all entries inside specified table tablename due to filter.

24.411.6 Return Values

Vendor-defined; see Source Notes.

24.411.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.411.8 Source Notes

- [acet-tbl-match (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettblmatch.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.412 Function Entry: ACET-TEMP-SEGMENT

24.412.1 Name

‘acet-temp-segment‘

24.412.2 Class

BricsCAD ExpressTools API Function

24.412.3 Syntax

`(acet-temp-segment p1 p2 p3 highlight)`

24.412.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.412.5 Description

Draws temporary line segments [p1,p2] and [p2,p3] on actual viewport; if highlight is not NIL, the line segments are highlighted, otherwise drawn normally. Calling (acet-temp-segment ...) clears any previously drawn temporary line segments.

24.412.6 Return Values

Vendor-defined; see Source Notes.

24.412.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.412.8 Source Notes

- [acet-temp-segment (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettempsegment.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.413 Function Entry: ACET-TEXT2MTEXT

24.413.1 Name

‘acet-text2mtext‘

24.413.2 Class

BricsCAD ExpressTools API Function

24.413.3 Syntax

```
(acet-text2mtext entity)
```

24.413.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.413.5 Description

Creates a MText entity from given Text, AttDef or Attribute entity; the original entity is not deleted. Returns the ENAME of created MText entity, or NIL.

24.413.6 Return Values

Vendor-defined; see Source Notes.

24.413.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.413.8 Source Notes

- [acet-text2mtext (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-text2mtext.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.414 Function Entry: ACET-TJUST

24.414.1 Name

‘acet-tjust‘

24.414.2 Class

BricsCAD ExpressTools API Function

24.414.3 Syntax

`(acet-tjust ss justification)`

24.414.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.414.5 Description

This functions sets each text, mtext, attribute and derived entity in ss to the text justification; the justification string must be one of those as returned by (acet-tjust-keyword). Return the number of changes entities.

24.414.6 Return Values

Vendor-defined; see Source Notes.

24.414.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.414.8 Source Notes

- [acet-tjust (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettjust.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.415 Function Entry: ACET-TJUST-KEYWORD

24.415.1 Name

‘acet-tjust-keyword‘

24.415.2 Class

BricsCAD ExpressTools API Function

24.415.3 Syntax

```
(acet-tjust-keyword entgetlist)
```

24.415.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.415.5 Description

Returns the text justification of the entity as one of the strings below; the entgetlist must be the entity definition list (as returned by (entget) function of a Text, MText, Attribute and derived entity - at least, the DXF group code 72 and 73 data must be contained. Possible justification strings are : START, CENTER, RIGHT, ALIGNED, MIDDLE, FIT BL, BC, BR ML, MC, MR TL, TC, TR If the entgetlist does not contain the 72 + 73 DXF group codes, or does not specify a valid entity, an empty string is returned.

24.415.6 Return Values

Vendor-defined; see Source Notes.

24.415.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.415.8 Source Notes

- [acet-tjust-keyword (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acettjustkeyword.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.416 Function Entry: ACET-UCS-CMD

24.416.1 Name

‘acet-ucs-cmd‘

24.416.2 Class

BricsCAD ExpressTools API Function

24.416.3 Syntax

`(acet-ucs-cmd cmdList)`

24.416.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.416.5 Description

Runs the each item in cmdList with the _UCS command.

24.416.6 Return Values

Vendor-defined; see Source Notes.

24.416.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.416.8 Source Notes

- [acet-ucs-cmd (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetucscmd.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.417 Function Entry: ACET-UCS-GET

24.417.1 Name

‘acet-ucs-get‘

24.417.2 Class

BricsCAD ExpressTools API Function

24.417.3 Syntax

```
(acet-ucs-get from)
```

24.417.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.417.5 Description

Returns the coordinate system defined by from as list [Origin xAxis yAxis]. If from is an entity ENAME, it returns the the entity coordinate system, otherwise it returns the list of [UCSORG UCSXDIR UCSYDIR].

24.417.6 Return Values

Vendor-defined; see Source Notes.

24.417.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.417.8 Source Notes

- [acet-ucs-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetucsget.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.418 Function Entry: ACET-UCS-SET

24.418.1 Name

‘acet-ucs-set‘

24.418.2 Class

BricsCAD ExpressTools API Function

24.418.3 Syntax

`(acet-ucs-set points)`

24.418.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.418.5 Description

Sets the current UCS to be defined by 3 points contained (Origin xAxis yAxis)

24.418.6 Return Values

Vendor-defined; see Source Notes.

24.418.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.418.8 Source Notes

- [acet-ucs-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetucsset.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.419 Function Entry: ACET-UCS-SET-Z

24.419.1 Name

‘acet-ucs-set-z‘

24.419.2 Class

BricsCAD ExpressTools API Function

24.419.3 Syntax

`(acet-ucs-set-z zVec)`

24.419.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.419.5 Description

Sets the current UCS as defined by zVec argument (zAxis).

24.419.6 Return Values

Vendor-defined; see Source Notes.

24.419.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.419.8 Source Notes

- [acet-ucs-set-z (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetucssetz.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.420 Function Entry: ACET-UCS-TO-OBJECT

24.420.1 Name

‘acet-ucs-to-object‘

24.420.2 Class

BricsCAD ExpressTools API Function

24.420.3 Syntax

`(acet-ucs-to-object entity)`

24.420.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.420.5 Description

Sets the current UCS to the entity coordinate system.

24.420.6 Return Values

Vendor-defined; see Source Notes.

24.420.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.420.8 Source Notes

- [acet-ucs-to-object (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetucstoobject.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.421 Function Entry: ACET-UI-ENTSEL

24.421.1 Name

'acet-ui-entsel'

24.421.2 Class

BricsCAD ExpressTools API Function

24.421.3 Syntax

```
(acet-ui-entsel sellst)
```

24.421.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.421.5 Description

This function provides entity selection, like (entsel), with much more comfort. The sellst argument controls the behaviour of entity selection, and must have this format : (msg flags keys useWindow filter allowLocked) msg : user prompt flags : bit values as used by (initget) function (integer) keys : keyword string as used by (initget) function (string) useWindow : allow to switch to "crossing window" selection, if no entity is selected filter : filter condition, as used by (ssget "C" ... filter) when switching to "crossing window" selection allowLocked : if T, selection of entity/entities on locked layers is allowed; NIL to disallow Each argument can be set as NIL, which uses default values (or none) in that case.

24.421.6 Return Values

Vendor-defined; see Source Notes.

24.421.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.421.8 Source Notes

- [acet-ui-entsel (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuientsel.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.422 Function Entry: ACET-UI-FENCE-SELECT

24.422.1 Name

‘acet-ui-fence-select‘

24.422.2 Class

BricsCAD ExpressTools API Function

24.422.3 Syntax

`(acet-ui-fence-select)`

24.422.4 Description

This functions allows to specify a fence polygon (suitable for later fence selection). Returns a list of fence points.

24.422.5 Return Values

Vendor-defined; see Source Notes.

24.422.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.422.7 Source Notes

- [acet-ui-fence-select (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuifenceselect.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.423 Function Entry: ACET-UI-GET-LONG-NAME

24.423.1 Name

‘acet-ui-get-long-name‘

24.423.2 Class

BricsCAD ExpressTools API Function

24.423.3 Syntax

`(acet-ui-get-long-name msg)`

24.423.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.423.5 Description

Wrapper for (getstring T msg); returns the string as entered by user.

24.423.6 Return Values

Vendor-defined; see Source Notes.

24.423.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.423.8 Source Notes

- [acet-ui-get-long-name (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuigetlongname.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.424 Function Entry: ACET-UI-GETCORNER

24.424.1 Name

‘acet-ui-getcorner‘

24.424.2 Class

BricsCAD ExpressTools API Function

24.424.3 Syntax

`(acet-ui-getcorner refPoint)`

24.424.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.424.5 Description

Uses (getcorner refPoint ”Other corner: ”) to query the corner point and returns the point specified by user.

24.424.6 Return Values

Vendor-defined; see Source Notes.

24.424.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.424.8 Source Notes

- [acet-ui-getcorner (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuigetcorner.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.425 Function Entry: ACET-UI-GETFILE

24.425.1 Name

‘acet-ui-getfile‘

24.425.2 Class

BricsCAD ExpressTools API Function

24.425.3 Syntax

`(acet-ui-getfile title deffile defext dlgname flags)`

24.425.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.425.5 Description

Opens the file selection dialog, similar to standard (getfiled) function. dlgname : is a string, specifying the ”dialog name”, can be any arbitrary string, or even an empty string.

24.425.6 Return Values

Vendor-defined; see Source Notes.

24.425.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.425.8 Source Notes

- [acet-ui-getfile (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuigetfile.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.426 Function Entry: ACET-UI-M-GET-NAMES

24.426.1 Name

'acet-ui-m-get-names'

24.426.2 Class

BricsCAD ExpressTools API Function

24.426.3 Syntax

```
(acet-ui-m-get-names argumentList)
```

24.426.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.426.5 Description

This function provides extended user-input functionality similar to (getkword), but allows multiple-keyword selection. argumentList is a single list providing detailed parameters: (allowSpace prompt keywords).

allowSpace T / NIL to allow input of space character(s) by the user (same as in (initget) function).

prompt string to be displayed at the commandline (same as in (getstring) function).

keywords list of keywords to be used (similar to (initget) function, which uses a flat string instead).

Returns a list of matching keywords, always in uppercase notation. If no matching key is found, a message is printed to the commandline and the input query is repeated; empty input always finishes the query, returning NIL. The user can enter multiple keywords, separated by "," (comma); wildcards can be used with each key in the answer string. This function then checks each entered keyword against the keyword list, and matching keyword(s) are added to the result list.

Examples:

```
(acet-ui-m-get-names (list nil "
* aaa bbb ccc : " lst))
  => space input finishes input
  => input : a*       returns '("AAA")
  => input : aaa      returns '("AAA")
  => input : *        returns '("AAA" "BBB" "CCC")
  => input : a*,bbb   returns '("AAA" "BBB")

(acet-ui-m-get-names (list T "
* a1 a2 a3 : " lst))
  => space input is allowed
  => input : a*       returns '("A1" "A2" "A3")
  => input : *3       returns '("A3")
```

24.426.6 Return Values

Vendor-defined; see Source Notes.

24.426.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.426.8 Source Notes

- [acet-ui-m-get-names (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuimgetnames.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.427 Function Entry: ACET-UI-POLYGON-SELECT

24.427.1 Name

‘acet-ui-polygon-select‘

24.427.2 Class

BricsCAD ExpressTools API Function

24.427.3 Syntax

`(acet-ui-polygon-select mode)`

24.427.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.427.5 Description

This functions allows to specify a selection polygon (suitable for later polygon selection). If mode is T or 1, the polygon is highlighted, otherwise it is displayed normally. Returns a list of fence points.

24.427.6 Return Values

Vendor-defined; see Source Notes.

24.427.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.427.8 Source Notes

- [acet-ui-polygon-select (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuipolygonselect.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.428 Function Entry: ACET-UI-PROGRESS-DONE

24.428.1 Name

‘acet-ui-progress-done‘

24.428.2 Class

BricsCAD ExpressTools API Function

24.428.3 Syntax

`(acet-ui-progress-done)`

24.428.4 Description

Closes the statusbar progress meter. Always returns NIL.

24.428.5 Return Values

Vendor-defined; see Source Notes.

24.428.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.428.7 Source Notes

- [acet-ui-progress-done (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuiprogressdone.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.429 Function Entry: ACET-UI-PROGRESS-INIT

24.429.1 Name

‘acet-ui-progress-init‘

24.429.2 Class

BricsCAD ExpressTools API Function

24.429.3 Syntax

```
(acet-ui-progress-init message limit)
```

24.429.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.429.5 Description

Initialises the statusbar progress meter. message the text to be displayed for the statusbar progress meter limit the maximum value of range Returns T on success, NIL otherwise

24.429.6 Return Values

Vendor-defined; see Source Notes.

24.429.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.429.8 Source Notes

- [acet-ui-progress-init (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuiprogressinit.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.430 Function Entry: ACET-UI-PROGRESS-SAFE

24.430.1 Name

‘acet-ui-progress-safe‘

24.430.2 Class

BricsCAD ExpressTools API Function

24.430.3 Syntax

```
(acet-ui-progress-safe messageOrPosition)
```

24.430.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.430.5 Description

If messageOrPosition is a string, the new message text is displayed for the progress meter, and position is reset as 0; if messageOrPosition is an integer number, the progress meter is updated to specified position :positive value positions the progress meter to the absolute position, negative values are meant is increments. Returns T/NIL if message text is updated, or the actual position value.

24.430.6 Return Values

Vendor-defined; see Source Notes.

24.430.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.430.8 Source Notes

- [acet-ui-progress-safe (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuiprogresssafe.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.431 Function Entry: ACET-UI-SINGLE-SELECT

24.431.1 Name

‘acet-ui-single-select‘

24.431.2 Class

BricsCAD ExpressTools API Function

24.431.3 Syntax

```
(acet-ui-single-select filter allowLocked)
```

24.431.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.431.5 Description

This function provides single entity selection, like (entsel), with some customisation. filter : filter condition, as used by (ssget ... filter) to filter entity as required allowLocked : if T, selection of entity on a locked layers is allowed; NIL to disallow Each argument can be set as NIL, which uses default values (or none) in that case.

24.431.6 Return Values

Vendor-defined; see Source Notes.

24.431.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.431.8 Source Notes

- [acet-ui-single-select (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetuisingleselect.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.432 Function Entry: ACET-UI-TABLE-NAME-GET

24.432.1 Name

‘acet-ui-table-name-get‘

24.432.2 Class

BricsCAD ExpressTools API Function

24.432.3 Syntax

```
(acet-ui-table-name-get lstParams)
```

24.432.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.432.5 Description

This function allows to select one or more Table or Dictionary entries, using a dialog or the commandline; it calls (acet-ui-table-name-get-dlg) or (acet-ui-table-name-get-cmd) internally. if CMDDIA is set to 0, or if there is any other ”outer” command running incl. script file, then the commandline version is used. lstParams : (list) list of parameters which specify the behaviour of (acet-ui-table-name-get) function lstParams parameters list : (title default tableName|dictionaryName [flags [filter [helpfile [helptopic]]]]) title (string) the title of the dialog (if the dialog is used; not used in commandline version) default (string) the default value being selected in the dialog’s list tableName or dictionaryName (string) the name of the Table (like Layer, Block, Linetype ...), or the name of a root dictionary flags integer bit values controlling the behaviour, default is 0 0 must select existing name 1 allows to specify a new name, not existing yet 2 disable selection of external references (applies to symbol table names only) 4 enable filtering of entries that match the provided filter 8 disable object selection option (pick button or the ”=” option at commandline prompt) 16 - allow ”ByLayer” and ”ByBlock” entries (if applicable, otherwise ignored) 32 enable selection of multiple entries/objects 64 ignore locked layers (for LAYER table only) 128 ignore frozen layers (for LAYER table only) 256 ignore off layers (for LAYER table only) filter helpfile the help file to be used on ”Help” button, and F1 key; same as in (help) functions helptopic the help topic to be used on ”Help” button, and F1 key; same as in (help) functions see (acet-ui-table-name-get-dlg) and (acet-ui-table-name-get-cmd) functions

24.432.6 Return Values

Vendor-defined; see Source Notes.

24.432.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.432.8 Source Notes

- [acet-ui-table-name-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-ui-table-name-get.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.433 Function Entry: ACET-UI-TABLE-NAME-GET-CMD

24.433.1 Name

‘acet-ui-table-name-get-cmd‘

24.433.2 Class

BricsCAD ExpressTools API Function

24.433.3 Syntax

```
(acet-ui-table-name-get-cmd lstParams)
```

24.433.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.433.5 Description

This function allows to select one or more Table or Dictionary entries using the commandline version. lstParams : (list) list of parameters which specify the behaviour of (acet-ui-table-name-get-cmd) function lstParams parameters list : (title default tableName|dictionaryName [flags [filter [helpfile [helptopic]]]]) see (acet-ui-table-name-get) for more details on the lstParams parameters

24.433.6 Return Values

Vendor-defined; see Source Notes.

24.433.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.433.8 Source Notes

- [acet-ui-table-name-get-cmd (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-ui-table-name-get-cmd.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.434 Function Entry: ACET-UI-TABLE-NAME-GET-DLG

24.434.1 Name

‘acet-ui-table-name-get-dlg‘

24.434.2 Class

BricsCAD ExpressTools API Function

24.434.3 Syntax

`(acet-ui-table-name-get-dlg lstParams)`

24.434.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.434.5 Description

This function allows to select one or more Table or Dictionary entries, using a dialog; it calls (acet-ui-table-name-get-cmd) internally if CMDDIA is set to 0, or if there is any other "outer" command running incl. script file, under such conditions, the commandline version is automatically used. lstParams : (list) list of parameters which specify the behaviour of (acet-ui-table-name-get-dlg) function lstParams parameters list : (title default tableName|dictionaryName [flags [filter [helpfile [helptopic]]]]) see (acet-ui-table-name-get) for more details on the lstParams parameters

24.434.6 Return Values

Vendor-defined; see Source Notes.

24.434.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.434.8 Source Notes

- [acet-ui-table-name-get-dlg (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acet-ui-table-name-get-dlg.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.435 Function Entry: ACET-UNDO-BEGIN

24.435.1 Name

'acet-undo-begin'

24.435.2 Class

BricsCAD ExpressTools API Function

24.435.3 Syntax

`(acet-undo-begin)`

24.435.4 Description

Runs (command "$_{\text{undo}}$" "$_{\text{begin}}$").

24.435.5 Return Values

Vendor-defined; see Source Notes.

24.435.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.435.7 Source Notes

- [acet-undo-begin (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetundobegin.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.436 Function Entry: ACET-UNDO-END

24.436.1 Name

‘acet-undo-end‘

24.436.2 Class

BricsCAD ExpressTools API Function

24.436.3 Syntax

`(acet-undo-end)`

24.436.4 Description

Runs (command ”$_{\text{undo}}$” ”$_{\text{end}}$”).

24.436.5 Return Values

Vendor-defined; see Source Notes.

24.436.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.436.7 Source Notes

- [acet-undo-end (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetundoend.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.437 Function Entry: ACET-UTIL-VER

24.437.1 Name

‘acet-util-ver‘

24.437.2 Class

BricsCAD ExpressTools API Function

24.437.3 Syntax

`(acet-util-ver)`

24.437.4 Description

Returns the version of ExpressTools API as double value (1.38 currently).

24.437.5 Return Values

Vendor-defined; see Source Notes.

24.437.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.437.7 Source Notes

- [acet-util-ver (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetutilver.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.438 Function Entry: ACET-VAR-GETVAR

24.438.1 Name

‘acet-var-getvar‘

24.438.2 Class

BricsCAD ExpressTools API Function

24.438.3 Syntax

```
(acet-var-getvar varname savemode)
```

24.438.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.438.5 Description

Loads a custom variables from several storage places as specified by savemode. saveMode can be a combination (bitflags) of : 1 = dictionary 2 = registry 4 = environment; if saveMode is NIL, the variable with its value is loaded from dictionary and registry (saveMode = 3); if the variable is successfully retrieved from one place, the other potential storage places are not examined (if savemode specifies more than 1 place).

24.438.6 Return Values

Vendor-defined; see Source Notes.

24.438.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.438.8 Source Notes

- [acet-var-getvar (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetvargetvar.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.439 Function Entry: ACET-VAR-SETVAR

24.439.1 Name

'acet-var-setvar'

24.439.2 Class

BricsCAD ExpressTools API Function

24.439.3 Syntax

```
(acet-var-setvar varname varval savemode)
```

24.439.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.439.5 Description

Saves a custom variable varname with value varval to several storage places as specified by savemode. saveMode can be a combination (bitflags) of : 1 = dictionary 2 = registry 4 = environment; if saveMode is NIL, the variable with its value is stored in dictionary and registry

24.439.6 Return Values

Vendor-defined; see Source Notes.

24.439.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.439.8 Source Notes

- [acet-var-setvar (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetvarsetvar.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.440 Function Entry: ACET-VIEWPORT-FROZEN-LAYER-LIST

24.440.1 Name

‘acet-viewport-frozen-layer-list‘

24.440.2 Class

BricsCAD ExpressTools API Function

24.440.3 Syntax

```
(acet-viewport-frozen-layer-list vpename)
```

24.440.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.440.5 Description

Returns the list of viewport-frozen layers (strings) for the viewport specified by vpename. If vpename is not the ename of a modelspace viewport, or if no layers are viewport-frozen, NIL is returned.

24.440.6 Return Values

Vendor-defined; see Source Notes.

24.440.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.440.8 Source Notes

- [acet-viewport-frozen-layer-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetviewportfrozenlayerlist.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.441 Function Entry: ACET-VIEWPORT-LOCK-SET

24.441.1 Name

‘acet-viewport-lock-set‘

24.441.2 Class

BricsCAD ExpressTools API Function

24.441.3 Syntax

```
(acet-viewport-lock-set vpename setlocked)
```

24.441.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.441.5 Description

Locks or Unlocks the viewport specified by vpename; setlocked is NIL to unlock, any other value locks the viewport. Returns vpename when locking status was changed, or NIL when the viewport is already in desired mode.

24.441.6 Return Values

Vendor-defined; see Source Notes.

24.441.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.441.8 Source Notes

- [acet-viewport-lock-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetviewportlockset.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.442 Function Entry: ACET-VIEWPORT-NEXT-PICKABLE

24.442.1 Name

‘acet-viewport-next-pickable‘

24.442.2 Class

BricsCAD ExpressTools API Function

24.442.3 Syntax

`(acet-viewport-next-pickable)`

24.442.4 Description

Returns the pickable modelspace viewport as list (id msg); if there is no pickable modelspace viewport, id is NIL, and msg contains an error message; otherwise, id is a valid viewport ID number, and msg will be NIL.

24.442.5 Return Values

Vendor-defined; see Source Notes.

24.442.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.442.7 Source Notes

- [acet-viewport-next-pickable (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetviewportnextpickable.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.443 Function Entry: ACET-WMFIN

24.443.1 Name

‘acet-wmfin‘

24.443.2 Class

BricsCAD ExpressTools API Function

24.443.3 Syntax

```
(acet-wmfin wmffile point xscale yscale rotation)
```

24.443.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.443.5 Description

Inserts a WMF/EMF file as BlockDefinition and BlockReference into the current drawing. wmffile (string) specifies the WMF/EMF file to be imported; if specified as NIL, the user is asked to select or specify the file name point (2d/3d point) specifies the insertion point for the BlockReference; if specified as NIL, the user is asked to define the insertion point xscale (double) specifies the X scale for the BlockReference; if specified as NIL, the user is asked to define the X scale yscale (double) specifies the Y scale for the BlockReference; if specified as NIL, the user is asked to define the Y scale rotation (double) specifies the rotation (in actual UCS XY plane) for the BlockReference; if specified as NIL, the user is asked to define the rotation

24.443.6 Return Values

Vendor-defined; see Source Notes.

24.443.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.443.8 Source Notes

- [acet-wmfin (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetwmfin.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.444 Function Entry: ACET-XDATA-GET

24.444.1 Name

‘acet-xdata-get‘

24.444.2 Class

BricsCAD ExpressTools API Function

24.444.3 Syntax

```
(acet-xdata-get xkeylist)
```

24.444.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.444.5 Description

Returns a list of XData attached to an entity. The xkeylist argument list must have the format : (ename AppId dataKey).

24.444.6 Return Values

Vendor-defined; see Source Notes.

24.444.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.444.8 Source Notes

- [acet-xdata-get (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetxdataget.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.445 Function Entry: ACET-XDATA-SET

24.445.1 Name

‘acet-xdata-set‘

24.445.2 Class

BricsCAD ExpressTools API Function

24.445.3 Syntax

`(acet-xdata-set xkeylist)`

24.445.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.445.5 Description

Assigns XData to an entity; The xkeylist argument list must have the format : (ename AppId dataList), where dataList must use the format (dataKey dataVal dxfKey); dataKey is a application provided key string dataVal is the isp data value to be stored dxfKey is the XData DXF group code appropriate for the dataVal (the DXF group code >= 1000)

24.445.6 Return Values

Vendor-defined; see Source Notes.

24.445.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.445.8 Source Notes

- [acet-xdata-set (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetxdataset.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.445.9 ExpressTools API (acet:: namespace) (8 entries)

24.446 Function Entry: ACET::ACOS

24.446.1 Name

‘acet::acos‘

24.446.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.446.3 Syntax

`(acet::acos value)`

24.446.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.446.5 Description

Returns the arccosine of the given value.

24.446.6 Return Values

Vendor-defined; see Source Notes.

24.446.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.446.8 Source Notes

- [acet::acos (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetacos.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.447 Function Entry: ACET::ARC-POINT-LIST

24.447.1 Name

‘acet::arc-point-list‘

24.447.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.447.3 Syntax

```
(acet::arc-point-list center startpoint endpoint maxangle alt)
```

24.447.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.447.5 Description

Returns a list of equidistant raster points along the specified arc. center (2d/3d point list) the center point of the arc startpoint (2d/3d point list) the start point of the arc endpoint (2d/3d point list) the end point of the arc maxangle (integer or double) maximum angle to scan. starting from startpoint alt (T/NIL) specifies the resolution mode : if NIL, the arc is rasterised using the default 9 raster points per full circle; if T, the number of raster points depends on the radius Scanning for raster points stops when endpoint or maxangle is reached, depending which happens first.

24.447.6 Return Values

Vendor-defined; see Source Notes.

24.447.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.447.8 Source Notes

- [acet::arc-point-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetarc-point-list.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.448 Function Entry: ACET::EXPANDFN

24.448.1 Name

‘acet::expandfn‘

24.448.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.448.3 Syntax

`(acet::expandfn)`

24.448.4 Description

(acet::expandfn- filename check) Returns the fully expanded name (string) of given filename (string); if check is specified as T, the existence of the file is verified.

24.448.5 Return Values

Vendor-defined; see Source Notes.

24.448.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.448.7 Source Notes

- [acet::expandfn (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetexpandfn.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.449 Function Entry: ACET::FILETYPE

24.449.1 Name

‘acet::filetype‘

24.449.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.449.3 Syntax

`(acet::filetype filename)`

24.449.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.449.5 Description

Returns the filetype (string) of specified filename (string).

24.449.6 Return Values

Vendor-defined; see Source Notes.

24.449.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.449.8 Source Notes

- [acet::filetype (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetfiletype.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.450 Function Entry: ACET::NAMEONLY

24.450.1 Name

‘acet::nameonly‘

24.450.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.450.3 Syntax

`(acet::nameonly filename)`

24.450.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.450.5 Description

Returns the base name (string) of the specified filename (string).

24.450.6 Return Values

Vendor-defined; see Source Notes.

24.450.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.450.8 Source Notes

- [acet::nameonly (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetnameonly.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.451 Function Entry: ACET::NORMALIZE-FILENAME

24.451.1 Name

‘acet::normalize-filename‘

24.451.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.451.3 Syntax

```
(acet::normalize-filename fileName)
```

24.451.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.451.5 Description

This function replaces the "\" respectively "/" characters in the fileName(string) by the path delimiter of the actual OS/Platform. As Windows accepts both path delimiters, no replacement is made under Windows; but path delimiter effectively happens on Linux and Mac. Therefore, Lisp application code can safely use this function on all platforms, the returned file name is always suitable for the running OS/Platform. Returns normalised file name as string.

24.451.6 Return Values

Vendor-defined; see Source Notes.

24.451.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.451.8 Source Notes

- [acet::normalize-filename (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetnormalize-filename.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.452 Function Entry: ACET::PATHONLY

24.452.1 Name

‘acet::pathonly‘

24.452.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.452.3 Syntax

```
(acet::pathonly filename)
```

24.452.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.452.5 Description

Returns the path (string) of the specified filename (string).

24.452.6 Return Values

Vendor-defined; see Source Notes.

24.452.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.452.8 Source Notes

- [acet::pathonly (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetpathonly.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.453 Function Entry: ACET::PL-POINT-LIST

24.453.1 Name

‘acet::pl-point-list‘

24.453.2 Class

BricsCAD ExpressTools API Function (acet:: namespace)

24.453.3 Syntax

```
(acet::pl-point-list polyline bulgePoints)
```

24.453.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.453.5 Description

Returns a list of definition points for the specified polyline. polyline (ename or vla-object) the polyline entity, as classic ENAME or VLA-Object; must be 2d/3d/Lightweight polyline (ACDB2DPOLYLINE ACDB3DPOLYLINE ACDBPOLYLINE) bulgePoints (integer) the number of points used to rasterise a bulge segment

24.453.6 Return Values

Vendor-defined; see Source Notes.

24.453.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.453.8 Source Notes

- [acet::pl-point-list (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/acetpl-point-list.htm)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.453.9 DOSLib for Linux + Mac (23 entries)

24.454 Function Entry: DOC$_{\text{CLIPBOARD}}$

24.454.1 Name

‘doc$_{\text{clipboard}}$‘

24.454.2 Class

BricsCAD DOSLib Function

24.454.3 Syntax

```
(doc_clipboard)
```

24.454.4 Description

dos$_{\text{clipboard}}$ Returns the current system clipboard text content or copies the specified.string to the system clipboard. string : if specified, the string to be copied to the system clipboard Returns the string if successful (when saving to clipboard, and when the text is properly retrieved from the clipboard); returns NIL in case of any error Examples : saves the text to the clipboard (dos$_{\text{clipboard}}$) returns ”My Sample Text”

24.454.5 Return Values

Vendor-defined; see Source Notes.

24.454.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.454.7 Source Notes

- [doc$_{\text{clipboard}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/doc_clipboard.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.455 Function Entry: $\mathrm{DOS}_{\mathrm{ACITORGB}}$

24.455.1 Name

‘$\mathrm{dos}_{\mathrm{acitorgb}}$‘

24.455.2 Class

BricsCAD DOSLib Function

24.455.3 Syntax

`(dos_acitorgb color)`

24.455.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.455.5 Description

Returns the Red/Green/Blue true color values for specified ACI color index as list (Red Green Blue). This function is redirected to implementation, for details see vle-aci2rgb.

24.455.6 Return Values

Vendor-defined; see Source Notes.

24.455.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.455.8 Source Notes

- [$\mathrm{dos}_{\mathrm{acitorgb}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_acitorgb.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.456 Function Entry: DOS$_{\text{COMMAND}}$

24.456.1 Name

'dos$_{\text{command}}$'

24.456.2 Class

BricsCAD DOSLib Function

24.456.3 Syntax

```
(dos_command command [mode])
```

24.456.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.456.5 Description

Runs the specified OS command (like when starting from a DOS box or terminal box. command :is a string specifying the OS command including all parameters mode :is an integer to control the behaviour of the OS shell box : 0 show the shell window in normal (last-used) mode 1 show the shell window in normal (last-used) mode, but is not activated 2 show the shell window in minimised, but inactive state 3 show the shell window in minimised, and active state 4 show the shell window in maximised state Returns the string executed in OS command shell, or NIL in case of an error. Examples : runs Notepad.exe with normal shell box runs Notepad.exe with minimised shell box runs control panel with normal shell box runs control panel with minimised shell box

24.456.6 Return Values

Vendor-defined; see Source Notes.

24.456.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.456.8 Source Notes

- [dos$_{command}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_command.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.457 Function Entry: $\mathbf{DOS_{COPY}}$

24.457.1 Name

‘dos_{copy}‘

24.457.2 Class

BricsCAD DOSLib Function

24.457.3 Syntax

```
(dos_copy sourceFileSpec destFileSpec)
```

24.457.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.457.5 Description

This function copies one or more source files to a target file resp. to a target folder; the source file specification may contain wildcard characters like "*" and "?". sourceFileSpec : (string) specifies an existing file (no wildcards used), or declares a file mask when using wildcards destFileSpec : (string) specifies the target file, if sourceFileSpec also specifies an existing file -OR- it specifies a folder to copy all source files to, if sourceFileSpec Returns T on success or NIL on any failure. Examples : T T T T (dos_{copy} "J:.txt" "J:\") NIL Note 1 : any existing target file will be overwritten ! Note 2 : missing target folder(s) will be automatically created Note 3 : when specifying a target folder, the string must end with "\" or "*" character to clearly declare it to be a folder ! Note 4 : both "\" and ""* path separators are allowed on all platforms (Windows, Linux, Mac; they will be adjusted automatically)

24.457.6 Return Values

Vendor-defined; see Source Notes.

24.457.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.457.8 Source Notes

- [dos_{copy} (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_copy.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.458 Function Entry: DOS$_{\text{DELTREE}}$

24.458.1 Name

‘dos$_{\text{deltree}}$‘

24.458.2 Class

BricsCAD DOSLib Function

24.458.3 Syntax

`(dos_deltree path)`

24.458.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.458.5 Description

Deletes the specified folder, and contained sub-folders and files. path : is the name of the folder to be deleted including all sub-folders and files Returns path if the folder exists and operation was successful, or NIL otherwise. Note 1 : on all platforms (Windows, Linux, Mac), both path delimiters "\" and "*" are supported Note 2 : the path can contain a final "\" or "*" path delimiter, but this is not required

24.458.6 Return Values

Vendor-defined; see Source Notes.

24.458.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.458.8 Source Notes

- [$\text{dos}_{\text{deltree}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_deltree.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.459 Function Entry: DOS$_{\textbf{DIR}}$

24.459.1 Name

‘dos$_{\text{dir}}$‘

24.459.2 Class

BricsCAD DOSLib Function

24.459.3 Syntax

```
(dos_dir [filespec [sort]])
```

24.459.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.459.5 Description

Returns a list of all file names matching the filespec as sorted or unsorted list, due to sort argument. filespec : specifies a file mask/filter, can contain path specification, and wildcards for the file name; if filespec does not contain a path, the current working directory is used; if filespec does not contain a mask for the file names, the default "." is used (resp. "**" for Linux/Mac) (the ".**" and "**" are automatically handled by the Lisp engine, so using ".**" and "**" are all valid, on all platforms) sort : specifies whether and how the file names are sorted 0 no sorting 1 alphabetically ascending 2 alphabetically descending 3 by date ascending 4 by date descending 5 by size ascending 6 by size descending Returns the list of directories and subdirectories, or NIL on error. Examples : (dos$_{\textbf{dir}}$ "*home/mint.*" 1)**

24.459.6 Return Values

Vendor-defined; see Source Notes.

24.459.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.459.8 Source Notes

- [dos$_{\text{dir}}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_dir.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.460 Function Entry: DOS_{DIRP}

24.460.1 Name

'dos_{dirp}'

24.460.2 Class

BricsCAD DOSLib Function

24.460.3 Syntax

`(dos_dirp path)`

24.460.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.460.5 Description

Verifies whether the given folder path exists. path : is the name of the folder to be verified Returns T if the folder exists, or NIL otherwise. Note 1 : on all platforms (Windows, Linux, Mac), both path delimiters "\" and "*" are supported Note 2 : the path can contain a final "\" or ""* path delimiter, but this is not required

24.460.6 Return Values

Vendor-defined; see Source Notes.

24.460.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.460.8 Source Notes

- [dos_{dirp} (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_dirp.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.461 Function Entry: DOS$_{\text{DIRTREE}}$

24.461.1 Name

'dos$_{\text{dirtree}}$'

24.461.2 Class

BricsCAD DOSLib Function

24.461.3 Syntax

```
(dos_dirtree [path])
```

24.461.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.461.5 Description

Returns a list of all directories and subdirectories for the specified path. path : is the name of an existing directory (as string); if not specified, the current directory is used Returns the list of directories and subdirectories, or NIL on error.

24.461.6 Return Values

Vendor-defined; see Source Notes.

24.461.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.461.8 Source Notes

- [dos$_{\text{dirtree}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_dirtree.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.462 Function Entry: DOS$_{\text{ENCRYPT}}$

24.462.1 Name

‘dos$_{\text{encrypt}}$‘

24.462.2 Class

BricsCAD DOSLib Function

24.462.3 Syntax

```
(dos_encrypt filename password)
```

24.462.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.462.5 Description

Encrypts and Decrypts the specified file filename using password; password is a string of 4…512 characters. Note 1 : without the correct password, an encrypted file can not be decrypted ! Note 2 : starting with V26, the existing (dos$_{\text{encrypt}}$) function uses a new encryption algorithm, compatible with original McNeill implementation; therefore, files encrypted using BricsCAD’s (dos$_{\text{encrypt}}$) in version V25 (and before) can not be decrypted anymore with V26 (dos$_{\text{encrypt}}$) ! To solve this problem, V26 introduces a new function (dos$_{\text{encryptv25}}$) which uses the ”old” V25 (and before) algorithm. Returns filename when successfully encrypted/decrypted, NIL in case of an error (file not found).

24.462.6 Return Values

Vendor-defined; see Source Notes.

24.462.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.462.8 Source Notes

- [dos$_{\text{encrypt}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_encrypt.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.463 Function Entry: $DOS_{ENCRYPTV25}$

24.463.1 Name

'$dos_{encryptv25}$'

24.463.2 Class

BricsCAD DOSLib Function

24.463.3 Syntax

```
(dos_encrypt_v25 filename password)
```

24.463.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.463.5 Description

Encrypts and Decrypts the specified file filename using password, with the "old" V25 (and before) encryption algorithm. password is a string of 4...256 characters. Note 1 : without the correct password, an encrypted file can not be decrypted ! Note 2 : starting with V26, the existing ($dos_{encrypt}$) function uses a new encryption algorithm, compatible with original McNeill implementation; therefore, files encrypted using BricsCAD's ($dos_{encrypt}$) in version V25 (and before) can not be decrypted anymore with V26 ($dos_{encrypt}$) ! To solve this problem, V26 introduces the new function ($dos_{encryptv25}$) which uses the "old" V25 (and before) algorithm. Returns filename when successfully encrypted/decrypted, NIL in case of an error (file not found).

24.463.6 Return Values

Vendor-defined; see Source Notes.

24.463.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.463.8 Source Notes

- [dos$_{\text{encrypt v25}}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_encrypt_v25.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.464 Function Entry: DOS$_{\text{FILEEX}}$

24.464.1 Name

‘dos$_{\text{fileex}}$‘

24.464.2 Class

BricsCAD DOSLib Function

24.464.3 Syntax

`(dos_fileex filepath)`

24.464.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.464.5 Description

Returns informations about the specified file. filename :is the file to retrieve file details for Returns a list of file details : (<path> <file name> <file size> <file creation time> <file modification time> <file last-access time> <file attributes>) <file attributes> : combination of bit flags : 1 read-only 2 hidden 4 system 8 archive 16 compressed 32 temporary Returns NIL in case of an error.

24.464.6 Return Values

Vendor-defined; see Source Notes.

24.464.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.464.8 Source Notes

- [dos$_{\text{fileex}}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_fileex.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.465 Function Entry: DOS$_{\text{GETDIR}}$

24.465.1 Name

'dos$_{\text{getdir}}$'

24.465.2 Class

BricsCAD DOSLib Function

24.465.3 Syntax

```
(dos_getdir [title [folder [info [allowNewFolder ]]]])
```

24.465.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.465.5 Description

Opens the system's FolderSelect dialog to let the user select a folder. title : is the FolderSelect dialog title; if omitted, a default title will be used folder : preset folder to be selected when the dialog opens; if omitted, no folder is preselected info : optionally an additional message to be shown at the dialog; if omitted, no info is displayed allowNewFolder : if T (default), it allows to create a new folder, otherwise only existing folder can be chosen Returns selected folder name when successful, or NIL if dialog was cancelled.

24.465.6 Return Values

Vendor-defined; see Source Notes.

24.465.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.465.8 Source Notes

- [dos$_{\text{getdir}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_getdir.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.466 Function Entry: DOS$_{\text{GETFILEM}}$

24.466.1 Name

‘dos$_{\text{getfilem}}$‘

24.466.2 Class

BricsCAD DOSLib Function

24.466.3 Syntax

```
(dos_getfilem title folder filter)
```

24.466.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.466.5 Description

Opens BricsCAD’s FileSelection dialog to let the user select one or multiple file(s). title : is the FileSelect dialog title folder : preset folder to be used for file selection; user can change to any other folder filter : a string containing of file specification tokens - a token has the form ”any description |*.xxx|”, multiple tokens can be combined; to indicate that the tokens sequence is finished, add an extra ”|” as last character. Note : under BricsCAD, a simplified filter can be used - simply combine all file type extensions by comma or semicolon : ”dwg;txt” Returns a list of selected files when successful, or NIL if dialog was cancelled or failed Note 1 : the first entry of the list is the folder name ! Note 2 : the number of files being selected is limited to ~6000 files ! Example : (dos$_{\text{getfilem}}$ ”Select Drawings or Text files” ”C:\” ”Drawing Files (dwg)|*.dwg|Text Files (**.txt)**|.txt||”) will return a list like (”C:\” ”entry.dwg” ”places.dwg” ”readme.txt”)

24.466.6 Return Values

Vendor-defined; see Source Notes.

24.466.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.

- Status: BricsCAD-only.

24.466.8 Source Notes

- [dos$_{getfilem}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_getfilem.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.467 Function Entry: $\mathrm{DOS}_{\mathrm{GUIDGEN}}$

24.467.1 Name

'$\mathrm{dos}_{\mathrm{guidgen}}$'

24.467.2 Class

BricsCAD DOSLib Function

24.467.3 Syntax

`(dos_guidgen)`

24.467.4 Description

Creates a GUID (Globally Unique IDentifier) in string format "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee"; all characters are uppercase. Returns NIL in case of an error. Example : ($\mathrm{dos}_{\mathrm{guidgen}}$) "0D4BBF8C-4621-4168-A7F1-D3C3C8ABBEF9"

24.467.5 Return Values

Vendor-defined; see Source Notes.

24.467.6 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.467.7 Source Notes

- [$\mathrm{dos}_{\mathrm{guidgen}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_guidgen.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.468 Function Entry: DOS$_{\text{HLSTORGB}}$

24.468.1 Name

'dos$_{\text{hlstorgb}}$'

24.468.2 Class

BricsCAD DOSLib Function

24.468.3 Syntax

`(dos_hlstorgb hue luminance saturation)`

24.468.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.468.5 Description

Returns the RGB color value for HLS color value specified by hue luminance saturation integer values.

24.468.6 Return Values

Vendor-defined; see Source Notes.

24.468.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.468.8 Source Notes

- [dos$_{\text{hlstorgb}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_hlstorgb.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.469 Function Entry: $\text{DOS}_{\text{MKDIR}}$

24.469.1 Name

‘$\text{dos}_{\text{mkdir}}$‘

24.469.2 Class

BricsCAD DOSLib Function

24.469.3 Syntax

`(dos_mkdir path)`

24.469.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.469.5 Description

Creates the specified folder path including all sub-folders. path : is the fully qualified name of the folder to be created Returns path if successful, NIL otherwise. Note 1 : on all platforms (Windows, Linux, Mac), both path delimiters "\" and "*" are supported Note 2 : the path can contain a final "\" or ""* path delimiter, but this is not required

24.469.6 Return Values

Vendor-defined; see Source Notes.

24.469.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.469.8 Source Notes

- [$\text{dos}_{\text{mkdir}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_mkdir.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.470 Function Entry: DOS$_{\text{POPUPMENU}}$

24.470.1 Name

‘dos$_{\text{popupmenu}}$‘

24.470.2 Class

BricsCAD DOSLib Function

24.470.3 Syntax

```
(dos_popupmenu menuitems [itemmodes])
```

24.470.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.470.5 Description

Shows a popup menu at actual cursor (mouse) position on screen. The user can dismiss the menu by ESCAPE key or clicking anywhere outside the menu. menuitems : list of strings, specifying the menu item labels; an empty string ”” creates a separator item itemmodes : optional argument; list of integers specifying the item state for each menu item 0 enabled 1 disabled 2 checked, enabled 3 checked, disabled each menu item defaults to state 0 (enabled, not checked) Return : the index of selected item, or NIL on error or escaped menu; Note : separator items do not count for the item indices ! Examples : (setq items ’(”Select” ”” ”First Entity” ”Second Entity”)) (setq modes ’(0 0 3 0)) (dos$_{\text{popupmenu}}$ items modes)

24.470.6 Return Values

Vendor-defined; see Source Notes.

24.470.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.470.8 Source Notes

- [dos$_{\text{popupmenu}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_popupmenu.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.471 Function Entry: DOS$_{\text{RGBTOACI}}$

24.471.1 Name

‘dos$_{\text{rgbtoaci}}$‘

24.471.2 Class

BricsCAD DOSLib Function

24.471.3 Syntax

`(dos_rgbtoaci rgb)`

24.471.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.471.5 Description

(dos$_{\text{rgbtoaci}}$ red green blue) Returns the closest ACI color index for specified RGB color value. These functions are redirected to implementation, for details see vle-rgb2aci.

24.471.6 Return Values

Vendor-defined; see Source Notes.

24.471.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.471.8 Source Notes

- [dos$_{\text{rgbtoaci}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_rgbtoaci.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.472 Function Entry: DOS$_{RGBTOHLS}$

24.472.1 Name

‘dos$_{rgbtohls}$‘

24.472.2 Class

BricsCAD DOSLib Function

24.472.3 Syntax

`(dos_rgbtohls rgb)`

24.472.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.472.5 Description

(dos$_{rgbtohls}$ red green blue) Returns the HLS color value (as list of integers) for RGB color value as specified by rgb or red,green,blue integer values.

24.472.6 Return Values

Vendor-defined; see Source Notes.

24.472.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.472.8 Source Notes

- [dos$_{rgbtohls}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_rgbtohls.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.473 Function Entry: DOS$_{\text{STRTOKENS}}$

24.473.1 Name

‘dos$_{\text{strtokens}}$‘

24.473.2 Class

BricsCAD DOSLib Function

24.473.3 Syntax

```
(dos_strtokens string splitters [NoEmptyParts])
```

24.473.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.473.5 Description

This function splits the given string into a list of tokens, based on specified splitters string; if any optional 3rd argument is given, the result list does not contain empty strings. string : (string) the string to be splitted into a list of tokens splitters : (string) specifies the characters used as delimiters [NoEmptyParts] : if a third argument is given, empty strings are not returned Examples : (dos$_{\text{strtokens}}$ ”Split this string of tokens and more strings; end.” ” ,;” T) (”Split” ”this” ”string” ”of” ”tokens” ”and” ”more” ”strings” ”end.”) (dos$_{\text{strtokens}}$ ”Split this string of tokens and more strings; end.” ” ,;” NIL) (”Split” ”this” ”string” ”of” ”tokens” ”and” ”more” ”strings” ”end.”) (dos$_{\text{strtokens}}$ ”Split this string of tokens and more strings; end.” ” ,;”) (”Split” ”this” ”string” ”of” ”” ”tokens” ”and” ”more” ”strings” ”” ”end.”)

24.473.6 Return Values

Vendor-defined; see Source Notes.

24.473.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.473.8 Source Notes

- [dos$_{\text{strtokens}}$ (BricsCAD V26 DevRef)](https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_strtokens.htm)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.474 Function Entry: DOS$_{\text{STRTRIM}}$

24.474.1 Name

‘dos$_{\text{strtrim}}$‘

24.474.2 Class

BricsCAD DOSLib Function

24.474.3 Syntax

```
(dos_strtrim string [delis])
```

24.474.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.474.5 Description

This function removes all leading and trailing characters if delis from the given string. string : (string) the string to be trimmed for leading and trailing characters delis : (string, optional) contains a set of characters to be removed; if omitted, NewLine, Tab and Space characters are removed

24.474.6 Return Values

Vendor-defined; see Source Notes.

24.474.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.474.8 Source Notes

- [dos$_{\text{strtrim}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_strtrim.htm`)
- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.475 Function Entry: DOS$_{\text{STRTRIMLEFT}}$

24.475.1 Name

‘dos$_{\text{strtrimleft}}$‘

24.475.2 Class

BricsCAD DOSLib Function

24.475.3 Syntax

`(dos_strtrimleft string [delis])`

24.475.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.475.5 Description

This function removes all leading characters if delis from the given string. a string : (string) the string to be trimmed for leading characters delis : (string, optional) contains a set of characters to be removed; if omitted, NewLine, Tab and Space characters are removed

24.475.6 Return Values

Vendor-defined; see Source Notes.

24.475.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.475.8 Source Notes

- [dos$_{\text{strtrimleft}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_strtrimleft.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

24.476 Function Entry: DOS$_{\text{STRTRIMRIGHT}}$

24.476.1 Name

‘dos$_{\text{strtrimright}}$‘

24.476.2 Class

BricsCAD DOSLib Function

24.476.3 Syntax

`(dos_strtrimright string [delis])`

24.476.4 Arguments and Values

Arguments are described in the vendor page; see Source Notes.

24.476.5 Description

This function removes all ltrailing characters if delis from the given string. string : (string) the string to be trimmed for trailing characters delis : (string, optional) contains a set of characters to be removed; if omitted, NewLine, Tab and Space characters are removed

24.476.6 Return Values

Vendor-defined; see Source Notes.

24.476.7 Availability

- AutoCAD 2026: not documented.
- BricsCAD V26: documented.
- Status: BricsCAD-only.

24.476.8 Source Notes

- [dos$_{\text{strtrimright}}$ (BricsCAD V26 DevRef)](`https://developer.bricsys.com/bricscad/help/en_US/V26/DevRef/source/dos_strtrimright.htm`)

- Status: documented (Phase 3 follow-up: body filled from the BricsCAD V26 per-symbol page on 2026-04-26).

25 25 Environment Profiles and Dialects

25.1 Overview

This chapter defines the environment-profile mechanism for ‘clautolisp‘.

25.2 Normative Rules

- CAD target behavior and native operating-system behavior are distinct concerns
- the runtime may expose a Windows-like profile on macOS or Linux
- strict/lax acceptance policy is separate from dialect selection
- environment profiles are selected by configuration or command-line options

25.3 Source Notes

- project architectural policy recorded in ‘AGENTS.md‘
- Status: implementation-defined for ‘clautolisp‘, consistent with project goals.

26 26 System Construction and Delivery

26.1 Overview

The implementation must support SBCL and CCL and must be able to produce a standalone executable.

26.2 Normative Rules

- buildability on SBCL is required
- buildability on CCL is required
- standalone delivery is a project requirement

26.3 Source Notes

- project policy from ‘AGENTS.md‘

27 27 Version Compatibility and Portability

27.1 Overview

Version drift is central to any real AutoLISP specification effort.

27.2 Normative Rules

Important milestones include:

- AutoCAD 2009 ‘initcommandversion‘
- AutoCAD 2012 ‘command-s‘ and ‘*push-error-*‘
- AutoCAD 2021 Unicode/LISPSYS transition
- AutoCAD 2024 LT AutoLISP support
- BricsCAD V12.2 VLE and optimizer
- BricsCAD V18.2 BLADE and namespace support
- BricsCAD V21+ expanding property and vertical APIs

27.3 Implementation Guidance

Every major behavior difference should eventually be tagged by:

- product
- version range
- platform
- strict/lax relevance

27.4 Source Notes

- [What’s New or Changed with AutoLISP](`https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85` `htm`)
- [New and Changed AutoLISP Functions Reference (2015)](`https://help.autodesk.com/cloudhelp/2015/ESP/AutoCAD-AutoLISP/files/GUID-037BF4D4-755E-4A5C-8136-80E85CCEDF3E.htm`)

- [What's New and What's More in BricsCAD Lisp](https://developer.bricsys.com/bricscad/help/en_US/V25/DevRef/source/WhatsNew.htm)

28 28 Glossary

28.1 AutoLISP

The Lisp dialect embedded in AutoCAD and emulated or extended by compatible systems such as BricsCAD.

28.2 Visual LISP

The extension layer that adds ‘vl-*‘, automation, reactor, namespace, and tooling facilities on top of AutoLISP.

28.3 Strict Mode

An acceptance mode that rejects under-specified, weakly portable, or ambiguous syntax and behavior.

28.4 Lax Mode

An acceptance mode that tolerates broader real-world syntax and compatibility-driven behavior.

28.5 Environment Profile

The host-environment model presented to Lisp code, independent of the actual operating system running ‘clautolisp‘.

28.6 Source Notes

- terms defined by this specification draft and project policy

29 Open Work

The following areas remain incomplete and are governed by ‘autolisp-spec/documentation/specification-resolution-plan.org‘:

- full dictionary coverage for all standardized names
- exact reader grammar beyond the core subset
- exact string escape repertoire
- detailed reactor semantics
- detailed packaging/compiled artifact semantics
- exhaustive special-form and function dictionary coverage

The structure above is intended to remain stable while those areas are filled in.